

“NOTHING TOO SERIOUS”:
CORPUS RESOURCES AND METHODS
FOR DATA-DRIVEN APPROACHES
TO POLARITY SENSITIVITY

A Dissertation

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This dissertation introduces the **Polar Bigrams Resource (PBR)** a large-scale corpus-based dataset designed to support data-driven investigations of polarity sensitivity. To address a major challenge for bottom-up processing—the lack of overt indicators of polarity environments—this work employs **polarity approximations**, paired with aggressive post-processing enabled by the statistical power of large-scale data.

A recurring theme of **asymmetry** emerges, seen in the empirical polarity landscape (positive polarity dominates negative 22:1 in PBR) and in the theoretical structure of polarity licensing (polarity-sensitive items require licensing, but not vice versa). These asymmetries motivate the use of **(backwards) asymmetric association** measures to quantify the polarity sensitivity of lexical units.

The **72 million** bigram tokens of PBR are drawn from dependency-parsed corpora, including the novel corpus, **Puddin** derived from a portion of *The Pile* (Gao *et al.* 2020). Contributing roughly 1.4 billion words, Puddin is largely responsible for PBR’s empirical scale. Bigram polarity is approximated by defining positive polarity as either *absence of a negative* or *presence of a positive*. A secondary sampling method balances polarity sizes by down-sampling positive samples. These methods

define four comparison spaces, offering flexibility in analytical design. To support transparency and replication, detailed descriptions of data-processing procedures are provided for both Puddin and PBR.

A detailed **case study of the adverb *exactly*** illustrates the practical application of PBR. It provides a fine-grained analysis of *exactly* across comparison spaces and relevant lexical units—*exactly* alone, bigrams containing *exactly*, and the most relevant adjectives considered independently—and includes PBR-sourced examples to elucidate the quantitative findings. The patterns uncovered suggest a preliminary theory in which pre-adjectival *exactly* is contingently polarity sensitive, depending on the scalar semantics of its adjective argument.

Together, these methodological contributions and findings offer a new empirical foundation for polarity research and demonstrate the power—and limits—of data-driven approaches to evaluating complex semantic and pragmatic phenomena.

BIOGRAPHICAL SKETCH

Andrea R. Hummel grew up in central Missouri before a National Merit scholarship propelled her to her undergraduate career at Texas A&M University in College Station, TX. She graduated *summa cum laude* in 2012, earning a B.S. in Psychology with minors in German and Linguistics. In 2013, after a year working at a bookstore, Andrea joined the Linguistics Ph.D. program at Cornell University in Ithaca, NY.

While completing the requisite theoretical linguistics coursework and qualifying papers, Andrea assisted in the instruction of numerous courses in linguistics, cognitive science, and communications, including several semesters of writing-intensive instruction. As an active member of her department, Andrea also served as President (1 year) of the Linguistics graduate student organization, the Cornell Linguistics Circle (CLC), and also on the CLC Social Committee as both a member (2 years) and Chair (1 year).

Andrea completed her first qualifying paper on the semantics and pragmatics of the polarity sensitive rhetorical device of *litotes* in 2014, under the guidance of her committee chair, Dr. Mats Rooth, and additional members, Dr. Dorit Abusch, Dr. John Bowers, and Dr. John Whitman. Shifting her subfield focus as required by the program, Andrea's second qualifying paper investigated the phonetic convergence of pitch contours in longitudinal speech data of cooperative map completion games collected as part of a graduate seminar taught by Dr. Sam Tilsen. This work, guided by Dr. Tilsen as chair, and Dr. Draga Zec and Dr. Rooth as additional members, precipitated her Cornell Linguistics M.A. and admission to Ph.D. candidacy in 2016.

Over the course of dissertating, Andrea had the opportunity to work with Dr.

Sue Hertz and colleagues at Synfonica LLC¹, a small NIH grant funded company specializing in computer (formant-based) speech synthesis software for emotive modes and child voices to assist speech impaired individuals. This experience sharpened Andrea’s technological acumen, particularly in Python development, data structures, and system management.

Advised by Dr. Rooth (committee chair), Dr. Abusch, and Dr. Sarah Murray (members), Andrea’s dissertation began as an extension of her first qualifying paper on the pragmatic ambiguity of scalar privation as exemplified by *litotes*. However, it soon expanded to encompass large-scale processing of linguistic data, and this endeavor, proving more complex and labor-intensive than anticipated, became the primary focus of her dissertation by necessity.

Her code portfolio can be found on GitHub under the moniker, *AndreaTheLinguist*². Two separate repositories, *puddin*³ and *sanpi*⁴, represent the development and current versions of the resources and methods presented in this dissertation.

¹<https://www.synfonicaspeech.com>

²<https://github.com/AndreaTheLinguist>

³<https://github.com/AndreaTheLinguist/puddin>

⁴<https://github.com/AndreaTheLinguist/sanpi>

To myself.

- To all my past and future selves, but also the ones destined to remain perfectly pristine but incorporeal, preserved in potentiality.
- To the one who never finished her kindergarten crafts, but whose March lion had the most impressively thick mane for the half that wasn't bald.
- To the one who pretended to tie her shoes during a timed quiz because she would rather appear oblivious of time limits than admit she didn't know the answers.
- To the one who was always the last to leave the classroom and missed the bus *home* more than once because she couldn't decide what she needed to take.
- To the one who started pulling all-nighter's in elementary school rather than risk earning anything less than full marks.
- To the one who learned to replace the pride of success with the expectation of it.
- To the one who has probed and perused and pondered and puzzled and percolated and posited so much for so long that she presumed she hadn't at all.
- To the one who has more ideas than she can ever convey outside her own skull, and more love than she knows how to share.
- To the one who finds constructing a canoe and oars by hand the easier means to cross a lake than "simply" swimming unassisted.
- To the one who was trying to ice skate in snow shoes, wondering all the while how everyone else seemed to glide by with ease.
- To the one who is flying instead of falling, fascinated instead of distracted, creative instead of inefficient, unburdened instead of without, liberated instead of lost, and thrilled by *possibility* instead of petrified by uncertainty.
- To the one who will forget things, who will not be able to hold on to every thought or connection indefinitely, but who will also find them again, if she truly needs to.
- To the one who cannot stand in an empty room, because *she* is in the room, who never lacks good company because she is her own best companion.
- To the one who, following fascination, found herself trapped under a mountain she was told she must climb, paralyzed by the enormity of the earth itself pressing down on her.
- To the one who realized she did not have to escape the cavern and scale the surface, but could follow that fascination further inward and upward to emerge at the summit, blinking in surprise at sudden sunlight and blue skies above the clouds.

This was a labor of self-love, even if I did not know it when I set out.
I am as much a product of this work—all of it, not just the glimpse that this document provides—as it is of me.

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Of course, I did need to reach the destination eventually, and setting aside that plethora of potential pursuits was difficult. I am especially grateful to Dr. Dorit Abusch for pointing out that what I had already accomplished was both *impressive* and—more importantly—**enough** for a dissertation, and to Dr. Sarah Murray for reminding me that the shape of my dissertation must reflect the path I have trod, not the one I—or anyone else—had originally envisioned. Without these insights, I might still be wandering the rabbit warren or adding *another* hundred pages to this tome.

I am also profoundly grateful to my mental health support team. I first started working with my ADHD dissertation coach, Laure Conklin-Kampe, in 2021, roughly 5 years after I had “started dissertating,” and I am thankful, in retrospect, that she did not outright laugh in my face when I told her I hoped to finish in a year or so. She has helped me make concrete goals, acknowledge my own process, needs, and limitations, as well as encouraging me to ask for the support I need. Her unfaltering presence along this last half of the journey has been an invaluable anchor and through-line. That I have reached this point is in no small part due to her.

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LIST OF ABBREVIATIONS

wNPI	weak negative polarity item
sNPI	strong negative polarity item
PBR	Polar Bigrams Resource
PSI	polarity sensitive item
PPI	positive polarity item
NPI	negative polarity item
AM	association measure
N	total number of tokens
LRC	Evert 's conservative log ratio (2022)
G2	Dunning 's log-likelihood (1993)
P1	backwards conditional probability
P2	forwards conditional probability
dP1	backwards ΔP
dP2	forwards ΔP
Nyt	New York Times Newswire
Apw	Associated Press Worldstream
News	selected Newswire Corpora, Nyt and Apw
f	joint frequency
f1	marginal frequency of first element
f2	marginal frequency of second element
hit	pattern-matching corpus token
Puddin	The Pile + Universal Dependencies and Decreased Internal Noise
CoCra Puddin	Common Crawl Puddin
BiRC	Bigram Restricted Corpus
ADV	adverb
ADJ	adjective
advmod	adverbial modification
bpmt	bigrams per million tokens

NEQ+	negative equivalent (down)sampling of positive set
ALL+	non-sampled/entire positive set
UD	Universal Dependencies
LDC	Linguistic Data Consortium

CHAPTER 1

INTRODUCTION

1.1 Theoretical Thrust of Polarity Sensitivity

Polarity refers to the entailment relationships between the components within a given linguistic scope. *Negative Polarity* environments invert the truth values produced by *Positive Polarity* environments. In short, polarity encapsulates whether an embedded proposition is true or false. In terms of stative predicates—the focus of this work—negative polarity environments indicate *privation* or “lack of” the indicated state. For example, (1.1), as positive polarity, indicates *lengthy(document)* is true, while its inverted counterpart in (1.2a), as negative polarity, indicates $\neg \textit{lengthy}(\textit{document})$ is true. In this case, *not* is the negative polarity *licensor* or *trigger*, the word or linguistic element responsible for the environment. The scope of a polarity environment depends on the position of the operator—only the portion of the sentence falling under the scope of the operator has negative polarity. The brackets in (1.2b) indicate the scope of the negative polarity environment for a phrasal predicate; only the word *lengthy* has negative polarity in (1.2a).

(1.1) POSITIVE POLARITY:

- a. This document is lengthy.
- b. Some dissertations are shorter.

(1.2) NEGATIVE POLARITY:

- a. This document is **not** lengthy.
- b. This document is **not** [*scope* all that lengthy for a dissertation].

The scope of the positive polarity environment in (1.1) is the entire sentence. This contrast illustrates an important feature of polarity. Positive polarity is the default case, whereas negative polarity is marked. That is, no explicit licenser is required to denote positive polarity—anything that is not negative polarity is necessarily positive polarity.

Polarity sensitivity is the related phenomenon where certain words or phrases are *sensitive* to the polarity of their environment. negative polarity items (NPIs) are lexical items that can only occur within negative polarity environments, and positive polarity items (PPIs) are restricted to positive environments⁵. The sentences in (1.3) illustrate the acceptability of the NPIs *any*, *in months*, and *ever* in and out of negative polarity.

(1.3) NPIs:

- a. ✓ I have **not** [read **any** related documents **in months**].
- b. # I have [read **any** related documents **in months**].
- c. ✓ **Nothing** [was **ever** proven conclusively].
- d. # Something [was **ever** proven conclusively].

⁵PPIs have received less attention in general, but have begun to garner more in recent years (Solt & Wilson (2021); Castroviejo & Gehrke (2022); Staniszweski (2021), a.o.). I do not discuss PPIs in detail.

- e. # Everything [was **ever** proven conclusively].

There are a vast array of environments more complicated than those illustrated in (1.3), as well as considerable variation in the forms that polarity sensitive items (PSIs) take. This variation spans grammatical categories, syntactic configurations, and specific semantic denotations, in both the set of PSIs and the set of environment triggers.

To complicate matters, additional optional elements can extend (e.g. the neg-raising phenomenon; (Gajewski (2007); Collins & Postal (2017); Horn (2020); Staniszweski (2021), a.o.)) or restrict (e.g. interference of additional quantifiers; (Gajewski (2011); Atlas (2021); Milica Denić *et al.* (2018), a.o.)) the scope of a polarity environment. Not all PSIs exhibit uniform distributions—or licensing conditions—across the myriad possible configurations, and it has been proposed that even fully implicit pragmatic enrichment impacts the acceptability of certain PSIs (Gajewski (2011), a.o.)

All of these moving parts, exacerbated by their intrinsic co-dependency, conspire to create a rich field of inquiry which has fueled decades of debate over the various aspects of polarity sensitivity. It is difficult to concretely define “what is or what makes something polarity sensitive” without relying on assumptions of “what polarity environments are” in the first place, and vice versa. Despite this veritable balancing act of defining characteristics, many sound theories have been proposed by theoretical accounts. See Linebarger (1987); Ladusaw (1979); Lahiri (1998); Zwarts (1998);

Israel (2011); Gajewski (2011); Chierchia (2013) for examples of theoretical accounts of polarity sensitivity. Homer (2020) provides a comprehensive overview of the theoretical polarity landscape.

Historically, polarity research has been dominated by theoretical approaches, relying on the in-depth analysis of a limited number of observed cases, bolstered by the inspection of fabricated examples contrived to prove particular theoretical points. While the label “armchair linguist” may be used with a degree of derision, these introspective methods are pivotal in the formulation of working theories to direct future investigation. Such work is vital when the object of inquiry is highly nuanced, making it harder to study empirically. Naturally, the features that make data-driven approaches difficult simultaneously lend themselves toward theoretical approaches.

1.2 Benefits of Data-Driven Approaches

While the bulk of polarity sensitivity research in the field of linguistics has been theoretical—with good reason—these robust theories would benefit greatly from large-scale empirical corroboration and exploration. Data-driven approaches can both (i) provide a necessary means of testing the predictive accuracy of proposed models and (ii) enable bottom-up exploratory work to potentially identify new PSIs and environments not previously considered in the crafting of polarity licensing theories, simply because their skewed distributions have not yet been noticed by

human inspection. In truth, the latter enhances the power of the former, as well as simply providing more elements to bear the essential scrutiny of theoretical linguists. That is, data-driven approaches should not and cannot stand fully apart from theoretical approaches. Rather, they should work in tandem, with each method of study *informing* and *being informed by* the other in turn.

Data-driven investigations are not entirely absent in the polarity literature, especially more recently. See [Jumelet *et al.* \(2021\)](#); [Fritzinger *et al.* \(2010\)](#); [Hoeksema \(2010\)](#); [Milica Denić *et al.* \(2018\)](#); [Bylinina & Tikhonov \(2022\)](#); [Agmon *et al.* \(2019\)](#); [Morales \(2016\)](#) for examples of data-driven advances. However, the breadth is nowhere near that of theoretical work, and many of these empirical works have unrealized potential for theoretical connections. There is a dearth of data-driven polarity research endeavoring to probe linguistic features flagged as relevant by theory, or otherwise bridge the gap between the two spheres.

1.3 Overcoming the Obstacles of Data-Driven Approaches

As previously alluded, polarity sensitivity is an exceedingly complex phenomenon that presents a number of obstacles to data-driven approaches. These must necessarily be addressed for data-driven methods to maximize their potential. In this dissertation, I present novel resources and methods with the goals of clearing these inherent hurdles and facilitating future work on polarity sensitivity informed by both theory and large-scale data.

1.3.1 Obstacles

The primary impediment to taking data-driven approaches to polarity sensitivity is the layered degrees of exponential variability present in both sensitive items and their environments, which are inextricably linked. Variability arises at a fundamental level within the relevant linguistic elements—PSIs and triggers—and is compounded at multiple levels within any empirical pipeline. In addition to the array of starting items and triggers, additional optional elements can interfere, either extending or restricting the polarity scope. Then there is the natural variation in specific realizations of these items by different speakers (alternate spellings, idiomatic usage, etc.). The transcription phase is also subject to variability both at the level of the speaker as errors and at the level of processing and parsing as inconsistency in sentence segmentation, tokenization, lemmatization, and part-of-speech tagging, which all decrease the reliability of automated structural parsing outcomes. These data processing issues can lead to non-uniform and unpredictable representations of the relationships between the already wide array of potentially relevant elements. This degree of variability extends all the way to the extreme of fully implicit elements, in the form of contextual and/or pragmatic enrichment.

These issues are natural features of language use and transformation for scholastic purposes, and affect many other highly nuanced and elusive semantic and pragmatic phenomena. Since the true object of study for semantics and pragmatics is the relationship between what is known, produced, comprehended, and believed—

iteratively—half of the puzzle is inherently inaccessible via direct, explicit avenues. This elusiveness is compounded by the lack of unified annotations for “what matters” (which varies by account), which means that massive amounts of data are needed for patterns to begin to emerge in existing annotation paradigms.

An additional complication for polarity sensitivity is that polarity environments are not always marked, beyond even the potential for implicit marking. The fact that positive polarity is the default case makes direct retrieval of known positive polarity environments difficult.

1.3.2 Contributed Solutions

To combat these hazards of data-driven methods, a lot of data is needed, and this data must also be constrained in some relevant way to make reasonable processing possible. It also requires automated processing paired with human validation and supervision. Heuristic methods must be employed to approximate polarity environments, and significant levels of data cleaning and filtration are required. Finally, meaningful quantitative measures, guided by theoretical expectations, are needed to mitigate the unsurprising representation discrepancies between the default case (positive polarity) and the marked case (negative polarity).

1.4 Research Aims and Scope

The primary aim of this dissertation is to present a set of resources and methods for studying polarity sensitivity in a data-driven manner. The purpose is not to argue for or against any particular theoretical account of polarity sensitivity, but rather to provide a framework for empirical investigation that can be used to test and refine existing theories. Likewise, I make no claim that this is the only way to go about creating the tools needed to study polarity sensitivity quantitatively—even if this work is used only as an example of what *not* to do, I will consider it a success. The goal is merely to share *one attempt* to overcome the impediments to data-driven approaches.

1.4.1 Creation of Corpus Resources

A central contribution of this work is the creation of the Polar Bigrams Resource (PBR), a large-scale corpus-based dataset. PBR is designed to support systematic, data-driven investigations into the logical properties that define polarity sensitivity, thereby offering a new empirical foundation for this area of linguistic inquiry. The newly parsed [CoCra Puddin](#) also contributes to the growing base of structurally annotated resources available to linguists.

1.4.2 Coverage of Methods: Comprehensive Transparency

The resources and methods presented in this dissertation are designed to be as transparent and reproducible as possible. The goal is to provide a clear and comprehensive overview of the data collection, processing, and analysis steps taken, allowing easy replication and extension of the work. This transparency is crucial for building trust in the findings and for facilitating further research in the field.

1.4.3 Overarching Considerations for Data-Driven Approaches to Approaches to Polarity Sensitivity

In the course of creating and exploring these tools, I have identified a number of overarching considerations that are important for any data-driven approach to polarity sensitivity. These include the need for heuristic approximations of polarity sensitivity, the innate representative discrepancy between real-world observations of positive and negative polarity (the preponderance of positive polarity/positive preponderance), and the increased importance of backwards asymmetric association measures for quantifying polarity sensitivity. These considerations are not unique to the resources and methods presented in this dissertation, but reflect the inherent challenges of studying polarity sensitivity in a data-driven manner.

1.5 Summary and Outline

In summary, this work aims to provide both a robust corpus-based resource and transparent documentation of large-scale data processing methods for the study of polarity sensitivity in natural language⁶. These contributions provide a robust empirical foundation to complement traditional theoretical approaches. In the following chapters, detailed methodological procedures, empirical results, and their theoretical implications are discussed further.

Chapter Outline

Chapter 2 Polar Bigrams Resource: design, polarity schema;
overarching considerations; preview of utility.

Chapter 3 Case Study: polarity patterns of the adverb *exactly*.

Chapter 4 Polarity Specific Methodology: polarity designation post-processing,
frequency tabulation, association measure calculation.

Chapter 5 Bigram Collection Methodology: preprocessing, extracting and cleaning
corpus queries.

Chapter 6 Corpora Descriptions and Methods: UD-parsing, BiRC pre-sampling.

Chapter 7 Conclusion and Future Work: summary and future directions.

⁶Specifically, English, though the methods reported can be adapted for corpora representative of other languages.

CHAPTER 2

PBR: THE POLAR BIGRAMS RESOURCE

2.1 Introducing the Polar Bigrams Resource

Polar Bigrams Resource (PBR)

a very large collection of adverb adjective bigram tokens (roughly 72 million) cleaned for enhanced consistency and reliability and categorized by environment polarity for the investigation of polarity sensitivity and related phenomena

Polar Bigrams Resource (PBR) is a collection of *bigrams*, or two linearly contiguous word tokens grounded within the context of a sentence observed in a linguistic corpus. The objects collected in PBR technically exceed this weakest “*two adjacent words*” sense of “bigram”, because they must satisfy additional requirements of category and structure. Although they may more technically be classified as *two-word adverbially modified adjective constituents*, I use *bigram* as a simplification for bigrams which also meet the following requirements.

Categorically, the first word of all PBR bigrams must belong to the adverb (ADV) category, and the second must belong to the adjective (ADJ) category.⁷ Structurally, each PBR bigram must also have an adverbial modification relationship. That is, the adverb modifies the adjective that follows it as encoded by an *adverbial modification*

⁷Both ADV and ADJ tokens are identified by their treebank part-of-speech tag, RB* and JJ* respectively.

(**advmod**) dependency relation in the Universal Dependencies (**UD**) framework. The linear, structural, and categorical requirements of all bigrams in PBR are outlined in (2.1) and represented visually in Figure 2.1 as exemplified by the bigram, *too serious*.

(2.1) Adverbial Modification Relationship of Bigrams in PBR

Each **ADV ADJ** bigram is a two word sequence—two nodes—where...

- the first word, **ADV**, has an adverb part-of-speech tag (**RB***)
- the second word, **ADJ**, has an adjective part-of-speech tag (**JJ***)
- ADV** immediately precedes **ADJ**, with no intervening word or punctuation tokens
- there is a direct **adverbial modification** (**advmod**) dependency relationship between the nodes, which is governed by the adjective node (i.e. **ADJ** is the “head” or “source”, and **ADV** is the “target”)

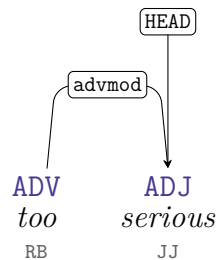


Figure 2.1: Visualization of Requirements for all Bigrams in PBR

Each bigram of PBR has a unique identifier, a *hit_id*, which indicates its corpus source, enabling retrieval of additional information if desired. The source sentence

is also provided in full⁸, as well as a *text window*⁹, which is only the portion of the sentence string centered on the bigram. Finally, each bigram is categorized by **environment polarity**—arguably the most distinctive contribution of PBR.

The motivations and details of the polarity designation schema for PBR are discussed at length in § 2.1.4 below and technical details of specification and collection can be found in Chapter 5. As a brief overview, the polarity classification as represented in PBR is a strict binary division: Any given bigram token is either positive or negative. There are no bigrams with mixed or missing polarity category designations—every bigram is assigned one and only one polarity sign, *positive* or *negative*. Within this binary partitioning, a bigram token may have optional membership in a more restrictive subset, referred to as the *mirror subset* or @*S*. The entirety of polarity classified bigrams in PBR is referred to as the *superset* or **S*. The four polarity designation sets of PBR are shown in Table 2.1 along with processing aliases.

The basis of polarity designation rests on the presence of a direct dependency relationship between the bigram¹⁰ and a polarity operator identified in the preceding

⁸Sentences are stored as both `sent_text`, with typical orthographic spacing conventions, and `token_str`, as space-delimited “tokens”. For example, the contraction *wasn’t* would appear as “wasn’t” in the `sent_text` field, but as “was n’t” in the `token_str` field because contractions are treated as separate tokens. Similarly, punctuation marks are immediately adjacent to the preceding or following word depending on convention in `sent_text`, but surrounded by whitespace in `token_str`.

⁹The text window is encoded in the `text_window` field.

¹⁰Technically, the dependency relationship is with the adjective node of the bigram. Since the

Table 2.1: Polarity Designation Sets of PBR

	Positive	Negative
superset, *S	Positive Superset, not-RBdirect	Negative Superset, RBdirect
subset, @S	Positive (Mirror) Subset, POSmirror	Negative (Mirror) Subset, NEGmirror

context. Negative Polarity classification—also indicated in this text with $(-)$, *neg*, E_{neg} , $\ominus$, and $E_{(-)}$ —is applied when the bigram has a direct dependency with a token linearly preceding it, which acts as a *negative polarity operator*¹¹ (i.e. a word¹² with the ability to induce a negative polarity environment). *Negative polarity operators* are defined as any word token with a lemma in the sets¹³ shown in Table 2.2.

Table 2.2: Designated **Negative** $(-)$ **Polarity** Operator Lemmas

<i>token set</i>	Negative Polarity Operator Lemmas
<i>Superset, *S: *N</i>	{ <i>not, no, nothing, none, nobody, neither, nor, hardly, scarcely, never, rarely, barely, seldom, without, few</i> ¹⁴ , <i>ain't, aint</i> ¹⁵ }
<i>Mirror Subset, @S: @N</i>	{ <i>no, nothing, none, nobody, neither, nor, hardly, scarcely, never, rarely, barely, seldom</i> } (i.e. all the <i>superset</i> trigger lemmas $\notin$ { <i>not, without, few, ain't, aint</i> })

structure of the bigram is fixed with the adjective necessarily governing the adverb, I will refer to this relationship as being with the bigram as a whole.

¹¹Other terms I will use to refer to this object is the *trigger*, *licenser*, or *marker*.

¹²For the present work, only single word units are considered, despite the potential for multi-word polarity operators.

¹³noted absences for this set which were determined later are: *nowhere* and *cannot*. Although *can't* is parsed as *ca n't* where the *n't* form has the lemma *not*, *cannot* is not parsed similarly, and instead receives the lemma *cannot*.

Table 2.3: Designated **Positive (+) Polarity** Operator Lemmas

token set	positive polarity operator lemmas
Superset, $*S$: $*C_N$	$\emptyset$ [unmarked]
Mirror Subset, $@S$: $@P$	$\{somebody, someone, everyone, everybody, something, everything, all, many, some, every, always, sometimes, often, either, both, or\}$

Positive Polarity—also indicated with (+), *pos*, E_{pos} , $\oplus$, $E_{(+)}$ —definitions differ more notably between the superset and mirror subset than those for Negative Polarity. In the superset, positive polarity bigrams are designated by the **absence** of a preceding **negative** trigger and do not necessarily have an explicit *positive* trigger. Conversely, the positive tokens within the mirror subset are defined by the **presence** of a preceding **positive** operator—tokens with a lemma which *mirrors* those lemmas identifying the negative subset tokens only with positive rather than negative quantification (the origin of the name, “*mirror* subset”). *Positive polarity operators* are defined as any word token with a lemma in the sets shown in [Table 2.3](#). As already indicated, [§ 2.1.4](#) covers the particulars of identifying—more accurately, *approximating*—polarity status in greater detail.

[PBR](#) is not organized like a typical linguistic corpus resource: rather than being stored as annotated sentences, documents, dialogues, etc., PBR is stored as individual bigram tokens in a tabular format. In this sense it is more like a hybrid between a linguistic corpus and a pre-processed set of retrieved pattern matches (also referred to as “hits” in this text). This format is essential for the current polarity classifications, since the scope of the indicated environments is not well-defined—the bigrams themselves are categorized for polarity, but the full extent of the polarity

environment is not available in the resource at this time.¹⁶

2.1.1 Superset Overview

Although PBR is not directly comparable to a classic text corpus due to differences in format and usage mechanics, a rough size comparison estimate by means of total bigram tokens places it in the upper ranks of current linguistic corpora. The superset of PBR contains a total of **71,961,373** bigram tokens (N)—separate observations of **2,118,583** unique combinations (i.e., bigram types) of **56,449** unique adverb types and **41,550** unique adjective types. As shown in Table 2.4, the most common adverb overall is *very*, appearing in 9.2 million bigrams. The most common adjective overall is *important*, appearing in 2.0 million bigrams. However, the most common bigram in all of PBR contains neither of these: the bigram *so many* (also represented as *so_many*) is the most frequently observed combination with 1.0 million occurrences.

PBR’s bigram total of 71,961,373 translates to *at least* 143,922,746 word tokens, since each bigram is composed of two words (one adverb and one adjective). It is hard to quantify the data size overall because extensive cleaning and filtering

¹⁶That is, “Jack had **not** been **fully solvent** in weeks.” would be marked as having a negative polarity bigram (“fully solvent”) in PBR but the current information gives no indication that the negative polarity environment extends to encompass “in weeks” as well. PBR could be reverted to a general dependency parsed corpus format with additional annotations pertaining to polarity applied to each bigram within their original context, making them accessible for queries on the sentence parse as a whole, but that remains a project for future work.

Table 2.4: Top Values in PBR Superset ($N = 71,961,373$)

	total unique	top value	with total tokens
adverbs	56,449	<i>very</i>	9,220,012
adjectives	241,550	<i>important</i>	2,001,942
bigrams	2,118,583	<i>so many</i>	1,049,935

are performed after collecting the bigrams, based both on general parameters (as described in [Chapter 5](#)) and polarity-specific concerns (as described in [Chapter 4](#)). However, the initial size of the search area in the creation of PBR was 28.8 billion¹⁷ tokens overall, and 2.7 billion¹⁸ tokens when only sentences containing a valid bigram token were considered (see [Chapter 6](#) for details). The same overall size for the data retained in PBR is unknown, but even without considering the full extent of PBR data origins, the minimum token estimate (ADV and ADJ tokens in bigrams only) of 144 million exceeds the entirety of many corpora.

For comparison, the BNC (British National Corpus) contains 111 million word and punctuation tokens ([Burnard 2007](#)), the TIME magazine corpus contains 106 million words, the SCOTUS corpus (Corpus of US Supreme Court Opinions) contains 129 million words, the Movies corpus contains 199 million words, and the SOAP corpus (Corpus of American Soap Operas) contains 101 million words as of March 2025 ([Davies n.d.](#)). A simple ADV ADJ search¹⁹ of iWeb (The Intelligent Web-based Corpus, hosted by english-corpora.org), which contains 14.1 billion words, yields a total of 68,610,836 tokens of 45,654 unique bigrams. This is nearing the total

¹⁷precisely 28,805,717,900 (word or punctuation) tokens

¹⁸precisely 2,692,793,781

¹⁹without structural modification vetting

number of bigrams collected in PBR (71,961,373). The same simplified ADV ADJ part-of-speech designated bigram search performed on the NOW Corpus (News on the Web; also hosted by english-corpora.org) yields a total of 84,918,669 tokens of 46,817 unique bigrams. While this number does exceed the total bigrams retained in PBR, it does not exceed the *raw* total count of bigrams collected prior to all additional cleaning and filtration applied for PBR’s final form—95,486,374 bigrams overall—and the size of the search area (i.e. NOW corpus) is much larger than any other (external) corpus mentioned with 20.4 billion words as of March 5, 2025, which only falls short of PBR’s origin corpora by 8.4 billion. None of the above corpora contain explicit information identifying polarity environments.

These bigrams are collected from **65,619,972** unique sentences as quantified by sentence IDs, ~ 6.3 million fewer than the bigram total. The source of the difference is two-fold. First, a single sentence may contain more than one distinct bigram token, and second, not all sentences searched have unique strings. More discussion on the latter can be found in [Chapter 5](#), but in short, this is an intentional choice to permit unique sentence IDs (separate instances of them) even if identical strings, so long as they are short enough to have a reasonable expectation of natural production. For example, the sentence “*I’m not **exactly sure***” appears many times, but it is short enough that each instance could plausibly be an organic—albeit far from creative—utterance. It is therefore kept as a valid of natural language exemplar, rather than a carbon copy as might be seen for boiler plate text, reprinted news articles, or quoted excerpts.

2.1.1.1 Lexical Frequency Distributions within PBR

Despite enhanced cleaning procedures—both exclusion and translation—(see [Chapter 5](#) for discussion) applied to the bigrams selected for PBR inclusion, irregular forms do remain. In general, these are either alternate spellings or spelling errors of actual words (speaker “production errors”), or collection or word-tokenization errors (corpus “comprehension errors”). Most of these (and certainly the most egregious) are very rare—many occur only once or twice in the entire dataset—as the more common alternate forms were identified and handled in post-processing. However, it should be noted that their presence in PBR *at all* means they are included in the marginal frequency vectors. This increases the total number of unique values and affects the shape of the distributions by extending the low-frequency “tail”.

Table 2.5: Descriptive Statistics of PBR Superset Frequencies

	adverb counts	adjective counts	bigram counts
<i>mean</i>	1,275	298	34
<i>std</i>	70,099	9,885	1,546
<i>median</i>	1	1	1
<i>75 %</i>	2	3	3
<i>max</i>	9,220,012	2,001,942	1,049,935
<i># unique</i>	56,449	241,550	2,118,583

Correspondingly, the descriptive statistics (displayed in [Table 2.5](#)) are also affected to an extent: the mean, median, etc. values are lower than they would be if all such irregular forms were excluded or accurately grouped with their canonical form, thereby reducing the sparsity of the data by eliminating some of the rarely

observed word types. The median and third quartile or 75th percentile provide a particularly clear indication of the prevalence of low-frequency word types: For all three lexical units considered—ADV, ADJ, and bigrams—at least half (50%) of their respective unique forms appear only once²⁰. An additional quarter of unique adverb types are observed only twice (75th percentile of 2), and an additional quarter of adjective and bigram types appear only thrice (75th percentile of 3). At least three quarters of unique word type representations considered (or word combination types in the case of bigrams) appear no more than 3 times in the 72 million tokens²¹ present in PBR. This relationship is illustrated graphically by the box plot in [Figure 2.2](#), highlighting that the **mean** frequency for each lexical unit’s unique types clearly exceeds the classic definition of “outliers.”²²

That being said, there are still many rarely occurring word types that are not merely irregular or erroneous representations, as is expected for word frequencies per Zipf’s Law, which states that the frequency any given word is inversely proportional to its relative frequency rank ([Zipf 1935, 1949](#); [Piantadosi 2014](#); [Powers 1998](#)). [Equation 2.1](#) provides the original representation of Zipf’s Law, where $f(r)$ is the frequency of the word at rank r and α is a constant which approximates 1 in most

²⁰The the minimum and second quartile/25th percentile are omitted from [Table 2.5](#) because they are also, necessarily, 1, given the median of 1.

²¹The bigram, ADV, and ADJ categories all have the same total of 71,961,373 since each bigrams contains exactly one ADV token and one ADJ token..

²²The classic definition is the sum of the third quartile (2 or 3 in this case) and the inter-quartile range (1 or 2 in this case). This is confirmed by the outlier limit for adverbs (2 + 1) aligning with the third quartile—or the top of the “box”—for adjectives and bigrams (3).

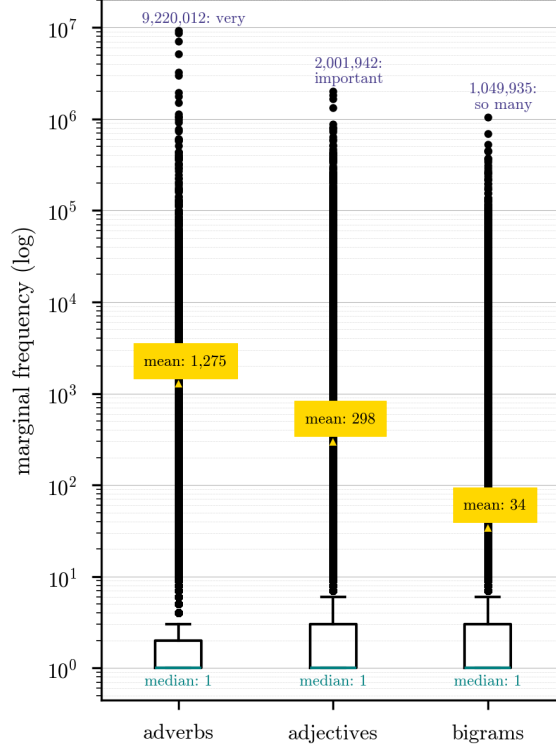


Figure 2.2: Frequency Distribution of Lexical Units in PBR

languages. Equation 2.2 provides a derived generalization, Zipf-Mandelbrot’s Law, where $\alpha \approx 1$ and $\beta \approx 2.7$ (Piantadosi 2014). Figure 2.3 provides an illustration of both equations on logarithmic axes.

$$\text{Zipf : } f(r) \propto \frac{1}{r^\alpha} \approx \frac{1}{r} \quad (2.1)$$

$$\text{Zipf-Mandelbrot : } f(r) \propto \frac{1}{(r + \beta)^\alpha} \approx \frac{1}{(r + 2.7)} \quad (2.2)$$

The plots in Figures 2.4, 2.5, and 2.6 illustrate the relative frequencies across unique lexical type representations for all three lexical units, with a dual logarithmic

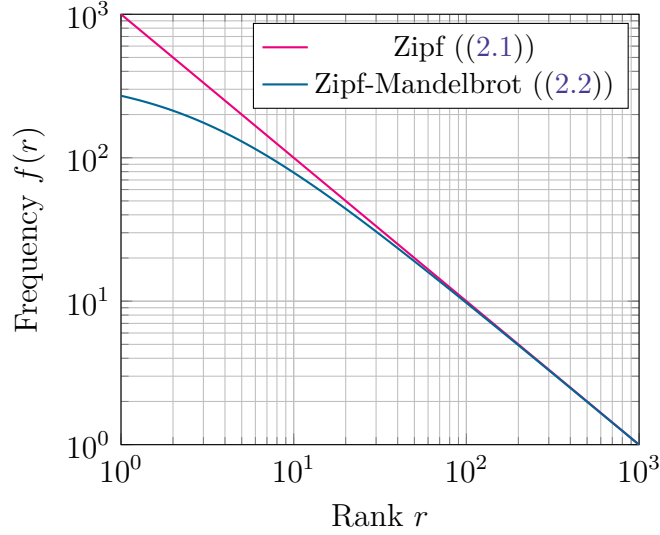


Figure 2.3: Illustration of Zipfian Equations on Logarithmic Axes

frequency-rank plot parallel to [Figure 2.3](#) on the left and a log transformed histogram of square root transformed marginal frequencies on the right. The visual similarity between the frequency-rank plots to the theoretical illustration of Zipf's Law is as expected for natural language frequencies. The flat line at or nearing 1 on the frequency-rank plots in the lowest ranks correspond to the the leftmost bin in the paired histogram, which is the most populated by far for all units.

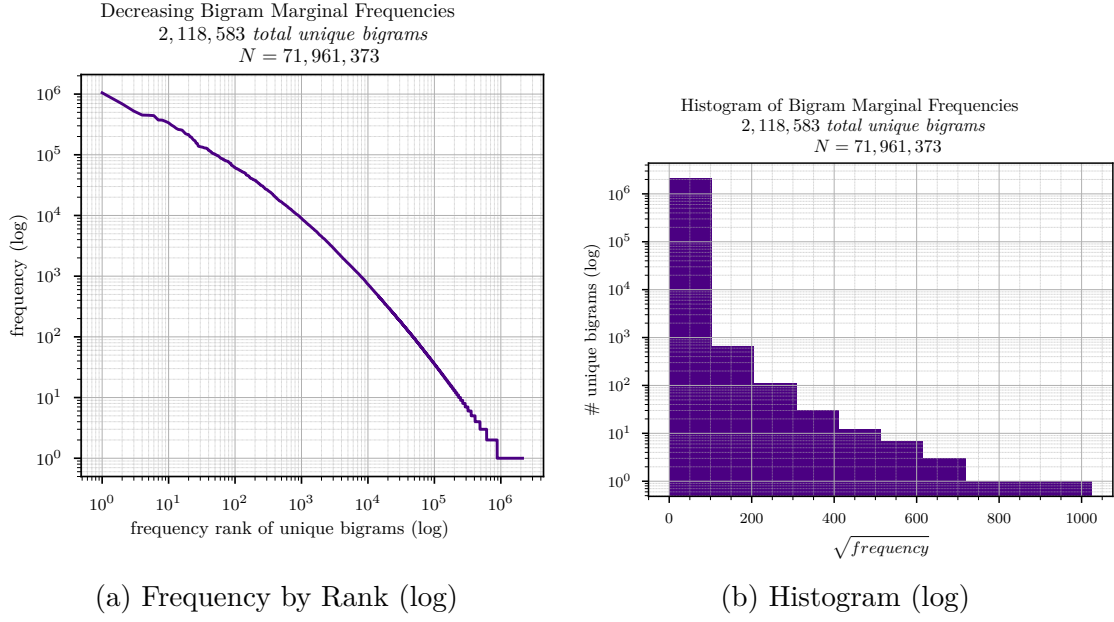


Figure 2.4: Bigram Marginal Frequencies

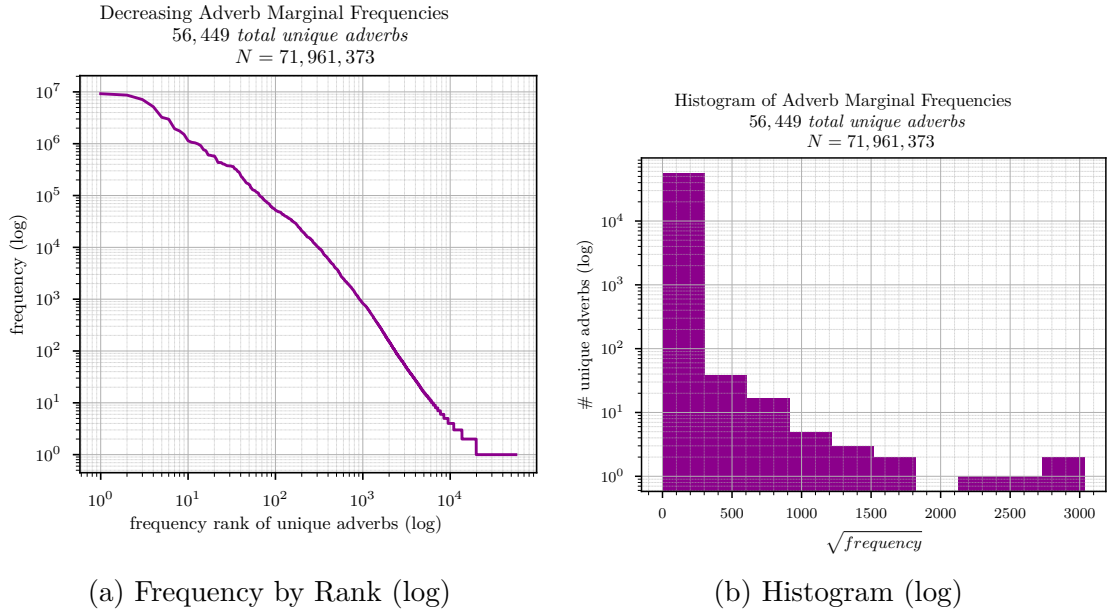


Figure 2.5: Adverb Marginal Frequencies

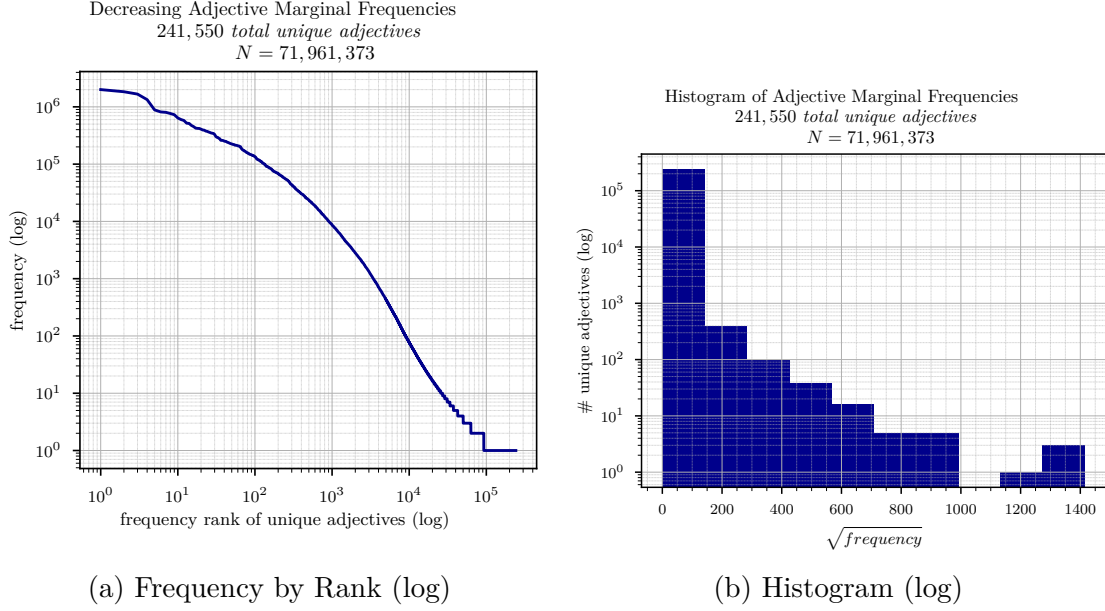


Figure 2.6: Adjective Marginal Frequencies

Table 2.6: Top Lexical Frequencies in PBR

#	Adverb	Frequency	Adjective	Frequency	Bigram	Frequency
1	<i>very</i>	9,220,012	<i>important</i>	2,001,942	<i>so many</i>	1,049,935
2	<i>more</i>	8,659,678	<i>many</i>	1,835,842	<i>most important</i>	690,094
3	<i>most</i>	7,120,428	<i>good</i>	1,670,122	<i>so much</i>	524,372
4	<i>so</i>	5,173,538	<i>much</i>	1,336,122	<i>more likely</i>	451,270
5	<i>as</i>	3,231,640	<i>likely</i>	884,149	<i>very good</i>	447,797
6	<i>too</i>	2,976,740	<i>more</i>	815,328	<i>too much</i>	441,932
7	<i>really</i>	1,945,388	<i>different</i>	802,192	<i>most popular</i>	373,799
8	<i>much</i>	1,760,618	<i>popular</i>	765,023	<i>as many</i>	372,946
9	<i>pretty</i>	1,507,900	<i>difficult</i>	729,385	<i>too many</i>	354,640
10	<i>less</i>	1,138,786	<i>available</i>	639,462	<i>very important</i>	336,610

Table 2.6 provides the top ten frequencies observed for each unit, which correspond to the leftmost values in the frequency-rank plots and the rightmost values in the histograms. One interesting observation about these “most frequent” lexical

units are that *more* and *much* appear in the top set for both adverbs and adjectives, although they only appear in the top bigrams as ADV and ADJ respectively. An illustration of the generally higher marginal frequencies seen for adverbs comes in the fact that *much* actually appears more often as an adverb²³, yet ranks fourth for all adjectives but eighth for all adverbs.

2.1.1.2 Versatility within PBR

Another way to quantify the bigrams contained in PBR is to examine the **versatility** ($\mathbb{V}$) of the adverbs and adjectives of their composition. I use *versatility* to refer to how many unique partners any given bigram component is observed with, both as simple counts and percentages ($\mathbb{V}^{\%}$). Table 2.7 provides descriptive statistics for *trivial versatility*—the number of unique partners²⁴ with *any* tokens at all—and *strict versatility*—the number of unique partners with *five or more* tokens. Note that the total words quantified by each vector is dependent on the versatility threshold type: since strict versatility excludes pairings with fewer than five tokens, all words that only participate in bigrams with frequencies below this floor cannot be quantified. This applies necessarily to all words with a marginal frequency below five, but also to words that, despite having an independent count over five, still appear with no single unique partner word more than five times. Such low frequency lexemes are considered to have an implicit strict versatility of zero. Put another way, *strict versatility* is equivalent to the versatility of only those bigrams with five or more tokens. With

²³*Much* has −424,496 fewer tokens as an adjective

²⁴unique adverb modifiers for adjectives and unique adjective arguments for adverbs

such a frequency restriction in place, the total unique adverbs is reduced by $\sim 92\%$ to **4,307** unique types, and the total unique adjectives is reduced by $\sim 91\%$ to **21,704** unique types.

Table 2.7: Versatility of PBR (superset)

	Presentation	Versatility	mean	std	min	25%	50%	75%	max
ADV	Raw Counts	<u>strict</u>	95.0	477.4	1	1	3	20	12,707
		trivial	37.5	526.4	1	1	1	2	61,082
	% of Potential	<u>strict</u>	0.0%	0.2%	0%	0%	0%	0%	5.3%
		trivial	0.0%	0.2%	0%	0%	0%	0%	25.3%
	<i>Total Unique Adverbs:</i>			<u>strict = 4,307</u>		trivial = 56,449			
ADJ	Raw Counts	<u>strict</u>	18.9	47.3	1	1	3	13	865
		trivial	8.8	52.1	1	1	1	2	3,425
	% of Potential	<u>strict</u>	0.0%	0.1%	0%	0%	0%	0%	1.5%
		trivial	0.0%	0.1%	0%	0%	0%	0%	6.1%
	<i>Total Unique Adjectives:</i>			<u>strict = 21,704</u>		trivial = 241,550			

The effect of different frequency floors is quantified in Table 2.8, where the number heading each column indicates the minimum bigram frequency required for a token to be considered, and the rows indicate the unique types for each lexical unit. Figure 2.7 provides a visualization of the impact observed for each increase of the bigram frequency floor. Note that the impact on total unique types is more extreme for the floor increase from 1 to 2 (+1) than it is for the increase from 5 to 10 (+5), despite the frequency difference being $5\times$ larger in the latter case.

Table 2.8: Unique Values by Minimum Frequency Floor

bigram f floor	1	2	3	4	5	10
bigram	2,118,583	879,409	610,686	484,912	409,231	247,504
adverb	56,449	10,357	6,404	5,041	4,307	2,986
adjective	241,550	53,775	33,096	25,652	21,704	14,161

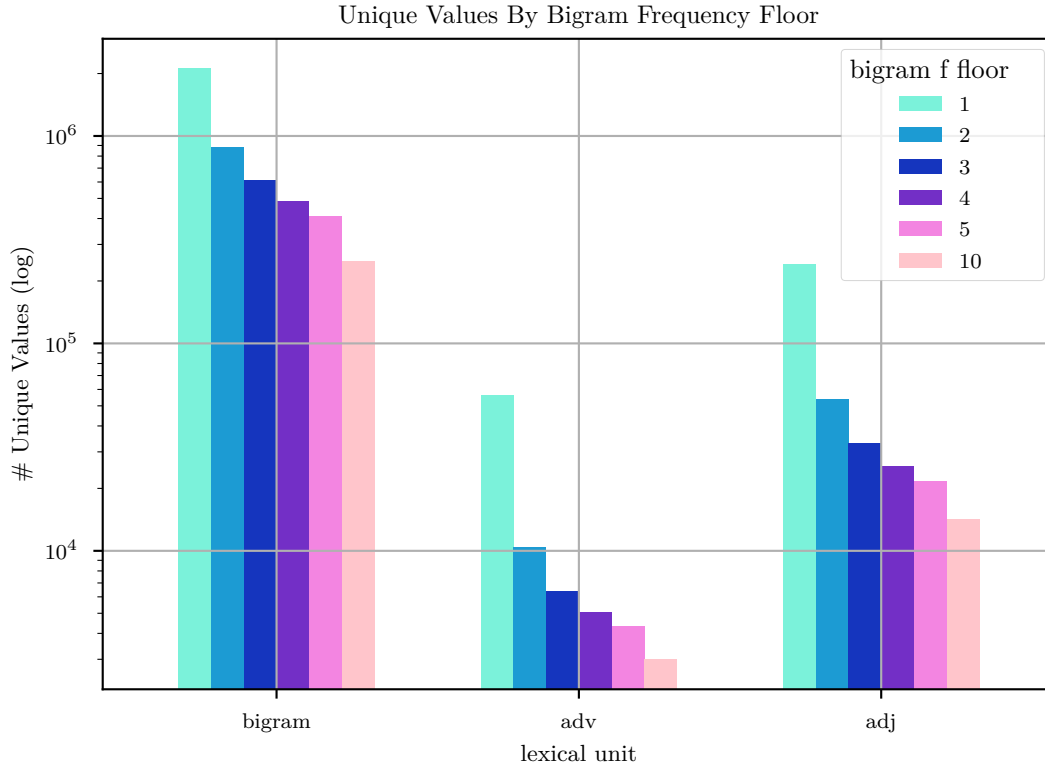


Figure 2.7: Impact of Bigram Frequency Floor on Unique Types

Table 2.9 shows the adverbs with the highest strict versatility—the adverbs that appear with the most distinct adjective types five or more times. Table 2.10 shows the same for adjectives—those adjectives which appear with the most distinct adverb types five or more times. The marginal frequency as well as the percentage of

potential partners²⁵.

Table 2.9: Most Versatile Adverb Forms

Adverb	versatility, $\mathbb{V}$		% of unique ADJ		Marginal Frequency
	strict	trivial	strict	trivial	
<i>more</i>	12,707	61,082	58.5	25.3	8,659,678
<i>most</i>	9,433	39,681	43.5	16.4	7,120,428
<i>very</i>	8,828	38,305	40.7	15.9	9,220,012
<i>so</i>	6,909	24,205	31.8	10.0	5,173,538
<i>as</i>	6,049	18,300	27.9	7.6	3,231,640
<i>less</i>	5,308	18,150	24.5	7.5	1,138,786
<i>too</i>	4,958	15,528	22.8	6.4	2,976,740
<i>also</i>	4,444	16,916	20.5	7.0	1,064,426
<i>pretty</i>	4,443	13,838	20.5	5.7	1,507,900
<i>really</i>	4,287	14,789	19.8	6.1	1,945,388
<i>almost</i>	4,141	15,013	19.1	6.2	433,607
<i>often</i>	4,129	13,386	19.0	5.5	312,584
<i>even</i>	4,125	14,639	19.0	6.1	1,040,332
<i>quite</i>	4,043	10,961	18.6	4.5	988,755
<i>somewhat</i>	3,945	11,077	18.2	4.6	286,573

The most versatile adverb, *more*, is observed with 61.1k different adjectives at least once, 25.3% of all unique adjectives in PBR (241,550). However, under strict versatility, it is only observed with 12.7k distinct adjectives ($f \geq 5$), which is only 5.3% of the unique adjectives found in all bigrams with 5+ tokens (21,704).

The discrepancy between strict and trivial versatility is less severe for the most versatile adjective, *good*, perhaps because the trivial total is lower to start²⁶. The

²⁵The total unique—the potential partners—is dependent on the bigram frequency floor or *strict* status. These values are noted in Table 2.7

²⁶It is worth noting that the most versatile adverbs appear to have an almost functional nature, so it is unsurprising that they have such a wide repertoire. The adjectives on the other hand lack

Table 2.10: Most Versatile Adjective Forms

Adjective	versatility, V		% of unique ADV		Marginal Frequency
	strict	trivial	strict	trivial	
<i>good</i>	865	3,425	20.1	6.1	1,670,122
<i>high</i>	750	2,454	17.4	4.3	518,631
<i>beautiful</i>	701	2,254	16.3	4.0	292,172
<i>different</i>	684	2,081	15.9	3.7	802,192
<i>bad</i>	604	2,043	14.0	3.6	425,797
<i>funny</i>	582	1,814	13.5	3.2	122,146
<i>important</i>	576	1,813	13.4	3.2	2,001,942
<i>low</i>	574	1,701	13.3	3.0	277,389
<i>difficult</i>	567	1,679	13.2	3.0	729,385
<i>close</i>	563	1,653	13.1	2.9	407,539
<i>simple</i>	559	1,618	13.0	2.9	394,988
<i>strong</i>	550	1,668	12.8	3.0	329,848
<i>clear</i>	528	1,606	12.3	2.8	346,404
<i>more</i>	526	2,457	12.2	4.4	815,328
<i>wrong</i>	520	1,547	12.1	2.7	146,437

adjective *good* is modified by 3.4k distinct adverbs at least once, 6.1 % of all unique adverbs in PBR (56,449). With the strict threshold in place, it appears with 865 different adverbs five times or more, which is only 1.5 % of the unique adverbs seen in those bigrams with 5+ tokens (4,307).

Note that for both ADV and ADJ, the most **versatile** word is not the most **frequent** overall: *very* makes more appearances than *more*, and *important* is more common than *good*.

the functional capacity available adverbs and thus the most versatile are simply those which are most semantically prevalent or, perhaps, the most semantically bleached.

2.1.1.3 Comparing Versatility and Frequency

Comparing versatility with marginal frequencies illuminates differences in individual word distributions. The concept is similar to *dispersion* in classic corpus linguistics, a term which refers to the quantification of how widespread certain token types are across an entire corpus (Stefanowitsch 2020). Instead of evaluating the dispersion across documents/portions of the corpus, *versatility* quantifies the “dispersion” of one modifier type across the other. A word with a high marginal frequency but low versatility will necessarily participate in high frequency bigrams, while one with a high marginal frequency but high versatility is not obligated to have *any* corresponding high frequency bigrams. There’s a good chance that it will have some high frequency bigrams, but it is mathematically possible for the marginal frequency to be the sum of many low frequency pairs. Tables 2.11, 2.12, 2.13, and 2.14²⁷ juxtapose adverbs and adjectives with simultaneously similar and divergent values across frequency and versatility.

Similar versatility, but divergent frequency. While the adverbs *even* and *often* (Table 2.11) and the adjectives *important* and *funny* (Table 2.12) are similarly versatile, appearing at least five times with 1.7% of all adjectives and 1.0% of all adverbs respectively, their overall frequencies diverge noticeably. This means the bigram frequencies observed will also necessarily be higher on average for *even* and

²⁷The values in these for versatility juxtaposition tables are calculated from the full superset of PBR, with $N = 71,961,373$.

important, than for *often* and *funny*. While this might be suspected simply from the marginal frequencies, the similar versatility values support it.

Table 2.11: Adverbs *even* and *often*
($N = 71,961,373$)

		<i>even</i>	<i>often</i>
<i>frequency</i>	marginal f	1,040,332	312,584
	% of N	$\sim 1.5\%$	$\sim 0.4\%$
<i>versatility</i>	strict versatility	4,125	4,129
	% of unique ADJ	$\sim 1.7\%$	$\sim 1.7\%$
		1,040,332	312,584
mean bigram f estimate		$\div 4,125$	$\div 4,129$
		~ 252	~ 76

Table 2.12: Adjectives *important* and *funny*
($N = 71,961,373$)

		<i>important</i>	<i>funny</i>
<i>frequency</i>	marginal f	2,001,942	122,146
	% of N	$\sim 2.8\%$	$\sim 0.2\%$
<i>versatility</i>	strict versatility	576	582
	% of unique ADV	$\sim 1.0\%$	$\sim 1.0\%$
		2,001,942	122,146
mean bigram f estimate		$\div 576$	$\div 582$
		$\sim 3,476$	~ 210

Similar frequency, but divergent versatility. On the other hand, the adverbs *far* and *truly* (Table 2.13) and the adjectives *recent* and *successful* (Table 2.14) have similar frequencies (which more apparent via the normalized frequency proportions of PBR than from the raw counts), but differ in versatility. In these cases, word

with lower versatility will have higher bigram frequencies on average, as indicated by the final column, despite having roughly equivalent marginal frequencies. Of course, word frequency data is never expected to have a flat or even normal distribution, so in reality a large percentage of the partners indicated by the versatility count will have relatively low joint frequencies (i.e. much closer to 5 than to the mean bigram frequency estimate), but versatility as a descriptive feature can still inform linguistic theory.

Table 2.13: Adverbs *far* and *truly*
 $N = 71,961,373$

		<i>far</i>	<i>truly</i>
<i>frequency</i>	marginal f	402,768	386,904
	% of N	$\sim 0.6\%$	$\sim 0.5\%$
<i>versatility</i>	strict versatility	794	3,197
	% of unique ADJ	$\sim 0.3\%$	$\sim 1.7\%$
		402,768	386,904
mean bigram f estimate		$\div 794$	$\div 3,197$
		~ 507	~ 121

Table 2.14: Adjectives *recent* and *successful*
 $N = 71,961,373$

		<i>recent</i>	<i>successful</i>
<i>frequency</i>	marginal f	373,138	364,190
	% of N	$\sim 0.5\%$	$\sim 0.5\%$
<i>versatility</i>	strict versatility	73	492
	% of unique ADJ	$\sim 0.1\%$	$\sim 0.9\%$
		373,138	364,190
mean bigram f estimate		$\div 73$	$\div 492$
		$\sim 5,111$	~ 740

2.1.2 Mirror Subset Overview

As illustrated in Figure 2.8, the mirror subset of PBR is much smaller than the superset, accounting for only 2.34% of the total bigrams. Using different terminology, the odds are roughly 41.8 to 1 that any bigram token in PBR is *not* in the mirror subset. However, it is by no means “small”, as it still contains **1,680,633 total bigram tokens**. The fact that this is less than 3% of the whole simply emphasizes the overall size of PBR.

When filtered by polarity status, the ratio of mirror to non-mirror (subset to non-subset) tokens remains nearly identical for positive polarity, but shifts somewhat for negative polarity. The proportion of tokens contained in the mirror subset goes from 2.34% overall to 2.02% of the positive polarity token subtotal (Figure 2.9b), but 9.19% of the negative polarity token subtotal (Figure 2.9a). This jump appears more striking when described by odds rather than proportions: Compared to the

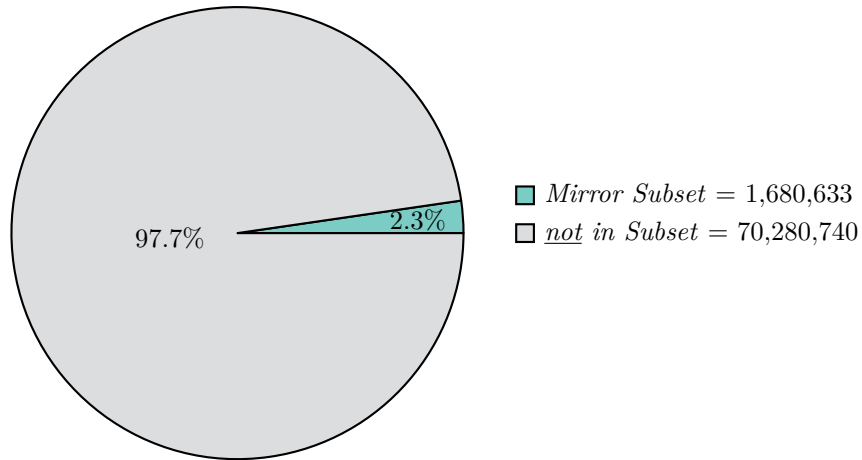


Figure 2.8: Distribution of all bigram tokens in PBR by polarity approximation method

overall odds of approximately 42:1, a positive token has approximately 49:1 odds of *not* being in mirror subset, while the odds are “only” ~ 10 to 1 that a negative token is *not* in mirror subset. Table 2.15 shows the total size of the superset and subset split by polarity and combined, as well as the size of the difference between each set (as the “remainder”). A more detailed discussion of the differences between the $*S$ superset and $@S$ subset can be found in § 2.1.4.

Table 2.15: Token Totals by Evaluation Set and Polarity

	Positive	Negative	Total
superset	68,787,692	3,173,681	71,961,373
<i>mirror subset</i>	1,388,898	291,735	1,680,633
remainder ($\notin$ subset)	67,398,794	2,881,946	70,280,740

The 1.6 million bigrams in the mirror subset are sourced from **1,645,372** unique sentences and span **197,589** unique compositions—different combinations attested

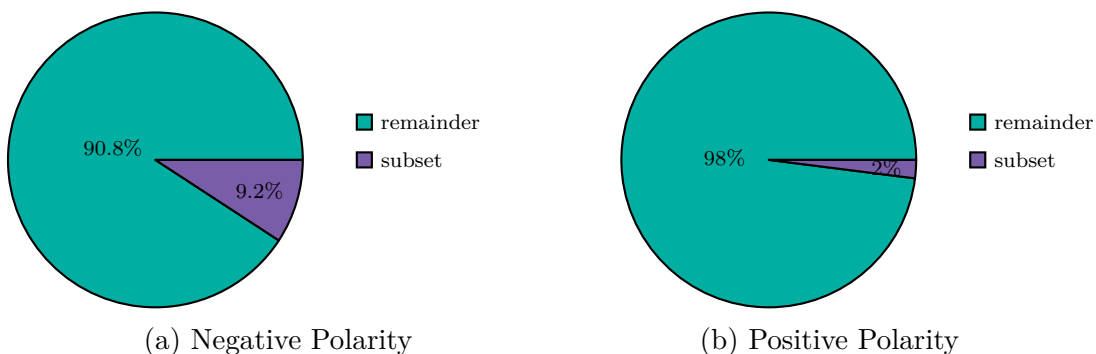


Figure 2.9: PBR Superset–Subset Size Discrepancies by Polarity Approximation

Table 2.16: Top Values in PBR Mirror Subset ($N = 1,680,633$)

	total unique	top value	with total tokens
adverbs	1,680,633	<i>more</i>	285,773
adjectives	1,680,633	<i>important</i>	43,351
bigrams	1,680,633	<i>more important</i>	23,142

from **5,760** unique adverbs and **23,314** unique adjectives. The most frequent adverb is *more* with **285,773** appearances, and the most frequent adjective is *important*, which appears **43,351** times in the subset. Unlike the superset, the most common bigram type does follow the most frequent adverb and adjective: *more important* with a total token count of **23,142**. Simple descriptive statistics for the most frequent form of each category are shown in [Table 2.16](#).

As with the superset, the mirror subset word frequency distribution also clearly follows Zipf’s Law. This can be observed in the quartile values for each of the three lexical units described in [Table 2.17](#). Note however, that the total size of the frequency vector is negatively correlated with upper quartile values: the fewer unique values, the larger the 3rd, and—to a lesser extent—the 2nd quartile (i.e. the 75th

percentile, and the 50th percentile/median). Since there is a 1:1 relationship between tokens and each of the lexical units—each (1) token or *hit* has exactly one adverb, one adjective, and one bigram—a dimension with fewer categories/unique values (e.g. adverb types) will have higher counts in general than a dimension with more possible categories (e.g. bigram types). This is true of the superset distribution as well (adverb: Q3=3 with 56,450 unique vs. adjective, bigram: Q3=2 with 241,556 and 2,118,583 unique respectively), but is more pronounced for the subset. This makes sense though, since the mirror subset has far fewer tokens overall (only 2.34% of the superset), which leads to far fewer unique values *observed* (at least once) for each of the lexical units being counted. If the 0 counts (i.e. non-attested word types) had been included in the marginal frequency vectors, the 3rd quartile values for the subset would be *lower* (than currently shown below, but also lower) than seen for the superset, due to the generally lower frequencies; e.g. *important* is the most frequently occurring adjective in both superset and subset, but only 43,351 of its total 2,001,942 bigram tokens in the superset (< 2.2%) are included in the subset. The superset bigram quartiles would also be lower if all non-attested pairs were included in the calculations above.

Table 2.17: Mirror Subset Frequency Descriptive Statistics

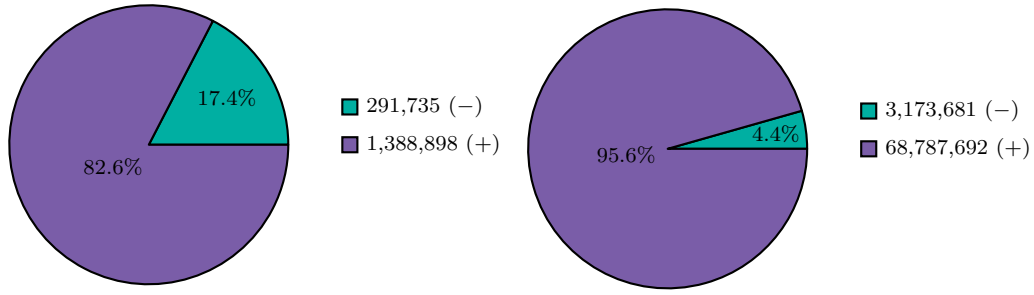
	adverb counts	adjective counts	bigram counts
<i>count</i>	5,760	23,314	197,589
<i>mean</i>	291.8	72.1	8.5
<i>std</i>	5,639.7	686.9	110.7
<i>Q1= 25 %</i>	1	1	1
<i>median</i>	2	1	1
<i>Q3= 75 %</i>	8	6	2
<i>max</i>	285,773	43,351	23,142
<i>unique</i>	5,760	23,314	197,589

2.1.3 Preponderance of Positive Polarity

The fact that the overall subset percentage nearly matches the the percentage for positive polarity alone, despite the observed divergence in the negative polarity subset ratio, suggests a weight disparity between the different polarities. The close resemblance between *all* tokens and the positive polarity tokens alone can be attributed to the distribution of bigram tokens into polarity categories²⁸.

Similar to the large difference seen in subset and superset cardinality, the overwhelming preponderance of bigram tokens in [PBR](#) are observed with positive polarity. An remarkable 95.6 % of all of [PBR](#) bigram tokens are classified as having positive polarity, with only 4.4 % categorized as negative polarity, as illustrated by the pie chart in [Figure 2.10b](#). Thus the odds of any bigram being categorized as positive is roughly 22 to 1.

²⁸in both the superset and subset; however, I first discuss the superset alone



(a) Mirror Subset: Present Positive (b) PBR Superset: Absent Negative

Figure 2.10: Polarity Distribution of Tokens by Polarity Approximation Method

Within the mirror subset alone, the odds of a token being positive are reduced to roughly 5 to 1, as illustrated in Figure 2.10a. The specific implications of the mirror subset selection are discussed in greater detail in § 2.1.4, but the key observation here is that even though 1:5 are *better* odds for negative polarity than 1:22, even in the subset, positive polarity is five times as common as negative polarity. These numerical relationships are summarized in Table 2.18.

Table 2.18: Total Bigram Tokens By Polarity Approximation

Polarity Approximation		*S : Superset	@S : Mirror Subset
Frequency	N	71,961,373	1,680,633
	POSITIVE	68,787,692	1,388,898
	NEGATIVE	3,173,681	291,735
Proportion	POSITIVE	95.59 %	82.64 %
	NEGATIVE	4.41 %	17.36 %
Odds of being POSITIVE		~21.7 :1	~4.8 :1

This observed preponderance of positive polarity, regardless of approximation method (subset membership), is a naturally occurring phenomenon with significant

implications for any approach to the polarity inquisition which relies on corpus co-occurrence frequencies. Given the size of this dataset, it cannot simply be an effect of inadequate sample size—if it were, we would not expect the sizes to be *closer* for the much smaller subset. Additionally, the extensive cleaning (described in [Chapter 5](#)) and post-processing applied to enhance the reliability of polarity classification (described in [Chapter 4](#)), suggest that it is not simply an effect of noise, either. In fact, in a prior version with fewer post-processing checks, of the 83 million bigram tokens retained in this earlier version of PBR (83,284,343 total bigrams), only 3.8% of bigram tokens were negated (3,147,999 total negated tokens), while the remaining 96.2% were not negated (80,136,344). While this previous version did exhibit stronger divergence between the polarity proportions, it is not as much as would be expected if it were purely the result of noise, which is minimized in the current version.

Future research is required to determine whether an elimination of this discrepancy is fully desired or not—although it presents complications for typical corpus evaluation methods that rely on frequency calculations, it is also the reality of polarity usage in natural language. Positive—*unmarked*—utterances **are** simply that much more common than negatively marked ones.

That being said, special consideration is required in analyzing co-occurrence when one element—environment polarity—is a binary choice with wildly disparate marginal frequencies. The basic premise at the root of all corpus frequency analysis is that co-occurrence indicates an underlying *association* or syntactic or

semantic connection between the considered items/features. To accurately interpret co-occurrence, marginal frequencies of the observation types compared must be controlled for, else patterns of association between less common forms/features would be obscured and impossible to detect. This is true even when none of the comparisons categories present a binary choice.

When one comparison category is a binary choice, as in the case of environment polarity and the positive preponderance of the superset provides a 96 to 4 split of the data as a default, “chance” or statistical independence is 96/4 rather than the expected 50/50. Any distribution of tokens adhering closely to the overall marginal frequency ratio does not provide any evidence of divergence from what would be expected by chance. That is, for any element to show a positive association or *attraction* to positive polarity environments, it must have *more* than 96 % of its tokens occurring within positive polarity—even if 96 % of its tokens are positive, this does not deviate from “chance”. On the flip side, for *negative* polarity, any proportion of tokens with negative polarity exceeding even 5 % of its total count begins to demonstrate a positive association. Counterintuitively, this observed preponderance of positive polarity tokens means that any bigram with an *equal* number of positive polarity and negative polarity tokens (i.e., 50 % positive and 50 % negative) would be evaluated as exhibiting a positive association with (attracted to/preference for) *negative* polarity, since these percentages differ from the baseline distribution by $\pm 46\%$. The precise impact of this baseline expectation is discussed in more detail in § 2.2.3 below.

To provide a more comprehensive assessment of polarity patterns, this effect can be ameliorated by examining an additional comparison space where the marginal frequencies or sample size of positive and negative polarity have been equalized. In practice, since the negative polarity marginal frequency is always lower, such a comparison space/perspective will consider the full set of negative tokens, alongside a random selection of positive tokens where the sample size is equivalent to the negative subtotal or implicit sample size. That is, the positive superset has a **negative equivalent (down)sampling of positive set (NEQ+)**, defined in Equation 2.3, with the same number of tokens as the negative superset (3,173,681). Likewise, the positive mirror subset has a (random) NEQ+ sample with the same number of tokens as the negative mirror subset (291,735). In this comparison space, where the polarity category sizes have been controlled for, the benchmark for statistical independence is reset to 50/50, rather than 96/4 (for full the superset) or 83/17 (for the full mirror subset).

$$f(E_{(+)}) = f(E_{(-)}) = \frac{N}{2} \quad (2.3)$$

2.1.4 Polarity Approximation

In addition to the specific information collected for specific linguistic nodes, every bigram token can be analyzed based on which patterns it has matched. These top-level categorizations are then used to approximate the polarity of the bigram token’s environment.

Because neither weak **negative** polarity environments nor positive polarity environments can be succinctly captured by patterns describing positive data (i.e. features overtly and systematically *present* in the context), environment polarity can only be estimated via imperfect approximations.

I evaluated two different approaches to approximate bigram token’s environment polarity, presented below and summarized in [Table 2.19](#) and [Table 2.20](#). These methods are primarily distinguished by whether the non-negative set membership is determined by positive or negative evidence of polarity trigger. This distinction leads to different implications for the resultant categorizations, as well as inherent sample size discrepancies. An in depth discussion of these implications can be found in [Appendix A](#).

The first approximation method, corresponding to the superset of PBR or $*S$ is the **negative evidence** approach. This approach defines the negative polarity environment, $*N$, as having a (strong²⁹) negative polarity trigger, including the lemma **not**, in the context preceding the bigram, and relies on the *absence* of negative triggers as evidence of positive polarity. That is, $*S$ positive polarity originates as the boolean *complement* of the $*N$ set, and is therefore labeled $*C_N$. Its membership consists of cases where a $*N$ negation trigger was *not* found preceding the bigram token. Note that the polarity environment of $*C_N$ is more technically “**not** (strong) **negative** polarity”, and may be either actual positive polarity **or** *weakly negative*—

²⁹This is an approximation. There may be triggers that are not included, and there are definitely dependency configurations that are not included. But this set *attempts* to include all possible strong negative polarity triggers

i.e., weak negative polarity triggers that are not explicitly negative, such as questions, conditionals, comparatives, etc. These known weak negative polarity environments are not included in the pattern specifications due to complexity of identification.

The second approximation method, corresponding to the Mirror Subset or @S is the **positive evidence** approach. It is based on *presence* of positive markers as evidence of positive polarity. Negative polarity in this approximation, @N, is designated by a *specific* (strong) negative polarity trigger in the preceding environment. The greatest distinction, however, is the positive polarity category, @P, which is likewise designated by a *specific* positive polarity trigger preceding the bigram (with a valid dependency relation between them).

Table 2.19: Negative Evidence:
Positive Polarity as the Absence of a Negative Marker

Superset Polarity Approximation	negative	comparison $\approx$ “not (strongly) negated”
Environment Type	*N	* $\mathcal{L}_N$
Subtypes	$\{N_{not}, N_{no}, N_{nobody}, N_{nothing}, \dots\}$	weakly negative or positive
Defining Relation	<i>dependency with strong negative trigger</i>	<i>no such dependency—the COMPLEMENT (inferred results)</i>
Processing Label	RBdirect	not-RBdirect (= RBXadj $\notin$ RBdirect)

Figure 2.11 illustrates the “positive evidence” approach of the Mirror subset, where the positive trigger *somebody* is “present” in relation to the bigram *definitely responsible*. In contrast, Figure 2.12 illustrates again a bigram where negation

Table 2.20: Positive Evidence:
Positive Polarity as the Presence of a Positive Marker

Mirror Subset Approximation	negative	comparison $\approx$ <i>positive</i>
Environment Type	@ <i>N</i>	@ <i>P</i>
Subtypes	$\{N_{no}, N_{nobody}, N_{nothing}, \dots\}$ ³⁰	$P_{\{some, every\}} \times \{\emptyset, body, thing, one, \dots\}$
Defining Relation	<i>dependency with negative "mirror" trigger</i>	<i>dependency with positive "mirror" trigger</i>
Processing Label	NEGmirror	POSmirror

triggers are absent. Both of these sentences would be categorized as positive polarity in the superset, but only Figure 2.11 would be included in the mirror subset.³¹

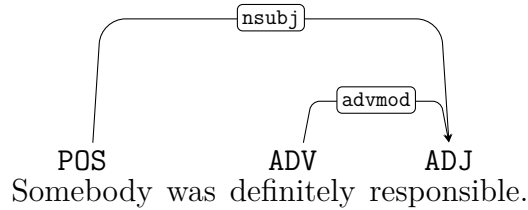


Figure 2.11: Bigram with **positive** trigger, *somebody*, present

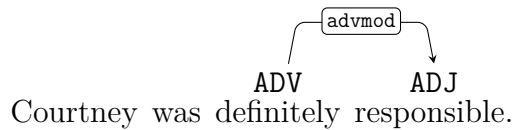


Figure 2.12: Bigram with **negative** trigger absent

The primary difference in polarity sensitivities expected between the approaches, as shown in Table 2.21, is whether the comparison set is hospitable for weak negative

³¹It could of course be argued that *Courtney* also indicates existence, but without assessing the semantics of every potential noun or subject, existential status cannot be assumed.

polarity items (**wNPIs**). When the comparison set is defined by the presence of a positive operator ($@P$), **wNPIs** *should not* be licensed within the positive environment, but when it is defined by the absence of a strong negative operator ($*C_N$), **wNPIs** should have the potential be licensed.³² **wNPIs** should be licensed in both negative polarity approximations, since strong negative polarity item (**sNPI**) environments license both **sNPIs** *and* **wNPIs**.

Table 2.21: Polarity Licensing Expectations By Approximation Method

approach	approximation	defining status	Env, E	<i>wNPI</i>	<i>sNPI</i>	<i>PPI</i>
Superset	negated	$*N$ trigger present	$*N$	✓	✓	⊘
	comparison $\approx$ <i>not negated</i>	$*N$ trigger absent	$*C_N$	✓	⊘	✓
MirrorSubset	negated	$@N$ trigger present	$@N$	✓	✓	⊘
	comparison $\approx$ <i>positive</i>	$@P$ trigger present	$@P$	⊘	⊘	✓

To summarize, superset $*S$ associations can illuminate *strong negative polarity items* (**sNPIs**), while mirror subset $@S$ associations can potentially illuminate *weak negative polarity items* (**wNPIs**) and—in theory—**PPIs**. Positive evidence detected **NPIs** within $@S$ may be either strong or weak, and although the contexts examined are strong, **wNPIs** are more likely to be detected via the $@S$ data than the $*S$ data. **sNPIs** can theoretically be detected by both, but the effect sizes may be weaker in the $@S$ data due to two factors. The first is the inclusion of **wNPIs**—“sharing the

³²This expectation is weaker for the $@P$ set, since comparisons, questions, conditionals, and hypotheticals are not strictly prevented from appearing in the pattern matches. However, this same fact only enhances the likelihood of **wNPI** licensing in the set $*C_N$.

stage” so to speak—but also the potential that **wNPIs** have higher overall frequencies, enhancing their association scores in this approach (i.e. when not licensed by both groups). The second factor is the “narrowing of the gap” between the sample sizes seen in @*S*, which raises the relative bar for detection.

2.1.5 Comparison Spaces as Differing Frequency Perspectives

In summary, PBR contains four separate comparison spaces which represent distinct frequency perspectives and polarity approximations. These correspond to the four possible combinations of the binary features of shown in (2.2) and (2.3).

(2.2) *polarity approximation method* (see § 2.1.4)

Options:

- a. **RBdirect** superset (**S*), where “positive polarity” is designated by the **absence of a negative** trigger (negative evidence basis)
- b. **mirror** subset (@*S*), where “positive polarity” is designated by the **presence of a positive** trigger (positive evidence basis), and “negative polarity” is restricted to those negation triggers which have a clear “mirror” of the selected “positive triggers”³³

(2.3) *positive token selection*

Options:

- a. **non-sampled/entire positive set (ALL+)** , where all available positive tokens for the given approximation method are considered in the association calculations.

³³This essentially just excludes sentential negation, but that amounts to a significant reduction in the total quantity.

- b. **negative equivalent (down)sampling of positive set (NEQ+)** (“negative equivalent”), where the positive token set is limited to a random sample of the same size as the corresponding negative token set.³⁴ This results in a total combined sample that is evenly divided, with 1:1 odds of any bigram token occurring in a positive or negative environment (however that is approximated).

Each of the four possible combinations of these features has a positive set and a negative set, as dictated by the feature values. The four combination cases and final eight (4×2 polarities each) potential polarity environment representations are consolidated in Table 2.22. All quantitative evaluations of polarity association (sensitivity) are performed *within* one of the four “comparison spaces” or “frequency perspectives” indicated by the first two columns, with the calculations made *between* the positive and negative pair therein.

Table 2.22: Overview of the evaluated polarity environment configurations

polarity approx	(+) polarity sampling	size (N)	polarity	env tokens per million	polarity subtotal	processing directory
Superset	ALL+	71,961,373	(+)	955,897	68,787,692	not-RBdirect
			(−)	44,103	3,173,681	RBdirect
	NEQ+	6,347,362	(+)	500,000	3,173,681	not-RBdirect
			(−)	500,000	3,173,681	RBdirect
Subset	ALL+	1,680,633	(+)	826,414	1,388,898	POSmirror
			(−)	173,586	291,735	NEGmirror
	NEQ+	583,470	(+)	500,000	291,735	POSmirror
			(−)	500,000	291,735	NEGmirror

Even though the NEQ+ samples are not truly different datasets, in terms of what they represent, I will treat them as novel *perspectives*, thus if the NEQ+sets are

³⁴Note that the *negative token set remains the same* regardless, but the positive set cardinality affects the quantification of environment association.

considered, there are a total of four data perspectives for examining co-occurrence patterns with polarity in PBR. Due to the observed distribution in each token set, the NEQ+ collection accounts for approximately 8 % of the full superset and approximately 34 % of the full (ALL+) @S subset. The relative proportions of each comparison space in PBR are summarized in [Table 2.23](#).

Table 2.23: Frequency Perspectives Overview

	ALL+ (unbalanced)			NEQ+ (balanced)		
Superset	ALL+ Superset: <i>100% of PBR</i>			NEQ+ Superset: <i>8.82% of PBR</i>		
	N=71,961,373	95.6% (+)	4.4% (-)	N=6,347,362	50% (+)	50% (-)
Mirror	ALL+ Mirror: <i>2.34% of PBR</i>			NEQ+ Mirror: <i>0.81% of PBR</i>		
	N=1,680,633	82.6% (+)	17.4% (-)	N=583,470	50% (+)	50% (-)

2.2 Interpreting the Polar Bigrams Resource

All interpretation and analysis methods will depend on specifics of the active inquiry, but the most readily available path to quantitative evaluation of polarity sensitivity using [PBR](#) is via frequency of co-occurrence with polarity categories. These frequencies—simple counts for different contingencies of the examined categories—can be illustrative on their own. However, as previously mentioned, accurate effect size or statistical significance cannot be discerned from raw frequencies alone. A wide array of association measures ([AMs](#)) have been proposed in the literature ([Pecina 2005](#); [Evert 2017](#); [Uhrig et al. 2018](#)) to quantify effect size and statistical significance of co-occurrence frequencies. A small selection of these are presented with reference to PBR.

§ 2.2.1 provides an overview of the raw frequency observations for the different comparison spaces or perspectives of PBR. § 2.2.3 then introduces the specific association measures that will be used in the following discussion, as well as a brief preview of some top values. A more detailed discussion of particular lexemes and their polar associations as well as potential theoretical underpinnings appears in Chapter 3.

2.2.1 Raw Polar Frequencies

This section is intended foremost as an illustration of the different polarity categories. PBR is of course, too large to cover all joint frequencies here, but the highest joint frequencies for the four polarity designations (summarized in Table 2.1) and lexical forms are given to elucidate the overall format of the dataset and ground the interpretation of AMs. Keep in mind that the *informativity* of the values given in this section is limited—no conclusions can be drawn from raw joint frequencies without considering the overall frequency distribution of other forms. Contextualizing frequencies to make them theoretically interpretable is the purview of AMs (covered in § 2.2.3).

2.2.1.1 Superset Positive Polarity: 69 million bigrams

The preponderance of positive polarity observed in PBR was introduced in § 2.1.3: 69 million of the 72 million total bigram tokens in the superset occur in positive

polarity classified environments. While the preceding discussion focused only on the total token count for each polarity case, this section provides more specific information regarding the distribution, and an extended comparison of frequencies in different spaces. The following discussion pertains to adverbs, adjectives, and bigrams across the 62,719,255 unique sentences with positive polarity bigrams in the superset of PBR.

Table 2.24: Descriptive Statistics for *Positive Superset* ($\ast\mathcal{L}_N$) Marginal Frequencies

$f(\ast\mathcal{L}_N, \text{unit})$	adverb counts	adjective counts	bigram counts
<i>mean</i>	1,241	291	33
<i>std</i>	68,632	9,587	1,535
<i>median</i>	1	1	1
<i>Q3= 75 %</i>	2	3	3
<i>max</i>	9,030,189	1,952,845	1,047,841
<i># unique</i>	55,421	236,230	2,067,878

Table 2.24 presents descriptive statistics for the lexical frequencies observed in the Superset with Positive Polarity classification. Note that the bigram counts indicated are the *attested* bigrams: There are adverb–adjective pairs with no occurrences observed. The total unique bigrams that *could* be found—“could” in a linguistically uninformed sense, that is, because some combinations are prohibited by their intrinsic semantics, and therefore are not truly potential combinations—is over **13 billion**³⁵. The 2.1 million³⁶ forms that are actually observed in the positive polarity category account for only 0.02 % of these potential combinations. However, of the unique

³⁵(236,230 $\times$ 55,421 = 13,092,102,830)

³⁶2,067,878 unique bigrams are observed in (superset) positive polarity at least once.

bigrams observed in general (regardless of polarity), 97.6% occur in a positive polarity environment at least once³⁷.

The frequencies for the adverbs, adjectives, and bigrams that occur most often with positive polarity classification in PBR are given in Table 2.25 and displayed as horizontal bar charts in Figure 2.13. The horizontal axis is on a logarithmic scale and the width and limits are identical across all three charts to facilitate comparison.

Table 2.25: **PBR Superset** Top Positive Polarity ($\ast\mathcal{C}_N$) Adverb, Adjective, & Bigram Frequencies

#	ADV	total (+)	ADJ	total (+)	bigram	total (+)
1	<i>very</i>	9,030,189	<i>important</i>	1,952,845	<i>so many</i>	1,047,841
2	<i>more</i>	8,497,762	<i>many</i>	1,821,912	<i>most important</i>	689,158
3	<i>most</i>	7,101,224	<i>good</i>	1,539,168	<i>so much</i>	502,631
4	<i>so</i>	4,829,392	<i>much</i>	1,287,855	<i>more likely</i>	449,316
5	<i>as</i>	2,699,909	<i>likely</i>	874,621	↑ <i>too much</i>	429,404
6	<i>too</i>	2,667,071	<i>more</i>	806,911	↓ <i>very good</i>	417,705
7	<i>really</i>	1,847,878	<i>different</i>	757,054	<i>most popular</i>	373,189
8	<i>much</i>	1,707,596	<i>popular</i>	747,995	<i>as many</i>	369,742
9	<i>pretty</i>	1,505,944	<i>difficult</i>	700,004	<i>too many</i>	348,592
10	<i>less</i>	1,125,336	<i>available</i>	584,259	<i>very important</i>	334,825

Unsurprisingly, given the preponderance of positive polarity data already discussed at length, the top frequency ranks observed for the positive superset (up to ten) are nearly identical to those observed overall. While the frequencies values themselves differ, the only change in rank observed in Table 2.25 compared to the parallel information provided earlier in Table 2.6 is for the bigrams *too much* and *very good*: *too much* has surpassed *very good* in joint (+) frequency rank.

³⁷per these data, noting that there is a non-zero error rate in categorization

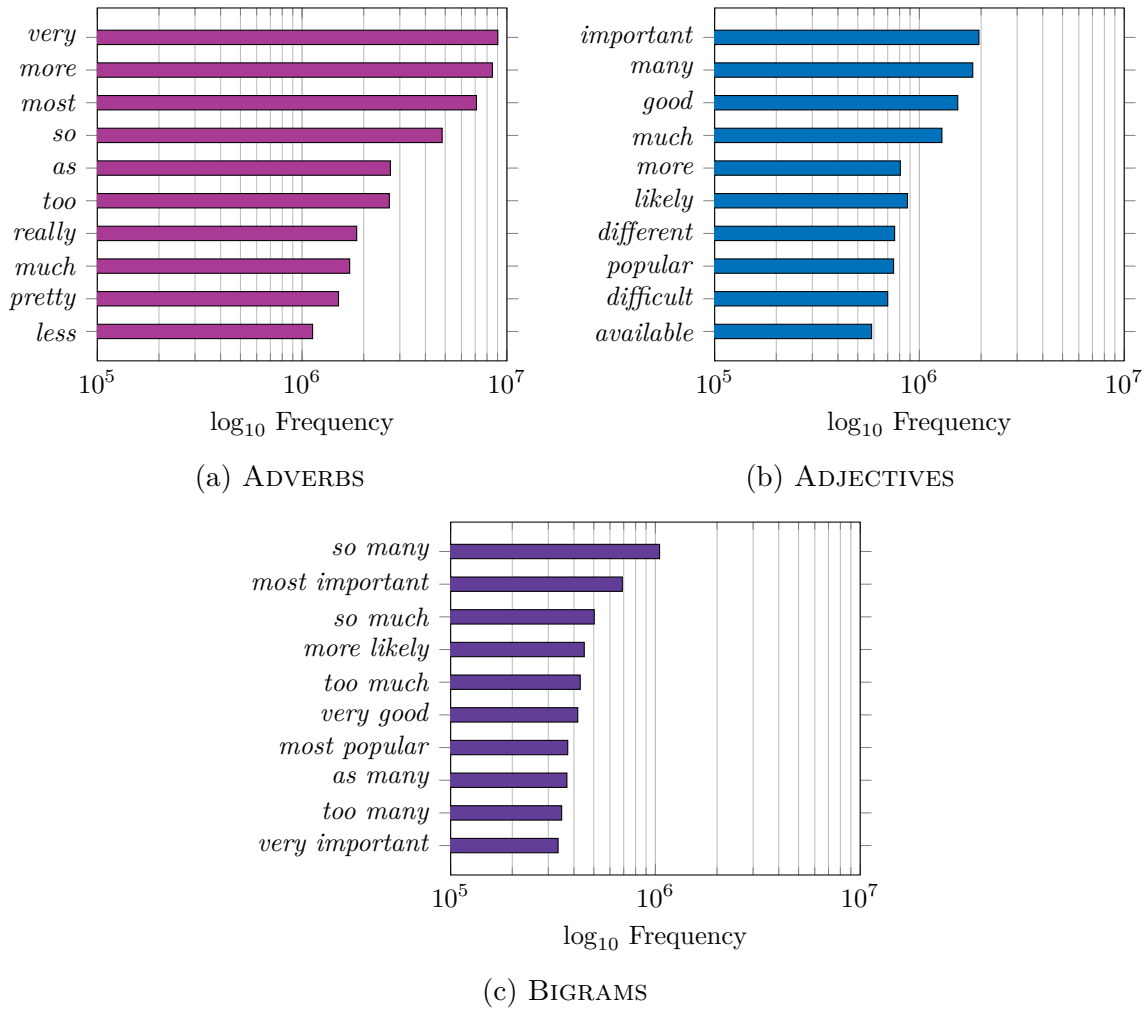


Figure 2.13: Top *Positive Superset* ($*C_N$) Lexical Frequencies

The discrepancy between the joint $*S(+)$ ($= *C_N$) frequencies of *too much* and *very good*—11,699 tokens—is almost twice that of their marginal frequencies—5,865 tokens. This can be attributed to *very good* “losing” more than twice as many tokens: $\frac{30,092}{12,528} = 2.4$. An overly simplistic conclusion would be that *too much* exhibits a preference for positive polarity (or at least a greater preference than *very good* does),

or that *very good* shows a preference for negative polarity. No verdict can be reached however without taking all other contingency factors into account. Realistically, neither seems well motivated, even with the differences observed, since they are still the **fifth** and **sixth** *most common* bigrams observed in the positive superset.

Versatility within the *Positive Superset*. In addition to joint frequencies for specific pairings, it can also be informative to examine the number of unique bigram types are attested for each adverb or adjective. Table 2.26 shows the most versatile adverbs within negative-absent defined positive polarity, and Table 2.27 the most versatile adjectives.

Compared to the versatility rankings³⁸ for the entirety of PBR regardless of polarity (indicated by $\#V(\pm)$ and $V(\pm)$), *almost* and *often* have surpassed *really*, and *somewhat* and *still* have surpassed *quite* (not in top 15 for Positive Polarity). All other versatility rankings remain the same for positive polarity adverb observations. The relative versatility for the positive superset—the percentage of the $V(\pm)$ still present in $V(+)$, denoted by $V\%(+)$ —shows a more nuanced pattern, but these variations do not consistently impact the ordering observed.

Applying the same comparison for adjectives, Table 2.27 shows that *low* has surpassed both *funny* and *important*, *more* has surpassed *clear*, and *large* has surpassed *wrong* (not in top 15 for Positive Polarity). As with the adverbs, $V\%(+)$

³⁸Rankings based on strict versatility counts, or the number of distinct adjectives that appear with each given adverb at least five times (at all or in the positive superset in this case.)

Table 2.26: Most Versatile Adverbs under Superset *Positive Polarity* Compared with Overall Versatility ($\mathbb{V} : * \mathcal{L}_N$ vs. $* S$)

<i>ADV</i>	$\# \mathbb{V}(+)$	$\# \mathbb{V}(\pm)$	$\mathbb{V}(+)$	$\mathbb{V}(\pm)$	$\mathbb{V}^{\%}(+)$
<i>more</i>	1	1	12,632	12,707	99.4
<i>most</i>	2	2	9,425	9,433	99.9
<i>very</i>	3	3	8,723	8,828	98.8
<i>so</i>	4	4	6,640	6,909	96.1
<i>as</i>	5	5	5,595	6,049	92.5
<i>less</i>	6	6	5,274	5,308	99.4
<i>too</i>	7	7	4,740	4,958	95.6
<i>also</i>	8	8	4,435	4,444	99.8
<i>pretty</i>	9	9	4,433	4,443	99.8
<i>almost</i>	10	11	4,136	4,141	99.9
<i>often</i>	11	12	4,120	4,129	99.8
<i>really</i>	12	10	4,020	4,287	93.8
<i>even</i>	13	13	3,968	4,125	96.2
<i>somewhat</i>	14	15	3,942	3,945	99.9
<i>still</i>	15	16	3,936	3,939	99.9

shows more variation in the total adverbs preserved in the $\mathbb{V}(+)$ values, despite these differences not changing the relative versatility rank. For example, the 95.5% retention of *bad* is very nearly surpassed by the 99.7% retention of *low*—which does surpass *funny* and *important*—but the difference of the starting $\mathbb{V}(\pm)$ values (604 and 574 respectively) is enough that the adjective *low* does not rise an additional place.

Table 2.27: Most Versatile Adjectives under Superset *Positive Polarity* Compared with Overall Versatility ($\mathbb{V} : * \mathbb{L}_N$ vs. $* S$)

<i>ADJ</i>	$\mathbb{V}(+)$ Rank	$\mathbb{V}(\pm)$ Rank	$\mathbb{V}(+)$	$\mathbb{V}(\pm)$	$\mathbb{V}^{\%}(+)$
<i>good</i>	1	1	854	865	98.7
<i>high</i>	2	2	742	750	98.9
<i>beautiful</i>	3	3	692	701	98.7
<i>different</i>	4	4	673	684	98.4
<i>bad</i>	5	5	577	604	95.5
<i>low</i>	6	8	572	574	99.7
<i>funny</i>	7	6	570	582	97.9
<i>important</i>	8	7	568	576	98.6
<i>difficult</i>	9	9	558	567	98.4
<i>close</i>	10	10	552	563	98
<i>simple</i>	11	11	546	559	97.7
<i>strong</i>	12	12	539	550	98
<i>more</i>	13	14	524	526	99.6
<i>clear</i>	14	13	512	528	97
<i>large</i>	15	16	511	518	98.6

2.2.1.2 Mirror Subset *Positive Polarity*: 1.4 million bigrams

When restricted to only the positive mirror subset ($@S(+) = @P$)—only 2.02% of positive superset ($*S(+) = * \mathbb{L}_N$) tokens—the total counts are much lower, as discussed in § 2.1.2. Below, Table 2.28 presents descriptive statistics for the joint frequency vectors of each lexical unit with a positive trigger present.

For positive polarity, the actual trigger frequencies are only relevant for the mirror subset, since the superset positive designation is based on absence of negative triggers instead. Each of the 1.4 million bigram tokens in the positive mirror subset occurs with one of the 16 attested trigger lemmas. Table 2.29 presents general frequency

Table 2.28: Descriptive Statistics for *Positive Mirror Subset* (@P) Marginal Frequency Vectors

$f(@P, \text{unit})$	adverbs	adjectives	bigrams
<i>mean</i>	261	65	8
<i>std</i>	4,633	579	90
<i>25%</i>	1	1	1
<i>median</i>	2	1	1
<i>75%</i>	9	6	2
<i>max</i>	207,448	34,359	14,910
<i># unique</i>	5,322	21,483	178,916

distribution information for the trigger lemma versus the trigger’s case-normalized form. While the lemma parameter is constrained to the set dictated by the positive mirror pattern specifications, the case-normalized form is not directly mandated—any form that appears with an approved lemma is included.³⁹ Since bigram token environment identification is based on explicit lemma strings, trigger exploration can default to lemma values more reliably, but all other lexical units (for [ADV](#), [ADJ](#), and bigrams) primarily refer to the case-normalized form (**_lower*).⁴⁰⁴¹

The most common lexical forms appearing with positive-present defined positive polarity are shown in [Table 2.30](#) and illustrated as logarithmic-scaled bar charts in [Figure 2.14](#).⁴² The lexical forms in [Table 2.30](#) are annotated to indicate comparison

³⁹Approved positive trigger lemmas are listed in [Table 2.3](#). For additional details of pattern specifications, see [Chapter 5](#).

⁴⁰For more detail on the distinction between the lemma and form values, and the motivations for using case-normalized forms in most cases, as opposed to lemmas across the board, see [Chapter 5](#).

⁴¹Frequencies for individual trigger forms for the designated @P triggers can be found in [Appendix A](#).

⁴²The limits as well as the width of x-axis in these plots is controlled to facilitate comparison

Table 2.29: Descriptive Statistics for *Positive Mirror Subset* (@P) Trigger Frequencies

@P Trigger	lemma	lowercase form
<i>unique</i>	16	35
<i>mean</i>	87,270	39,895
<i>std</i>	104,293	82,105
<i>min</i>	334	1
<i>25%</i>	22,695	2
<i>50%</i>	44,761	18
<i>75%</i>	88,813	39,017
<i>max</i>	309,632	309,483
<i>CV%</i>	120	206

with the Positive Superset frequency ranks for each unit. Forms which did not appear in the top ten for the superset are preceded by +, those which have risen in rank compared to the superset are preceded by $\uparrow$, and those which have fallen are preceded by $\downarrow$. Bold formatting indicates a rise in rank compared to the positive superset, regardless of whether they appear in the top ten for the positive superset or not. Since the set of top ten lexical units **overall** (by marginal frequency) have the same membership as the top ten set for the positive superset (by joint frequency with positive polarity), these annotations are synonymous with comparison to overall marginal frequency ranks⁴³.

As a result of the additions indicated in Table 2.30, multiple top Positive Superset between unit frequencies.

⁴³The only deviation between the marginal frequency top ranks and the positive superset top ranks appeared in the relative ranking of the bigrams *too much* and *very good*, which exchanged places. Since the relative ordering between the positive mirror subset places both of these lower than in either prior set, the annotations are valid for both comparisons.

Table 2.30: **PBR Mirror Subset** Top *Positive Polarity* Adverb, Adjective, & Bigram Frequencies

#	ADV	total (+)	ADJ	total (+)	bigram	total (+)
1	↑ more	207,448	↑ different	34,359	+ as simple	14,910
2	↓ <i>very</i>	175,004	↓ <i>important</i>	28,594	+ more important	10,941
3	↑ too	111,487	<i>good</i>	22,716	+ completely different	8,340
4	<i>so</i>	104,330	+ simple	22,546	<i>more likely</i>	7,713
5	<i>as</i>	76,202	+ wrong	15,357	+ too easy	7,307
6	+ even	58,341	↑ difficult	13,932	+ too familiar	7,028
7	<i>really</i>	47,487	+ easy	13,640	+ very different	6,204
8	+ quite	34,160	+ special	13,453	↑ very important	6,103
9	+ just	27,770	+ better	12,500	+ even better	6,089
10	↓ <i>pretty</i>	24,525	+ familiar	12,064	↓ <i>very good</i>	5,883

($\ast\mathcal{C}_N$) forms are excluded from the top Positive Subset ($\mathcal{P}$) forms. These include adverbs (*most*, *much*, and *less*), adjectives (*many*, *much*, *more*, *likely*, *popular*, and *available*), and bigrams (*so many*, *most important*⁴⁴, *so much*, *too much*, *most popular*, *as many*, and *too many*⁴⁵).

⁴⁴Strangely enough, the rank occupied by *most important* in the previous orderings (2^{nd}) is now assumed by its semantic neighbor, *more important*.

⁴⁵Although further investigation is required to formulate a well-motivated and informed theory, it is intriguing to note the lowered frequency ranking of the “functional” adverbs—*most*, *much*—and adjectives—*many*, *much*, *more*—from this more restrictive selection’s most common forms. Whether this is connected to polarity sensitivity and scalar semantics or simply results from other selection criteria (e.g., syntactic restrictions) remains a question for future research.

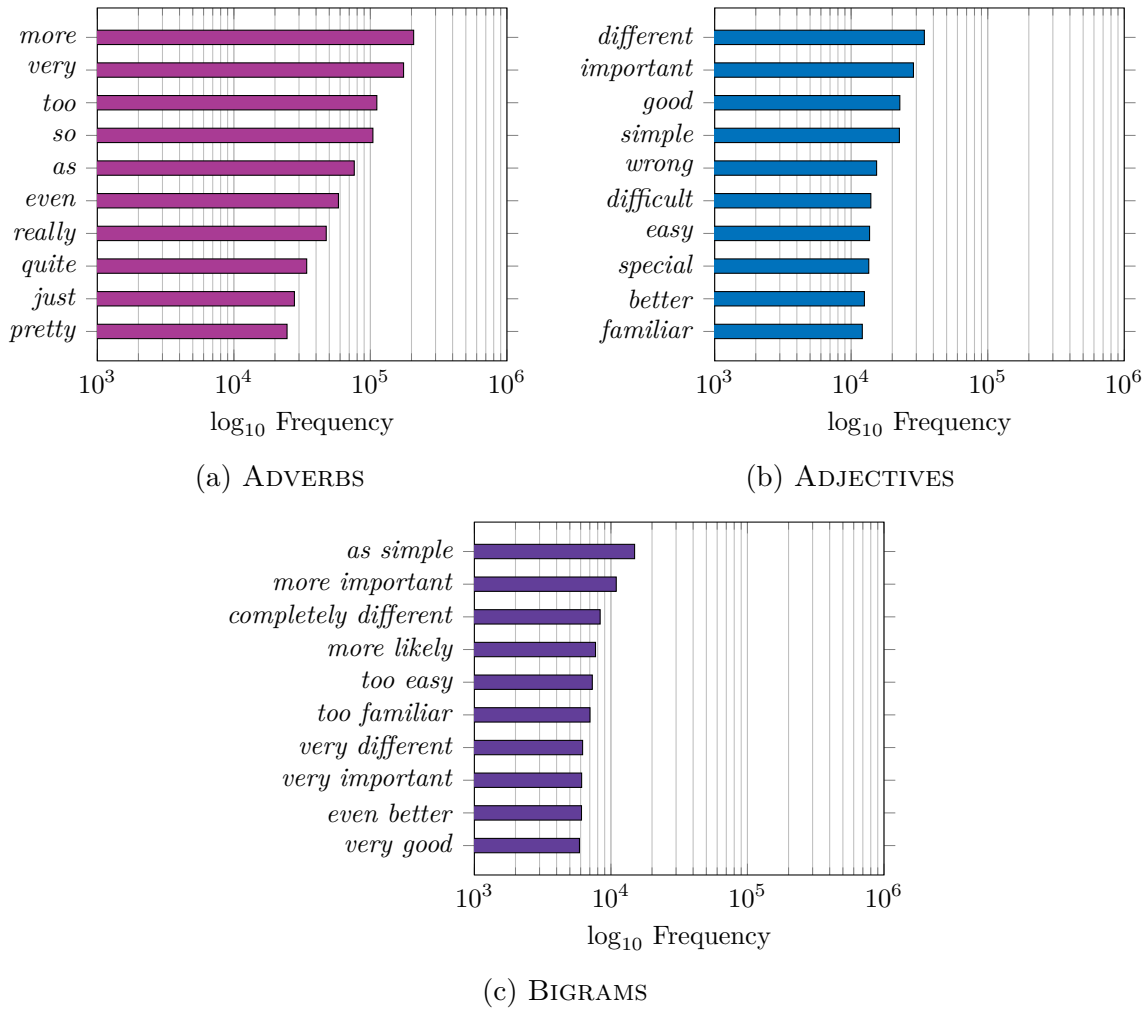


Figure 2.14: Top Positive Mirror Subset Lexical Frequencies

The adjective frequency ranks of the positive mirror subset in Table 2.30 provide an illustrative example of one key consideration in the interpretation of PBR: Since, as previously discussed, ADV and ADJ are both quantified by case-normalized forms rather than lemmas, both the stem or base form *good* and its comparative derivation *better* rank in the ten adjectives most commonly occurring with positive triggers. If

lemmas had been used instead the total joint frequency for the adjective (lemma) *good* would be higher (up to 35,216⁴⁶), and might surpass both *different* and *important*.

This independent representation of morphological comparatives also raises the concept of the comparative forms that are not synthetic and instead depend on the adverb *more*. For example, if all comparatives were uniformly constructed in English, the second most commonly observed bigram (with a positive trigger), *more important*, would be evaluated the same as *better*, and the semantic assertion of *even more important* would be evaluated the same as the ninth place bigram *even better*. Essentially, the 207k tokens observed for the first place adverb, *more*, are the sum of all non-synthetic comparatives. Of course, this highlights a more general concern as well—the irregularity of composition, categorical distinctions, and “word-hood” must all be considered when evaluating “word” frequencies as a means of investigating linguistic patterns.

Trigger Frequencies for Positive Mirror Subset. [Table 2.31](#) ranks the positive trigger lemmas observed in the Mirror Subset by frequency, both raw and as the percentage of all positive triggered tokens (i.e. the positive mirror subset). In addition, token totals for each trigger lemma are split by the governor of the trigger dependency, or whether (the adjective node of) the bigram or the trigger node is the head of the relation. While the majority of tokens are governed by the bigram, there are still a notable portion of tokens headed by the trigger node, especially for

⁴⁶Due to lemmatization inaccuracies, it is not a given that all tokens with the lowercase form *better* have been assigned the lemma *good*. See [Chapter 5](#) for more details.

Table 2.31: *Positive Mirror Subset (@P)* Trigger Lemma Frequencies By Dependency Head

Lemma	Dependency Head		TOTAL	% @P
	BIGRAM	TRIGGER		
<i>something</i>	19,715	289,917	309,632	22.2
<i>or</i>	304,920	51	304,971	21.8
<i>all</i>	242,683	1,279	243,962	17.5
<i>some</i>	105,476	2,158	107,634	7.7
<i>often</i>	82,474	65	82,539	5.9
<i>always</i>	82,213	25	82,238	5.9
<i>sometimes</i>	64,092	58	64,150	4.6
<i>both</i>	49,170	148	49,318	3.5
<i>many</i>	39,151	1,053	40,204	2.9
<i>everything</i>	35,500	2,348	37,848	2.7
<i>everyone</i>	30,705	1,139	31,844	2.3
<i>someone</i>	8,264	19,364	27,628	2.0
<i>everybody</i>	7,646	248	7,894	0.6
<i>either</i>	3,254	401	3,655	0.3
<i>somebody</i>	1,032	1,441	2,473	0.2
<i>every</i>	328	6	334	<0.1
TOTAL	1,076,623	319,701		

the lemmas, *something*, *someone*, and *somebody*, each of which actually has more trigger-headed tokens than bigram-headed.

The total frequency of each positive trigger lemma is displayed as bars on a log scale in Figure 2.15⁴⁷.

⁴⁷The token frequencies by trigger *form* (rather than lemma) are available in Appendix A, Table A.2

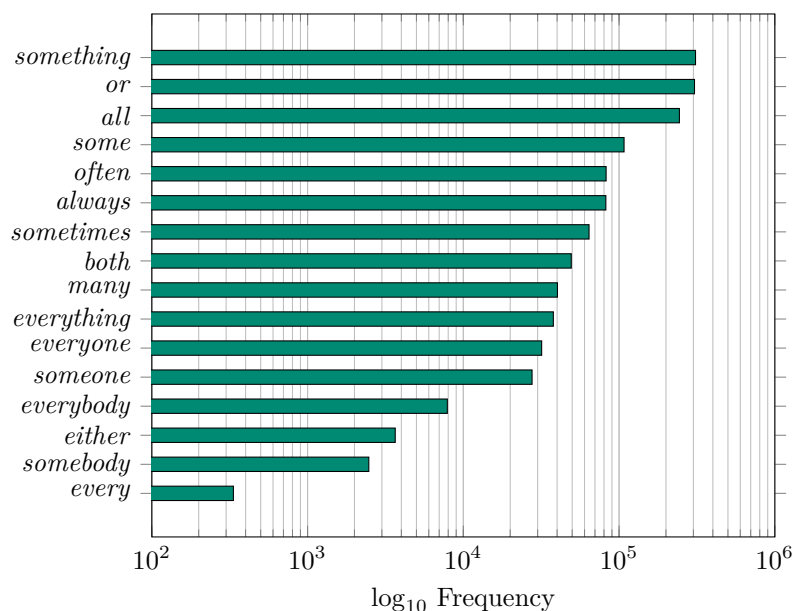


Figure 2.15: Positive Mirror Subset (@P) TRIGGER LEMMAS

2.2.1.3 Negative Superset: 3.2 million bigrams

In the whole of [PBR](#)—the **S* Superset—there are 3,084,940 unique sentence strings containing (**N*) negative polarity trigger tokens with dependencies to 3,173,681 negative polarity bigram tokens.

Table 2.32: Descriptive Statistics for *Negative Superset* (**N*) Marginal Frequencies

<i>*N</i> ⁺	adverb counts	adjective counts	bigram counts
<i>mean</i>	569	113.1	16.9
<i>std</i>	10,864.9	1,773.3	297
<i>median</i>	1	1	1
<i>75%</i>	6	5	3
<i>max</i>	531,731	130,954	44,441
<i>unique</i>	5,578	28,054	187,831

Table 2.33 provides descriptive statistics for the Negative Superset trigger lemmas and forms. As stated in § 2.2.1.2, the triggers are a far more restricted set of values. This means they necessarily have much higher frequencies than any of the other lexical units, but a similar relationship between lower form and lemma can be seen for the negative triggers as for the positive mirror triggers above: even case-normalized forms show much more variation than the explicitly dictated lemmas. The exact breakdown of *N trigger forms per lemma can be found in Appendix A, but for the most part, trigger forms will be ignored, diverging from the treatment of bigram and its components.

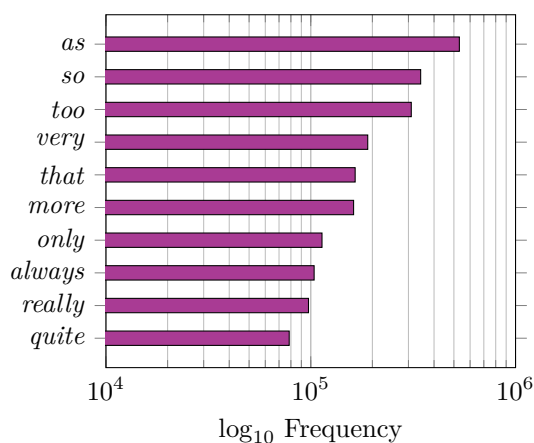
Table 2.33: Descriptive Statistics for *Negative Superset* (*N) Trigger Frequencies

*N ⁺ Trigger	lemma	lowercase form
<i>unique</i>	15	54
<i>mean</i>	212,553	59,043
<i>std</i>	736,488	294,681
<i>min</i>	796	1
<i>25%</i>	4,382	1
<i>50%</i>	6,528	13
<i>75%</i>	23,689	1,847
<i>max</i>	2,871,807	2,005,270
<i>CV%</i>	346	499

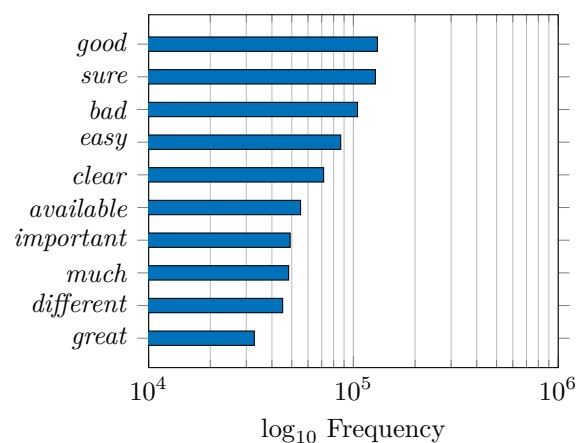
Table 2.34 show the lexical forms with the top ten joint frequencies with *N⁺ triggers, and Figure 2.16 provides bar charts to facilitate visual comparison. As with Table 2.30, annotations indicate how each form compares to its positive superset joint frequency rank (see Table 2.25).

Table 2.34: **PBR Superset** Top *Negative Polarity* Adverb, Adjective, & Bigram Frequencies

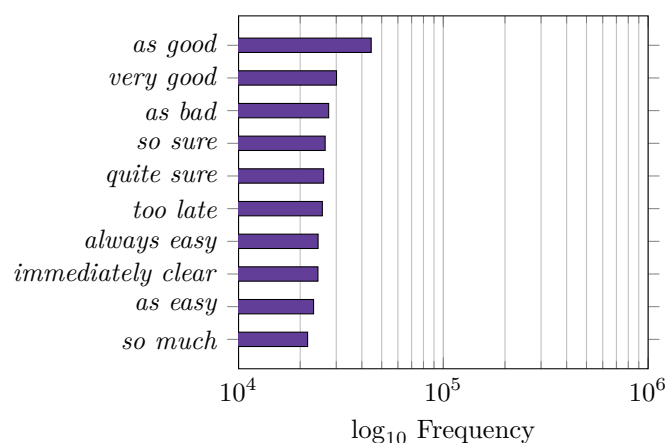
#	ADV	total (-)	ADJ	total (-)	bigram	total (-)
1	↑ as	531,731	↑ good	130,954	+ as good	44,441
2	↑ so	344,146	+ sure	128,062	↑ very good	30,092
3	↑ too	309,669	+ bad	104,709	+ as bad	27,548
4	↓ very	189,823	+ easy	86,698	+ so sure	26,480
5	+ that	164,768	+ clear	71,543	+ quite sure	26,026
6	↓ more	161,916	↑ available	55,203	+ too late	25,656
7	+ only	113,502	↓ important	49,097	+ always easy	24,459
8	+ always	103,883	↓ much	48,267	+ immediately clear	24,416
9	↓ really	97,510	↓ different	45,138	+ as easy	23,250
10	+ quite	78,500	+ great	32,815	↓ so much	21,741



(a) ADVERBS



(b) ADJECTIVES



(c) BIGRAMS

Figure 2.16: Top Negative *S Superset Lexical Frequencies

Where the positive mirror top ranks exchanged three adverbs, six adjectives, and seven bigrams, the negative superset exchanged four adverbs, five adjectives, and eight bigrams. It is worth noting that there is a decent amount of overlap in the positive superset/overall top frequency ranked forms that fall from the top ranks in the positive mirror subset and the negative superset. The adverbs are the same (*most*, *much*, *less*) plus the additional exclusion of *pretty*. Four of the adjectives excluded are the same (*many*, *likely*, *more*, *popular*), with an additional exclusion of *difficult*, but the retention of *much* (descended to eighth) and *available* (ascended to sixth). Six of the same bigrams have fallen out of the top ten (*so many*, *most important*, *too much*, *most popular*, *as many*, *too many*), with the additional exclusions of *more likely* and *very important*, but the retention of *so much* (descended to tenth).

The lexical forms which have risen to the upper *N joint frequency ranks show more deviation from the @P additions. Only the adverb *quite* and the adjective *easy* overlap in the newly risen ranks. While *even* and *just* rose into the @P adverb joint frequency ranks, *that*, *only*, and *always* have risen in the *N adverb ranks. Rather than the adjectives *simple*, *wrong*, *special*, *better*, *familiar* which ascended into the top values for positive mirror subset frequencies, *sure*, *bad*, *clear*, and *great* have joined the negative superset top ten most common adjectives. Finally, the bigrams which joined the top ten for the positive mirror subset were *as simple*, *more important*, *completely different*, *too easy*, *too familiar*, *very different*, and *even better*, but the new additions for the negative superset are *as good*, *as bad*, *so sure*, *quite sure*, *too late*, *always easy*, *immediately clear*, and *as easy*.

Trigger Frequencies for Negative Superset, *N. Table 2.35 splits the frequencies observed for each trigger lemma by the governance of its dependency with the bigram. Bigram governed trigger dependencies are much more common for all lemmas except *nothing*, which has nearly twice as many trigger-headed tokens as bigram-headed, even exceeding the total tokens for the dominant negative trigger lemma, *not*, in the trigger-headed set. The totals for each trigger lemma are visualized as the log-scaled bar chart in Figure 2.17.

Table 2.35: *Negative Superset (*N)* Trigger Lemma Frequencies By Dependency Head

Lemma	Dependency Head		TOTAL	% *N ⁺
	BIGRAM	TRIGGER		
<i>not</i>	2,863,597	8,210	2,871,807	90.1
<i>never</i>	111,026	63	111,089	3.5
<i>nothing</i>	34,067	66,446	100,513	3.2
<i>none</i>	21,502	9,951	31,453	1.0
<i>nor</i>	15,925	0	15,925	0.5
<i>without</i>	13,339	1,486	14,825	0.5
<i>no</i>	12,343	95	12,438	0.4
<i>neither</i>	6,436	92	6,528	0.2
<i>nobody</i>	5,827	538	6,365	0.2
<i>hardly</i>	5,517	8	5,525	0.2
<i>rarely</i>	4,464	10	4,474	0.1
<i>few</i>	4,136	153	4,289	0.1
<i>barely</i>	1,231	0	1,231	<0.1
<i>seldom</i>	1,034	5	1,039	<0.1
<i>scarcely</i>	795	1	796	<0.1
TOTAL	3,101,239	87,058		

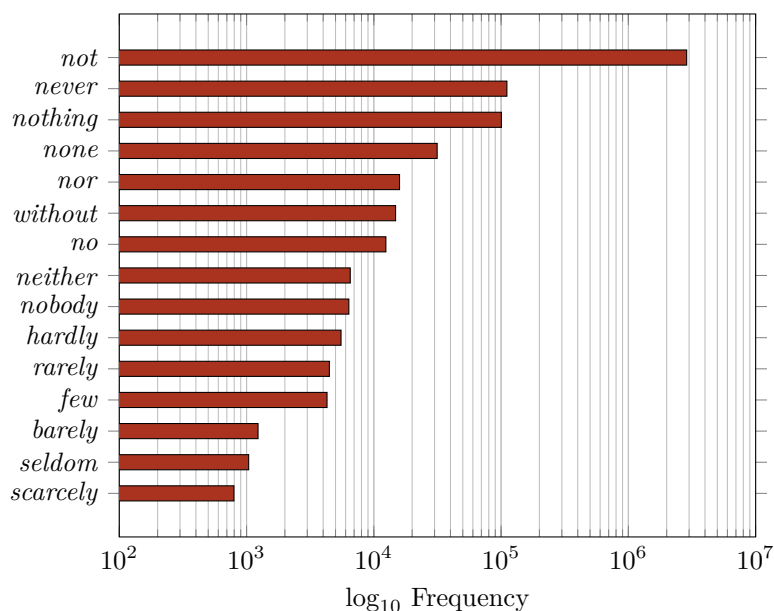


Figure 2.17: Negative Superset TRIGGER LEMMAS

Versatility within the Negative Supeset, *N. The versatility of adverbs and adjectives appearing in negative polarity environments is shown in Table 2.36 and Table 2.37, respectively. Annotations similar to the preceding tables have been applied to indicate comparison with overall (strict) versatility scores.

Of particular note, Table 2.36 shows adverbs which did not rank in even the fifty most versatile adverbs overall, but which have risen into the top fifteen when only negative classified tokens are considered: *necessarily*, *exactly*, and *that*. This suggests that the broad applicability of these adverbs is not impeded by negative polarity to the same extent as many other adverbs. More extensive discussion of these differences will follow in § 2.2.2.

Table 2.37 shows the most versatile adjectives with *N operators. While similar

Table 2.36: Most Versatile Adverbs under Superset *Negative Polarity* Compared with Overall Versatility ($\mathbb{V} : *N$ vs. $*S$)

<i>ADV</i>	$\mathbb{V}(-)$ Rank	$\mathbb{V}(\pm)$ Rank	$\mathbb{V}(-)$	$\mathbb{V}(\pm)$	$\mathbb{V}^{\%}(-)$
<i>as</i>	1	5	2,989	6,049	49.4
<i>so</i>	2	4	2,413	6,909	34.9
<i>only</i>	3	23	1,922	3,186	60.3
<i>more</i>	4	1	1,711	12,707	13.5
<i>too</i>	5	7	1,634	4,958	33
<i>just</i>	6	19	1,366	3,542	38.6
<i>very</i>	7	3	1,352	8,828	15.3
<i>really</i>	8	10	1,064	4,287	24.8
<i>particularly</i>	9	26	945	3,033	31.2
<i>necessarily</i>	10	82	921	1,128	81.6
<i>exactly</i>	11	101	848	920	92.2
<i>always</i>	12	30	835	2,959	28.2
<i>that</i>	13	93	822	1,002	82
<i>entirely</i>	14	43	683	2,218	30.8
<i>completely</i>	15	17	660	3,852	17.1

rank shuffling occurs, none make so drastic a leap as seen in the adverb ranks, especially that of *exactly*.

Table 2.37: Most Versatile Adjectives under Superset *Negative Polarity* Compared with Overall Versatility ($\mathbb{V} : *N$ vs. $*S$)

<i>ADJ</i>	$\mathbb{V}(-)$ Rank	$\mathbb{V}(\pm)$ Rank	$\mathbb{V}(-)$	$\mathbb{V}(\pm)$	$\mathbb{V}^{\%}(-)$
<i>good</i>	1	1	120	865	13.9
<i>available</i>	2	36	118	455	25.9
<i>different</i>	3	4	117	684	17.1
<i>bad</i>	4	5	111	604	18.4
<i>wrong</i>	5	15	110	520	21.2
<i>enough</i>	6	56.5	100	386	25.9
<i>interested</i>	7	60	99	380	26.1
<i>aware</i>	8	27.5	98	479	20.5
<i>able</i>	9	65.5	94	376	25
<i>possible</i>	10	58	94	384	24.5
<i>true</i>	11	31	93	461	20.2
<i>related</i>	12	32	92	460	20
<i>clear</i>	13	13	92	528	17.4
<i>better</i>	14	17	89	514	17.3
<i>concerned</i>	15	84	88	341	25.8

2.2.1.4 *Negative Mirror*: 0.29 million bigrams

As is the case for Positive Polarity within the Mirror Subset ($@P$; discussed in § 2.2.1.2), the Negative Polarity Mirror Subset ($@N$) cardinality of 291,735 is much smaller than that of the Negative Superset⁴⁸. However, the proportion of bigrams with negative polarity triggers is much higher, at 9.2% of the negative bigram tokens in the superset.

Descriptive statistics for lexical unit frequency vectors are shown in Table 2.38 and for $@N$ trigger marginal frequencies in Table 2.39.

⁴⁸There are 3,173,681 bigram tokens in $*N$

Table 2.38: Descriptive Statistics for Negative Mirror Subset Marginal Frequencies

	adverb counts	adjective counts	bigram counts
<i>mean</i>	144.9	36.2	7.1
<i>std</i>	2,266.5	295.9	97.2
<i>25%</i>	1	1	1
<i>median</i>	2	2	1
<i>75%</i>	6	8	2
<i>max</i>	78,325	14,757	12,201

Table 2.39: Descriptive Statistics for Negative Mirror Subset Trigger Frequencies

	@N Trigger	lemma	lowercase form
<i>unique</i>		12	20
<i>mean</i>		24,455	14,673
<i>std</i>		38,879	32,027
<i>min</i>		797	1
<i>25%</i>		3,671	5
<i>50%</i>		6,426	1,144
<i>75%</i>		19,770	7,162
<i>max</i>		111,102	111,102
<i>CV%</i>		159	218

Trigger Frequencies for Negative Mirror Subset. The marginal frequencies Negative Mirror subset trigger lemmas are shown in [Table 2.40](#), and [Figure 2.19](#) illustrates the relative proportion of @N bigrams each trigger is responsible for, while [Figure 2.18](#) depicts the total trigger marginal frequencies on a logarithmic scale.

Table 2.40: Mirror Negative Trigger Lemma Frequencies By Dependency Head

Lemma	Dependency Head		TOTAL	% @N
	BIGRAM	TRIGGER		
<i>never</i>	111,039	63	111,102	37.9
<i>nothing</i>	33,760	66,284	100,044	34.1
<i>none</i>	21,319	9,929	31,248	10.6
<i>nor</i>	15,944	0	15,944	5.4
<i>no</i>	9,081	94	9,175	3.1
<i>neither</i>	6,400	91	6,491	2.2
<i>nobody</i>	5,823	538	6,361	2.2
<i>hardly</i>	5,519	8	5,527	1.9
<i>rarely</i>	4,470	10	4,480	1.5
<i>barely</i>	1,245	0	1,245	0.4
<i>seldom</i>	1,037	5	1,042	0.4
<i>scarcely</i>	796	1	797	0.3
TOTAL	216,433	77,023		

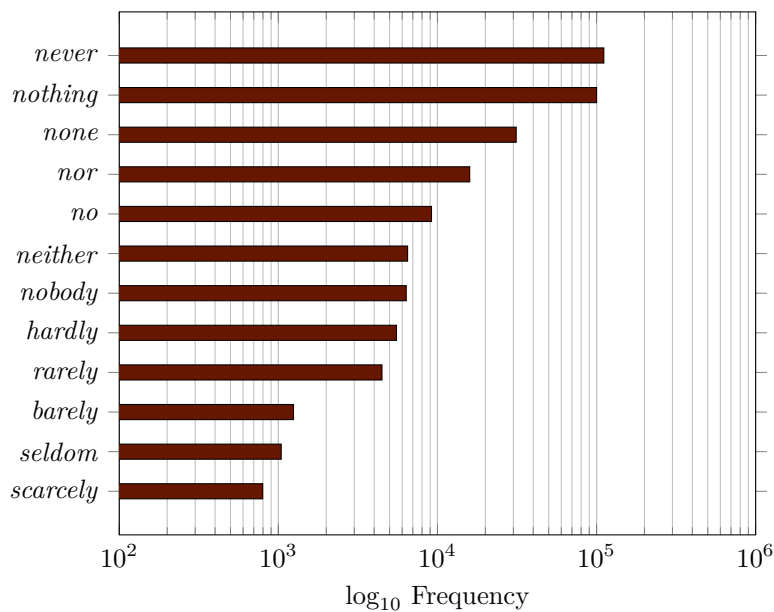


Figure 2.18: Negative Mirror Subset (@N) TRIGGER LEMMAS

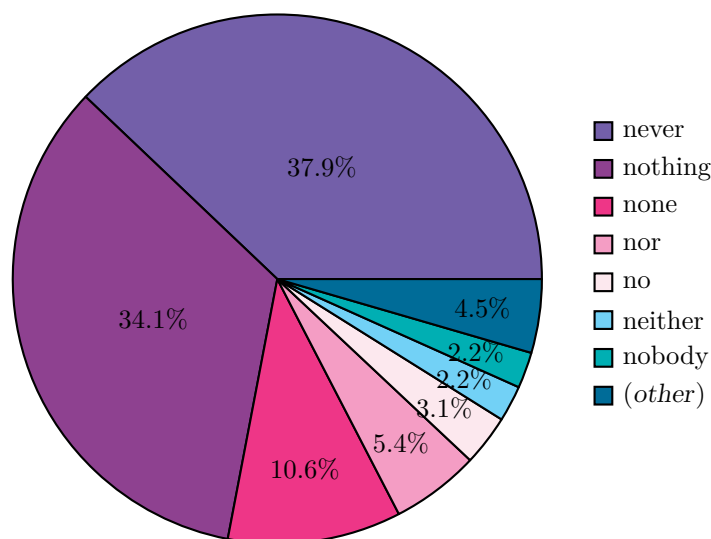


Figure 2.19: Distribution of *Negative Subset* (@N) Operators

These values are very close to those observed in the Negative Superset, as is expected. Some variation in the totals seen for @N and *N are expected due to slight differences in processing procedures, as well as direct limitations imposed on collection pattern specifics. The primary difference is simply the absence of **not**, permitting the remaining triggers much more responsibility in environment licensing.

Lexical Case—Normalized Form Frequencies for Negative Mirror Subset, @N. The most frequent values for Negative Polarity environments within the Mirror Subset are visualized as bar charts in Figure 2.20. The actual frequency values, as well as juxtaposed comparisons with the three polarity categories already covered, are presented in Tables 2.41 (ADV), 2.42 (ADJ), and 2.43 (bigrams).

As the Negative @S Mirror Subset (@N) is directly related to both the Negative

Table 2.41: **PBR Mirror Subset** Top Negative Polarity Adverb Frequencies

#	$f(@N, x)$	ADV, x	$*N^+ \Delta$	$@P \Delta$	$PBR \Delta$
1	78,325	<i>more</i>	$\blacktriangle^5$	=	$\blacktriangle^1$
2	46,648	<i>too</i>	$\blacktriangle^1$	$\blacktriangle^1$	$\blacktriangle^6$
3	31,169	<i>as</i>	$\blacktriangledown^2$	$\blacktriangle^2$	$\blacktriangle^1$
4	25,091	<i>so</i>	$\blacktriangledown^2$	=	=
5	11,325	<i>really</i>	$\blacktriangle^4$	$\blacktriangle^2$	$\blacktriangle^2$
6	9,243	<i>particularly</i>	+	+	+
7	8,914	<i>very</i>	$\blacktriangledown^3$	$\blacktriangledown^5$	$\blacktriangledown^5$
8	6,254	<i>quite</i>	$\blacktriangle^2$	=	+
9	4,865	<i>less</i>	+	+	$\blacktriangle^1$
10	4,709	<i>ever</i>	+	+	+

Superset, $*N^+$ —its direct superset—and the Positive Mirror Subset, $@P$ —its complement, slightly different annotations have been adopted to clearly convey multiple comparisons simultaneously. For completeness, comparisons with PBR overall are also indicated in the final column. Change in rank from each indicated comparison set is denoted by an upwards triangle or downwards triangle with the difference in rank indicated as a superscript. A plus sign indicates that the form was not in the top frequency ranks for the comparison set, and an equals sign indicates the ranking is identical.

Table 2.42: **PBR Mirror Subset** Top Negative Polarity Adjective Frequencies

#	$f(@N, x)$	ADJ, x	$\Delta * N^+$	@P Δ	PBR Δ
1	14,757	<i>important</i>	$\blacktriangle^6$	$\blacktriangle^1$	=
2	11,049	<i>late</i>	+	+	+
3	8,745	<i>good</i>	$\blacktriangledown^2$	=	+
4	5,772	<i>sure</i>	$\blacktriangledown^2$	+	=
5	5,370	<i>wrong</i>	+	=	+
6	5,175	<i>early</i>	+	+	+
7	5,057	<i>easy</i>	$\blacktriangledown^3$	=	+
8	3,444	<i>bad</i>	$\blacktriangledown^5$	+	+
9	3,435	<i>happy</i>	+	+	+
10	2,903	<i>exciting</i>	+	+	+

Table 2.43: **PBR Mirror Subset** Top Negative Polarity Bigram Frequencies

#	$f(@N, x)$	BIGRAM, x	$\Delta * N^+$	@P Δ	PBR Δ
1	12,201	<i>more important</i>	+	$\blacktriangle^1$	+
2	10,998	<i>too late</i>	$\blacktriangle^4$	+	+
3	5,156	<i>too early</i>	+	+	+
4	3,443	<i>as good</i>	$\blacktriangledown^3$	+	+
5	2,946	<i>quite sure</i>	=	+	+
6	2,615	<i>so easy</i>	+	+	+
7	2,345	<i>too old</i>	+	+	+
8	1,884	<i>more frustrating</i>	+	+	+
9	1,567	<i>very good</i>	$\blacktriangledown^7$	$\blacktriangle^1$	+
10	1,513	<i>inherently wrong</i>	+	+	+

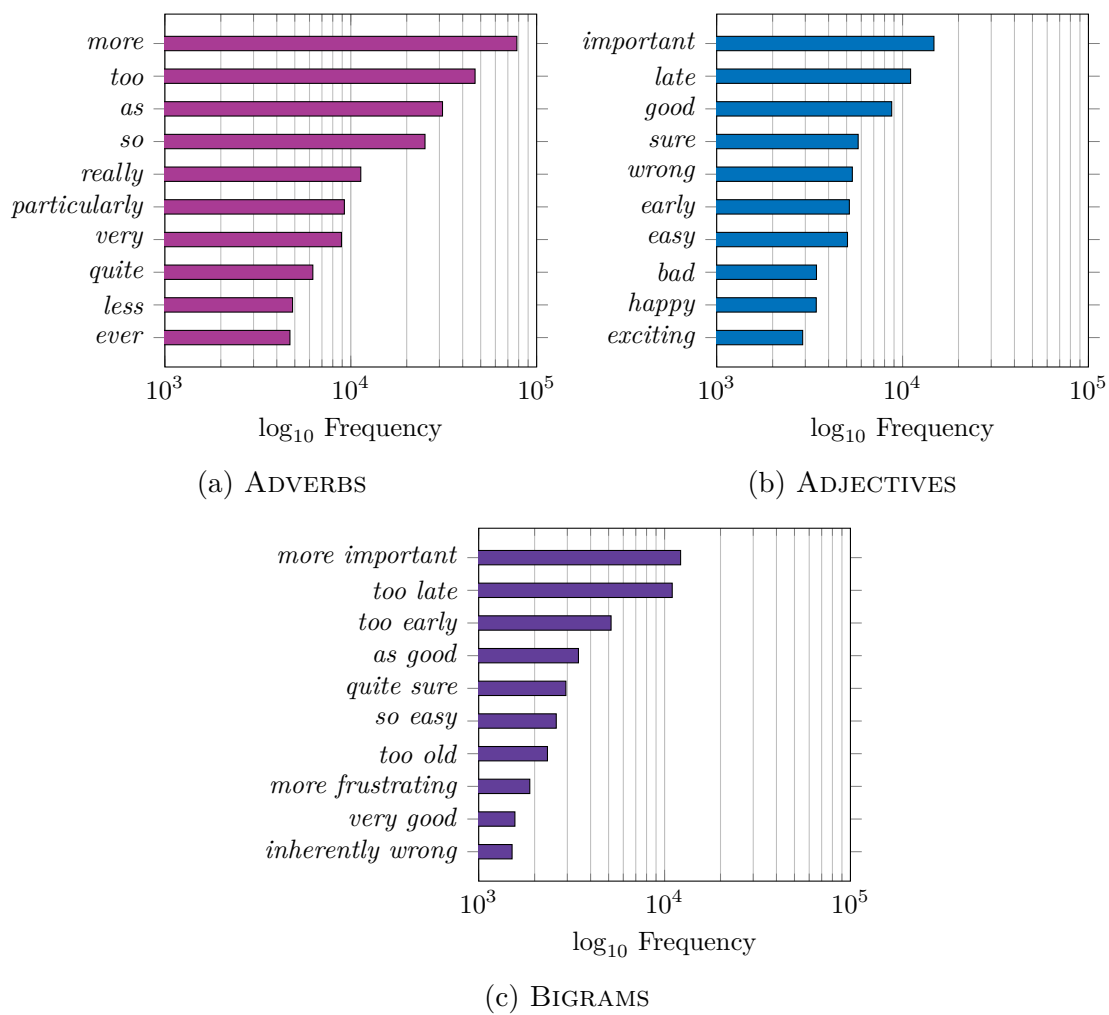


Figure 2.20: Top Negative Mirror Subset Lexical Frequencies

Truly, frequency ranking is not the most accurate means of analyzing lexical prevalence, since it greatly decreases the granularity and obscures the discrepancies *between* ranks. These tables provide only a glimpse of differences that might arise in the different comparison spaces, and more rigorous analysis is required to form any firm conclusions. The primary observation is simply that the negative subset appears to bear resemblance to both the negative superset and the positive subset—it is just a matter of which elements are considered.

2.2.2 Brief Frequency Comparisons of Polarity Perspectives

All of the following comparisons refer only to **S* superset values. Mirror Subset comparisons are not included in this discussion.

2.2.2.1 Normalized Frequency Comparison

Comparing the frequency values for specific lexical unit types between polarity categories does indicate the relative occurrence rates, but in terms of identifying patterns of association (polarity sensitivity), it is quite uninformative. As previously alluded, no patterns can truly be ascertained without considering the broader frequency *context* or what is *expected* given the marginal frequency distributions and the assumption of independence.

Take the bigram *very good* for example. This bigram places in the top 10 for

both positive and negative polarity, but the counts are obviously different. There are 30,092 instances of *very good* with negative markers, coming in second overall for bigrams occurring with negative polarity, but 417,705 observations *without* negative markers (i.e. in the positive superset), despite only ranking as sixth most frequent for positive environments. From the perspective of rank alone, the raw polarized frequencies might indicate that *very good* has an association with negative polarity. But if the actual frequency itself is considered more important, the conclusion might be that it is associated with positive polarity, since 93.3% of all *very good* tokens occur with positive polarity classification.

In this specific case, it is “frequent” with both polarities, highlighting the unreliability of accepting either verdict at face value. For many other cases, this may not be so, making erroneous conclusions based on raw frequency even more likely. This example illustrates the impact of the category size disparity and the importance of controlling for it when assessing sensitivity. Given that PBR, on the whole (superset), is 95.6% positive polarity and only 4.4% negative, an item with “independence” or “by chance” occurrences—i.e. no sensitivity either way—will have similar proportions. Thus, compared to this benchmark, the 93.3% positive proportion is an indication that *very good* “underperformed” in positive environments. By simple chance, roughly 96% of its appearances would be without negative operators, thus the observed 93% is 3% less than expected.

But does this mean *very good* is a negative polarity sensitive item? Such a conclusion hardly seems warranted with over 400k counter-examples. However, for

a less common item (with a lower marginal frequency), it becomes less obvious how important a 3% discrepancy really is. In order to quantify these differences or ascertain patterns with any accuracy or reliability, we must turn to AMs in § 2.2.3.

To provide an overview of relative polar frequencies for each lexical unit, Tables 2.44 (bigrams), 2.45 (adverbs), and 2.46 (adjectives) show the forms with the highest percentage of their total tokens occurring within each polarity in all of PBR⁴⁹. A minimum frequency threshold was applied for all three lexical units excluding bigrams with fewer than 500 tokens, adverbs and adjectives with fewer than 1,000 tokens. When the relevant percentage is 100% (or otherwise identical), ranking defaults to marginal frequency.

Table 2.44: Bigrams with Highest **Negative & Positive Polarity** Percentage (500+ tokens)

Note: Explicitly negative bigrams excluded

		Marginal f	* $\mathcal{L}_{N+}$	* N^+	$f\%(+)$	$f\%(-)$
#	Bigram	$f(\pm)$	$f(+)$	$f(-)$	$\frac{f(-)}{f(\pm)}\%$	$\frac{f(+)}{f(\pm)}\%$
(+) 1	<i>increasingly difficult</i>	18,107	18,107	0	100.00	0.00
	2 <i>most incredible</i>	10,137	10,137	0	100.00	0.00
	3 <i>relatively minor</i>	8,679	8,679	0	100.00	0.00
	4 <i>now defunct</i>	7,390	7,390	0	100.00	0.00
	5 <i>pretty little</i>	7,021	7,021	0	100.00	0.00
	6 <i>most remote</i>	7,007	7,007	0	100.00	0.00
	7 <i>almost daily</i>	6,372	6,372	0	100.00	0.00
	8 <i>relatively modest</i>	6,089	6,089	0	100.00	0.00
	9 <i>more holistic</i>	6,016	6,016	0	100.00	0.00
	10 <i>forever grateful</i>	5,911	5,911	0	100.00	0.00
(-) 1	<i>yet final</i>	640	0	640	0.00	100.00
	2 <i>that uncommon</i>	804	2	802	0.25	99.75

Continued on next page

⁴⁹These values are all for the PBR superset, without any additional sampling (ALL+). Similar rankings of Mirror Subset, and NEQ+ data have been omitted due to space concerns.

Table 2.44: Bigrams with Highest Negative & Positive Polarity Percentage (500+ tokens)

#	Bigram	Marginal f	$*C_{N+}$	$*N^+$	$f\%(+)$	$f\%(-)$
3	<i>exactly sure</i>	8,840	46	8,794	0.52	99.48
4	<i>necessarily easy</i>	914	5	909	0.55	99.45
5	<i>exactly easy</i>	1,072	6	1,066	0.56	99.44
6	<i>necessarily indicative</i>	1,397	8	1,389	0.57	99.43
7	<i>exactly cheap</i>	695	4	691	0.58	99.42
8	<i>yet clear</i>	10,471	65	10,406	0.62	99.38
9	<i>that surprising</i>	1,142	9	1,133	0.79	99.21
10	<i>yet certain</i>	873	7	866	0.80	99.20

Table 2.45: Adverbs with Highest **Negative & Positive Polarity** Percentage (1,000+ tokens)

Note: Explicitly negative adverbs excluded

		Marginal f		$*N^+$	$f\%(+)$	$f\%(-)$
#	Adverb	$f(\pm)$	$f(+)$	$f(-)$	$\frac{f(-)}{f(\pm)}\%$	$\frac{f(+)}{f(\pm)}\%$
(+) 1	<i>yearly</i>	2,474	2,474	0	100.00	0.00
	2 <i>legendarily</i>	1,007	1,007	0	100.00	0.00
	3 <i>albeit</i>	15,666	15,663	3	99.98	0.02
	4 <i>increasingly</i>	373,633	373,560	73	99.98	0.02
	5 <i>preferably</i>	4,782	4,781	1	99.98	0.02
	6 <i>alternately</i>	3,895	3,894	1	99.97	0.03
	7 <i>presumably</i>	7,233	7,230	3	99.96	0.04
	8 <i>briefly</i>	2,455	2,454	1	99.96	0.04
	9 <i>unseasonably</i>	1,928	1,927	1	99.95	0.05
	10 <i>relatively</i>	582,383	582,065	318	99.95	0.06
(-) 1	<i>necessarily</i>	48,641	6,046	42,595	12.43	87.57
	2 <i>that</i>	206,801	42,033	164,768	20.33	79.68
	3 <i>exactly</i>	56,109	12,296	43,813	21.91	78.09
	4 <i>immediately</i>	91,746	35,647	56,099	38.85	61.15
	5 <i>yet</i>	94,755	42,888	51,867	45.26	54.74
	6 <i>any</i>	32,161	16,777	15,384	52.17	47.83
	7 <i>remotely</i>	15,394	9,733	5,661	63.23	36.77
	8 <i>terribly</i>	58,529	40,580	17,949	69.33	30.67
	9 <i>only</i>	432,612	319,110	113,502	73.76	26.24
	10 <i>altogether</i>	17,819	13,251	4,568	74.36	25.64

Table 2.46: Adjectives with Highest **Negative & Positive Polarity** Percentage (1,000+ tokens)

		Marginal f	$*C_{N+}$	$*N^+$	$f\%(+)$	$f\%(-)$
#	Adjective	$f(\pm)$	$f(+)$	$f(-)$	$\frac{f(-)}{f(\pm)}\%$	$\frac{f(+)}{f(\pm)}\%$
(+) 1	<i>evolving</i>	2,953	2,953	0	100.00	0.00
	2 <i>earliest</i>	1,929	1,929	0	100.00	0.00
	3 <i>northerly</i>	1,792	1,792	0	100.00	0.00
	4 <i>chopped</i>	1,553	1,553	0	100.00	0.00
	5 <i>southerly</i>	1,339	1,339	0	100.00	0.00
	6 <i>away</i>	1,157	1,157	0	100.00	0.00
	7 <i>marrow</i>	1,017	1,017	0	100.00	0.00
	8 <i>later</i>	2,263	2,262	1	99.96	0.04
	9 <i>own</i>	87,772	87,729	43	99.95	0.05
	10 <i>unpublished</i>	3,003	3,001	2	99.93	0.07
(–) 1	<i>shabby</i>	7,069	1,540	5,529	21.79	78.22
	2 <i>dissimilar</i>	4,598	2,087	2,511	45.39	54.61
	3 <i>sure</i>	256,817	128,755	128,062	50.14	49.87
	4 <i>farfetched</i>	1,296	732	564	56.48	43.52
	5 <i>rosy</i>	5,355	3,282	2,073	61.29	38.71
	6 <i>far-fetched</i>	5,056	3,101	1,955	61.33	38.67
	7 <i>clear-cut</i>	3,574	2,212	1,362	61.89	38.11
	8 <i>flashy</i>	4,470	2,862	1,608	64.03	35.97
	9 <i>binding</i>	2,293	1,639	654	71.48	28.52
	10 <i>hasty</i>	2,125	1,541	584	72.52	27.48

As already stated, this comparison of relative polarity proportions of a forms marginal frequency is insufficient as evidence of polarity sensitivity, but some general preliminary observations can be made.

While the forms with the highest relative positive polarity frequency ($f\%(+)$) are almost all 100% positive, the top relative negative polarity frequencies are not as uniform or extreme. No adverb or adjective appears solely in negative ($*N$) environments. The highest negative proportions for adverbs range from 87.6 % for *necessarily* to 25.6 % for *altogether*. The highest negative proportions for adjectives

range from 78.2% for *shabby* to 27.5% for *hasty*. Although the negative proportions for bigrams are higher overall than for adverbs or adjectives independently, and there is one bigram with 100% of their tokens in negative environments—*yet final*—the tenth highest relative negative polarity frequency is less than 100% with *yet certain* at 99.2%.

Of course, it must be noted that the marginal frequencies for the most negative bigrams are also much lower than those of the most positive. Only *exactly sure* and *yet clear* would still qualify if the bigram marginal frequency floor were raised to 5 k, but the positive selection would be unaltered. This discrepancy in bigram marginal frequencies is unsurprising given the disparity between the environment marginal frequencies.

The bigrams with the largest positive polarity proportions are for a variety of combinations containing eight distinct adverbs (two adverbs appear twice: *most incredible* and *most remote*; *relatively minor* and *relatively modest*). In contrast, the top negative bigrams include only four distinct adverbs—*yet*, *exactly*, *that*, and *necessarily*. These account for 4 of the top 5 adverbs, whereas the adverbs appearing in the top positive bigrams do not appear in the top positive adverb set at all. There is little overlap between the adjectives appearing in the top bigrams and the independent ranking of adjectives (for either polarity). As a final note, both *farfetched* and *far-fetched* rank in most negative (i.e. highest proportion negative tokens) adjectives; 40%—occurring in negative environments, which would have little impact on their relative position.

2.2.2.2 Versatility Comparison

Another avenue for comparison based on simple counts is to compare unique types instead of unique observations. Table 2.47 juxtaposes the versatility values with frequencies, marginal and polarity normalized. The adverbs shown are the union of the top **ten** for a number of comparisons—overall frequency, frequency in the Positive superset, frequency in the Negative Superset, overall strict versatility, strict versatility in the Positive Superset, and strict versatility in the Negative superset—and the top **five** adverbs with the highest relative polarity frequency ($f\%(-)$ and $f\%(+)$).

The first three columns of Table 2.47 indicate the marginal and polarity joint frequency for each adverb, the second three columns indicate the number of unique adjectives that occurred in the given polarity with the given adverb at least 5 times. Note that the $\mathbb{V}(\pm)$ column indicates the unique adverbs that appeared 5 times across both polarities. This means that if there were an adjective with 4 appearances under each environment, it would be excluded from the $\mathbb{V}(-)$ and $\mathbb{V}(+)$ value ($4 \not\geq 5$), but would be included in $\mathbb{V}(\pm)$ since the undifferentiated total would be 8 ($8 \geq 5$). The $\mathbb{V}(+)$ and $\mathbb{V}(-)$ columns need not sum to $\mathbb{V}(\pm)$, in part due to this frequency limitation (where an adjective numbered in the total column does not appear in the polar columns), but also because adjectives can be counted in the versatility score for both polarities.

The final three columns display manipulations of the preceding versatility values:

$\mathbb{V}^{\%}(+) = \frac{\mathbb{V}(+)}{\mathbb{V}(\pm)} \times 100$ and $\mathbb{V}^{\%}(-) = \frac{\mathbb{V}(-)}{\mathbb{V}(\pm)} \times 100$. The $\mathbb{V}^{\%}$ values are not expected to sum to 100 for each word for the same reasons stated above. Words are ranked by the final column, $\frac{\mathbb{V}(-)}{\mathbb{V}(+)}$, the “(Negative) Polar Odds” of any potential pairing being attested in negative environments as opposed to positive. Values of 1 indicate equal versatility values—i.e.; the same number of unique adjectives appear with the adverb in negative polarity as in positive polarity—while values < 1 indicate there are more pairings observed with positive polarity, and values > 1 indicate more distinct pairings observed in negative polarity than in positive polarity.

Table 2.47: Polar Comparison of Adverb Frequencies & Versatility

Adverb	Marginal f	$f\%(+)$	$f\%(-)$	$\mathbb{V}(\pm)$	$\mathbb{V}(+)$	$\mathbb{V}(-)$	$\mathbb{V}^{\%}(+)$	$\mathbb{V}^{\%}(-)$	$\frac{\mathbb{V}(-)}{\mathbb{V}(+)}$
<i>exactly</i>	56,109	21.91	78.09	920	110	848	12.0	92.2	7.71
<i>necessarily</i>	48,641	12.43	87.57	1,128	254	921	22.5	81.6	3.63
<i>that</i>	206,801	20.33	79.68	1,002	434	822	43.3	82.0	1.89
<i>only</i>	432,612	73.76	26.24	3,186	2,182	1,922	68.5	60.3	0.88
<i>as</i>	3,231,640	83.55	16.45	6,049	5,595	2,989	92.5	49.4	0.53
<i>just</i>	597,938	90.87	9.13	3,542	3,079	1,366	86.9	38.6	0.44
<i>so</i>	5,173,538	93.35	6.65	6,909	6,640	2,413	96.1	34.9	0.36
<i>yet</i>	94,755	45.26	54.74	1,284	1,050	367	81.8	28.6	0.35
<i>too</i>	2,976,740	89.60	10.40	4,958	4,740	1,634	95.6	33.0	0.34
<i>particularly</i>	511,734	89.15	10.85	3,033	2,821	945	93.0	31.2	0.33
<i>always</i>	604,097	82.80	17.20	2,959	2,741	835	92.6	28.2	0.30
<i>really</i>	1,945,388	94.99	5.01	4,287	4,020	1,064	93.8	24.8	0.26
<i>immediately</i>	91,746	38.85	61.15	574	534	129	93.0	22.5	0.24
<i>quite</i>	988,755	92.06	7.94	4,043	3,915	644	96.8	15.9	0.16
<i>very</i>	9,220,012	97.94	2.06	8,828	8,723	1,352	98.8	15.3	0.15
<i>much</i>	1,760,618	96.99	3.01	2,967	2,867	412	96.6	13.9	0.14
<i>more</i>	8,659,678	98.13	1.87	12,707	12,632	1,711	99.4	13.5	0.14
<i>less</i>	1,138,786	98.82	1.18	5,308	5,274	556	99.4	10.5	0.11
<i>most</i>	7,120,428	99.73	0.27	9,433	9,425	537	99.9	5.7	0.06
<i>also</i>	1,064,426	99.82	0.19	4,444	4,435	71	99.8	1.6	0.02
<i>pretty</i>	1,507,900	99.87	0.13	4,443	4,433	46	99.8	1.0	0.01
<i>almost</i>	433,607	99.91	0.09	4,141	4,136	9	99.9	0.2	0.00

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Table 2.47: Polar Comparison of Adverb Frequencies & Versatility

Adverb	Marginal f	$f\%(+)$	$f\%(-)$	$V(\pm)$	$V(+)$	$V(-)$	$V\%(+)$	$V\%(-)$	$\frac{V(-)}{V(+)}$
<i>increasingly</i>	373,633	99.98	0.02	2,957	2,957	1	100.0	0.0	0.00
<i>preferably</i>	4,782	99.98	0.02	178	178	0	100.0	0.0	0.00
<i>albeit</i>	15,666	99.98	0.02	632	632	0	100.0	0.0	0.00
<i>legendarily</i>	1,007	100.00	0.00	28	28	0	100.0	0.0	0.00
<i>yearly</i>	2,474	100.00	0.00	84	84	0	100.0	0.0	0.00

In a similar vein, Table 2.48 presents comparative polarity relative versatility values for adjectives in PBR. The same selection criteria were employed for the adjectives shown as for the adverbs included in Table 2.47.

Table 2.48: Polar Comparison of Adjective Frequencies & Versatility

Adjective	Marginal f	$f\%(+)$	$f\%(-)$	$V(\pm)$	$V(+)$	$V(-)$	$V\%(+)$	$V\%(-)$	$\frac{V(-)}{V(+)}$
<i>farfetched</i>	1,296	56.5	43.5	24	22.0	9.0	92	38	0.41
<i>sure</i>	256,817	50.1	49.9	196	177.0	69.0	90	35	0.39
<i>dissimilar</i>	4,598	45.4	54.6	67	58.0	20.0	87	30	0.34
<i>enough</i>	147,805	85.4	14.6	386	363.0	100.0	94	26	0.28
<i>rosy</i>	5,355	61.3	38.7	71	63.0	17.0	89	24	0.27
<i>available</i>	639,462	91.4	8.6	455	438.0	118.0	96	26	0.27
<i>interested</i>	259,727	91.0	9.0	380	369.0	99.0	97	26	0.27
<i>able</i>	210,786	93.4	6.6	376	358.0	94.0	95	25	0.26
<i>likely</i>	884,149	98.9	1.1	197	184.0	48.0	93	24	0.26
<i>possible</i>	238,712	91.5	8.5	384	366.0	94.0	95	24	0.26
<i>wrong</i>	146,437	89.6	10.4	520	496.0	110.0	95	21	0.22
<i>aware</i>	228,623	91.6	8.4	479	464.0	98.0	97	20	0.21
<i>bad</i>	425,797	75.4	24.6	604	577.0	111.0	96	18	0.19
<i>clear</i>	346,404	79.3	20.7	528	512.0	92.0	97	17	0.18
<i>different</i>	802,192	94.4	5.6	684	673.0	117.0	98	17	0.17
<i>easy</i>	578,041	85.0	15.0	496	477.0	78.0	96	16	0.16
<i>popular</i>	765,023	97.8	2.2	434	424.0	68.0	98	16	0.16
<i>much</i>	1,336,122	96.4	3.6	219	213.0	31.0	97	14	0.15
<i>difficult</i>	729,385	96.0	4.0	567	558.0	79.0	98	14	0.14
<i>good</i>	1,670,122	92.2	7.8	865	854.0	120.0	99	14	0.14

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Table 2.48: Polar Comparison of Adjective Frequencies & Versatility

Adjective	Marginal f	$f\%(+)$	$f\%(-)$	$V(\pm)$	$V(+)$	$V(-)$	$V\%(+)$	$V\%(-)$	$\frac{V(-)}{V(+)}$
<i>important</i>	2,001,942	97.5	2.5	576	568.0	78.0	99	14	0.14
<i>great</i>	306,251	89.3	10.7	447	435.0	58.0	97	13	0.13
<i>close</i>	407,539	92.9	7.1	563	552.0	64.0	98	11	0.12
<i>funny</i>	122,146	91.9	8.1	582	570.0	61.0	98	10	0.11
<i>high</i>	518,631	96.3	3.7	750	742.0	75.0	99	10	0.10
<i>shabby</i>	7,069	21.8	78.2	52	50.0	5.0	96	10	0.10
<i>more</i>	815,328	99.0	1.0	526	524.0	39.0	100	7	0.07
<i>beautiful</i>	292,172	97.6	2.4	701	692.0	49.0	99	7	0.07
<i>low</i>	277,389	99.0	1.0	574	572.0	36.0	100	6	0.06
<i>many</i>	1,835,842	99.2	0.8	409	407.0	15.0	100	4	0.04
<i>evolving</i>	2,953	100.0	0.0	30	30.0	0.0	100	0	0
<i>chopped</i>	1,553	100.0	0.0	5	5.0	0.0	100	0	0
<i>northerly</i>	1,792	100.0	0.0	4	4.0	0.0	100	0	0
<i>southerly</i>	1,339	100.0	0.0	4	4.0	0.0	100	0	0
<i>earliest</i>	1,929	100.0	0.0	3	3.0	0.0	100	0	0

Although these values are not necessarily any more informative than joint frequencies are without the additional contextualization supplied by AMs, this comparison can help elucidate the potential source of any potential association patterns. For example, despite 61% of the observed tokens for *immediately* occurring in negative environments, of the 574 unique adjectives that occur with *immediately* at least 5 times regardless of polarity, only 22.5% of those occur with *immediately* under negation 5 or more times. Although this is not sufficient to make any firm conclusion, it suggests there could be a handful of *immediately* bigrams are responsible for the frequency results observed. While the bigrams responsible may be polarity sensitive, it would be unwarranted in such a case to say the same of *immediately* as an independent unit.

An examination of the individual bigram frequencies corroborates this suspicion (table below). Of the nine bigrams accounting for more than 0.5% of the negated total, only four of them represent a larger percentage of the negative *immediately* tokens (56,099) than they do of the positive token total (35,647). And of those four, the two most common, *immediately clear* and *immediately available*, account for over half ($\sim 55\%$) of all *immediately* observations just between themselves, with *clear* providing 28% and *available* contributing 27%. It appears that these two specific bigrams are the primary cause of the observed 61% negative status of *immediately* overall. Given that they represent only 0.35% of the unique adjectives *immediately* is observed modifying, assigning this polarity propensity to *immediately* in general seems unsupported.

Table 2.49: *Immediately* Polar Frequencies vs. Component Representation

<i>Immediately</i> ...	Total	(+)	(-)	(+)%	(-)%	% of <i>I</i>	% of <i>I</i> ₍₊₎	% of <i>I</i> ₍₋₎
	<i>I</i>	<i>I</i> ₍₊₎	<i>I</i> ₍₋₎					
MARGIN	91,746	35,647	56,099	38.9	61.1	100.0	100.0	100.0
clear	26,002	1,586	24,416	6.1	93.9	28.3	4.4	43.5
available	24,769	3,691	21,078	14.9	85.1	27.0	10.4	37.6
<i>obvious</i>	3,915	1,677	2,238	42.8	57.2	4.3	4.7	4.0
<i>apparent</i>	4,864	2,849	2,015	58.6	41.4	5.3	8.0	3.6
possible	1,042	42	1,000	4.0	96.0	1.1	0.1	1.8
able	980	354	626	36.1	63.9	1.1	1.0	1.1
<i>visible</i>	1,133	699	434	61.7	38.3	1.2	2.0	0.8
<i>evident</i>	1,369	950	419	69.4	30.6	1.5	2.7	0.7
<i>successful</i>	522	232	290	44.4	55.6	0.6	0.7	0.5
<i>recognizable</i>	1,716	1,544	172	90.0	10.0	1.9	4.3	0.3

Similar coverage for the adjective *shabby* is given in Table 2.50. An even larger proportion of negated *shabby* tokens are attributed to a single bigram than negated *immediately* tokens. A full 97.5% of all negated *shabby* are due to the bigram *too*

shabby—unsurprising given the near lexicalized status of the phrase “not too shabby”. This discredits potential explanation that *shabby* alone is negative polarity sensitive, especially given the over 1.5k counter examples where *shabby* appears without a negative (*N) trigger.

Table 2.50: *Shabby* Polar Frequencies vs. Component Representation

... <i>Shabby</i>	Total	(+)	(-)	(+)%	(-)%	% of <i>S</i>	% of <i>S</i> ₍₊₎	% of <i>S</i> ₍₋₎
MARGIN	<i>S</i>	<i>S</i> ₍₊₎	<i>S</i> ₍₋₎					
	7,069	1,540	5,529	21.79	78.22	100.00	100.00	100.00
<i>too</i>	5,482	90	5,392	1.64	98.36	77.55	5.84	97.52
<i>so</i>	146	73	73	50.00	50.00	2.07	4.74	1.32
<i>exactly</i>	23	0	23	0.00	100.00	0.33	0.00	0.42
<i>as</i>	41	33	8	80.49	19.51	0.58	2.14	0.14
<i>that</i>	7	0	7	0.00	100.00	0.10	0.00	0.13
<i>very</i>	158	155	3	98.10	1.90	2.24	10.06	0.05
<i>more</i>	71	69	2	97.18	2.82	1.00	4.48	0.04
<i>also</i>	8	6	2	75.00	25.00	0.11	0.39	0.04
<i>altogether</i>	4	2	2	50.00	50.00	0.06	0.13	0.04
<i>terribly</i>	2	0	2	0.00	100.00	0.03	0.00	0.04

2.2.3 Association Measures

Association Measures (AMs) are commonly used for “collocation identification”, or locating and evaluating lexicalization of multi-word expressions, as defined by contiguous phrases or structural relationships (Pecina 2005). Since one of the primary defining characteristics of such “collocations” is *non-compositionality* (Pecina 2005), evaluations of AM efficacy have focused on precision and recall of predefined sets of known non-compositional collocations (Evert *et al.* (2017); Pecina (2005); Uhrig *et al.* (2018), a.o.). This means that the “best” AM is the one that can best identify when

a linguistic unit conveys a meaning that could not be derived from its constituent words.

This is not, strictly speaking, the full story for polarity sensitivity. While full NPIs and PPIs (if they exist), may qualify as lexicalized in some sense, or at least not *fully compositional*, the complexity of polarity sensitivity complicates matters.

This may raise the question of whether it is methodologically or theoretically sound to employ AMs to evaluate a phenomenon with no such expectation of non-compositionality. I argue that the calculations themselves are still a reliable assessment of the relationships sought: It is the interpretation of said calculations that changes to reflect the target inquiry—different AMs excel at identifying different types of lexicographical collocations (Uhrig *et al.* (2018), a.o.) so it easily follows that AMs in general can be harnessed to assess things besides non-compositional collocations. AMs are simply ways of manipulating relative probabilities as extracted from linguistic frequency data (Evert 2022), and the values (or “scores”) they yield represent a particular relationship between observations. As long as the relationship each represents is carefully considered, extension to compositional linguistic phenomena is sound, especially for AMs deemed well-motivated by mathematical and statistical theory.

AMs operate on the basic input of a *frequency signature* (Evert 2005), or the observed frequencies of all “contingencies” for the lexemes or linguistic units in question, as represented in a classic *contingency table*. The following excerpt from Evert *et al.* (2017) illustrates, on the right, the observed frequency signature collected

expected frequencies			observed frequencies			
	collocate	¬collocate		collocate	¬collocate	
node	$E_{11} = \frac{R_1 C_1}{N}$	$E_{12} = \frac{R_1 C_2}{N}$	node	O_{11}	O_{12}	$= R_1$
¬node	$E_{21} = \frac{R_2 C_1}{N}$	$E_{22} = \frac{R_2 C_2}{N}$	¬node	O_{21}	O_{22}	$= R_2$
				$= C_1$	$= C_2$	$= N$

O_{ij} = contingency table of observed frequencies
 O_{11} = observed co-occurrence frequency
 E_{ij} = contingency table of expected frequencies
 E_{11} = expected co-occurrence frequency
 R_i = row sums of the contingency table
 R_1 = marginal frequency of node
 C_j = column sums of the contingency table
 C_1 = marginal frequency of collocate
 N = sample size

Figure 2.21: Contingency Table Schema (Evert *et al.* 2017)

for every co-occurrence possibility in the typical contingency table format. The table on the left shows the *expected* frequencies for the different contingencies: expected frequencies are calculated from the observed *marginal* frequencies—the subtotal frequencies for the different category contingencies relegated to the margins of the table—as well as the total number of tokens (N).

The expected frequencies essentially represent the null hypothesis: They are the values that should be observed if the elements in question are statistically

independent of one another. Thus a very simple calculation for superficially gauging the *lack* of independence is merely the difference between the expected and observed joint frequency for the selected forms. The larger the magnitude of the difference (positive or negative), or the more *unexpected* the observed frequency is, the more likely it is that the items in question affect one another. However, to know the real strength or significance of this association, additional information must be considered. The vast array of association measures are all attempts to achieve this.

Although there are some “heuristic” measures that are not directly grounded in theory (Evert 2022), many AMs draw on mathematical or statistical principles to highlight or verify different aspects of the observed frequencies.

Hypothesis test AMs, such as log-likelihood, (G^2) (Dunning 1993) and chi-squared (χ^2) evaluate statistical significance of co-occurrence, and are biased toward high-frequency words (Evert 2022). Effect size AMs, such as relative risk/*LogRatio* (Hardie (2014) cited by Evert (2022)), indicate the association strength as indicated by relative frequencies, and are biased toward very low-frequency words (Evert 2022). Evert (2022) proposed a new measure: **LRC** or **conservative log ratio**⁵⁰, “a conservative estimate of LogRatio” (p. 203). She presents this score as a compromise between the approaches, providing theoretically motivated indicators of both effect size and significance, while balancing out the aforementioned low and high frequency biases.

⁵⁰“LRC” is the first abbreviation for “conservative log ratio” that I came across and adopted, despite its divergence from the English translation. I have since seen it abbreviated as “CLR” or “clr”, but I have not adopted this reordering of the initialism.

Despite differences in statistical motivation and the type of indicator returned, the AMs in the preceding paragraph are *symmetric*: they provide a single score for each potential case/pair evaluated. *Asymmetric* measures, on the other hand, provide separate values based on direction, forward (l_1 to l_2) or backward (l_2 to l_1), and therefore provide transparency to the directionality of association (Gries & Durrant 2020). That is, given a bigram of form, *adv adj*, where *adv* = R and *adj* = J , if R “prefers” J , but J has no such tendency, the strong association of $R \rightarrow J$ may be overshadowed by the lack a corresponding $R \leftarrow J$ association. If the former is strong enough, a symmetric AM may still indicate the association, but the same value could just as well represent an association that is mutual/bidirectional, $R \leftrightarrow J$, or occurring in the opposite direction, $R \leftarrow J$. For example, the bigrams *upwardly mobile* and *well intentioned* exhibit such one-sided preferences: *upwardly* rarely occurs with anything but *mobile*, while *mobile* is commonly modified by many other adverbs. In the reverse, *well* appears modifying a vast array of adjectives, while *intentioned* almost always follows *well*.

The specific linguistic inquiry determines the importance of directionality in assessing potential association. In the current work, directionality is far more important for evaluating the environment of bigram tokens than for evaluating the composition of the tokens themselves. Furthermore, I argue in § 2.2.3.1 that, due to the data landscape realities of PBR and theoretical expectations of polarity sensitivity, backwards asymmetric association is the most indicative of polarity sensitivity.

Regardless of the active inquiry, the best approach for evaluating association is to consider multiple alongside each other. Only a limited number of AMs can reasonably be presented and discussed without sacrificing coherence, so the remainder of this work focuses on the four specific AMs, two asymmetric and two symmetric. The asymmetric inclusions are **conditional probability**—backwards conditional probability (P1) and forwards conditional probability (P2)—for its intuitive illustrative capacity, and **delta P**—backwards ΔP (dP1) and forwards ΔP (dP2)—primarily *backwards* ($\Delta P(1|2)$) when evaluating environment polarity—for its null-hypothesis-normalized interpretability. The symmetric AM that will be utilized most is **Evert’s conservative log ratio (2022) (LRC)** as a clear indicator of statistical significance. **Dunning’s log-likelihood (1993) (G2)** will also be provided alongside LRC and the asymmetric probability AMs, primarily to serve as a comparison point, since it is widely used and considered one of the most robust AMs (Pecina 2005; Uhrig *et al.* 2018; Evert *et al.* 2017).⁵¹ More detailed discussions of the probability AMs in relation to polarity sensitivity follow in § 2.2.3.2 and § 2.2.3.3.

As a preview of AMs in action, the bigrams shown in Table 2.51 are limited to those with a joint frequency of 500 or more ($f \geq 500$). That is, these values are calculated on the frequencies observed for the different contingencies between ADV and ADJ types, with polarity of environment entirely obscured.

⁵¹For my purposes, it is the least informative of the selected AMs, as it is highly sensitive to overall magnitudes and seems to obscure strong patterns which occur in words that are simply less common on the whole. This could be due to its “high frequency data” bias, which is exacerbated by the overall size of the dataset, making it even more likely that relevant patterns are drowned out.

Table 2.51: Top ALL+ Superset Context–Blind Bigrams by Assorted AMs, by descending raw joint frequency

Bigram	f	LRC	dPavg	dP1	P1	dP2	P2	G2	f1	f2	unexpF	unexp%	Top 5 For
<i>so many</i>	1,049,935	4.1	0.35	0.51	0.57	0.19	0.20	3,329,624	5,173,538	1,835,842	917,950	87.4	f; G2; unexpF
<i>most important</i>	690,094	2.2	0.17	0.25	0.35	0.08	0.10	926,677	7,120,428	2,001,942	492,006	71.3	f; unexpF
<i>so much</i>	524,372	3.0	0.21	0.33	0.39	0.09	0.10	1,132,044	5,173,538	1,336,122	428,314	81.7	f; G2; unexpF
<i>more likely</i>	451,270	2.9	0.22	0.40	0.51	0.05	0.05	812,777	8,659,678	884,149	344,873	76.4	f; unexpF
<i>very good</i>	447,797	1.3	0.09	0.14	0.27	0.03	0.05	240,463	9,220,012	1,670,122	233,814	52.2	f
<i>too much</i>	441,932	3.5	0.22	0.30	0.33	0.14	0.15	1,250,614	2,976,740	1,336,122	386,662	87.5	G2; unexpF
<i>much more</i>	309,604	4.6	0.26	0.36	0.38	0.17	0.18	1,292,288	1,760,618	815,328	289,656	93.6	G2
<i>much better</i>	254,524	4.9	0.27	0.40	0.42	0.14	0.15	1,122,880	1,760,618	600,470	239,833	94.2	G2
<i>very own</i>	86,224	8.4	0.43	0.86	0.98	0.01	0.01	339,895	9,220,012	87,772	74,978	87.0	P1
<i>very least</i>	74,083	9.1	0.44	0.86	0.99	0.01	0.01	296,979	9,220,012	74,810	64,498	87.1	P1
<i>partly cloudy</i>	54,239	12.2	0.77	0.84	0.84	0.69	0.69	734,391	78,766	64,456	54,168	99.9	dPavg
<i>crystal clear</i>	6,881	7.6	0.50	0.02	0.02	0.98	0.98	72,298	7,009	346,404	6,847	99.5	dP2; P2
<i>scantily clad</i>	4,638	15.0	0.85	0.71	0.71	0.99	0.99	90,347	4,682	6,562	4,638	100.0	LRC; dPavg; dP2; P2
<i>most wanted</i>	2,810	8.3	0.44	0.89	0.99	<0.01	<01	12,622	7,120,428	2,846	2,528	90.0	P1
<i>proof positive</i>	1,395	8.5	0.49	0.01	0.01	0.97	0.97	16,534	1,440	167,067	1,392	99.8	dP2; P2
<i>thinly veiled</i>	1,333	16.7	0.77	0.78	0.78	0.76	0.76	28,017	1,746	1,705	1,333	100.0	LRC; dPavg; unexp%
<i>chock full</i>	1,209	9.4	0.51	0.01	0.01	1.00	1.00	16,142	1,210	90,742	1,207	99.9	dP2; P2
<i>far away</i>	1,150	12.5	0.50	0.99	0.99	<0.01	<01	11,845	402,768	1,157	1,144	99.4	dP1; P1
<i>bone marrow</i>	993	19.8	0.89	0.98	0.98	0.81	0.81	22,805	1,222	1,017	993	100.0	LRC; dPavg; dP1; unexp%
<i>sparsely populated</i>	900	14.3	0.47	0.35	0.35	0.58	0.58	16,679	1,559	2,558	900	100.0	unexp%
<i>super duper</i>	626	12.8	0.50	0.99	1.00	<0.01	<01	6,886	290,876	627	623	99.6	dP1; P1
<i>above reproach</i>	575	17.5	0.56	0.96	0.96	0.17	0.17	11,355	3,402	599	575	100.0	LRC; dP1; unexp%
<i>nerve wracking</i>	527	20.6	0.99	0.98	0.98	0.99	0.99	13,363	531	538	527	100.0	LRC; dPavg; dP1; dP2; P2; unexp%

The following tables (Tables 2.52–2.59) show the union of the three strongest values across each of the given AMs ($\{\text{dP1, P1, LRC, G2}\}$) for each polarity within each lexical category—*bigram*, *adverb*, or *adjective*. Considerations were restricted to those lexical units (i.e. 12 values) that do not match an “explicitly negative” regular expression, **and** have at least (i.e., $f2 \geq$) 1,000 tokens for $*S:ALL+$, 500 tokens for both $*S:NEQ+$ & $@S:ALL+$, and 250 tokens for $@S:NEQ+$. Additional comparisons are available in Appendix A.

Table 2.52: $*S:ALL+$ Polar Bigram AMs for Union of Top

Metric	f		P1		dP1		LRC		G2		f2	
	Polarity	neg	pos	neg	pos	neg	pos	neg	pos	neg		pos
<i>so many</i>		2,094	1,047,841	0.00	1.00	−0.04	0.04	−4.30	4.30	−78,025	78,025	1,049,935
<i>most important</i>		936	689,158	0.00	1.00	−0.04	0.04	−4.74	4.74	−54,069	54,069	690,094
<i>most popular</i>		610	373,189	0.00	1.00	−0.04	0.04	−4.40	4.40	−28,507	28,507	373,799
<i>most recent</i>		74	292,307	0.00	1.00	−0.04	0.04	−6.39	6.39	−25,511	25,511	292,381
<i>very little</i>		62	217,044	0.00	1.00	−0.04	0.04	−6.13	6.13	−18,860	18,860	217,106
<i>partly cloudy</i>		3	54,236	0.00	1.00	−0.04	0.04	−5.93	5.93	−4,848	4,848	54,239
<i>so sure</i>		26,480	7,647	0.78	0.22	0.73	−0.73	6.09	−6.09	129,884	−129,884	34,127
<i>always easy</i>		24,459	2,356	0.91	0.09	0.87	−0.87	7.58	−7.58	137,118	−137,118	26,815
<i>immediately clear</i>		24,416	1,586	0.94	0.06	0.90	−0.90	8.11	−8.11	140,793	−140,793	26,002
<i>increasingly difficult</i>			18,107		1.00		0.04		4.72		1,634	18,107
<i>yet clear</i>		10,406	65	0.99	0.01	0.95	−0.95	10.57	−10.57	64,207	−64,207	10,471
<i>most incredible</i>			10,137		1.00		0.04		3.88		915	10,137
<i>exactly sure</i>		8,794	46	0.99	0.01	0.95	−0.95	10.64	−10.64	54,348	−54,348	8,840
<i>relatively minor</i>			8,679		1.00		0.04		3.66		783	8,679
<i>yet ready</i>		7,501	74	0.99	0.01	0.95	−0.95	9.98	−9.98	46,016	−46,016	7,575
<i>necessarily indicative</i>		1,389	8	0.99	0.01	0.95	−0.95	9.13	−9.13	8,574	−8,574	1,397
<i>exactly easy</i>		1,066	6	0.99	0.01	0.95	−0.95	8.87	−8.87	6,581	−6,581	1,072

Table 2.53: *S:NEQ+ Polar Bigram AMs for Union of Top AMs

Metric Polarity	f		P1		dP1		LRC		G2		f2
	neg	pos	neg	pos	neg	pos	neg	pos	neg	pos	
<i>so many</i>	2,094	48,117	0.04	0.96	-0.46	0.46	-4.30	4.30	-52,538	52,538	50,211
<i>most important</i>	936	31,659	0.03	0.97	-0.47	0.47	-4.75	4.75	-36,845	36,845	32,595
<i>so sure</i>	26,480	357	0.99	0.01	0.49	-0.49	5.69	-5.69	33,518	-33,518	26,837
<i>quite sure</i>	26,026	329	0.99	0.01	0.49	-0.49	5.77	-5.77	33,102	-33,102	26,355
<i>immediately clear</i>	24,416	60	1.00	0.00	0.50	-0.50	7.50	-7.50	33,184	-33,184	24,476
<i>most popular</i>	610	17,339	0.03	0.97	-0.47	0.47	-4.42	4.42	-19,602	19,602	17,949
<i>even sure</i>	14,459	12	1.00	0.00	0.50	-0.50	7.98	-7.98	19,900	-19,900	14,471
<i>most recent</i>	74	13,425	0.01	0.99	-0.50	0.50	-6.44	6.44	-17,824	17,824	13,499
<i>yet clear</i>	10,406	5	1.00	0.00	0.50	-0.50	7.93	-7.93	14,363	-14,363	10,411
<i>very little</i>	62	10,033	0.01	0.99	-0.49	0.49	-6.19	6.19	-13,255	13,255	10,095
<i>exactly sure</i>	8,794	3	1.00	0.00	0.50	-0.50	7.87	-7.87	12,154	-12,154	8,797
<i>partly cloudy</i>	3	2,506	0.00	1.00	-0.50	0.50	-6.05	6.05	-3,433	3,433	2,509
<i>relatively short</i>	1	909	0.00	1.00	-0.50	0.50	-4.81	4.81	-1,246	1,246	910
<i>increasingly difficult</i>		853		1.00		0.50		4.88		1,183	853
<i>necessarily indicative</i>	1,389		1.00		0.50		5.60		1,926		1,389
<i>that common</i>	1,207		1.00		0.50		5.39		1,673		1,207
<i>too surprising</i>	4,014		1.00		0.50		7.14		5,567		4,014

Table 2.54: @S:ALL+ Polar Bigram AMs for Union of Top

Metric Polarity	f		P1		dP1		LRC		G2		f2
	neg	pos	neg	pos	neg	pos	neg	pos	neg	pos	
<i>more important</i>	12,201	10,941	0.53	0.47	0.36	-0.36	2.28	-2.28	15,173	-15,173	23,142
<i>too late</i>	10,998	1,911	0.85	0.15	0.68	-0.68	4.53	-4.53	28,743	-28,743	12,909
<i>completely different</i>	24	8,340	0.00	1.00	-0.17	0.17	-4.51	4.51	-2,944	2,944	8,364
<i>too familiar</i>	31	7,028	0.00	1.00	-0.17	0.17	-4.06	4.06	-2,396	2,396	7,059
<i>even better</i>	12	6,089	0.00	1.00	-0.17	0.17	-4.52	4.52	-2,195	2,195	6,101
<i>too early</i>	5,156	234	0.96	0.04	0.79	-0.79	6.08	-6.08	16,295	-16,295	5,390
<i>too common</i>	14	4,905	0.00	1.00	-0.17	0.17	-4.11	4.11	-1,730	1,730	4,919
<i>quite sure</i>	2,946	42	0.99	0.01	0.81	-0.81	7.04	-7.04	9,916	-9,916	2,988
<i>even impossible</i>		1,270		1.00		0.17		3.29		484	1,270
<i>even higher</i>		959		1.00		0.17		2.88		366	959
<i>really sure</i>	905	13	0.99	0.01	0.81	-0.81	6.20	-6.20	3,040	-3,040	918
<i>far greater</i>		621		1.00		0.17		2.24		237	621
<i>too fancy</i>	528	14	0.97	0.03	0.80	-0.80	5.35	-5.35	1,725	-1,725	542

Table 2.55: @S:NEQ+ Polar Bigram AMs for Union of Top

Metric	f		P1		dP1		LRC		G2		f2
	neg	pos	neg	pos	neg	pos	neg	pos	neg	pos	
<i>more important</i>	12,201	2,304	0.84	0.16	0.35	-0.35	2.19	-2.19	7,581	-7,581	14,505
<i>too late</i>	10,998	391	0.97	0.03	0.47	-0.47	4.33	-4.33	12,580	-12,580	11,389
<i>too early</i>	5,156	52	0.99	0.01	0.49	-0.49	5.44	-5.44	6,682	-6,682	5,208
<i>quite sure</i>	2,946	8	1.00	0.00	0.50	-0.50	5.99	-5.99	3,999	-3,999	2,954
<i>completely different</i>	24	1,787	0.01	0.99	-0.49	0.49	-4.55	4.55	-2,261	2,261	1,811
<i>too familiar</i>	31	1,452	0.02	0.98	-0.48	0.48	-4.04	4.04	-1,758	1,758	1,483
<i>too pleased</i>	1,289	8	0.99	0.01	0.49	-0.49	4.78	-4.78	1,703	-1,703	1,297
<i>even better</i>	12	1,240	0.01	0.99	-0.49	0.49	-4.50	4.50	-1,603	1,603	1,252
<i>too common</i>	14	1,030	0.01	0.99	-0.49	0.49	-4.13	4.13	-1,301	1,301	1,044
<i>quite different</i>	4	663	0.01	0.99	-0.49	0.49	-4.12	4.12	-876	876	667
<i>too real</i>	3	497	0.01	0.99	-0.49	0.49	-3.79	3.79	-657	657	500
<i>even impossible</i>		273		1.00		0.50		3.33		379	273
<i>entirely sure</i>	364		1.00		0.50		3.76		505		364
<i>ever easy</i>	368		1.00		0.50		3.78		510		368

Table 2.56: *S ALL+ Polar Adverb AMs for Union of Top

Metric	f		P1		dP1		LRC		G2		f2
	neg	pos	neg	pos	neg	pos	neg	pos	neg	pos	
<i>very</i>	189,823	9,030,189	0.02	0.98	-0.03	0.03	-1.11	1.11	-167,026	167,026	9,220,012
<i>more</i>	161,916	8,497,762	0.02	0.98	-0.03	0.03	-1.26	1.26	-185,605	185,605	8,659,678
<i>most</i>	19,204	7,101,224	0.00	1.00	-0.05	0.05	-4.03	4.03	-525,721	525,721	7,120,428
<i>as</i>	531,731	2,699,909	0.16	0.84	0.13	-0.13	2.08	-2.08	727,573	-727,573	3,231,640
<i>relatively</i>	318	582,065	0.00	1.00	-0.04	0.04	-5.96	5.96	-49,292	49,292	582,383
<i>almost</i>	397	433,210	0.00	1.00	-0.04	0.04	-5.25	5.25	-35,325	35,325	433,607
<i>only</i>	113,502	319,110	0.26	0.74	0.22	-0.22	2.92	-2.92	242,355	-242,355	432,612
<i>increasingly</i>	73	373,560	0.00	1.00	-0.04	0.04	-7.00	7.00	-32,850	32,850	373,633
<i>that</i>	164,768	42,033	0.80	0.20	0.75	-0.75	6.36	-6.36	831,670	-831,670	206,801
<i>exactly</i>	43,813	12,296	0.78	0.22	0.74	-0.74	6.19	-6.19	216,170	-216,170	56,109
<i>necessarily</i>	42,595	6,046	0.88	0.12	0.83	-0.83	7.14	-7.14	230,466	-230,466	48,641
<i>albeit</i>	3	15,663	0.00	1.00	-0.04	0.04	-4.71	4.71	-1,374	1,374	15,666
<i>yearly</i>		2,474		1.00		0.04		2.56		223	2,474
<i>legendarily</i>		1,007		1.00		0.04		1.26		91	1,007

Table 2.57: *S:NEQ+ Polar Adverb AMs for Union of Top

Metric	f		P1		dP1		LRC		G2		f2
Polarity	neg	pos	neg	pos	neg	pos	neg	pos	neg	pos	
<i>as</i>	531,731	124,952	0.81	0.19	0.35	-0.35	2.06	-2.06	300,341	-300,341	656,683
<i>very</i>	189,823	416,707	0.31	0.69	-0.21	0.21	-1.11	1.11	-95,938	95,938	606,530
<i>more</i>	161,916	391,566	0.29	0.71	-0.23	0.23	-1.25	1.25	-107,333	107,333	553,482
<i>too</i>	309,669	123,480	0.71	0.29	0.23	-0.23	1.30	-1.30	88,563	-88,563	433,149
<i>most</i>	19,204	327,788	0.06	0.94	-0.47	0.47	-4.04	4.04	-348,425	348,425	346,992
<i>that</i>	164,768	1,912	0.99	0.01	0.50	-0.50	6.25	-6.25	214,472	-214,472	166,680
<i>exactly</i>	43,813	565	0.99	0.01	0.49	-0.49	5.95	-5.95	55,764	-55,764	44,378
<i>necessarily</i>	42,595	321	0.99	0.01	0.50	-0.50	6.63	-6.63	55,995	-55,995	42,916
<i>relatively</i>	318	26,629	0.01	0.99	-0.49	0.49	-5.96	5.96	-34,010	34,010	26,947
<i>almost</i>	397	20,201	0.02	0.98	-0.48	0.48	-5.28	5.28	-24,695	24,695	20,598
<i>increasingly</i>	73	17,139	0.00	1.00	-0.50	0.50	-7.03	7.03	-22,964	22,964	17,212
<i>albeit</i>	3	737	0.00	1.00	-0.50	0.50	-4.82	4.82	-987	987	740

Table 2.58: *S:ALL+ Polar Adjective AMs for Union of Top

Metric	f		P1		dP1		LRC		G2		f2
Polarity	neg	pos	neg	pos	neg	pos	neg	pos	neg	pos	
<i>many</i>	13,930	1,821,912	0.01	0.99	-0.04	0.04	-2.52	2.52	-89,076	89,076	1,835,842
<i>likely</i>	9,528	874,621	0.01	0.99	-0.03	0.03	-1.99	1.99	-33,382	33,382	884,149
<i>more</i>	8,417	806,911	0.01	0.99	-0.03	0.03	-2.05	2.05	-31,852	31,852	815,328
<i>bad</i>	104,709	321,088	0.25	0.75	0.20	-0.20	2.79	-2.79	210,059	-210,059	425,797
<i>clear</i>	71,543	274,861	0.21	0.79	0.16	-0.16	2.46	-2.46	119,592	-119,592	346,404
<i>sure</i>	128,062	128,755	0.50	0.50	0.46	-0.46	4.40	-4.40	459,579	-459,579	256,817
<i>first</i>	121	145,454	0.00	1.00	-0.04	0.04	-5.07	5.07	-11,931	11,931	145,575
<i>own</i>	43	87,729	0.00	1.00	-0.04	0.04	-5.40	5.40	-7,446	7,446	87,772
<i>least</i>	68	74,742	0.00	1.00	-0.04	0.04	-4.72	4.72	-6,082	6,082	74,810
<i>shabby</i>	5,529	1,540	0.78	0.22	0.74	-0.74	6.04	-6.04	27,252	-27,252	7,069
<i>dissimilar</i>	2,511	2,087	0.55	0.45	0.50	-0.50	4.45	-4.45	9,530	-9,530	4,598
<i>evolving</i>		2,953		1.00		0.04		2.71		266	2,953
<i>earliest</i>		1,929		1.00		0.04		2.10		174	1,929
<i>northerly</i>		1,792		1.00		0.04		1.99		162	1,792

Table 2.59: *S:NEQ+ Polar Adjective AMs for Union of Top

Metric	f		P1		dP1		LRC		G2		f2
Polarity	neg	pos	neg	pos	neg	pos	neg	pos	neg	pos	
<i>sure</i>	128,062	5,996	0.96	0.04	0.47	-0.47	4.31	-4.31	139,261	-139,261	134,058
<i>bad</i>	104,709	15,068	0.87	0.13	0.38	-0.38	2.73	-2.73	76,707	-76,707	119,777
<i>many</i>	13,930	83,846	0.14	0.86	-0.36	0.36	-2.51	2.51	-56,266	56,266	97,776
<i>clear</i>	71,543	12,415	0.85	0.15	0.36	-0.36	2.45	-2.45	46,592	-46,592	83,958
<i>likely</i>	9,528	40,467	0.19	0.81	-0.31	0.31	-1.99	1.99	-20,759	20,759	49,995
<i>more</i>	8,417	37,353	0.18	0.82	-0.32	0.32	-2.05	2.05	-19,896	19,896	45,770
<i>first</i>	121	6,677	0.02	0.98	-0.48	0.48	-5.08	5.08	-8,216	8,216	6,798
<i>shabby</i>	5,529	84	0.99	0.01	0.49	-0.49	5.21	-5.21	6,913	-6,913	5,613
<i>own</i>	43	4,118	0.01	0.99	-0.49	0.49	-5.46	5.46	-5,292	5,292	4,161
<i>least</i>	68	3,471	0.02	0.98	-0.48	0.48	-4.75	4.75	-4,236	4,236	3,539
<i>dissimilar</i>	2,511	98	0.96	0.04	0.46	-0.46	3.89	-3.89	2,782	-2,782	2,609
<i>broadier</i>	16	848	0.02	0.98	-0.48	0.48	-4.00	4.00	-1,039	1,039	864

2.2.3.1 Polarity Sensitivity as Backwards Asymmetric Association

Asymmetric measures yield two values indicating *forward* and *backward* association separately. **Forward association** is the extent to which the preceding element associates with or “*is attracted to*” the following element, and **backward association** is the extent to which the following element associates with or “*is attracted to*” the preceding element. **Symmetric measures** collapse these perspectives into a single value intended to convey the overall association. When the “liking” is mutual, symmetric and asymmetric measures are equally informative, but when it is “unrequited”, some information is lost. Whether that loss is impactful depends largely on the specific inquiry.

Measure symmetry is an especially meaningful distinction for evaluating polarity sensitivity, at least for the current approach⁵². Specifically, polarity sensitivity relationships are defined by backward association, and are relatively uninformed by forward association. This is because neither the data composition nor the theoretical concepts underlying the phenomenon suggest that association *should* be observed in both directions.

Table 2.60 illustrates the relative uninformativity of forwards association— $P(x | E)$ and $\Delta P(x | E)$ —in polarity sensitivity investigations. The pairings shown are “top” values for the indicated AMs. First, note that the values for these columns

⁵²I have ideas for other ways to partition the environments that might affect the frequency disparities at least.

are very low, especially compared to those seen for context-blind evaluation above. Second, the dP2 and P2 top selections do not coincide with any of the other selected AMs. Third, the P2 and dP2 columns seem to correlate quite strongly with both the joint (joint frequency (f)), and bigram marginal frequencies (marginal frequency of second element (f2)). That is, these measures are a mark of how common a bigram is in general, rather than indicating an association with the environment.

Table 2.60: Relative **Un**informativity of Forwards Asymmetric AMs for Polarity Investigations

#	Polar Bigram	Top For	f	dP2	P2	f2	LRC
1	(+) <i>so many</i>	dP2; P2	1,047,841	0.015	0.015	1,049,935	4.4
2	(+) <i>most important</i>	dP2; P2	689,158	0.010	0.010	690,094	4.8
3	(-) <i>as good</i>	dP2; P2	44,441	0.012	0.014	197,963	2.6
4	(-) <i>so sure</i>	G2	26,480	0.008	0.008	34,127	6.1
5	(-) <i>always easy</i>	G2	24,459	0.008	0.008	26,815	7.6
6	(-) <i>immediately clear</i>	G2	24,416	0.008	0.008	26,002	8.2
7	(+) <i>increasingly difficult</i>	P1	18,107	<0.001	<0.001	18,107	5.4
8	(-) <i>yet clear</i>	LRC	10,406	0.003	0.003	10,471	10.8
9	(+) <i>most incredible</i>	P1	10,137	<0.001	<0.001	10,137	4.6
10	(-) <i>exactly sure</i>	LRC	8,794	0.003	0.003	8,840	10.9
11	(+) <i>relatively minor</i>	P1	8,679	<0.001	<0.001	8,679	4.4
12	(-) <i>yet ready</i>	LRC	7,501	0.002	0.002	7,575	10.2
13	(-) <i>yet final</i>	dP1	640	<0.001	<0.001	640	9.5
14	(-) <i>yet eligible</i>	dP1	448	<0.001	<0.001	448	8.9
15	(-) <i>yet official</i>	dP1	352	<0.001	<0.001	352	8.6

First of all, as an unavoidable reality, intrinsic aspects of the feature categories examined here entail that the l_1 category—the binary environment classification—must have vastly greater marginal frequencies for one or both categories⁵³. With

⁵³If the discrepancy between negative and positive environments were even *greater*, it would technically be possible for an adverb’s marginal frequency to reach a similar level.

these disparate marginals, it is almost impossible to perceive any forward association, because such calculations are basically asking the following question.

- (2.4) Does this 1-of-2 item with millions of observed tokens exhibit a *special* “preference” for a/any single 1-of-thousands/millions item?

For a non-linguistic comparison of this concept, consider Seattle and rain. Although Seattle is commonly associated with rainy weather, it is strange if not outright logically unsound to say that the relationship is bidirectional.

- (2.5) a. ✓ If you’re in Seattle, there’s a good chance it’s rainy.
b. ?? If it’s rainy, there’s a good chance you’re in Seattle.

The second claim, that rainy weather associates with Seattle, falls short of convincing because Seattle accounts for a relatively minuscule percentage of the area where rain falls globally and it’s not even the “rainiest” place in North America, much less the entirety of rain’s “domain”. In this analogy based on different frequencies and proportions of the search space, negative environments are the rain and individual adverbs/adjectives are Seattle/other specific localities.

From the theoretical angle, even when polarity sensitivity is uncontroversially present, the licensee must associate with its licenser, but the licenser is under no such compunction to license the licensee. To capture the “need-based” relationship between polarity environments and polarity items, a more accurate metaphor is to

say that negation is the ocean, while particular adverbs/adjectives are all the different species of sea creatures. Tuna is “licensed” by the ocean, but the ocean will not stop existing if tuna are fished?/hunted? to extinction.

- (2.6) a. ✓ If you’re a tuna fish, there’s a good chance you’re in the ocean.
b. ?? If you’re in the ocean, there’s a good chance you’re a tuna fish.

The ridiculousness of such reasoning is illustrated to comedic effect here, but the point is that, given the frequency discrepancies, the same is true of polarity environments (the ocean) compared to individual adverbs, adjectives, or bigrams (the tuna fish), even for polarity items that are uncontested in the literature.

Still the fact that the relationship is one-directional does not prevent the association from existing—oceans and tuna, and Seattle and rain are still conceptually linked.

Unidirectional association can still be identified by looking at the symmetric relationship because it is *half* of what is considered after all. However, the threshold required for a one-sided attraction to come through in an *overall* measure is much higher than that required when both directions show moderate association. For example, {5, 5}, {4, 6}, {2, 8}, and {1, 9} all have a mean of 5. For some inquiries, this average is still the most informative measure, but in the current inquiry of polarity licensing, measuring the backwards association is more efficient, at least for effect size. In these data, as already discussed, similar/balanced directional associations—e.g. {5, 5} and {4, 6}—are essentially impossible, while

the disparate/unbalanced directional associations—e.g. $\{2, 8\}$ and $\{1, 9\}$ —are the only possible configuration. The symmetric measurements would be identical for each of these, but the asymmetric scores—or rather, **one** of the values calculated by the asymmetric measures—would be able to detect the underlying values, making them better suited to identifying unbalanced association.

Since any polarity environment association is *expected* to be a one-way *backwards only* association—where the lexical unit (adverb, adjective, or composite bigram) associates with the polar environment, but not vice versa—(backwards) asymmetric measures are the most efficient for identifying polarity sensitivity. Specifically, backwards ΔP —**dP1** or $\Delta P(1|2)$ —is best suited to identify strength of sensitivity when the marginal frequency of first element (**f1**) values (environment marginal frequencies) differ greatly as in the **ALL+** data selection. Interestingly, when the **f1** values are equal, as in the random sampled **NEQ+** evaluation, backwards conditional probability—**P1** or $P(1|2)$ —is more informative. This can be attributed to the delta P “adjustment” no longer being needed. When there are just as many negative polarity tokens considered as there are positive polarity tokens, the simple ratio of co-occurrence is no longer misleading. It is when there is a large discrepancy in the number of tokens belonging to each polarity that the adjustment is needed. For more details in the math underlying this, see below.

The table above also clearly illustrates the lack of informativity of the *forward* association (all 0 or close thereto), confirming that the extent that the environments (ocean) associate with *exactly* (tuna), has a negligible effect on the overall association,

as it is perceived or as it is measured via the symmetric AMs. Since asymmetric measures only capture effect size, symmetric measures are important for hypothesis testing/significance, as well as providing a meaningful comparison point to the more commonly used measures (e.g. the “favorite”, G2) but asymmetric measures provide a clearer estimation of the target phenomenon, polarity sensitivity.

2.2.3.2 Conditional Probability, P

Representations: $P(1|2)$ and $P(2|1)$; P1 and P2 (backward and forward, respectively)

The information conveyed by the conditional probability (P1 and P2) is straightforward: the probability that x given $y = \frac{f(x \wedge y)}{f(y)}$, or the proportion of y tokens that are x .

In a given comparison space, the P1 values convey the ratio of tokens that fall in one environment versus the other. That is, P1 is the ratio of *exactly* tokens that occur in each environment. For example, $P(E|exactly) = \frac{f}{f_{exactly}}$ (i.e. joint E & *exactly* divided by the *exactly* subtotal or marginal frequency in the comparison space; i.e. $P1 = \frac{f}{f_2}$). This is directly equivalent to the “normalized” frequencies presented in § 2.2.2: values referred to previously as $f\%(-)$ are equal to $P(E_{(-)}|x) \times 100$ where x is the form being quantified. Because the environments are all binary partitions of the token set, these values will always sum to 1. The P2 values, however, are all close to zero as mentioned previously. This is because, in polarity evaluations, $P(l_2|l_1)$ represents the the proportion of the *environment subtotal* where the co-occurring

adverb is *exactly*.

This case is the first row in tables above; relevant values repeated below in Equation 2.6 and Equation 2.8.

environment (E)	ALL+	
	$*N^+$	$*C_{N^+}$
polarity	(-)	(+)
E & <i>exactly</i> (f)	43,813	12,296
E & * (f1)	3,173,681	68,787,692
	(+/−)	
* & <i>exactly</i> (f2)	56,109	
N	71,961,373	

Equation 2.4 defines P1 and Equation 2.5 defines P2. The following lines then provide provide and example calculations using observed frequencies for the adverb *exactly* and negative polarity environments in the (ALL+) superset. Equation 2.6 and Equation 2.7 illustrate P1 while Equation 2.8 and Equation 2.9 illustrate P2.

$$P(1|2) = P(l_1|l_2) = \frac{f(l_1 \wedge l_2)}{f(l_2)} \quad (2.4)$$

$$P(2|1) = P(l_2|l_1) = \frac{f(l_1 \wedge l_2)}{f(l_1)} \quad (2.5)$$

$$P(1|2) = P(*N|exactly) = \frac{f(*N \wedge exactly)}{f(exactly)} \quad (2.6)$$

$$= \frac{\mathbf{f}}{\mathbf{f}2} = \frac{43,813}{56,109} = 0.7809 \quad (2.7)$$

$$P(2|1) = P(exactly|*N) = \frac{f(*N \wedge exactly)}{f(*N)} \quad (2.8)$$

$$= \frac{\mathbf{f}}{\mathbf{f}1} = \frac{43,813}{3,173,681} = 0.0138 \quad (2.9)$$

The values indicate that 78.09% of *exactly* ADJ tokens (in the ALL+ superset evaluation space) are negated, while only 1.38% of the negated (ALL+:*N) environment tokens contain *exactly*—*Tuna, meet Ocean. Ocean, meet Tuna*. Overall, this supports my suggestion that backwards association (in this case, P1) is far more informative in evaluating polarity sensitivity than forward association (P2).

2.2.3.3 Delta P, an adjusted conditional probability

ΔP (Gries 2013) (pronounced “delta p”) is quite similar to conditional probability, but contain an adjustment term that highlights the deviation from typical distributions. As Equation 2.10 indicates, the “delta” in question is the discrepancy between the conditional probability of x with y and that of x without y .

$$\Delta P(x|y) = P(x|y) - P(x|\neg y) \quad (2.10)$$

Since ΔP in an asymmetric [AM](#) just like the conditional probability it is derived from, there are always two separate values for any given pair of elements. *Backwards delta P* is represented in this work as both $\Delta P(1|2)$ and [dP1](#), where “1” represents the leftmost/first element and “2” represents the rightmost/second element. Correspondingly, forwards delta P is represented as both $\Delta P(2|1)$ and [dP2](#), with the same designations.

This is an effect size measure essentially, which indicates the divergence from all other relevant observations, an estimation of “chance”. Concerning *exactly* and polarity specifically, $\Delta P(E_{(-)}|exactly)$ represents how much the probability of negation preceding *exactly* diverges from what is normal for “the rest”, represented by the probability of any other adverb type being negated. Likewise, $\Delta P(E_{(+)}|exactly)$ indicates how much the probability of *exactly* being positive diverges from any other adverb being positive.

dP1, “Delta P” of ALL + super : NEG $\sim$ *exactly*

$$= \Delta P(*N|exactly)$$

$$= P(*N|exactly) - P(*N|any other adverb)$$

$$P(*N|exactly) = 0.7809$$

$$\begin{aligned} & P(*N|any other adverb) \\ = & \frac{f(*N) - f(*N \wedge exactly)}{N - f(exactly)} \\ = & \frac{3,173,681 - 43,813}{71,961,373 - 56,109} \\ = & \frac{3,129,868}{71,905,264} \\ = & 0.0435 \end{aligned}$$

$$\Delta P(*N|exactly)$$

$$= 0.7809 - 0.0435$$

$$= \mathbf{0.7374}$$

The dP2 value works the same, just reversed: Since the environment category is a binary choice, $\neg *N$ is equivalent to the non-negated complement $*C_N$ $\Delta P(exactly|*N) = P(exactly|*N) - P(exactly|*C_N)$; i.e., $\frac{\# \text{ negative exactly}}{\text{total } \# \text{ negative}} - \frac{\# \text{ positive exactly}}{\text{total } \# \text{ positive}}$. However, both of these values are close to zero, and the difference remains essentially the

same. The “adjustment” is far more drastic for the positive cases in the ALL+ data selections: Because the positive environments account for such a large proportion of tokens, the magnitude of the negative term is often larger than that of the positive term, leading to many ΔP values, whether forward *or* backward, that are zero or less. Another result of the polarities being a binary division of the tokens is that the magnitude of the ΔP value for one environment is the opposite of the other. That is, where this feature results in P values that sum to 1 for any given lexical form in a comparison space, the ΔP values sum to 0 (the same magnitude but different signs). The calculations below confirm this.

$$\begin{aligned}
& \Delta P(E_{(+)}|x) \\
&= -1 \times \Delta P(\neg E_{(+)}|x) \\
&= -1 \times \Delta P(E_{(-)}|x)
\end{aligned}$$

$$\begin{aligned}
& \Delta P(*N|exactly) \\
&= \frac{f_{(*N,exactly)}}{f_{exactly}} - \frac{f_{*N} - f_{(*N,exactly)}}{N - f_{exactly}} \\
&= 0.7374
\end{aligned}$$

$$\begin{aligned}
& \Delta P(*C_N|exactly) = (f_{(*C_N,exactly)} \div f_{exactly}) \\
& \quad - [(f_{*C_N} - f_{(*C_N,exactly)}) \div (N - f_{exactly})] \\
&= (12,296 \div 56,109) - [(68,787,692 - 12,296) \div (71,961,373 - 56,109)] \\
&= 0.2191 - [68,775,396 \div 71,905,264] \\
&= 0.2191 - 0.9565 \\
&= -0.7374
\end{aligned}$$

The reason for the additive inverse relationship of **dP1** in binary polarity evaluations is a consequence of the binary division of all tokens between the two polarity environments (in any given comparison space), as shown in (2.7). This circumstance means that the backward conditional probability values must sum to 1 or 100%, as verified by (2.8). Since delta P is built on conditional probability—and

specifically, **dP1** is built on **P1**—the subtraction of the second term means the **dP1** values must sum to 0, which is subsequently verified in (2.9). This same logical necessity is seen in several other measures as well, because they depend on similar mechanisms to compare the frequencies of the possible contingencies.

(2.7) Joint frequencies of lexical unit, x , within binary environments, $\ominus$ and $\oplus$, sum to $f(x)$.

$$f(\ominus \wedge x) + f(\oplus \wedge x) = f(x)$$

$$f(\ominus \wedge x) = f(x) - f(\oplus \wedge x)$$

(2.8) Backwards conditional probabilities of lexical unit, x , within binary environments, $\ominus$ and $\oplus$, sum to 1.

Proof.

$$P(E|x) = \frac{f(E \wedge x)}{f(x)}$$

$$P(\oplus|x) = \frac{f(\oplus \wedge x)}{f(x)}$$

$$P(\ominus|x) = \frac{f(\ominus \wedge x)}{f(x)}$$

$$P(\ominus|x) = \frac{[f(x) - f(\oplus \wedge x)]}{f(x)}$$

$$P(\ominus|x) \times f(x) = f(x) - f(\oplus \wedge x)$$

$$\frac{P(\ominus|x) \times f(x)}{f(x)} = \left[\frac{f(x)}{f(x)} \right] - \left[\frac{f(\oplus \wedge x)}{f(x)} \right]$$

$$P(\ominus|x) = 1 - \left[\frac{f(\oplus \wedge x)}{f(x)} \right]$$

$$P(\ominus|x) = 1 - P(\oplus|x)$$

$$\therefore P(\ominus|x) + P(\oplus|x) = 1$$

□

(2.9) Backwards delta P values of lexical unit, x , within binary environments, $\ominus$ and $\oplus$, sum to 0.

Proof.

$$x' = \neg x$$

$$\Delta P(E|x) = P(E|x) - P(E|\neg x)$$

$$\Delta P(\ominus|x) = P(\ominus|x) - P(\ominus|x')$$

$$\Delta P(\oplus|x) = P(\oplus|x) - P(\oplus|x')$$

$$\Delta P(\ominus|x) + \Delta P(\oplus|x)$$

$$[P(\ominus|x) - P(\ominus|x')] + [-P(\oplus|x) + P(\oplus|x')]$$

$$[P(\ominus|x) + P(\oplus|x)] - [P(\ominus|x') + P(\oplus|x')]$$

$$1 - 1 = 0$$

$$\therefore \Delta P(\ominus|x) + \Delta P(\oplus|x) = 0$$

□

A final point on ΔP is that the adjustment it incorporates is unnecessary when the marginal frequencies of binary environments are equal, as is the case for **NEQ+** perspectives. In such evaluations, the backwards conditional probabilities are on a comparable scale already and “chance”, or the null hypothesis of independence is no longer extremely unbalanced as seen in the **ALL+** perspectives. Instead, “chance” or

independence is equivalent to conditional probabilities of 0.5 for both environments. Both $P(\ominus|x) = 0.5$ and $P(\oplus|x) = 0.5$ indicate that x occurs exactly as expected given the marginal frequencies of the environments and x , thereby exhibiting no deviation from the null hypothesis assumption of independence.

The range of magnitudes possible for backwards Delta P calculated on binary polarities is directly dependent on the disparity observed in the environment marginal frequencies. Since ΔP essentially measures the deviation from the “typical” probability of co-occurrence seen for each polarity, and disparity between mean or even maximum lexical (i.e. adverb, adjective, or bigram) marginal frequencies is so vast, the “adjustment” term (probability of environment given any type **besides** x) nears the probability of the environment overall: $P(E|\neg x) \approx P(E) (= \frac{f(E)}{N})$. It may be slightly less than $P(E)$, but $P(E)$ is the maximum deduction that can be applied in any calculation of $\Delta P(E|x)$. Therefore, while the possible range for $P(E|x)$ is always $[0, 1]$, the possible range for $\Delta P(E|x)$ is $[-(1 - P(E)), +(1 - P(E))]$. The maximum magnitude for $\Delta P(\ominus|x)$ is $1 - P(\ominus)$ which is equal to $P(\oplus)$. Similarly, the maximum magnitude possible for $\Delta P(\oplus|x)$ is $P(\ominus)$. In the $*S :ALL+$ space, the negative ($*N$) proportion of tokens is 0.04,⁵⁴ while the positive ($*L_N$) proportion is 0.96,⁵⁵ meaning the negative **dP1**, $\Delta P(\ominus|x)$ (i.e., $\Delta P(*N|x)$), maximum magnitude of 0.96 dwarfs that of the positive, $\Delta P(\oplus|x)$ (i.e., $\Delta P(*L_N|x)$), which is 0.04. In the $@S :ALL+$ space, the maximum negative polarity dP1 ($\Delta P(\ominus|x) = \Delta P(@N|x)$) magnitude is 0.83 while the maximum magnitude for positive polarity

⁵⁴3,173,681 of all 71,961,373 $*S$ tokens are categorized as negative polarity.

⁵⁵68,787,692 of all 71,961,373 $*S$ tokens are categorized as positive polarity.

dP1 ($\Delta P(\oplus|x) = \Delta P(@\mathbf{P}|x)$) is 0.17. As already previewed, the maximum dP1 magnitude for **either** polarity environment in the NEQ+ spaces is 0.5.

CHAPTER 3

NOT *EXACTLY* COMPREHENSIVE RESULTS

3.1 Introduction to Polarity Sensitivity Inquiries using PBR

This chapter provides an in depth case study of the adverb “exactly” as informed by PBR. This case study is intended to illustrate the potential application and usefulness of PBR and the associated methods by providing fine-grained inspections of the many different perspectives and comparisons that PBR makes possible, and demonstrating how such careful considerations can interface with linguistic theory. That is, PBR should enable a myriad of potential applications for linguistic theory, including both exploration and verification research aims (i.e., open-ended questions to generate preliminary hypotheses and develop theories, or specific research questions to test existing hypotheses/theories), both *informing* and being *informed by* linguistic theory. In this specific exploration, PBR is utilized in generating a preliminary theory for the underlying nature of pre-adjectival “exactly” with regards to polarity environment—that the data patterns observed in PBR suggest that “exactly” is contingently (negative) polarity sensitive depending on the compositional semantics of specific scalar features of the adjective being modified.

3.2 Exposition of the Polar Propensities of *Exactly*

The adverb *exactly* exhibits a positive association with (approximated) negative polarity environments⁵⁶, based on both quantitative and qualitative evidence from corpus data described in [Chapter 6](#) and methods described in [Chapters 4](#) and [5](#).

[§ 3.2.1](#) presents quantitative evidence in the form of direct raw frequency comparisons between polarities across different data selections and polarity approximation methods (in [§ 3.2.1.1](#)) and calculated association metrics, considering both symmetric and asymmetric measures (in [§ 3.2.1.2](#)).

[§ 3.2.2](#) supports this with qualitative evidence in the evaluation of sentences observed in the corpora. This includes the obvious—examples where *exactly* * bigrams appear in negative environments (see [§ 3.2.2.1](#)). But it also includes fabricated alternate versions of these perfectly acceptable negative polarity *exactly* bigram examples (both superset *N and subset @N), which consistently degrade with the removal of the element contributing the negative environment (see [§ 3.2.2.2](#) for details).

⁵⁶see [Chapter 2](#) for details of polarity approximation methods

3.2.1 Exactly Association Quantified by the Corpus

3.2.1.1 Raw Frequency Evidence

Table 3.1 presents raw polar frequency comparisons for *exactly* across all four comparison spaces: the entire token total (N), adverb *exactly* token subtotal (f2), polarity environment token subtotal (f1), and observed co-occurrence (i.e. joint) frequency of *exactly* with the given environment across the four distinct comparison classes (f). The final column, fu, or “f values unexpected”, indicates the difference of the observed joint frequency (f) and the expected joint frequency ($f_{observed} - f_{expected}$) for the given marginal frequencies and the assumption of independence—the null hypothesis: under-performance (fewer joint tokens than expected) is indicated by values < 0 and over-performance (more joint tokens than expected) is indicated by values > 0 .

Table 3.1: Polarity Env~*exactly* **Raw Frequencies** Across Comparison Spaces

Set	Sampling	Polar	f1	f2	N	f	fu
*E	ALL+	(+)	68,787,692	56,109	71,961,373	12,296	−41,338
		(−)	3,173,681	56,109	71,961,373	43,813	41,338
	NEQ+	(+)	3,173,681	44,378	6,347,362	565	−21,624
		(−)	3,173,681	44,378	6,347,362	43,813	21,624
@E	ALL+	(+)	1,388,898	1,031	1,680,633	218	−634
		(−)	291,735	1,031	1,680,633	813	634
	NEQ+	(+)	291,735	862	583,470	49	−382
		(−)	291,735	862	583,470	813	382

Across all four evaluated comparison spaces, *exactly* occurred more often in

environments identified as negative. This holds regardless of whether polarity environments were selected via the absence of negative triggers (i.e. $*S$ superset), or the presence of positive triggers (i.e., $@S$ mirror subset). The total examined volume also does not change this fact. Table 3.2, which presents the frequencies normalized by comparison space size as tokens per million⁵⁷, indicates *exactly* has more negated tokens per million when the representation of the positive environment was restricted to a random sample the size of the corresponding negative environment—the NEQ+ frequency perspective.

Table 3.2: Polarity ENV $\sim$ *exactly* **Normalized Frequencies** (tokens/million) Across Comparison Spaces

Set	Sampling	Polar	f (tpm)	f1 (tpm)	f2 (tpm)	fu (tpm)
$*S$	ALL+	(+)	171	955,897	780	−574
		(−)	609	44,103	780	574
	NEQ+	(+)	89	500,000	6,992	−3,407
		(−)	6,903	500,000	6,992	3,407
$@S$	ALL+	(+)	130	826,414	613	−377
		(−)	484	173,586	613	377
	NEQ+	(+)	84	500,000	1,477	−655
		(−)	1,393	500,000	1,477	655

Figure 3.1 provides a visualization of the relative comparison space sizes (N), and the *exactly* marginal frequencies ($f2$), both raw and normalized (tokens per million), and Figure 3.2 depicts the observed and unexpected frequency values for each space and polarity classification.

⁵⁷“tokens per million [tokens]” is equivalent to $\frac{f}{N} \times 10^6$

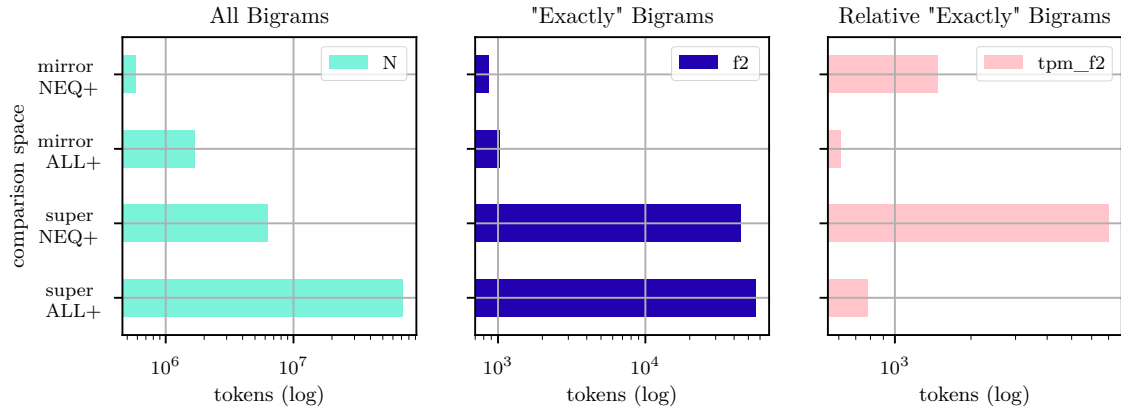


Figure 3.1: Comparison Space Frequencies for *Exactly*: N & marginal frequency

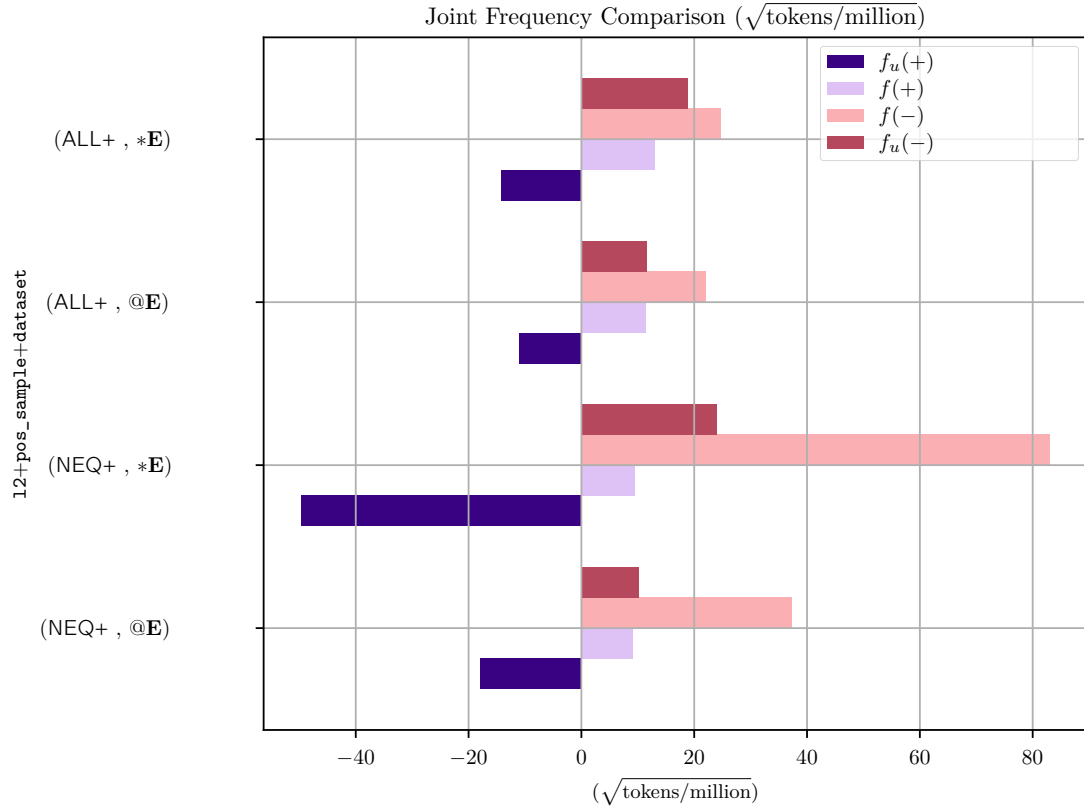


Figure 3.2: Observed vs. Unexpected Joint Frequencies for *Exactly*

The more interesting fact is that the the raw count is *also* higher when the positive environment representation is left unlimited. As discussed in [Chapter 2](#), the odds of any given token occurring in a positive environment are 22:1 in the superset $*S$ token set. The mirror subset $@S$ reduces this discrepancy significantly, but it is still 5:1 in favor of a positive environment in the mirror subset. These relationships are reiterated in [Table 3.3](#).

Table 3.3: Overall (unrestricted/ALL+) Environment Comparisons

	$*S : \textit{Superset}$		$@S : \textit{Mirror Subset}$	
	# Tokens	% of N	# Tokens	% of N
Positive	68,787,692	95.6%	1,388,898	82.6%
Negative	3,173,681	4.4%	291,735	17.4%
N (combined)	71,961,373	100.0%	1,680,633	100.0%
Odds of being Positive:	22:1		5:1	

Despite the overwhelming odds overall that tokens will be *positive*, the odds are 3.6:1 that an *exactly* bigram will be *negative* in $*S$ and 3.7:1 that it will be negative in $@S$. The [NEQ+](#) comparison spaces simply highlight the disparity of *exactly*’s negative trend, with odds of negative occurrences surpassing 16:1 for $@S$ [NEQ+](#) and **77:1** for $*S$ [NEQ+](#). This illustrates that the joint frequency of *exactly* in positive environments describes a much sparser collection than *exactly* in negative environments. These *exactly* specific relationships are summarized in [Table 3.4](#).

Table 3.4: *Exactly* specific comparisons

sampling	dataset	(-) odds	polarity	$\frac{f}{N}$ %	$\frac{f1}{N}$ %	$\frac{f2}{N}$ %
ALL+	*S	3.6:1	(+) (-)	0.02 0.06	95.6 4.4	0.08
	@S	3.7:1	(+) (-)	0.01 0.05	82.6 17.4	0.06
NEQ+	*S	77.5:1	(+) (-)	0.01 0.69	50	0.70
	@S	16.6:1	(+) (-)	0.01 0.14	50	0.15

f = joint frequency of polarity & *exactly* ADJ
 $f1$ = marginal frequency of polarity
 $f2$ = marginal frequency of *exactly* ADJ

The total number of unique *exactly* * bigrams attested (more than five times; $f \geq 5$) is also highly skewed toward negative environments, as illustrated in Table 3.5. This is a rephrasing of the strict versatility introduced in Chapter 2 (Table 2.47) with the addition of versatility calculations based on NEQ+ and mirror subset @S comparison spaces as well⁵⁸. Essentially, even if the total frequencies were closer, the fact that *exactly* is so much more *versatile* within negative environments motivates further investigation.

⁵⁸Prior versatility calculations (presented in Chapter 2) evaluated only the ALL+ *S comparison space

Table 3.5: Strict Versatility—Unique *Exactly* * bigrams attested ($f \geq 5$)

sampling	dataset	unique overall $\mathbb{V}(\pm)$	polarity	unique per polarity $\mathbb{V}$	proportionally $\mathbb{V}\%$
ALL+	* <i>S</i>	920	(+)	110	12.0%
			(−)	848	92.2%
	@ <i>S</i>	24	(+)	10	41.7%
			(−)	19	79.2%
NEQ+	* <i>S</i>	854	(+)	15	1.8%
			(−)	848	99.3%
	@ <i>S</i>	20	(+)	2	10.0%
			(−)	19	95.0%

3.2.1.2 Association Measure Evidence

The reduced “density” suggested by the disparity between the ALL+ and NEQ+ comparisons is an important feature in interpreting corpus frequency data. To quantify this importance more formally, association measures take into account frequencies of all contingencies for the elements being compared. Table 3.6—showing the $E \sim \textit{exactly}$ scores for LRC, dP1, P1, and G2 alongside raw joint frequency, normalized joint frequency, and normalized unexpected frequency—confirms the pattern of association implied by the raw frequency data covered in § 3.2.1.1. For all of the included measures with negative values, these indicate a “negative correlation” or “repulsion” between exactly and the specified environment, while positive values indicate the opposite—a “positive correlation” or “attraction”.⁵⁹

⁵⁹While these distinctions are commonly referred to as positive or negative association, depending on the sign of the numerical value, the terms are inconveniently similar to positive and negative polarity references. I will default to “attraction” and “repulsion” for the sake of clarity.

Although the strength of the association varies by metric and comparison space, all cases suggest *exactly* has an attraction with negative polarity environments and a repulsion from positive polarity environments. The AMs are also illustrated with bar charts in Figure 3.3. For the sake of transparency, Table 3.7 shows the forwards asymmetric dP2 and P2, despite their relative *uninformativity* (as discussed at length in § 2.2.3.1).

Table 3.6: Compiled AMs for *exactly*
(tpm=“tokens per million”)

Set	Sampling	Polarity	dP1	LRC	P1	G2	f	f (tpm)	fu (tpm)
*S	ALL+	(+)	−0.74	−6.2	0.219	−216,170	12,296	171	−574
		(−)	0.74	6.2	0.781	216,170	43,813	609	574
	NEQ+	(+)	−0.49	−6.0	0.013	−55,764	565	89	−3,407
		(−)	0.49	6.0	0.987	55,764	43,813	6,903	3,407
@S	ALL+	(+)	−0.62	−3.6	0.211	−1,868	218	130	−377
		(−)	0.62	3.6	0.789	1,868	813	484	377
	NEQ+	(+)	−0.44	−3.0	0.057	−820	49	84	−655
		(−)	0.44	3.0	0.943	820	813	1,393	655

Table 3.7: Asymmetric AMs for *Exactly*

Set	Sampling	Polarity	dP1 <i>backward</i>	dP2 <i>forward</i>	P1 <i>backward</i>	P2 <i>forward</i>	f	f1	f2
*E	ALL+	(−)	0.737	0.014	0.781	0.014	43,813	3,173,681	56,109
		(+)	−0.737	−0.014	0.219	0.000	12,296	68,787,692	56,109
	NEQ+	(−)	0.491	0.014	0.987	0.014	43,813	3,173,681	44,378
		(+)	−0.491	−0.014	0.013	0.000	565	3,173,681	44,378
@E	ALL+	(−)	0.615	0.003	0.789	0.003	813	291,735	1,031
		(+)	−0.615	−0.003	0.211	0.000	218	1,388,898	1,031
	NEQ+	(−)	0.444	0.003	0.943	0.003	813	291,735	862
		(+)	−0.444	−0.003	0.057	0.000	49	291,735	862

The descriptive statistics in general illustrate the divergent score distributions. The rows in Table 3.8 correspond to the different divisions of all *exactly* bigrams in all

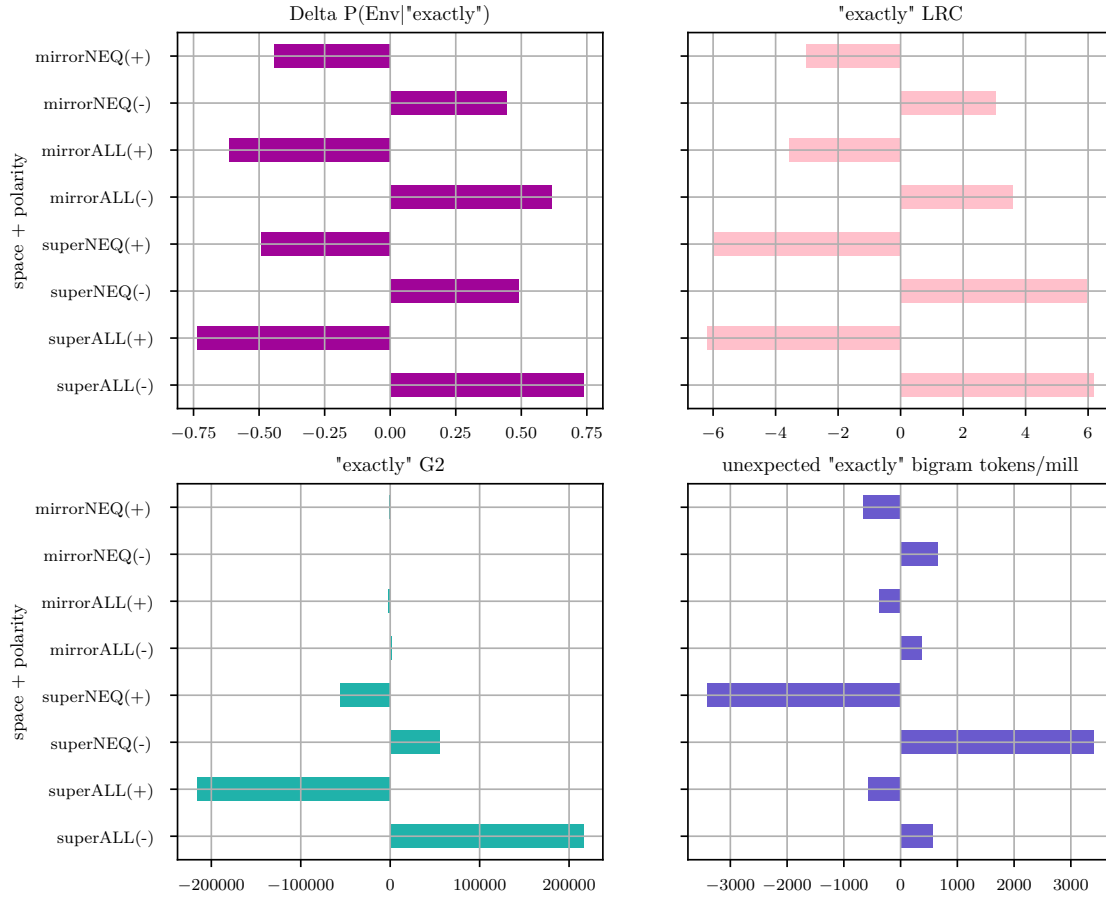


Figure 3.3: Environment Associations of *Exactly*: $\Delta P(E|exactly)$, LRC, G2, and unexpected tokens per million

Table 3.8: Descriptive Statistics for all *exactly* * bigram evaluations (s.t. $f \geq 5$)

metric	<i>exactly</i> bigrams	count	mean	std	min	25%	median	75%	max
dP1	positive only	137	-0.37	0.34	-0.95	-0.72	-0.29	-0.08	0.17
	negative only	1,810	0.69	0.23	-0.09	0.50	0.50	0.96	0.96
	ALL	1,947	0.61	0.36	-0.95	0.50	0.50	0.96	0.96
LRC	positive only	137	-1.85	2.58	-10.93	-3.40	-0.24	0.00	0.00
	negative only	1,810	1.47	2.06	0.00	0.00	0.11	2.67	10.93
	ALL	1,947	1.24	2.27	-10.93	0.00	0.00	2.33	10.93

different comparison spaces that met the frequency minimum of 5 occurrences (i.e. the LRC and dP1 values for each of the unique bigram types indicated in Table 3.5), with “**negative** only” corresponding to the negative polarity subset, “**positive** only” corresponding to the positive polarity subset, and “*all*” indicating all 1,947 unique $E \sim \textit{exactly}$ ADJ pairings across both polarities. Clearly, the bulk of polar preferences found within the unique bigrams are with negative polarity environments. The LRC floor and ceiling for each subset encapsulate this neatly: no individual *exactly* * bigram was found to have a significant positive association with positive polarity. Looking at the negative only subset, at least a quarter of the unique *negative*~*bigram* pairs have significant positive LRC scores and are 96% more likely to appear with negation than any other bigram (with any adverb or adjective) is, and a full three quarters of the unique *exactly* bigrams are 50% more likely to appear in negative polarity environments compared with any other bigram type.

The example of *immediately*⁶⁰ in Chapter 2 raised the potential of an adverb’s general polarity distribution being carried by a single or small number of bigrams. This is not the case for *exactly*, as the versatility values provided in Table 3.5 already suggest. The *exactly* bigrams shown in Table 3.9—just the 25 most strongly negative prone—indicate that a large variety of bigrams show a similar propensity for negative environments. No polarity filtering was forced on this selection—there simply are no *exactly* bigrams in PBR which yield LRC values > 0 for positive polarity.

This selection is determined by meeting the preliminary *maximum* bigram marginal frequency threshold of 50⁶¹ as well as ranking in the top 13 of the polar mean across all comparison spaces of either LRC or dP1, and then ranked by descending polar mean LRC (*LRCm*). That is, each value shown is the mean of the four values calculated for $E \sim \textit{exactly ADJ}$ across the the four comparison spaces. Calculating the mean for positive and negative polarity AMs would result in zero values, since the negative polarity LRC and dP1 values are additive inverses of the positive polarity calculations for the same evaluation. For example, the calculations for the *LRCm* and *dP1m* values in Table 3.9 for $((-) \textit{exactly}) \textit{cheap}$ are illustrated in Table 3.10.⁶²

⁶⁰Table 2.49 indicated that over 60

⁶¹To be considered as a top "polar sensitive" adjective argument for *exactly*, the maximum bigram subtotal (i.e. superset ALL+ f2) must be at least 50 tokens.

⁶²The mean values for dP1 (and P1) are less informative than for LRC because the range of values—and their interpretation differ between comparison spaces: The maximum dP1 magnitude possible in the NEQ+ spaces is 0.5 while the maximum in the ALL+ spaces is 1. However, since all bigrams are impacted, the relative mean values are still valid for ranking purposes. See ?? and ?? for detailed discussion.

Table 3.9: Top 25 Most Polarity Sensitive *Exactly* * Bigrams—All Negative selected by cross-space means of LRC (LRCm) & backwards Delta P (dP1m)

#	(-) <i>Exactly...</i>	LRCm	dP1m	P1m	G2m	fm	f2m
1	<i>sure</i>	7.10	0.694	0.999	16,806	4,471	4,483
2	<i>clear</i>	5.27	0.691	0.995	3,309	899	908
3	<i>new</i>	4.67	0.684	0.989	2,601	700	705
4	<i>easy</i>	4.44	0.694	0.999	2,039	543	545
5	<i>cheap</i>	3.56	0.694	0.998	1,311	349	350
6	<i>surprising</i>	3.40	0.695	0.999	843	223	223
7	<i>happy</i>	3.23	0.692	0.997	824	221	222
8	<i>ideal</i>	3.11	0.691	0.995	779	211	213
9	<i>true</i>	3.01	0.561	0.866	2,444	730	784
10	<i>subtle</i>	2.97	0.694	0.998	503	135	135
11	<i>great</i>	2.93	0.689	0.994	573	156	158
12	<i>conductive</i>	2.91	0.696	1	399	105	105
13	<i>fair</i>	2.90	0.360	0.496	486	130	131
14	<i>friendly</i>	2.74	0.696	1	321	85	85
15	<i>shocking</i>	2.66	0.696	1	289	76	76
16	<i>pleasant</i>	2.62	0.696	1	272	72	72
17	<i>revolutionary</i>	2.57	0.696	1	240	65	65
18	<i>famous</i>	2.55	0.696	1	252	67	67
19	<i>difficult</i>	2.53	0.696	1	242	64	64
20	<i>shy</i>	2.51	0.696	1	241	64	64
21	<i>impressive</i>	2.33	0.696	1	188	50	50
22	<i>uncommon</i>	2.25	0.696	1	169	45	45
23	<i>straightforward</i>	2.19	0.696	1	159	42	42
24	<i>unexpected</i>	2.19	0.696	1	160	43	43
25	<i>welcome</i>	2.13	0.696	1	146	39	39

Table 3.10: Polar Mean Calculations Across All Comparison Spaces for *exactly cheap*

		<i>exactly cheap</i>					
<i>comparison</i>		<i>LRC</i>		<i>dP1</i>		<i>f</i>	
<i>space</i>		(−)	(+)	(−)	(+)	(−)	(+)
@ <i>S</i>	ALL+	0.00	0.00	0.83	−0.83	7	0
	NEQ+	0.00	0.00	0.50	−0.50	7	0
* <i>S</i>	ALL+	9.11	−9.11	0.95	−0.95	691	4
	NEQ+	5.12	−5.12	0.50	−0.50	691	1
polar mean		3.56	−3.56	0.69	−0.69	349	1.3
mean		0.00		0.00		175.1	

3.2.2 *Exactly* Association Qualified by the Corpus

The following examples of *exactly* occurring “in the wild” provide a better sense of what the above values actually mean. Since the objective right now is only to illustrate the basic concepts, these examples were intentionally limited by their complexity.

3.2.2.1 Simple Negative Polarity Examples: “Not exactly”

The quantitative evidence that *exactly* occurs in negated contexts presented in the preceding section can be corroborated by qualitative inspection. *Exactly* appears modifying a broad array of adjectives following the lemma *not* in the following sentences extracted from [Puddin](#) and [News](#) corpora.

Table 3.11: Examples of *Exactly* * Bigrams with the Negative Trigger Lemma, **Not**

Adjective	Corpus Example Utterance	bigram_id
acceptable	<i>It's happened three times already, and isn't exactly acceptable for a team of Ferrari's caliber to mess up like this.</i>	pcc_eng_18_031 .6854_x0496656_23 :11-12
alike	<i>Bradley... and therefore my teaching approaches can not be exactly alike for every student.</i>	pcc_eng_19_014 .4354_x0217054_16 :11-12
	<i>And the labels are very similar... but not exactly alike.</i>	pcc_eng_20_054 .1180_x0858085_20 :10-11
	<i>While school and home environments may not be exactly alike, schools can still promote general safety strategies and ease parental concerns.</i>	pcc_eng_18_030 .8118_x0482529_11 :09-10
alive	<i>The city of Osgiliath was never rebuilt after the War and many long years later it is the home of one who is not exactly alive, but not exactly dead either.</i>	pcc_eng_19_096 .7181_x1547451_03 :25-26
amicable	<i>Ben has been a negative in CHI, and the way he left the Pistons wasn't exactly amicable.</i>	pcc_eng_08_065 .1434_x1039017_092 :18-19
apologetic	<i>Notes: Oakley, asked prior to Tuesday night's game about punching McInnis a month ago, wasn't exactly apologetic.</i>	apw_eng_20010103 _0245_17 :21-22
appetizing	<i>but the display of seaweeds in O'Shea's kitchen was not exactly appetizing.</i>	nyt_eng_19970311 _0560_40 :12-13
brilliant	<i>As a panel we weren't exactly brilliant at playing the game—it really made you appreciate how good at it the likes of Gilbert Harding were as they'd get down to the nub of the issue in as few questions as possible compared with our flappy approach.</i>	pcc_eng_08_089 .9052_x1439196_06 :7-8
certain	<i>He says he's not exactly certain just how many signs and other automotive related artifacts he has, but conservatively guesses the total to be in the 400-500 item range.</i>	pcc_eng_06_094 .0669_x1505280_43 :6-7
	<i>The problem lies in the fact that many would be investors aren't exactly certain what specific types of investment they wish to make.</i>	pcc_eng_12_026 .9751_x0420335_2 :14-15

Continued on next page

Table 3.11: *Exactly* * Bigrams with the Negative Trigger Lemma, **Not** (cont'd)

Adjective	Corpus Example Utterance	bigram_id
cheap	<i>The little robot can talk to children, and play with them, but at Rs. 19,000 it's not exactly cheap.</i>	pcc_eng_08_047 .2145_x0748051_04 :21-22
clean	<i>This adorable young fox cub made a not exactly clean escape from a spot of bother when he was hauled from a building site caked in mud.</i>	pcc_eng_19_102 .3482_x1638512_01 :09-10
clear	<i>It will be free to download on the PC and iOS, and it's not yet exactly clear how it will make money.</i>	pcc_eng_20_098 .3925_x1573540_090 :19-20
	<i>it is not exactly clear what Gao was exporting to China after she returned to the United States.</i>	apw_eng_20060123 _1186_21 :4-5
	<i>I suppose it's not exactly clear whether candy corn is an ingredient in a smoothie titled "candy corn smoothie", and can get confusing when the recipe claims to be healthy.</i>	pcc_eng_06_094 .2360_x1508034_03 :6-7
cold	<i>"The weather isn't exactly cold; it's cooler than it has been," said Chris Schachte, a trader at GSC Energy Corp. in Atlanta.</i>	nyt_eng_19980116 _0526_5 :6-7
conductive	<i>Of his 29 Premier League appearances at the time of writing, he's only completed the 90 minutes three times—a situation which isn't exactly conductive to his long-term development.</i>	pcc_eng_08_030 .5855_x0478963_30 :28-29
	<i>it's true that the atmosphere under the new Israeli government wasn't exactly conductive to such a "We Are the World" project, but the show's groundwork had been laid years earlier and most people felt that "Sesame Street's" brand of Muppet love was needed now more than ever.</i>	nyt_eng_19980325 _0365_23 :14-15
	<i>The cabins on our boat weren't exactly conductive to humping.</i>	pcc_eng_08_010 .3113_x0150539_37 :8-9
controversial	<i>There isn't even a token effort to deal with a social issue other than the not exactly controversial position that genocide is bad.</i>	pcc_eng_18_061 .3552_x0976950_05 :18-19
correct	<i>I took the not exactly correct train to the not exactly correct location—but!</i>	pcc_eng_06_091 .0017_x1455722_08 :11-12

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Table 3.11: *Exactly* * Bigrams with the Negative Trigger Lemma, **Not** (cont'd)

Adjective	Corpus Example Utterance	bigram_id
	<i>First, it isn't exactly correct.</i>	pcc_eng_20_047 .8764_x0757285_09 :6-7
culinary	<i>If there's a category "cake," it's not exactly culinary and it's not exactly linguistic.</i>	pcc_eng_08_013 .6554_x0204493_21 :13-14
deadly	<i>it's not exactly deadly, of course, but it can still be harmful.</i>	nyt_eng_19990513 _0089_2 :4-5
difficult	<i>it's not exactly difficult.</i>	apw_eng_20041006 _0407_8 :4-5
	<i>It isn't exactly difficult to figure out that singer John Legend is seriously talented.</i>	pcc_eng_20_064 .1015_x1019478_03 :4-5
	<i>The benefits of these apps are not exactly difficult to find, though some of them may certainly be unexpected.</i>	pcc_eng_12_065 .3428_x1040113_23 :8-9
dirty	<i>They're not exactly dirty but they're not exactly clean either.</i>	pcc_eng_08_048 .0065_x0760944_28 :4-5
dry	<i>Taking its inspiration from the Welsh Brecon Beacons mountain range, known for its not exactly dry uphill trail routes, the Seal Skinz Brecon Gloves offer complete water and windproofing to help keep you moving through the very worst weather conditions.</i>	pcc_eng_20_027 .3272_x0425642_23 :16-17
	<i>Don't you notice that mine aren't exactly dry?</i>	pcc_eng_06_107 .6511_x1725156_07 :09-10
easy	<i>the strike was not exactly easy on Bonds, who lost some \$1 million of his \$4.2 million salary.</i>	nyt_eng_19950408 _0039_24 :5-6
	<i>Though it wasn't exactly easy to do, given his appearance, Tulvir sought to keep his Ice Wing heritage hidden from his fellow Fire Wings.</i>	pcc_eng_08_028 .6081_x0447095_011 :5-6

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Table 3.11: *Exactly* * Bigrams with the Negative Trigger Lemma, **Not** (cont'd)

Adjective	Corpus Example Utterance	bigram_id
	<i>It's not exactly easy to get here!</i>	pcc_eng_12_040 .2110_x0634181_61 :4-5
economical	<i>That's why replacing a \$26 bottle of dry shampoo every few weeks isn't exactly economical.</i>	pcc_eng_08_030 .5863_x0478979_26 :17-18
endangered	<i>beyond reuniting John Cleese, Michael Palin, Jamie Lee Curtis and the ever-nimble Kevin Kline (and giving these actors new roles that make this an encore rather than a sequel), "Fierce Creatures" takes aim at a not exactly endangered species: the corporate predator.</i>	nyt_eng_19970123 _0392_3 :44-45
equal	<i>"It would permit, while not exactly equal, more similar treatment among everybody," he told The Associated Press in an interview.</i>	apw_eng_20011207 _1383_3 :8-9
	<i>if your contributions were not exactly equal, then you won't be able to claim exclusions totaling \$500,000, says IRS spokesman Larry Wright.</i>	nyt_eng_19991003 _0711_47 :6-7
	<i>Although Feuer said he has enjoyed the season, soccer in the United States isn't exactly equal to the sport in other countries.</i>	pcc_eng_18_041 .0286_x0647713_46 :17-18
	<i>The Royals have discussed Wil Myers in trades for Rays right-hander James Shields and Red Sox left-hander Jon Lester, but Ken Rosenthal of FOX Sports points out that the two hurlers aren't exactly equal.</i>	pcc_eng_08_069 .2828_x1105752_10 :37-38
fair	<i>Elections there are not exactly fair.</i>	pcc_eng_08_108 .6175_x1741169_45 :5-6
	<i>While this is fair in some cases, this sort of ham-fisted critique is not exactly fair to Kickstarter studios that DO a ton of testing and solid development of their games.</i>	pcc_eng_12_087 .8104_x1402837_57 :16-17
fancy	<i>While not exactly fancy, Betty's car is worthy of the Preppy Handbook.</i>	pcc_eng_18_028 .9851_x0452753_06 :3-4
far-fetched	<i>While all these may have come off as wishful thinking at best not so long ago, it is not exactly far-fetched at this point in time given the tech-inspired revolution that appears to be imminent, if not already upon us.</i>	pcc_eng_06_084 .7956_x1355122_03 :21-22

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Table 3.11: *Exactly* * Bigrams with the Negative Trigger Lemma, **Not** (cont'd)

Adjective	Corpus Example Utterance	bigram_id
fun	<i>Dealing with the dreaded bites isn't exactly fun—though every family has their own homemade ways of dealing with them—but there are definitely tried and true methods to prevent bites in the first place.</i>	pcc_eng_18_003 .9659_x0048060_03 :8-9
	<i>There are some great add-ons in XBMC that allow you to stream video from the web, but digging through the Add-On section isn't exactly fun.</i>	pcc_eng_06_054 .6134_x0867409_38 :29-30
gentle	<i>While the ride up isn't exactly gentle and could lead to a few scratches, Jancowski doesn't think anyone will complain.</i>	pcc_eng_20_085 .9699_x1372835_18 :7-8
good	<i>and caramel corn and Cinnebons aren't exactly good for you.</i>	nyt_eng_19950820 _0085_48 :8-9
	<i>While the beaches and water aren't exactly good for swimming, if you want to hang out with a bunch of local Indians, head to a beach in Northwest Mumbai on a Sunday evening.</i>	pcc_eng_06_093 .7396_x1499963_21 :8-9
	<i>No, it's not exactly good, but at least he's still heading in the right direction.</i>	pcc_eng_06_105 .0156_x1682703_029 :6-7
	<i>Now a new 10-year auction has been held by the Treasury and the result was not exactly good.</i>	pcc_eng_08_005 .1434_x0067061_03 :19-20
	<i>"We have several cement buildings stacked up on a hillside—not exactly good for wireless," Moy said.</i>	pcc_eng_19_040 .8794_x0643657_23 :14-15
great	<i>While they may still have the drive to hunt, they aren't all exactly great at it.</i>	pcc_eng_18_072 .2705_x1154300_07 :15-16
handsome	<i>i grew up in a house a bit like an E.M. Forster house, red brick, not exactly handsome.</i>	nyt_eng_19970203 _0348_36 :19-20
happy	<i>and the Russians, if not exactly happy, were less confrontational about the prospect of NATO expanding eastward and signing up new members from the former Warsaw Pact.</i>	apw_eng_19960604 _1295_4 :7-8

Continued on next page

Table 3.11: *Exactly* * Bigrams with the Negative Trigger Lemma, **Not** (cont'd)

Adjective	Corpus Example Utterance	bigram_id
	<i>if players aren't exactly happy with the system at the moment...no press is bad press.</i>	apw_eng_20080827 _1341_33 :5-6
	<i>but Battier, who grew up in Birmingham, Mich., wasn't exactly happy about that.</i>	nyt_eng_19990328 _0287_39 :14-15
	<i>But some days, I'm not exactly happy about that.</i>	pcc_eng_06_080 .8075_x1290558_101 :8-9
	<i>Skindred are now talking about their exit from the Rock on the Range and Rocklahoma festivals, revealing that they're not exactly happy about how things went down.</i>	pcc_eng_20_085 .5352_x1365821_03 :23-24
	<i>Had he not done this, it seems he would have not have had to do the pre-emptive strike of outing himself in advance of a political scandal, and everyone would be, if not exactly happy, at least keeping their jobs.</i>	pcc_eng_18_071 .1995_x1136995_08 :37-38
hard	<i>We have videos and GIFs on the web, so it's not exactly hard to make images move.</i>	pcc_eng_18_020 .9012_x0322057_14 :14-15
helpful	<i>I tried asking over on the Facebook page, but I didn't get a whole lot of responses or 'likes' and half of the comments I got weren't exactly helpful.</i>	pcc_eng_06_053 .3583_x0846984_291 :33-34
hypothetical	<i>to put the Republican's initial stumbles into perspective, it helps to step back a bit and imagine a not exactly hypothetical situation.</i>	nyt_eng_19950113 _0434_11 :22-23
identical	<i>The product you receive may not be exactly identical to the one pictured.</i>	pcc_eng_19_024 .6742_x0382179_40 :8-9
impressed	<i>Not surprisingly, the owner of the car, not to mention Jade's father, Hugh (Bruce Greenwood, Star Trek Into Darkness) aren't exactly impressed.</i>	pcc_eng_06_002 .5707_x0025593_27 :29-30
keen	<i>He wasn't exactly keen on technology just waltzing around freely, but he wasn't against the idea either, knowing just how precise machinery could be and had faith in its capabilities to keep order within the world around them.</i>	pcc_eng_18_100 .2921_x1608256_058 :4-5

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Table 3.11: *Exactly* * Bigrams with the Negative Trigger Lemma, **Not** (cont'd)

Adjective	Corpus Example Utterance	bigram_id
large	<i>Thankfully, it is possible to purchase electronic appliances that fit the shape and size of your home, even if it's not exactly large.</i>	pcc_eng_12_095 .7585_x1531570_15 :25-26
	<i>Secondly, their limits on NFL sides and totals are not exactly large in the online sports betting world.</i>	pcc_eng_20_025 .2206_x0391637_058 :12-13
	<i>The architects consider their work to be “more than a hut but less than a house,” since it’s footprint isn’t exactly large or comfortable enough for full-time living.</i>	pcc_eng_20_090 .5915_x1447429_10 :26-27
likable	<i>Freed paints a portrait of a not exactly likable character, but as a woman of her day refusing to be dependent on a man for her happiness and self-worth, Agnes deserves admiration.</i>	pcc_eng_18_043 .0800_x0680878_46 :8-9
natural	<i>However, if you have ever looked at those ingredients on those gels they are isn’t exactly natural.</i>	pcc_eng_20_079 .8754_x1274336_40 :17-18
new	<i>“This isn’t exactly new... but they didn’t do it as well 20 years ago,” Johnson said.</i>	nyt_eng_19960409 _0560_10 :5-6
	<i>While NFL-themed weddings are isn’t exactly new, we can’t imagine they’re a common sight in West Sussex, England, where one Miami Dolphins superfan just got married.</i>	pcc_eng_20_105 .9286_x1695555_04 :8-9
	<i>Strangely, the idea for a 20-cent coin was not exactly new when the first one appeared in 1875.</i>	pcc_eng_18_049 .0803_x0778183_003 :13-14
original	<i>The setup isn’t exactly original.</i>	pcc_eng_18_010 .3729_x0151613_20 :5-6
parallel	<i>7. Once all of the grab irons are in place you might notice that a few are not exactly parallel.</i>	pcc_eng_18_060 .5273_x0963618_10 :20-21
	<i>Fighter aircraft, “Spooky” and “Spectre”, and Vietnamese A-1E’s destroyed over ten thousand trucks, about a half-million tons of supplies, and broke at least three pipelines running close to, but not exactly parallel to the trail, dozens of times.</i>	pcc_eng_06_082 .3806_x1316035_342 :41-42

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Table 3.11: *Exactly* * Bigrams with the Negative Trigger Lemma, **Not** (cont'd)

Adjective	Corpus Example Utterance	bigram_id
	<i>The guide was not exactly parallel after taking in all the errors that arose when I drilled the screws to affix it to the plank below.</i>	pcc_eng_12_025 .8644_x0402289_142 :5-6
	<i>The We Are Hunted and Tunigo deals aren't exactly parallel, since Twitter used We Are Hunted to build a brand-new music app, and Spotify doesn't need one of those.</i>	pcc_eng_06_030 .6212_x0479286_08 :10-11
	<i>Their circumstances are not exactly parallel, but they are both women at very different points in their lives whose stories involve dilemmas with life-changing outcomes.</i>	pcc_eng_06_059 .9981_x0954641_20 :5-6
perpendicular	<i>The legs are a tiny bit too short, so you have to put the switch in at a not exactly perpendicular angle to the front panel.</i>	pcc_eng_12_003 .4983_x0040384_03 :21-22
pleased	<i>he wouldn't be exactly pleased, I don't think."</i>	nyt_eng_19990113 _0320_18 :5-6
	<i>"We're not exactly pleased with this idea," Sara Lempa said.</i>	apw_eng_20000924 _0042_41 :5-6
	<i>Carlisle isn't angry that Carson and Raybon are trying to steal his thunder with country listeners, but he's not exactly pleased either.</i>	apw_eng_19970607 _0859_38 :23-24
psyched	<i>Some of you might remember that Hank wasn't exactly psyched to head home before he'd finished physical therapy.</i>	pcc_eng_12_093 .6323_x1497161_06 :10-11
realistic	<i>It's something that people in real life could admire even if they acknowledge that some of his views on life aren't exactly realistic.</i>	pcc_eng_19_077 .3068_x1232665_11 :24-25
restrictive	<i>Every break is greeted with a stadium-sized cheer, every song transmuted into an anthem, at times the community choir is louder than the not exactly restrictive guitar sound—Bloodbuzz Ohio and Mistaken For Strangers were particularly good for this.</i>	pcc_eng_20_063 .6969_x1012964_52 :28-29
revolutionary	<i>While it's tempting to label Esmail's new series as groundbreaking simply for its limited run time, portraying drama in shorter bursts than what network television has groomed us to accept isn't exactly revolutionary.</i>	pcc_eng_08_035 .8109_x0563777_07 :36-37

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Table 3.11: *Exactly* * Bigrams with the Negative Trigger Lemma, **Not** (cont'd)

Adjective	Corpus Example Utterance	bigram_id
right	<i>Because we understand that not every purchase is exactly right, we've made returns easy.</i>	pcc_eng_06_105 .4360_x1689347_08 :09-10
	<i>That wouldn't be exactly right.</i>	pcc_eng_19_092 .7152_x1482569_066 :5-6
rosy	<i>The red splotches on the jolly old elf's cheeks were not exactly rosy.</i>	pcc_eng_19_042 .9749_x0677653_049 :13-14
round	<i>Then again, even the roundest looking freshwater pearls may not be exactly round when measured.</i>	pcc_eng_20_078 .7433_x1256126_075 :13-14
scary	<i>The scene wasn't exactly scary, but Lois Smith's voice and Marnie's whole off-kilter personality had me uneasy enough that I got a little creeped out.</i>	pcc_eng_19_036 .8190_x0578229_13 :5-6
shy	<i>I pondered over the idea for a second before remembering Harry wasn't exactly shy when it came to these sort of intimate topics.</i>	pcc_eng_06_073 .2489_x1168669_135 :14-15
	<i>Put simply, they don't like you—and they're not exactly shy about it.</i>	pcc_eng_06_094 .9459_x1519477_05 :14-15
	<i>Swift has churned out hit song after hit song, and she's not exactly shy about the fact that she has been inspired by her relationships when songwriting.</i>	pcc_eng_18_006 .5777_x0090374_23 :15-16
simple	<i>of course, that task is not exactly simple either: The Bills have the Colts at home Sunday.</i>	nyt_eng_19991227 _0014_19 :8-9
small	<i>Gilbert likewise has a business in the city, although it isn't exactly small.</i>	pcc_eng_19_094 .3184_x1508634_126 :14-15
	<i>As for who'd be inclined to send the mobster such a dangerous gift, executive producer Josh Berman admits the pool of suspects isn't exactly small.</i>	pcc_eng_19_057 .0878_x0905244_06 :26-27
	<i>The Regal isn't exactly small, but it will be the smallest Buick in some time.</i>	pcc_eng_18_070 .3262_x1122744_22 :5-6

Continued on next page

Table 3.11: *Exactly* * Bigrams with the Negative Trigger Lemma, **Not** (cont'd)

Adjective	Corpus Example Utterance	bigram_id
subtle	<i>the message is not exactly subtle to be liberated in Japan is to become someone else but at the end Suo, with characteristic optimism, has Aoki-san casting off his wig and becoming, at last, his true self.</i>	nyt_eng_19970701 _0508_12 :5-6
	<i>and while the satire is n't exactly subtle, "Bad Company" is amusingly unpredictable.</i>	apw_eng_20081003 _0045_34 :7-8
	<i>Of course, Canadian geese are n't exactly subtle about their presence, as they have a distinct tendency to leave droppings all over the place.</i>	pcc_eng_19_039 .4449_x0620298_52 :8-9
	<i>not exactly subtle, but not in your face too hard either.</i>	pcc_eng_12_070 .3657_x1120574_097 :2-3
	<i>The designs are n't exactly subtle and you can expect to pay about \$70 per pair.</i>	pcc_eng_18_046 .0510_x0729113_3 :5-6
successful	<i>You see, her now ex Julian (Jon Dore) has a not exactly successful panache for gambling ("housing markets, cockroach races, sunsets, who's going to die next"), creating an unstable home.</i>	pcc_eng_12_091 .1254_x1456178_26 :15-16
sunny	<i>it's possible U.S. foreign policy will survive the intensifying War of Presidential Succession, but the prognosis is not exactly sunny.</i>	nyt_eng_19950526 _0129_1 :21-22
sure	<i>I'm not exactly sure why but as the friendly, reptile exhibit person asked my child if she wanted to touch a snake my mouth opened in shock and horror.</i>	pcc_eng_18_033 .2784_x0522544_39 :4-5
	<i>But I'm not exactly sure any of it garnered the respect or attention it deserved.</i>	pcc_eng_12_033 .8260_x0531235_09 :5-6
	<i>The doctors were n't exactly sure how she pulled through, but she did, and we had turkey the following Thursday.</i>	pcc_eng_20_059 .2399_x0940743_095 :5-6
	<i>Well, guys, we're not exactly sure if this tactic is creepy or straight-up romantic, but it's definitely one way to grab a girl's attention.</i>	pcc_eng_20_024 .6427_x0382325_02 :8-9
	<i>I'm not exactly sure when I realized that not everyone was as enamored with the act of reading as I was, but sometime during my teenage life, I learned to keep silent about my enthusiasm.</i>	pcc_eng_19_044 .1962_x0697451_05 :4-5

Continued on next page

Table 3.11: *Exactly* * Bigrams with the Negative Trigger Lemma, **Not** (cont'd)

Adjective	Corpus Example Utterance	bigram_id
	<i>If he'd been able to think about the whole question rationally, he wouldn't have been exactly sure what he'd done to deserve such punishment, but he no doubt would have been certain that he must have done something.</i>	pcc_eng_18_104 .5923_x1678356_0034 :19-20
	<i>The first introduction is always the hardest and as I approached mine, I wasn't exactly sure what I'd say.</i>	pcc_eng_08_052 .7176_x0837292_021 :17-18
	<i>I'm not exactly sure how this works.</i>	pcc_eng_19_092 .8934_x1485518_07 :4-5
	<i>There are many more and I'm not exactly sure what they all entail, but we have the opportunity to create new traditions and be inventive with how we spend our time.</i>	pcc_eng_18_007 .6805_x0108156_15 :09-10
	<i>These events sell out quickly so it is best to get your reservation in even if you aren't exactly sure where and when you can attend.</i>	pcc_eng_06_058 .9346_x0937483_66 :20-21
tech-friendly	<i>In the meantime, we'd be curious what kind of demands the famously recalcitrant and not exactly tech-friendly Assange might have made.</i>	pcc_eng_12_080 .4530_x1283691_16 :18-19
tiny	<i>the merger of those Japanese banks will create, by far, the world's largest bank—with \$819 billion in assets, it will have more than three times the assets of the not exactly tiny Citicorp—and 100 times the assets of Bill Gates.</i>	nyt_eng_19950331 _0322_12 :38-39
trivial	<i>Not strictly relevant to what I want to talk about but not exactly trivial, either.</i>	pcc_eng_08_004 .9511_x0063914_07 :13-14
true	<i>although Taylor said he has been waived by the Giants, that's not exactly true.</i>	nyt_eng_19950509 _0330_39 :15-16
	<i>This is not exactly true in the case of those women who regularly apply makeup to their own faces.</i>	pcc_eng_18_003 .9338_x0047542_15 :4-5
ubiquitous	<i>While Turkish firearms are not exactly ubiquitous on our side of the pond, the Turks have an excellent regional reputation for quality firearms.</i>	pcc_eng_06_010 .0987_x0147050_014 :6-7

Continued on next page

Table 3.11: *Exactly* * Bigrams with the Negative Trigger Lemma, **Not** (cont'd)

Adjective	Corpus Example Utterance	bigram_id
unexpected	<i>But since Dwayne Johnson will definitely have some drawing power here at the box office, it's not exactly unexpected.</i>	pcc_eng_06_080 .3126_x1282497_07 :20-21
unpopular	<i>They do not dominate the African-American community, but they are not exactly unpopular either.</i>	pcc_eng_19_091 .5922_x1464279_09 :15-16
wearable	<i>perhaps McQueen's hooded jackets festooned with lighted circuit-board patterns weren't exactly wearable, but at least McQueen is not asleep at the wheel.</i>	nyt_eng_19990408 _0187_12 :13-14
well-versed	<i>he's played hours of softball in our back yard with friends and family, but he's never actually stepped onto a Little League field and he's not exactly well-versed in the intricacies of the game</i>	nyt_eng_19990607 _0110_7:31-32

As indicated in § 2.2.1, the lemma *not* is active in $\sim 90.1\%$ of all (superset) negative bigram tokens. However, *exactly* bigrams are also observed within that remaining $\sim 9.9\%$ of tokens appearing with a different negative operator. A sample of these is presented Table 3.12.

Table 3.12: *Exactly* * Bigrams with the Negative Trigger Lemma **other than Not**

Negative Trigger	Adjective	Corpus Example Utterance	hit_id
few	<i>woeful</i>	<i>Things were grim in their early losing run but few of those games were exactly woeful when considered individually; the two worst games came at the end.</i>	pcc_eng_11_066 .4711_x1059663_42 :10-15-16
	<i>solvable</i>	<i>In 2D far fewer are exactly solvable, the simplest being a rectangle with Dirichlet boundary conditions.</i>	pcc_eng_04_001 .4775_x0007747_3 :4-6-7

Continued on next page

Table 3.12: *Exactly* * Bigrams with the Negative Trigger Lemma **other than Not** (cont'd)

Negative Trigger	Adjective	Corpus Example Utterance	hit_id
neither	handsome	<i>And as she grew, she grew beautiful, which mystified both her parents because neither of them were exactly handsome, to put it lightly.</i>	pcc_eng_22_071 .8904_x1145773_062 :16-20-21
	right	<i>Of the two months' worth of refunds, neither were exactly right.</i>	pcc_eng_29_098 .2271_x1570854_054 :10-12-13
	wrong	<i>Or put another way, my description of my mother would differ from her own, but neither of us are exactly wrong or right.</i>	pcc_eng_03_030 .9282_x0484596_39 :18-22-23
never	classy	<i>The characters in the early series were never exactly classy—despite their ancestors—but had a certain snooty glamour and allure.</i>	pcc_eng_18_009 .1168_x0131341_67 :08-09-10
	clear	<i>Morale is one of those terms you hear a lot, but it's never exactly clear what some users mean when they use it.</i>	pcc_eng_27_053 .4085_x0847120_09 :15-16-17
no	alike	<i>Woodland scenes, no two exactly alike, drawn to music and sounds of nature.</i>	pcc_eng_29_091 .1900_x1456711_35 :4-6-7
		<i>Given that still-life portraiture allows the painter or photographer to exert maximal design control, it is also worth noting that the naturalism of flowers—no two exactly alike—injects a wild note into the most controlled studio setting.</i>	pcc_eng_20_086 .2909_x1378010_02 :29-31-32
nobody	sure	<i>Nobody is exactly sure of its age but the menhir slots somewhere between the late Neolithic and early Bronze Age periods:</i>	pcc_eng_22_072 .3352_x1152912_06 :1-3-4
	wrong	<i>A war where nobody is exactly wrong is refreshing in Anime.</i>	pcc_eng_11_095 .6594_x1532489_088 :4-6-7
none	certain	<i>An emergency staff meeting was called to brainstorm ideas to improve turnover, which wasn't easy since none of us were exactly certain what the business actually did.</i>	pcc_eng_19_099 .8706_x1598689_11 :23-24
	radical	<i>none of his four points is exactly radical, and all sound sensible enough.</i>	nyt_eng_20060809 _0248_33 :1-7-8

Continued on next page

Table 3.12: *Exactly* * Bigrams with the Negative Trigger Lemma **other than Not** (cont'd)

Negative Trigger	Adjective	Corpus Example Utterance	hit_id
	<i>right</i>	<i>If none of these fittings are exactly right for you Ellantha still offers custom fittings if you don't have the time or inclination to mess with the edit menu.</i>	pcc_eng_27_108 .02420_x1735360_06 :2-7-8
	<i>spectacular</i>	<i>None of this is exactly spectacular, but there's something about it that just makes it all interesting enough to watch.</i>	pcc_eng_04_047 .1129_x0745292_21 :1-5-6
nor	<i>alike</i>	<i>Nor are any two cornfields truly exactly alike, despite Monsanto's best efforts.</i>	pcc_eng_03_092 .4722_x1481139_28 :1-7-8
	<i>free</i>	<i>Nor are ocean-dwellers exactly free from initiating conflict amongst themselves.</i>	pcc_eng_08_105 .6196_x1693943_19 :1-4-5
	<i>sure</i>	<i>Nor, it seems (according to material released by the Crown), were they exactly sure who the Prime Minister was.</i>	pcc_eng_20_033 .6389_x0527732_10 :01-17-18
nothing	<i>definite</i>	<i>A knowing glance here, a friendly touch there, but nothing exactly definite until-</i>	pcc_eng_09_096 .9861_x1553002_031 :12-13-14
	<i>new</i>	<i>Nothing they do is exactly new, but how they do it is completely unique.</i>	pcc_eng_08_043 .4970_x0687784_23 :1-5-6
without	<i>bad</i>	<i>WITHOUT BEING exactly bad, Soundgarden shows little spiritual, emotional or musical growth.</i>	pcc_eng_04_079 .3832_x1266180_40 :1-3-4
	<i>sure</i>	<i>We get what he's saying, without being exactly sure.</i>	pcc_eng_13_009 .1863_x0132036_04 :08-10-11

3.2.2.2 A comparison with positive polarity

The examples provided in Table 3.11 and Table 3.12 illustrate *exactly* tokens with negative environments. This indicates, at the very least, *exactly* is certainly not a *positive* polarity item, but beyond the positive evidence of negated occurrences, many of these attested sentences become infelicitous when their negative element is removed. The examples in (3.1) demonstrate the phenomenon by comparing a sample of the attested (negative) sentences with their fabricated positive polarity counterparts. Not only does the infelicity arise quite consistently when negation is omitted, but the selection spans a variety of adjectives and syntactic configurations⁶³.

(3.1) Positive Alterations across Syntactic Configurations

NOTE: The † character below indicates that the following sentence is attested in the corpus data, as a whole or partial utterance.

- a. [*it*] *was (not) exactly good*
inanimate singular SUBJECT with negated predicate, past tense
 - i. † Now a new 10-year auction has been held by the Treasury and the result was **not exactly good**.
 - ii. # ...and the result was **exactly good**.
- b. *we were(n't) exactly brilliant*
first person plural subject with negative contraction on copula, past tense
 - i. † As a panel we weren't **exactly brilliant** at playing the game...
 - ii. # ...we were **exactly brilliant** at playing the game...
- c. *a (not) exactly endangered species*
prenominal adjective following indefinite determiner, present tense
 - i. † ...“Fierce Creatures” takes aim at a **not exactly endangered** species: the corporate predator.

⁶³Note that the syntactic descriptions provided are intended only to convey the range of differences observed. They are not strictly formal classifications.

- ii. # ...“Fierce Creatures” takes aim at an **exactly endangered** species...
- d. *the (not) exactly {tiny,restrictive}* NP
 prenominal adjective following definite determiner, present tense
 - i. † with \$819 billion in assets, it will have more than three times the assets of the **not exactly tiny** Citicorp
 - ii. # ... it will have more than three time the assets of the **exactly tiny** [corporation]...
 - iii. † ...at times the community choir is louder than the **not exactly restrictive** guitar sound...
 - iv. # ... [it] is louder than the **exactly restrictive** guitar sound...
- e. *are (not) exactly fair*
 inanimate plural subject, with present tense negated predicate
 - i. † Elections there are **not exactly fair**.
 - ii. ? Elections there are **exactly fair**.
- f. *we’re not exactly pleased*
 first person plural subject, with present tense negated predicate
 - i. † “We’re **not exactly pleased** with this idea,” Sara Lempa said.
 - ii. # We’re **exactly pleased** with this idea...
- g. *[they] are(n’t) exactly subtle*
 inanimate plural subject, with present tense contracted negation of predicate
 - i. † The designs are **n’t exactly subtle** and you can expect to pay about \$70 per pair.
 - ii. # The designs are **exactly subtle** and you can expect to pay about \$70 per pair.
- h. *they’re (not) exactly shy*
 third person plural subject, with present tense negated predicate
 - i. † Put simply, they don’t like you—and they’re **not exactly shy** about it.
 - ii. # ... they’re **exactly shy** about it.

3.3 Evaluating Explanations for *Exactly*’s Polar Propensities

The preceding section (§ 3.2) established there is indeed a connection between *exactly* as observed in adverb-adjective bigrams in PBR. However, the source of this relationship is not a given, since multiple components and interactions must be considered. Three potential explanations for the polarity associations observed for *exactly* independent of other specifics are summarized below (3.2). The remainder of the chapter will discuss each option in turn, appealing to specific evidence from PBR for each case.

(3.2) Pre-adjectival *exactly* is...

a. ***universally*** *polarity sensitive*.

It is polarity sensitive in all cases, regardless of other factors.

b. ***vacuously*** *polarity sensitive*.

It is not itself polarity sensitive, but inherits an appearance of polarity sensitivity from consistent association with polarity sensitive adjective arguments.

c. ***contingently*** *polarity sensitive*.

It is polarity sensitive when contingencies of semantic composition with its adjective argument are met.⁶⁴ That is, *some* but *not all* bigrams containing *exactly* are polarity sensitive, and this status depends on the semantics of the modified adjective.

The next two sections (§ 3.4 and § 3.5) contest (3.2a) and (3.2b), respectively.

⁶⁴An additional consideration is whether these *exactly* bigrams have compositional or non-compositional semantics. It might be plausible to question whether *exactly* merely appears in specific bigrams that are polarity sensitive as a non-compositional collocation. I take it as a given in this work that the relevant *exactly* bigrams have compositional semantics, as that seems most plausible, but this nuanced distinction may benefit from a similarly thorough investigation in future work

First, § 3.4 **rejects** ~~universal sensitivity~~ (3.2a) on the grounds of significant counter-examples, as indicated by both qualitative evidence of literal corpus examples and quantitative evidence of raw frequencies and AMs. Then § 3.5 **rejects** ~~vacuous sensitivity~~ (3.2b) on the grounds that adjectives associated with *exactly* do not consistently show attraction to negative polarity environments when evaluated independently of *exactly* (or any other adverb).

After the former possibilities are discarded, § 3.6 argues that **contingent sensitivity** (3.2c) is the best explanation for the patterns observed in § 3.2. This final section goes on to suggest a preliminary classification schema for adjectives that yield polarity sensitive *exactly* bigrams—the (POLARITY) SENSITIVE class—and those that do not: the (POLARITY) IMPERVIOUS class.

3.4 *Exactly* is Not Universally Polarity Sensitive

The first possible explanation is simply that *exactly* is ubiquitously—in all contexts and with all arguments (specifically, in this particular study, adjectives)—sensitive to (licensed by) negative polarity environments. If the adverb *exactly* is universally polarity sensitive, regardless of argument, it should interact with polarity environments in a systematic and predictable way regardless of any other element or property. All bigrams containing *exactly* should also show a propensity for negative polarity, and all cases where *exactly* occurs outside of negative environments should be infelicitous or severely marked.

In short, *exactly* cannot be fully, independently, universally negative polarity sensitive, because it does felicitously occur outside of negative polarity environments. Although the strong association with negative environments observed for *exactly* on the whole illustrates a global trend, it obscures detailed patterns that consider context and argumentation. Both manual inspection of (a sample of) positive cases and association measures for $E\sim$ bigram contingencies confirm that *exactly* can indeed appear outside of negative environments.

3.4.1 Felicitous Examples of *Exactly* Bigrams in Positive Polarity Environments

The most obvious argument against the explanation that *exactly* is obligatorily negative polarity sensitive is simply that of counter-example: *Exactly* appears (felicitously) outside of negative polarity environments (i.e. under positive polarity). Bigrams containing *exactly* are attested in both unmarked utterances (where default positive quantification is assumed), and utterances where positive polarity is explicitly marked via a non-negated positive quantifier or positively quantified adverb. The following corpus examples, (3.3–3.5), illustrate the variety of environments where *exactly* can grammatically/felicitously modify an adjective outside of negative polarity environments (strong or weak).

Corpus Examples of Positive Polarity *Exactly*

(3.3) NEGATIVE TRIGGER ABSENT

- a. Downtown Dorset has two major landmarks, Peltier’s Store and, across the street, the **exactly 200-year-old** Dorset Inn.
(nyt_eng_19961011_0632_5:18-19)
- b. We’ll dress **exactly alike**.
(pcc_eng_20_016.7917_x0255028_142)
- c. But almost every bill in Congress has a title to it which is **exactly opposite** of what it does.
(pcc_eng_12_104.2152_x1668202_191:14-15)

(3.4) POSITIVE EXISTENTIAL TRIGGER PRESENT

$\exists x : [exactly\ ADJ](x)$

- a. When I hear that **something** is “**exactly right**,” I expect mathematical precision.
(pcc_eng_07_015.6498_x0237106_59:10-16-17)
- b. [W]hen researchers analyzed the antibodies, they “were surprised to discover that **some** of these antibodies were **exactly identical** to antibodies that were known to also protect mice from bacterial infections,” Silverman said.
(pcc_eng_00_044.4965_x0702889_30:09-12-13)

(3.5) POSITIVE UNIVERSAL TRIGGER PRESENT

$\forall x : [exactly\ ADJ](x)$

- a. **All** of the events of the preceding paragraph are **exactly parallel** to Thomas Wolfe’s life.
(pcc_eng_15_055.8417_x0886350_040:01-10-11)
- b. The angles, the foam, the design—**everything** is **exactly right**.
(pcc_eng_25_107.1034_x1717046_17:10-12-13)
- c. The assets controlled by an organization are **always exactly equal** to the claims on those assets held by its creditors and owners.
(pcc_eng_23_022.9183_x0354068_129:08-09-10)

The additional example sentences below are coherent for native English speakers

attested in the corpus. These examples are grouped by adjective, rather than environment features. The first set in (3.6-3.15) feature adjectives that frequently appear modified by *exactly* in positive contexts and are sorted by their prevalence. Note that some of these adjectives, despite their rate of appearance with positive *exactly*, may have examples which are degraded to some speakers (myself included); however, they were attested and are not obvious language errors, so I have included them for the sake of transparency. The remainder are then presented alphabetically by adjective, with some more dubious examples of positive *exactly* bigrams in (3.47).

(3.6) *exactly equal*

- a. together, they can accommodate about 900 tourists __ a number almost **exactly equal** to the year-round population of the island.
nyt_eng_19950619_0404_7:13-14
- b. Activase and Retavase are “**exactly equal** in price, safety and efficacy,” Yost said. nyt_eng_19970317_0333_15:6-7
- c. In the circumstances where the result after the adjustment of the handicap line is **exactly equal** to the betting line, then all bets on this offer will be declared void. pcc_eng_18_040.3822_x0637289_152:15-16
- d. The size of the patty should be **exactly equal** to the size of the halved bun in diameter. pcc_eng_18_103.7349_x1664403_056:8-9
- e. On that score, you are **exactly equal** to all other drivers.
pcc_eng_18_023.4738_x0363554_021:7-8
- f. Hence the reason we assume that the days and nights are **exactly equal**, however this assumption may be simple to make, but incorrect.
pcc_eng_19_032.9666_x0515962_18:12-13
- g. In July 1, 1959, the United States and the Commonwealth of Nations agreed in defining one pound to be **exactly equal** to 453.59237 grams.
pcc_eng_19_056.5117_x0895927_2:22-23
- h. The Constitution guarantees that the rights of the irreligious are **exactly equal** to the rights of the aggressively pious.
pcc_eng_20_024.9358_x0387046_15:11-12

(3.7) *exactly right*

- a. “He’s **exactly right**,” Obama said at a rally Saturday.
apw_eng_20070930_0101_7:4-5
- b. They are **exactly right**. pcc_eng_06_018.7536_x0287125_054:3-4
- c. Well you’re **exactly right**, if you go back to 9/11 you can see a lot of failures by the FBI and the CIA. pcc_eng_06_031.6269_x0495450_132:4-5
- d. Anna Jones Crabtree has it **exactly right**. pcc_eng_12_041.6040_x0656601_45:6-7
- e. “Power tends to corrupt and absolute power corrupts absolutely,” said Lord Acton, being **exactly right**⁶⁵. pcc_eng_12_063.0307_x1003027_12:18-19
- f. Or maybe you just proved that Andy Warhol was **exactly right** about the future of celebrity. pcc_eng_20_010.1362_x0147182_26:10-11
- g. When I mentioned that to Stephen, he just kind of paused and said, “That’s **exactly right**.” pcc_eng_20_017.8787_x0272581_150:19-20

(3.8) *exactly parallel*

- a. that’s **exactly parallel** to human relations. nyt_eng_19950811_0405_48:3-4
- b. To use them, two straight or extension ladders must be placed **exactly parallel** to one another. pcc_eng_06_046.1979_x0731227_129:13-14
- c. This struck me as **exactly parallel** to, say, walking around the centre of Sydney (or Melbourne) on a Friday night just after 5; or cruising an airport business lounge. pcc_eng_08_003.5298_x0040937_012:5-6
- d. Go low so that your hips are **exactly parallel** to the floor.
pcc_eng_08_033.4648_x0525677_037:8-9
- e. The tyrannical tiger was **exactly parallel** to the tyrannical landlord.
pcc_eng_08_065.9827_x1052668_3:5-6
- f. “What I am doing [with the doctrine of justification], often enough, is **exactly parallel**, in terms of method, to what Martin Luther did. . . .
pcc_eng_08_066.6395_x1063296_03:18-19
- g. “Parallelism” occurs when two adjacent stems or branches are **exactly parallel** to each other. pcc_eng_12_008.1756_x0116074_128:12-13
- h. All of the precision attachments in a partial denture are positioned so that they are **exactly parallel** to each other. pcc_eng_19_074.3947_x1185714_47:16-17
- i. And this treasure is a central theme in God’s Word, **exactly parallel** with all the many expressions of growth, life, and fruit.
pcc_eng_12_085.5057_x1365520_455:12-13
- j. I know my NASA history as well as any aerospace geek and could come up with an almost **exactly parallel** parody article.
pcc_eng_12_053.7649_x0853021_011:19-20

⁶⁵somewhat strange wording to my ears

(3.9) *exactly zero*⁶⁶

- a. The chance of his winning nomination and election is **exactly zero**.”
pcc_eng_19_076.1562_x1214113_13:10-11
- b. " The odds on anyone. . . finding a magic solution to the computer security problems are **exactly zero**. pcc_eng_12_105.5826_x1690464_17:17-18
- c. There is now **exactly zero** possibility for any of the global destructive plans of the Cabal to come to fruition. pcc_eng_08_081.7270_x1307253_36:4-5
- d. More than 20 years of investigations, accusations, smears, and innuendo have resulted in **exactly zero** charges. pcc_eng_06_062.6810_x0998204_245:17-18
- e. In roundup because it seemed like there were **exactly zero** discussions happening anywhere on the social media site.
pcc_eng_06_040.1286_x0632784_05:09-10
- f. But I see **exactly zero** reason why economics is different, in that light, from management science or history, for example.
pcc_eng_12_083.2823_x1329530_138:4-5
- g. There’s also **exactly zero** noise. pcc_eng_06_084.5458_x1351062_20:5-6
- h. And, most importantly, because these wives know that their husbands love them and not me, and that I constitute **exactly zero** threat to their marriages. pcc_eng_20_108.07207_x1742501_24:23-24
- i. Six is n’t that many, but it’s conspicuously more than I would have heard on a similar ride 15 years ago, which would have been **exactly zero**.
pcc_eng_08_002.5298_x0024864_08:29-30
- j. The fact that Nevada is No. 1 on this list is surprising to **exactly zero** people, we know. pcc_eng_12_017.4926_x0266856_39:14-15
- k. We had **exactly zero** problem with that. pcc_eng_08_045.0787_x0713372_10:3-4

⁶⁶While *zero* is often parsed as an adjective—and I believe it truly is in some cases—this status is questionable in certain contexts. For example, when it could be replaced by a specific numeral, say, “3” without impacting acceptability, the categorization as adjective becomes shakier. Compare the following examples (from the corpus). Both use the phrase “exactly zero” as a way of indicating total absence and are felicitous as such, but *zero* seems to be an exceptional case and may merit further investigation.

- (i) “We had exactly zero problem with that”
 - a. We had exactly 3 {#problem, problems} with that
 - b. We had absolutely no problem with that.
- (ii) “The chance of his winning nomination and election is exactly zero.”
 - a. The chance of his winning nomination and election is exactly 3 (percent).
 - b. The chance of his winning nomination and election is absolutely {#no, none}.

(3.10) *exactly alike*

- a. But while it is feasible that two snowflakes can appear to be **exactly alike** on the outside, they are never identical on an atomic level.
pcc_eng_19_074.1506_x1181727_11:13-14
- b. We'll dress **exactly alike**. pcc_eng_20_016.7917_x0255028_142:4-5
- c. They are **exactly alike**, as you see; same size, same shape.
pcc_eng_12_009.5564_x0138439_147:3-4
- d. They were both sent to Leavenworth Prison at the same time, in 1903, and after some confusion, the staff understood they had two different prisoners with the exact same name, who looked **exactly alike**.
pcc_eng_06_094.5984_x1513818_2:37-38
- e. Several different vehicles of the same manufacturer may look **exactly alike**, but they can all have different towing capacities.
pcc_eng_12_021.1752_x0326402_25:10-11
- f. So far as could be made out, they are **exactly alike** in plan and ornament.
pcc_eng_12_041.9698_x0662529_328:11-12
- g. A common culprit that keeps screenwriters from making their work studio quality material is characters that all sound **exactly alike**.
pcc_eng_20_018.4667_x0282158_136:19-20

(3.11) *exactly opposite*

- a. Kramnik's opinion on the game course was **exactly opposite**.
apw_eng_20061012_1620_12:09-10
- b. People low on the trait is classified as Introverts, who tend to be **exactly opposite** to [extroverts]. pcc_eng_06_096.1444_x1538842_074:15-16
- c. The inputs and outputs are **exactly opposite**!
pcc_eng_08_092.7228_x1484787_2:6-7
- d. The attitude of the two is **exactly opposite** to the conventional love scenes on the Western stage, where the maiden refuses and is hard-hearted, while the lover tries to gain her favour. pcc_eng_06_103.2696_x1654525_08:7-8
- e. Rather it was through an **exactly opposite** strategy.
pcc_eng_06_105.5922_x1691936_071:6-7

(3.12) *exactly correct*

- a. It turns out we were **exactly correct**. pcc_eng_06_056.2074_x0893182_15:6-7
- b. What you said is **exactly correct**. pcc_eng_20_075.3313_x1201017_11:5-6
- c. "That's **exactly correct**, your Honor, because there is the bigger issue of the—" the attorney stated. pcc_eng_18_004.2191_x0052218_29:4-5
- d. And, Clay Travis is **exactly correct** with his assertion.
pcc_eng_06_055.4625_x0881157_41:6-7

- e. Because if that is what they are trying to communicate, they would be **exactly correct**. pcc_eng_06_031.5780_x0494694_005:15-16
- f. That is **exactly correct** and is why the bill should be opposed. pcc_eng_06_042.6345_x0673364_33:3-4
- g. I think you are **exactly correct** about the vets in this case. pcc_eng_18_039.9459_x0630147_60:5-6
- h. “Rangel was **exactly correct** to call Romer on that,” said Rep. Martin Frost of Texas, chairman of the Democratic Congressional Campaign Committee. nyt_eng_19981026_0348_29:4-5
- i. “So in this particular case, I think the choice was **exactly correct**.” apw_eng_20031114_0622_16:13-14
- j. You are **exactly correct**, this was better than 2014’s Godzilla. pcc_eng_06_008.4117_x0119729_74:3-4
- k. We’ve had several poems submitted recently, and I think you are **exactly correct** - speaking from the heart is a powerful way to address this topic. pcc_eng_20_094.9502_x1517782_16:14-15

(3.13) ?? *exactly same*⁶⁷

⁶⁷I am not convinced *same* should be parsed as an adjective as much as it has been (if at all). Most of the *same* examples I have looked at read as speech errors to me. I feel they are missing a determiner entirely or it precedes *exactly* when it should have followed. Compare *(the) exactly same* with *exactly the same*: The latter is perfectly felicitous while the former (which is how the bulk of tokens appear) are parsable, but feel distinctly degraded.

The following examples juxtapose *same* with synonyms or near synonyms with noncontroversial adjective status: *alike*, *equal*, and *identical*. In all three cases, *same* fails to have the same distribution.

- (i) compared with *alike*
 - a. They say that we look **exactly** {**alike**, ***same**} in our uniforms.
 - b. They say that we look *exactly* **the** {*same*, **alike*} in our uniforms.
 - c. No two situations are **exactly** {**alike**, ***same**}.
 - d. No two situations are *exactly* **the** {*same*, **alike*}.
- (ii) compared with *equal*
 - a. Their measurements were **exactly** {**equal**, ***same**}.
 - b. Their measurements were *exactly* **the** {*same*, **equal*}.
- (iii) compared with *identical*
 - a. I swear they are **exactly** {**identical**, ***same**}.
 - b. I swear they are *exactly* **the** {*same*, #*identical*}.

- a. There are schools with the **exactly same** demographic that have created a culture at the school which allows an academic environment.
pcc_eng_08_038.1698_x0601696_044:6-7
 - b. His objectives for the third stage were the **exactly same**.
pcc_eng_18_093.8303_x1503455_17:09-10
 - c. Everything **exactly same**, minimum 290 people dead.
pcc_eng_08_106.0266_x1700594_12:2-3
- (3.14) ?? *exactly sure*
- a. You gotta be **exactly sure** of what kinda of build you wanna have.
pcc_eng_06_016.9485_x0258019_37:4-5
 - b. Even the leading products have been tested so as to be **exactly sure** of what to expect. pcc_eng_18_067.2727_x1073176_058:12-13
 - c. Next, you will want to be **exactly sure** of your water meter size.
pcc_eng_18_087.2705_x1397044_09:8-9
 - d. Make **exactly sure** of what they are saying. pcc_eng_19_105.4247_x1688454_30:2-3
- (3.15) ?? *exactly ideal*
- a. “This is an **exactly ideal** model that you can have in agriculture production in the worst conditions in the world. pcc_eng_08_104.4633_x1675275_16:5-6
 - b. We have now had time to thoroughly associate ourselves with her and perhaps more importantly her with us, and by all observations her personality seems to be **exactly ideal**. pcc_eng_18_051.3747_x0815585_10:29-30

The examples (3.16–3.46) are not as commonly occurring as the prior examples, but present a random sampling of the wide variety of adjectives that may occur with *exactly* modification outside of negative environments. Once again, there are some dubious examples—especially (3.47)—that, although attested in the corpora, have degraded acceptability by my own judgments. Some of these are clear cases of parsing or scope errors, but I have included them for the sake of transparency and as a reminder that the reported frequencies encompass such erroneous or questionable cases.

- (3.16) exactly analogous

- a. *It is **exactly analogous** to using a calculator instead of pencil and paper to add a long list of figures.* pcc_eng_20_103.1800_x1651011_278:3-4
 - b. *“It’s **exactly analogous** to how a regular steam or combustion engine pushes the pistons.”* pcc_eng_19_015.4810_x0233685_16:4-5
- (3.17) exactly average
*Matthew Vaughn walked into making 2014’s “Kingsman: The Secret Service” like the kid that everyone expected would get **exactly average** on the midterm.*
pcc_eng_20_041.6349_x0656559_04:23-24
- (3.18) exactly backasswards
*Recently it is clear what is going on here and it is **exactly backasswards**.*
pcc_eng_12_015.9880_x0242480_46:13-14
- (3.19) exactly backward
*“This analysis is **exactly backward**,” Boot wrote.*
pcc_eng_19_077.4261_x1234639_06:5-6
- (3.20) exactly central
*To make the image **exactly central** I go to magnify settings and then right click > fit to screen.*
pcc_eng_19_020.6650_x0317453_013:5-6
- (3.21) exactly commensurate
*Bush, by all accounts, seems to regard DeLay with a respect **exactly commensurate** with the power he wields in Congress.*
nyt_eng_19991009_0113_6:14-15
- (3.22) exactly compelling (enough)
*He’s just less relentlessly knowing than LMFAO, less calculated than One Direction, less divisive than dubstep, and **exactly compelling** enough to entertain everybody.* pcc_eng_18_072.3923_x1156271_45:22-23
- (3.23) exactly congruent
*One’s capacity to suffer and one’s capacity to love are **exactly congruent**.*
pcc_eng_19_002.0805_x0017480_37:13-14
- (3.24) exactly consistent
*McCurry said Pena’s meetings with the president and foreign minister were “**exactly consistent**” with U.S. policy on Taiwan, on which both Beijing and Taipei had been briefed.* apw_eng_19941214_0415_16:14-15
- (3.25) exactly contradictory
*Beside the cultural impacts it glazes past, this ordinary Oscar bait ends up # Fake News, **exactly contradictory** to its supposed attempts.*
pcc_eng_19_051.6960_x0818175_055:19-20
- (3.26) exactly contrary

- a. “The actions that the ‘Zoe’s Ark members’ have done are **exactly contrary** to why the other NGOs are here, which is basically to aid and protect the refugees.” apw_eng_20071031_0718_10:15-16
 - b. Given the flexibility of the Paris Agreement, exiting this agreement would run **exactly contrary** to US interests. pcc_eng_12_076.1703_x1214687_35:14-15
 - c. But, as a general fact, we find that those regions towards the north are the best supplied with moisture, and therefore best wooded—**exactly contrary** to the character of those regions which receive their moisture from the Gulf of Mexico. pcc_eng_06_102.2296_x1637520_266:28-29
- (3.27) exactly divisible
A year was a leap year of 366 days if it was **exactly divisible** by 4.
pcc_eng_20_103.3784_x1654252_16:13-14
- (3.28) exactly east(-to-west)
“At this time of year, the winds are **exactly east** to west and at 130,000 feet”40,000 meters" they can go clear around the globe, ” he said Thursday by telephone from control headquarters in Alice Springs.
apw_eng_19981226_0821_12:11-12
- (3.29) exactly edge-on
- a. It is of particular scientific interest because it is very similar to our own Milky Way Galaxy but is seen almost **exactly edge-on**.
pcc_eng_20_077.3211_x1233135_18:22-23
 - b. It reports that Jupiter’s point in relationship to ours an event called “Mutual Phenomenon” occurs where the Jovian moons will occult and/or eclipse each other due to our two worlds being almost **exactly edge-on** with each other.
pcc_eng_08_072.9226_x1164400_31:38-39
- (3.30) exactly enough⁶⁸

⁶⁸It is debatable whether *enough* should be parsed as an adjective (in these examples specifically, but also elsewhere). At first glance, it seems plausible to say *enough* modifies the following material, *time*, *work*, *money*, etc., or an implicit material inherited from the context, as the in the “*exactly enough [fuel]*” example (where I added the “[fuel]” to improve clarity.) Adjectives can indicate quantity of a substance, and the adjective “*sufficient*”—and we know this is an adjective, because its adverb derivation is morphologically transparent: *sufficiently*—can often be used as a substitute for *enough*. If *exactly* were not present in these sentences, *sufficient* could felicitously replace *enough* (or put another way, it would replace both *exactly* **and** *enough*, not just *enough*). If *exactly* remains, though, *sufficient* cannot substitute for *enough*, as shown below. This can be considered a failure of the substitution constituency test, suggesting that, (a) [*exactly enough*] is more likely a constituent than [*enough {time, money, etc.}*], and (b) categorizing *enough* as a simple adjective is questionable at best.

- (i) Comparing *enough*, *exactly enough*, and *sufficient*

- a. *But even if an ET revolution will be difficult, it is still achievable: “We have **exactly enough** time, starting now,” said Friedman.*
pcc_eng_19_023.0215_x0355684_31:19-20
- b. *At the same time, you had also better hope that you’re constantly producing **exactly enough** translation work to keep all of your translators busy.*
pcc_eng_20_062.7326_x0997404_37:16-17
- c. *It confidently predicts that we have fuel for either: - (1) 100 miles more than we need; 30% of tank used. (2) **exactly enough** [fuel] to get 5 miles from the marina; 35% of tank used. or (3) We will run out 100 miles short.*
pcc_eng_18_035.9530_x0565752_184:25-26
- d. *We need to form it even faster, because humanity, as Dana Meadows said, has “**exactly enough** time —starting now.”*
pcc_eng_12_101.3671_x1621965_135:19-20
- e. *“Dole’s campaign will spend **exactly enough** money in California to avoid the story that they’ve written it off,” Popkin said.* nyt_eng_19960909_0542_43:7-8

(3.31) exactly equivalent

*The complexity of the part and the robot’s limitation to generate a path **exactly equivalent** to the part edge make it difficult for the deburring operations to meet overall quality goals.* pcc_eng_18_035.4784_x0558115_29:15-16

(3.32) exactly halfway

*officially, Kansas farmers are expected to harvest 364 million bushels of wheat this year, which is almost **exactly halfway** between the disappointing 255 million bushels harvested last year and the state record 472 million reaped in 1990.*
nyt_eng_19970628_0178_41:20-21

(3.33) exactly identical

- a. *The structure of the Roman Catholic Church, furthermore, is **exactly identical** to the political structure of the old Roman Empire.*
pcc_eng_18_028.9626_x0452364_09:12-13
- b. *It’s always useable though and looks **exactly identical** to the edges in all your photos.* pcc_eng_20_107.08955_x1727447_04:8-9
- c. *All rooms were **exactly identical**. and the hotel appeared to have been built, Lego-like, from modular blocks; only the ground floor differed, and it housed a walk-through lobby, a continental-breakfast bar with lunchroom style seats,*

-
- a. We have ~~exactly enough~~ sufficient time.
 - b. [It] will spend ~~exactly enough~~ sufficient money to avoid...
 - c. # We have exactly ~~enough~~ sufficient time.
 - d. # [It] will spend exactly ~~enough~~ sufficient money to avoid...

- a business center, a little onsen and locker room, and the ubiquitous vending machines.* pcc_eng_19_091.0498_x1455512_11:4-5
- d. *Each piece is unique and we cannot guarantee you will receive an item **exactly identical** to that shown online.* pcc_eng_06_035.1367_x0552166_3:14-15
- (3.34) exactly inverse
- a. *Comfort is about finding just the right speed, where the suspension rebounds with a sine wave **exactly inverse** relative to the road, counteracting the oscillations.* pcc_eng_08_076.6285_x1224489_077:18-19
- (3.35) exactly massless
- We start by proving explicitly that, in contrast to claims made previously in literature, Nambu-Goldstone bosons of spontaneously broken global color symmetry remain **exactly massless** even after imposing the color neutrality constraint.* pcc_eng_12_040.8381_x0644297_08:27-28
- (3.36) exactly overhead
- Sun would be **exactly overhead** of the equator.* pcc_eng_12_100.5776_x1609119_04:4-5
- (3.37) exactly perfect
- a. *She is **exactly perfect** for us.* pcc_eng_08_057.6320_x0917148_12:3-4
- b. *we've published a lot of books and journals where you put all the work into making sure it's **exactly perfect** up front.* nyt_eng_19970314_0344_13:21-22
- c. *"Political enemies would attack whoever the person was for whatever reason unless it was just **exactly perfect**."* pcc_eng_18_107.3200_x1722468_33:17-18
- (3.38) exactly perpendicular
- a. *It is oriented **exactly perpendicular** to the sun, which is, of course, the source of the lens flare.* pcc_eng_19_054.8574_x0869138_12:4-5
- b. *This line is **exactly perpendicular** to the cycloid, and if you drew a circle around this point* pcc_eng_06_058.9136_x0937134_099:4-5
- c. *There is a reason modern tanks have angled side walls, bullets cannot hit with nearly as much energy when they hit an angled surface, a projectile hits with the most force when **exactly perpendicular** to the surface it strikes.* pcc_eng_20_010.4805_x0152677_033:35-36
- d. *Use the builder's square to position the rail **exactly perpendicular** with the face frame of the cabinet.* pcc_eng_20_083.3782_x1331116_020:10-11
- (3.39) exactly plumb
- this will create a chalked line on the wall that will be **exactly plumb**.* nyt_eng_19991115_0402_9:13-14
- (3.40) exactly representative

- a. *This is **exactly representative** about how I feel about my present and my anticipated future.* pcc_eng_20_009.2135_x0132515_38:3-4
 - b. *He is, however, **exactly representative** of how I myself and many other Americans feel about our country and our government right now so I could care less what letter is next to his name on the voting card.*
pcc_eng_06_106.1976_x1701806_20:6-7
- (3.41) exactly reproducible
*The wave length thus provided an absolute and **exactly reproducible** standard of length.* pcc_eng_19_063.6002_x1010742_07:09-10
- (3.42) exactly similar
- a. *The other four axioms tell us what we can conclude if parts or wholes are indistinguishable (indiscernible/equal/**exactly similar**) in some way.*
pcc_eng_18_066.5038_x1060717_018:22-23
 - b. *These results were almost **exactly similar** to Kevin's, as mentioned earlier.*
pcc_eng_20_045.8498_x0724556_50:5-6
 - c. *The process of finding out information about the enemy while one is dressed in civil clothes is called "spying". the **exactly similar** process when one is dressed in uniform is called "reconnoitring" or "scouting"*
pcc_eng_20_053.3133_x0845186_04:24-25
 - d. *There is also an almost **exactly similar** project to the California high speed railway about to begin construction in Republican dominated Texas.*
pcc_eng_18_020.3106_x0312567_10:6-7
- (3.43) exactly square
*They were **exactly square** for the first time in 22/23o during November/December 1992, and they were **exactly square** for the second and last time in 23o in July 1993.* pcc_eng_08_088.2335_x1412216_162:3-4
- (3.44) exactly true
*As it turns out, pretty much everything you've suspected about conservatism and racism being correlated with being scientifically proven stupid is **exactly true**.*
pcc_eng_18_040.9980_x0647245_06:24-25
- (3.45) exactly unchanged
*indeed, prices received by American producers of finished goods were **exactly unchanged** over the past 12 months.* nyt_eng_19940712_0190_46:12-13
- (3.46) exactly wrong
- a. *When she started having problems, I'm ashamed to say that each and every time I misinterpreted the issue and then handled it **exactly wrong**.*
pcc_eng_06_077.2037_x1232492_05:25-26

- b. *Any counter-terrorism experts out there are welcome to correct me, but this strategy seems **exactly wrong**.* pcc_eng_08_014.5577_x0219287_04:16-17
- c. *“Reducing our military power to a level appropriate to the Serbians is **exactly wrong**.”* apw_eng_19950717_0579_8:14-15
- d. *One fraught morning, I attempted to answer a question of textual detail in Derrida to which Spivak responded, “**exactly wrong**.”*
pcc_eng_06_040.7249_x0642333_005:22-23

(3.47) **Some Dubious Examples**

The “!” character is used to indicate a potential misparse; i.e., cases where the structural requirements of [ADV ADJ] bigrams defined in [Chapter 2](#) are not actually met.

- a. ?? *Just when you were getting **exactly comfortable** with the vertical location of your center of gravity and the correlations between arm location and spin velocity, suddenly it all starts changing.*
pcc_eng_12_005.7573_x0076516_13:6-7
- b. ! *How **exactly important** it is, is the subject of a recent admission by Belarusian economist Georgi Gryts.* pcc_eng_19_001.0881_x0001410_17:2-3
- c. ?? *My oldest niece told me how **exactly much** she is digging working on a paper this weekend for her literature class.*
pcc_eng_12_095.0342_x1519902_12:7-8
- d. ?? *So thank you all and make sure you are **exactly obedient** to get blessings.*
pcc_eng_19_058.9547_x0935483_34:10-11
- e. ! *Every time I am in a supermarket I wonder how is it **exactly possible** that they sell this amount of pastries each day.*
pcc_eng_06_076.9332_x1228201_19:13-14
- f. ?? *“Everyone wants things to run **exactly smooth** the first year, but the good news is we’re pressing on.”* pcc_eng_20_043.9071_x0693217_45:7-8
- g. ! *When people ask me what **exactly wellness** means, my go -to response is: “Doctors know what keeps your body functioning optimally based on research, but only you know what makes your whole being feel good.”*
pcc_eng_08_067.4686_x1076447_02:6-7

3.4.2 Polar Frequency Distribution of *Exactly* Bigrams

Positive polarity usage is not merely possible for *exactly* * bigrams, but accounts for over one fifth (21.9 %) of all *exactly* bigram tokens observed—non-negative environments account for 12,296 of the total 56,109 bigram tokens where the adverb is *exactly*. While this is only 22 % of the *exactly* bigram tokens appearing in the final results—hence the clear indication of negative preference for *exactly* generalized over individual adjectives—it is still notable. Table 3.13 juxtaposes relevant frequencies for first the full superset and subset comparison spaces followed by Table 3.14 which does the same for their negative equivalent sampled counterparts⁶⁹. *Exactly* bigram token distribution between the polar environments is not as extreme as would be expected for universal adverb polarity sensitivity.

Even with concerns for messy data sources with inconsistent parsing, orthography, and annotations, this quantity should not be dismissed out of hand, especially since many steps have been taken to ameliorate the impact on the results (see Chapters 4–6 for discussion). Less than 2% (1.77 %) of the non-negated tokens (218 of 12,296; 0.39 % of all 56,109 *exactly* tokens) occur with explicit positive markers (i.e. in the mirror subset), but this is not peculiar given the sampling differences. As the table below illustrates, this is in keeping with the overall size discrepancy between the **S* superset and @*S* subset: The @*S* subset as a whole is 2.34 % of the **S* superset and the negative and positive portions of the subset are 9.19% (–) and 2.02% (+) of their

⁶⁹In these tables, $f(E \wedge \neg \textit{exactly})$ indicates the polar remainder, or the bigram tokens in *E* that have adverbs other than *exactly*.

Table 3.13: **ALL+** : **Entire** Superset $*S$ and Mirror Subset $@S$ Comparison Spaces

ALL+	S:	@S: Mirror Subset		*S: Superset	
N	N: set total	1,680,633		71,961,373	
	N% of N^{*S}	2.34 %		100 %	
	ΔN	-70,280,740			
f2	$f2 = f(exactly)$	1,031		56,109	
<i>exactly</i>	% of $f2^{*S}$	1.84 %		100 %	
subtotal	$\Delta f2$	-55,078			
polarity	environment, E	$(-):@N$	$(+):@P$	$(-):*N$	$(+):*\mathbb{C}_N$
f1	$f1 = f(E)$	291,735	1,388,898	3,173,681	68,787,692
polar	$f1\%$ of $f1^{*S}$	9.19 %	2.02 %	100 %	100 %
subtotal	$f1\%$ of N	17.36 %	82.64 %	4.41 %	95.59 %
	$f1\%$ of N^{*S}	0.41 %	1.93 %		
f	$f = f(E \wedge exactly)$	813	218	43,813	12,296
polar	$f\%$ of f^{*S}	1.86 %	1.77 %	100 %	100 %
<i>exactly</i>	$f\%$ of f2	78.86 %	21.14 %	78.09 %	21.91 %
	$f\%$ of $f2^{*S}$	1.45 %	0.39 %		
E remainder: $f(E \wedge \neg exactly)$		290,922	1,388,680	3,129,868	68,775,396

superset counterparts respectively. Furthermore, the relative approximated polarity distributions are also consistent—within 1% of each other—with 78.1 % of superset and 78.9 % of subset *exactly* bigrams occur in negative environments (+0.8% from superset to subset) and 21.9 % of the superset and 21.1 % of the subset tokens occur in non-negated/positive environments (−0.8% from superset to subset). The differences in positive vs. negative representation of *exactly* in the subset are actually smaller than the difference between the totals and *exactly* subtotals between the superset and subset: 4.4 % of superset and 17.4 % subset bigram tokens overall are categorized as

Table 3.14: **NEQ+** : **Negative Equivalent Sampled** Superset $*S$ and Mirror Subset $@S$ Comparison Spaces

NEQ+	S:	@E mirror subset		*S superset	
N	N: set total	583,470		6,347,362	
	N% of N *S	9.19 %		100 %	
	ΔN	-5,763,892			
f2	f2 = $f(exactly)$	862		44,378	
<i>exactly</i>	% of f2 *S	1.94 %		100 %	
subtotal	$\Delta f2$	-43,516			
polarity	environment, E	(-):@N	(+):@P	(-):*N	(+):*C _N
f1	f1 = $f(E)$	291,735	291,735	3,173,681	3,173,681
polar	f1% of f1 *S	9.19 %	9.19 %	100 %	100 %
subtotal	f1% of N	50.00 %	50.00 %	50.00 %	50.00 %
	f1% of N *S	4.60 %	4.60 %		
f	f = $f(E \wedge exactly)$	813	49	43,813	565
polar	f% of f *S	1.86 %	8.67 %	100 %	100 %
<i>exactly</i>	f% of f2	94.32 %	5.68 %	98.73 %	1.27 %
	f% of f2 *S	1.83 %	0.11 %	100 %	100 %
E remainder: $f(E \wedge \neg exactly)$		290,922	291,686	3,129,868	3,173,116

negative (+13% from superset to subset).

The relative polarity proportions are displayed as pie charts in Figures 3.4 and 3.5. While there is a clear difference between the the polar proportions for the superset and subset overall shown in Figure 3.4 in stark contrast, the *exactly* tokens in Figure 3.4 are near identical between the superset and subset.

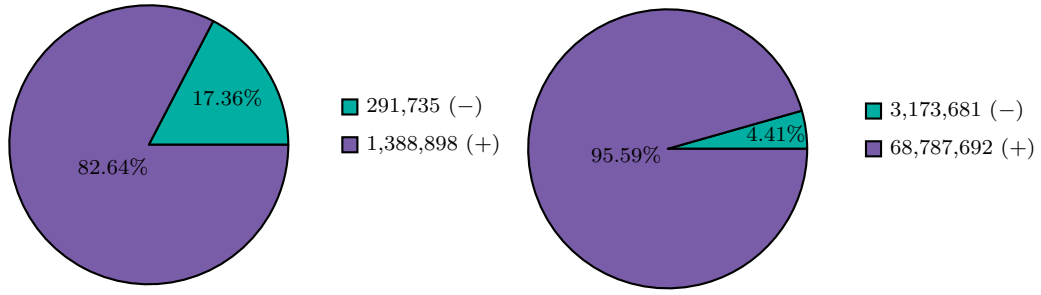


Figure 3.4: Polarity Distribution of **All** Bigrams by Polarity Approximation Method

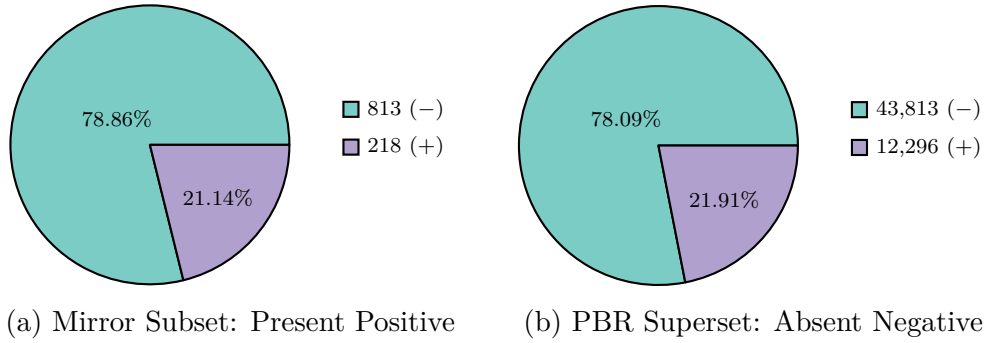


Figure 3.5: Polarity Distribution of *Exactly* Bigrams by Polarity Approximation Method

3.4.3 LRC Designations of Polarity Insensitive *Exactly* Bigram

Many *exactly ADJ* bigram forms do not exhibit significant LRC measures of polarity sensitivity⁷⁰. Over 88% of 16,758 evaluations of all 3,995 unique *exactly ** bigrams do not reach significant LRC values for association with polarity environments. Even when minimum frequency thresholds are applied, over 10% of unique bigram

⁷⁰LRC $\neq 0$ indicates statistical significance ($p \leq 0.001$).

p value controlled by `alpha` parameter for LRC calculations.

Table 3.15: Proportions of *Exactly* Bigrams with **Insignificant** LRC by Comparison Space ($E \sim \text{bigram}$ LRC= 0)

	ALL+	NEQ+	<i>overall</i>
Superset	81.7 % of 3,995	93.9 % of 3,755	87.6% of 7,750
Mirror Subset	97.5 % of 321	98.7 % of 308	97.7% of 629
overall	82.9% of 4,316	94.3% of 4,063	88.4% of 8,379

evaluations fail to reach significant LRC values with polarity environments (i.e., LRC is 0). When the cross-space mean bigram subtotal ≥ 10 , **42%** of 1,850 evaluations (of 306 unique bigrams) are insignificant. When the individual bigram subtotal ≥ 20 , **13%** of 1,216 evaluations (of 302 unique bigrams) are insignificant. Table 3.15 breaks down the proportion of *exactly* * bigrams that are insignificant (LRC=0) by space.

In fact, a few cases even show a propensity for **positive** polarity as measured by the asymmetric backwards ΔP ; despite none of these cases reaching significance (per the symmetric LRC), their existence is stronger evidence than simple insignificance (LRC of 0) that *exactly* alone is insufficient to explain the negative prone patterns observed overall.

To further corroborate the qualitative inspection of examples and the top-level generalized frequency comparison, the **association scores calculated for specific exactly bigrams do not always favor negative environments**. While many *exactly* bigrams show a significant attraction to negative polarity environments, and none are significantly repelled by negative polarity (attracted to positive polarity)⁷¹, a

⁷¹Emphasis on *significantly*—some bigram evaluations do show backwards delta P values

majority of the unique combinations show no significant association with negative environments. Although this does not support the contrary case where *exactly* has some potential for *positive* polarity licensing, this distribution of bigram sensitivities suggests that *exactly* is not **universally** negative polarity sensitive.

When no limitation is placed on bigram marginal frequencies, over 88% of all evaluations demonstrate no significance, as presented in the table below. The minimum insignificant proportion observed in any of the four comparisons spaces individually is still roughly 82% (in the ALL+ superset comparison space).

Table 3.16 presents the direction of polarity sensitivity of all unique *exactly* bigrams observed in PBR, as measured by LRC—evaluations where LRC==0 are labeled as *insignif(icant)*, those where LRC> 0 are labeled as *attract(ion)*, and those where LRC< 0 are labeled as *repel(led)*.

Table 3.16: Polarity Sensitive Significance *Exactly* Bigram Evaluations

Polar sensitivities across 16,758 evaluations							
<i>unique bigrams</i> = 3,995							
<i>minimum bigram subtotal in any space</i> = 1							
<i>minimum bigram subtotal in superset</i> = 1							
<i>minimum mean bigram subtotal</i> = 0							
Polarity	Direction	mirror		super		total	ratio
		ALL+	NEQ+	ALL+	NEQ+		
neg	attract	8	4	732	228	972	0.116
	insignif	313	304	3,263	3,527	7,407	0.884
	repel	0	0	0	0	0	0
pos	attract	0	0	0	0	0	0
	insignif	313	304	3,263	3,527	7,407	0.884
	repel	8	4	732	228	972	0.116

indicating an attraction to positive polarity, despite their LRC failing to reach significance.

Of course, the total proportions presented above are greatly inflated by low frequency bigrams—such cases simply do not have the sample size required to achieve significance. The distribution of significant *exactly* bigrams across relative bigram subtotals (f2) (represented as percentiles (“f2 Percentile”) in Tables 3.17 and 3.18) illustrates the extent of low frequency interference.⁷² Beyond the significance counts displayed, note that the percentiles themselves indicate the prevalence of low frequency token types⁷³ with 50 % of all bigram forms across the four comparison spaces having only a single token, and 80 % having no more than five appearances in the given comparison space (in any environment). It is no wonder then that over 80% of all bigram forms do not show significant LRC values for polarity interactions, as 5 observations is hardly a sufficient sample size. However, the fact there are at least 187 bigram evaluations⁷⁴ fail to reach significance despite having $\geq 8,840$ tokens (in either polarity environment of that comparison space) suggests low frequency bigrams are not the only source of LRC values indicating polarity insensitivity.

Table 3.18 is included primarily to emphasize that the rate of nonzero LRC values by bigram subtotal is relatively consistent regardless of the comparison space they occur in. Although only eight and four unique bigram show significant negative polarity sensitivity in the ALL+ and NEQ+ mirror subset comparison spaces respectively, this is out of 321 and 308 possible—only 8.0 % of unique ALL+ superset bigrams and 8.2 % of unique NEQ+ superset bigrams are represented in the mirror

⁷²The percentile definitions are the same for both Table 3.17 and Table 3.18, but a shorthand has been applied in Table 3.18 to conserve space.

⁷³as expected per Zipf’s Law

⁷⁴There are four potential evaluations per unique bigram form; one for each of the comparison spaces.

Table 3.17: *Exactly* * Bigrams with Significant Negative Polarity Association by Bigram Marginal Frequency (f2) Percentile

f2 Percentile		attraction (LRC > 0)	insignificance (LRC = 0)
[0, 50]%	$f2 = 1$		4,543
(50, 60]%	$f2 = 2$		1,133
(60, 70]%	$f2 = 3$		561
(70, 80]%	$4 \leq f2 \leq 5$		540
(80, 90]%	$6 \leq f2 \leq 13$	335	443
(90, 100]%	$14 \leq f2 \leq 8,840$	637	187

Table 3.18: *Exactly* * Bigrams with Significant Negative Polarity Association by Comparison Space and Bigram Marginal Frequency (f2) Percentile

f2 Percentile		attraction (LRC > 0)				insignificance (LRC = 0)			
		mirror		super		mirror		super	
		ALL+	NEQ+	ALL+	NEQ+	ALL+	NEQ+	ALL+	NEQ+
50%	1					227	221	2,106	1,989
60%	2					50	49	533	501
70%	3					15	12	276	258
80%	5					7	8	270	255
90%	13	1		334		12	11	54	366
100%	8,840	7	4	398	228	2	3	24	158

subset.

However, limiting the scope of investigation to only unique bigrams that average 10 or more tokens across all four comparison spaces leads to a reduction of $-7,454$ negative environment evaluations (and $-3,689$ unique bigram forms). As shown below, this does increase the polarity sensitive percentage from 12 % of all negative polarity evaluations to 58 %, but 42 % remain insignificant.

Restricting by *mean* bigram subtotal, however, means that the smaller comparison spaces are still included for any unique bigram with sufficient tokens in

Table 3.19: Significant Polarity Sensitive *Exactly* * Bigram Evaluations with a cross-space mean subtotal ≥ 10

Polar sensitivities across 1,850 evaluations <i>unique bigrams</i> = 306 <i>minimum bigram subtotal in any space</i> = 1 <i>minimum bigram subtotal in superset</i> = 19 <i>minimum <u>mean</u> bigram subtotal</i> = 10							
Polarity	Direction	mirror		super		total	ratio
		ALL+	NEQ+	ALL+	NEQ+		
neg	attract	8	4	295	228	535	0.578
	insignif	152	149	11	78	390	0.422
	repel	0	0	0	0	0	0
pos	attract	0	0	0	0	0	0
	insignif	152	149	11	78	390	0.422
	repel	8	4	295	228	535	0.578

the ALL+ superset data. That means the “insignificant” count above includes `env~exactly_ADJ` evaluations where the bigram has a very low marginal frequency. Since the bigram subtotal is, in essence, the sample size for the polarity probe of that comparison space, infrequent bigrams will necessarily fail to attain significant LRC values. When the evaluations are restricted by individual bigram subtotals *per comparison space*, rather than the mean across all four spaces, the proportion of evaluations indicating polarity sensitivity nearly invert from the unrestricted distribution.

Table 3.20 shows the evaluation significance totals for only those bigram evaluations which have ≥ 20 bigram tokens in the given comparison space, eliminating those that were included above due to one or more comparison spaces containing sufficient bigram tokens to reach a mean cross-space subtotal of 10 or more. Instead of 12 % of evaluations showing evidence of polarity sensitivity (attraction to negative polarity

environments), only 13% *fail* to show evidence of polarity sensitivity. Despite this reversal, though, the fact that 78 evaluations of a ≥ 20 token bigram samples do not attain significance warrants further investigation, especially since 64 of those arise in comparison spaces where the natural preponderance of positive polarity tokens has been mitigated via the negative equivalent sampling of positive tokens (NEQ+ spaces).

Table 3.20: Significant Polarity Sensitive *Exactly* * Bigram Evaluations with a space specific subtotal ≥ 20

Polar sensitivities across 1,216 evaluations							
<i>unique bigrams</i> = 302							
<i>minimum bigram subtotal in any space</i> = 20							
<i>minimum bigram subtotal in superset</i> = 20							
<i>minimum mean bigram subtotal</i> = 10							
Polarity	Direction	mirror		super		total	ratio
		ALL+	NEQ+	ALL+	NEQ+		
neg	attract	7	4	291	228	530	0.872
	insignif	0	3	11	64	78	0.128
	repel	0	0	0	0	0	0
pos	attract	0	0	0	0	0	0
	insignif	0	3	11	64	78	0.128
	repel	7	4	291	228	530	0.872

3.4.4 Polarity Sensitivity of *Exactly* Bigrams Exhibiting Strongest Context-Blind Association

Not all *exactly* * bigrams showing the most context-blind bigram-internal cohesion show significant polarity sensitivity. While some do, if *exactly* were *universally* polarity sensitive, those bigrams with the greatest likelihood of occurring in the

first place should also be polarity sensitive. That is, even when bigrams with low overall frequency or with no particular context-blind association are ignored, there are still cases that fail to show significant polarity sensitivity.

Tables 3.21 and 3.22 display the context-blind association metrics for the 25 adjectives with the strongest symmetric—represented by LRC—bigram-internal association with the adverb *exactly*. Since bigram cohesion does not depend on context, these were designated via the ALL+ Superset data alone, which is shown in the table below. A minimum joint frequency restriction was also employed, requiring at least 50 *exactly* * bigram tokens for consideration ($f \geq 50$, which entails $f_2 \geq 50$). Note that the f column in this $\text{adv} \sim \text{adj}$ AM table corresponds to the f_2 attribute for the $\text{env} \sim \text{bigram}$ evaluations.

Table 3.21: Adjectives with strongest context-blind associations with *exactly* based on ALL+ superset only; ranked by LRC
 $\{f_2, f\} \geq 50$

#	<i>Exactly...</i>	LRC	ΔP avg	$\Delta P(x \text{ADJ})$	$\Delta P(\text{ADJ} x)$	G2	unexp_r	f	f2
1	<i>alike</i>	6.53	0.051	0.087	0.014	5,879	0.991	776	8,804
2	<i>right</i>	5.75	0.077	0.044	0.110	39,336	0.982	6,269	141,227
3	<i>opposite</i>	5.68	0.031	0.054	0.008	3,087	0.986	467	8,461
4	<i>sure</i>	5.40	0.094	0.034	0.154	51,389	0.977	8,840	256,817
5	<i>zero</i>	4.76	0.019	0.032	0.005	1,701	0.976	306	9,343
6	<i>parallel</i>	4.56	0.017	0.031	0.004	1,123	0.976	204	6,404
7	<i>stellar</i>	4.49	0.017	0.031	0.003	950	0.976	172	5,349
8	<i>analogous</i>	4.24	0.017	0.033	0.002	573	0.977	102	3,024
9	<i>conducive</i>	4.09	0.013	0.022	0.004	1,014	0.966	208	8,990
10	<i>ideal</i>	4.05	0.013	0.018	0.007	1,894	0.958	424	22,613
11	<i>perpendicular</i>	3.88	0.020	0.040	0.001	299	0.981	50	1,228
12	<i>equal</i>	3.63	0.011	0.013	0.008	1,896	0.943	491	35,927
13	<i>correct</i>	3.52	0.011	0.011	0.012	2,539	0.934	705	59,301
14	<i>cheap</i>	3.48	0.011	0.011	0.012	2,470	0.933	695	59,936

Continued on next page

Table 3.21: Adjectives with strongest context-blind associations with *exactly* based on *ALL+* superset only; ranked by *LRC* ($\{f2, f\} \geq 50$)

#	<i>Exactly...</i>	LRC	ΔP avg	$\Delta P(x \text{ADJ})$	$\Delta P(\text{ADJ} x)$	G2	unexp_r	f	f2
15	<i>forthcoming</i>	3.04	0.008	0.014	0.002	422	0.947	106	7,257
16	<i>equivalent</i>	3.04	0.007	0.012	0.002	545	0.938	147	11,634
17	<i>true</i>	2.95	0.016	0.006	0.026	4,454	0.892	1,647	227,128
18	<i>same</i>	2.94	0.007	0.008	0.007	1,277	0.911	422	48,249
19	<i>revolutionary</i>	2.81	0.007	0.011	0.002	429	0.934	120	10,183
20	<i>thrilled</i>	2.64	0.006	0.008	0.003	615	0.909	206	24,075
21	<i>new</i>	2.52	0.013	0.005	0.021	3,069	0.858	1,387	252,270
22	<i>clear</i>	2.46	0.016	0.004	0.027	3,737	0.848	1,779	346,404
23	<i>comparable</i>	2.39	0.005	0.008	0.002	343	0.914	111	12,202
24	<i>reassuring</i>	2.34	0.006	0.010	0.001	261	0.927	77	7,225
25	<i>novel</i>	2.15	0.005	0.010	0.001	219	0.924	66	6,416

In addition to the *ALL+* superset *AM* scores, Table 3.22 shows the context-blind values for the most relevant *AM*s across all comparison spaces. Both the above and below tables illustrate the difference between ΔP values calculated for polar environment comparisons ($E \sim \{\text{bigram}, \text{adv}, \text{adj}\}$) and those calculated for context-blind comparisons ($\text{exactly} \sim \text{ADJ}$). While backwards ΔP is especially informative for polarity investigations (and forward ΔP becomes near meaningless; see Chapter 2), when the sets being compared are equal in cardinality as is the case for the context-blind evaluations, forward and backward ΔP have similar levels of informativity. They serve to illustrate the directionality of symmetric scores like *LRC*, but the distribution of values is far less skewed, and thus less starkly informative overall. That is, while backwards ΔP is highly informative overall and forwards ΔP is near meaningless for polar investigations, entailing a large gap in informativity and very different distributions, the distributions of forwards and backwards ΔP values

are much more similar for context-blind evaluations, but the informativity of either is far less overall than that of backwards ΔP in the polar inquiries.

Table 3.22: Strongest context-blind associations with *exactly* (Selection based on ALL+ superset only)

#	<i>Exactly...</i>	$\Delta P(\text{ADJ} x)$				$\Delta P(x \text{ADJ})$				LRC			
		mirror		super		mirror		super		mirror		super	
		ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ
1	<i>alike</i>	0.08	0.06	0.01	0.00	0.20	0.37	0.09	0.14	7.35	6.85	6.53	3.69
2	<i>right</i>	0.13	0.08	0.11	0.02	0.02	0.03	0.04	0.03	4.36	3.18	5.75	2.25
3	<i>opposite</i>	0.01	0.00	0.01	0.00	0.02	0.03	0.05	0.04	0.00	0.00	5.68	0.49
4	<i>sure</i>	0.14	0.16	0.15	0.18	0.02	0.02	0.03	0.06	4.29	3.21	5.40	3.20
5	<i>zero</i>	0.00	0.00	0.01	0.00	0.03	0.05	0.03	0.04	0.00	0.00	4.76	0.00
6	<i>parallel</i>	0.00		0.00	0.00	0.02		0.03	0.09	0.00		4.56	2.19
7	<i>stellar</i>	0.00	0.00	0.00	0.00	0.04	0.08	0.03	0.21	0.00	0.00	4.49	4.32
8	<i>analogous</i>			0.00	0.00			0.03	0.08			4.24	1.61
9	<i>conductive</i>	0.00	0.00	0.00	0.00	0.01	0.02	0.02	0.10	0.00	0.00	4.09	3.28
10	<i>ideal</i>	0.00	0.00	0.01	0.01	0.01	0.03	0.02	0.12	0.00	0.00	4.05	3.78
11	<i>perpendicular</i>			0.00	0.00			0.04	0.13			3.88	0.23
12	<i>equal</i>	0.01	0.01	0.01	0.00	0.01	0.02	0.01	0.02	0.07	0.00	3.63	0.84
13	<i>correct</i>	0.01	0.00	0.01	0.00	0.01	0.00	0.01	0.02	0.00	0.00	3.52	1.49
14	<i>cheap</i>	0.01	0.01	0.01	0.01	0.00	0.01	0.01	0.10	0.00	0.00	3.48	3.63
15	<i>forthcoming</i>			0.00	0.00			0.01	0.04			3.04	1.68
16	<i>equivalent</i>	0.00	0.00	0.00	0.00	0.02	0.01	0.01	0.04	0.00	0.00	3.03	0.82
17	<i>true</i>	0.01	0.01	0.03	0.03	0.00	0.00	0.01	0.03	0.00	0.00	2.95	2.34
18	<i>same</i>	0.01	0.00	0.01	0.00	0.05	0.09	0.01	0.01	0.54	0.00	2.94	0.00
19	<i>revolutionary</i>	0.01	0.01	0.00	0.00	0.02	0.04	0.01	0.09	0.00	0.00	2.81	2.86
20	<i>thrilled</i>	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.03	0.00	0.00	2.64	1.61
21	<i>new</i>	0.02	0.03	0.02	0.03	0.00	0.01	0.00	0.06	0.00	0.00	2.52	2.98
22	<i>clear</i>	0.05	0.05	0.03	0.03	0.01	0.01	0.00	0.01	2.03	1.82	2.46	1.34
23	<i>comparable</i>			0.00	0.00			0.01	0.03			2.39	1.20
24	<i>reassuring</i>			0.00	0.00			0.01	0.07			2.34	2.09
25	<i>novel</i>	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.07	0.00	0.00	2.15	1.97

The polar environment associations seen for these *internally-cohesive* bigrams indicate that some of the most probable adjective arguments for *exactly* are not obligatorily polarity sensitive. The following table shows the LRC values for both

these and the 25 adjectives which produce *exactly ADJ* bigrams with the strongest indication of polarity sensitivity. These sets have a non-empty intersection with eight adjectives—*cheap*, *clear*, *conducive*, *ideal*, *new*, *revolutionary*, *sure*, and *true*—ranking in the top 25 for both context-blind association with *exactly* and polarity sensitivity when modified by *exactly*, thus a total of 42 adjectives will be evaluated for the remaining investigation of *exactly*’s polarity sensitivity.

The LRC values in Table 3.23 demonstrate that, while the 25 adjectives exhibiting the strongest evidence of polarity sensitivity under *exactly* modification (*polar* and *both* row groups) clearly have significant LRC values, the remaining 17 with strong context-blind attraction to *exactly* do not necessarily form bigrams (i.e., *exactly ADJ*) that exhibit a tendency toward negative polarity (as a unit).

Table 3.23: LRC for Polarity & *Exactly* Bigrams with the Most Relevant Adjectives

Top	Adv Polarity Dataset Pos Sample Adj	<i>exactly</i>							
		neg				pos			
		mirror		super		mirror		super	
		ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ
BLIND	<i>alike</i>	0.47	0.00	1.39	0.75	−0.47	0.00	−1.39	−0.75
	<i>analogous</i>			1.66	0.39			−1.66	−0.39
	<i>comparable</i>			4.71	1.83			−4.71	−1.83
	<i>correct</i>	0.00	0.00	3.02	2.02	0.00	0.00	−3.02	−2.02
	<i>equal</i>	0.00	0.00	0.79	0.13	0.00	0.00	−0.79	−0.13
	<i>equivalent</i>	0.00	0.00	1.51	0.09	0.00	0.00	−1.51	−0.09
	<i>forthcoming</i>			7.00	2.57			−7.00	−2.57
	<i>novel</i>	0.00	0.00	6.25	1.81	0.00	0.00	−6.25	−1.81
	<i>opposite</i>	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	<i>parallel</i>	0.00		1.18	0.05	0.00		−1.18	−0.05
	<i>perpendicular</i>			0.00	0.00			0.00	0.00
	<i>reassuring</i>			6.49	2.06			−6.49	−2.06

Continued on next page

Table 3.23: LRC for Polarity & *Exactly* Bigrams with the Most Relevant Adjectives

Top	Adv	<i>exactly</i>							
	Polarity	neg				pos			
	Dataset	mirror		super		mirror		super	
	Pos Sample	ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ
Adj									
	<i>right</i>	0.30	0.00	0.96	0.82	−0.30	0.00	−0.96	−0.82
	<i>same</i>	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	<i>stellar</i>	0.00	0.00	7.29	3.29	0.00	0.00	−7.29	−3.29
	<i>thrilled</i>	0.00	0.00	7.41	3.56	0.00	0.00	−7.41	−3.56
	<i>zero</i>	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
BOTH	<i>cheap</i>	0.00	0.00	9.11	5.12	0.00	0.00	−9.11	−5.12
	<i>clear</i>	3.94	1.69	8.98	6.47	−3.94	−1.69	−8.98	−6.47
	<i>conducive</i>	0.00	0.00	8.03	3.60	0.00	0.00	−8.03	−3.60
	<i>ideal</i>	0.00	0.00	8.07	4.38	0.00	0.00	−8.07	−4.38
	<i>new</i>	2.61	0.69	9.25	6.12	−2.61	−0.69	−9.25	−6.12
	<i>revolutionary</i>	0.34	0.00	7.19	2.76	−0.34	0.00	−7.19	−2.76
	<i>sure</i>	5.58	3.33	11.00	8.49	−5.58	−3.33	−11.00	−8.49
	<i>true</i>	0.00	0.00	6.74	5.30	0.00	0.00	−6.74	−5.30
POLAR	<i>difficult</i>	0.00	0.00	7.27	2.84	0.00	0.00	−7.27	−2.84
	<i>easy</i>	2.22	0.00	9.54	6.00	−2.22	0.00	−9.54	−6.00
	<i>fair</i>			7.66	3.93			−7.66	−3.93
	<i>famous</i>	0.00	0.00	7.32	2.88	0.00	0.00	−7.32	−2.88
	<i>friendly</i>	0.00	0.00	7.70	3.27	0.00	0.00	−7.70	−3.27
	<i>great</i>	0.00	0.00	7.53	4.18	0.00	0.00	−7.53	−4.18
	<i>happy</i>	0.00	0.00	8.23	4.70	0.00	0.00	−8.23	−4.70
	<i>impressive</i>	0.00	0.00	6.88	2.44	0.00	0.00	−6.88	−2.44
	<i>pleasant</i>	0.00	0.00	7.45	3.02	0.00	0.00	−7.45	−3.02
	<i>shocking</i>	0.00	0.00	7.54	3.11	0.00	0.00	−7.54	−3.11
	<i>shy</i>	0.00	0.00	7.24	2.81	0.00	0.00	−7.24	−2.81
	<i>straightforward</i>	0.00	0.00	6.60	2.16	0.00	0.00	−6.60	−2.16
	<i>subtle</i>	0.00	0.00	7.95	3.95	0.00	0.00	−7.95	−3.95
	<i>surprising</i>	0.00	0.00	8.89	4.71	0.00	0.00	−8.89	−4.71
	<i>uncommon</i>	0.00	0.00	6.71	2.28	0.00	0.00	−6.71	−2.28
	<i>unexpected</i>	0.00	0.00	6.60	2.16	0.00	0.00	−6.60	−2.16
	<i>welcome</i>	0.00	0.00	6.47	2.04	0.00	0.00	−6.47	−2.04

Four of these adjectives—*opposite*, *same*, *perpendicular*, and *zero*—form bigrams

that show no significant polarity sensitivity in any of the (four) comparison spaces⁷⁵. Many others, even those selected as the most strongly polarity sensitive, are insignificant or unattested in the mirror data despite reaching significance in at least the full (ALL+) superset comparison. Only the bigrams where *exactly* modifies *alike*, *right*, *clear*, *new*, *revolutionary*, *sure*, and *easy* show evidence of LRC-defined polarity sensitivity in the mirror data⁷⁶. Bigrams with *exactly* modifying *forthcoming*, *analogous*, *comparable*, *perpendicular*, *reassuring*, and *fair* do not appear in the Mirror Subset at all. Notably, *exactly parallel* is insignificant in the full mirror subset, but unattested in the negative equivalent sample. This fact is telling because the NEQ+ sample is identical for all negative polarity tokens, thus all *exactly parallel* bigram tokens in the ALL+ Mirror comparison space must necessarily occur with positive polarity (@P) designation.

3.4.5 Asymmetric AM Clashes

Quantifying polarity sensitivity on nonzero LRC alone clashes with asymmetrical effect size in some cases. **While insignificant LRC values do indicate polarity insensitivity, significant LRC values alone are insufficient for identifying true polarity sensitivity.** Whether any of the insignificant cases are due to low frequency (i.e., simply not enough observations to be significant) or weak/non-existent effect

⁷⁵That is, the bigrams either have LRC value of 0 or are unattested in the given comparison space

⁷⁶And only *clear*, *new*, and *sure* have an LRC > 0 in both positive samples

size is unclear given LRC values alone. The corresponding ΔP values, however, can elucidate this. If sample size (low frequency) alone prevents significance, backwards ΔP values will be high (or higher) where $LRC == 0$, but if effect size (deviation from typical distribution) is the cause.

It is important to consider the source of null LRC values because conflict between superset and subset outcomes which are due simply to frequency differences is not particularly interesting or informative: The subset (especially the NEQ+ selection) is, as discussed previously, so much smaller that large discrepancies in bigram subtotals are not unsurprising. Insignificance resulting from a difference in *effect size* between the superset and subset evaluations, however, could indicate a difference in the *class* of polarity sensitivity. That is, the superset polarity approximation method—reliance on the absence of a negative trigger to indicate positive polarity—means that it will include *wNPI* triggers that are not explicitly negative (e.g., questions, comparatives, hypotheticals, etc.) and thus show weaker effect sizes for potential *wNPIs* than it will for *sNPI*. However, the subset’s polarity approximation method—requiring the presence of a positive trigger—will not include such weakly negative environments in the positive class, and thus will not show smaller effect sizes for any weak NPIs, and may in fact show *larger* effect sizes for such cases, as the weakly triggered tokens will no longer offset the tokens found in negative environments.

That being said, none of the *exactly* bigrams discussed here as being significantly negative prone in the superset but insignificant in the subset show much change in

effect size between mirror subset and superset. There are cases of *exactly* * bigrams exhibiting this conflict between polarity approximations—nonzero (significant) LRC calculations for the negative environments in the superset, but insignificant (zero) LRC values in the mirror subset—where the effect size, as indicated by dP1, is much smaller within the mirror subset, even to the extent of becoming negative (i.e. *repel* instead of *attract*), but there are also cases where the effect size goes up slightly.

However, the transparent meaning of dP1 must also be considered before polarity sensitivity can be concluded. Consider Table 3.24, which juxtaposes polar AMs and frequency values for *exactly right* across the comparison spaces. Despite $LRC > 0$ for negative contexts, $\Delta P(*N(-)|\textit{exactly right}) = 0.06$ in the ALL+ superset comparison space. This indicates that the bigram *exactly right* is only 6% more likely to be negated⁷⁷ than other bigrams are likely to be negated. This does not appear to be a distribution indicative of true polarity sensitivity, which should require most if not all of its tokens have the requisite environment. Similarly *exactly equal*, as presented in Table 3.25 has a dP1 (in the full superset) of 0.1, which indicates it is only 10% more likely to be negated than other bigram forms.

The asymmetric AMs $\Delta P(E|\textit{bigram})$ and $P(E|\textit{bigram})$ are not impacted by the (expected) lack of bidirectional association between environment and bigram. As previously discussed in Chapter 2, polarity sensitivity is inherently an asymmetric phenomenon, and thus, despite the lack of *symmetric* significance, the following

⁷⁷In this case, “negated” is defined as having a valid dependency relationship with a *N trigger lemma.

Table 3.24: Polarity sensitivity of *exactly right*

Dataset	mirror				super			
Pos Sample	ALL		NEQ		ALL		NEQ	
Adj Total	5,535		2,019		141,227		21,217	
F2	133		68		6,269		877	
Direction	attract	repel	insignif		attract	repel	attract	repel
Polarity	neg	pos	neg	pos	neg	pos	neg	pos
P1	0.39	0.61	0.76	0.24	0.10	0.90	0.72	0.28
dP1	0.22	−0.22	0.26	−0.26	0.06	−0.06	0.22	−0.22
LRC	0.30	−0.30	0	0	0.96	−0.96	0.82	−0.82
unexp_r	0.56	−0.36	0.35	−1	0.56	−0.06	0.31	−0.80

Table 3.25: Polarity sensitivity of *exactly equal*

Dataset	mirror				super			
Pos Sample	ALL		NEQ		ALL		NEQ	
Adj Total	1,236		329		35,927		3,051	
F2	13		7		491		89	
Direction	insignif		insignif		attract	repel	attract	repel
Polarity	neg	pos	neg	pos	neg	pos	neg	pos
P1	0.23	0.77	0.43	0.57	0.14	0.86	0.80	0.20
dP1	0.06	−0.06	−0.07	0.07	0.10	−0.10	0.30	−0.30
LRC	0	0	0	0	0.79	−0.79	0.13	−0.13
unexp_r	0.25	−0.07	−0.17	0.13	0.70	−0.12	0.37	−1

values should be considered for polarity sensitivity inquiries. Although receiving insignificant **LRC** values for polarity environment within the mirror subset, the bigrams *exactly equal*, *exactly opposite*, and *exactly parallel* have backwards delta P values indicating they are *less* likely to occur in negative environments than other bigrams are to occur in negative environments in general. These effects are weak and do not indicate a need for positive polarity licensing, but support polar insensitivity as the source of the insignificant LRC values observed.

Table 3.26: Polarity sensitivity of *exactly opposite*

Dataset	mirror				super			
Pos Sample	ALL		NEQ		ALL		NEQ	
Adj Total	279		74		8,461		684	
F2	7		2		467		32	
Direction	insignif		insignif		insignif		insignif	
Polarity	neg	pos	neg	pos	neg	pos	neg	pos
P1	0	1	0	1	0.03	0.97	0.41	0.59
dP1	-0.17	0.17	-0.50	0.50	-0.02	0.02	-0.09	0.09
LRC	0	0	0	0	0	0	0	0
unexp_r	-1	0.17	-1	0.50	-0.58	0.02	-0.23	0.16

Table 3.27: $\Delta P(\text{ENV}|\text{exactly ADJ})$ for Polarity & *Exactly* Bigrams with the Most Relevant Adjectives

		<i>exactly</i>							
		neg				pos			
		mirror		super		mirror		super	
		ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ
Top	Adv Polarity Dataset Pos Sample Adj								
BLIND	<i>alike</i>	0.30	0.22	0.13	0.32	-0.30	-0.22	-0.13	-0.32
	<i>analogous</i>			0.29	0.47			-0.29	-0.47
	<i>comparable</i>			0.75	0.48			-0.75	-0.48
	<i>correct</i>	0.13	0.50	0.32	0.42	-0.13	-0.50	-0.32	-0.42
	<i>equal</i>	0.06	-0.07	0.10	0.30	-0.06	0.07	-0.10	-0.30
	<i>equivalent</i>	0.08	0.50	0.23	0.39	-0.08	-0.50	-0.23	-0.39
	<i>forthcoming</i>			0.96	0.50			-0.96	-0.50
	<i>novel</i>	0.83	0.50	0.96	0.50	-0.83	-0.50	-0.96	-0.50
	<i>opposite</i>	-0.17	-0.50	-0.02	-0.09	0.17	0.50	0.02	0.09
	<i>parallel</i>	-0.17		0.18	0.37	0.17		-0.18	-0.37
	<i>perpendicular</i>			0.16	0.27			-0.16	-0.27
	<i>reassuring</i>			0.96	0.50			-0.96	-0.50
	<i>right</i>	0.22	0.26	0.06	0.22	-0.22	-0.26	-0.06	-0.22
	<i>same</i>	0.08	0.00	0.04	0.18	-0.08	0.00	-0.04	-0.18
	<i>stellar</i>	0.83	0.50	0.94	0.50	-0.83	-0.50	-0.94	-0.50

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Table 3.27: $\Delta P(\text{ENV}|\textit{exactly} \text{ ADJ})$ for Polarity & *Exactly* Bigrams with the Most Relevant Adjectives

Top	Adv Polarity Dataset Pos Sample Adj	<i>exactly</i>							
		neg				pos			
		mirror		super		mirror		super	
		ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ
BOTH	<i>thrilled</i>	0.83	0.50	0.94	0.50	−0.83	−0.50	−0.94	−0.50
	<i>zero</i>	0.49	0.50	0.00	0.11	−0.49	−0.50	0.00	−0.11
	<i>cheap</i>	0.83	0.50	0.95	0.50	−0.83	−0.50	−0.95	−0.50
	<i>clear</i>	0.83	0.50	0.94	0.50	−0.83	−0.50	−0.94	−0.50
	<i>conductive</i>	0.83	0.50	0.96	0.50	−0.83	−0.50	−0.96	−0.50
	<i>ideal</i>	0.83	0.50	0.94	0.50	−0.83	−0.50	−0.94	−0.50
	<i>new</i>	0.79	0.50	0.94	0.50	−0.79	−0.50	−0.94	−0.50
	<i>revolutionary</i>	0.83	0.50	0.96	0.50	−0.83	−0.50	−0.96	−0.50
	<i>sure</i>	0.83	0.50	0.95	0.50	−0.83	−0.50	−0.95	−0.50
	<i>true</i>	0.49	0.42	0.83	0.49	−0.49	−0.42	−0.83	−0.49
POLAR	<i>difficult</i>	0.83	0.50	0.96	0.50	−0.83	−0.50	−0.96	−0.50
	<i>easy</i>	0.83	0.50	0.95	0.50	−0.83	−0.50	−0.95	−0.50
	<i>fair</i>			0.94	0.50			−0.94	−0.50
	<i>famous</i>	0.83	0.50	0.96	0.50	−0.83	−0.50	−0.96	−0.50
	<i>friendly</i>	0.83	0.50	0.96	0.50	−0.83	−0.50	−0.96	−0.50
	<i>great</i>	0.83	0.50	0.93	0.50	−0.83	−0.50	−0.93	−0.50
	<i>happy</i>	0.83	0.50	0.94	0.50	−0.83	−0.50	−0.94	−0.50
	<i>impressive</i>	0.83	0.50	0.96	0.50	−0.83	−0.50	−0.96	−0.50
	<i>pleasant</i>	0.83	0.50	0.96	0.50	−0.83	−0.50	−0.96	−0.50
	<i>shocking</i>	0.83	0.50	0.96	0.50	−0.83	−0.50	−0.96	−0.50
	<i>shy</i>	0.83	0.50	0.96	0.50	−0.83	−0.50	−0.96	−0.50
	<i>straightforward</i>	0.83	0.50	0.96	0.50	−0.83	−0.50	−0.96	−0.50
	<i>subtle</i>	0.83	0.50	0.95	0.50	−0.83	−0.50	−0.95	−0.50
	<i>surprising</i>	0.83	0.50	0.95	0.50	−0.83	−0.50	−0.95	−0.50
	<i>uncommon</i>	0.83	0.50	0.96	0.50	−0.83	−0.50	−0.96	−0.50
	<i>unexpected</i>	0.83	0.50	0.96	0.50	−0.83	−0.50	−0.96	−0.50
	<i>welcome</i>	0.83	0.50	0.96	0.50	−0.83	−0.50	−0.96	−0.50

In summary, the first option—that *exactly* itself is solely responsible for its observed “negative tilt”—must be rejected on the grounds that it is insufficient to

predict polarity sensitivity.

3.5 *Exactly* is Not Vacuously Polarity Sensitive

The purpose of this section is to entertain the possibility that, although *exactly* appears to have a propensity for negation, the adverb itself is vacuous in this regard, and “inherits” this feature from other elements entirely: the particular adjectives it associates with.

For this to be the case, the adjectives most associated with *exactly* independent of environment (strongest **exactly~ADJ** AM scores) and the adjectives which, when paired with *exactly*, create the most polarity sensitive bigrams/bigrams most attracted to negative environments (strongest $\ominus\sim\textit{exactly}$ ADJ AM scores), should show indication of *negative* polarity sensitivity when evaluated **independent of adverb** ($\ominus\sim\text{ADJ}$ AM scores > 0).

If adjectives alone were responsible for the negative lean observed, then the following observations should apply. Firstly, a majority of the adjectives showing attraction with *exactly* **independent of environment** would need to exhibit a similar attraction to negative environments **independent of adverb**. This means the adjectives with high context-blind association with *exactly* (**exactly~ADJ**) should have high $\ominus\sim\text{ADJ}$ scores. Secondly, any *exactly* ADJ bigram showing a strong association with negative environments ($\ominus\sim\text{exactly_ADJ}$) should have an adjective

form which shows a strong association with negative environments, **independent of adverb** ($\ominus \sim \text{ADJ}$). That is, adjectives that, when combined with *exactly*, exhibit strong negative polarity sensitivity should also exhibit negative polarity sensitivity on their own.

However, neither case is observed in the data.

For *exactly* to inherit its ostensible negative polarity inclination from its adjective argument, the most common arguments must have such a negative inclination to inherit. However, many of the most relevant adjectives for *exactly*—the forty-two adjectives which either exhibit the strongest context-blind attraction⁷⁸ or form bigrams which exhibit the strongest polarity environment attraction⁷⁹—are not independently negative polarity sensitive.

Table 3.28 shows the polarity environment sensitivity totals—quantified as significant LRC values—of this adjective selection.

Just over half of the environment evaluations for these adjectives—88 of 168 (per environment polarity), $\approx 52\%$ —show a significant attraction ($\text{LRC} > 0$) to negative environments. Over a third of the evaluations—59 of 168, $\approx 35\%$ —do not exhibit significant interactions with environment (of either polarity), and the remaining evaluations—21 of 168, $\approx 13\%$ —show significant attraction to *positive* environments (are “repelled” by negative environments). This serves as a solid

⁷⁸25 total

⁷⁹25 total, 8 of which are also in the top 25 context-blind attraction set

Table 3.28: Polar sensitivities of Most Relevant Adjectives

unique adjective forms in selection = 42
 42×4 comparison spaces $\times 2$ polarities
= 336 evaluations (168 per polarity)
minimum adjective subtotal = 1,228

Polarity	Direction	mirror		super		total	ratio
		ALL+	NEQ+	ALL+	NEQ+		
neg	attract	17	13	29	29	88	0.524
	insignif	20	24	7	8	59	0.351
	repel	5	5	6	5	21	0.125
pos	attract	5	5	6	5	21	0.125
	insignif	20	24	7	8	59	0.351
	repel	17	13	29	29	88	0.524

illustration that *exactly*'s arguments are not consistently negative-prone regardless of adverbial modification.

Table 3.29 shows these most relevant adjectives separated by their relevance factor, with BLIND indicating strong bigram-internal, context-blind cohesion with *exactly*, POLAR indicating participation in strong (negative) polarity sensitive *exactly* bigrams, and BOTH indicating the intersection of these sets. Evaluations where $LRC > 0$ (attractive environments) are highlighted in green, and those where $LRC < 0$ (repellent environments) are highlighted in brown/tan. If all adjectives were negative-prone independent of adverb, no cells on the right half (under the **pos** heading) would be green, and no cells on the left half (under the **neg** heading) would be brown/tan.

Notably, even those adjectives which form polarity sensitive bigrams with *exactly* do not exhibit independent polarity sensitivity when the modifying adverb is ignored.

If the negative polarity prone *exactly* * bigrams were inheriting their disposition from the adjectives, the 25 adjectives participating in the most strongly polarity sensitive bigrams should all exhibit an adverb-independent negative polarity preference.

While over half of the adjectives most relevant for *exactly* (64.3%, 27/42) show a significant (and non-conflicted) attraction to negative polarity environments in at least one of the four comparison spaces⁸⁰ the remaining 15 (35.7%, 15/42) either have no significant polarity sensitivity or are actually *repelled* by negative environments (i.e., attracted to positive polarity environments).

Just over one-tenth of the adjectives selected (11.9%, 5/42) show no significant association either way with polarity environment in any comparison space: *equivalent*, *opposite*, *parallel*, *perpendicular*, and *shy*. Three adjectives—*famous*, *subtle*, and *unexpected*—have conflicting values. The adjective *famous* shows a negative lean in the superset, but a positive lean in the subset (ALL+ at least), while both *subtle* and *unexpected* exhibit a negative lean in the subset but a positive lean in the superset. The remaining sixth (16.7%, 7/42)—*equal*, *same*, *zero*, *new*, *difficult*, *friendly*, and *welcome*—show significant (and non-conflicted, though weak in some cases, especially *new*) attraction to positive polarity in at least one comparison space. This evidence of positive polarity attraction is not consistently attributed to either polarity approximation, with four adjectives (*same*, *zero*, *new*, and *welcome*) only

⁸⁰13 adjectives are significant in all 4 comparisons: *comparable*, *forthcoming*, *novel*, *thrilled*, *clear*, *conductive*, *revolutionary*, *sure*, *true*, *easy*, *great*, *happy*, and *surprising*
 3 are significant in 3/4: *reassuring*, *fair*, and *straightforward*
 11 are significant in 2/4: *alike*, *analogous*, *correct*, *right*, *stellar*, *cheap*, *ideal*, *impressive* (very weakly), *pleasant*, *shocking*, and *uncommon*

showing a positive polarity disposition in the superset, but two (*equal* and *friendly*) only showing a positive disposition in the subset. Only *difficult* exhibits a significant attraction to positive polarity environments in all four comparison spaces. From this illustrative selection alone, there is no discernible pattern between polarity sensitivity or internal bigram cohesion of the adjective’s distribution under *exactly* modification and its adverb independent polarity sensitivity: The aforementioned cases are scattered across the three divisions labeled in Table 3.29 (POLAR, BOTH, and BLIND).

Table 3.29: $E \sim \text{ADJ LRC}$ for Adjectives Relevant to *Exactly*

TOP	Polarity Dataset Approx Meth Pos Sample adjective	(-)				(+))			
		mirror		super		mirror		super	
		present (+)		absent (-)		present (+)		absent (-)	
		ALL+	NEQ+	ALL+	NEQ+	ALL+	NEQ+	ALL+	NEQ+
BLIND	<i>alike</i>	0	0	0.54	0.29	0	0	-0.54	-0.29
	<i>analogous</i>	0	0	0.47	0.26	0	0	-0.47	-0.26
	<i>comparable</i>	0.79	0.33	1.84	1.64	-0.79	-0.33	-1.84	-1.64
	<i>correct</i>	0	0	1.37	1.26	0	0	-1.37	-1.26
	<i>equal</i>	-0.61	-0.48	0	0	0.61	0.48	0	0
	<i>equivalent</i>	0	0	0	0	0	0	0	0
	<i>forthcoming</i>	1.04	0.39	2.81	2.47	-1.04	-0.39	-2.81	-2.47
	<i>novel</i>	0.59	0.11	0.89	0.66	-0.59	-0.11	-0.89	-0.66
	<i>opposite</i>	0	0	0	0	0	0	0	0
	<i>parallel</i>	0	0	0	0	0	0	0	0
	<i>perpendicular</i>	0	0	0	0	0	0	0	0
	<i>reassuring</i>	0.19	0	1.07	0.93	-0.19	0	-1.07	-0.93
	<i>right</i>	0	0	1.35	1.32	0	0	-1.35	-1.32
	<i>same</i>	0	0	-2.33	-2.29	0	0	2.33	2.29
	<i>stellar</i>	0	0	0.98	0.70	0	0	-0.98	-0.70
	<i>thrilled</i>	0.75	0.38	2.27	2.08	-0.75	-0.38	-2.27	-2.08

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Table 3.29: $E \sim \text{ADJ LRC}$ for Adjectives Relevant to *Exactly* (cont'd)

Top	Polarity		(-)				(+)			
	Dataset		mirror		super		mirror		super	
	Pos	Sample	ALL+	NEQ+	ALL+	NEQ+	ALL+	NEQ+	ALL+	NEQ+
	L2									
BOTH	<i>zero</i>		0	0	-1.37	-1.26	0	0	1.37	1.26
	<i>cheap</i>		0	0	0.50	0.45	0	0	-0.50	-0.45
	<i>clear</i>		1.24	1.13	2.46	2.45	-1.24	-1.13	-2.46	-2.45
	<i>conductive</i>		0.48	0.38	1.98	1.73	-0.48	-0.38	-1.98	-1.73
	<i>ideal</i>		0	0	1.16	0.97	0	0	-1.16	-0.97
	<i>new</i>		0	0	-0.02	0	0	0	0.02	0
	<i>revolutionary</i>		0.69	0.30	0.49	0.25	-0.69	-0.30	-0.49	-0.25
	<i>sure</i>		4.56	4.25	4.40	4.31	-4.56	-4.25	-4.40	-4.31
	<i>true</i>		0.96	0.81	1.40	1.31	-0.96	-0.81	-1.40	-1.31
POLAR	<i>difficult</i>		-0.41	-0.39	-0.09	-0.06	0.41	0.39	0.09	0.06
	<i>easy</i>		0.69	0.65	1.90	1.87	-0.69	-0.65	-1.90	-1.87
	<i>fair</i>		0.10	0	1.38	1.29	-0.10	0	-1.38	-1.29
	<i>famous</i>		0.29	0	-1.30	-1.28	-0.29	0	1.30	1.28
	<i>friendly</i>		-1.56	-1.43	0	0	1.56	1.43	0	0
	<i>great</i>		0.28	0.25	1.33	1.28	-0.28	-0.25	-1.33	-1.28
	<i>happy</i>		0.59	0.51	0.63	0.61	-0.59	-0.51	-0.63	-0.61
	<i>impressive</i>		0	0	0.09	0.03	0	0	-0.09	-0.03
	<i>pleasant</i>		0	0	0.97	0.91	0	0	-0.97	-0.91
	<i>shocking</i>		0	0	0.43	0.32	0	0	-0.43	-0.32
	<i>shy</i>		0	0	0	0	0	0	0	0
	<i>straightforward</i>		0.05	0	0.93	0.90	-0.05	0	-0.93	-0.90
	<i>subtle</i>		-0.81	-0.76	0.37	0.30	0.81	0.76	-0.37	-0.30
	<i>surprising</i>		0.97	0.83	2.65	2.51	-0.97	-0.83	-2.65	-2.51
	<i>uncommon</i>		0	0	2.68	2.43	0	0	-2.68	-2.43
	<i>unexpected</i>		-0.84	-0.73	0.77	0.65	0.84	0.73	-0.77	-0.65
	<i>welcome</i>		0	0	-0.14	-0.04	0	0	0.14	0.04

Table 3.30 shows the backwards asymmetric $\Delta P(\text{polarity}|\text{adjective})$ for the same evaluations. Despite their lack of significance indicators, these values are most relevant for polarity sensitivity investigations as argued in § 2.2.3.1. Since this is an effect size measure rather than a hypothesis testing measure, it does not

encode statistical significance (Evert 2022). However, it does elucidate the adjective propensities for either environment polarity, without the potential obfuscation of also considering the forward association of the environment propensities for particular adjectives.

These values corroborate the general observation relevant to this discussion: neither the tendency of creating a negative polarity prone *exactly* * bigram nor a strong association with *exactly* regardless of environment appears to pattern consistently with adverb-independent associations between polarity environments and adjective.

For those cases where significance ($\text{LRC} \neq 0$) was not reached in any comparison space—*equivalent*, *opposite*, *parallel*, *perpendicular*, and *shy*⁸¹—or unconflicted but partial positive polarity attraction was observed—*equal*, *same*, *zero*, *new*, *friendly*, *welcome*⁸²—dP1 values mostly suggest that the direction of effect, even if insignificant per LRC calculation, is a preference for positive polarity. The exceptions to this arise in the very weak attraction seen for $*N \sim \textit{equivalent}$, $*N \sim \textit{shy}$, and $*N \sim \textit{friendly}$.

⁸¹color coded as purple text

⁸²color coded as blue text

Table 3.30: Independent Backwards Association, $\Delta P(E|\text{ADJ})$, for Adjectives Relevant to *Exactly*

TOP	Polarity	(-)				(+)			
	Dataset	mirror		super		mirror		super	
	Approx Meth	present (+)		absent (-)		present (+)		absent (-)	
	Pos Sample	ALL+	NEQ+	ALL+	NEQ+	ALL+	NEQ+	ALL+	NEQ+
	adjective								
BLIND	<i>alike</i>	0.02	0.05	0.035	0.14	-0.02	-0.05	-0.035	-0.14
	<i>analogous</i>	0.08	0.08	0.044	0.19	-0.08	-0.08	-0.044	-0.19
	<i>comparable</i>	0.25	0.27	0.117	0.31	-0.25	-0.27	-0.117	-0.31
	<i>correct</i>	0.03	0.05	0.070	0.23	-0.03	-0.05	-0.070	-0.23
	<i>equal</i>	-0.10	-0.23	-0.003	-0.02	0.10	0.23	0.003	0.02
	<i>equivalent</i>	-0.10	-0.26	0.003	0.01	0.10	0.26	-0.003	-0.01
	<i>forthcoming</i>	0.32	0.31	0.230	0.39	-0.32	-0.31	-0.230	-0.39
	<i>novel</i>	0.21	0.23	0.056	0.20	-0.21	-0.23	-0.056	-0.20
	<i>opposite</i>	-0.11	-0.27	-0.006	-0.03	0.11	0.27	0.006	0.03
	<i>parallel</i>	-0.05	-0.08	-0.008	-0.07	0.05	0.08	0.008	0.07
	<i>perpendicular</i>	-0.07	-0.10	-0.009	-0.03	0.07	0.10	0.009	0.03
	<i>reassuring</i>	0.15	0.21	0.065	0.24	-0.15	-0.21	-0.065	-0.24
	<i>right</i>	0.02	0.02	0.066	0.23	-0.02	-0.02	-0.066	-0.23
	<i>same</i>	-0.08	-0.14	-0.038	-0.37	0.08	0.14	0.038	0.37
	<i>stellar</i>	0.15	0.16	0.063	0.22	-0.15	-0.16	-0.063	-0.22
	<i>thrilled</i>	0.20	0.22	0.153	0.34	-0.20	-0.22	-0.153	-0.34
	<i>zero</i>	-0.07	-0.22	-0.034	-0.32	0.07	0.22	0.034	0.32
BOTH	<i>cheap</i>	-0.04	-0.09	0.023	0.11	0.04	0.09	-0.023	-0.11
	<i>clear</i>	0.19	0.23	0.163	0.36	-0.19	-0.23	-0.163	-0.36
	<i>conductive</i>	0.24	0.34	0.134	0.32	-0.24	-0.34	-0.134	-0.32
	<i>ideal</i>	0.08	0.07	0.061	0.21	-0.08	-0.07	-0.061	-0.21
	<i>new</i>	-0.01	-0.01	-0.003	-0.02	0.01	0.01	0.003	0.02
	<i>revolutionary</i>	0.20	0.22	0.031	0.12	-0.20	-0.22	-0.031	-0.12
	<i>sure</i>	0.69	0.47	0.456	0.47	-0.69	-0.47	-0.456	-0.47
	<i>true</i>	0.15	0.19	0.069	0.23	-0.15	-0.19	-0.069	-0.23
POLAR	<i>difficult</i>	-0.05	-0.11	-0.004	-0.02	0.05	0.11	0.004	0.02
	<i>easy</i>	0.10	0.14	0.107	0.30	-0.10	-0.14	-0.107	-0.30
	<i>fair</i>	0.10	0.12	0.072	0.24	-0.10	-0.12	-0.072	-0.24
	<i>famous</i>	0.09	0.11	-0.028	-0.23	-0.09	-0.11	0.028	0.23
	<i>friendly</i>	-0.13	-0.30	0.002	0.01	0.13	0.30	-0.002	-0.01
	<i>great</i>	0.06	0.10	0.063	0.22	-0.06	-0.10	-0.063	-0.22
	<i>happy</i>	0.09	0.12	0.025	0.12	-0.09	-0.12	-0.025	-0.12

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Table 3.30: Independent Backwards Association, $\Delta P(E|\text{ADJ})$, for Adjectives Relevant to *Exactly*, (cont'd)

Top	Polarity	(-)				(+))			
	Dataset	mirror		super		mirror		super	
	Pos Sample L2	ALL+	NEQ+	ALL+	NEQ+	ALL+	NEQ+	ALL+	NEQ+
	<i>impressive</i>	0.01	0.02	0.006	0.03	-0.01	-0.02	-0.006	-0.03
	<i>pleasant</i>	0.01	0.03	0.046	0.19	-0.01	-0.03	-0.046	-0.19
	<i>shocking</i>	0.05	0.07	0.022	0.10	-0.05	-0.07	-0.022	-0.10
	<i>shy</i>	-0.04	-0.07	<0.001	0.01	0.04	0.07	>-0.001	-0.01
	<i>straightforward</i>	0.06	0.09	0.044	0.18	-0.06	-0.09	-0.044	-0.18
	<i>subtle</i>	-0.10	-0.23	0.018	0.09	0.10	0.23	-0.018	-0.09
	<i>surprising</i>	0.17	0.21	0.190	0.37	-0.17	-0.21	-0.190	-0.37
	<i>uncommon</i>	-0.03	-0.02	0.208	0.38	0.03	0.02	-0.208	-0.38
	<i>unexpected</i>	-0.11	-0.24	0.038	0.16	0.11	0.24	-0.038	-0.16
	<i>welcome</i>	-0.05	-0.08	-0.009	-0.05	0.05	0.08	0.009	0.05

Finally, the LRC values for *exactly*~*ADJ* (BLIND), $(-)\sim\text{ADJ}$ (ADJ), and $(-)\sim\text{exactly ADJ}$ (BIGRAM) are juxtaposed in Table 3.31, ranked by descending ALL+ *N ~ADJ values. Of particular note is the fact that the only adjective which shows a ubiquitous positive polarity propensity, *difficult*, appears to be repelled by *exactly* when polarity is ignored in the superset. All other adjectives in the selected set either have no significant association with *exactly* or show an attraction to the adverb. Regardless, the array of LRC values below illustrates that the adverb-independent polarity sensitivities of these most relevant adjectives cannot be sufficient to explain the negative polarity propensity exhibited by adjective-independent *exactly*.

Table 3.31: Juxtaposed LRC for Relevant ADJ across context-blind, bigram polarity, and adjective polarity

Polarity	?				neg							
L2 Unit	BLIND				ADJ				BIGRAM			
Dataset	mirror		super		mirror		super		mirror		super	
Pos Sample	ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ
Adj												
<i>sure</i>	4.29	3.21	5.40	3.20	4.56	4.25	4.40	4.31	5.58	3.33	11.00	8.49
<i>forthcoming</i>			3.04	1.68	1.04	0.39	2.81	2.47			7.00	2.57
<i>uncommon</i>	0.00	0.00	2.01	0.75	0.00	0.00	2.68	2.43	0.00	0.00	6.71	2.28
<i>surprising</i>	0.00	0.00	2.47	1.24	0.97	0.83	2.65	2.51	0.00	0.00	8.90	4.71
<i>clear</i>	2.03	1.82	2.46	1.34	1.24	1.13	2.46	2.45	3.94	1.69	8.98	6.47
<i>thrilled</i>	0.00	0.00	2.64	1.61	0.75	0.38	2.27	2.08	0.00	0.00	7.41	3.56
<i>conductive</i>	0.00	0.00	4.09	3.28	0.48	0.38	1.98	1.73	0.00	0.00	8.03	3.60
<i>easy</i>	0.00	0.00	0.91	0.16	0.69	0.65	1.90	1.87	2.22	0.00	9.54	6.00
<i>comparable</i>			2.39	1.20	0.79	0.33	1.84	1.64			4.71	1.83
<i>true</i>	0.00	0.00	2.95	2.34	0.96	0.81	1.40	1.31	0.00	0.00	6.74	5.30
<i>fair</i>			2.20	1.76	0.10	0.00	1.38	1.29			7.66	3.93
<i>correct</i>	0.00	0.00	3.52	1.49	0.00	0.00	1.37	1.26	0.00	0.00	3.02	2.02
<i>right</i>	4.36	3.18	5.75	2.25	0.00	0.00	1.35	1.32	0.30	0.00	0.96	0.82
<i>great</i>	0.00	0.00	0.00	0.00	0.28	0.25	1.33	1.28	0.00	0.00	7.53	4.18
<i>ideal</i>	0.00	0.00	4.05	3.78	0.00	0.00	1.16	0.97	0.00	0.00	8.07	4.38
<i>reassuring</i>			2.34	2.09	0.19	0.00	1.07	0.93			6.49	2.06
<i>stellar</i>	0.00	0.00	4.49	4.32	0.00	0.00	0.98	0.70	0.00	0.00	7.29	3.29
<i>pleasant</i>	0.00	0.00	0.80	0.63	0.00	0.00	0.97	0.91	0.00	0.00	7.45	3.02
<i>straightforward</i>	0.00	0.00	0.00	0.00	0.05	0.00	0.93	0.90	0.00	0.00	6.60	2.16
<i>novel</i>	0.00	0.00	2.15	1.97	0.59	0.11	0.89	0.66	0.00	0.00	6.25	1.81
<i>unexpected</i>	0.00	0.00	0.42	0.35	-0.84	-0.73	0.77	0.65	0.00	0.00	6.60	2.16
<i>happy</i>	0.00	0.00	0.00	0.00	0.59	0.51	0.63	0.61	0.00	0.00	8.23	4.70
<i>alike</i>	7.35	6.85	6.53	3.69	0.00	0.00	0.54	0.29	0.47	0.00	1.39	0.75
<i>cheap</i>	0.00	0.00	3.48	3.63	0.00	0.00	0.50	0.45	0.00	0.00	9.11	5.12
<i>revolutionary</i>	0.00	0.00	2.81	2.86	0.69	0.30	0.49	0.25	0.34	0.00	7.19	2.76
<i>analogous</i>			4.24	1.61	0.00	0.00	0.47	0.26			1.66	0.39
<i>shocking</i>	0.00	0.00	1.50	1.61	0.00	0.00	0.43	0.32	0.00	0.00	7.54	3.11
<i>subtle</i>	0.00	0.00	2.05	2.21	-0.81	-0.76	0.37	0.30	0.00	0.00	7.95	3.95
<i>impressive</i>	0.00	0.00	0.00	0.00	0.00	0.00	0.09	0.03	0.00	0.00	6.88	2.44
<i>equivalent</i>	0.00	0.00	3.03	0.82	0.00	0.00	0.00	0.00	0.00	0.00	1.51	0.09
<i>parallel</i>	0.00		4.56	2.19	0.00	0.00	0.00	0.00	0.00		1.18	0.05
<i>opposite</i>	0.00	0.00	5.68	0.49	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
<i>equal</i>	0.07	0.00	3.63	0.84	-0.61	-0.48	0.00	0.00	0.00	0.00	0.79	0.13
<i>perpendicular</i>			3.88	0.23	0.00	0.00	0.00	0.00			0.00	0.00

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Table 3.31: Juxtaposed LRC for Relevant ADJ across context-blind, bigram polarity, and adjective polarity

Polarity	?				neg							
L2 Unit	BLIND				ADJ				BIGRAM			
Dataset	mirror		super		mirror		super		mirror		super	
Pos Sample	ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ
Adj												
<i>shy</i>	0.00	0.00	1.09	1.54	0.00	0.00	0.00	0.00	0.00	0.00	7.24	2.81
<i>friendly</i>	0.00	0.00	0.00	0.25	-1.56	-1.43	0.00	0.00	0.00	0.00	7.70	3.27
<i>new</i>	0.00	0.00	2.52	2.98	0.00	0.00	-0.02	0.00	2.61	0.69	9.25	6.12
<i>difficult</i>	0.00	0.00	-1.30	-0.95	-0.41	-0.39	-0.09	-0.06	0.00	0.00	7.27	2.84
<i>welcome</i>	0.00	0.00	0.00	0.00	0.00	0.00	-0.14	-0.04	0.00	0.00	6.47	2.04
<i>famous</i>	0.00	0.00	0.00	0.00	0.29	0.00	-1.30	-1.28	0.00	0.00	7.32	2.88
<i>zero</i>	0.00	0.00	4.76	0.00	0.00	0.00	-1.37	-1.26	0.00	0.00	0.00	0.00
<i>same</i>	0.54	0.00	2.94	0.00	0.00	0.00	-2.33	-2.29	0.00	0.00	0.00	0.00

The absence of independent negative bias for the adjectives found in the most negatively associated *exactly* bigrams makes it unlikely that *exactly*'s pattern of negative association overall is inherited from specific adjectives it modifies. For example, despite *exactly new* showing a very strong preference for negative polarity environments (Table 3.32) the adjective *new* considered independently of adverb (Table 3.33) shows no conclusive evidence of sensitivity to any of the four comparison spaces. Although the LRC does show a weak attraction to the positive superset ($*C_N \sim new$) it is countered by the $dP1$ value of 0.00, indicating that the polarity distribution for *new* bigrams does not diverge from those with other adjectives. The same can be said for *shy* despite the polar associations of *exactly shy*, as illustrated by the values in Table 3.35 and Table 3.34.

Table 3.32: Polarity sensitivity of *exactly new*

Dataset	super				mirror			
(+) Sample	NEQ		ALL		NEQ		ALL	
Adj Total	21,548		252,270		4,393		12,793	
F2	1,372		1,387		29		30	
Direction	repel	attract	repel	attract	repel	attract	repel	attract
Polarity	(+)	(-)	(+)	(-)	(+)	(-)	(+)	(-)
F	1	1,371	16	1,371	0	29	1	29
Env~Exactly New								
P1	0	1	0.01	0.99	0	1	0.03	0.97
dP1	-0.50	0.50	-0.94	0.94	-0.50	0.50	-0.79	0.79
LRC	-6.12	6.12	-9.25	9.25	-0.69	0.69	-2.61	2.61
unexp_r	-1	0.50	-1	0.96	-1	0.50	-1	0.82

Table 3.33: Independent polarity sensitivity of *new*

L2	new							
Dataset	super				mirror			
(+) Sample	NEQ		ALL		NEQ		ALL	
Direction	insignif		repel	attract	insignif		insignif	
F2	21,548		252,270	252,270	4,393		12,793	
Polarity	(+)	(-)	(-)	(+)	(+)	(-)	(+)	(-)
F	11,160	10,388	10,388	241,882	2,253	2,140	10,653	2,140
$E \sim New$								
P1	0.52	0.48	0.04	0.96	0.51	0.49	0.83	0.17
dP1	0.02	-0.02	0.00	0.00	0.01	-0.01	0.01	-0.01
LRC	0.00	0.00	-0.02	0.02	0.00	0.00	0.00	0.00
unexp_r	0.03	-0.04	-0.07	0.00	0.03	-0.03	0.01	-0.04

Table 3.34: Polarity sensitivity of *exactly shy*

L2		exactly shy							
Dataset		super				mirror			
(+) Sample		NEQ		ALL		NEQ		ALL	
Adj Total		3,071		35,069		359		1,147	
F2		124		124		4		4	
Direction		repel	attract	repel	attract	insignif		insignif	
Polarity		(+)	(-)	(+)	(-)	(+)	(-)	(+)	(-)
F		0	124	0	124	0	4	0	4
<i>E ~ Exactly Shy</i>									
P1		0	1	0	1	0	1	0	1
dP1		-0.50	0.50	-0.96	0.96	-0.50	0.50	-0.83	0.83
LRC		-2.81	2.81	-7.24	7.24	0	0	0	0
unexp_r		-1	0.50	-1	0.96	-1	0.50	-1	0.83

Table 3.35: Independent polarity sensitivity of *shy*

L2		shy							
Dataset		mirror				super			
(+) Sample		ALL		NEQ		ALL		NEQ	
Direction		insignif		insignif		insignif		insignif	
F2		1,147		359		35,069		3,071	
Polarity		(-)	(+)	(-)	(+)	(-)	(+)	(-)	(+)
F		153	994	153	206	1,557	33,512	1,557	1,514
<i>E ~ Shy</i>									
P1		0.13	0.87	0.43	0.57	0.04	0.96	0.51	0.49
dP1		-0.04	0.04	-0.07	0.07	0	0	0.01	-0.01
LRC		0	0	0	0	0	0	0	0
unexp_r		-0.30	0.05	-0.17	0.13	0.01	0	0.01	-0.01

3.6 *Exactly* Polarity Sensitivity as Compositional Contingencies of Adjectival Semantics

The best explanation—the only one remaining of those posited—is that *exactly*’s propensity for negative environments, as observed in these data⁸³, arises due to the compositional semantics that arise when *exactly* modifies **specific** adjectives. It is neither a global phenomenon of *exactly* alone, nor a highly localized event due to lexicalized meaning of particular phrases. Syntactic configurations do not appear highly influential, either, although it may affect the ease with which the requisite semantics are composed.

This can be posited primarily because there are good arguments **against** all other plausible conclusions, but there is also an argument **for** this explanation. The specific adjective arguments that lead to polarity sensitive *exactly* bigrams can be described—and predicted—based on their semantics.

3.6.1 The Polarity Sensitive Contingencies of *Exactly*: Polarity Sensitive & Polarity Impervious Adjectives

There are two relevant groups of adjectival arguments for *exactly*: those adjectives that yield a polarity sensitive bigram with *exactly* modification, and those that do

⁸³restricted to an adjectival modifier in a linearly contiguous *ADV ADJ* bigram

not. More specifically, the observed polar effects are that the bigram will either be sensitive to or licensed by negative polarity environments or not sensitive to environment polarity at all—no adjectives were observed with *positive* polarity association when modified by *exactly*.⁸⁴ I will refer to the two groups of adjectives as *polarity **sensitive*** (by which I really mean, “*negative* polarity sensitive”) and *polarity **impervious***, (or *sensitive* and *impervious* as shorthand).

In broad strokes, the polarity impervious adjectives have scalar semantics with *precise* and *concrete* (or measurable) thresholds of evaluation. These requirements can be met by either intrinsic aspects/entailments of the adjective’s denotation, or contextually supplied/pragmatically enriched addenda.

The polarity sensitive adjectives are all those that do not meet these requirements. This includes non-scalar adjectives, as well as scalar adjectives which are either **vague** in general (i.e., open scale/no fixed threshold of evaluation/Kennedy’s “relative gradable” adjectives (2007)), or non-vague in that they have fixed *conceptual* thresholds, but nonetheless **abstract** such that they *cannot be measured* concretely (or at least not easily).

Even without referring to specific counts, the level of restriction for impervious status and the resultant *lack* of restriction for sensitivity strongly suggest that the cardinality of the sensitive set greatly exceeds that of the impervious set. In fact, this may be a large underlying factor in why *exactly* on the whole displays such a decided

⁸⁴I would also theorize that any adjectives which are positive polarity prone independently of adverbial modification would be far less likely to felicitous with *exactly*, even in positive contexts.

Table 3.36: ADJ Polarity Sensitivity Contingency Classes

Polarity Class	exemplars
POLARITY IMPERVIOUS	{ <i>alike, average, contrary, double, enough, equal, equivalent, flat, identical, many, opposite, parallel, perpendicular, right, same, similar, wrong, zero</i> }
POLARITY SENSITIVE	{ <i>certain, cheap, clear, comfortable, conducive, easy, fair, free, friendly, fun, great, happy, hard, healthy, ideal, new, original, pleased, popular, shocking, stellar, subtle, sure, surprising, thrilled</i> }

negative propensity in the first place. If the impervious and sensitive classes were closer in total membership/composite frequency, *exactly* as an adverb *independent of its adjective argument* would likely not exhibit such strong association with negative contexts.

The impervious class has been previewed above, especially in § 3.4.1. Similarly, the sensitive adjectives have debuted in negated examples as well. Each class is presented more conclusively and juxtaposed in Table 3.36.

These adjectives were selected on the basis of the LRC and dP1 values for the corresponding *exactly* bigram and negative or positive polarity. The impervious class consists of those showing the weakest attraction to negative polarity/strongest attraction to positive polarity (though none actually meet that threshold), and the sensitive class is comprised of those with the strongest attraction to negative polarity environments. Both class sets listed in Table 3.36 are non-exhaustive—there are many, many other adjectives not listed that belong to the sensitive class, as well as other impervious adjectives.

(3.48) **Selection Criteria for Adjective Exemplar Designation**

- a. For all considerations: $\max(f(\text{bigram})) \geq 50$
That is, only bigram types with at least 50 tokens in the PBR superset $*S$ are considered, but local bigram subtotal varies by class.
- b. IMPERVIOUS class: 18 adjectives
Selection Criteria:
 - i. $f(\text{bigram}) \geq 25$ (at least 25 bigram tokens across both polarities)
 - ii. $\geq 80^{th}$ percentile for Positive Polarity⁸⁵ for each of:
 - LRC for any comparison space
and
 - (dP1 for ALL+ comparison space
or P1 for NEQ+ comparison space)
- c. SENSITIVE class: 25 adjectives
Selection Criteria:
 - i. $f(\text{bigram}) > 150$ (at least 150 bigram tokens across both polarities)
 - ii. $\geq 90^{th}$ percentile for Negative Polarity) for each of:
 - LRC for any comparison space
and
 - (dP1 for ALL+ comparison space
or P1 for NEQ+ comparison space)

Table 3.37: Percentile Definitions for Polarity Sensitivity Class Designations

class	f2 floor	polar percentile	LRC	dP1	P1
IMPERVIOUS	≥ 25	$[\leq](-) 20^{th}$	1.629	0.494	0.918
		$[\geq](+) 80^{th}$	-1.629	-0.494	0.082
SENSITIVE	≥ 150	$[\geq](-) 90^{th}$	7.184	0.956	1.000

Maximum bigram marginal frequency (f2) ≥ 50 for all evaluations

⁸⁵This is equivalent to $\leq 20^{th}$ percentile for Negative Polarity: The adjective selections are identical.

3.6.1.1 Impervious Adjective AMs—the least negative *exactly* bigrams

Table 3.38 shows the AM scores for *exactly* * bigrams identified (in any comparison space) as having the least association with negative environments. It is sorted by adjective to promote comparison between comparison spaces and polarities. All evaluations with fewer than 10 bigram tokens across both polarities have been omitted.

Table 3.38: IMPERVIOUS Adjective Class Exemplars

			f		f2		dP1		P1		LRC		
			Sampling	ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ
<i>Exactly...</i>	Dataset	Polarity											
<i>perpendicular</i>	<i>*S</i>	(−)	10	10	50	13	0.16	0.27	0.20	0.77	0.00	0.00	
		(+)	40	3	50	13	−0.16	−0.27	0.80	0.23	0.00	0.00	
<i>flat</i>	<i>*S</i>	(−)	29	29	50	33	0.54	0.38	0.58	0.88	2.65	0.00	
		(+)	21	4	50	33	−0.54	−0.38	0.42	0.12	−2.65	0.00	
<i>double</i>	<i>*S</i>	(−)	4	4	51	6	0.03	0.17	0.08	0.67	0.00	0.00	
		(+)	47	2	51	6	−0.03	−0.17	0.92	0.33	0.00	0.00	
<i>many</i>	<i>*S</i>	(−)	3	3	55	6	0.01	0.00	0.05	0.50	0.00	0.00	
		(+)	52	3	55	6	−0.01	0.00	0.95	0.50	0.00	0.00	
<i>contrary</i>	<i>*S</i>	(−)	0	0	58	2	−0.04	−0.50	0.00	0.00	0.00	0.00	
		(+)	58	2	58	2	0.04	0.50	1.00	1.00	0.00	0.00	
<i>average</i>	<i>*S</i>	(−)	6	6	73	9	0.04	0.17	0.08	0.67	0.00	0.00	
		(+)	67	3	73	9	−0.04	−0.17	0.92	0.33	0.00	0.00	
<i>equivalent</i>	<i>*S</i>	(−)	41	41	147	46	0.23	0.39	0.28	0.89	1.51	0.09	
		(+)	106	5	147	46	−0.23	−0.39	0.72	0.11	−1.51	−0.09	
<i>parallel</i>	<i>*S</i>	(−)	45	45	204	52	0.18	0.37	0.22	0.87	1.18	0.05	
		(+)	159	7	204	52	−0.18	−0.37	0.78	0.13	−1.18	−0.05	
<i>enough</i>	<i>*S</i>	(−)	42	42	253	51	0.12	0.32	0.17	0.82	0.65	0.00	
		(+)	211	9	253	51	−0.12	−0.32	0.83	0.18	−0.65	0.00	
<i>identical</i>	<i>@S</i>	(−)	5	5	14	6	0.18	0.33	0.36	0.83	0.00	0.00	
		(+)	9	1	14	6	−0.18	−0.33	0.64	0.17	0.00	0.00	
	<i>*S</i>	(−)	80	80	278	88	0.24	0.41	0.29	0.91	2.05	0.97	
		(+)	198	8	278	88	−0.24	−0.41	0.71	0.09	−2.05	−0.97	
<i>zero</i>	<i>*S</i>	(−)	14	14	306	23	0.00	0.11	0.05	0.61	0.00	0.00	
		(+)	292	9	306	23	0.00	−0.11	0.95	0.39	0.00	0.00	
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Table 3.38: IMPERVIOUSAdjective Class Exemplars

Exactly...	Dataset	Sampling Polarity	f		f2		dP1		P1		LRC	
			ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ
<i>similar</i>	<i>*S</i>	(−)	47	47	333	67	0.10	0.20	0.14	0.70	0.47	0.00
		(+)	286	20	333	67	−0.10	−0.20	0.86	0.30	−0.47	0.00
<i>same</i>	<i>*S</i>	(−)	36	36	422	53	0.04	0.18	0.09	0.68	0.00	0.00
		(+)	386	17	422	53	−0.04	−0.18	0.91	0.32	0.00	0.00
<i>opposite</i>	<i>*S</i>	(−)	13	13	467	32	−0.02	−0.09	0.03	0.41	0.00	0.00
		(+)	454	19	467	32	0.02	0.09	0.97	0.59	0.00	0.00
<i>equal</i>	@S	(−)	3	3	13	7	0.06	−0.07	0.23	0.43	0.00	0.00
		(+)	10	4	13	7	−0.06	0.07	0.77	0.57	0.00	0.00
	<i>*S</i>	(−)	71	71	491	89	0.10	0.30	0.14	0.80	0.79	0.13
		(+)	420	18	491	89	−0.10	−0.30	0.86	0.20	−0.79	−0.13
<i>wrong</i>	@S	(−)	21	21	28	21	0.58	0.50	0.75	1.00	0.98	0.07
		(+)	7	0	28	21	−0.58	−0.50	0.25	0.00	−0.98	−0.07
	<i>*S</i>	(−)	178	178	569	198	0.27	0.40	0.31	0.90	2.58	1.56
		(+)	391	20	569	198	−0.27	−0.40	0.69	0.10	−2.58	−1.56
<i>alike</i>	@S	(−)	39	39	83	54	0.30	0.22	0.47	0.72	0.47	0.00
		(+)	44	15	83	54	−0.30	−0.22	0.53	0.28	−0.47	0.00
	<i>*S</i>	(−)	133	133	776	162	0.13	0.32	0.17	0.82	1.39	0.75
		(+)	643	29	776	162	−0.13	−0.32	0.83	0.18	−1.39	−0.75
<i>right</i>	@S	(−)	52	52	133	68	0.22	0.26	0.39	0.76	0.30	0.00
		(+)	81	16	133	68	−0.22	−0.26	0.61	0.24	−0.30	0.00
	<i>*S</i>	(−)	634	634	6,269	877	0.06	0.22	0.10	0.72	0.96	0.82
		(+)	5,635	243	6,269	877	−0.06	−0.22	0.90	0.28	−0.96	−0.82

Although there are some bigrams with LRC values that suggest significant positive association with negative environments, these cases—*alike*, *equivalent*, *flat*, *straight*, and, at the very cusp, *square* and *parallel*—can still be safely categorized as polarity impervious. For starters, all of the aforementioned divergent LRC values arise from the unrestricted, and highly unbalanced ALL+ superset frequency perspective. Identification of (positive) association with positive polarity is the least accurate in this particular comparison space, because it exhibits the greatest

discrepancy between the f1 (environment subtotals) values. As discussed previously, the odds of any token in the ALL+ superset occurring in a positive environment for this set is $\approx 96:4$ or $\approx 22:1$, meaning that only lexical items with **over 96%** of their tokens categorized as positive polarity have any chance of reaching significant indication of attraction to positive polarity.

The second reason that these aberrant cases are not a concern is that impervious status is *not* based on positive polarity attraction/licensing, but on the absence of association with **either** polarity. Such weak preferences are not sufficient grounds for relegating these cases to the sensitive class. And finally, they may also be influenced by word-specific usage quirks. These might include hackneyed phrases which happen to include negation—e.g., “No two x are exactly *alike*”—or alternate word senses which cannot be precisely measured and actually *are* polarity sensitive—e.g. *straight* in the sense of *heterosexual* or *law-abiding* or *square* in the sense of (a person who is) *uptight*, *lame*, etc.

To reiterate, the adjectives designated as impervious are not positive polarity sensitive. Even when only the NEQ+ comparisons are considered, no *exactly* bigram shows indication of a significant propensity for positive polarity. As [Table 3.39](#) shows, *opposite* is the closest any adjective comes to “preferring” positive polarity, but LRC is still 0 (indicating $p > 0.001$), and the [dP1](#) value indicates that *exactly opposite* is only 9% more likely to occur with positive contexts than any other bigram is.

Table 3.39: Strongest Positive NEQ+ *exactly* bigrams

polar approx	ADJ	f	dP1	P1	LRC	G2
*S, (-) absent	<i>opposite</i>	19	0.09	0.59	0	1.13
*S, (-) absent	<i>zero</i>	9	-0.11	0.39	0	-1.10
*S, (-) absent	<i>same</i>	17	-0.18	0.32	0	-6.97
*S, (-) absent	<i>similar</i>	20	-0.20	0.30	0	-11.20
@S, (+) present	<i>alike</i>	15	-0.22	0.28	0	-11.05
*S, (-) absent	<i>enough</i>	9	-0.32	0.18	0	-23.17

3.6.1.2 Sensitive Adjective AMs—the most negative *exactly* bigrams

Table 3.40 shows AM values for all *exactly* * bigram tokens with LRC, $\Delta P(E|bigram)$ for ALL+ , or $P(E|bigram)$ for NEQ+ values above the respective 90th percentile for each respectively across all environments where it appeared with *exactly* 150 or more times.

Table 3.40: SENSITIVE Adjective Class Exemplars

			f		f2		dP1		P1		LRC	
Sampling			ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ
<i>Exactly...</i>	Dataset	Polarity										
<i>shocking</i>	<i>*S</i>	(−)	151	151	151	151	0.96	0.50	1.00	1.00	7.54	3.11
		(+)	0	0	151	151	−0.96	−0.50	0.00	0.00	−7.54	−3.11
<i>free</i>	<i>*S</i>	(−)	156	156	161	156	0.92	0.50	0.97	1.00	6.77	3.16
		(+)	5	0	161	156	−0.92	−0.50	0.03	0.00	−6.77	−3.16
<i>original</i>	<i>*S</i>	(−)	160	160	162	160	0.94	0.50	0.99	1.00	7.20	3.20
		(+)	2	0	162	160	−0.94	−0.50	0.01	0.00	−7.20	−3.20
<i>friendly</i>	<i>*S</i>	(−)	167	167	167	167	0.96	0.50	1.00	1.00	7.70	3.27
		(+)	0	0	167	167	−0.96	−0.50	0.00	0.00	−7.70	−3.27
<i>stellar</i>	<i>*S</i>	(−)	170	170	172	170	0.94	0.50	0.99	1.00	7.29	3.29
		(+)	2	0	172	170	−0.94	−0.50	0.01	0.00	−7.29	−3.29
<i>certain</i>	<i>*S</i>	(−)	169	169	172	169	0.94	0.50	0.98	1.00	7.13	3.28

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Table 3.40: SENSITIVE Adjective Class Exemplars

<i>Exactly. . .</i>	Dataset	Sampling	Polarity	f		f2		dP1		P1		LRC	
				ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ
			(+)	3	0	172	169	-0.94	-0.50	0.02	0.00	-7.13	-3.28
<i>pleased</i>	<i>*S</i>		(-)	172	172	174	173	0.94	0.49	0.99	0.99	7.31	3.06
			(+)	2	1	174	173	-0.94	-0.49	0.01	0.01	-7.31	-3.06
<i>popular</i>	<i>*S</i>		(-)	171	171	174	171	0.94	0.50	0.98	1.00	7.15	3.30
			(+)	3	0	174	171	-0.94	-0.50	0.02	0.00	-7.15	-3.30
<i>comfortable</i>	<i>*S</i>		(-)	187	187	190	188	0.94	0.49	0.98	0.99	7.29	3.19
			(+)	3	1	190	188	-0.94	-0.49	0.02	0.01	-7.29	-3.19
<i>healthy</i>	<i>*S</i>		(-)	192	192	194	192	0.95	0.50	0.99	1.00	7.48	3.48
			(+)	2	0	194	192	-0.95	-0.50	0.01	0.00	-7.48	-3.48
<i>thrilled</i>	<i>*S</i>		(-)	203	203	206	203	0.94	0.50	0.99	1.00	7.41	3.56
			(+)	3	0	206	203	-0.94	-0.50	0.01	0.00	-7.41	-3.56
<i>hard</i>	<i>*S</i>		(-)	203	203	207	203	0.94	0.50	0.98	1.00	7.28	3.56
			(+)	4	0	207	203	-0.94	-0.50	0.02	0.00	-7.28	-3.56
<i>conductive</i>	<i>*S</i>		(-)	208	208	208	208	0.96	0.50	1.00	1.00	8.03	3.60
			(+)	0	0	208	208	-0.96	-0.50	0.00	0.00	-8.03	-3.60
<i>fun</i>	<i>*S</i>		(-)	224	224	226	224	0.95	0.50	0.99	1.00	7.71	3.71
			(+)	2	0	226	224	-0.95	-0.50	0.01	0.00	-7.71	-3.71
<i>fair</i>	<i>*S</i>		(-)	260	260	264	260	0.94	0.50	0.98	1.00	7.66	3.93
			(+)	4	0	264	260	-0.94	-0.50	0.02	0.00	-7.66	-3.93
<i>subtle</i>	<i>*S</i>		(-)	263	263	265	263	0.95	0.50	0.99	1.00	7.95	3.95
			(+)	2	0	265	263	-0.95	-0.50	0.01	0.00	-7.95	-3.95
<i>great</i>	<i>*S</i>		(-)	308	308	316	308	0.93	0.50	0.97	1.00	7.53	4.18
			(+)	8	0	316	308	-0.93	-0.50	0.03	0.00	-7.53	-4.18
<i>ideal</i>	<i>*S</i>		(-)	417	417	424	418	0.94	0.50	0.98	1.00	8.07	4.38
			(+)	7	1	424	418	-0.94	-0.50	0.02	0.00	-8.07	-4.38
<i>surprising</i>	<i>*S</i>		(-)	440	440	441	440	0.95	0.50	1.00	1.00	8.89	4.71
			(+)	1	0	441	440	-0.95	-0.50	0.00	0.00	-8.89	-4.71
<i>happy</i>	<i>*S</i>		(-)	439	439	445	439	0.94	0.50	0.99	1.00	8.23	4.70
			(+)	6	0	445	439	-0.94	-0.50	0.01	0.00	-8.23	-4.70
<i>cheap</i>	<i>*S</i>		(-)	691	691	695	692	0.95	0.50	0.99	1.00	9.11	5.12
			(+)	4	1	695	692	-0.95	-0.50	0.01	0.00	-9.11	-5.12
<i>easy</i>	<i>@S</i>		(-)	20	20	20	20	0.83	0.50	1.00	1.00	2.22	0.00
			(+)	0	0	20	20	-0.83	-0.50	0.00	0.00	-2.22	0.00
	<i>*S</i>		(-)	1,066	1,066	1,072	1,066	0.95	0.50	0.99	1.00	9.54	6.00
			(+)	6	0	1,072	1,066	-0.95	-0.50	0.01	0.00	-9.54	-6.00
<i>new</i>	<i>@S</i>		(-)	29	29	30	29	0.79	0.50	0.97	1.00	2.61	0.69
			(+)	1	0	30	29	-0.79	-0.50	0.03	0.00	-2.61	-0.69

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Table 3.40: SENSITIVE Adjective Class Exemplars

			f		f2		dP1		P1		LRC	
		Sampling	ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ	ALL	NEQ
<i>Exactly...</i>	Dataset	Polarity										
	*S	(−)	1,371	1,371	1,387	1,372	0.94	0.50	0.99	1.00	9.25	6.12
		(+)	16	1	1,387	1,372	−0.94	−0.50	0.01	0.00	−9.25	−6.12
clear	@S	(−)	52	52	52	52	0.83	0.50	1.00	1.00	3.94	1.69
		(+)	0	0	52	52	−0.83	−0.50	0.00	0.00	−3.94	−1.69
	*S	(−)	1,746	1,746	1,779	1,747	0.94	0.50	0.98	1.00	8.98	6.47
		(+)	33	1	1,779	1,747	−0.94	−0.50	0.02	0.00	−8.98	−6.47
sure	@S	(−)	148	148	148	148	0.83	0.50	1.00	1.00	5.58	3.33
		(+)	0	0	148	148	−0.83	−0.50	0.00	0.00	−5.58	−3.33
	*S	(−)	8,794	8,794	8,840	8,797	0.95	0.50	0.99	1.00	11.00	8.49
		(+)	46	3	8,840	8,797	−0.95	−0.50	0.01	0.00	−11.00	−8.49

For all of those with 5 or more positive polarity tokens—*free*, *clear*, *easy*, *great*, *happy*, *ideal*, *new*, and *sure*—the AM values (< 0) nonetheless indicate negative association (repulsion) with the positive contexts. The token counts in these cases can easily be explained by simple error rates, but there are instances of these adjectives appearing felicitously without explicit negation. Manual inspection of a sample confirms that there are errors in the supposedly positive polarity examples⁸⁶, there are also cases that truthfully not explicitly negated, but which are implicitly negative nonetheless (i.e. where negation or the threat/expectation of negative quantification—are contributed by the context). Other accurately parsed cases include those where additional modifiers result in the normally sensitive adjective becoming impervious⁸⁷.

⁸⁶primarily uncaught negations due to unexpected orthography, parsing errors with incorrect constituency, or just general sentence garbling

⁸⁷These are cases of modification which provides a concrete and precise threshold of evaluation or replaces the scale provided by the (vague or abstract) adjective with its own precise, concrete

Sample of Sensitive Adjectives in Supposed Positive Contexts

(3.49) *exactly clear*

- a. † It is **exactly clear** what kind of activity was not effective.⁸⁸
- b. † Getting back to the topic, the purpose of your website needs to be **exactly clear** in your head.⁸⁹
 $\Rightarrow$ *because if it is not...*
- c. † Now, he made it **exactly clear** how important this game was.⁹⁰
 $\Rightarrow$ *so you cannot say he did not...*
- d. † This suggestion is particularly telling with regards to the priorities of “developed-world” economists, and advocating for lowering the cost of employment is so patently pro-ownership and anti-worker as to make **exactly clear** where the article’s economic allegiances rest.⁹¹
- e. † Available to customers the choice of language to help them feel safe, **exactly clear** how to trade, with whom transactions.⁹²
- f. † “Let me be **exactly clear** about what healthcare reform means to you,” the president told residents of the Garden State.⁹³
- g. † “But as a result of the fire on Wednesday [Oct. 31], there was somewhat of a jolt to our confidence and more of an awareness on our level that we needed to be **exactly clear** on what the contractor intended on doing.”⁹⁴
- h. ? On the Leipzig show, I must confess that I’m **exactly clear** on whether Libby liked it or not.⁹⁵

(3.50) *exactly easy*

- a. ? Despite his sunny smile and outlook, Cooper admits the watching and knowing he is headed for a third reconstructive knee surgery and rehab it’s **exactly easy**.⁹⁶

scale

⁸⁸ pcc_eng_23_056.7929_x0901779_19:3-4
⁸⁹ pcc_eng_07_004.5291_x0056969_46:15-16
⁹⁰ pcc_eng_24_065.6237_x1045489_065:6-7
⁹¹ pcc_eng_17_015.9913_x0242407_17:36-37
⁹² pcc_eng_18_030.5299_x0477946_27:14-15
⁹³ pcc_eng_04_046.7632_x0739628_35:5-6
⁹⁴ pcc_eng_11_063.2384_x1007231_17:38-39
⁹⁵ pcc_eng_01_032.9047_x0515709_16:12-13
⁹⁶ pcc_eng_16_066.1204_x1054086_49:27-28

(3.51) *exactly great*

- a. † But instead of rambling on and on about how **exactly great** our company is, let's focus on who you are and why you should rent a portable restroom.⁹⁷
- b. !:[bad constituent] [Where **exactly**] [DP **great** whites] go to mate and give birth is a mystery.⁹⁸
- c. !:[uncaught neg] Those two had demos that ~~were~~ [weren't] **exactly great** either but if you like either of those titles you'll love MGR.—⁹⁹
- d. ? Each and every teen should have a nice floral skirt for the summer months and high waisted and low rise shorts are **exactly great** also.¹⁰⁰

(3.52) *exactly happy*

- a. † So if you are **exactly happy** with where you're at, then I'm not speaking to you.¹⁰¹
- b. !:[uncaught neg] New research from the University of Missouri finds that the vegetables in your salad bowl ~~arena~~ ~~EUR~~ ~~(TM)~~ ~~t~~ [aren't] **exactly happy**.¹⁰²

(3.53) *exactly ideal*

- a. † So, it helps a lot that you will get these expectations as your guide in ensuring that the system you end up with is **exactly ideal** for you and for your trading goals.¹⁰³
- b. !:[uncaught neg] To say the least Lawson s experiences with bush ~~haven~~ ~~t~~ [haven't] been **exactly ideal**.¹⁰⁴

(3.54) *exactly new*

Many of these cases are all questions (explicit or implicit) which introduce the proposition that *x is not exactly* [*ADJ*]

- a. † And what **exactly new** will the second season have to offer, now that the Panthers have made it to the state finals?¹⁰⁵

⁹⁷ pcc_eng_17_012.7450_x0189828_11:10-11

⁹⁸ pcc_eng_01_106.2451_x1700940_29:2-3

⁹⁹ pcc_eng_03_020.1630_x0309946_13:7-8

¹⁰⁰ pcc_eng_27_008.5390_x0121368_20:23-24

¹⁰¹ pcc_eng_20_087.6873_x1400437_140:5-6

¹⁰² pcc_eng_23_094.7538_x1515391_12:22-23

¹⁰³ pcc_eng_07_001.1137_x0001804_063:26-27

¹⁰⁴ pcc_eng_10_026.7468_x0416064_12:13-14

¹⁰⁵ pcc_eng_04_003.6693_x0043310_15:3-4

- b. † To try an operational system **exactly new**, exactly that this demands moments of continued formation of our professors, that this demands investments and public politics significant and focadas in the digital fluency of its professors and educators of a general form, so that has a true appropriation of this system.¹⁰⁶
- c. † This one has me scratching my head as to how it is **exactly new**.¹⁰⁷
- d. † What **exactly new** is going to happen ?¹⁰⁸
- e. † I like the idea but I'm wondering how many are **exactly new**.¹⁰⁹

(3.55) *exactly sure*

- a. † The camp is, I'm 100% **exactly sure**, like a prison.¹¹⁰
- b. † To be **exactly sure** about which one is used in your home, it is a good idea to take your old thermostat to the store you are about to buy a new thermostat.¹¹¹
- c. † Only complete the real form when you are **exactly sure** what is the best thing to say.¹¹²
- d. † It is difficult to say whether [...] but that all that is happening in Ukraine, is aimed against Russia—he is **exactly sure**.¹¹³; Abridged for clarity.
- e. !:[uncaught neg] My boyfriend and I ~~weren~~ ~~and~~ ~~# 39~~ ; ~~t~~ [weren't] **exactly sure** what to expect about this trip, but we were more than pleasantly surprised!¹¹⁴
- f. !:[uncaught neg] This was going to be a trial run of going from Class D right into Class C and I wasnt [was not] **exactly sure** how smooth it was going to go.¹¹⁵
- g. !:[uncaught neg] “Whilst we cannot be **exactly sure** what the labyrinths were used for, they were clearly a symbol of the Christian way, representing the path of the soul through life.” [3]¹¹⁶

¹⁰⁶ pcc_eng_05_018.0041_x0275184_08:6-7

¹⁰⁷ pcc_eng_06_085.7432_x1370533_35:13-14

¹⁰⁸ pcc_eng_11_017.7195_x0270507_59:2-3

¹⁰⁹ pcc_eng_15_021.7278_x0334953_38:12-13

¹¹⁰ pcc_eng_16_077.2073_x1233193_07:09-10

¹¹¹ pcc_eng_22_063.1457_x1004711_17:3-4

¹¹² pcc_eng_09_024.5855_x0381920_142:09-10

¹¹³ pcc_eng_13_032.6782_x0512237_03:37-38

¹¹⁴ pcc_eng_08_043.2212_x0683312_033:11-12

¹¹⁵ pcc_eng_00_065.2623_x1038947_2:21-22

¹¹⁶ pcc_eng_25_018.1977_x0277874_001:6-7

- h. !:[uncaught neg] An interesting position, especially given the fact that the authors of this particular quote admit, “we cannot be **exactly sure** what the labyrinths were used for...”¹¹⁷
- i. !:[uncaught neg] Counter to the statement “we cannot be **exactly sure** what the labyrinths were used for” is a wealth of literature, some easy to obtain, others that should be kept hidden on dusty shelves.¹¹⁸
- j. ?? To quit and that’s why Manny took Dennis **exactly sure**–alliance apart by AT&T W I live it’s available at your rival android Blackberry device for a coupon AT&T AT&T forgy.¹¹⁹
- k. ? **Exactly sure**, it is very important go with a design which you exceedingly are certain to get connected to, and this may end in mentioning your favorite attitude.¹²⁰
- l. ? **Exactly sure** of a timetable.¹²¹

3.6.2 Polarity Partitioned Context–Blind *Exactly*~ADJ Association

The values shown in the previous sections which quantify the polarity association of bigrams in the context of exactly modification. Context–blind AM values—the bigram internal cohesion of different ADV and ADJ pairings—were presented previously (e.g., Table 3.21), but only for the full set of tokens. Table 3.41 and Table 3.42 show context–blind AMs *exactly* and the different adjectives when polarity of environment is considered. The polar subsets used for calculation were taken from the *S ALL+ superset data, which includes all negation triggers, approximates positive polarity by the absence of explicit negation, and has no additional sampling

¹¹⁷ pcc_eng_25_018.1977_x0277874_004:22-23

¹¹⁸ pcc_eng_25_018.1977_x0277874_027:09-10

¹¹⁹ pcc_eng_23_087.7582_x1402128_043:10-11

¹²⁰ pcc_eng_21_010.6826_x0156293_04:1-2

¹²¹ pcc_eng_06_088.5188_x1415346_05:1-2

performed on the positive polarity representation.

Table 3.41 shows the top 25 bigrams, as ranked by LRC calculated for only the tokens occurring in negative polarity contexts.¹²² While there is some overlap in the pairs selected by the AM scores for the truly context-blind evaluations¹²³ or the explicit negative polarity evaluations¹²⁴, there is not a 1:1 relationship with either. The strength of these scores do not inherently suggest a propensity for negative polarity. Rather they are an indication of general attraction with *exactly* when polarity of context is controlled for.

Table 3.41: Top “Blind” LRC for **Negative** bigrams where PBR bigram subtotal ≥ 40

163 unique adjectives where Negative *exactly*~ADJ $f \geq 10$
& overall *exactly*~ADJ $f \geq 40$
exactly marginal frequency = 43,813

Negative <i>exactly</i> ...	f	LRC	dPavg	G2	dP1	P1	dP2	P2	f2	unexp f	unexp r
<i>stellar</i>	170	4.19	0.14	770	0.28	0.30	0.004	0.004	574	162	0.95
<i>cheap</i>	691	3.56	0.09	2,324	0.16	0.17	0.015	0.016	4,040	635	0.92
<i>ideal</i>	417	3.50	0.09	1,421	0.16	0.18	0.009	0.010	2,379	384	0.92
<i>alike</i>	133	3.30	0.09	477	0.18	0.19	0.003	0.003	694	123	0.93
<i>new</i>	1,371	3.22	0.07	3,924	0.12	0.13	0.028	0.031	10,388	1,228	0.90
<i>revolutionary</i>	120	2.92	0.07	380	0.14	0.16	0.003	0.003	768	109	0.91
<i>conductive</i>	208	2.82	0.06	585	0.12	0.13	0.004	0.005	1,597	186	0.89
<i>parallel</i>	45	2.70	0.09	163	0.18	0.19	0.001	0.001	231	42	0.93
<i>unknown</i>	72	2.41	0.06	207	0.12	0.13	0.001	0.002	536	65	0.90
<i>sure</i>	8,794	2.31	0.11	15,824	0.06	0.07	0.163	0.201	128,062	7,026	0.80
<i>mainstream</i>	75	2.31	0.06	204	0.11	0.12	0.002	0.002	603	67	0.89
<i>unheard</i>	65	2.27	0.06	180	0.11	0.13	0.001	0.001	509	58	0.89
<i>subtle</i>	263	2.20	0.04	529	0.07	0.08	0.005	0.006	3,119	220	0.84
<i>normal</i>	135	2.13	0.04	295	0.08	0.09	0.003	0.003	1,453	115	0.85
<i>reassuring</i>	77	1.95	0.04	175	0.08	0.10	0.002	0.002	786	66	0.86

Continued on next page

¹²²For a similar table restricted to only the positive polarity $\ast\mathcal{C}_N$ set, see Appendix B.

¹²³association between *exactly* and following adjective regardless of context

¹²⁴association of *exactly* bigrams with negative polarity environments

Table 3.41: Top “Blind” LRC for **Negative** bigrams

<i>exactly...</i>	f	LRC	dPavg	G2	dP1	P1	dP2	P2	f2	unexp_f	unexp_r
<i>novel</i>	66	1.94	0.05	157	0.09	0.10	0.001	0.002	639	57	0.87
<i>commonplace</i>	46	1.93	0.05	122	0.11	0.12	0.001	0.001	383	41	0.89
<i>comforting</i>	100	1.91	0.04	208	0.07	0.09	0.002	0.002	1,141	84	0.84
<i>true</i>	1,447	1.89	0.03	1,975	0.04	0.06	0.025	0.033	25,531	1,095	0.76
<i>identical</i>	80	1.87	0.04	172	0.08	0.09	0.002	0.002	874	68	0.85
<i>shy</i>	124	1.86	0.03	237	0.07	0.08	0.002	0.003	1,557	103	0.83
<i>analogous</i>	34	1.78	0.06	95	0.11	0.13	0.001	0.001	265	30	0.89
<i>shocking</i>	151	1.64	0.03	239	0.05	0.07	0.003	0.003	2,302	119	0.79
<i>same</i>	36	1.63	0.05	92	0.10	0.11	0.001	0.001	318	32	0.88
<i>legal</i>	126	1.63	0.03	207	0.05	0.07	0.002	0.003	1,851	100	0.80

Table 3.42 juxtaposes the different context-blind LRC values for the union of the 42 adjectives previously identified as relevant for *exactly* and the top 20 blind attractions with *exactly* across the different (ALL+) selections—entire $*S$ (either polarity), negative only $*N$, and positive only $*C_N$.

Since *exactly* can only occur in positive environments with impervious adjectives (concretely precise scalars), the divergence seen in this table between the $*C_N, (+)$ column and the others is unsurprising. All of the negative LRC values indicating repulsion from *exactly* under positive polarity are for adjectives belonging to the sensitive class, even if they are not listed in the exemplars provided in Table 3.36¹²⁵ (*new, easy, friendly, happy, famous, impressive, great, and difficult*). The adjectives showing an attraction within the positive polarity restriction correspond with the impervious class exemplars, with the exception of *analogous, correct, true, and comparable*, which all have arguments for meeting the specified semantic criteria.

¹²⁵Different frequency limitations were employed for collecting exemplars than for inclusion in this table, and the exemplars do not include all members of the previously used set of “relevant” adjectives.

Table 3.42: “Blind” LRC for *Exactly*~ADJ Across Polarity Partions

ADJ	*N, (−)	*S, (±)	*C _N , (+)
<i>alike</i>	3.46	6.64	8.66
<i>opposite</i>	0.00	5.82	8.07
<i>right</i>	1.35	5.79	7.95
<i>sure</i>	2.33	5.42	0.00
<i>zero</i>	1.48	4.94	7.14
<i>parallel</i>	2.99	4.78	6.69
<i>stellar</i>	4.33	4.73	0.00
<i>analogous</i>	2.11	4.58	6.33
<i>perpendicular</i>	1.67	4.41	6.51
<i>conducive</i>	2.94	4.31	0.00
<i>ideal</i>	3.58	4.20	0.00
<i>equal</i>	1.04	3.76	5.80
<i>correct</i>	1.09	3.63	5.29
<i>cheap</i>	3.62	3.59	0.00
<i>forthcoming</i>	1.34	3.37	0.00
<i>equivalent</i>	1.41	3.30	5.12
<i>revolutionary</i>	3.08	3.11	0.00
<i>same</i>	1.96	3.08	5.18
<i>true</i>	1.93	3.02	2.02
<i>thrilled</i>	1.21	2.86	0.00
<i>reassuring</i>	2.15	2.74	0.00
<i>comparable</i>	1.01	2.70	2.16
<i>surprising</i>	0.68	2.62	0.00
<i>new</i>	3.26	2.60	−0.12
<i>novel</i>	2.17	2.59	0.00
<i>clear</i>	0.69	2.53	0.00
<i>identical</i>	2.06	2.52	4.25
<i>fair</i>	1.50	2.39	0.00
<i>uncommon</i>	0.48	2.38	0.00
<i>contrary</i>	0.00	2.31	4.85
<i>subtle</i>	2.30	2.24	0.00
<i>wrong</i>	0.00	1.96	3.75
<i>unheard</i>	2.49	1.76	0.00
<i>shocking</i>	1.78	1.76	0.00
<i>average</i>	0.00	1.59	3.89
<i>mainstream</i>	2.52	1.44	0.00

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Table 3.42: “Blind” LRC for *Exactly*~ADJ Across Polarity Partions

ADJ	$*N, (-)$	$*S, (\pm)$	$*C_N, (+)$
<i>shy</i>	2.01	1.38	0.00
<i>pleasant</i>	0.60	1.07	0.00
<i>easy</i>	0.00	0.99	−1.88
<i>unexpected</i>	0.54	0.80	0.00
<i>normal</i>	2.28	0.69	0.00
<i>friendly</i>	0.62	0.14	−0.90
<i>happy</i>	0.00	0.04	−1.56
<i>unknown</i>	2.62	0.00	0.00
<i>famous</i>	0.78	0.00	−1.82
<i>welcome</i>	0.70	0.00	0.00
<i>straightforward</i>	0.00	0.00	0.00
<i>impressive</i>	0.00	0.00	−1.43
<i>great</i>	−0.22	0.00	−0.84
<i>difficult</i>	−1.18	−1.50	−3.50

CHAPTER 4

METHODS FOR INTERPRETING CORPUS POLARITY

The preceding chapters introduced PBR (Chapter 2) and analyzed some of the patterns therein (Chapter 3). With the *what* and *why* addressed, this chapter turns to *how*, covering the methodological specifics of both how PBR was built and how the aforementioned evaluations (based on co-occurrence frequencies and AMs) were performed.

First, the general construction of PBR from undifferentiated corpus matches is reported in § 4.1—for the specifics of how the initial corpus matches were gathered, see Chapter 5. Then § 4.2 describes additional polarity-informed post-processing steps—exclusions which are specific to first negative polarity designations (*N and @N), then positive polarity designations (*L_N and @P). § 4.3 presents the methods for collecting co-occurrence frequency data for a comparison space, and § 4.4 covers the calculation of association measures for the given frequency data.

4.1 General Construction of PBR

PBR is constructed from a cleaned set of bigram tokens retrieved from UD-parsed corpora via pattern matching. For more detailed information on how pattern matches—or *hits*—are retrieved from corpora and processed, see Chapter 5, and for more details on the corpora selection, see Chapter 6. The hits are stored in a

tabular format, with each row representing a single bigram token.

Each hit is assigned a unique identifier—its `hit_id`. The `hit_id` is composed of the unique identifier of the source sentence—the `sent_id`—and hyphen-delimited identifiers of the nodes collected—the `match_id`—joined by a colon. The formats of these identifiers are summarized in (4.1).¹²⁶ Since the initial collection of bigrams is context-blind and there is no expectation that any trigger be present in the preceding text, there are only two collected nodes for basic bigram tokens—the `ADV` and `ADJ` nodes. Context-informed patterns, however, collect a third node, the trigger. Therefore, to enable comparison, an additional identifier—the `bigram_id`—is assigned which excludes the trigger node ID. For a basic bigram hit with no trigger node information, the `hit_id` is identical to the `bigram_id`.

Table 4.1 exemplifies the information collected for basic bigrams (i.e., the preliminary collection of basic `ADV ADJ` bigrams without additional contextualization). Table 4.2 provides the original parse information for the hit described in Table 4.1.¹²⁷

¹²⁶Node IDs start at 1, in keeping with the values encoded in the original corpus information. An additional value, the `adv_index`, is stored in the hit tables which enables direct indexing of the `token_str` in `python`. It is simply the ID of the `ADV` node minus 1.

¹²⁷Note that this sentence—“*It was **so loud** there was nothing else but the seemingly interminable howling*”—actually contains two unique bigram tokens: not only the token demonstrated, *so loud*, but also the later bigram, *seemingly interminable*. The latter case is not retained in PBR, because it does not have a valid direct dependency with a negative trigger—*nothing* does not have the required relationship with *interminable*—but still fails “potential negation” filter applied to $*C_N$ in post-processing, as described below.

- (4.1) general formats
- `sent_id`: unique identifier for source sentence¹²⁸
 - `match_id`: “-” delimited node IDs
 - `hit_id`: `[sent_id]:[match_id]`
- (4.2) example from Table 4.1
- `sent_id`: `pcc_eng_val_1.3687_x06009_17`
 - `match_id`: `3-4`
 - `hit_id`: `pcc_eng_val_1.3687_x06009_17:3-4`
 - `token_str`: `It was so loud there was nothing else but the seemingly interminable howling .`

Table 4.1: Context-Blind Bigram Token Information—“*so loud*”

<i>so loud</i>	
<code>hit_id</code>	<code>pcc_eng_val_1.3687_x06009_17:3-4</code>
<code>sent_id</code>	<code>pcc_eng_val_1.3687_x06009_17</code>
<code>match_id</code>	<code>3-4</code>
<code>adv_form</code>	<i>so</i>
<code>adj_form</code>	<i>loud</i>
<code>sent_text</code>	<i>It was so loud there was nothing else but the seemingly interminable howling.</i>
<code>text_window</code>	<code>it was so loud there was nothing else but the</code>
<code>token_str</code>	<code>It was so loud there was nothing else but the seemingly interminable howling .</code>
<code>adv_lemma</code>	<code>so</code>
<code>adj_lemma</code>	<code>loud</code>
<code>adv_index</code>	<code>2</code>
<code>utt_len</code>	<code>14</code>
<code>adv_form_lower</code>	<code>so</code>
<code>adj_form_lower</code>	<code>loud</code>
<code>bigram_lower</code>	<code>so_loud</code>
<code>window_len</code>	<code>10</code>
<code>part</code>	<code>PccVa</code>

¹²⁸The sentence ID encodes the corpus, corpus part, document, and ordinal sentence position within document. See § 6.1 for more details.

Table 4.2: Abbreviated Sentence Parse Corresponding to Table 4.1

sent_id = pcc_eng_val_1.3687_x06009_17
text = *It was so loud there was nothing else but the seemingly interminable howling.*

ID	form	lemma	UPOS	XPOS	head	deprel
1	<i>It</i>	it	PRON	PRP	4	nsubj
2	<i>was</i>	be	AUX	VBD	4	cop
3	<i>so</i>	so	ADV	RB	4	advmod
4	<i>loud</i>	loud	ADJ	JJ	0	root
5	<i>there</i>	there	PRON	EX	6	expl
6	<i>was</i>	be	VERB	VBD	4	ccomp
7	<i>nothing</i>	nothing	PRON	NN	6	nsubj
8	<i>else</i>	else	ADJ	JJ	7	amod
9	<i>but</i>	but	CCONJ	CC	13	case
10	<i>the</i>	the	DET	DT	13	det
11	<i>seemingly</i>	seemingly	ADV	RB	12	advmod
12	<i>interminable</i>	interminable	ADJ	JJ	13	amod
13	<i>howling</i>	howling	NOUN	NN	7	conj
14	.	.	PUNCT	.	4	punct

In PBR, these context-blind bigrams are compared with context-specific pattern matches representing different polarity categories, in order to construct comprehensive sets of polarity categorized bigrams. Each of the context-specific patterns collects an additional node: the trigger or lexical operator responsible for creating the target polarity environment¹²⁹. In the negative superset (***N**) and subset (**@N**) patterns, **RB-direct** and **NEGmirror** respectively, this node is labeled **NEG**, while in the positive subset (**@P**) pattern, **POSmirror**, the third collected node is labeled **MIR**. The dependency relation between these extra nodes and the bigram (i.e. **ADJ** node) are labeled **neg** and **mir** respectively. All triggers, regardless of collection label, are unified in the **trigger_*** parameters (**trigger_lower**, the lowercase-normalized

¹²⁹The lemmas selected for each pattern/polarity approximation set are given in Chapter 2 as Table 2.2 and Table 2.3

form, and `trigger_lemma`, the lemma of the designated trigger node.)

Examples of the information stored for context-specific hits, split into two groups based on the direction of governance between the trigger (NEG or MIR node) and the bigram (ADJ node), can be found in [Appendix C](#). “Governance” refers to which of the two nodes connected via a valid dependency relation is the *head* or *source*, and each contextualized pattern collects both cases. The context-blind patterns contain no such variation, because the bigram relationship (labeled as `mod`) is obligatorily adjective governed per its pattern specifications.

4.1.1 Initial Token Selections for Superset Polarities

The negative superset ($*N$) is already initialized as the pattern matches corresponding to the superset negation pattern specifications, as encoded in `sanpi/Pat/RBdirect/*.pat`.

As covered in [Chapter 2](#), membership of the positive polarity superset $*C_N$ is determined by the absence of negative triggers, or those bigrams which are *not explicitly negated*. To identify these tokens, first the full context-blind bigram set is compared with the negative hits, and then additional filtration is applied to remove any lingering cases of negation. The initial positive superset consists of any bigram token included in the clean context-blind set which does not satisfy the negative superset pattern specifications. This is achieved via a simple comparison of bigram ids encoded as text file separated by line breaks. The shell script

`collect_com_ids.sh`¹³⁰ achieves this on a part by part basis using the built-in bash utility, `comm`. Note that this is performed on the *bigram_ids*, not *hit_ids*, to maintain comparability between the two node hits in the context-blind matches and the three node hits in the context-specific matches.

Since the positive superset is constructed from the cleaned index, those exclusions have already been applied for the positive tokens. The Negative superset, however, is initialized as direct pattern matches, and therefore contains bigrams deemed unacceptable by the cleaning filters. Rather than redo all of the cleaning processing, those which dictate only exclusion versus inclusion can be applied by filtering the negated hits to only those in the cleaned context-blind set. This is done in a similar fashion to how the negative complement (positive superset) is designated: the shell script `collect_updatedNeg_ids.sh` also utilizes `comm` to retrieve an updated ID index for each corpus part.

After both the negative and positive polarities for the superset are limited to only those bigrams which passed the cleaning filters, the cleaning steps which *modified* the information encoded for the hits must be replicated as well. This primarily consists of orthographic adjustments to the adverb and adjective forms to be evaluated. (See [Chapter 5](#) for details on these modifications.) When the general adjustments have been performed, polarity specific filters, which are described below in § 4.2, are applied to both groups.

As the results for each corpus part are processed, they are double-checked for

¹³⁰`com` stands for *complement*

duplication. Duplicate cases within a single part are dealt with at the context-blind level, so this is primarily a sanity check. When the parts are combined, however, a final duplicate sweep must be performed to catch duplication *across*—rather than *within*—the different parts. All processing prior to this concatenation is performed on the separate parts due to size limitations of computing resources. The full input dataset is much too large to be manipulated efficiently as a single object, and must therefore be handled piecemeal.

The negative superset (`RBdirect`) is handled prior to the positive superset (`not-RBdirect`), so that the same program which cleans and compiles the complement can also pull the `NEQ+` sample at the same time—as a *negative equivalent* sample, the size of the negative set must be known prior to sampling.

The PBR subset is processed after the superset is finalized. Due to the subset relation between them, it is not necessary to repeat all steps applied to the superset. Instead, in a fashion similar to the update applied to the negative superset, the subset polarity categories are limited to only those remaining in the final version of their superset counterpart. Orthographic modifications are also applied when processing the mirror subset results, as well as sanity check duplication sweeps.

All environment-informed pattern results (i.e., those containing a trigger node: negative superset `*N`, negative subset `@N`, and positive subset `@P`) are processed via two consecutively run python programs: `update_env_hits.py` and `compile_env_from_parts.py`. The positive superset (`*CN`) is processed via a unified python program, `compile_com_from_parts.py`, which both applies the needed

adjustments and compiles the hits from all corpus parts in one go.

4.2 Additional Pattern & Category-Specific Filtration Methods

As already established in [Chapter 2](#), the approach taken in PBR is to *approximate* the polarity of any given bigram’s environment. This starts with the trigger identification as specified by the different queries¹³¹, but these have been known to be incomplete coverage from their instantiation. In an effort to enhance the accuracy, beyond the more general cleaning applied at the context-blind level (as described in [Chapter 5](#)), pattern informed methods are applied to each contextualized bigram collection. These are dependent on the presence of a third node/positive evidence of a trigger in general (§ 4.2.1), as well as the specific polarity of the tokens (Negative Polarity in § 4.2.2, and Positive Polarity in § 4.2.3).

4.2.1 For all triggered Environments: *N, @N, and @P

For triggered hits for **either** polarity, negative superset or subset and positive superset, tokens with long text windows (the number of tokens between trigger and bigram) are excluded. Excessively long “sentences” are excluded before any triggered pattern matches are considered (see [Chapter 5](#) for more information.) When triggers are introduced, the distance between the trigger and the bigram can also be

¹³¹The § 5.2

evaluated, quantified as the dependency span or **dep_distance**—the total number of tokens preceding the bigram, starting at the trigger¹³². That is, the minimum **dep_distance** is one, because even a linearly adjacent trigger itself counts towards the dependency span.

With only the two nodes of the bigram present and no trigger, this dependency span is irrelevant as well as impossible to calculate. The addition of third node, the trigger, which is *not* required to be linearly adjacent (though it must linearly precede) the bigram, means that an excessive number of tokens in the dependency span—here defined as > 18 tokens—indicates a long-distance dependency. Since such cases are much less likely to accurately denote the intended semantic environment, they are excluded from further evaluation. The following snippet shows examples of hits that were removed by this filter for negative superset hits in one corpus part. While a close inspection shows that some are actually cases where the trigger does place the bigram within a negative polarity environment, there are more which fail to do so, and the total removal barely makes a dent in the total sample size, removing less than a third of a tenth of a percent.

In addition to excluding tokens outright that exceed this higher threshold, tokens with a **dep_distance** exceeding a much lower threshold of 3 are evaluated for dependency relations which have been flagged for quarantine (marginal dependency relations; e.g. the catchall **dep** relation. See § 5.2 for more details). Quarantine

¹³²**dep_distance** = **adv_index** - **neg_index** for negative polarity cases (negative superset and mirror subset), and **dep_distance** = **adv_index** - **mir_index** for the positive mirror subset cases. (The positive superset cannot have a **dep_distance** since these data do not include a trigger/trigger dependency.)

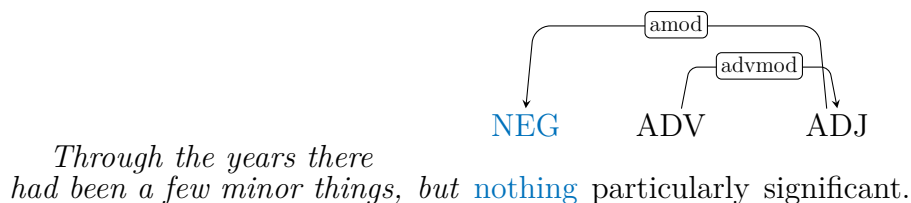
- Dropping 28 hits with more than 18 tokens from trigger to bigram
 ## Example of hits exceeding 18 tokens

hit_id	distance	text_window	dependency relation
pcc_eng_25_091 .7995_x1469317 _29:01-20-21	19	<i>Nor can the odd chance that the real firepower underscoring the strength of our pro-gun lobbyists and legislators is really more about making money than</i>	cc
pcc_eng_25_046 .2671_x0732851 _2:01-21-22	20	<i>Nothing groundbreaking here , but some good ideas for those of you working on such projects yourself - it 's certainly helpful to me ; I</i>	nsubj
pcc_eng_25_064 .8025_x1033543 _04:31-51-52	20	<i>n't funny enough — nor , given how many American actors it features , and given the artificiality of its settings , is it particularly Continental .</i>	cc
pcc_eng_25_101 .9059_x1633051 _69:04-25-26	21	<i>No she 's not , in modern day terms its similar to someone who has only just turned 16 getting married , it is very close to being unacceptable and there</i>	advmod
pcc_eng_25_017 .5138_x0266928 _38:02-24-25	22	<i>or not as this is the only thing i can see which can be fraudulent , as the finger prints being identical is very rare .</i>	advmod
pcc_eng_25_080 .9491_x1294225 _12:01-26-27	25	<i>Nothing else to do then than to purchase their two major label albums , Head Down and Pressure And Time (2011) , both equally good , in my opinion</i>	nsubj
pcc_eng_25_042 .2330_x0667458 _09:01-34-35	33	<i>Few of the available titles like Fruity Friends , Starburst , Black Jack , Gonzo's Quest , Spinata Grande , Roulette Pro , Go Bananas , Monte Carlo , Mighty Sphinx will be always ready to launch and have</i>	nsubj
pcc_eng_25_074 .8999_x1196700 _22:01-34-35	33	<i>Few if any artists with Colorado ties have earned a larger place in art history than Bayer , and a gallery devoted to the museum 's ample collection of his work has been long overdue .</i>	advmod

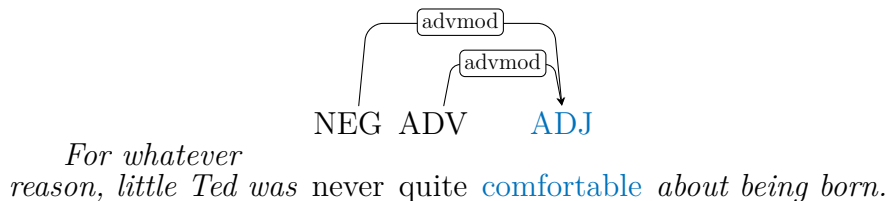
Literal 4.1: For Corpus Part Pcc25, 28 of 94,107 (0.03%) remaining RBdirect hits were dropped due to this filter.

status was determined by manual inspection of sample of pattern matches, and then extended to unexpected relations that showed up in the full processing. These are the cases that were not consistent enough—in regards to the resulting polarity environment of the bigram—to hard-code into the pattern specifications as prohibited or to keep as a representation of the target environment—i.e., cases where both true and false matches were captured. A sanity-check was also applied to ensure that no dependency relations which were intended to be entirely **prohibited** were present in the final results. Both relation sets, quarantined and prohibited, are sensitive to the direction of governance—whether the trigger–bigram dependency is headed by the trigger node or the adjective node (necessarily the head/governor/source of the bigram dependency). That is, different grammatical relations are seen and expected for the following sentences, repeated from the examples above.

Figure 4.1: Direction of Governance



(a) Trigger Governed



(b) Adjective Governed

While the `amod` dependency relation shown in the first case is consistent with

a negative polarity environment for the bigram, *particularly significant*, an **amod** dependency with the ADJ node as head fails to produce a negative environment (and such parses are rare, whereas **amod** is one of the most common relation types for trigger headed dependencies.)

The excerpt in 4.2 shows all of the hits removed due to quarantined dependency relations in the same corpus part (Pcc25). Again, while some cases may indeed provide a valid negative polarity environment, it cannot be predicted on the basis of the available dependency information alone, and its impact is even less significant for the whole than seen for the unqualified length limitation discussed above.

Quarantined dependency relations are irrelevant for patterns which capture only the bigram nodes—the relation between them is restricted to **advmod** (adverb modification) by definition, and no other relations are collected for such patterns. At an earlier development stage, however, the concept of the dependency quarantine was important for the superset positive polarity designation: Since superset positive polarity designation relies on *absence of negative trigger*, and therefore is equivalent to *every bigram token that is not in the negative superset*, dependency relations marked as prohibited for the negative pattern—and therefore never collected—were implicitly categorized as positive cases. Permitting a relation type to be collected (i.e. not explicitly excluded) means that it will still be **initially** grouped with the negative tokens. The quarantined cases are only removed from the negative set *after* its complement has been collected, meaning that those dropped at this point are included in the data sample for *neither* polarity. However, the later addition of

processing applied for all positive polarity hits described below in § 4.2.3 effectively nullified this impact on the positive superset ($\ast\mathcal{C}_N$).

- * Dropping 6 hits with more than 3 tokens from trigger to bigram and questionable deprel type

hit_id	distance	text_window	relation	head
pcc_eng_25_036 .4948_x0574597 _22:24-28-29	4	<i>the escalators were too few for me , too few !</i>	advmod	NEG
pcc_eng_25_039 .1880_x0618124 _10:17-22-23	5	<i>' the puzzles ar few and gentle at the very best , and also the</i>	obl	NEG
pcc_eng_25_013 .2368_x0197929 _05:07-12-13	5	<i>advertised that there is nothing to which man is not related ; that every thing</i>	acl	NEG
pcc_eng_25_052 .4360_x0832744 _045:3-8-9	5	<i>There is nothing that this VPN is not good at .</i>	acl	NEG
pcc_eng_25_017 .6967_x0269930 _10:08-16-17	8	<i>Whistleblower " knew practically NOTHING in that those ridiculous charges were far more dramatic & wrong , just</i>	acl	NEG
pcc_eng_25_096 .9314_x1552545 _072:07-15-16	8	<i>a mental quiet — none of the continuous self -assessment , being completely certain of failure , being</i>	acl	NEG

- * 94,072 remaining hits after validating dependencies
- * 100.0% of pre-validation. (99.96%)
- * (35 dropped.)

Literal 4.2: Examples of quarantined dependency relations

4.2.2 For Negative Environments: *N and @N

All negative polarity hits are filtered to exclude known cases where lexical items interact such that the resultant environment is not negative: the “*not only P*” construction and positively quantified or determinate *few*. Both adjustments are applied to each corpus part individually via the script `update_env_hits.py` mentioned above.

All hits containing *not only P*—either as trigger and adverb, or as trigger followed by an intervening *only* before the adverb of the bigram—are excluded from the negative polarity token set. Despite the logical negation of this phrase, the denial at hand is that the immediately following predicate *P* is “the **only** thing that is true”, but *P* is, in fact, true. The scope of negation is necessarily narrow when immediately followed by *only*. This is by no means a novel observation—*only* is a well-known focus operator with unique operations over alternatives.

The formal semantics of *only* is not within the purview of this dissertation, but the following example is presented to illustrate the motivation for excluding *not only* cases from the negative polarity category. In (4.3) “*Ted is only sad*,” indicates that `sad(Ted) = True`, but the continuations in (4.3b) or (4.3c) are infelicitous. Informally, “*Ted is **not** only sad*” is equivalent to “*Ted is sad + [something else/worse]*”. This suggests that the scope is “[*not only*] *sad*” rather than “*not [only sad]*”, meaning that the adjective *sad* isn’t negated at all.

(4.3) Ted is **only** sad.

- a. $\models$ Ted is sad.
- b. # “*Sad*” is the **worst** thing Ted can be.
- c. # Ted is *sad* **and** *broke* (and this is relevant).

In addition to this fabricated example in (4.3), (4.4) provides two examples of *not only* found in the corpus data (as described in the remaining chapters), but excluded from PBR’s negative polarity category. Note that all exclusions are dropped entirely from PBR—they do not simply change polarity classification and join the positive set. All *not only* cases examined manually (a sample of those removed), present a similar structure, where the first argument (whether a proposition, phrase, or a single predicate), is immediately followed by a parallel claim that is “also true”.

(4.4) Corpus Examples of “*not only*”

- a. *only* intervening between *not* and bigram

pcc_eng_00_001 **Not only** are our distributors
.5549...2 extremely good at what they do, but they are
:1-6-7 also local.

Claim 1: our distributors are extremely good at what they do

Claim 2: our distributors are also local

- b. *only* as part of bigram

pcc_eng_00_002 According to Naik, the infographic is **not**
 .1940...25 **only applicable** to Coca-Cola, but to all
 :08-09-10 caffeinated fizzy drinks.

Claim 1: the infographic is applicable to Coca-Cola

Claim 2: the infographic is applicable to all
 caffeinated fizzy drinks.

The second adjustment made to negative polarity hits specifically is to account for the difference between positive *few* (with determiner) and negative *few* (without determiner). Again, this is not a novel claim, but something that has been observed in the polarity literature (Atlas (2021), a.o.). The utterance, *A few are exceptionally bright* indicates “a small number”, $\approx$ “one”, “a couple”¹³³, etc. exceed expectations of brightness (be that intelligence or luminosity). The same utterance without the determiner—*Few are exceptionally bright*—indicates instead that “many/most are **not** (exceptionally bright)”. This pattern is robust: when *few* has a determiner, the negative polarity environment that it would otherwise create is not generated. Therefore, in the creation of PBR, the regular expression shown in Literal 4.3 was used to identify and exclude hits where `trigger_lemma=='few'` but the polarity of the resultant environment is not truly negative. The precluded cases then include *{the, those, these, their, a, his, her, my, your, our}* preceding the trigger *few*, with an optional single intervening word. For example, both “*the few [who applied were highly qualified]*” and “*these **last** few [weeks have been absolutely brutal]*” as well as the prior example “*a few [are exceptionally bright]*”.¹³⁴

Of course, *only* is still aberrant and can complicate matters: **Only** *[a few] are*

¹³³Despite the similarity in meaning of “a few” and “a couple”, “couple” cannot appear without a determiner: *couple are exceptionally bright* can only be understood as an implicit “a couple” where

```

POS_FEW_REGEX = re.compile(
    r'\b(?:th[oe](?:s?e?|i?r?)|a|h[ie][rs]|their|my|y?our '
    r')\b ?[a-z]* few\b'
)

```

Literal 4.3: Regular Expression for Positive “few”

exceptionally bright may be the explicit/original version of bare *few*. Using the [wNPI](#) *ever* as a diagnostic, there is clear evidence that *only a few* patterns differently, as show below in (4.5).

- (4.5) a. *Few* of my friends ever eat snails.
 b. # *A Few* of my friends ever eat snails.
 c. **Only** *a few* of my friends ever eat snails.

Indiscriminately rejecting “a few” as non-negative may be an oversimplification as *only* can cause interference and checking for its presence is required to make this fully accurate. However, both this removal step, and that of the previously described *not only* cases, were done **after** the “not negated” complement ID set (the positive superset) was calculated. Thus, any removal at this stage excludes the token from evaluation *at all*, rather than allowing it to be counted *against* the negative observations.

Another important factor in these exclusions—and most if not all others—is the

“a” has simply not been vocalized. It has no meaning that parallels that of bare *few*.

¹³⁴Note that the POS_FEW_REGEX expression is matched against lowercase-normalized token strings.

sheer mass of data under consideration. Working with so-called “Big Data” means it is functionally impossible to manually check for the problem case exhaustively. All that is available is an imperfect approximation that doesn’t *quite* deal with every case perfectly, *but* **because** there is a lot of data, the errors that fall out from this predicament aren’t sufficient to truly impact the overall results—large datasets can tolerate more noise.

4.2.3 For *Positive* Environments: *C_N and @P

As with Negative Polarity, to improve the accuracy of the approximation methods employed for positive polarity, additional adjustments were made in the creation of PBR from the original collection of hits described in § 4.1. The extent of exclusion is more significant, but simpler in nature: Any token that is *potentially negated* was excluded from further processing. This was done using a regular expression (defined in Literal 4.4) search of the text preceding the bigram.

```

NEG_REGEX = re.compile(
    # matching between word boundaries, `b` case-insensitive
    (r"\b(?P<neg>"
    # no, nor
    r"nor?"
    # not, n't, none, nobody, nothing, nowhere
    # ([and technically:] n'ne, n'body, etc.)
    r"[n[o']](?P<no_endings>t|ne|body|thing|where)"
    # rare, scarce, seldom, rarely, scarcely, seldomly
    # ([and technically:] rarel, rarey, etc.)
    r"|(?P<optional_ly>rare|scarce|seldom)l?y?"
    # hardly, barely
    r"|(?P<obligatory_ly>hard|bare)ly"
    # ain't, aint, neither, without, never, few, doubt
    r"|ain'?t|neither|without|never|few|doubt"
    r")\b"),
    # case-insensitive
    flags=re.IGNORECASE)

```

Note that this is stronger than the original negation pattern, as it includes words like:

1. forms not captured at all in the initial patterns (via `lemma`): *nowhere* and *doubt*
2. forms only captured if lemmatized as adverb derivation, with the suffix *-ly*:
scarce, *rare*, and *seldom*,
3. (unexpected) “contraction” forms only captured if lemmatized as spelled-out:
n’ne, *n’body*, *n’thing*, *n’where*

It **does not** capture forms like:

cannot, *w/o*, *nae*, *ne’er*, *nah*, *nuffin*, *doubtful*, *doubts*, *noooo*, *nooot*, *nooooobody*, etc.

Literal 4.4: Regular Expression for Potential Negation

Technically, this is too strong a filter, as negative words can precede a bigram but not scope over it, leaving the polarity of its environment unaffected. Relying fully on text alone make it almost impossible to accurately identify the true scope of negated environments, regardless of whether regular expressions are utilized. Structural information is absolutely necessary to reliably determine the range of

polarity environments.

However, as covered in [Chapter 2](#), the reality of the data landscape is that the total “*possibly* positive” dataset is so large—and ***so much larger than what is available for the negative***—that erring on the side of “too strong” in data exclusions is likely to have better results. That is, counting something as positive that should be negative is far worse than not counting it. The same is true of the filters applied to the negative set—better to exclude any questionable token than risk including it in the *wrong* set. Across the entirety of the complement of the negation superset, a total of 13,967,776 bigram tokens were dropped due to a potential negation in the preceding text.

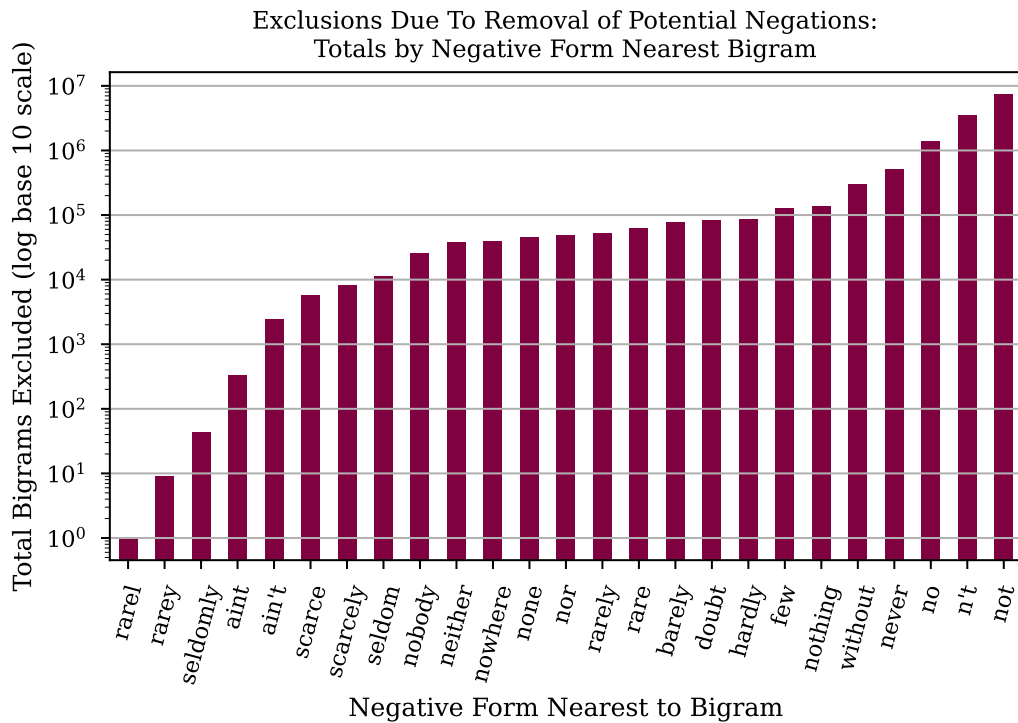


Figure 4.2: Nearest Potential Negation Frequencies (Superset)

4.3 Collecting Co-Occurrence Frequencies for Polarity Categories:

Raw Frequencies

All quantitative measures of bigram, adverb, adjective, and polarity interactions are extrapolated from co-occurrence frequencies—how often each word¹³⁵ or word pair (bigram¹³⁶ occurs with different polarity categorizations or each other. These co-

¹³⁵represented as `*_form_lower`

¹³⁶represented as `bigram_lower`

Table 4.3: Overview of Comparison Spaces

polarity approximation	frequency perspective	N	polarity	processing directory
superset	ALL+	71,961,373	(-) (+)	RBdirect/ not-RBdirect/
	NEQ+	6,347,362	(-) (+)	RBdirect/ not-RBdirect/
mirror subset	ALL+	1,680,633	(-) (+)	NEGmirror/ POSmirror/
	NEQ+	583,470	(-) (+)	NEGmirror/ POSmirror/

occurrence frequencies, contextualized via contingency tables, serve as the primitive observations for all analysis.

The processing described in § 4.1 and § 4.2 yields full data tables for each hit in the final selection for each category—allowing for additional inquiries for the specific selections, and thereby enabling both quality assurance and further analysis, both quantitative and qualitative. In addition to this more comprehensive output, co-occurrence frequencies for all observed adverb and adjective pairings are collected for each polarity category across all four *comparison spaces*, all combinations of *frequency perspective* type (ALL+ vs. NEQ+) and *polarity approximation* method (superset/“absence of negative” distinction vs. mirror subset/“presence of positive” designation).

The bigram marginal frequencies¹³⁷, are extracted for each of the six unique

¹³⁷The *marginal* frequency for any given bigram is identical to the *joint* frequency of its components; e.g., $f(\text{“too serious”})(=5,953) \equiv f(\text{“too”} \wedge \text{“serious”})(=5,953)$

combinations in the eight rows in the above table, repeated from [Chapter 2](#) for convenience. Note that there are **six distinct frequency records** instead of eight, because the negative polarity frequencies are the same regardless of frequency perspective—NEQ+ and ALL+ only differ in their representation of positive polarity and have identical representations of negative polarity. Each frequency record is stored in a tab-delimited format, with columns pertaining to joint frequency (f), adverb (`adv_form_lower`; l_1), and adjective (`adj_form_lower`; l_2) from left to right, and rows sorted by descending frequency, then alphabetically. The initial (highest joint frequencies) and final rows of all six `freq_tsv/*/*final-freq.tsv` files, grouped by dataset, are shown in [Literal 4.5](#) and [Literal 4.6](#).

Note that the lowest frequency included in these tables is **1**. While only attested combinations are explicitly recorded, non-occurring combinations are not stored because they can be inferred. The single token combinations shown are also a testament to both the questionable nature of some collected bigram parses and the difficulty of weeding out such cases that lack clear orthographic cues.

1. {ALL+ , NEQ+ } : *N

```
'results/freq_tsv/RBdirect/AdvAdj_ALL_RBdirect_final-freq.tsv'
44441 as good
30092 very good
27548 as bad
26480 so sure
... ...
1 zoned residential
1 zowielala amazing
1 zzz surprising
```

2. ALL+ : *C_N

```
'results/freq_tsv/RBdirect/AdvAdj_ALL_not-RBdirect_final-freq.tsv'
1047841 so many
689158 most important
502631 so much
449316 more likely
... ...
1 zyuganov more
1 zyuganov worse
1 zzz boring
```

3. NEQ+ : *C_N

```
'results/freq_tsv/RBdirect/AdvAdj_NEQ_not-RBdirect_final-freq.tsv'
48117 so many
31659 most important
23184 so much
20811 more likely
... ...
1 zone balanced
1 zoogoers behind-the-scene
1 zoonify dependable
```

Literal 4.5: Previews of Distinct Frequency Records for PBR **Superset**

1. {ALL+ , NEQ+ } : @N

'results/freq_tsv/mirror/AdvAdj_ALL_NEGmirror_final-freq.tsv'		
12201	more	important
10998	too	late
5156	too	early
3443	as	good
...	...	...
1	youthfully	callow
1	zapatero	elected
1	zowielala	amazing

2. ALL+ : @P

'results/freq_tsv/mirror/AdvAdj_ALL_POSmirror_final-freq.tsv'		
14910	as	simple
10941	more	important
8340	completely	different
7713	more	likely
...	...	...
1	zealously	spiritual
1	zombies	uninterested
1	zoologically	correct

3. NEQ+ : @P

'results/freq_tsv/mirror/AdvAdj_NEQ_POSmirror_final-freq.tsv'		
3162	as	simple
2304	more	important
1787	completely	different
1619	more	likely
...	...	...
1	yougube	wrong
1	youthfully	earnest
1	zanily	defensive

Literal 4.6: Previews of Distinct Frequency Records for PBR **Mirror Subset**

For each comparison space, the designated negative and positive frequency records are concatenated—without reference to source file (i.e. polarity)—to provide the basis of the context-blind evaluation of all attested adverb and adjective forms. These are illustrated in Literal 4.7 and Literal 4.8, which show the highest frequency values for each comparison space, sans polarity indicators (though still separate by polarity; i.e. these values are *not* the total joint frequencies). These have the same format, but any attested pair may have more than 1 row, with different frequencies indicated, if it occurs in both polarities.

For example, in Literal 4.8, *more important* appears on both the second and fourth rows in the third excerpt (ALL+ : @E). Comparing with the single-polarity values above, it is clear that row 2 (12201 more important) originates in the negative mirror frequencies, {ALL+ ,NEQ+ } : @N, and row 4 (10941 more important) in the unrestricted positive mirror frequencies, ALL+ : @P. This and any other repeated form combinations are summed in the next stage of processing.

1. ALL+ : *S

```
'results/freq_tsv/ANYdirect/AdvAdj_ALL_any-direct_final-freq.tsv'  
1047841 so      many  
689158  most    important  
502631  so      much  
449316  more    likely  
429404  too     much  
417705  very    good  
373189  most    popular  
369742  as      many  
...     ...     ...
```

2. NEQ+ : *S

```
'results/freq_tsv/ANYdirect/AdvAdj_NEQ_any-direct_final-freq.tsv'  
48117   so      many  
44441   as      good  
31659   most    important  
30092   very    good  
27548   as      bad  
26480   so      sure  
26026   quite   sure  
25656   too     late  
...     ...     ...
```

Literal 4.7: Previews of Concatenated Frequency Records for PBR **Superset**

1. ALL+ : @S

```
'results/freq_tsv/ANYmirror/AdvAdj_ALL_any-mirror_final-freq.tsv'  
14910  as      simple  
12201  more     important  
10998  too      late  
10941  more     important  
8340   completely different  
7713   more     likely  
7307   too      easy  
7028   too      familiar  
...    ...      ...
```

2. NEQ+ : @S

```
'results/freq_tsv/ANYmirror/AdvAdj_NEQ_any-mirror_final-freq.tsv'  
12201  more     important  
10998  too      late  
5156   too      early  
3443   as       good  
3162   as       simple  
2946   quite    sure  
2615   so       easy  
2345   too      old  
...    ...      ...
```

Literal 4.8: Previews of Concatenated Frequency Records for PBR **Mirror Subset**

While the initial frequency retrieval (the six separate files) is performed by the same program as the processing described in § 4.1 and § 4.2, the ANY* context-blind concatenation is done via a bash script, `script/cat_tsvs.sh` (and the conglomerate script, `cat_every_tsv.sh` which calls `script/cat_tsvs.sh` for each of the four comparison spaces). This is exemplified in Literal 4.9.

```
bash cat_every_tsv.sh
# is equivalent to:
( bash script/cat_tsvs.sh
  && bash script/cat_tsvs.sh -s
  && bash script/cat_tsvs.sh -p mirror
  && bash script/cat_tsvs.sh -p mirror -s )
```

Literal 4.9: Concatenating Frequency .tsv's

4.4 Association Measures

Once the frequency records of the previous section, § 4.3 have been collected, the association measures laid out in Chapter 2 can be calculated. The first subsection, § 4.4.1, covers the calculation methods, using UCS¹³⁸ and then a python module, `association-measures`¹³⁹, both developed by Stephanie Evert and colleagues (Evert 2005, 2010; Evert *et al.* 2017; Heinrich & Opolka 2024; Uhrig *et al.* 2018; Evert 2022). The second subsection, § 4.4.2, provides a brief look at the outcomes (more detailed

¹³⁸For installation, see <http://www.collocations.de/software.html#UCS> (Evert 2005, 2010)

¹³⁹For installation, see <https://pypi.org/project/association-measures/>;
pip command: `pip install association-measures`

dissection can be found in [Chapter 3](#), and additional sample tables were provided in [Chapter 2](#)).

4.4.1 Calculating Association Measures

The entirety of the AM calculation process is performed via the single python script, `polar_assoc.py`, for a single set of parameters, but the compilation shell script, `run_assoc.sh`, can be used to directly run the program the full array of arguments. The tabular-formatted output files include polar environment comparisons for each of the three lexical units¹⁴⁰—bigram, adverb, and adjective—as well as the context-blind `adv~adj` comparison represented by the `ANY*` records, run across an array of minimum frequency thresholds, which determines the rows (evaluated combinations) visible in each table. This shell script does not run all four comparison spaces automatically, but they can be indicated via simplified arguments, as suggested in [Literal 4.10](#).

The first step performed by `polar_assoc.py`, in preparation for UCS processing, is the conversion of the bigram frequencies (housed in their separate files split by polarity) into a format that directly indicates polarity classification. The simple concatenation performed to generate the composite context-blind frequencies retains no such indication. To do this, each row is simply adjusted to indicate the designated polarity label (as indicated in the table provided in the preceding section), in addition

¹⁴⁰all represented by the lowercase-normalized form

```

# ensure the specified log directory exists
mkdir -p logs/associate
# Superset, ALL & NEQ
( bash script/run_assoc.sh
  && bash script/run_assoc.sh -E
) > logs/associate/run_direct_AM.$(date +%Y%m%d-%I%P).log 2>&1
# Mirror Subset, ALL & NEQ
( bash script/run_assoc.sh -P mirror
  && bash script/run_assoc.sh -P mirror -E
) > logs/associate/run_mirror_AM.$(date +%Y%m%d-%I%P).log 2>&1

```

Literal 4.10: To calculate all AM scores using `run_assoc.sh`

```

==> "./results/freq_tsv/RBdirect/AdvAdj_ALL_RBdirect_final-freq.tsv" <==
30092      very      good
==> "./results/freq_tsv/RBdirect/AdvAdj_ALL_not-RBdirect_final-freq.tsv" <==
417705     very      good

```

translates to

30092	NEGATED	very_good
417705	COMPLEMENT	very_good

Literal 4.11: Converting Context–Blind Frequencies by Pattern Results to Polarized Comparison Frequencies for Comparison

to the selected lexical unit. If said unit is the full bigram, then the adverb and adjective strings are joined via an underscore; e.g. `more<TAB>important` becomes `more_important`. The result of this process is illustrated in [Literal 4.11](#) with the co-occurrence of *very* modifying *good* in the full superset (`ALL+ : *S`) frequencies¹⁴¹.

Each polarity of each comparison space is stored in a separate file within the

¹⁴¹Tab size has been normalized for reading clarity.

Table 4.4: Frequency Profile for *very good* in each of the polar comparisons

<code>../env_prepped_tsv/*.tsv</code>	<i>f</i>	<i>E</i>	bigram
<code>complement_bigram_ALL-direct.tsv</code>	417,705	COMPLEMENT	very_good
<code>negated_bigram_ALL-direct.tsv</code>	30,092	NEGATED	very_good
<code>complement_bigram_NEQ-direct.tsv</code>	19,394	COMPLEMENT	very_good
<code>negated_bigram_NEQ-direct.tsv</code>	30,092	NEGATED	very_good
<code>posmir_bigram_ALL-mirror.tsv</code>	5,883	POSMIR	very_good
<code>negmir_bigram_ALL-mirror.tsv</code>	1,567	NEGMIR	very_good
<code>posmir_bigram_NEQ-mirror.tsv</code>	1,219	POSMIR	very_good
<code>negmir_bigram_NEQ-mirror.tsv</code>	1,567	NEGMIR	very_good

`results/ucs/env_prepped_tsv/` directory, as listed in [Table 4.4](#)

The python script `polar_assoc.py` connects the required inputs with the UCS tools which generate an initial set of AM scores for each of the specified comparisons. Each distinct frequency threshold applied to 16 separate tables of AM calculations: $4 \text{ comparison spaces} \times 4 \text{ unit selections (3 polarized units + 1 context-blind)}$. These are illustrated for the minimum joint frequency threshold of 5,000 in [Literal 4.12](#).

Each comma-delimited table also has a parquet version. Only tables that are small enough (i.e. where the frequency cutoff has eliminated enough rows) are stored as a `.csv` as well.

The python script, `polar_assoc.py`, calls the `ucs` command, `ucs-make-tables`, to create the initial AM table from the frequency tsv's for a given comparison space. The `ucs` tool calculates all contingency frequencies, storing them as a *frequency signature*. This is essentially a flattened contingency table, where **f** is the joint frequency, **f1** and **f2** are the marginal frequencies of **l1** and **l2** respectively, and **N** is

```

$ tree -P '_min5000x_' --prune -I "meta-info/extra" \
  "results/assoc_df/results/assoc_df/"
+-- adv_adj
|   +-- ANYdirect
|   |   +-- AdvAdj_ALL_any-direct_final-freq_min5000x.csv
|   |   +-- AdvAdj_NEQ_any-direct_final-freq_min5000x.csv
|   +-- ANYmirror
|   |   +-- AdvAdj_ALL_any-mirror_final-freq_min5000x.csv
|   |   +-- AdvAdj_NEQ_any-mirror_final-freq_min5000x.csv
+-- polar
|   +-- RBdirect
|   |   +-- adj
|   |   |   +-- polarized-adj_ALL-direct_min5000x.csv
|   |   |   +-- polarized-adj_NEQ-direct_min5000x.csv
|   |   +-- adv
|   |   |   +-- polarized-adv_ALL-direct_min5000x.csv
|   |   |   +-- polarized-adv_NEQ-direct_min5000x.csv
|   |   +-- bigram
|   |   |   +-- polarized-bigram_ALL-direct_min5000x.csv
|   |   |   +-- polarized-bigram_NEQ-direct_min5000x.csv
|   +-- mirror
|   |   +-- adj
|   |   |   +-- polarized-adj_ALL-mirror_min5000x.csv
|   |   |   +-- polarized-adj_NEQ-mirror_min5000x.csv
|   |   +-- adv
|   |   |   +-- polarized-adv_ALL-mirror_min5000x.csv
|   |   |   +-- polarized-adv_NEQ-mirror_min5000x.csv
|   |   +-- bigram
|   |   |   +-- polarized-bigram_ALL-mirror_min5000x.csv
|   |   |   +-- polarized-bigram_NEQ-mirror_min5000x.csv
12 directories, 16 files

```

Literal 4.12: All AM calculations for $f \geq 5,000$

the sum of all frequencies provided. The frequency assignments are illustrated using the bigram “*super duper*” (as quantified in the ALL+ Superset comparison space) as an example in ??, and the initial UCS table output (as text) is shown in Literal 4.13.

Table 4.5: Contingency Frequency Definitions, exemplified by bigram, “*super duper*”

variable	contingency represented	with example, <i>super duper</i>	frequencies
011	joint frequency (=f)	<i>super</i> and <i>duper</i> .	626
012	# tokens where 11 appears without 12	<i>super</i> , but not <i>duper</i> .	290,250
021	# tokens where 12 appears without 11	not <i>super</i> , but <i>duper</i> .	1
R1	# tokens with 11, (=f1)	all bigrams with (adverb) <i>super</i> (marginal frequency).	290,876
R2	# tokens without 11, (=N-f1)	all bigrams without (adverb) <i>super</i> .	71,670,497
C1	# tokens with 12, (=f2)	all bigrams with (adjective) <i>duper</i> (marginal frequency).	627
C2	# tokens without 12, (=N-f2)	all bigrams without (adjective) <i>duper</i> .	71,960,746

The python code then runs the bash program, `transform_ucs.sh`, on the initialized “ucs table”. This shell script adds columns for the *expected* frequencies (labeled E11), given the observed frequency signature, and multiple built-in AMs, including log likelihood, G^2 ¹⁴². In addition to the existing measures, it also calculates custom defined measures for forwards and backwards *delta P* (ΔP), and conditional probability, (P). These custom definitions are shown in Literal 4.14. Literal 4.15

¹⁴²labeled `am.log.likelihood` by `ucs`, and converted to `am_log_likelihood` by `polar_assoc.py`, and eventually abbreviated to `G2`

l1	l2	f	f1	f2	N
super	duper	626	290876	627	71961373
far	away	1150	402768	1157	71961373
nerve	wracking	527	531	538	71961373

Literal 4.13: Initial UCS table

provides an example of the updated output after all UCS (built-in and custom definitions) have been applied.

1. ΔP (Adjusted Conditional Probability)

```
am.p1.given2 := (%011% / %C1%) - (%012% / %C2%)
am.p2.given1 := (%011% / %R1%) - (%021% / %R2%)
```

2. (Simple) Conditional Probability

```
am.p1.given2.simple := (%011% / %C1%)
am.p2.given1.simple := (%011% / %R1%)
```

Literal 4.14: Custom AM defined for UCS processing

At this point in processing, most of the [AMs](#) selected for coverage in [Chapter 2](#) are available, but *conservative log ratio (LRC)* is still absent. The python script, `polar_assoc.py`, uses the module `association-measures`, another tool developed by Evert and colleagues, to calculate LRC values ([Heinrich & Opolka 2024](#)). Calculation parameters for LRC were set to evaluate a two-tailed hypothesis, to an alpha of 0.005¹⁴³, 0 values were replaced with 1, and Bonferroni vocabulary correction

¹⁴³or $\alpha = 0.001$ for later calculations, including those discussed at length in [Chapter 3](#)

l1	l2	f	E11
super	duper	626	2.534
far	away	1150	6.476
nerve	wracking	527	0.004
> log.likelihood p1.given2 p2.given1			
> -----			
>	6886.22715	0.9944	0.00215
>	11844.58694	0.9884	0.00286
>	13362.52510	0.9796	0.99247
> p1.given2.simple p2.given1.simple			
> -----			
>	0.99841		0.00215
>	0.99395		0.00286
>	0.97955		0.99247

Note: E11 indicates the expected joint frequency, and text has been wrapped for display purposes.

Literal 4.15: UCS Table Transformed with AM Scores (abbreviation & rounding applied)

was applied.

Some additional descriptive values were also calculated using `pandas` dataframe operations within `polar_assoc.py`. These include the “unexpected” frequency, or the *observed* frequency minus the *expected* frequency, $f - E_{11}$. The unexpected ratio (or percentage if multiplied by 100) is the ratio of the observed tokens which were “unexpected”, $(f - E_{11})/f$. For example, *far away* has an observed joint frequency of 1,150, and an expected frequency of ~ 6 . This means that 1,144 of the 1,150 observed tokens for *far away* were unexpected, $\sim 99.5\%$. Although intuitive, this metric has the potential to mislead when the observed and expected diverge more than their own values (i.e. when $\text{abs}(f - E_{11}) > f$). Finally, some attempt is made to merge the distinct ΔP values into a single score: this takes the form of an additional column containing the mean of the two scores, labeled `deltaP_mean`¹⁴⁴.

4.4.2 Association Samples

The final AM tables contain all of the information shown the example table provided in Table 4.6. The entries selected for this illustration include context-blind evaluations from the ALL+ Superset comparison space above the frequency floor of $f = 1,000$ with the highest LRC (*thinly veiled*), average ΔP (*scantily clad*), and the minimum ΔP for each direction within the top 20 LRC values (*far away* and *upwardly mobile*). The latter two bigrams illustrate the differences between symmetric and

¹⁴⁴Also abbreviated as `dPavg`

asymmetric AMs—although the overall symmetric association is similar, the sources are reversed in these cases. All following example tables have been abbreviated for viewing clarity.

Table 4.6: Example of final AM tables, in full

key	top LRC <i>thinly~veiled</i>	top dP_mean <i>scantily~clad</i>	top 20 LRC, min dP2 <i>far~away</i>	top 20 LRC, min dP1 <i>upwardly~mobile</i>
f	1,333	4,638	1,150	1,893
exp_f	0.04	0.43	6.48	0.56
G2	28,016.71	90,347.15	11,844.59	30,316.40
odds_r_disc	5.79	6.59	4.44	4.83
dP1	0.78	0.71	0.99	0.09
dP2	0.76	0.99	<0.01	0.95
P1	0.78	0.71	0.99	0.09
P2	0.76	0.99	<0.01	0.95
f1	1,746	4,682	402,768	2,002
f2	1,705	6,562	1,157	20,114
N	71,961,373	71,961,373	71,961,373	71,961,373
Fyy(O11)	1,333	4,638	1,150	1,893
Fyn(O12)	413	44	401,618	109
Fny(O21)	372	1,924	7	18,221
Fnn(O22)	71,959,255	71,954,767	71,558,598	71,941,150
Fy_(R1)	1,746	4,682	402,768	2,002
Fn_(R2)	71,959,627	71,956,691	71,558,605	71,959,371
F_y(C1)	1,705	6,562	1,157	20,114
F_n(C2)	71,959,668	71,954,811	71,960,216	71,941,259
exp_12	1,745.96	4,681.57	402,761.52	2,001.44
exp_21	1,704.96	6,561.57	1,150.52	20,113.44
exp_22	71,957,922.04	71,950,129.43	71,557,454.48	71,939,257.56
t	36.51	68.10	33.72	43.50
dice	0.77	0.82	0.01	0.17
MI	4.51	4.04	2.25	3.53
deltaP_min	0.76	0.71	<0.01	0.09
deltaP_max	0.78	0.99	0.99	0.95
deltaP_abs_max	0.78	0.99	0.99	0.95
deltaP_mean	0.77	0.85	0.50	0.52
unexp_f	1,332.96	4,637.57	1,143.52	1,892.44

Continued on next page

Table 4.6: Example of final AM tables, in full

key	<i>thinly~veiled</i>	<i>scantily~clad</i>	<i>far~away</i>	<i>upwardly~mobile</i>
unexp_r	1	1	0.99	1
LRC	16.71	14.96	12.48	11.67
LRC_001	16.68	14.95	12.40	11.66
LRC_01	16.72	14.96	12.52	11.67
LRC_05	16.74	14.97	12.61	11.68
LRC_nc	16.94	15.07	13.45	11.77
LRC_dv	16.76	14.98	12.69	11.69
f_sqrt	36.51	68.10	33.91	43.51
f1_sqrt	41.79	68.43	634.64	44.74
f2_sqrt	41.29	81.01	34.01	141.82

4.4.2.1 Context-Blind Association Examples

This section shows the context-blind associations for four distinct comparison spaces for randomly selected bigrams. The same bigrams are shown for the different positive sampling perspectives of each dataset, but different between dataset. While most of these bigrams (*physically able*, *really happy*, *yet final*, *more enjoyable*, *particularly good*, *so beautiful*, and *that simple*) show at least a slight “positive association” or *attraction* between the words in each pair, *so clear* shows indication of a “negative association” or *repulsion* in the superset examples (both [ALL+](#) in [Table 4.7](#) and [NEQ+](#) in [Table 4.8](#)).

Although these tables do not examine polarity directly, the frequency perspectives do indirectly provide some indication of polar dispositions. Note that *yet final* has exactly the same joint frequency, 640, in both [ALL+](#) ([Table 4.7](#)) and [NEQ+](#) ([Table 4.8](#)), while the other examples have their joint frequencies halved (or even

more severely restricted). While it’s possible some of those tokens are in positive environments, the explanation could be that most of the tokens are in negative environments, as the NEQ+ perspective retains 100 % of negative tokens. This means the differences seen for the other examples indicate that *at least* that many tokens are positive for each case. For example, *at least* 32,529 of “*really happy*”’s total 35,529 tokens ($\approx 91.56\%$) are positive, as they were excluded from the NEQ+ sample. These tables are provided purely to illustrate the outcomes of the processing described above See [Chapter 2](#) and [Chapter 3](#) for in depth discussions of AMs and their implications.

Table 4.7: ALL+ Superset: Context-Blind AM Sample

key	physically~able	really~happy	so~clear	yet~final
f	1,955	35,529	10,704	640
f1	63,922	1,945,388	5,173,538	94,755
f2	210,786	420,680	346,404	5,736
N	71,961,373	71,961,373	71,961,373	71,961,373
expF	187.237	11,372.571	24,904.114	7.553
unexpF	1,767.763	24,156.429	−14,200.114	632.447
unexpR	0.904	0.680	−1.327	0.988
P1	0.009	0.084	0.031	0.112
P2	0.031	0.018	0.002	0.007
dP1	0.008	0.058	−0.041	0.110
dP2	0.028	0.013	−0.003	0.007
dPavg	0.018	0.035	−0.022	0.058
LRC	3.207	1.686	−1.201	6.220
G2	5,700.842	34,398.429	−10,983.304	4,494.502
LogR	3.397	1.731	−1.281	6.575

Table 4.8: NEQ+ Superset: Context-Blind AM Sample

key	physically~able	really~happy	so~clear	yet~final
f	522	3,000	3,583	640
f1	7,930	182,968	566,272	53,779
f2	22,885	47,131	83,958	1,212
N	6,347,362	6,347,362	6,347,362	6,347,362
expF	28.591	1,358.590	7,490.208	10.269
unexpF	493.409	1,641.410	-3,907.208	629.731
unexpR	0.945	0.547	-1.090	0.984
P1	0.023	0.064	0.043	0.528
P2	0.066	0.016	0.006	0.012
dP1	0.022	0.035	-0.047	0.520
dP2	0.062	0.009	-0.008	0.012
dPavg	0.042	0.022	-0.027	0.266
LRC	3.865	1.047	-1.005	6.586
G2	2,087.796	1,545.067	-2,756.497	4,447.596
LogR	4.223	1.196	-1.136	7.033

Table 4.9: ALL+ Mirror Subset: Context-Blind AM Sample

key	more~enjoyable	particularly~good	so~beautiful	that~simple
f	1,241	475	1,660	529
f1	285,773	12,946	129,421	5,465
f2	2,315	31,461	7,912	25,344
N	1,680,633	1,680,633	1,680,633	1,680,633
expF	393.640	242.346	609.282	82.412
unexpF	847.360	232.654	1,050.718	446.588
unexpR	0.683	0.490	0.633	0.844
P1	0.536	0.015	0.210	0.021
P2	0.004	0.037	0.013	0.097
dP1	0.367	0.008	0.133	0.018
dP2	0.004	0.018	0.009	0.082
dPavg	0.185	0.013	0.071	0.050
LRC	2.181	0.618	1.458	2.363
G2	1,603.586	180.052	1,394.482	1,120.099
LogR	2.496	0.983	1.670	2.709

Table 4.10: NEQ+ Mirror Subset: Context-Blind AM Sample

	key	more~enjoyable	particularly~good	so~beautiful	that~simple
f		488	416	464	487
f1		121,848	10,020	47,033	4,538
f2		827	13,484	2,894	7,580
N		583,470	583,470	583,470	583,470
expF		172.705	231.562	233.283	58.954
unexpF		315.295	184.438	230.717	428.046
unexpR		0.646	0.443	0.497	0.879
P1		0.590	0.031	0.160	0.064
P2		0.004	0.042	0.010	0.107
dP1		0.382	0.014	0.080	0.057
dP2		0.003	0.019	0.005	0.095
dPavg		0.193	0.016	0.043	0.076
LRC		1.926	0.481	0.736	2.771
G2		569.009	124.677	198.539	1,267.945
LogR		2.447	0.867	1.124	3.132

4.4.2.2 Adverb Polarity Association Examples

Table 4.11: ALL+ Superset: Adverb Polarity AM Sample

polarity	(+)		(−)	
key	COM~easily	COM~legally	NEG~easily	NEG~legally
f	114,031	28,913	7,638	3,213
f1	68,787,692	68,787,692	3,173,681	3,173,681
f2	121,669	32,126	121,669	32,126
N	71,961,373	71,961,373	71,961,373	71,961,373
expF	116,303.085	30,709.161	5,365.915	1,416.839
unexpF	−2,272.085	−1,796.161	2,272.085	1,796.161
unexpR	−0.020	−0.062	0.297	0.559
P1	0.937	0.900	0.063	0.100
P2	0.002	<0.001	0.002	0.001
dP1	−0.019	−0.056	0.019	0.056

Continued on next page

Table 4.11: ALL+ Superset: Adverb Polarity AM Sample

polarity key	(+) COM~easily COM~legally		(-) NEG~easily NEG~legally	
	COM~easily	COM~legally	NEG~easily	NEG~legally
dP2	-0.001	-0.001	0.001	0.001
dPavg	-0.010	-0.028	0.010	0.028
LRC	-0.444	-1.119	0.444	1.119
G2	-895.694	-1,777.368	895.694	1,777.368
LogR	-0.538	-1.268	0.538	1.268

Table 4.12: NEQ+ Superset: Adverb Polarity AM Sample

polarity key	(+) COM~easily COM~legally		(-) NEG~easily NEG~legally	
	COM~easily	COM~legally	NEG~easily	NEG~legally
f	5,357	1,339	7,638	3,213
f1	3,173,681	3,173,681	3,173,681	3,173,681
f2	12,995	4,552	12,995	4,552
N	6,347,362	6,347,362	6,347,362	6,347,362
expF	6,497.500	2,276	6,497.500	2,276
unexpF	-1,140.500	-937	1,140.500	937
unexpR	-0.213	-0.700	0.149	0.292
P1	0.412	0.294	0.588	0.706
P2	0.002	<0.001	0.002	0.001
dP1	-0.088	-0.206	0.088	0.206
dP2	-0.001	-0.001	0.001	0.001
dPavg	-0.044	-0.103	0.044	0.103
LRC	-0.379	-1.022	0.379	1.022
G2	-403.285	-795.476	403.285	795.476
LogR	-0.512	-1.262	0.512	1.262

Table 4.13: ALL+ Mirror Subset: Adverb Polarity AM Sample

polarity key	(+) POS~entirely POS~even		(-) NEG~entirely NEG~even	
	POS~entirely	POS~even	NEG~entirely	NEG~even
f	7,863	58,341	2,415	2,838
f1	1,388,898	1,388,898	291,735	291,735

Continued on next page

Table 4.13: ALL+ Mirror Subset: Adverb Polarity AM Sample

polarity key	(+) POS~entirely POS~even		(-) NEG~entirely NEG~even	
	POS~entirely	POS~even	NEG~entirely	NEG~even
f2	10,278	61,179	10,278	61,179
N	1,680,633	1,680,633	1,680,633	1,680,633
expF	8,493.879	50,559.159	1,784.121	10,619.841
unexpF	-630.879	7,781.841	630.879	-7,781.841
unexpR	-0.080	0.133	0.261	-2.742
P1	0.765	0.954	0.235	0.046
P2	0.006	0.042	0.008	0.010
dP1	-0.062	0.132	0.062	-0.132
dP2	-0.003	0.032	0.003	-0.032
dPavg	-0.032	0.082	0.032	-0.082
LRC	-0.376	1.972	0.376	-1.972
G2	-250.363	9,472.915	250.363	-9,472.915
LogR	-0.548	2.110	0.548	-2.110

Table 4.14: NEQ+ Mirror Subset: Adverb Polarity AM Sample

polarity key	(+) POS~entirely POS~even		(-) NEG~entirely NEG~even	
	POS~entirely	POS~even	NEG~entirely	NEG~even
f	1,596	12,140	2,415	2,838
f1	291,735	291,735	291,735	291,735
f2	4,011	14,978	4,011	14,978
N	583,470	583,470	583,470	583,470
expF	2,005.500	7,489	2,005.500	7,489
unexpF	-409.500	4,651	409.500	-4,651
unexpR	-0.257	0.383	0.170	-1.639
P1	0.398	0.811	0.602	0.189
P2	0.005	0.042	0.008	0.010
dP1	-0.103	0.319	0.103	-0.319
dP2	-0.003	0.032	0.003	-0.032
dPavg	-0.053	0.175	0.053	-0.175
LRC	-0.368	1.949	0.368	-1.949
G2	-169.570	6,373.538	169.570	-6,373.538

LogR	-0.597	2.097	0.597	-2.097
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4.4.2.3 Adjective Polarity Association Examples

Table 4.15: ALL+ Superset: Adjective Polarity AM Sample

polarity key	(+) COM~gorgeous COM~toxic		(-) NEG~gorgeous NEG~toxic	
	COM~gorgeous	COM~toxic	NEG~gorgeous	NEG~toxic
f	32,156	28,448	700	675
f1	68,787,692	68,787,692	3,173,681	3,173,681
f2	32,856	29,123	32,856	29,123
N	71,961,373	71,961,373	71,961,373	71,961,373
expF	31,406.966	27,838.601	1,449.034	1,284.399
unexpF	749.034	609.399	-749.034	-609.399
unexpR	0.023	0.021	-1.070	-0.903
P1	0.979	0.977	0.021	0.023
P2	<0.001	<0.001	<0.001	<0.001
dP1	0.023	0.021	-0.023	-0.021
dP2	<0.001	<0.001	>-0.001	>-0.001
dPavg	0.012	0.011	-0.012	-0.011
LRC	0.777	0.647	-0.777	-0.647
G2	497.376	363.664	-497.376	-363.664
LogR	1.083	0.958	-1.083	-0.958

Table 4.16: NEQ+ Superset: Adjective Polarity AM Sample

polarity key	(+) COM~gorgeous COM~toxic		(-) NEG~gorgeous NEG~toxic	
	COM~gorgeous	COM~toxic	NEG~gorgeous	NEG~toxic
f	1,500	1,314	700	675
f1	3,173,681	3,173,681	3,173,681	3,173,681
f2	2,200	1,989	2,200	1,989
N	6,347,362	6,347,362	6,347,362	6,347,362
expF	1,100	994.500	1,100	994.500
unexpF	400	319.500	-400	-319.500

Continued on next page

Table 4.16: NEQ+ Superset: Adjective Polarity AM Sample

polarity key	(+) COM~gorgeous COM~toxic		(-) NEG~gorgeous NEG~toxic	
unexpR	0.267	0.243	-0.571	-0.473
P1	0.682	0.661	0.318	0.339
P2	<0.001	<0.001	<0.001	<0.001
dP1	0.182	0.161	-0.182	-0.161
dP2	<0.001	<0.001	>-0.001	>-0.001
dPavg	0.091	0.080	-0.091	-0.080
LRC	0.743	0.592	-0.743	-0.592
G2	297.786	209.040	-297.786	-209.040
LogR	1.099	0.960	-1.099	-0.960

Table 4.17: ALL+ Mirror Subset: Adjective Polarity AM Sample

polarity key	(+) POS~free POS~scared		(-) NEG~free NEG~scared	
f	3,896	1,681	889	396
f1	1,388,898	1,388,898	291,735	291,735
f2	4,785	2,077	4,785	2,077
N	1,680,633	1,680,633	1,680,633	1,680,633
expF	3,954.389	1,716.461	830.611	360.539
unexpF	-58.389	-35.461	58.389	35.461
unexpR	-0.015	-0.021	0.066	0.090
P1	0.814	0.809	0.186	0.191
P2	0.003	0.001	0.003	0.001
dP1	-0.012	-0.017	0.012	0.017
dP2	>-0.001	>-0.001	<0.001	<0.001
dPavg	-0.006	-0.009	0.006	0.009
LRC	<0.001	<0.001	<0.001	<0.001
G2	-4.892	-4.122	4.892	4.122
LogR	-0.120	-0.167	0.120	0.167

Table 4.18: NEQ+ Mirror Subset: Adjective Polarity AM Sample

polarity key	(+) POS~free POS~scared		(-) NEG~free NEG~scared	
	POS~free	POS~scared	NEG~free	NEG~scared
f	799	354	889	396
f1	291,735	291,735	291,735	291,735
f2	1,688	750	1,688	750
N	583,470	583,470	583,470	583,470
expF	844	375	844	375
unexpF	-45	-21	45	21
unexpR	-0.056	-0.059	0.051	0.053
P1	0.473	0.472	0.527	0.528
P2	0.003	0.001	0.003	0.001
dP1	-0.027	-0.028	0.027	0.028
dP2	>-0.001	>-0.001	<0.001	<0.001
dPavg	-0.014	-0.014	0.014	0.014
LRC	<0.001	<0.001	<0.001	<0.001
G2	-4.815	-2.356	4.815	2.356
LogR	-0.154	-0.162	0.154	0.162

4.4.2.4 Bigram Polarity Association Examples

Table 4.19: ALL+ Superset: Bigram Polarity AM Sample

polarity key	(+) COM~gorgeous COM~toxic		(-) NEG~gorgeous NEG~toxic	
	COM~gorgeous	COM~toxic	NEG~gorgeous	NEG~toxic
f	32,156	28,448	700	675
f1	68,787,692	68,787,692	3,173,681	3,173,681
f2	32,856	29,123	32,856	29,123
N	71,961,373	71,961,373	71,961,373	71,961,373
expF	31,406.966	27,838.601	1,449.034	1,284.399
unexpF	749.034	609.399	-749.034	-609.399
unexpR	0.023	0.021	-1.070	-0.903
P1	0.979	0.977	0.021	0.023
P2	<0.001	<0.001	<0.001	<0.001

Continued on next page

Table 4.19: ALL+ Superset: Bigram Polarity AM Sample

polarity key	(+) COM~gorgeous COM~toxic		(−) NEG~gorgeous NEG~toxic	
	COM~gorgeous	COM~toxic	NEG~gorgeous	NEG~toxic
dP1	0.023	0.021	−0.023	−0.021
dP2	<0.001	<0.001	>−0.001	>−0.001
dPavg	0.012	0.011	−0.012	−0.011
LRC	0.777	0.647	−0.777	−0.647
G2	497.376	363.664	−497.376	−363.664
LogR	1.083	0.958	−1.083	−0.958

Table 4.20: NEQ+ Superset: Bigram Polarity AM Sample

polarity key	(+) COM~gorgeous COM~toxic		(−) NEG~gorgeous NEG~toxic	
	COM~gorgeous	COM~toxic	NEG~gorgeous	NEG~toxic
f	1,500	1,314	700	675
f1	3,173,681	3,173,681	3,173,681	3,173,681
f2	2,200	1,989	2,200	1,989
N	6,347,362	6,347,362	6,347,362	6,347,362
expF	1,100	994.500	1,100	994.500
unexpF	400	319.500	−400	−319.500
unexpR	0.267	0.243	−0.571	−0.473
P1	0.682	0.661	0.318	0.339
P2	<0.001	<0.001	<0.001	<0.001
dP1	0.182	0.161	−0.182	−0.161
dP2	<0.001	<0.001	>−0.001	>−0.001
dPavg	0.091	0.080	−0.091	−0.080
LRC	0.743	0.592	−0.743	−0.592
G2	297.786	209.040	−297.786	−209.040
LogR	1.099	0.960	−1.099	−0.960

Table 4.21: ALL+ Mirror Subset: Bigram Polarity AM Sample

polarity key	(+) POS~free POS~scared		(−) NEG~free NEG~scared	
	POS~free	POS~scared	NEG~free	NEG~scared
f	3,896	1,681	889	396

Continued on next page

Table 4.21: ALL+ Mirror Subset: Bigram Polarity AM Sample

polarity key	(+) POS~free POS~scared		(-) NEG~free NEG~scared	
	POS~free	POS~scared	NEG~free	NEG~scared
f1	1,388,898	1,388,898	291,735	291,735
f2	4,785	2,077	4,785	2,077
N	1,680,633	1,680,633	1,680,633	1,680,633
expF	3,954.389	1,716.461	830.611	360.539
unexpF	-58.389	-35.461	58.389	35.461
unexpR	-0.015	-0.021	0.066	0.090
P1	0.814	0.809	0.186	0.191
P2	0.003	0.001	0.003	0.001
dP1	-0.012	-0.017	0.012	0.017
dP2	>-0.001	>-0.001	<0.001	<0.001
dPavg	-0.006	-0.009	0.006	0.009
LRC	<0.001	<0.001	<0.001	<0.001
G2	-4.892	-4.122	4.892	4.122
LogR	-0.120	-0.167	0.120	0.167

Table 4.22: NEQ+ Mirror Subset: Bigram Polarity AM Sample

polarity key	(+) POS~free POS~scared		(-) NEG~free NEG~scared	
	POS~free	POS~scared	NEG~free	NEG~scared
f	799	354	889	396
f1	291,735	291,735	291,735	291,735
f2	1,688	750	1,688	750
N	583,470	583,470	583,470	583,470
expF	844	375	844	375
unexpF	-45	-21	45	21
unexpR	-0.056	-0.059	0.051	0.053
P1	0.473	0.472	0.527	0.528
P2	0.003	0.001	0.003	0.001
dP1	-0.027	-0.028	0.027	0.028
dP2	>-0.001	>-0.001	<0.001	<0.001
dPavg	-0.014	-0.014	0.014	0.014
LRC	<0.001	<0.001	<0.001	<0.001
G2	-4.815	-2.356	4.815	2.356
LogR	-0.154	-0.162	0.154	0.162

CHAPTER 5

METHODS FOR RETRIEVING AND CLEANING

5.1 Introduction

The aim of this chapter is to describe how the input data for PBR was collected and processed. It is broken into three main parts, which correspond to § 5.2–§ 5.4. The end product of these steps is the input data for the methodology discussed in Chapter 4—the creation of PBR.

§ 5.2 discusses the universal dependency relationships collected by pattern specifications. § 5.3 describes the methodology—or processing pipeline—for retrieving relevant pattern-matching corpus tokens ([hits](#)) and initial processing and consolidation efforts. Finally, § 5.4 presents the cleaning steps applied to the basic context-blind bigram set.¹⁴⁵

5.2 Pattern Specifications

The relevant linguistic data was collected from corpora containing Universal Dependency annotations, stored in the CoNLL-U format as `.conllu` files. More information on the specifics of the selected corpora can be found in Chapter 6.

¹⁴⁵All post-processing cleaning procedures dependent on specific identified contexts are discussed in Chapter 4.

Relevant tokens are identified by `grew-match` (henceforth shortened to `grew`; Guillaume (2021, 2019)), a program for querying UD-parsed corpora. “Relevance” is defined by pattern specifications constructed following `Grew-match` syntax¹⁴⁶, denoting (word) node and dependency relation features of interest required for selection. While multiple different relation types can be stipulated or prohibited for a single dependency relationship or *tie* in the `grew` pattern syntax, it is not possible to collect data on fully optional ties/nodes, variably defined node features, or unspecified governance relationships. Therefore, the patterns¹⁴⁷, are split primarily into categories based on the required node definitions. Generally speaking, these correspond to the polarity approximation categories that depend on positive evidence of an environment trigger—`*N`, the negative superset, `@P`, the positive subset, and `@N`, the negative subset—or the general context-blind `ADV ADJ` bigram selection—where no environment information is sought. Within each category¹⁴⁸ there is a separate pattern for the two opposing governance relationships between a specified trigger node and the adjective node (or bigram). That is, there are two separate patterns defined: one where the adjective is the head and one where the environment trigger is. The basic context-blind bigram pattern has only one pattern¹⁴⁹, because the ADJ node is obligatorily the head of the dependency with the ADV node.

¹⁴⁶`Grew` pattern syntax follows `OCaml` for regular expressions.

¹⁴⁷The pattern specifications are stored as `.pat` files

¹⁴⁸All pattern categories, which are synonymous with the previously defined polarity approximation categories, are represented by directories in which the corresponding pattern specification (`.pat`) files are stored.

¹⁴⁹And therefore only one `.pat` file

The most basic pattern, for basic (context-blind) bigrams, is presented in Literal 5.1. It specifies no relationships beyond the confines of the ADV ADJ bigram itself. There are only three requirements: First, every node must have the indicated part of speech¹⁵⁰. Second, the adverb node must immediately precede the adjective. And finally, there must be an adverbial modification (*advmod*) dependency relationship between them, of the form *ADV* $\xleftarrow{\text{advmod}}$ *ADJ*, where the *ADJ* is the head node of the dependency and the *ADV* is the target node. These specifications are displayed in *grew-match* syntax in Literal 5.1.

```
pattern {
  ADJ [xpos=re"JJ.?" ] ;
  ADV [xpos=re"RB.?" ] ;
  mod: ADJ -[advmod]-> ADV ;
  ADV < ADJ
}
```

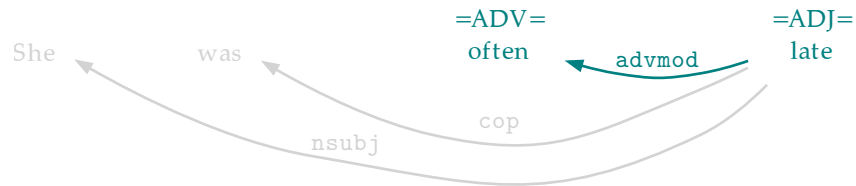
Literal 5.1: Basic Context-Blind Bigram Pattern Specification

Note that this pattern requires a specific linear order as well as specific part-of-speech categories and dependency relationship between them: The adverb node must immediately precede the adjective, and no other configurations are collected. For example, this pattern matches the “often late” in (5.1), but matches neither “late often” in (5.2), nor “Often...late” in (5.3). It does match “extremely late” in (5.4)

¹⁵⁰CoNLL-U formatted UD includes one or both of (a) universal part-of-speech tags, abbreviated as UPOS, or (b) Tree-Bank style part-of-speech tags, abbreviated as XPOS. While the majority of the corpora contains both, a small portion only contains XPOS, so the described patterns define the requisite grammatical categories using these.

and (5.5), but excludes “often” in both. Finally, to round out the picture, (5.6) does not match the pattern at all, as *extremely* and *often* are both adverbs and follow the adjective *late*, rather than precede it.

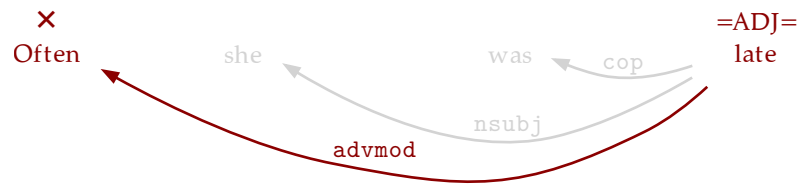
(5.1) She was often_M late_A.



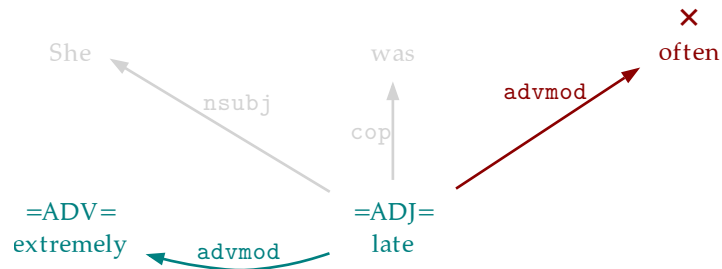
(5.2) She was late_A often_M.



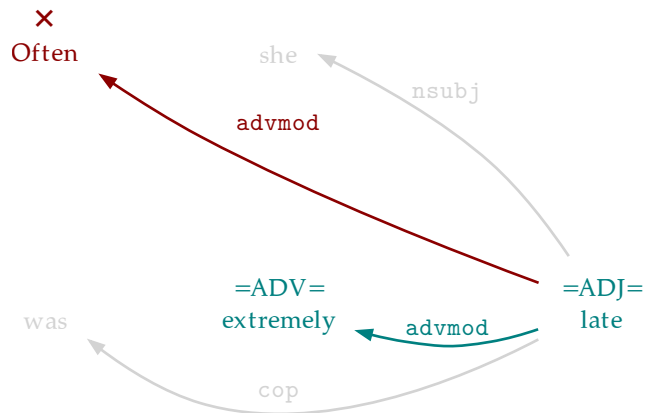
(5.3) Often_M she was late_A.



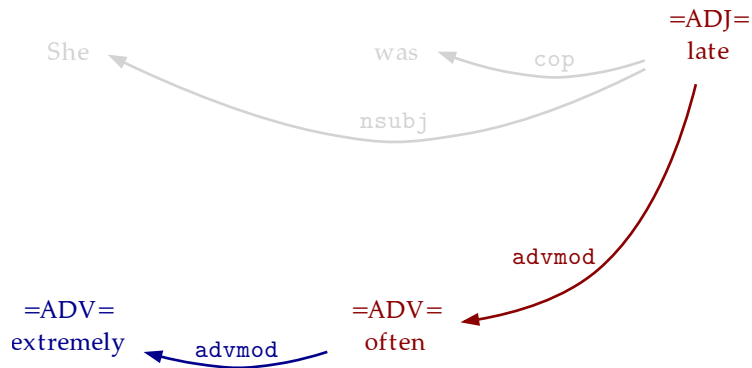
(5.4) She was often_M extremely_M late_A.



(5.5) Often_M she was extremely_M late_A.

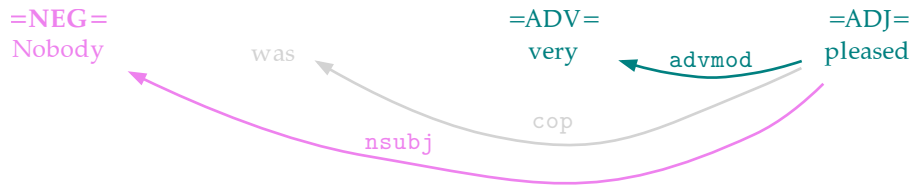


(5.6) She was late_A extremely_M often_M.

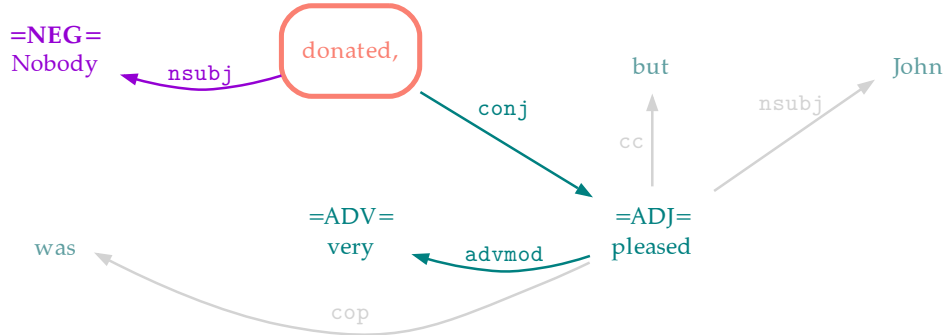


The remainder of this section describes the methodology for identifying tokens relevant to the desired phenomenon, which, in this dissertation is the negation or *privation* of *ADV ADJ* bigrams. Examples (5.7) and (5.8) illustrate the need for structural information—in this case, dependency parses.

(5.7) **Negated:** Nobody was very pleased.



(5.8) **Not Negated:** Nobody donated, but John was very pleased.



Example (5.8) emphasizes the importance of the dependency relation, rather than just the presence of the node itself. Although there is a valid negation trigger in the sentence, it (*nobody*) does not have a dependency relationship with *pleased* that places the bigram in a negative polarity environment. It is only by the *lack* of a negation trigger node with a valid dependency relation to the adjective, *pleased*, that these sentences can be identified as “positive”.

A key consideration for contextualized pattern specifications is that each node (i.e. word) in a sentence has only **one** *head* node, but can have **any number** of child/*target* nodes (including 0). This is important because it directly affects how the token collection patterns must be structured: Scope information is not directly encoded with dependencies—there is nothing equivalent to w_i *c-commands* w_j . Such relationships must be determined via chained *ties* between node pairs with optional

filtration based on *dependency relation* types (*relations*) for each tie. For most relations, a direct dependency tie is indicative to a c-command relationship, but that is not true universally. For example, the dependency relation of “*parataxis*” explicitly subverts this structural hierarchy. To say that *B is under the scope of A*, a minimum requirement for many semantic calculations, *B* needs to have a *dependency path* to *A* such that the relation of every consecutive tie/link is one which keeps all past/lower nodes under the scope of the *head* node in that tie.

The dependency structure is straightforward for the majority of tokens to be collected: *classical/sentential/predicate* negation as conferred by *not*¹⁵¹, negative subjects (e.g., “**nobody**, **nothing** was very helpful.”), or modification by a negative adverb are all denoted with a **direct dependency** relationship between the privation (negative trigger) element and the bigram. That is, the universal dependency parse directly connects the adjective (ADJ) node of the bigram (always the second position, following the adverb (ADV)) and the node of the linguistic element/operator that creates a privative (negative) environment—what would be called the *operator*, *licenser*, or *trigger* in polarity sensitivity literature. All identified triggers have this direct dependency. For mirror subset retrieval specifications, the positive trigger has the same structure: a direct dependency between the positive operator node and the adjective (bigram).

As already alluded to, the **relation type** of this single tie between ADJ and

¹⁵¹That is, the **lemma**, **not**. Corresponding forms include “*not*” and “*-n’t*”—encompassing both standard contractions and substandard, “*ain(’)t*”

TRIGGER must also be evaluated. There are relations that consistently indicate that ADJ (and therefore the bigram as a unit) is under the scope of TRIGGER, ones that consistently indicate that ADJ is **not** under the scope of TRIGGER, and even some that are inconsistent in the scope relation they represent. Therefore, the different dependency relation types must be taken into consideration to evaluate whether the bigram is actually under the scope of the trigger (i.e. within the polarity environment it produces).

Another consideration is the **direction** of the dominance or governance relationship between the nodes. In the diagrams and inline dependency representations, the governor/head/source node is the “source” of the arrow which points to the “target” node. The predominant trigger dependency is that where the ADJ node is the head, i.e. ADJ governs NEG in the majority of ties between TRIGGER and ADJ nodes. This is encoded as an arrow pointing away from the head node; e.g., TRIGGER $\leftarrow$ ADJ. However, in a subset of cases, the direction of governance is reversed, and TRIGGER is the head of the dependency: TRIGGER $\rightarrow$ ADJ. In this case, ADJ is the target node.

The basic varieties of dependency paths specified for collection differ on governance direction and the specific relation type of the dependency. Simplified examples of the direct dependency relationships collected are represented graphically below.

5.2.1 ADJ-headed Dependencies

The pattern specifications for negative dependencies governed by the ADJ node—or the bigram—are shown in Literal 5.2. Then (5.9) gives example matches for this pattern.

```
pattern {
  ADJ [xpos=re"JJ.?"];
  ADV [xpos=re"RB.?"];
  NEG [lemma="not"|"hardly"|"scarcely"|"never"|"rarely"|"barely"
        |"seldom"|"no"|"nothing"|"none"|"nobody"|"neither"
        |"without"|"few"|"nor"|"ain't"|"aint"];
  mod: ADJ - [advmod] -> ADV;
  neg: ADJ - [~advcl|amod|discourse|prep|obl|obl:npm|
        |parataxis|dislocated|rel]-> NEG;
```

Literal 5.2: Pattern Specifications for ADJ-headed Negative Dependencies

(5.9) NEG $\leftarrow$ ADJ (bigram)

PREDICATIVE ADJECTIVE

a. *sentential negation*:

His talk was [**not particularly interesting**].

b. *simple negatively quantified subject*:

[**None** [of the talks today]] were [**particularly interesting**].

c. *coordinating conjoined APs*:

The claims were [**neither**_{NEG₁} **entirely novel**_{ADJ₁}]¹
 [**nor**_{NEG₂} **particularly interesting**_{ADJ₁}]²

ATTRIBUTIVE ADJECTIVE

d. *directly negated AP*:

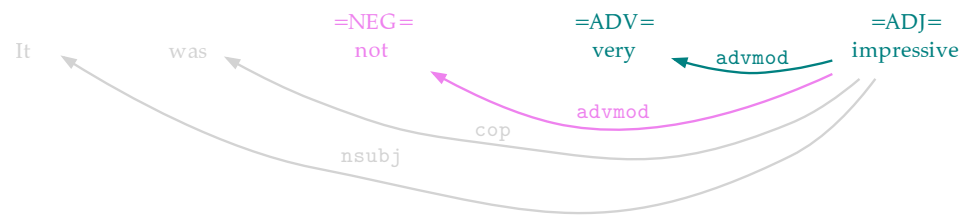
It was [a [[**not entirely new**] problem]].

The following examples (5.10–5.18) provide textual and visual representations of

different potential cases.

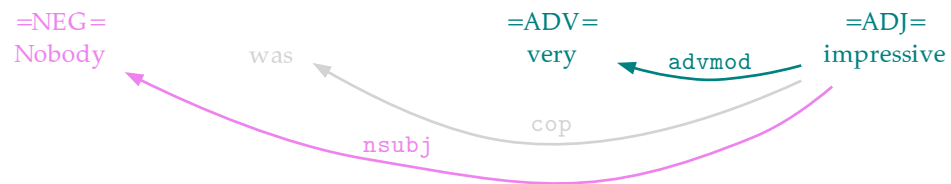
(5.10) SENTENTIAL NEGATION

It was **not** very impressive.



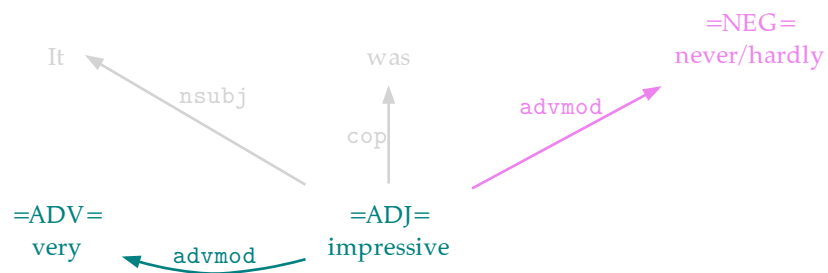
(5.11) NEGATIVE SUBJECT

Nobody was very impressive.



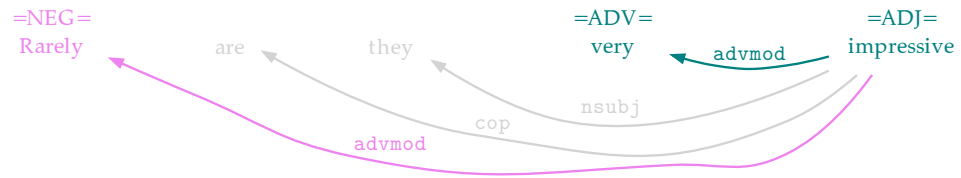
(5.12) NEGATIVE ADVERB

It was **never/hardly** very impressive.



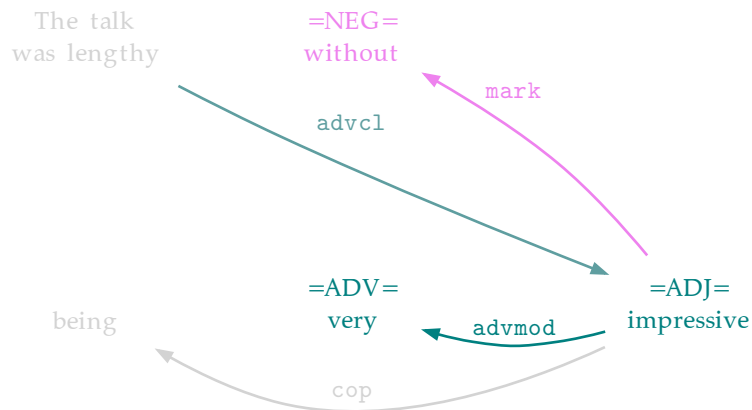
(5.13) FRONTED NEGATIVE ADVERB

Rarely are they very impressive.



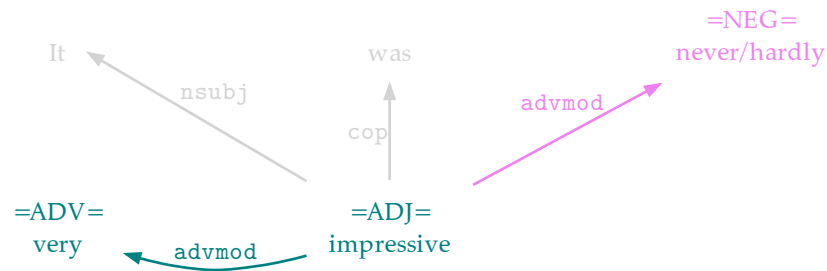
(5.14) NEGATIVE PREPOSITION (*without*)

The talk was lengthy **without** being **very impressive**.



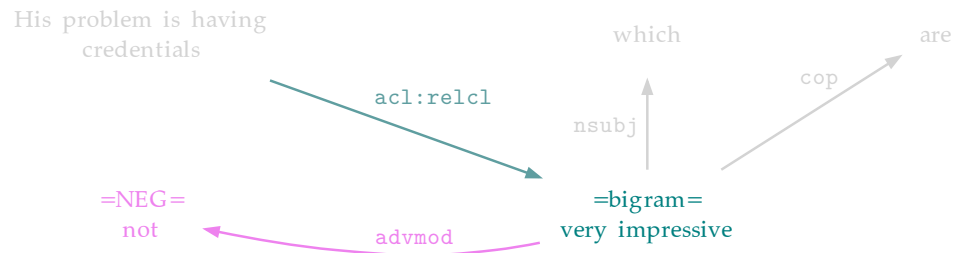
(5.15) NEGATIVE PRENOMINAL MODIFIER PHRASE

His problem is having [_{NP}[_{NegP} **not very impressive**] credentials].



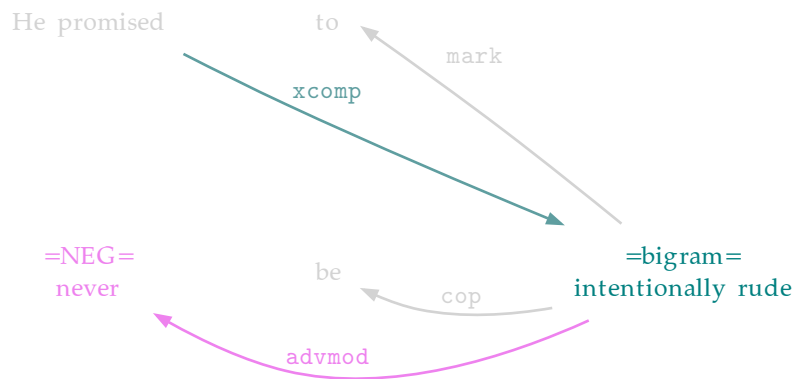
(5.16) NEGATIVE RELATIVE CLAUSE (*essentially negated predicate*)

His problem is having [credentials [which are [**not very impressive**]]].



(5.17) INFINITIVE

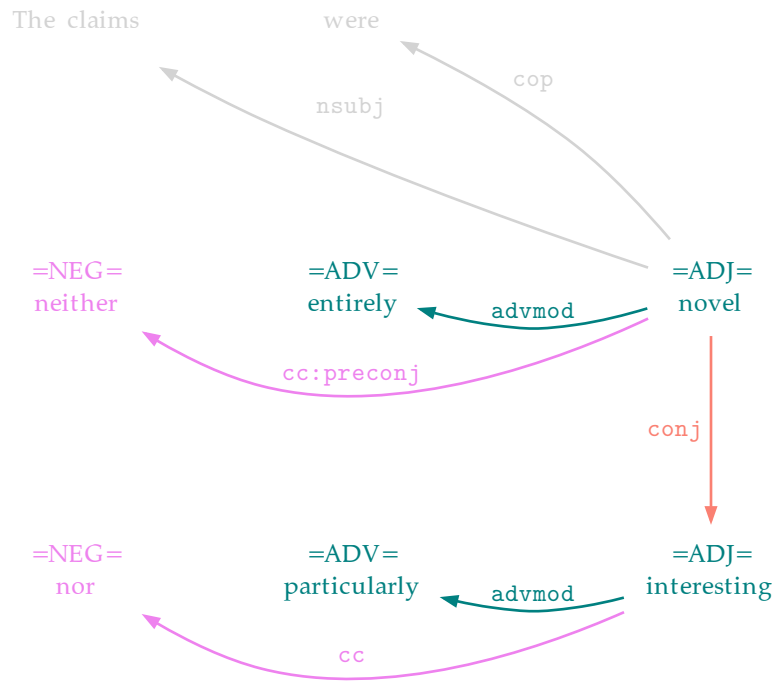
He promised[to **never** be intentionally **rude**]



(5.18) COORDINATING (NEGATIVE) CONJUNCTION(S)

The claims were [**neither** entirely **novel**]₁

[**nor** particularly **interesting**]₂.



5.2.2 TRIGGER-headed Dependencies

The pattern specifications for negative dependencies governed by the NEG node—the negative trigger—are shown in Literal 5.3. (5.19) gives example matches for this pattern.

```

pattern {
  ADJ [xpos=re"JJ.?"];
  ADV [xpos=re"RB.?"];
  mod: ADJ - [advmod] -> ADV;
  NEG [lemma="not"|"hardly"|"scarcely"|"never"|"rarely"|"barely"
      |"seldom"|"no"|"nothing"|"none"|"nobody"|"neither"
      |"without"|"few"|"nor"|"ain't"|"aint"];
  neg: NEG - [^acl:relcl|advcl|appos|conj|nmod|parataxis] -> ADJ;
  ADV < ADJ;
  NEG << ADV
}

```

Literal 5.3: Pattern Specifications for NEG-headed Negative Dependencies

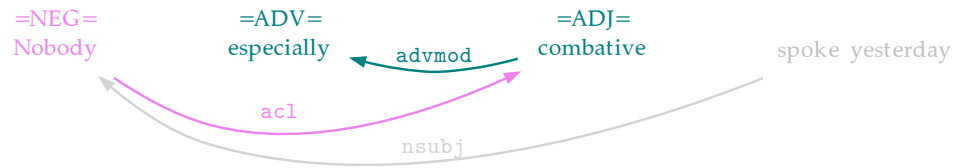
- (5.19) NEG $\longrightarrow$ ADJ (bigram)
attributive adjective modifying negative pronoun as clausal subject or expletive object;
 e.g.:
- [**Nobody** [particularly important]] made an appearance.
 - [**Nothing** [especially offensive]] was said.
 - It's [nothing [too serious]].
 - He had [nothing [particularly relevant]] to add.

In the following sections, examples (5.20–5.25) provide textual and visual representations of different potential cases.

5.2.2.1 Post-modified Negative Pronoun as Subject

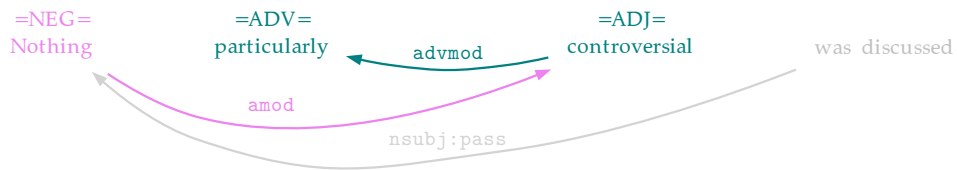
- (5.20) AS ACTIVE SUBJECT

Nobody especially combative spoke yesterday.



(5.21) AS PASSIVE SUBJECT

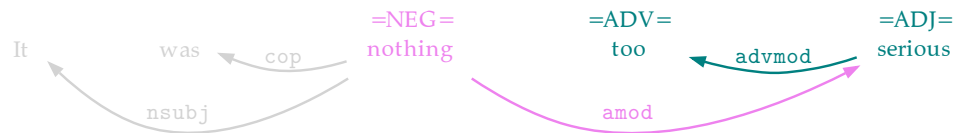
Nothing particularly **controversial** was discussed.



5.2.2.2 Post-modified Negative Pronoun as Predicate

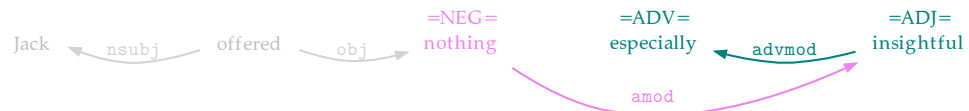
(5.22) AS SIMPLE PREDICATE

It was **nothing** too **serious**.



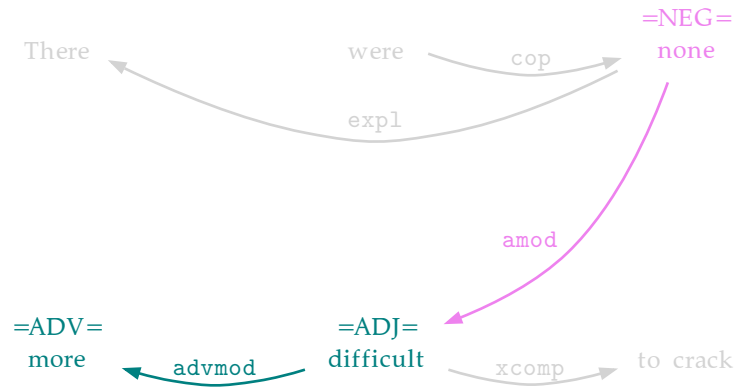
(5.23) AS DIRECT OBJECT

Jack offered **nothing** especially **insightful**.



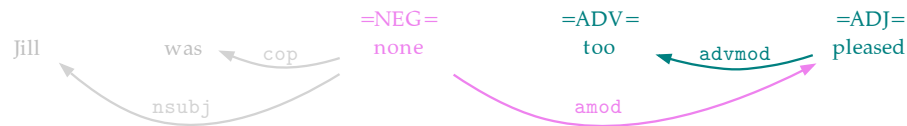
(5.24) FOLLOWING EXPLETIVE SUBJECT

There were **none** **more** **difficult** to crack.



While *nothing* and *nobody* are more likely to appear as dependency heads, there are less productive/semi-idiomatic constructions that put other negative triggers as the head; namely, *none too* {*pleased, happy, cheap, ...*}, as shown in (5.25).

(5.25) Jill was **none** **too** **pleased**.



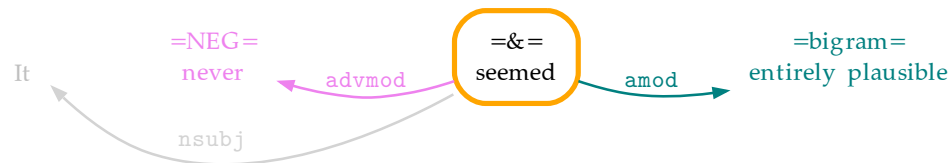
5.2.3 Indirect Dependency Relationships

It should be noted that, although the aforementioned (direct) dependencies account for the majority of negated bigram tokens, direct relationships do not capture all possible cases of negative polarity environments. Certain cases elude capture due to

at least one node—an additional dependency tie—intervening between the ADJ and TRIGGER nodes. This includes a broad array of syntactic structures—such as the neg-raising phenomenon, negated subject NPs, and prenominal modification of negated objects—but the exponential complexity of evaluating such formations paired with their relative infrequency made the cost of pursuing them prohibitive. Addressing this deficit remains a project for future work, but the following illustrations are provided for the sake of transparency: None of these cases appear in the results discussed in Chapters 2–4. However, the methods for approximating polarity status (initial designation and post-processing) as described in the preceding chapter (Chapter 4), ensure that such cases do not pollute the positive polarity token sets. The cases exemplified below are therefore entirely exempt from the final data for the current iteration.

(5.26) NEG-RAISING

It never seemed entirely plausible



(5.27) NEGATIVELY QUANTIFIED ENTITY

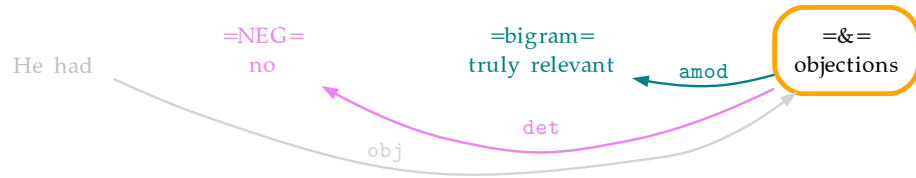
a. Subject:

No one/exhibit was incredibly remarkable



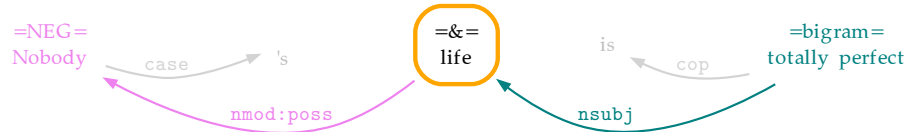
b. Object:

He had no truly relevant objections



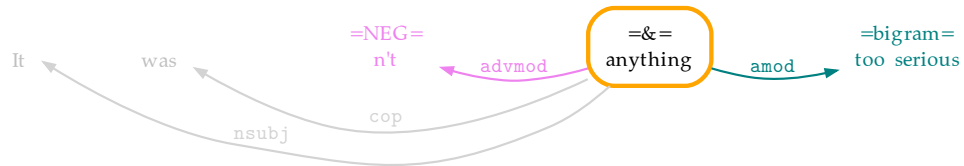
c. Possessive Negative Pronoun:

Nobody's life is totally perfect.



(5.28) NEGATED POST-MODIFIED PRONOUN¹⁵²

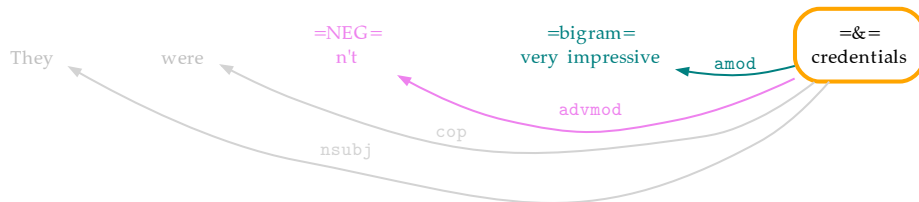
It wasn't anything too serious.



(5.29) PRENOMINAL MODIFICATION

a. negated simple predicate

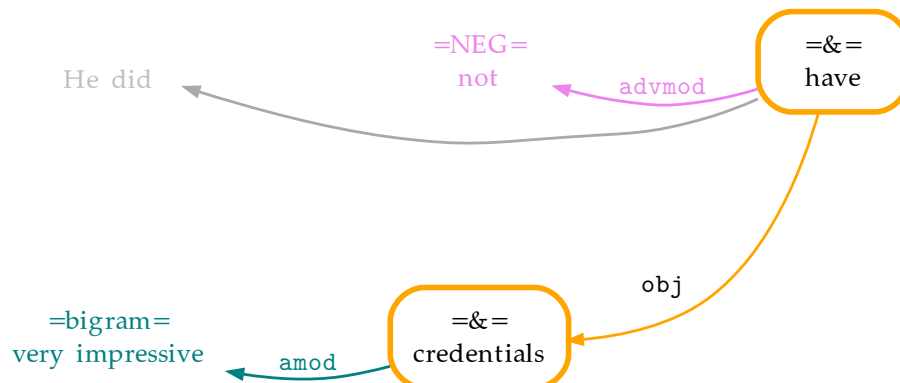
They weren't very impressive credentials.



b. negated direct object

He did not have very impressive credentials

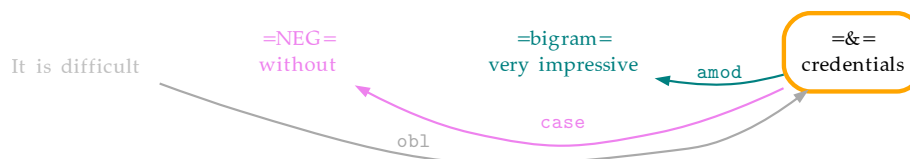
¹⁵²which is not itself inherently negative



(5.30) EMBEDDED IN NEGATIVE PREPOSITIONAL PHRASE

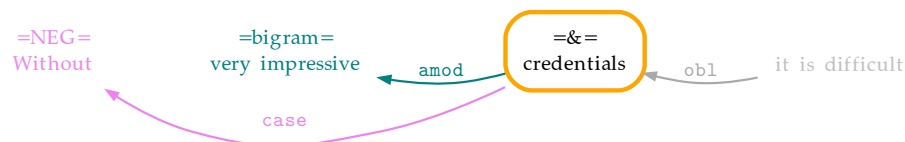
a. PRENOMINAL MODIFICATION OF OBJECT OF PREPOSITION

It is difficult without very impressive credentials.



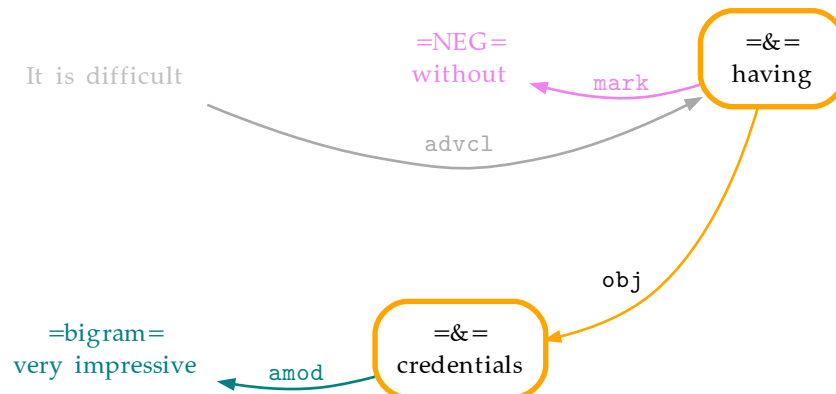
b. Note that the placement of the without clause does not affect the presence of an intervening node.

Without very impressive credentials it is difficult.

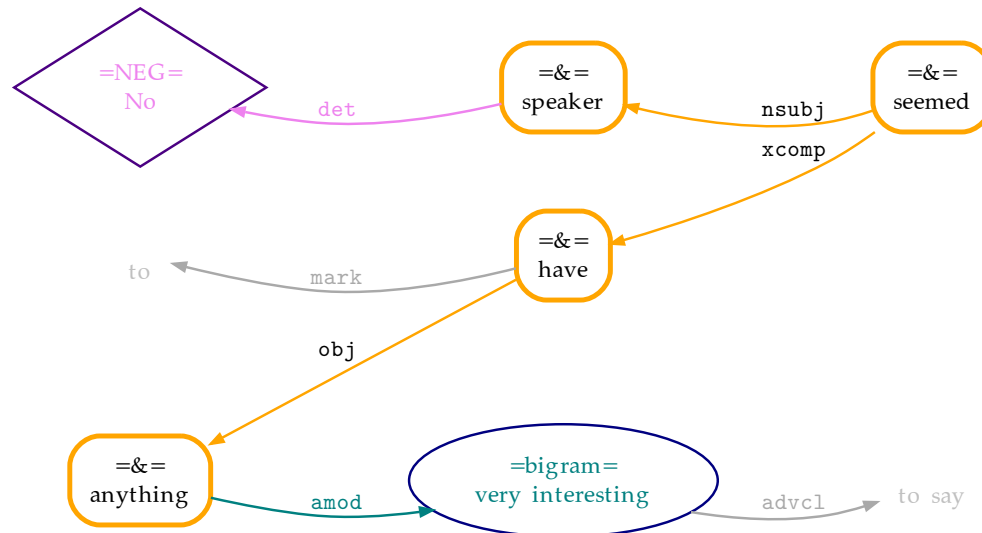


c. EMBEDDED EVEN DEEPER: WITHIN GERUND OBJECT OF PREPOSITION

It is difficult without having¹ very impressive credentials²



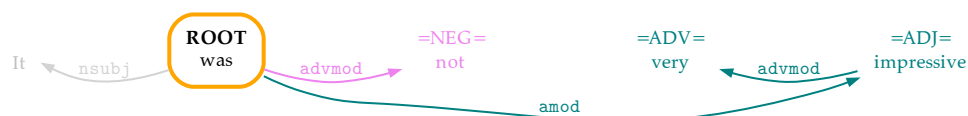
- (5.31) COMBINATION OF CASES (4 INTERVENING NODES/TOKENS)
No speaker¹ seemed² to have³ anything⁴ very impressive to say.



The particular parser used impacts the fundamental structure. In investigating other parsers and patterns, I determined that the dependency parser included in `spacy` produces an indirect dependency relationship even for sentential negation of predicate adjectives (and their modifier adverb). It sets the copula *be* as the “root” of

the sentence, and relates both the adjective and the negation to that. This means the sentence, “*It was not very impressive*,” has the relationships *was* $\xrightarrow{\text{acomp}}$ *impressive*¹⁵³ and *was* $\xrightarrow{\text{neg}}$ *not*¹⁵⁴, giving it the structure shown in (5.32). This does not impact the current work (which was parsed with `stanza`), but it does further motivate future work to extend the pattern specifications to include desired indirect structures.

(5.32) It was not very impressive.



5.2.4 Mirror Subset Patterns

The patterns described above—the superset polarity approximation approach—identify *some* negated bigrams. Although post-processing does ensure that unidentified negation cases—those with indirect dependencies to the bigram—do not count toward the positive polarity cases, that processing does not include cases where the trigger itself is more difficult to identify. That is, the posited weak NPI triggers of questions, comparatives (including *before*-clauses), and hypotheticals (*if*-clauses) were not identified using the current approach.

As discussed in [Chapter 2](#), an alternate polarity approximation method was introduced to account for this discrepancy, where active pattern matches are used

¹⁵³*was* -[acomp]-> *impressive*

¹⁵⁴*was* -[neg]-> *not*

to designate positive polarity environments. The *mirror* patterns are essentially the same as the patterns used to designate direct negation dependencies, but differ in that (a) there are 2 sets—one for each polarity—and (b) the pattern specifications employ **WITHOUT** clauses to (attempt to) avoid matching cases with *both* types of triggers present. The negative mirror triggers are limited to those with a clear “positive” counterpart, and the positive mirror triggers are defined as the aforementioned counterparts. Full lists of these are provided in [Chapter 2](#).

Table 5.1: Summary of Patterns

category (directory)	description	text representation	*.pat file(s) ¹⁵⁵	example
RBXadj	bigram only pattern	ADV ADJ	rb-bigram	<i>particularly important</i>
RBdirect	ADJ dependency with superset of negative triggers	NEG * ADV ADJ	direct-adj-head direct-neg-head	<i>not... , none of them were... , nothing...</i>
NEGmirror	ADJ dependency with mirror subset of negative triggers	NEG * ADV ADJ	neg-mirror-L neg-mirror-R	<i>none of them were... , nothing...</i>
POSmirror	ADJ dependency with (mirror) positive triggers	MIR * ADV ADJ	pos-mirror-L pos-mirror-R	<i>some of them were... , something...</i>

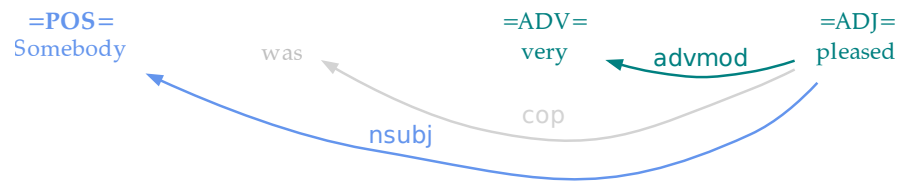
In the mirror corpus query approach, the patterns directly identify positive trigger elements preceding the bigram token. The positive triggers selected are all linguistic operators that can be interpreted as semantic “mirrors” of negative triggers. That is, they have generally the same syntactic restrictions, categorical and semantic selection requirements, and are used in a similar manner to convey a “positive” *existence*

semantics—rather than a “negative” *privation* semantics—for the bigram in question. For example, *somebody* and *everybody* are positive *mirrors* of the negation trigger, *nobody*.

(5.33) “MIRRORED” SENTENCES

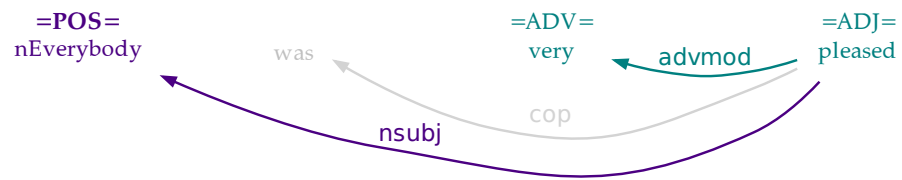
a. Somebody: $\exists x$

Somebody was very pleased.



b. Everybody: $\forall x$

Everybody was very pleased.



c. Nobody: $\neg\exists x$

Nobody was very pleased.

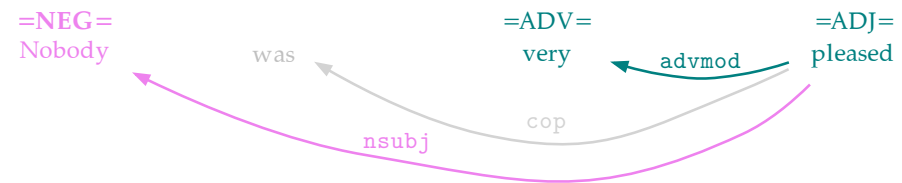
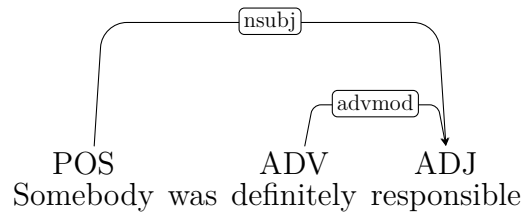


Figure 5.1: Bigram with positive trigger, *somebody*, present

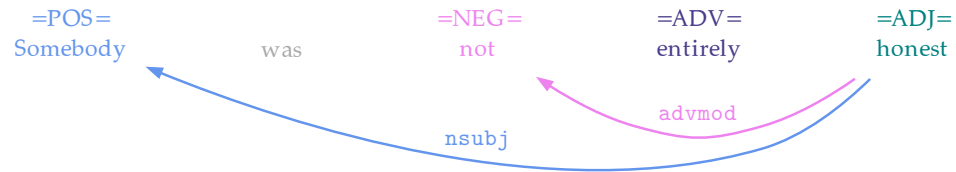


This increases the accuracy of polarity status evaluation, because you can inherently *know you found something* with more certainty than you can ever *know your **failure to find something** means it isn't there.*

In addition to this direct (rather than indirect) identification of positive polarity environments, the mirror patterns, in both their positive and negative instantiations, also employ **without** clauses to prohibit other identified negation patterns occurring elsewhere in the environment. In addition to positive search terms (what *includes* a token in the match set), they also include negative search terms (what *precludes* a token from the match set).

For example, the pattern for a direct dependency between the positive environment trigger indicates the permitted positive trigger lemmas, and what kind of dependency relation must exist with the bigram. However, it also excludes any token where a negation trigger is also found in a prohibited relationship with the given bigram token. The sentence below matches the existence search term, with the **nsubj** dependency relation between *honest* and *somebody* would fail to match the positive mirror pattern due to the conflicting **advmod** relation between *honest* and *not*.

(5.34) *Somebody was not entirely honest*



The mirror patterns also restrict the triggers to only those with a corresponding trigger of the opposite polarity. That is, they also impose mirrored limitations on the triggers selected as representing each polarity. With the inferred positive polarity method, any and all negation triggers were sought, in all possible syntactic configurations where they might meaningfully appear¹⁵⁶.

The primary pattern specification—sans the aforementioned **without** clauses—for mirror tokens is shown below. Full pattern specifications can be found in [Appendix D](#)¹⁵⁷.

¹⁵⁶or that was the intended goal anyway—in reality, the current data represents only those identified in a direct dependency relation with the bigram

¹⁵⁷The indicated pattern specification .pat files are also available in the **sanpi** GitHub repository—<https://github.com/AndreaTheLinguist/sanpi>.

1. pos-mirror-R.pat:

```
pattern {
  ADJ [xpos=re"JJ.?"];
  ADV [xpos=re"RB.?"];
  ADV < ADJ;
  mod: ADJ -[advmod]-> ADV;
  MIR [lemma="somebody"|"someone"|"everyone"|"everybody"|"something"
       |"everything"|"all"|"many"|"some"|"every"|"always"|"sometimes"
       |"often"|"either"|"both"|"or"];
  mir: ADJ -[~advcl|amod|discourse|prep|obl|obl:npm|parataxis
        |dislocated|rel|nsubjpass|attr|punct|csubj|ccomp
        |reparandum|obj|compound|expl|partmod|dep|conj]-> MIR;
  MIR << ADV
}
```

2. neg-mirror-R.pat:

```
pattern {
  ADJ [xpos=re"JJ.?"];
  ADV [xpos=re"RB.?"];
  ADV < ADJ;
  mod: ADJ -[advmod]-> ADV;
  NEG [lemma="nobody"|"nothing"|"no"|"none"|"never"|"rarely"
       |"seldom"|"hardly"|"scarcely"|"barely"|"neither"|"nor"];
  neg: ADJ -[~advcl|amod|discourse|prep|obl|obl:npm|parataxis
        |dislocated|rel|nsubjpass|attr|punct|csubj|ccomp
        |reparandum|obj|compound|expl|partmod|dep|conj]-> NEG;
  NEG << ADV
}
```

Literal 5.4: Bigram Governed Patterns for Mirror Subset

1. pos-mirror-L.pat:

```
pattern {
  ADJ [xpos=re"JJ.?"];
  ADV [xpos=re"RB.?"];
  ADV < ADJ;
  mod: ADJ -[advmod]-> ADV;
  MIR [lemma="somebody"|"someone"|"everyone"|"everybody"|"something"
       |"everything"|"all"|"many"|"some"|"every"|"always"|"sometimes"
       |"often"|"either"|"both"|"or"];
  mir: MIR -[~acl:relcl|advcl|appos|conj|nmod|parataxis|acl|obl
        |advmod|acomp|nsubj|attr|dobj|list|partmod|prep]-> ADV;
  MIR << ADV
}
```

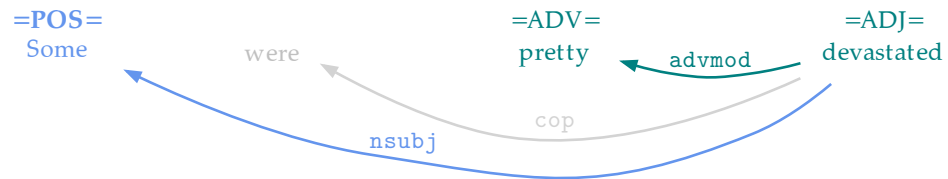
2. neg-mirror-L.pat:

```
pattern {
  ADJ [xpos=re"JJ.?"];
  ADV [xpos=re"RB.?"];
  ADV < ADJ;
  mod: ADJ -[advmod]-> ADV;
  NEG [lemma="nobody"|"nothing"|"no"|"none"|"never"|"rarely"
       |"seldom"|"hardly"|"scarcely"|"barely"|"neither"|"nor"];
  neg: NEG -[~acl:relcl|advcl|appos|conj|nmod|parataxis|acl|obl
        |advmod|acomp|nsubj|attr|dobj|list|partmod|prep]-> ADV;
  NEG << ADV
}
```

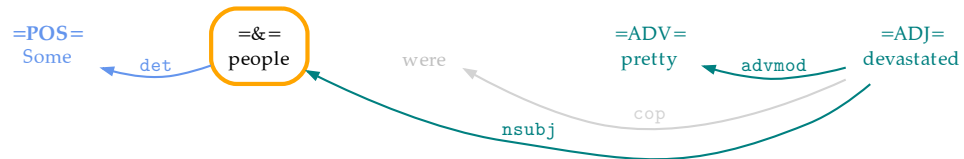
Literal 5.5: Trigger Governed Patterns for Mirror Subset

5.2.5 Examples of Positive Trigger Dependency Structures

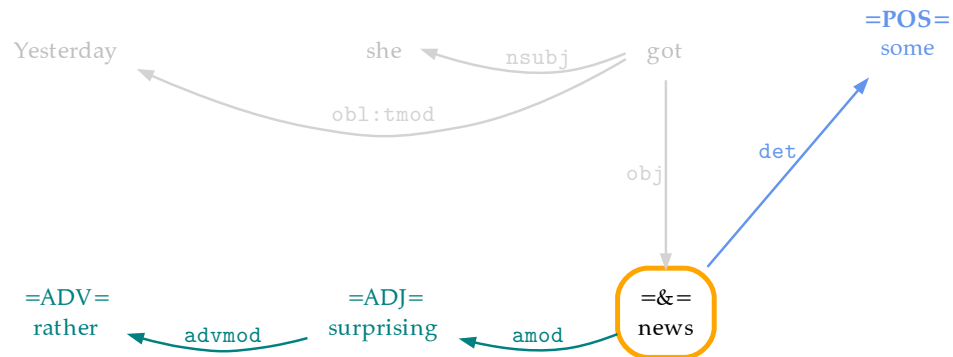
(5.35) *Some were pretty devastated.*



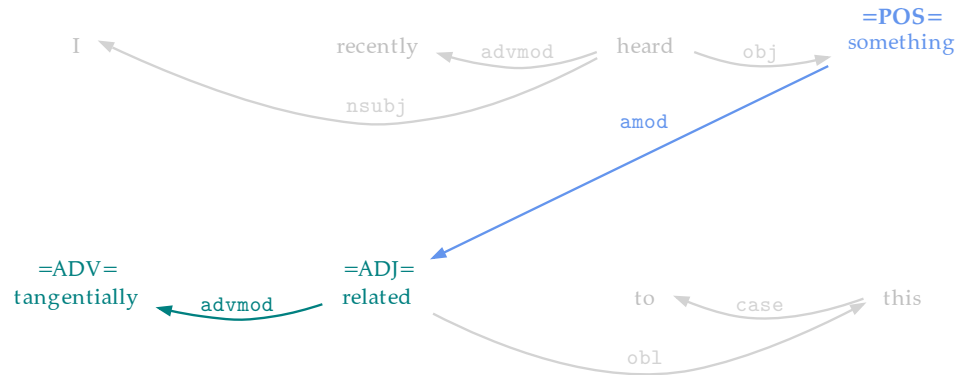
(5.36) *Some people were pretty devastated.*



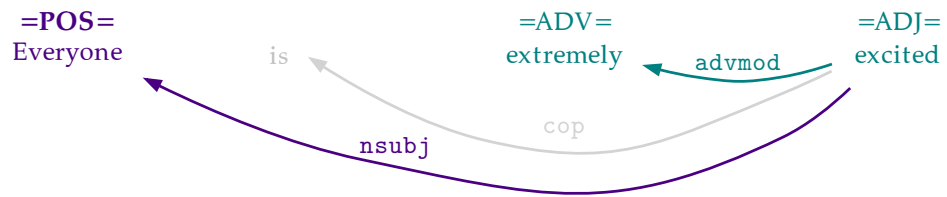
(5.37) *Yesterday she got some rather surprising news.*



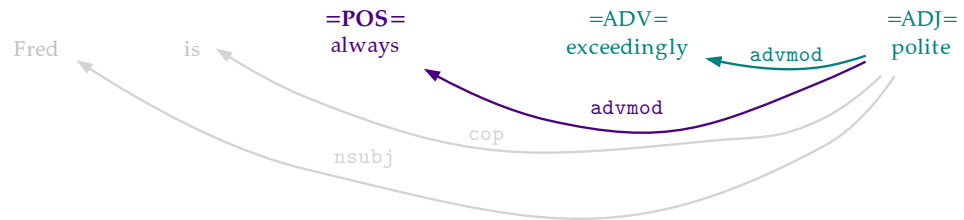
(5.38) *I heard something tangentially related to this today.*



(5.39) *Everyone is extremely excited.*



(5.40) *Fred is always exceedingly polite.*



5.2.5.1 Prohibited and Quarantined Dependency Relation Types

Each pattern specification directly indicates dependency relation types which are prohibited for the trigger dependency. In addition to these strict exclusions, later

Table 5.2: Quarantined and Prohibited Dependency Relations by Head

severity	left (<i>trigger-governed</i>)	right (<i>adjective-governed</i>)
quarantine	acl, acomp, advmod, attr, dobj, list, nsubj, obl, partmod, prep	attr, ccomp, compound, conj csubj, dep, expl, nsubjpass, obj, partmod, punct, reparandum
prohibit	acl:relcl, advcl, appos, conj, nmod, parataxis	advcl, amod, discourse, dislocated, obl, obl:npmmod, parataxis, prep, rel

post-processing filters—discussed in [Chapter 4](#)—are applied based on *quarantined* relations. These are specific to each pattern specification and are listed in [Table 5.2](#).

5.3 Gather and Consolidate Initial Hit Information

Once the search area is delineated and patterns have been specified to identify relevant linguistic tokens, the next step is to collect the actual tokens and do some general pre-processing and consolidation of the data. The python program `run_pipeline.py`, is a “glue” script which runs all of the following parts—defined in their own modules—together for the specified data and query parameters.

5.3.1 Identify Pattern Matches in Selected Corpora

The first python module run is `grew_search.py`, which uses `grew` commands to search for pattern matches in `.conllu` files, given the name of a subdirectory in `./Pat/` which contains one or more `.pat` file(s) specifying a `pattern` in `grew` syntax; this string (name of directory) is the pattern `category`, and the path to corpus directory containing `.conllu` files; the name of the directory is the `corpus part`. It uses `multiprocessing` module to search these files in parallel, and yields one `.raw.json` file per searched `.conllu` file per `.pat` file. The file output directory can be specified, but defaults to `'[DATA_DIR]/sanpi/1_json_grew-matches/'`.

Files are organized using input path identifiers. Specifically, the top subdirectory name—for all the outputs of the current job—is the same as the pattern category (the directory name given as an argument)¹⁵⁸. The next level subdirectory name follows the template: `[CORPUS_PART].[PATTERN_STEM]`, with the parts defined as follows.

- (5.41) a. `CORPUS_PART`: the corpus part (corpus directory name containing the `.conllu` files being searched)¹⁵⁹ The subset `.conllu` files themselves are prepended with a subset identifier (e.g., `BIGRAM.`) ensuring the source can be correctly identified.

¹⁵⁸The pattern category can include more than 1 pattern (`.pat` file), and this top level output directory will contain the hits for all of them.

¹⁵⁹Subset corpus files are saved in a subdirectory within their *superset* directory. The `CORPUS_PART` string for these files is still defined as the superset directory name.

- b. `PATTERN_STEM`: the filestem of the pattern specification file (`.pat`)

The output file(s) are then given the same stem as the `.conllu` file that was searched (i.e. the source of the match info contained in the file), with the suffixes, `.raw.json`.

```
[DATA_DIR]
... 1_json_grew-matches
    ... [PATTERN_CATEGORY]
        ... [CORPUS_PART].[PATTERN_STEM]
            ... [SOURCE_CORPUS_FILE].raw.json
```

Literal 5.6: `DATA_DIR` template structure

For example, given a pattern category of `RBXadj` (selecting directory `./Pat/RBXadj/`, which contains the file `rb-bigram.pat`), and the path to the corpus directory, `../puddin/PccVa.conll/` contains the subdirectory `subset_bigram/`, containing the files `BIGRAM.pcc_eng_val-01.conllu`, `BIGRAM.pcc_eng_val-02.conllu`, and `BIGRAM.pcc_eng_val-03.conllu`, the `grew` pattern match results are saved with the following structure shown in [Literal 5.7](#).

```
1_json_grew-matches/
...RBXadj/
    ...bigram-PccVa.rb-bigram/
        ...BIGRAM.pcc_eng_val-02.raw.json
        ...BIGRAM.pcc_eng_val-01.raw.json
        ...BIGRAM.pcc_eng_val-03.raw.json
```

Literal 5.7: Initial `grew` Results Directory Structure

The information collected by the `grew` included identifiers for sentence, document,

nodes, and ties¹⁶⁰. That is, it points to specific locations in the searched `.conllu` file, but it does not include any language data. It says *where* the hit is, but not *what* the match is.

5.3.2 Retrieve Language Data for Hits

This step uses the location information provided by `grew` to collect lexical information for each hit. The python script, `fill_match_info.py`, uses the `pyconll` and `json` modules to pull the desired language data for each hit identified by `grew_search.py`, given the path to directory where `raw.json` files indicating hit locations are stored, and the path to directory housing source `.conllu` files. Output directory defaults to the same as the input directory, but can be specified as something else. Output filenames are the same as the input `.raw.json` file with `.raw` removed. Continuing the sample above, this yields the structure shown in Literal 5.8.

```
1_json_grew-matches
...RBXadj
...bigram-PccVa.rb-bigram
...BIGRAM.pcc_eng_val-01.json
...BIGRAM.pcc_eng_val-01.raw.json
...BIGRAM.pcc_eng_val-02.json
...BIGRAM.pcc_eng_val-02.raw.json
...BIGRAM.pcc_eng_val-03.json
...BIGRAM.pcc_eng_val-03.raw.json
```

Literal 5.8: Updated Output Directory

¹⁶⁰`grew` refers to these internally as “edges”, but I feel this term misrepresents the structural relationships in the given conceptual metaphor

This step yields one `.json` file per `.raw.json` file in given `1_json_grew-matches/` subdirectory. At this point in the data gathering phase, there has been no compilation of data across source corpus files. All hit data is separated by source file and pattern.

5.3.3 Tabulate Hit Information

The python script, `tabulate_hits.py`, uses `pandas` to begin compiling the data collected (and stored in `.json` format) into a tabular representation, given the path to a directory containing `.json` file(s) with filled/processed hit information. This module performs two functions: For each pattern, **compile results** from separate corpus (`.conllu`) files into single table for each corpus part (directory). The output directory for this module defaults to `/share/compling/data/sanpi/2_hit_tables/`. The script yields one `pkl.gz` (+ other formats) dataframe file per subcorpus directory—35 `*_hits.pkl.gz` files of varying sizes. For example, the directories `bigram-PccVa.rb-bigram/(*.json)` and `bigram-PccVa.rb-bigram/(*.json)` containing processed `.json` files with full language data for each hit are re-configured as tables and saved as `bigram-PccTe_rb-bigram_hits.pkl.gz` and `bigram-PccVa_rb-bigram_hits.pkl.gz` respectively in this step.

Because data has now been compiled across different sources, it is also possible at this point to **drop duplicate matches** within each corpus part. That is, some documents and sentences are repeated verbatim within a corpus. Only one instance

of each sentence text (and any associated bigram tokens) is retained.

All flagged duplicate hits are saved separately and in the original `1_json_grew-matches/` subdirectory, with one `[STEM].duplicates.csv` table per input `.json` file. The resulting file structure¹⁶¹ is shown in Literal 5.9

```
[DATA_DIR]
... 1_json_grew-matches
... RBXadj
... bigram-PccTe.rb-bigram
... BIGRAM.pcc_eng_test-01.duplicates.csv
... BIGRAM.pcc_eng_test-01.json
... BIGRAM.pcc_eng_test-02.duplicates.csv
... BIGRAM.pcc_eng_test-02.json
... BIGRAM.pcc_eng_test-03.duplicates.csv
... BIGRAM.pcc_eng_test-03.json
... bigram-PccVa.rb-bigram
... BIGRAM.pcc_eng_val-01.duplicates.csv
... BIGRAM.pcc_eng_val-01.json
... BIGRAM.pcc_eng_val-02.duplicates.csv
... BIGRAM.pcc_eng_val-02.json
... BIGRAM.pcc_eng_val-03.duplicates.csv
... BIGRAM.pcc_eng_val-03.json
... 2_hit_tables
... RBXadj
... bigram-PccTe.rb-bigram_hits.pkl.gz
... bigram-PccVa.rb-bigram_hits.pkl.gz
```

Literal 5.9: Final File Structure for Initial Processing

¹⁶¹*.raw.json files not included for clarity, but they are not removed by `tabulate_hits.py`

5.3.4 Prohibit Bigram “Double-Dipping”

This does not apply to the general context-blind bigram pattern, but for any pattern which contains a trigger node, additional steps are required to ensure no bigram token is accounted for more than once. Immediately after retrieval, such patterns are limited to only one hit per bigram (one hit per potential trigger node found per bigram): Only the nearest trigger is kept, ignoring the token farther from the bigram. To illustrate, the following sentences in (5.42) match the direct negative dependency pattern (RBdirect) twice.

(5.42) sentences matching the direct negative dependency pattern (RBdirect) twice

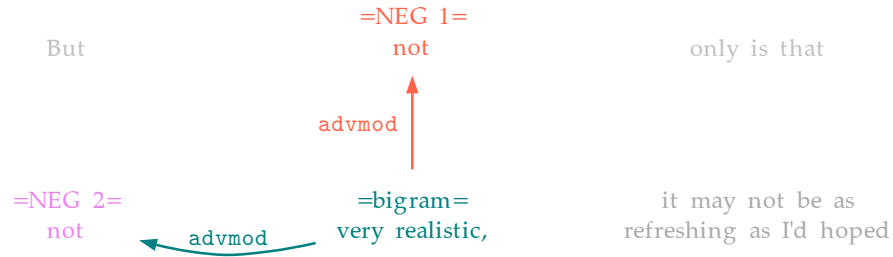
a. *Not* that we weren't still uncomfortable¹⁶²



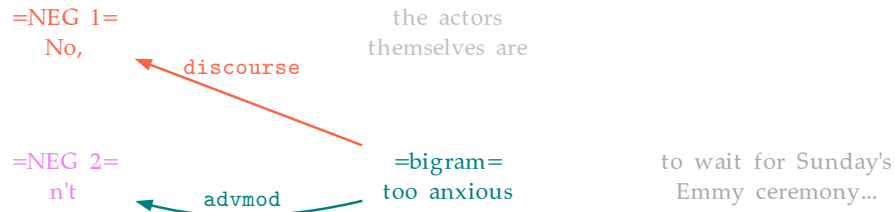
b. *But not only is that not very realistic, it may not be as refreshing as I'd hoped*¹⁶³

¹⁶²hit_id='pcc_eng_14_107.04938_x1721388_132:1-6-7'

¹⁶³hit_id='pcc_eng_11_108.0163_x1731235_46:2-7-8'



- c. *No*, the actors themselves aren't too anxious to wait for Sunday's Emmy ceremony – but their places are being held by large cardboard cards that bear their photos.



Although these examples differ in the polarity environment they create for the target bigram, their dependency structure and *collected* forms are near identical. If such cases were to be teased apart, it would require assessment for the formulaic structures used, namely “not that x is not y ” and “not only is x not y ”. At the point when this processing decision was made, these lexical patterns had not been identified, and there are many cases flagged as duplicate triggers on the same bigram token that are simply due to errors (in production/transcription or parsing¹⁶⁴). Therefore, the decision was made to keep the hit with the nearer trigger node, rather than throw out all bigram tokens with multiple triggers as “double negation”. Dealing

¹⁶⁴In fact, parsing some of the flagged cases at a later date—with updated parsing software/language models—yielded dependency structures with no overlap.

with these cases in a more nuanced fashion could be a future improvement to the data collection process.

5.3.5 Interim Summary

At the completion of the gather phase of the pipeline, all data selected for evaluation is contained in (up to) 35 dataframes (tables) per pattern specification (i.e., target structure/dependency configuration)¹⁶⁵, stored in compressed pickle format (file suffix `.pkl.gz`). For the context-blind bigram baseline, this results in 35 separate tables always, but for the triggered pattern categories will have *up to* 35 tables. The context-blind bigram-only collection is restricted to only 1 dependency shape, and therefore only one pattern specification. The triggered/context-seeking pattern categories, on the other hand, have two patterns each: The direction of governance in the dependency between trigger node and adjective node is not fixed as it is for the bigram relationship. The adjective may be the head of this dependency (the more common case by far) or the trigger may be the head.

¹⁶⁵This includes “bigram only” dataframes, which might appear redundant at first glance, but which (a) are necessary for normalizing frequencies and deducing the “non-negated” results and also (b) might differ from the entire `subset_bigram` compilation due to an additional part of speech restriction (`ADV` node was restricted to adverb part of speech only, where it was previously defined only by immediate precedence of `ADJ` with an `advmod` dependency relation) added to the “all bigram tokens” pattern after the subset’s creation. The time and resources required to generate the subset outweighed the inconsistency between the subset corpus files and the resultant “all bigram” hits data.


```

> 87,711,277 starting hits
= unique forms in initial hits
* ADV_FORM      :    72,396
* ADV_FORM_LOWER :    64,498
* ADJ_FORM      :   298,923
* ADJ_FORM_LOWER :   278,732

```

Literal 5.10: Unique Initial Counts Log Message

Table 5.3: **Adverb** Count Comparisons

	adv_form	adv_lemma	adv_form_lower
count	72,396	64,448	64,498
mean	1,211.5	1,361	1,359.9
std	69,462.5	77,087.4	75,877.3
min	1	1	1
25%	1	1	1
50%	1	1	1
75%	3	2	2
max	9,840,502	10,165,940	10,165,939

Table 5.4: **Adjective** Count Comparisons

	adj_form	adj_lemma	adj_form_lower
count	298,923	283,021	278,732
mean	293.4	309.9	314.7
std	10,766.8	12,015.2	11,249.4
min	1	1	1
25%	1	1	1
50%	1	1	1
75%	3	3	3
max	2,224,082	2,957,797	2,236,489

5.4 Post-Processing Bigram Tokens

After hits have been identified, relevant linguistic information has been retrieved, and all have been consolidated into tabular-formatted data files¹⁶⁶, additional post-processing steps are taken to increase representation accuracy of the collection. Some of these adjustments are methods that could have been applied during the initial retrieval process, but were applied separately as an artifact of the development process.

All cleaning, filtering, and other removal steps that apply regardless of pattern/context specifics are applied only once, to the all-bigrams/bigram only/baseline/unspecified context data(super)set (e.g. anything that applies based on the sentence text or bigram token values). Then the exclusions/filters are applied to the corresponding data points in the context-specific pattern hits¹⁶⁷.

This is done for three main reasons. First, applying these operations redundantly would be a waste of resources. Second, some operations cannot be completed successfully without having *all* relevant data present, and all of the context specific sets are subsets of the bigram superset. Finally, applying the operation only once increases consistency in the resultant exclusions.

¹⁶⁶such that there is one “hit table” per pattern per corpus part—35 separate tables for the context-blind bigrams, and up to 70 tables for each triggered pattern category.

¹⁶⁷One exception to this is the preliminary removal of duplicate sentences within a single corpus part. Because this is performed in the initial tabulation step, it minimizes the size and memory required for that initial tabular file: the hits for theses within-directory duplicate sentences are never saved in a tabular format, and thus never impact subsequent file requirements (e.g. storage space, working memory)

The post-processing applied is split into two parts: There was an initial sweep by the python program `count_bigrams.py` which applied to all 35 parts combined.

(5.43) CASE-NORMALIZED EXAMPLE OF “SENTENCE” WITH 250+ TOKENS

So I want you to keep that in mind and that's why lighting shouldn't be scary it doesn't have to be a formula it doesn't have to be a grid has to be you feel it and it becomes part of the whole shoot so what I want to encourage it is keep learning all the rest of it because this is just one pretty big part which is nice because that's a posing intimidates you well you've got the lighting down because you practice and now he can focus on that posing and then you get the posing down then you're not worried about the lighting or the posing

This initial method required a prohibitive amount of memory—the successful job requested 115 gigabytes, and many failed due to insufficient memory prior to that—so when the decision was made to apply more stringent cleaning methods, it was designed such that anything that could be applied in smaller chunks would be. Thus a second program devoted to post-processing bigram tokens was developed, `clean_bigrams_by_part.py`, which applied the filters described below to separate chunks of each of the 35 hit tables (1 per corpus part).

5.4.1 Orthographic Normalization: Filtration and Manipulation

Although orthographic variation could be studied, it is not the target of the current project. Therefore, it acts only as a barrier to identifying patterns impervious to

spelling choices or errors. In this section, I describe the different ways in which the string information collected for individual bigram tokens is manipulated and evaluated in order to minimize orthographic interference. This primarily involves case normalization, selection of word type representations, inferred standard forms from substandard, removal of extraneous punctuation characters, and filtration based on word types with abnormal characters.

5.4.1.1 Word Type Representations: Case-Normalized Forms

The default representations supplied by the corpora and collected in the initial hit tables are the literal *form* and the *lemma* assigned to each form. An example of these is provided in [Table 5.5](#).

Table 5.5: Distinction Between Form and Lemma

hit_id	nyt_eng_20040505_0023_33:7-8-9
adv_lemma	that
adv_form	“THAT”
adj_lemma	expensive
adj_form	“expensive”
sent_text	(<i>Jackman: “I’m not THAT expensive.”</i>)

These parameters bear distinct information, but also distinct problems. The form contains the full variation of orthographic choice and error, making it less useful as a classifier, but the lemma obscures morphologically supplied distinctions and—in the corpora searched—is inconsistent, making it unreliable and potentially misleading. Therefore, a (lower) **case normalized** form attribute is used as the word

type parameter for each node, as illustrated by the added *_form_lower rows below. With the forms all converted to lower case, the case variations, e.g., **that**, **THAT**, and **That**, are normalized and counted as the same adverb word type. Note that this adaptation alone is not sufficient to group other orthographic variations, e.g. **thaaaat**, with **that**. Attempts to account for this type of orthographic style are discussed in a later section.

Table 5.6: Case-Normalized Forms

hit_id	nyt_eng_20040505_0023_33:7-8-9
adv_form_lower	<i>that</i>
adj_form_lower	<i>expensive</i>

Importantly, this means that morphological derivations will be treated separately from their stems. For example, tokens with the case-normalized form, *better*, including the substandard forms of **Better**, **BETTER**, etc., are treated separately from tokens with the (lower) form **good**, or **best**, even though they *should* all have the lemma, **good**.

Table 5.7 hints at the inconsistency of lemma assignment found in the parses, which is part of the reasoning for using a case-normalized form parameter for grouping tokens, rather than the lemma. In addition to the unreliability of the lemma assignment, PBR is intended for use in semantic investigations, which are strongly influenced by scalarity (i.e. polarity sensitivity). The primary morphological difference seen across adjectives are the comparative and superlative morphemes. There will still be cases where the adjective form will not indicate

Table 5.7: Examples of Inconsistent Lemmatization of *Better*

lowercase form	lemma assigned	text_window
better	better	<i>our finishes might not have been any better than they were</i>
		<i>life ca n't get much better for Martin Truex</i>
	good	<i>work conditions are n't usually any better .</i>
		<i>the momentum could n't be any better for us .</i>
		<i>second half was n't much better , and Ike</i>
		<i>county office were not much better , either ,</i>
		<i>, which is n't much better .</i>

Table 5.8: Degree Modification of the lemma, *good*

	<i>less</i>	<i>extremely</i>	ADJ form
The cake was. . .	✓	✓	<i>good</i>
	??	??	<i>better</i>
	#	#	<i>best</i>

its comparative/superlative status, because it is instead modified by *more/most*, but in assessing form rather than lemma, this status will remain transparent for synthetic cases. Such derivations actually result in a different scalar semantics, which Table 5.8 previews: the degree argument is saturated by the comparative or superlative, making it infelicitous with further degree modification¹⁶⁸.

¹⁶⁸This scalar saturation also means that privation of comparative or superlative forms is not ambiguous, and does not take part in negative strengthening phenomena.

5.4.1.2 Substandard Orthography Adjustment and Exclusion

In addition to case-normalization, attempts were made to account for other unexpected characters within the adverb and adjective type representations. A first sweep of this targeted the *lemma* string: Any hits with an adjective or adverb *lemma* containing non-word characters were dropped, as described in more detail below. A total of 48,340 hits were removed in this first sweep due to abnormal lemma orthography.

That is, hits with adverb or adjective lemmas matching any of the criteria in (5.44) were deemed abnormal and excluded.

- (5.44)
- a. start with -, &, or .
 - b. end with - or &
 - c. or contains any character **not** in the set
{-, &, a-z, A-Z, 0-9, _, .} which is followed by a regular letter
[a-zA-Z])

This lemma-based filter was applied at an earlier stage in development, when the cleaning methods were less stringent. At that point, the primary filtering mechanism was based on word frequency, and many of the irregular forms were automatically excluded due to infrequency. Later in development, issues arising from irregular forms were identified and the cleaning methods were increased in strength and number. This included orthographic evaluation of the `ad?_form_lower` values, in addition to lemma, though the lemma filters were implicitly incorporated via the use of existing `hit_id` index files. Instead of simply using the forms as selection criteria, string

manipulations were also performed. This included the translation of certain known alternate spellings: specifically, variations on *very*, *ok*, *especially*, and *definitely* as indicated below.

The most numerous change due to this set of filters was the merging of the adjective forms “*okay*” and “*ok*”, and it was actually the observation of variations of “*ok*” which was the impetus for this method. Although this list is (clearly) far from conclusive—e.g., the example suggested in previous section, “*thaaat*” is not normalized to **that**—few if any of the other targeted words had alternate spellings occurring with enough tokens to be noticeable in their own right, meaning the impact on their totals is likely minimal at most. A more thorough treatment of such substandard cases might be attempted in future work¹⁶⁹. In addition to inferred standard form adjustments, any non-alphanumeric characters at word boundaries are trimmed. This includes punctuation used as emphasis markers or formatting syntax (e.g. ‘***very***’ to ‘**very**’), and cases of word-adjacent punctuation not successfully identified and tokenized by the word segmentation of the source corpora.

¹⁶⁹While abbreviations need to be hard-coded, the use of duplicated letters as an indicator of prosodic emphasis/focus (e.g. **r’v+e+r+y+** → “*verrrry*”) could be anticipated and applied more broadly. In a different vein, more complex methods relying on machine learning or natural language processing models could be employed to deal with the indicated variations. Although, it is also possible that all such orthography based cleaning mechanisms might become obsolete if more advanced language models lead to more accurate parsing and lemmatization of the corpora data itself, since such improvements might provide lemmas and derivational morphology tags (i.e. indicating comparative, superlative, etc.) that are reliable enough to use in place of the lowercase-normalized forms used here.

After these string manipulations have been made, salvaging cases that might otherwise be lost, any remaining hits with adverb or adjective representations containing orthographic abnormalities are excluded from the dataset. This includes any `ad*_form_lower` that contains no alphabetic characters/letters (e.g. `more [...]` is parsed as a bigram multiple times with “`/.../`” as the supposed adjective), or any entirely unexpected characters, such as `=`, `\`, `@`, etc. Although single-character words do exist in English, they are functional, and not used as adverbs or adjectives; thus, any single-character forms (even single alphabetic characters) are excluded as erroneous parses. As a final regular expression assessment, any hit was dropped if its `bigram_lower` string matched the pattern in Literal 5.11.

```
MISC_ORTH_REGEX = re.compile(
    (r"(?P<IrregularChar>[^\a-z0-9_'\-])"
     r"| (?P<MedialNumeral>[^\d_]+\d[^\d_]+)"
     r"| (?P<MultiHyphen>-(?:[^\-]+-){3,})"),
    re.IGNORECASE)
```

Literal 5.11: Odd Orthography Remnants

Only alphanumeric characters, hyphens, and apostrophes¹⁷⁰ are permitted at all, and some additional filters are applied to these based on placement¹⁷¹. As already mentioned, any `-` or `'` characters at the edge will be removed, but word medial cases

¹⁷⁰i.e. `r"[[a-z][A-Z][0-9]'\-]"`

¹⁷¹This means that “`.`” included for the purpose of abbreviation (e.g. `p.c.`, `l.a.-based`, `u.s.-trained`), and “`&`” used in compounds (e.g., `r&b`, `s&m`, `b&b`, `r&d`), and accented letters in borrowings (e.g., `naïve`, `naïvely`, `puréed`, `sautéd`, `risqué`, `passé`, `naïve`, `blasé`) are also excluded, an outcome that may be more controversial. The filtration could be modified to cover such cases in the future, but for the current work, they are infrequent enough that the impact is minimal.

are unchanged. Any forms with numeric characters fully embedded—word medial—will be flagged for exclusion. Numeric characters are only permitted at the beginning or end; e.g. *1st*, *3d*, *6-foot-11*, *2-to-1*, etc. are accepted, but *sym0athetic*, *conducibe*, *a0vigilant*, *repor4ezh*, *m5ca*¹⁷², etc. are prohibited, even if a human reading them in context could figure them out (e.g. *sympathetic*, *conductive*, *vigilant*, etc.). Finally, this pattern also matches any form with four or more hyphens.

Both clear-cut and questionable hyphenation exclusions are exemplified below. While some excluded cases are certainly parsable as complex adverbs or adjectives, it is unclear whether they *should* be treated the same as simple cases. Regardless, they are so rare—almost entirely single-occurrence forms—that their exclusion is negligible. Bigram forms excluded due to excessive hyphenation are shown below in (5.45)–(5.47).

(5.45) Obvious Error Cases

- a. *now* 0-for-june-july-august-september
- b. *also* sci-body-parts-list-nyt
- c. gon-suh-lee-no al-may-duh
- d. *as* bc-michael-ovitz-newsweek-nytsf

(5.46) Creative Phrase-as-a-Modifier Adverbs

- a. beat-you-over-the-head *bad*
- b. spit-out-the-spoilt-milk *bad*
- c. three-victories-in-16-games *bad*
- d. fists-pumping-in-football-stadiums *big*
- e. still-waiting-for-the-punch-line *bizarre*
- f. is-he-genuinely-losing-it *crazy*
- g. three-fries-short-of-a-happy-meal *dumb*
- h. stick-to-your-ribs french-canadian

¹⁷²All in this list were observed forms which led to hit exclusion

- i. snort-root-beer-out-your-nose *funny*
- j. get-rid-of-the-coach *happy*
- k. right-out-of-the-oven *hot*
- l. start-sweating-the-second-you-step-outside *hot*
- m. figure-it-out-as-you-go *nimble*
- n. captain-of-the-football-team *popular*
- o. rip-your-hair-out-at-night *difficult*
- p. you-could-hear-a-pin-drop *silent*
- q. anvil-dropping-on-the-head *obvious*
- r. too-many-men-on-the-ice *obvious*
- s. knock-you-off-your-stool *potent*
- t. close-your-eyes-and-think-of-england *prim*
- u. take-it-to-the-bank *solid*
- v. under-the-heat-lamp-too-long *stale*
- w. can-you-believe-my-match-last-ed-10-hours *tired*
- x. coyotes-dressed-as-mariachis-falling-out-of-a-pinata *weird*

(5.47) Creative Phrase-as-a-Modifier Adjectives

- a. *absolutely* throw-up-on-your-shoes
- b. *completely* sit-on-the-edge-of-your-seat
- c. *cringingly* all-for-one-and-one-for-all
- d. *just* i-hope-your-internet-goes-out-at-a-crucial-moment-and-on-your-way-to-a-nearby-coffee-shop-to-use-their-wi-fi-you-get-stuck-in-traffic
- e. *just* not-in-my-backyard-ism
- f. *just* super-double-ultra-plus-confirmed
- g. *more* lock-your-doors-and-pray-for-mercy
- h. *plain* black-cat-walking-under-a-ladder
- i. *presumably* music-centric-if-not-outright-musical
- j. *remotely* how-to-be-a-terrorist-sounding
- k. *seemingly* hard-wired-in-the-dna
- l. *so* duchess-of-windsor-on-acid
- m. *strikingly* civil-war-era-soldier-looking
- n. *strictly* fight-them-on-the-beaches
- o. *strictly* use-it-or-lose-it
- p. *vaguely* tea-dance-at-the-plaza

- q. *very* mark-zuckerberg-is-in-his-twenties-who-grew-up-in-a-world-where-you-want-to-be-liked
- r. *very* un-rock-and-roll-like

Additional cases in (5.48) show the use of hyphens to indicate emphasis. While these are also intelligible to a human, normalizing them reliably is less than straightforward, due to the variation employed, even within the bounds of an obvious strategy. In just these examples, the hyphen is used to separate letters, syllables, morphemes, and even less standard units, both with and without duplication of segments, and even one case of phrase hyphenation combined with spell-out orthography (i.e. *now w-a-y-t-o-o-l-o-n-g*) This strongly indicates that accurately reassigning such forms into standard orthography would be a complex affair.

- (5.48) Hyphenated emphasis strategies
- a. *even* re-re-re-re-revised¹⁷³
 - b. *simply* p-e-r-f-e-c-t
 - c. *so* s-t-r-e-t-c-h-y
 - d. *completely* a-w-e-s-o-m-e
 - e. *seriously* d-e-l-i-c-i-o-u-s
 - f. *so* q-u-i-e-t
 - g. *simply* r-e-j-e-c-t-e-d
 - h. *most* ex-as-per-at-ing
 - i. *simply* un-be-lie-va-ble
 - j. *de-fin-itely* be-a-utiful
 - k. *now* w-a-y-t-o-o-l-o-n-g
 - l. *already* fa-a-a-a-a-ast
 - m. *so* soo-ooo-ooo-rrrr-y

¹⁷³more emphasis in a morpho-semantic sense than prosodic

- n. *so de-e-e-e-eep*
- o. *more c-c-r-r-i-sply*
- p. *so b-i-i-i-g*
- q. *so hi-hi-hi-hi-hi-hi-high*

As a final more manual filter instigated by observed cases, any tokens with an adverb form in the following set were dropped: $\{ie, is, etc, th\}$.

If the specifics of these adjustment and exclusion filters do not speak for themselves, this treatment of abnormal orthography is motivated by more than concerns for “true word identity”. Firstly, such forms have a much higher probability of being from uninterpretable or otherwise misparsed sentences: misparsed language data adds noise, and noisy data leads to murky conclusions. Secondly, unexpected orthography also has greater potential for processing errors. As a developer, I attempt to infer as much as possible from the data, so that reliance on hard-coded expectations is minimized, but even so, permitting any and every kind of string increases the likelihood of something eventually breaking due to an unexpected case. Thirdly, even when the form did not come from a misparsed sentence, if unexpected orthography is present, the likelihood of accurately grouping the token with other like forms is greatly reduced—in addition to programmatic tools failing to categorize them, there is a possibility that a human hearer/reader might not be able to understand them either. And finally, the number of forms collected is already high enough that not only are many of these exclusions negligible on an individual basis, but limitations to the “view” are almost *required* to interpret results. Therefore, if forms need to be dropped anyway, reductions should start with cases already marked

due to other reasons.

5.4.2 Sentence Length Restrictions

In addition to filtration based on the adverb and adjective type representations, restrictions are placed on the sentence or utterance length as well. Applying the very conservative ceiling of **250 tokens per sentence** resulted in the removal of **3,651 hits** across **1,220 “sentences”** with 250+ tokens. This is measured in tokens—words or punctuation—rather than “words”, simply because it is a more accessible unit than “words”. Note that this filter does not include those hits that are dropped from context-specific categories based on the span length of the trigger-bigram dependency: although dependency spans are limited by the length of an utterance, they are not directly related.

(5.49) CASE-NORMALIZED EXCERPT OF “SENTENCE” EXCLUDED DUE TO LENGTH:

Yeah, so if you haven't seen that movie, jerry maguire has a girlfriend played by kelly preston and they're both sales people and they're like killer sales people and they're both very successful and then his star starts to decline and she's, like doesn't know she could be with him anymore because they're all about appearances being successful, having a deal sticking to the deal um and and then the numbness numbness to her own feelings it's so great that you point that out is associated with this third communication style and then w...

Sentence or utterance length restrictions follow similar motivations as those

cited above for orthographic-based filtration. Primarily, exceedingly long “sentence” strings are more likely to be segmentation errors. Once a certain length has been reached, the longer a string of text is, the less likely it is that the text string is actually only one single sentence rather than multiple which have not been clearly identified¹⁷⁴. If the fundamental segmentation has not been accurately identified, the probability for parse errors within the segmented “sentence” also increases, as the parser attempts to determine connections (dependencies) between tokens where none exist. Even if they are not due to segmentation errors—truly just a super long sentence—the likelihood of parse errors is still greater. There is a direct relationship between the number of tokens and the number of possible parses that *could* be assigned: more options leads to more opportunities to select the *wrong* option.

In addition to these data validity concerns, there is also the impact on storage and processing resources. The sentence string is stored with each hit found in it. Longer strings require more resources and increase file sizes more than shorter sentences. This translates into more storage space required, but also more working memory to access the contents of the data file, and more time to perform operations on any tables where the relevant column is included. On top of that, the longer the sentence, the more likely it will contain multiple hits, so the size issues are compounded for each additional hit that needs to store a long string.

¹⁷⁴Sentence segmentation errors are more likely when the text lacks punctuation (or employs punctuation in unexpected ways).

5.4.3 Duplication Exclusions

The final cleaning process performed on the basic bigram collection (i.e. bigram hits with no information on environment via trigger node collection) is the removal of duplicate hits. There are two criteria for duplicates evaluated:

1. case-sensitive *sentences*
2. case-neutralized text *windows* or the substring of the sentence text containing the bigram in question.

In both definitions, the bigram forms must be the same as well, since multiple distinct bigram tokens can arise in a single sentence, or even a single window, depending on length and position within the containing sentence.

The former definition, where duplicates are identified by case-sensitive sentences, is a proper subset of those caught by the latter filter, where duplication is based on case-normalized (insensitive) windows. That is, the sentence-based duplication filter is not explicitly required, since its effect is subsumed under the window-based assessment. This redundancy arose due to the nature of exploratory processing—or processing data without full knowledge of what said data truly contain. It was only after the weaker filter was applied that the stronger was motivated.

5.4.3.1 Case-Sensitive Sentence Duplicates

The weaker sentence filter was applied first (in both development and the processing pipeline), after identical corpus documents were observed: The New York Times corpus, for example, includes contains article reprints as well as the original publication, leading to entire documents within the corpus which are near verbatim.

Some sentence duplication can arise from naturally occurring repetition, and such cases are not the intended target of this cleaning step. The likelihood of a sentence being naturally generated *verbatim* decreases with length. Therefore, to avoid removing natural cases of repetition, duplicates of sentences—as represented by the `token_str` attribute for each hit—containing fewer than 10 tokens (words and punctuation marks) is permitted. See below for a more thorough discussion of motivations for this.

Some duplicate sentences were dropped during the gathering phase prior to completing data tabulation. Duplicate bigrams with duplicate sentences (identical `token_str` values) within a single `.conllu` file were identified and excluded with only that smallest selection of sentences loaded into memory at a time. In order to drop additional duplicate hits *across* all corpus parts, the relevant data must be accessible at the same time for comparison, meaning every hit must be loaded into memory to perform this operation. This makes it very expensive in terms of resources, which led to alternate approaches for future processing.

5.4.3.2 Case-Neutralized Window Duplicates

After additional processing and manual inspection, it became clear that a case-sensitive sentence filter was insufficient to catch all cases of replication that human evaluation can identify. The case-sensitive sentence comparisons discussed above do not capture cases of the following types listed in (5.50).

(5.50) Problem Cases

- a. All the tokens are the same, but the capitalization differs, (even if it's only one character)
- b. All the words are the same, but the punctuation is different, whether it's due to the absence of punctuation (e.g. one string has an Oxford comma and the other doesn't) or different punctuation marks (e.g. one uses single quotes '...', while the other uses double quotes "...", or something else, such as -LDQ-...-RDQ-)
- c. One sentence string is a match of a *portion* of the other, including but not limited to cases of: (a) quotation, where one sentence (intentionally) quotes the other, and (b) alternate sentence segmentation, where the same string is segmented as a different number of sentences (e.g. as a result of different line breaks in the original text, different formatting that was stripped during processing, etc.)
- d. Sentences have overlapping text spans, but both contain tokens the other does not. This could also be the result of word tokenization discrepancies or the replacement of one or more tokens with an alternate (e.g. synonym variation or a template sentence used with a different name, object, etc.)

Observations of such cases drove the implementation of stronger case-normalized window comparison: rather than comparing the full `token_str` values, the lower-

case `text_window` strings were compared for repetition, for hits with the same bigrams (i.e., `bigram_lower` strings). It is not a perfect solution, since the window could fail to capture the shared portion, but the higher the percentage of identical strings, the higher the probability that the window comparison will identify this partial duplication of the sentence. Similar filtering in the future could introduce “punctuation normalization” as well—or the removal of all punctuation tokens within the evaluated strings to prevent punctuation-only divergence. This would need to be applied prior to retrieving the text window, however, since otherwise the number of tokens included in the window could differ.

In addition to changing the hit attribute used for comparison, this filtration step employed a different method of implementation as well. As previously indicated, the resource cost of performing assessments on all hits simultaneously are prohibitive. Rather than load all data and check for duplication as was done before, these checks are applied at intervals, first removing any duplicates on a single chunk of a part, then on the full part, and finally on all of the parts together. This way, the size of the data needing to be loaded into memory at every subsequent filtering pass is minimized.

Similar to the duplicate sweep based on the full case-sensitive sentence, a minimum sentence length threshold must be met for the hit to qualify for duplicate removal; however, the requirement imposed is increased 20 tokens (from 10) for this evaluation. This increased length restriction is intended to safeguard against exclusion of common template phrases within naturally produced utterances where

the text windows line up perfectly.

5.4.3.2.1 Samples of exclusions based on lower-case text_windows despite non-identical token_str values. *Specific token differences within each token_str are indicated by ~~strikethrough~~ (indicating absence in comparison(s)) & underline (indicating alternative values in comparison(s)) formatting.*

WITHIN A SINGLE CORPUS DOCUMENT

(5.51) *wind are , in many places , **actually cheaper** than simply operating an existing coal*

The only distinction between these sentences is the presence of double-quotes surrounding it. This could result from variable sentence segmentation, or from a highlighted quote (e.g. in a separate text box within the page) which was scraped along with the full article text.

- a. pcc_eng_04_047.4209_x0750265_02(:27-28) [39 tokens]
*~~"~~ Significantly , the cost of renewables have dropped so sharply , that building and operating both solar and wind are , in many places , **actually cheaper** than simply operating an existing coal or nuclear plant . ~~"~~*
- b. pcc_eng_04_047.4209_x0750265_20(:26-27) [37 tokens]
*Significantly , the cost of renewables have dropped so sharply , that building and operating both solar and wind are , in many places , **actually cheaper** than simply operating an existing coal or nuclear plant .*

(5.52) *has made things **more convenient** for us **albeit secure** enough for us to innovate something*

The only difference here is the transitional phrase employed to introduce

the clause. Note, that this duplication contains two bigrams, *more convenient* and *albeit secure*, so the removal of the single duplicated sentence removes two bigrams.

- a. pcc_eng_04_042.5872_x0672000_23(:11-12) [25 tokens]
~~Nowadays~~ , technology has made things **more convenient** for us **albeit secure** enough for us to innovate something especially when using our tools alone .
- b. pcc_eng_04_042.5872_x0672000_30(:12-13) [26 tokens]
~~Because indeed~~ , technology has made things **more convenient** for us **albeit secure** enough for us to innovate something especially when using our tools alone .

(5.53) " these new chairs are **absolutely beautiful** , comfortable , practical , clean

In this case, one sentence string includes the source of the quote, whereas the other does not.

- a. pcc_eng_06_046.3353_x0733508_13:6-7 [24 tokens]
 " These new chairs are **absolutely beautiful** , comfortable , practical , clean , pretty - and no gum under the seats _ "
- b. pcc_eng_06_046.3353_x0733508_40:6-7 [27 tokens]
 " These new chairs are **absolutely beautiful** , comfortable , practical , clean , pretty - and no gum under the seats _ " *~~Tobin said.~~*

WITHIN A SINGLE CORPUS PART, BUT DIFFERENT DOCUMENTS

(5.54) *i am selling : the house is **actually unique** , it is actually **very special***

In this example, 2 pairs of words are swapped for synonyms, suggesting production was not fully original. Note, again, that this duplication contains two bigrams, *actually unique* and *very special*, so the removal of the single duplicated sentence removes two bigrams.

- a. pcc_eng_05_006.8698_x0095399_40:14-15 [22 tokens]
*I possess passion about the product I am selling : the house is **actually unique** , it is actually **very special** .*

- b. pcc_eng_05_067.2735_x1072195_43:14-15 [22 tokens]
*I have passion concerning the product I am selling : the house is **actually unique** , it is actually **very special** .*

(5.55) *this festival of candy and cupids are **actually dark** .*

Rather than simple word alternates, in this case the first conjunct clause is changed. Length limits were imposed to increase the likelihood that such cases of duplicate windows are indeed too specific to be original productions.

- a. pcc_eng_05_032.3461_x0507819_27(:26-27) [28 tokens]
~~*Choose from cakes , cookies , tarts , truffles and many other chocolate recipes*~~ but the origins of this festival of candy and cupids are **actually dark** .
- b. pcc_eng_05_107.02070_x1717812_02(:34-35) [36 tokens]
~~*A florist , who abides by a strict five - date - limit with any man , finds herself wanting more with the new*~~ but the origins of this festival of candy and cupids are **actually dark** .

(5.56) *a waiting area with wireless internet are **all available** for your convenience.*

Here three sentences were replicated, but with differing levels of specificity in amenities provided. Note also that the capitalization on *internet/Internet* differs as well, meaning that, even if the list had been the same, (ii) would not have been flagged as a duplicate sentence under the case-sensitive approach.

- a. pcc_eng_06_094.8667_x1518191_07(:21-22) [26 tokens]
*Local shuttle service , ~~loaner-vehieles~~ , early bird drop off , and a waiting area with wireless Internet are **all available** for your convenience.*
- b. pcc_eng_06_096.8470_x1550239_110(:24-25) [29 tokens]
*Local shuttle service , ~~loaner-vehieles , rental-cars~~ , early bird drop off , and a waiting area with wireless internet are **all available** for your convenience .*
- c. pcc_eng_06_099.4105_x1591833_062(:18-19) [23 tokens]
Local shuttle service , early bird drop off , and a waiting area with

wireless *Internet* are **all available** for your convenience .

- (5.57) *domestic product , israel 's economy is **about equal** to egypt , syria , lebanon*

The only difference in this case is the presence or absence of the possessive suffix, 's, on *Jordan*.

- a. nyt_eng_19951019_0336_23(:11-12) [23 tokens] *measured by gross domestic product , Israel 's economy is **about equal** to Egypt , Syria , Lebanon and Jordan 's combined .*
- b. nyt_eng_19961107_0353_17(:11-12) [22 tokens]
*measured by gross domestic product , Israel 's economy is **about equal** to Egypt , Syria , Lebanon and Jordan combined .*

- (5.58) *vegas for 22 years , married some **absolutely charming** women and gave them all my*

The level of information provided about the quoted speaker is the distinguishing characteristic in this example.

- a. nyt_eng_19960502_0570_4(:53-54) [62 tokens]
~~*joking about his career , the English composer and star of the 1960s musicals “ Stop the World , I Want to Get Off ” and “ The Roar of the Greasepaint , the Smell of the Crowd ” cracked :*~~ *“ I went to Vegas for 22 years , married some **absolutely charming** women and gave them all my money .*
- b. nyt_eng_19990415_0331_37(:25-26) [34 tokens]
~~*summing up the previous two decades , he remarked from the stage ,*~~ *“ I went to Vegas for 22 years , married some **absolutely charming** women and gave them all my money .*

- (5.59) *“it is absolutely essential for the public , in the*

This case again illustrates a discrepancy in quoting style, where the 1997 article's sentence used a medial speaker reference and the 1999 version did not. The source of the bigram token—the quoted utterance—is unchanged.

- a. nyt_eng_19970708_0284_5(:4-5) [63 tokens]
*“ It is **absolutely essential** for the public , in the process of the criminal-justice system , to have confidence in the system , ~~”Reno told the Senate Governmental Affairs Committee,~~“ and you can not do that when there is conflict or an appearance of conflict in the person who is , in effect , the chief prosecutor . ”*
- b. nyt_eng_19990306_0066_8(:4-5) [54 tokens]
*“ It is **absolutely essential** for the public , in the process of the criminal justice system , to have confidence in the system , and you can not do that when there is conflict or an appearance of conflict in the person who is , in effect , the chief prosecutor . ”*

5.4.3.3 Motivations for Excluding Duplicates

The duplicate sentences excluded from analysis were presumed to be undesirable by-products of text collection—a degradation of the representation of natural human language, which, in turn, would reduce the validity of any conclusion. The method of using corpus data to investigate linguistic phenomenon is predicated on the premise that the data contained in the corpus is representative of natural human language use, or, more specifically, *production*. Whomever is being *addressed* is relevant to certain aspects of linguistic production, but the same is not true of anyone who *hears* (or reads) a particular utterance (i.e., It need not be the speaker’s intent for them to hear it.) The recipient must be considered by the speaker before/at the time of production to be relevant to that production.

To illustrate, if a live broadcast is recorded and then used as a corpus, the count attributed to each token/word for such a corpus would be the number of times the token was produced (written, said, etc.). It would **not** be the number of times it was

heard—either by being replayed, or by different hearers¹⁷⁵. This means that language “consumption”, “processing”, or “comprehension” is only indirectly represented in corpus data through the lens of language production.

However, language is a circular phenomenon. To say language production *alone* is represented in **any** form is to claim that production is not informed and influenced by exposure and comprehension. This is equivalent to postulating that humans are static entities who do not (cannot) respond to their surroundings—that humans cannot learn—which is laughably false¹⁷⁶.

Still, whatever experience and comprehension effects may be present through this indirect lens are untraceable for most corpora. The exception comes in the form of dialogue in corpora, where the objects of comprehension are explicitly encoded in the linguistic context, with utterances that are not only *experienced/consumed/comprehended*, but also *responded to*. Even when all the necessary information is present, it is still not truly available for analysis unless the corpora are intentionally designed to provide this information (e.g. The Switchboard Corpus), and/or such dialogue interactions are annotated. Therefore, comprehension or *hearing/reading* cannot be systematically studied via most corpora.

Any text duplication appearing in “carbon copies” is not *natural human language*

¹⁷⁵And this is not just because we don’t have that information. Even attempting such a thing seems wholly unmotivated, at least from a theoretical linguistic perspective; it could be more relevant from the perspective of sociolinguistics or communications research.

¹⁷⁶regardless of what some psychologists may *think* generative linguists are claiming with the “poverty of the stimulus” argument for innate language faculties

production. This includes things like multiple reprints of a single article in the **news!** (**news!**) corpora, and, in [Puddin](#), machine reproduced text, alias locations for the same article/webpage, links to a webpage, previews of full content, the same “announcement”/information banner/“disclaimer” appearing on multiple pages of a website, etc. These reproductions increase viewership/distribution—i.e. their purpose is to increase the number of people who see or *consume* and (hopefully) *comprehend* the utterance—but they are not novel (*natural* productions) of the utterance. To count them as such would be equivalent to claiming the number of times a speaker says “indubitably” in a podcast should be measured as the number of occurrences in the recording *multiplied by* the number of times any person hears that clip, including the same person replaying it. That is, they do not represent natural language production.

If they do not represent natural language production, the starting premise—that which *enables* the application of corpus methods—is false. At best, they degrade the representation of natural human language (by introducing noise or “false positives” in a sense). At worst, they introduce bias into the data, making some features seem more prevalent than *they* are in truth, but also more prevalent than *other things* are, minimizing the relative frequency of potentially conflicting features that have not been overrepresented. Thus their inclusion in a corpus decreases the validity of any claims postulated on that corpus. Ergo, all cases deemed “artificial reproductions” were excluded from evaluation.

That being said, there are some caveats and potential concerns for this processing

decision. Questions like those in (5.60) arise.

(5.60) Open Questions

- a. How are “artificial reproductions” classified as such?
- b. What about an author who quotes another source?
- c. If a cashier were required to ask, “Would you like to donate to St. Jude’s Children’s Hospital today?” to every customer, would those utterances be “natural productions”?

While the latter questions (5.60b-5.60c) are not within the scope of this discussion, (5.60a) can be addressed. The same sentence may be uttered many times, and natural duplications—i.e. naturally generated, but identical utterances—are valid tokens of linguistic production. A sentence like, “*I’m not exactly sure,*” is said many times verbatim. It is so general and widely applicable that it is highly plausible that all (or most) of these utterances were genuine human production.

However, the likelihood of this being true declines rapidly with the number and specificity of words included in the utterance. Along a similar vein, multiple sentences in a row being identical also increases the likelihood of its being an artificial copy. In a previous iteration, the preceding and following sentences were used to determine “duplicate” status, but as a result of the pre-selection of bigram-containing sentences as discussed in § 6.3.4, context sentences are no longer easily available for this task¹⁷⁷. An accurate measure of the general complexity or specificity of words contained in a sentence was also not feasible within the scope of this project, but sentence length

¹⁷⁷Future work could retrieve the context sentences for this evaluation—they still exist and are identifiable, but not readily accessible at this time.

Table 5.9: Example Sentence Token Count

1	2	3	4	5	6
I	'm	not	exactly	sure	.

was evaluated as an estimate for these attributes. If context sentences or a specificity gauge were available, sentence length would not need to be consulted, and any length sentence could be excluded (provided the following and preceding sentences were also verbatim). Since they were not, a sentence length filter was applied: In order to retain genuine verbatim productions, the limitations were placed on both filters.

For the case-sensitive full sentence evaluation, only identical bigrams with the exact same `token_str` value (i.e. not a simplified lemma representation, but the exact surface form appearing in the text) and **more than ten tokens** in that `token_str` were excluded as duplicate sentences. For the case-normalize window evaluations, similar limitations are imposed: The `token_str` must have more than 20 tokens, but the lower-case `text_window` is checked for equivalence.

To clarify the numerical representation of sentence length, the example given above—*I'm not exactly sure.*—would have 6 tokens, as indicated in [Table 5.9](#).

There is a possibility that this cleaning operation still excluded genuine verbatim productions, but one of the motivations for taking a “big data” approach is that the effect of overzealous exclusions can be overcome by just how much data still remains.

CHAPTER 6

CORPORA QUERIED

The methods and outcomes described in the preceding chapters are highly dependent on the input data. Specifically, these techniques were designed for a very large (text only) corpus of sentence tokens parsed as Universal Dependencies (UD). As described in Chapter 4 and Chapter 5 certain additional steps were also required to account for *informal* and/or *messy* text data, such as duplication checks, orthographic normalizations, and other filters to minimize the noise that naturally occurs with such sources. Although all exclusion and manipulation filters were directly motivated by specific issues found in these data, they are also largely predicated on the sheer size of the sample: More careful consideration of particular cases would be required, but also *more feasible*, for a smaller corpus.

This chapter provides detailed information for the three primary input corpora: both the smaller off-the-shelf **News** corpora—sourced from *The New York Times* and *Associated Press* news publications—and the much larger **(CoCra) The Pile + Universal Dependencies and Decreased Internal Noise (Puddin)** corpus created specifically for this work from a subset of *The Pile* (Gao *et al.* 2020). In addition to the original source corpora, this chapter also describes a pre-sampling approach motivated by the overwhelming size of the corpora being investigated. This yields a more relevant subset of sentence tokens, the **Bigram Restricted Corpus (BiRC)**, which serves as the actual search field for all of the methods previously described. To tie these details in with the work as a whole, the motivations, limitations, and

implications of the available data are also discussed, as well as the exact methods used for creating both (CoCra) [Puddin](#) and [BiRC](#).

The first two sections, § 6.1 and § 6.2 provide specifics for the indicated corpus resources, both retrieved and created. Then § 6.3 discusses the meta-concerns of motivations, limitations, and implications. Finally, § 6.4 describes detailed methodology of creating both [Puddin](#) as a whole (§ 6.4.1) and the pre-sampled [BiRC](#) (§ 6.4.2).

6.1 Overview of Corpora Selection

6.1.1 News overview

A portion of the data used in this dissertation come from the *Annotated English Gigaword* corpus ([Napoles et al. 2012b,a](#)). This Linguistic Data Consortium ([LDC](#)) resource is a filtered and parsed version of *English Gigaword Fifth Edition* ([Parker et al. 2011](#)), a corpus of six different English newswire streams compiled by the LDC over several years. The fact that these data were collected via live transmissions and archived means that duplication of documents is common, both with minor adjustments made and also as fully identical documents ([Parker et al. 2011](#)).

This research uses two of the sources available in the *Annotated English Gigaword* corpus (as retrieved from *English Gigaword Fifth Edition*): *Associated*

Press Worldstream (*Apw*) and *New York Times Newswire* (*Nyt*). *Annotated English Gigaword* excludes all sentences with more than 100 tokens (Napoles *et al.* 2012a)—as segmented by the open-source tool **Splitta** and tokenized by the **Penn-Treebank Tokenizer**. This exclusion amounts to less than 1% of all sentences across the entire corpus, and added Stanford typed dependency structures in the form of Penn-Treebank styled parses. These structural annotations were generated using the English parser described in (Huang *et al.* 2010).

I refer to the *Apw* and *Nyt* corpora jointly as the *News* corpus. Both are composed of multiple files encoding **UD** structures in the CoNLL-U format (stored as `.conllu` files). This standardized format is the **UD** adaptation of the CoNLL-X format set for the 10th Conference of Computational Natural Language Learning meeting (de Marneffe *et al.* 2024). The CoNLL-U format separates sentences by blank lines and individual tokens within each sentence by line breaks. Each token row then has the ten tab-delimited information columns shown in (6.1).

(6.1) ID FORM LEMMA UPOS XPOS FEATS HEAD DEPREL DEPS MISC

Many of these columns are self-evident. **ID** refers to the original word identification numeral—i.e., word ID. Note that this starts at 1 in the CoNLL-U format, while word *indexing* may start at 0 elsewhere. The **FORM** column encodes the literal text string present, while **LEMMA** indicates the linguistic stem associated with it. **UPOS** indicates universal part of speech tags, while **XPOS** records treebank specific tags. Only one part of speech tag is required, but any searches performed on a given corpus will

of course be limited to the information present. **FEATS** is an optional column to provide additional featural information available for each word, but the specifics of this column depends entirely on the data at hand.

The columns labeled **HEAD** and **DEPREL** are arguably the most important for encoding dependency structures: The **HEAD** is the ID of the word which governs the given word, with a dependency relationship type indicated by **DEPREL**. **DEPS** optionally encodes the **HEAD** and **DEPREL** information in a single string object of the form, `[HEAD]:[DEPREL]`. Empty categories are denoted by an underscore (`_`). Miscellaneous additional information may be provided under the heading **MISC**, which, in some variants includes the start and end character index for the form indicated on the given line, but such information is not an obligatory requirement of the CoNLL-U format.

6.1.1.1 Nyt: *New York Times Newswire*

The files in New York Times Newswire ([Nyt](#)) are organized and labeled by date of publication, starting in September 1994 and ending in December 2010. The abbreviations **Nyt1** and **Nyt2** are used in this text to refer to the separate decades represented, with **Nyt1** referring to the 65 separate files corresponding to publications from the 1990's, and **Nyt2** referring to the 127 files corresponding to publications from the 2000's.

The file breaks are determined by date of publication with one file per month of

publications, thus the size of each file is dependent on publication volume. The naming schema of the files indicates the contents via the numerical component, following the template in (6.2a). The document and sentence identification strings within the corpus are similarly transparent, and follow the templates shown in (6.2b–6.2c).

(6.2) *Nyt* Naming Schema¹⁷⁸

- a. file:
`nyt_eng_[year][month](.conllu)`
- b. doc_id:
`[filestem][day of month]_[document #]`
explicitly,
`nyt_eng_[year][month][day of month]_[document #]`
- c. sent_id:
`[doc_id]_[sentence # (within document)]`
explicitly,
`nyt_eng_[year][month][day of month]_[document #]_[sentence # (within document)]`

The example in (6.3a), indicates the file containing publications from February (the second month) of 2004. Within this file, the document ID (`doc_id`) in (6.3b) points to the 51st document (or group of sentences) published on Leap Day 2004 (February 29, 2004). The sentence ID (`sent_id`) in (6.3c) refers to the 20th sentence of the 51st document.

(6.3) *Nyt* ID example

- a. file: `nyt_eng_200402.conllu`
- b. doc_id = `nyt_eng_20040229_0051`

¹⁷⁸Both document and sentence numbering start at 1.

```

c. sent_id = nyt_eng_20040229_0051_20
d. text = Edwards finished his acceptance speech on an appropriately
    salty note .

```

The CoNLL-U formatted lines encoding the `sent_id`, `text` (the sentence itself), and UD parse of the object assigned to the `sent_id` in (6.3c) are shown in Literal 6.1¹⁷⁹. The meta information of `sent_id` and `text` are included as “comments” preceded by a hash character, `#`. By convention, document IDs are only explicitly included for the first sentence of each document, and appear on line preceding that first sentence, as indicated at the top of the literal block.¹⁸⁰ This same sentence is represented graphically in Figure 6.1. As indicated above, the `UPOS` (fourth) column, is not populated, as only `XPOS` part of speech tags are provided for `Nyt`. The columns denoting additional node features (`FEATS`) and miscellaneous information such as start and end characters (`MISC`) are also unpopulated.

¹⁷⁹While columns in CoNLL-U formatting are delimited by tabs (`\t`), spacing has been adjusted here for legibility.

¹⁸⁰This first sentence `sent_id` also illustrates that document numbers are zero-padded, but sentence numbers are not. That is, this first sentence is denoted by `_1` rather than `_01` despite containing (at least) 20 sentences, while the document is denoted by `_0051`, rather than simply `_51`

```

# newdoc id = nyt_eng_20040229_0051
# sent_id = nyt_eng_20040229_0051_1
# text = But they did their part in bringing prestigious industry recognition
↪ Sunday to their writer-director , Blake Edwards , who picked up an honorary
↪ Oscar for life achievement .
...

# sent_id = nyt_eng_20040229_0051_20
# text = Edwards finished his acceptance speech on an appropriately salty note .
1  Edwards      Edwards      -   NNP   -   2  nsubj   2:nsubj   -
2  finished     finish       -   VBD   -   0  root    0:root    -
3  his          he           -   PRP$  -   5  poss    5:poss    -
4  acceptance   acceptance   -   NN     -   5  nn      5:nn      -
5  speech       speech       -   NN     -   2  dobj    2:dobj    -
6  on           on           -   IN     -   2  prep    2:prep    -
7  an           a            -   DT     -  10  det     10:det    -
8  appropriately appropriately -   RB     -   9  advmod  9:advmod  -
9  salty        salty        -   JJ     -  10  amod    10:amod   -
10 note         note         -   NN     -   6  pobj    6:pobj    -
11 .            .            -   .      -   0  -       0:-       -

```

Literal 6.1: Nyl Sample CoNLL-U Entry

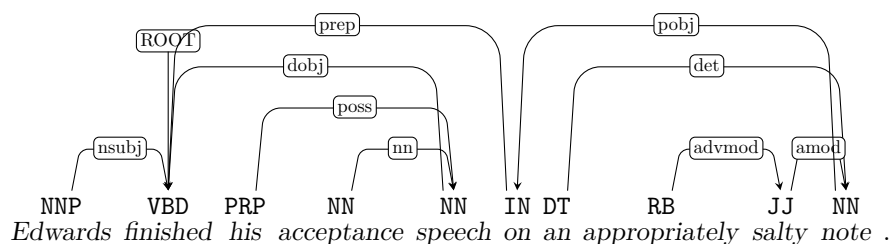


Figure 6.1: Nyl Sample UD Parse
nyt_eng_20040229_0051_20

The entirety of *Nyt* has 1.85 billion tokens¹⁸¹, distributed across 77.8 million total sentences, requiring 68.26 GB of storage space (uncompressed). Additional descriptive statistics for storage size, sentences, tokens, adverb tokens, adjective tokens, and bigram tokens across separate `.conllu` files are shown in Tables 6.1 and 6.2. Note that the “ADJ tokens” and “ADV tokens” columns here indicate the total number of (word) tokens in *Nyt* with an adjective or adverb classified part of speech tag, respectively.

Table 6.1: *Nyt* Summary Stats

	<i>Nyt</i> 1990s	<i>Nyt</i> 2000s	entire <i>Nyt</i>
<code>.conllu</code> files	65	127	192
storage size (GB)	31.9	36.4	68.3
SENTENCES	37,727,664	40,104,307	77,831,971
(word & punc) TOKENS	898,656,087	955,984,854	1,854,640,941
ADV ADJ BIGRAMS	3,209,958	3,889,200	7,099,158
BIGRAMS/million TOKENS	3,571	4,068	3,828

Table 6.2: *Nyt* Initial Stats across 192 `.conllu` files

	size (MB)	sentences	tokens	ADV tokens	ADJ tokens	bigram tokens
<i>mean</i>	355.5	405,375	9,659,588	303,534	555,328	36,975
<i>std</i>	155.7	191,411	4,562,546	125,105	237,693	15,057
<i>min</i>	10.3	11,599	275,113	8,558	16,212	1,105
25 %	257.8	279,446	6,644,232	227,853	410,176	27,489
<i>median</i>	351.7	373,685	8,899,287	310,282	555,464	37,607
75 %	456.2	543,575	12,971,749	392,471	704,888	48,345
<i>max</i>	693.6	820,724	19,532,928	570,729	1,076,937	69,407
CV%	43.8	47	47	41	43	41
TOTAL	68,257.3	77,831,971	1,854,640,941	58,278,503	106,622,956	7,099,158

¹⁸¹words mostly, but also punctuation markers

6.1.1.2 Apw: *Associated Press Worldstream*

Associated Press Worldstream ([Apw](#)) is a collection of articles published by the Associated Press from November 1994 to February 2010. As stated above, each sentence is parsed with Universal Dependencies and stored as multiple separate CoNLL-U formatted files (`.conllu`) Like [Nyt](#), the file breaks are determined by date of publication, with one file per month of publication transmissions. The naming schema again indicates this information in a similar fashion as well: e.g., `apw_eng_199606.conllu` contains publications from June of 1996. The documents and sentence IDs are similarly transparent, following the same template as [Nyt](#) and shown in (6.4).

(6.4) [Apw](#) Naming Schema

- a. file:
`apw_eng_[year][month](.conllu)`
- b. `doc_id`:
`[filestem][day of month]_[document #]`
explicitly,
`apw_eng_[year][month][day of month]_[document #]`
- c. `sent_id`:
`[doc_id]_[sentence # (within document)]`
explicitly,
`apw_eng_[year][month][day of month]_[document #]_[sentence # (within document)]`

The example in (6.5a) indicates the file containing publications from December (the twelfth month) of 2004. Within this file, the document ID (`doc_id`) in (6.5b) points to the 116th document (or group of sentences) published on Christmas Eve

2004 (December 24, 2004). The sentence ID (`sent_id`) in (6.5c) refers to the 5th sentence of the 116th document.

(6.5) [Apw](#) ID example

- a. file: `apw_eng_200412.conllu`
- b. `doc_id = apw_eng_20041224_0116`
- c. `sent_id = apw_eng_20041224_0116_5`
- d. text = `It was not immediately clear what started the blaze .`

The formatting of meta and parse information for the sentence referred to in (6.5) is the same as that for [Nyt](#) and is shown in [Literal 6.2](#)¹⁸². A visual representation of the parse information encoded therein is provided in [Figure 6.2](#).

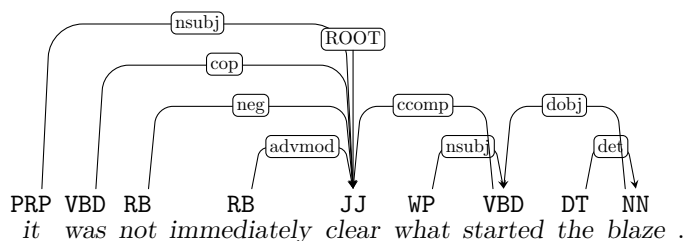


Figure 6.2: [Apw](#) Sample UD Parse
`apw_eng_20041224_0116_5`

¹⁸²While columns in CoNLL-U formatting are delimited by tabs, spacing has been adjusted here for legibility.

```

# newdoc id = apw_eng_20041224_0116
# sent_id = apw_eng_20041224_0116_1
# text = A Greek cargo ship caught fire off Sri Lanka but the 30-member crew
↳ extinguished the blaze and all aboard were safe , a rescue and salvage
↳ company said Friday .
...

# sent_id = apw_eng_20041224_0116_5
# text = It was not immediately clear what started the blaze .
1  it          it          - PRP - 5 nsubj 5:nsubj -
2  was         be          - VBD - 5 cop  5:cop  -
3  not         not         - RB  - 5 neg  5:neg  -
4  immediately immediately - RB  - 5 advmod 5:advmod -
5  clear       clear       - JJ  - 0 root  0:root  -
6  what        what        - WP  - 7 nsubj 7:nsubj -
7  started     start       - VBD - 5 ccomp 5:ccomp -
8  the         the         - DT  - 9 det  9:det  -
9  blaze       blaze       - NN  - 7 dobj  7:dobj  -
10 .          .          - .  - 0 -    0:-   -

```

Literal 6.2: Apw Sample CoNLL-U Entry

Smaller than Nyl, Apw has 1.45 billion tokens (approximately 0.4 billion fewer tokens than Nyl), across 61.1 million total sentences, and requires 53.71 GB of storage space (when uncompressed).

Table 6.3: **Apw** Summary Statistics

Apw Totals	
.conllu files	178
storage size (GB)	54
sentences	61,074,570
tokens	1,445,575,596
ADV ADJ bigrams	3,339,662
bigram tokens per million tokens	2,310

Table 6.4: **Apw** Initial Stats across 178 .conllu files

	file size (MB)	sentences	<i>all</i> tokens	ADV tokens	ADJ tokens	BIGRAMS
mean	301.7	343,116	8,121,211	189,195	470,498	18,762
std	115.1	150,304	3,571,605	68,871	172,235	7,180
min	1.2	1,147	27,017	840	1,809	77
25 %	214.5	234,477	5,533,845	135,132	346,318	12,910
median	284.4	305,207	7,204,914	175,909	458,215	17,008
75 %	402.9	479,052	11,387,052	256,489	633,998	25,998
max	525.9	751,043	17,745,454	317,547	775,489	32,508
CV%	38.2	44	44	36	37	38
Total	53,705.2	61,074,570	1,445,575,596	33,676,720	83,748,720	3,339,662

6.1.1.3 News Summary

Combined, selected Newswire Corpora, Nyt and Apw ([News](#)) contribute 370 separate CoNLL-U formatted corpus files. Across these, there are 139 million sentences containing 3.3 billion tokens. Of those tokens, 92 million and 190 million have [ADV](#) and [ADJ](#) part-of-speech tags, respectively, contributing to 10 million bigrams. This comes to an estimated 3,163 bigrams per million tokens ([bpmt](#)) and 75 bigrams

across every thousand sentences. These totals are displayed with more specificity in Table 6.5 and full descriptive statistics for the 370 separate files are shown in Table 6.6.

Table 6.5: News Totals

News Totals	
corpus parts	3
corpus files	370
storage size (MB)	121,962
sentences	138,906,541
(<i>all</i>) tokens	3,300,216,537
ADV tokens	91,955,223
ADJ tokens	190,371,676
NEG tokens	21,611,437
BIGRAMS	10,438,820
ADV per million tokens	27,863
ADJ per million tokens	57,685
BIGRAMS per million tokens	3,163
BIGRAMS per thousand sentences	75

Table 6.6: News Summary Statistics (370 files)

	size (MB)	sentences	(<i>all</i>) tokens	ADV tokens	ADJ tokens	BIGRAMS
min	1.2	1,147	27,017	840	1,809	77
25%	226.5	241,787	5,716,480	159,126	361,821	15,906
median	322.2	345,974	8,227,311	246,216	507,422	26,736
75%	432.1	503,977	11,997,738	313,945	671,684	38,220
max	693.6	820,724	19,532,928	570,729	1,076,937	69,407
mean	329.6	375,423	8,919,504	248,528	514,518	28,213
CV%	42.5	47	47	47	41	53
TOTAL	121,962.4	138,906,541	3,300,216,537	91,955,223	190,371,676	10,438,820

6.1.2 CoCra Puddin Overview

CoCra Puddin—*Common-Crawl of Pile + Universal Dependencies and Decreased Internal Noise*—is derived from the largely unfiltered text of the **Pile-CC** subcorpus of *The Pile* (Gao *et al.* 2020). The original source data files of *The Pile* are (a) in plain, **jsonlines** format and (b) mixed up into a heterogeneous “pile” with no separation between different corpus sets.

I manipulated this raw text into machine parsable form and used Stanford’s **stanza** parsing tool to add Enhanced Universal Dependency annotations (enhanced is the default output) stored as CoNLL-U formatted files (file extension **.conllu**). *The Pile* does not include any metadata for the original source of the texts beyond the subset which they belong to. This regrettably introduces a similar lack in **CoCra Puddin** (or any other potential flavor), but every sentence in **CoCra Puddin** has an ID which indicates its original source file in *The Pile*. No date, author, or website category is available. Rather than being organized by publication date as the News corpora from *Annotated English Gigaword* are, Puddin’s individual files are organized by the source file in *The Pile* and split into “slices” by simple order of appearance.

The Pile-CC dataset was partially cleaned by *The Pile* authors using the tool **JustText** (Endr dy & Nov k 2013) and presents an incredibly large informal dataset. However, even with this attempted filtration, the amount of messy text still included in this informal collection—the “internal noise”—does not belong in a syntactically annotated corpus intended for linguistic inquiries.

Therefore, the texts of the data set underwent string manipulations to make them as digestible as possible for the `stanza` parser. Then any texts (i.e. corpus “documents”) which still contained potentially problematic string patterns were excluded from the selection of data to be parsed. This precipitates a $\sim 50\%$ reduction in the size of the dataset, but even after dropping half of the documents/texts, the resulting, much cleaner dataset still dwarfs many other syntactically annotated informal corpora. For example, the parent corpus of *Nyt* and *Apw*, *Annotated English Gigaword*, contains 183,119,464 sentences in total (including those within *Nyt* and *Apw*) (Napoles *et al.* 2012b) while CoCra Puddin contains 819,256,183 sentences—an increase of roughly $\sim 447\%$.

Literal 6.3 illustrates the original formatting and rich text formatting of the first “Pile-CC” document in the source file `val.jsonl`, found on line 20. In Puddin, this single line becomes the roughly 96 lines of CoNLL-U formatted text shown in E.4. A full representation of this document with UD parsing in CoNLL-U formatting can be found in Appendix E as Literal E.4. An example of a different sentence from Puddin is shown in Literal 6.4 in as CoNLL-U formatted text and in Figure 6.3 as a visual representation of the parse information encoded therein.

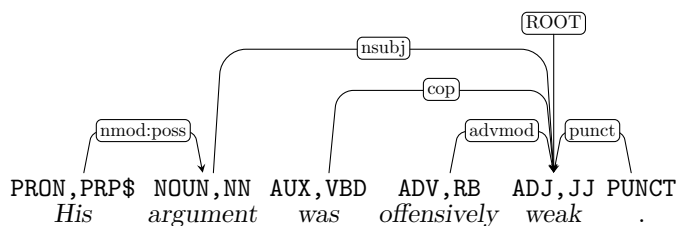


Figure 6.3: Puddin Sample UD Parse
pcc_eng_29_109.5887_x1749693_037

```
../pile$ grep -H -n -m 1 '"Pile-CC"' val.jsonl
val.jsonl:20:{
  "text": "A Podcast About Things\n\nPodTea\u2019d 010: The Varying Levels Of
  ↳ Audio Quality\n\nRIP Headphone Users. Recording from a phone or tablet mic
  ↳ is a very bad idea, and thus my audio quality is super duper loud at
  ↳ times. My apologies! Anyway, we discussed things that happened. I honestly
  ↳ don\u2019t remember what we talked about. Just go listen already, would
  ↳ you?", "meta": {"pile_set_name": "Pile-CC"}}
```

A Podcast About Things

PodTea'd 010: The Varying Levels Of Audio Quality

RIP Headphone Users. Recording from a phone or tablet mic is a very bad idea, and thus my audio quality is super duper loud at times. My apologies! Anyway, we discussed things that happened. I honestly don't remember what we talked about. Just go listen already, would you?

Literal 6.3: Sample Document from `val.jsonl` in *Pile-CC*

Summary statistics are provided for [Puddin](#) in [Table 6.7](#) and [Table 6.8](#). The storage size of [Puddin](#) relative to its parent is presented in [Table 6.9](#).

```

# newdoc id = pcc_eng_29_109.5887_x1749693
# sent_id = pcc_eng_29_109.5887_x1749693_001
# text = December 24, 2008
...

# sent_id = pcc_eng_29_109.5887_x1749693_037
# text = His argument was offensively weak.
1 His his PRON PRP$ Gender=Masc|Number=Sing| 2 nmod:poss _
    Person=3|Poss=Yes|
    PronType=Prs
2 argument argument NOUN NN Number=Sing 5 nsubj _
3 was be AUX VBD Mood=Ind|Number=Sing| 5 cop _
    Person=3|Tense=Past|
    VerbForm=Fin
4 offensively offensively ADV RB - 5 advmod _
5 weak weak ADJ JJ Degree=Pos 0 root _
6 . . PUNCT . - 5 punct _

```

Note: CoCra Puddin CoNLL-U files do include start and end characters for each token in a sentence under the *MISC* heading, but they have been trimmed from this example due to width restrictions. For the same reason, the *Feature* column has been wrapped manually.

Literal 6.4: CoCra Puddin Sample CoNLL-U Entry

Table 6.7: CoCra Puddin Summary Stats

CoCra Puddin Totals	
corpus parts	32
corpus files	3,247
storage size (MB)	1,240,354
1,300.445 sentences	819,256,183
(all) tokens	25,505,501,363
ADV tokens	673,513,143
ADJ tokens	1,055,173,456
BIGRAMS	85,047,554
ADV per million tokens	26,407
ADJ per million tokens	41,370
BIGRAMS per million tokens	3,334
BIGRAMS per thousand sentences	104

Table 6.8: CoCra Puddin Summary Statistics (3,247 files)

	size (MB)	sentences	(<i>all</i>) tokens	ADV tokens	ADJ tokens	BIGRAMS
min	260.3	173,016	435,945	138,338	221,985	17,084
25%	377.6	249,219	7,755,112	204,622	321,259	25,807
median	381.5	251,996	7,844,206	207,094	324,679	26,164
75%	385.6	254,840	7,936,450	209,804	328,025	26,539
max	437.5	290,337	9,334,512	238,165	373,124	30,002
<i>mean</i>	400.5	252,312	7,855,097	207,426	324,969	26,193
CV%	2.1	2	3	2	2	2
TOTAL	1,240,354.4	819,256,183	25,505,501,363	673,513,143	1,055,173,456	85,047,554

Table 6.9: Comparing CoCra Puddin to the unannotated source data—Storage Size extrapolated from [Gao *et al.* \(2020\)](#)

Component	Binary Size (GiB)	Metric Size (GB)	% of <i>The Pile</i>
<i>The Pile</i>	825	886	100%
Pile-CC	227	244	18%
CoCra Puddin	1,211	1,301	147%

6.1.3 Descriptive Stats/Size Comparisons

The entire starting corpora/composite corpus—News + Puddin—is 1.43 TB of data, when stored as uncompressed text files. It is split into 35 subcorpora or corpus parts/chunks, which take the form of directories. Each of these 35 directories contains 3 (**bigram-PccVa** & **bigram-PccTe**) to 178 (**bigram-Apw**) separate CoNLL-U formatted files. Overall, the starting data set is comprised of 3,617 individual parsed `.conllu` files, containing **28.8** billion **tokens**¹⁸³ distributed across **958** million **sentences**¹⁸⁴.

Note that “*tokens*” refers to all instances/occurrences of words and punctuation marks, *regardless of uniqueness*, while “*token forms*” refers to the strings of text corresponding to each token observations (i.e., the “form” the token takes in the sentence). To illustrate, the sentence displayed in [Table 6.10](#) contains 24 tokens (22 representing words and two representing punctuation), but only 22 unique token forms. This is because two (word) tokens—**that** and **to**—have forms identical to that of a preceding token.

This particular sentence also illustrates that tokens and lemmas are completely dependent on the tokenization of the sentence string. There are two hyphenated compound modifiers present in the sentence text, *well-written* and *badly-written*. Despite their parallel structure, the former is segmented into two tokens—**well** and

¹⁸³28,805,717,900 tokens overall

¹⁸⁴958,162,724 sentences overall

Table 6.10: Tokens Definitions Example

sent_id	pcc_eng_29_109.2328_x1744038_43
sent_text	<i>Everybody knows that you are not that stupid to be unable to discern between a well-written and badly-written dissertation paper.</i>
lemma_str	<i>everybody know that you be not that stupid to be unable to discern between a well -write and badly - write dissertation paper .</i>
token_str	Everybody knows that you are not that stupid to be unable to discern between a well -written and badly - written dissertation paper .
token_str <i>sans repeats</i>	Everybody knows that you are not _ stupid to be unable _ discern between a well -written and badly - written dissertation paper .
# tokens	24 = 22 words + 2 punctuation marks
unique token types	22 = 20 words + 2 punctuation marks

-written, which are classified as an adverb and verb, respectively—while the latter is segmented into three tokens—**badly**, **-**, **written**, which are similarly classified as an adverb and verb, but with the hyphen assigned to its own punctuation token. This prevents the *written* portion of these compounds from being treated as a single token type, as well as impacting lemmas assigned, *-write* vs. *write*. In this particular case, the resulting parse (illustrated in Figure 6.4) has not been adversely affected, but this is something that must be considered when working with large corpora. If one were searching for hyphenated adverb-verb pairs, word forms starting with “*wr*”, or all cases of the lemma “**write**”, the pattern specification would need to adjust to capture both *ADV-written* cases here.

Particular to the current line of inquiry, there are 765 million total adverb

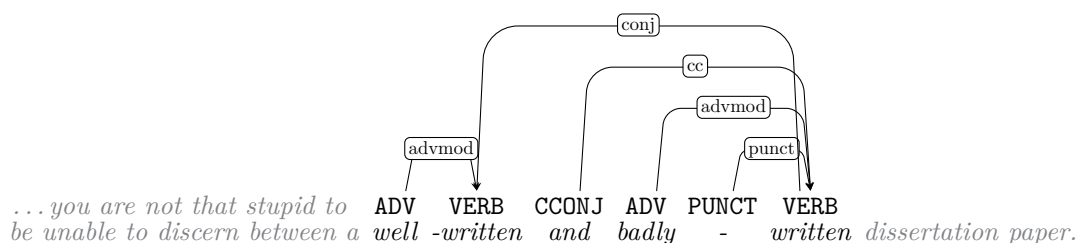


Figure 6.4: *Everybody knows that you are not that stupid to be unable to discern between a well-written and badly-written dissertation paper.*

tokens¹⁸⁵ and 1.25 billion total adjective tokens¹⁸⁶. Each individual corpus file (of the 3,617 total) has an average of 2.4k unique adverb lemmas, and an average of 16.1k unique adjective lemmas.¹⁸⁷

Tables 6.11–6.13 juxtapose descriptive statistics for the different source corpora, and Figure 6.5 illustrates the relative sizes and bigram contributions of *News* and *Puddin*. The relative bigram appearance rates shown below indicate that adverb tokens are more likely to participate in bigrams than adjective tokens are. That is, adjectives are more likely to appear without a preceding adverbial modifier than adverbs are to appear without an immediately following adjectival argument.

A curious observation is that ratio of *Puddin* counts to combined *News* counts stays approximately the same for both comparisons, whether *all* tokens or bigram tokens only. *Puddin* contains approximately 3,334 *bpmt*, and *News* contains

¹⁸⁵by part of speech tag: `xpos==RB*`

¹⁸⁶by part of speech tag: `xpos==JJ*`

¹⁸⁷This refers to lemmas as determined by automated parsing—errors are common, so this can only be used as a very loose approximation.

Table 6.11: Puddin & News Combined Totals

	Total
corpus parts	35
corpus files	3,617
storage size (MB)	1,362,317
sentences	958,162,724
(<i>all</i>) tokens	28,805,717,900
ADV tokens	765,468,366
ADJ tokens	1,245,545,132
NEG tokens	159,335,611
BIGRAMS	95,486,374
ADV per million tokens	26,573
ADJ per million tokens	43,240
BIGRAMS per million tokens	3,315
BIGRAMS per million ADV tokens	124,742
BIGRAMS per million ADJ tokens	76,662
BIGRAMS per thousand sentences	100

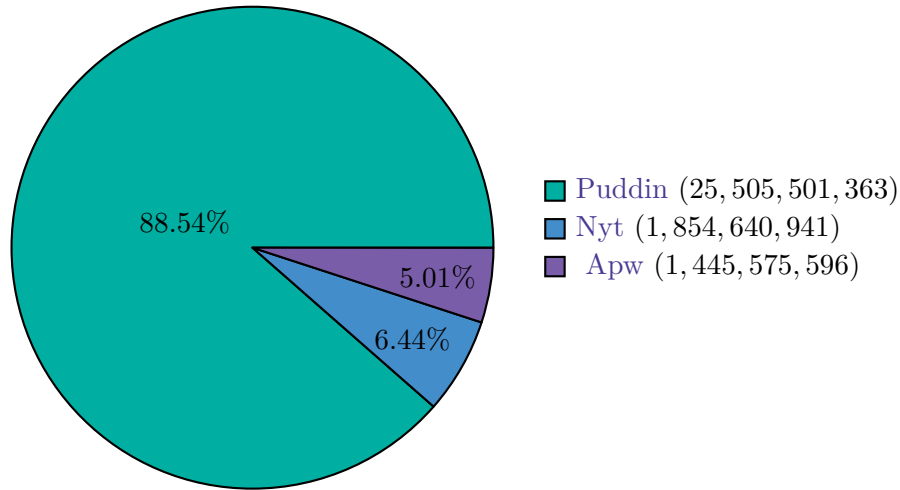
approximately 3,163 bpmt. Although the rate of bigram appearance is slightly lower for the News corpora, the magnitude of this difference is minimal given the granularity. This indicates that the prevalence of ADV ADJ bigrams is consistent between Puddin and News: neither corpus exhibits a notable divergence in the number of bigrams for its overall size (in tokens). However, the relative frequency of bigrams is less balanced between Nyt and Apw: Apw has a lower rate of bigram appearance (2,310 bpmt) than Nyt exhibits (3,828 bpmt).

Table 6.12: Puddin & News Combined Summary Statistics

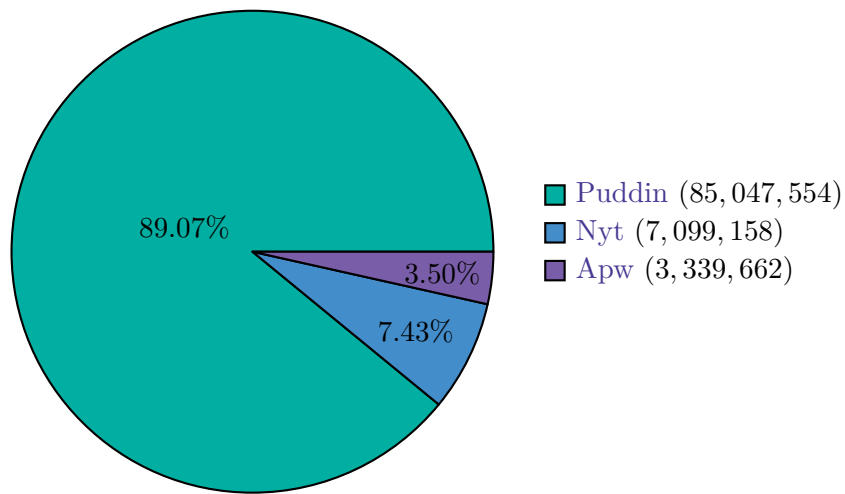
	size (MB)	sentences	(<i>all</i>) tokens	ADV tokens	ADJ tokens	BIGRAMS
<i>mean</i>	376.6	264,905	7,963,981	211,631	344,359	26,399
<i>std</i>	48.1	67,511	1,390,308	39,606	89,211	4,871
<i>min</i>	1.2	1,147	27,017	840	1,809	77
<i>25%</i>	376.9	249,191	7,746,686	204,460	321,416	25,764
<i>median</i>	381.3	252,229	7,844,694	207,234	325,172	26,167
<i>75%</i>	385.9	255,672	7,950,766	210,390	329,259	26,585
<i>max</i>	693.6	820,724	19,532,928	570,729	1,076,937	69,407
TOTAL	1,362,316.9	958,162,724	28,805,717,900	765,468,366	1,245,545,132	95,486,374

Table 6.13: Summary Totals: Puddin & News

corpus	# files	<i>all</i> tokens	sentences	size (GB)	BIGRAMS	<i>bpm</i>
Total	3,617	28,805,717,900	958,162,724	1,362	95,486,374	3,314.8
Puddin	3,247	25,505,501,363	819,256,183	1,240	85,047,554	3,334.5
News	370	3,300,216,537	138,906,541	122	10,438,820	3,163.1
(Apw)	178	1,445,575,596	61,074,570	54	3,339,662	2,310.3
(Nyt)	192	1,854,640,941	77,831,971	68	7,099,158	3,827.8



(a) ALL WORDS & PUNCTUATION



(b) BIGRAMS

Figure 6.5: Juxtaposed Token Distributions by Corpus—Total Bigrams vs. Total Overall

6.2 Bigram Restricted Corpus: BiRC

Due to the size of the search area presented by the corpora described above in § 6.1, I employed a preliminary selection filter to limit the active search area to only sentences meeting a minimum feature requirement. **BiRC** is a subcorpus limited to only the sentences in the original dependency parsed corpora that contain an $\text{ADV}^{\text{advmod}}\text{ADJ}$ bigram, as defined by the basic bigram pattern given in Chapter 5.

Rather than being an index of relevant sentences or a filter applied on the fly, **BiRC** is a separate collection containing copies of only the specified sentence tokens. This entails creating new file objects—`.conllu` files—with only the desired sentences. The structural organization of BiRC mirrors that of the input data, but the individual `.conllu` files themselves are much smaller. BiRC contains the bigram sentences from both the **News** and **Puddin** corpora, but the source is transparent—no sentence IDs are changed from the original, despite the full document/neighboring sentences no longer being present. The annotations are also identical to the original, as the only purpose is to increase the efficiency of more detailed searches over this predesignated space. See § 6.3 below for further discussion of motivations and impacts of using BiRC.

As an explicit example, the only sentence that is retained from the document provided above in § 6.1.2 `pcc_eng_val_1.0001_x00001`, is the fourth sentence, `pcc_eng_val_1.0001_x00001_4`, shown in (6.6). Of course, this particular sentence illustrates that the accuracy of identifying bigrams is not perfect, since *duper*’s status

as an adjective is questionable.

- (6.6) *Recording from a phone or tablet mic is a [bigram **very bad**] idea, and thus my audio quality is [bigram **super duper**] loud at times.*

6.2.1 BiRC Size Reduction

In terms of storage space, BiRC is a 87% reduction from the starting data, but it still contains 2.69 billion tokens (of any type) across 88.8 million sentences. In those 2.7 billion tokens, there are **95** million **bigram tokens**¹⁸⁸, with a mean of 26.4 k bigrams per .conllu, and a median of 26.2 k bigrams per .conllu. The maximum for a single file is 69.4 k bigrams found in BIGRAM.nyt_eng_199605.conllu. The minimum is 77 bigram tokens found in BIGRAM.apw_eng_199911.conllu. Note that the number of bigram tokens does *not* differ from those found in the entire corpus: since a bigram token's presence in a sentence was the only feature required for inclusion in BiRC, all bigram tokens are included in both collections.

BiRC's size reduction is quantified in Tables 6.14, 6.15 and 6.16 with overall values as well as a by-corpus breakdown for each unit of measure. Table 6.14 quantifies this comparison by sentence, the unit on which inclusion was based. Table 6.15 quantifies the reduction by storage size (in MB).

¹⁸⁸95,486,374 as counted from *raw* hits, prior to any exclusions

Table 6.14: BiRC Sentence Count Comparisons by Corpus & Overall

Corpus		TOTAL	<i>mean</i>	<i>smallest</i>	<i>largest</i>
News	BiRC	10,128,948	27,376	78	66,997
370	Full	138,906,541	375,423	1,147	820,724
-92.7%	Δ	-128,777,593	-348,048	-1,069	-753,727
Puddin	BiRC	78,719,122	24,244	15,733	27,883
3, 247	Full	819,256,183	252,312	173,016	290,337
-90.4%	Δ	-740,537,061	-228,068	-156,722	-262,454
Overall	BiRC	88,848,070	24,564	78	66,997
3, 617	Full	958,162,724	264,905	1,147	820,724
-90.7%	Δ	-869,314,654	-240,341	-1,069	-753,727

Table 6.15: BiRC File Size (MB) Comparisons by Corpus & Overall

Corpus		TOTAL	<i>mean</i>	<i>smallest</i>	<i>largest</i>
News	BiRC	11,846.1	32	0.1	76.5
370	Full	121,962.4	329.6	1.2	693.6
-90.3%	Δ	-110,116.3	-297.6	-1.1	-617.1
Puddin	BiRC	163,257.4	50.3	33.2	57.7
3, 247	Full	1,240,354.4	382	260.3	437.5
-86.8%	Δ	-1,077,097	-331.7	-226.9	-386.4
Overall	BiRC	175,103.6	48.4	0.1	76.5
3, 617	Full	1,362,316.9	376.6	1.2	693.6
-87.1%	Δ	-1,187,213.3	-328.2	-1.1	-617.1

Both Table 6.14 and Table 6.15 present the total sum as well as the mean, minimum, and maximum values seen across the pertinent files. The number of files pertaining to each set is indicated as $|\#files|$ within both tables. The labels “smallest” and “largest” are used in place of “min(imum)” and “max(imum)” because they are designated by *magnitude* rather than signed value. This is to keep the Δ columns intuitively aligned with the count columns, such that “smallest” refers to both the

smallest value observed in a single file, but also the smallest *change* observed for a single file and similarly for “largest”. Note that the number of files seen for each subcorpus and overall remains constant throughout—there are 3,617 files overall in both the original corpora and BiRC, with 370 (10.23 %) from News and the remaining 3,247 (89.77 %) from Puddin.

Table 6.16: BiRC Token Count Comparisons by Corpus & Overall

Corpus		<i>Any</i>	ADJ	ADV	NEG
News	Full	3,300,216,537	190,371,676	91,955,223	21,611,437
	BiRC	240,357,823	27,043,166	19,147,268	2,351,756
	Δ	−3,059,858,714	−163,328,510	−72,807,955	−19,259,681
	% Δ	−92.72	−85.79	−79.18	−89.12
Puddin	Full	25,505,501,363	1,055,173,456	673,513,143	137,724,174
	BiRC	2,452,435,958	204,032,545	168,130,974	25,311,753
	Δ	−23,053,065,405	−851,140,911	−505,382,169	−112,412,421
	% Δ	−90.38	−80.66	−75.04	−81.62
(11%N, 89%P)	Full	28,805,717,900	1,245,545,132	765,468,366	159,335,611
(9%N, 91%P)	BiRC	2,692,793,781	231,075,711	187,278,242	27,663,509
Overall	Δ	−26,112,924,119	−1,014,469,421	−578,190,124	−131,672,102
	% Δ	−90.65	−81.45	−75.53	−82.64

Table 6.16 quantifies the extent of restriction seen in BiRC by token counts, both overall (i.e., any word or punctuation token observed) and by specific categories, namely ADJ, ADV, and NEG. ADJ and ADV are defined by their part of speech tag—their counts encompass any token with a `xpos` matching `JJ*` or `RB*`, respectively. The NEG column, representing “negation tokens”, cannot be described by a single grammatical category, as it is a semantic, rather than syntactic group.

Rather, NEG membership is determined by a token’s `lemma`—if matching one of a predesignated set of “negative” lemmas, as introduced in [Chapter 2](#) and expanded on in [Chapter 4](#).

Despite [News](#) exhibiting a higher size reduction *rate* in the creation of [BiRC](#) than [Puddin](#) (respectively, -90.3% and -86.8% of storage size; -92.7% and -90.4% of sentences), the overall reduction of Puddin is much greater than that of News. The decrease in storage size differs by nearly 1 million MB ($1,077,097 - 110,116$), and the total number of removed sentences differs by 612 million ($740,537,061 - 128,777,593$). This is simply due to the huge size disparity of the original corpora.

Table 6.17: Reduction (BiRC – Full) Counts & Rates: Descriptive Stats

	mean	median	TOTAL for <i>removals</i>	difference of Totals	
fileMB	−328.2	−331.2	−1,187,213.3	‡	−1,187,213.3
sentences	−240,341.3	−228,003	−869,314,654	‡	−869,314,654
tokens	−7,219,498	−7,090,899	−26,112,924,119	‡	−26,112,924,119
ADVTokens	−159,853.5	−155,494	−578,190,124	‡	−578,190,124
ADJTokens	−280,472.6	−262,280	−1,014,469,421	‡	−1,014,469,421
NEGTokens	−36,403.7	−34,595	−131,672,102	‡	−131,672,102
tokPerSent	<0.1	<0.1	† 30	‡	0.2
ADVTokPerSent	1.3	1.3	† 0.7	‡	1.3
ADJTokPerSent	1.3	1.3	† 1.2	‡	1.3
NEGTokPerSent	0.1	0.2	† 0.2	‡	0.1
ADVTokPerMill	42,848.2	42,204.3	† 22,141.9	‡	42,974.4
ADJTokPerMill	42,792.1	41,888.6	† 38,849.3	‡	42,573.1
NEGTokPerMill	4,686.9	4,898.7	† 5,042.4	‡	4,741.8

Notes:

†: recalculated using totals of removed set

‡: the difference between the totals observed for BiRC and Full corpora

n/a values: cannot be summed informatively, as they convey unique values

6.3 Data Selection Motivations, Limitations, & Implications

In general, the motivations for the selections, limitations, and exclusions expounded on below are both necessitated and enabled by the sheer volume of data at the starting point. In having a lot of data, processing costs become cause for serious concern: beyond just being *slow* to process, certain operations become impossible¹⁸⁹ or at the very least grow significantly in overhead¹⁹⁰. Thus limiting the size of the data being processed is more impactful than it would be for a smaller dataset, in terms of processing costs. At the same time, however, having a very large data set means that pruning doesn't need to be as careful. It's like a giant, well-established forest as opposed to fresh saplings. You can trim a lot more without having detrimental side effects. Removing 5k sentences would be worrisome for a corpus that only contains 20k to start, but for a starting corpus that contains 5 million sentences, 5k is merely a drop in the bucket. This is not to say that said data pruning should not be motivated or that no care should be attempted, but it does create a larger margin for error.

¹⁸⁹None of this would have been possible without having access to the g2 server cluster.

¹⁹⁰I cannot even quantify how many hours I have spent in the pursuit of efficiency to avoid and remediation of errors due to insufficient processing resources.

6.3.1 Choosing Corpora for Investigation

A large part of the **corpus selection** was availability. I already had access to NYT and APW corpora with dependency parses. *The Pile*, while not parsed in the same way, was a large dataset which was freely available and in a format which, in its *lack* of annotations and meta data, was ripe for the “parsing”. Thus, CoCra Puddin was born.

The choice to build a corpus on **informal** language data, as opposed to more formal and potentially much cleaner data, was in part motivated by the sense that the **News** corpora preliminarily inspected might have its own register/style sheet/editing peculiarities. However, this was also motivated by size alone. The amount of (reasonably current) data that exists for a web-scrape compilation like the common crawl overshadows anything that could be procured from vetted, edited, published sources. More recently produced language data is also preferred, since phrasing and word selection are subject to temporal shifts: older patterns might not persist in current speakers’ idiolects, and new patterns might emerge which aren’t attested in less recent data. Although *The Pile* is intentionally scrambled and does not provide publication dates for The Pile-CC subcorpus, the content of sentences contained confirms that it is at least as recent as 2020, based on references to Covid-19 and the Coronavirus Pandemic.

- (6.7) a. *The economic fallout from the Covid-19 pandemic will be severe.*
sent_id = 'pcc_eng_val_3.00105_x34710_16'
b. *Due to the ongoing concerns with the Coronavirus as a Board we have*

decided that it's in the best interest to cancel the 2020 Arizona USATF outdoor track and field season.

`sent_id = 'pcc_eng_test_3.02409_x38277_3'`

Working within the ***Universal Dependencies*** framework (i.e., using corpora annotated with universal dependency parsing) was a result of simple inertia. This happened to be the format of the [News](#) corpora, and the formal structural representation which `grew` is designed for. There was not sufficient motivation to change to a different framework.

The CoNLL-U format permits cleanly storing and accessing large amounts of structurally annotated data. As the standard convention, many sophisticated tools exist for parsing and writing it as well.

6.3.2 Restricting the Scope of Investigation

In terms of processing methods, the search area was restricted to only **ADV ADJ bigrams** (as defined in [Chapter 2](#)) primarily due to feasibility concerns. A large amount of data is needed to illuminate the kinds of patterns we think might exist, but that means a corresponding increase in processing time, resources, and complexity. So the search field had to be narrowed for the search to be possible and/or for the results to be interpretable.

First of all, there are a lot of these particular constructions—it is a common linguistic unit in English. Secondly, in restricting the targeted unit to contiguous

words (immediate precedence relation), they are easier to find and process. Their being restricted to a single dependency relation and governance direction makes them even easier to find, as well as easier to combine with the necessary variation of contextualized dependency structures. If the bigrams were extended to include even adverbial *post*-modification of adjectives (i.e., `ADJ ADV`) that would at the very least double the number of search queries required to collect the necessary data, and it has the potential for expanding the queries even more, if the post-modification structure behaves in different ways or appears with different types of relationships with logical environment operators.

6.3.3 Motivations for BiRC

The reasons for creating BiRC are, again, primarily due to processing efficiency concerns. Multiple patterns have to be run on the `.conllu` files: `grew` runs on an entire `.conllu` file and every sentence object contained in the file is searched. Since this process is repeated for every different context configuration, sentences that would necessarily not match *any* pattern would still be searched repeatedly. The additional processing time for this adds up given the number of finalized patterns. On top of that, development of the patterns themselves requires its own processing as well. Therefore, removing all the data that doesn't meet the *minimum* requirement (having a bigram in the first place) means that these same irrelevant sentences are not included in every search iteration, needlessly increasing processing time and resource consumption. This also makes locating the original sentence objects easier,

should that be required, simply because smaller files are easier to manipulate, easier to search, and easier to open (due to lower memory costs).

6.3.4 Minor BiRC Fallout

The primary consequence of using BiRC is that it made context evaluation impossible. That is, to keep the subset files cleaner and truly prevent redundant searching, no context was maintained—preceding or following a sentence containing a bigram match was insufficient for inclusion. Some included sentences do have context (preceding or following sentences) as a coincidence: the surrounding sentences just happened to also have the target bigram construction as well. But many sentences in BiRC do not, because they were not retained solely for providing context.

Context is retrievable if absolutely needed. I kept a record of the context sentences (without parse information) separately, and the sentence indexing (IDs) did not change and are linearly ordered. This means the ID of any potential context sentence can be deduced from the current ID, and those IDs can be used to locate the original sentence object in the initial corpus files. However, the processing costs of retrieving this information has proven prohibitive. The only true impact is that of determining duplicate sentences (discussed in [Chapter 5](#)), but I believe I have employed a reasonable work-around.

In a similar vein, the sentence IDs are no longer continuously ordered, in terms of document ID order (i.e., the penultimate suffix of the sentence ID) and order

of sentence-in-document (i.e., the final suffix of the sentence ID). The documents themselves have been inverted in many cases: The final sentence appears first and the first appears last. That being said, no negative effects resulting from this quirk have appeared thus far.

6.4 Corpus Preparation Methods

6.4.1 (CoCra) Puddin Methods

6.4.1.1 Raw Data Input

The following methods could be used on any plain text dataset in the expected format, and the resultant corpus will necessarily depend on specifics of the input. The parsed corpus I created for this dissertation, **CoCra Puddin**, was made from a large informal data set, *Pile-CC* of *The Pile*, thus the outcome is also large and informal. However, if the process is applied to other language text data of different sizes, registers, or content areas, the results will reflect this.

6.4.1.2 Informal Data for Pragmatic Phenomena

It's important to have informal data for linguistic phenomena that may not be considered “correct” usage in a prescriptive sense. The target of my inquiry—

adverb adjective bigrams in privative contexts—falls solidly in this category: from a style perspective, indirect constructions like these, including *litotes*, have been viewed unfavorably and even explicitly disparaged by many authors or publishing “authorities” (Horn 2015a,b). Therefore, data that has not passed through such a prescriptive revision filter is far more likely to provide an accurate descriptive representation of naturally occurring usage.

Informal language data, like most data collected from natural rather than controlled settings, is inherently more irregular and unwieldy than formal language data. The uncontrolled nature of informal data does result in higher error rates, but this means both *descriptive* errors (i.e., that which the speaker actually didn’t intend to produce or inaccurate records of actual production) and *prescriptive* errors (i.e., that which the speaker intended, but was deemed unacceptable by external standards). In the case of many naturally linguistic phenomena, prescriptive “errors” are **desired**. More edited and stylized language data, such as that found in newspapers and books (with dialogue as a possible exception), is often intentionally pruned of the exact instances being sought.

6.4.1.3 Making Informal Data Useful

While additional data is essential to making any existing patterns visible, that additional data is only as useful as the representation of natural language that it provides. To investigate language features that are often considered prescriptive errors, informal data with higher “error” rates are required, but the descriptive

language errors and transcription and processing errors that pass through this insensitive error filter still have to be dealt with to make a dataset optimally useful for linguistic inquiries. Descriptive language errors are more difficult to flag using automatic methods, since they may *look* more like natural usage. However, cleaning and trimming the raw data before further processing—preemptively removing or altering the formatting of the problem data—can mitigate pollutant effects.

The Pile authors did perform some cleaning using the third-party tool **JustText** (Gao *et al.* 2020); however, per the tool’s intended usage (Endr dy & Nov k 2013), this primarily served to remove boiler plate text from the scraped website text. This did not ensure that the body of the webpage was actually natural English usage, so the original 244 GB of the Pile-CC subcorpus is rife with texts containing elements that aren’t even technically English (e.g., html, wikitext/wikimedia, and other formatting markups, as well as coding language snippets), and the full extent of unedited, error riddled documents and webpages with their separate elements (e.g., main text, side bars, banners, overlaid advertisements, etc.) are cobbled together—incoherently at times.

Since the dataset is so large, it is possible that this “noise” would make a negligible impact on some tasks. However, if the full level of noise is never calculated (and I doubt this was ever assessed given that it is machine generated and gigantic), it cannot be *assumed* that the noise has no impact on results. In fact, I would argue it does have *some* effect regardless—it just might not be *noticeable* given the sheer mass of it.

Therefore, the raw text of Pile-CC underwent cleaning and filtering steps on its way to becoming CoCra Puddin. The size comes in again here as well—it might be large enough that the instances of messy non-natural language would fade into the background, but that means it is *also so* large that **excluding** the least parsable sentences can improve accuracy and legibility of the data without nullifying the size increase that is the primary objective. Any documents that contain the flagged regular expressions are marked as exclusions and pulled before the final parsing step.

6.4.1.4 Software Used

While the methods for creating the CoCra Puddin corpus only requires a handful of tools, it requires a hefty resource pool. CoCra Puddin is generated via a Python program that relies primarily on the `stanza` (Qi *et al.* 2020) module and an English language NLP model which can be retrieved via `stanza` as well. It also calls on two helper modules in the preprocessing and cleaning phases. The `Unidecode` (?) module is used to translate unrecognized characters from their original codec to unicode for parsing consistency. The module `jsonlines` (Bolsterlee n.d.) is employed to facilitate iterating through the files of *The Pile* raw source data.

On the resource front, the Puddin recipe requires access to at least one GPU, but access to more than that speeds up processing significantly by enabling parallel processing of the source data. In this case, parallel processing was achieved via independent `slurm` (Yoo *et al.* 2003) jobs distributed across the G2 server cluster at Cornell University. However, the same can be replicated on a single server or local

machine if the sufficient GPUs are available. This could be done by employing `GNU Parallel` (Tange 2023) or the `multiprocessing` module in `python`, for example, both of which are used elsewhere in the current research.

6.4.1.5 Why Universal Dependencies?

While we definitely needed more data in general, to be fully useful, we also needed the additional data to have syntactic information included. As mentioned before, determining the scope of negation is very complicated. Having scope (i.e., structural/syntactic) information, makes it easier, and therefore increases accuracy of results and validity of conclusions.

We used dependency grammar, because dependencies are relatively easy to generate and contain all of the relational information that we needed to determine scope. Other grammars have greater overhead (require more input) for irrelevant additions to the output. The tools that exist for creating dependency parses are also well developed because dependency parses have become a standard way of annotating structural relationships in many NLP resources and large language data sets. That is, the popularity of universal dependencies, especially in NLP, increases the quality of the parsers available.

The choice of `stanza` (Qi *et al.* 2020) specifically was mostly due to coding language interface. The rest of the project was already using Python primarily and `stanza` is designed to work directly in Python.

6.4.1.6 Final Process

While the full development process is much more complicated¹⁹¹ the final processing pipeline is described below.

The first step was to acquire *The Pile* dataset from the host website, <https://pile.eleuther.ai/>. I used a very simple shell script (see Literal E.6 in Appendix E) Then the environment must be arranged such that data files are accessible to the Puddin source code, and all requisite dependencies must be installed. If the `slurm` resource manager is present on the system, run `sbatch puddin_slurm.sh` with desired resource allocation requests. Otherwise, directly run `python parse_pile.py` with the desired input files.

The structuring of all output data (excluding validation outputs) is exemplified in Literal 6.5, where the subdirectory `PccVa.con11/` corresponds to 1 of 32 initial subdirectories of *The Pile*. The same output structure is replicated for the other 31 corpus components.

¹⁹¹See <https://github.com/AndreaTheLinguist/puddin> for more details.

```
puddin/  
  PccVa.conll/  
    pcc_eng_val-01.conllu  
    pcc_eng_val-02.conllu  
    pcc_eng_val-03.conllu  
  info/  
    completed-puddin_meta-index.csv  
    completed-puddin_meta-index.pkl  
  pile_exclusions/  
    pile_val_Pile-CC_excl.pkl.gz  
  pile_tables/  
    master_all-slices_index.csv  
    tmp/  
      pile_val_Pile-CC_df.pkl.gz  
  raw/  
    pile_val_Pile-CC_df.pkl.gz  
  slices/  
    PccVa/  
      pile_val-1_Pile-CC_df.pkl.gz  
      pile_val-2_Pile-CC_df.pkl.gz  
      pile_val-3_Pile-CC_df.pkl.gz  
      tmp/  
        pile_val-1_Pile-CC_df.pkl.gz  
        pile_val-2_Pile-CC_df.pkl.gz  
        pile_val-3_Pile-CC_df.pkl.gz  
        slice-info_val.csv
```

Literal 6.5: Example of Final File Architecture of [CoCra Puddin](#):
PccVa sourced from `pile/val.jsonl`

6.4.1.7 Evaluation Metrics

Given the size of the final output, and the difficulties encountered in the creation process due to resource limitations, it is important to confirm that all is as expected. To preemptively mitigate any negative effects of uncaught errors, I programmatically compared contents of the initial dataset, the final output, and the exclusions dataset. All texts in the initial data from *The Pile* should be included in either the finalized `.conllu` files, or in the exclusions data, stored as pickled and compressed dataframes corresponding to each original `.jsonl` file in *The Pile*. The validation process also ensured that each `.conllu` file is properly formatted, by making sure it can be parsed by the python package `pyconll` and searched by `grew`. [Table 6.18](#) shows the total documents excluded from each copurs part of [Puddin](#).

More information on parsing and validation can be found in [Appendix E](#).

Table 6.18: CoCra Puddin Validation—All successful vs. excluded docs

corpus part	successful docs	excluded docs
Pcc02	1,080,804	667,197
Pcc09	1,082,487	669,367
Pcc08	1,078,421	667,840
Pcc06	1,079,734	667,303
Pcc07	1,081,773	666,772
Pcc05	1,080,599	667,271
Pcc00	1,081,093	667,632
Pcc04	1,080,734	665,253
Pcc01	1,081,008	666,245
Pcc03	1,079,740	668,730
Pcc11	1,078,906	667,341
Pcc14	1,081,710	667,029
Pcc12	1,081,995	666,851
Pcc16	1,079,720	667,409
Pcc15	1,081,993	667,570
Pcc17	1,081,358	667,054
Pcc19	1,081,367	667,466
Pcc18	1,079,637	669,858
Pcc13	1,081,029	666,006
Pcc10	1,080,557	667,205
Pcc21	1,082,171	666,888
PccTe	32,697	20,093
PccVa	32,612	20,180
Pcc20	1,081,894	666,688
Pcc22	1,079,918	665,890
Pcc26	1,080,409	667,073
Pcc28	1,080,442	666,737
Pcc24	1,080,672	668,030
Pcc27	1,080,664	667,664
Pcc23	1,079,667	665,915
Pcc25	1,079,621	667,772
Pcc29	1,083,529	667,711

6.4.2 BiRC Methods

The creation of BiRC actually necessitated creating new file objects—.conllu files—with only the desired sentences. This was achieved using `grewpy`, the python module sibling of `grew` (Guillaume n.d., 2021), which was run in parallel (using `GNU-parallel` (Tange 2023)) on the `g2-cluster` using `slurm` processing management software (Yoo *et al.* 2003).

The process of creating BiRC is relatively simple. Each initial `.conllu` file was searched for bigram matches¹⁹². Any sentence that contained a match was added to the bigram subset/subcorpus. The selected sentences parses were saved to a new `.conllu` file, in a subdirectory of their source data. More technical information is included below.

In addition to collecting the limited dataset and creating those objects, the program also compiled some stats for the initial corpus object and the subset object. Those files are saved in the `info/` subdirectories as `.counts.json`. The program created an “index” for the file paths corresponding to each initial `.conllu` file:

- (6.8) a. the path to the initial `./*.conllu`
- b. the path to the `./subset_bigram/*.conllu`
- c. the path to the initial counts
 (`./info/*counts.json`)
- d. the path to the subset counts
 (`./subset_bigram/info/*counts.json`)

¹⁹²as identified by the `RBXadj` pattern provided in [Chapter 5](#)

The file structure of the bigram subset outputs follows the template in [Literal 6.6](#).

```
[CORPUS_PART]
...*.conllu
...[INIT_STEM].conllu
...info/...
...subset_bigram
    ...BIGRAM.*.conllu
    ...BIGRAM.[INIT_STEM].conllu
    ...info
        ...BIGRAM.*.counts.json
        ...BIGRAM.[INIT_STEM].counts.json
        ...subset-bigram_path-index.[CORPUS_PART].csv
```

Literal 6.6: BiRC file architecture

Specific details on how to create a corpus subset such a BiRC are located in [Appendix E](#).

CHAPTER 7

CONCLUSION & FUTURE WORK

This dissertation set out to bridge the gap between formal accounts of polarity licensing and large-scale corpus methods by introducing the Polar Bigrams Resource (PBR), a 72 million-token dataset of adverb–adjective constructions annotated for positive and negative environments. Chapter 1 motivated the need for data-driven approaches to polarity sensitivity; Chapter 2 presented the design and extent of PBR; Chapter 3 applied asymmetric association measures to reveal compositional dependencies of the adverb *exactly*; Chapter 4 described additional polarity-informed post-processing; Chapter 5 detailed the retrieval and cleaning pipeline; and Chapter 6 outlined the corpora preparation and sampling methods. This concluding chapter summarizes the key takeaways and charts directions for future investigation.

To conclude, I have presented PBR, a novel linguistic corpus resource designed for polarity investigations, including detailed coverage of all computational methods—from preparing UD–parsed corpora (i.e.; CoCra Puddin), retrieving relevant tokens using pattern match queries, cleaning the results and assigning approximated polarity categories (which undergo their own specific post-processing based on polarity sign), to the calculation of quantitative association between polarity and various lexical units. I have also provided a nuanced inspection of the adverb *exactly* to illustrate the quantitative power and methodological flexibility of PBR in practice. The potential applications of PBR and the methods discussed far exceed the glimpse this case study provides. Many alternate investigations are possible within the existing version of

PBR and feasible extensions to the current resource would enable many more.

Beyond this vast potential, however, the work described here serves as an example of how other resources might be developed and manipulated to inform linguistic theory, not only on polarity sensitivity but other complicated semantics and pragmatic phenomena that are similarly elusive in their nuance. This work is especially useful as a cautionary tale—highlighting the potential methodological pitfalls and vital interpretative considerations in such approaches. PBR is not perfect, and I have no doubt that future attempts will greatly improve the process and final outcomes, but hopefully my hindsight and detailed coverage of processing and investigation methods will accelerate that process.

Alongside the resources—[CoCra Puddin](#), [BiRC](#), and [PBR](#)—and methodological techniques contributed, my work has precipitated three overarching observations that must be taken into consideration for all data-driven approaches to the polarity sensitivity inquisition, as listed in (7.1).

(7.1) a. **Polarity Approximation:**

Environment polarity designations must be approximated via theory-informed heuristics

b. **Positive Preponderance:**

The overwhelming majority of natural language¹⁹³ has positive polarity, ranging from 5× to 22× as frequent as negative polarity.

¹⁹³In so far as the 72 million English bigrams of PBR provide an accurate representation of natural language more generally.

c. **Polarity Sensitivity as Backwards Asymmetric Association:**

Polarity sensitivity is best quantified as a lexeme’s attraction to an environment considered independently from the environment’s attraction to the lexeme.

The first consideration for data-driven polarity research—the need for **polarity approximation**—is due to the complex and even contested nature of polarity licensing. For starters, there is no single uncontested theory defining what polarity environments are or what creates them (Homer 2020). Secondly, even if there were a single “best” model, no possible definition of polarity environments and/or what enables them can have a simplistic, easy-to-retrieve surface form. Polarity environments will always be affected by implicit factors, whether syntactic movement/differences between surface structure and underlying structure or pragmatic effects that are never explicitly encoded rendering them impossible to retrieve directly from text data. That is, there is a large gap between capturing even one of the many proposed licensing theories formally and effectively specifying such cases in corpus queries.

This means that, in a data-driven approach to polarity sensitivity, polarity of a lexeme’s environment can only be approximated¹⁹⁴ via imperfect heuristics. In this work, I have employed two distinct polarity approximations methods (as detailed in Chapter 2) that diverge primarily in how positive polarity is designated. The **S* Superset defines positive polarity as the absence of a negative marker, while the *@S* subset restricts positive polarity to overt positive markers and their negative

¹⁹⁴At this stage; Making this possible is a worthwhile goal.

counterparts or “mirrors” (i.e., lemmas with existential or universal quantification—*somebody*, *everybody*—that parallel lemmas with privative semantics—*nobody*). The full theoretical import of this distinction requires further investigation, but the different perspectives enabled by this dualistic approach enhance the empirical robustness of PBR.

Even with the strictest designations of positive polarity and negative polarity, with additional enforcement measures dropping all cases with *potential* negative markers preceding the bigram, tokens with positive environments grossly outnumber those with negative environments, ranging from 5:1 to 22:1 odds of being positive as quantified by PBR. This feature of the polarity data landscape, which I have dubbed the **positive preponderance**, both necessitates and enables certain processing and investigation methods. All evaluations—be they simple co-occurrence frequency comparisons, more sophisticated [AMs](#), or even qualitative evaluations of observed examples—must be interpreted with this significant disparity of representation in mind.

In particular, the asymmetry presented by the positive preponderance exacerbates the inherent asymmetry of theoretical expectations for polarity licensing: a polarity environment is required for the appearance of a [PSI](#), but not vice versa. This combined practical and theoretical asymmetry underscores the importance of *directionality* of association. That is, backwards asymmetric [AMs](#) have much greater weight when evaluating association between lexical items and polarity of environment. This elevated importance persists even when the positive preponderance is

mitigated (i.e., in the [NEQ+](#) comparison spaces), as the marginal frequencies of binary categorization must necessarily obliterate those of a parameter with tens of thousands (adverbs) to millions (bigrams) of unique types. Even so, the striking size discrepancy between polarity categories themselves adds an additional layer of complexity to the interpretation of these measures.

My in-depth inspection of *exactly* illustrates the reality of these considerations and exemplifies potential methods to address them. This case study elucidates the potential of PBR from a descriptive standpoint, as well as presenting a preliminary theory informed by the empirical findings of PBR, showcasing the path from large scale exploratory data to theoretical implications.

Notably, this work does not make claims regarding the underlying nature of polarity sensitivity or its interaction with scalarity. These concepts are employed for descriptive purposes with the goal of adding to the arsenal of future linguists investigating polarity sensitivity, especially through a semantic or pragmatic lens.

7.1 Future Directions

Future theoretical inquiries which I hope to facilitate fall outside the scope of this dissertation, but there are many potential extensions and enhancements that are more directly tied to the data processing aspects reported. These fall into two main groups: extensions to the evaluation and interpretation of PBR as it currently stands,

and enhancements or alterations of the resource itself (i.e., “PBR 2.0”).

7.1.1 Extended Interpretation Methods

The most obvious extension of PBR lies in the nuanced inspection of other lexical type representations, especially adverbs following the approach expounded on in [Chapter 3](#) for *exactly*. For example, PBR could shed light on the empirical landscape of other negative polarity prone adverbs such as *that* and *necessarily*, positive prone adverbs such as *increasingly*, *almost*, and *relatively* (as indicated by AMs presented in [Chapter 2](#)). A fine-grained analysis of noncontroversial NPIs like *yet* or *ever* within PBR could also increase our understanding of polarity environments or licensing conditions themselves, especially if particular triggers and/or dependency relationships were considered in addition to the generalized binary polarity categories. This directly connects to the potential theoretical implications of polarity approximation method. Rather than exploring distribution patterns across environments as a means of identifying novel polarity sensitive items (i.e., what requires licensing), this vein of inquiry could yield worthwhile evidence distinguishing different levels or aspects of polarity environments.

Calculating association based on all possible angles for multiple different items concurrently could also prove fruitful. This would likely require the creation of composite scores unique to the dataset to provide a “summary” or unified indicator of many different values collapsed along the chosen axis, but the choice of axis could

encompass lexical categories, environment aspects, or something else entirely. For example, the bigram tokens within PBR could be labeled with additional meta-information representing scalarity features. These could be hard-coded designations as identified in the literature (e.g. scalar class examples pulled from [Kennedy \(2007\)](#), [Tessler & Franke \(2019\)](#), [Solt \(2015\)](#), [Castroviejo *et al.* \(2018\)](#), [Solt & Wilson \(2021\)](#), [Geurts \(2009\)](#), [Morzycki \(2012\)](#), [Beltrama & Xiang \(2013\)](#), etc.), or derived from co-occurrence rates with diagnostic adverbs similarly identified or from degree (i.e. synthetic comparative or superlative adjectives) or negation (e.g., *un-* vs. *non-*) morphology. Composite meta-categories could also be derived for bigrams based on the combination of features of their adverb and adjective independently.

7.1.2 Enhanced Resource Construction

In addition to assessing different perspectives already present in the current instantiation of PBR, alternate formulations of PBR or something akin to it could be developed. While these may not inherently be an “enhancement” alternate attempts would be edifying if only for the purpose of eliminating blind spots.

One such iteration could reconfigure the designation of polarity approximations. Rather than two overlapping datasets of binary polarity designations that are evaluated independently—distinct [AM](#) calculations for each of $*S$ and $@S$, despite all tokens of $@S$ also being accounted for in $*S$ —a quaternary polarity designation could be applied to a unified set. That is, each lexical form would have four distinct

joint frequencies for the four possible combinations of polarity sign ((+) or (-)) and subset membership ($\in @S$ or $\notin @S$): $\{@P, @N, *N - @N, *C_N - @P\}$, where the third and fourth categories are all negative and positive tokens excluded from the subset respectively. This manipulation could enable a more transparent investigation of the differences between the polarity approximation methods by providing a comparisons where one set does not subsume the other. It would also have direct implications for the interpretation of **AMs**: Since the environment would no longer be a binary choice, **LRC**, **dP1**, and **P1** values for any single environment (of four) cannot be inferred from the corresponding value for only one other environment (of four). That is, in this configuration, $\Delta P(@P|exactly) \neq -1 \times \Delta P(@N|exactly)$.

That being said, there is no guarantee that this would be an improvement. While it could elucidate the differences between polarity approximations, the designations themselves are motivated more by practical concerns than natural classes and comparing groupings that lack a theoretical basis could simply occlude overall patterns, the typical objective of polarity research.

More fundamental changes to data processing can be as generic but extensive as running the entire processing pipeline on different corpora, or adapting the pipeline to identify both bigrams and polarity environments (i.e. relationship to polarity operators) from constituency parsed—rather than **UD** parsed—corpora. The current PBR does not directly encode scope of polarity environment. Explicit constituency annotations might make such a task more feasible. Alternatively, as a first though incomplete step in this direction using the existing **UD** parsing, the classifications

collected in PBR could be added to the original CoNLL-U formatted corpora as extra features for the indicated nodes. While this would not completely delineate environment scope, it would assist further processing by enabling additional pattern matching based on polarity features. Along these same lines, dependency parses could be evaluated directly for logical entailment patterns (Chen & Gao 2021).

Finally, a full reprocessing from start to finish would provide the opportunity to resolve the various issues illuminated by trial and error during the course of development more gracefully than done by the workarounds used in the current version. In addition to avoiding pitfalls, all cleaning filters that were eventually applied could be preemptively applied at the beginning of processing, and the “clean” bigram base (the end product of Chapter 5’s methods) could serve as the search area for the context-informed polarity patterns, rather than needing to update the selections after the fact. The more relevant search area might also facilitate the development of more nuanced patterns to adequately account for the indirect relations detailed in Chapter 5 which were excluded from the current iteration, such as negated subjects and neg-raising, as well as weaker licensing conditions such as conditionals, comparatives, and hypotheticals.

7.2 Concluding Remarks

Ultimately, this dissertation has delivered a transparent, adaptable pipeline for extracting and quantifying polarity environments across multiple analytical layers

in large UD-parsed corpora, and has shown—through the *exactly* case study—how multi-layer comparison can uncover the nuanced interactions underlying generalized polarity sensitivity patterns. With the Polar Bigrams Resource now available for further exploration, future research can leverage these methods and data to deepen our understanding of polarity licensing and its interplay with lexical semantics.

APPENDIX A

CHAPTER 2 APPENDIX

This appendix pertains to [Chapter 2](#).

A.1 Polarity Approximation Implications

The method of polarity approximation affects the theoretical implications of any patterns found in the results. The primary differences are due to (a) the total *number of tokens* identified for each environment and (b) the specific *polarity licensing profiles* being compared.

A.1.1 Sample Size Implications

The total bigrams analyzed—per environment and overall—is greatly reduced when polarity approximation relies on positive evidence only, since there are far more cases that are either unmarked entirely (the norm for positive polarity environments) or which are marked by a trigger not explicitly specified in the collection pattern. That is, there are far fewer bigram tokens that match **any positive evidence**, “specific trigger present”, `@N∪@P, mirror (ANYmirror195)` environment specification than are included in the entire bigram baseline (regardless of the frequency filter applied). Restricting both the negative and comparison datasets vastly reduces the number of tokens considered.

To be precise, the Mirror Subset of PBR has a combined sample size, N , of 1.7 million compared to the 72 million bigram tokens in the PBR Superset (which includes every baseline token). This is only 2.34 % of the total bigrams identified, meaning 97.66 % of the bigram tokens do not have a direct dependency relationship with a preceding trigger in the specified set.

In addition to this overall reduction to N , the comparison/positive portion is impacted to a much greater extent, which has the additional—potentially beneficial—effect of reducing the size discrepancy between the environment subtotals.

¹⁹⁵The conglomerate “trigger present” token set, consisting of tokens matching *either* `POSmirror` or `NEGmirror` specifications, is referred to as `ANYmirror` in later discussion and data processing.

Both approximated polarity sets are reduced. The @N (NEGmirror) specifications exclude negative polarity triggers that do not have a clear positive polarity mirror (e.g. exclude *not*; includes *nothing*) while the @P (POSmirror) specifications require the presence of one of those positive polarity mirror triggers (e.g., *something*, mirror of *nothing*). The @P set (POSmirror results) accounts for only $\sim 2\%$ of the potential non-negated tokens in $*C_{N+}$, and @N set (NEGmirror results) accounts for only $\sim 9\%$ of the $*N^+$ tokens.

The effect of polarity approximation method on the relative polarity proportions may seem negligible at first glance. PBR as a whole—the superset, relying on negative evidence to designate positive polarity—is 95.59 % positive polarity ($*C_{N+}$) while the Mirror Subset is 82.64 % positive polarity (@P). A difference of $\sim 13\%$ does not seem very striking.

However, this should be contextualized by the overall decrease in N , the reduction in total bigram tokens from the Superset to the Subset. Of the 72 million bigram tokens in PBR (of either polarity designation), only 1.7 million tokens (2.3 %) remain in the Mirror Subset, a $\sim 97.7\%$ reduction. Of the 70 million¹⁹⁶ tokens dropped, $\sim 95.9\%$ were members of the comparison set ($*C_{N+}$).

That is, the size discrepancy between the approximated polarities is *reduced* by 98.3 % going from a difference of 66 million¹⁹⁷ when relying on negative evidence (superset) to a difference of “only” 1.1 million¹⁹⁸ when relying on positive evidence (Mirror Subset). The distribution is far from balanced for either approximation approach, but is much *more* balanced when the approximated “positive polarity” set is designated by positive evidence. This was demonstrated by Figure 2.10 in § 2.1.3.

A.1.2 Polarity Licensing Implications

As mentioned above, how the polarity of a bigram token’s environment is approximated affects the precise licensing profile represented by each category. While the objective of both approaches is to sort the collected tokens into the binary categories of either (1) “*occurring in a **negative** polarity environment*”, or (2) “*occurring in a **positive** polarity environment*”, these are still imperfect designations. This leads to comparisons of binary categories representing environments with slightly different semantic—and syntactic—properties.

This difference is most apparent in where wNPIs are licensed, as discussed previously.

¹⁹⁶A total of 70,280,740 tokens are excluded from the Mirror Subset.

¹⁹⁷Polar discrepancy in Superset = 65,614,011

¹⁹⁸Polar discrepancy in Mirror Subset = 1,097,163

The negative category licenses **wNPIs** regardless of approximation method: both are defined by relationships with known indicators of strong negative polarity licensing, which subsumes weak negative polarity licensing. The comparison category (“not negated” or “positive”) is where the approaches differ on the property of **wNPI** licensing: the “not negated” $*C_{N+}$ data are more likely to include weakly negative environments than the “positive” $@P$ data. This is a hedged statement because there are environments known to license **wNPIs** that were not explicitly captured and as a result not excluded from either comparison category.

In fact, there are cases that can license *strong* NPIs that were not captured in the $*N^+$ data—certain cases (neg-raising and negated subjects) create strong negative polarity environments (Gajewski 2007, 2011; Horn 2020), but do not have a *direct* dependency with the predicate (where a **sNPI** would be licensed). That is, the relationship between the strong negative trigger and the bigram is comprised of one or more dependencies, such that the properties required by **sNPIs** are not blocked/still reach the predicate structure. These cases are much less frequent than the direct negation dependency, but it should be noted that they will show up—erroneously—in the $*C_{N+}$ data. The $@P$ specifications, however, explicitly reject any tokens with negative triggers potentially affecting the polarity of the bigram’s environment.

At this point, I do not have the means to track other weak NPI licensing environments which are defined by more complex syntactic configurations (questions, hypotheticals, conditionals), morphological features (comparative and superlative constructions), or even pragmatic effects (rhetorical questions or “leading” statements, contextually supplied or accommodated negative implicatures or presuppositions). As a result, neither approximation approach treats these with finesse—any such case will be categorized as $*C_{N+}$ or $@P$, depending on method. However, the explicit inclusion of positive quantification in the $@P$ specifications reduces the likelihood of co-existing **wNPI** licensing environments, because (unfocused¹⁹⁹) existential quantification (e.g. *something*) is dis-preferred in such contexts, if not outright infelicitous—English *any** is the preferred quantifier in such cases.²⁰⁰

Although accuracy could be enhanced, the explicit exclusion of negative trigger lemmas and explicit *inclusion* of positive trigger lemmas in the **mirror**/positive evidence based/trigger-present approach does partially mitigate the potential appearance of NPI

¹⁹⁹**some**~F~ seems more felicitous in questions and conditionals than a neutral *some* or reduced *s’m*. This makes sense theoretically since it has been proposed that the distinction between the semantics of *any* and the semantics of *some* is the level of emphasis/exhaustification. I.e., $[\text{some}_F] \simeq [\text{any}]$

²⁰⁰I believe this is true for both positive existential and positive universal quantification, but I can’t claim either with full certainty. And any cases where focus is active but not encoded in the text representation will slip under the radar as well.

licensing environments in the @*P* data. Since no such attempts were made with the * $\mathcal{L}_{N+}$ data, I expect it to contain far more environments with the ability to license weak (if not strong) NPIs relative to @*P*.

As a final note, the restriction of both @*N* and @*P* to mirroring negative and positive trigger elements, respectively, introduces the potential for a syntactic selection effect. The most common negation trigger lemma seen in * $\mathcal{L}_{N+}$ is *not* (encapsulating case-normalized forms of both *not* and *n't*), but there is no parallel operator to indicate the logical inverse of the same syntactic configuration—privation and existence are most often distinguished by marking the privative case while existence, as the default interpretation, is left unmarked.

- (A.1) a. Your cat is [$\oslash$ [on the roof]].
 b. Your cat is [**not** [on the roof]].
 c. The tin roof is [$\oslash$ [quite hot]].
 d. The tin roof is [**not** [quite hot]].²⁰¹

To keep @*N* and @*P* as parallel as possible, only operators that had both a negative and positive representative were included. For example:

- (A.2) a. + **Some** cat is [$\oslash$ [on the roof]].
 b. + **Every** cat is [$\oslash$ [on the roof]].
 c. – **No** cat is [$\oslash$ [on the roof]].
 (A.3) a. + **Somebody** is [$\oslash$ [on the roof]].
 b. + **Everybody** is [$\oslash$ [on the roof]].
 c. – **Nobody** is [$\oslash$ [on the roof]].

²⁰¹This specific bigram, *quite hot*, illustrates an additional difficulty in applying corpus methods to semantic inquiries: homophony or lexical ambiguity, yes, but this goes beyond that. The adverb *quite* can occur in both positive and negative environments—so it doesn't fit the classification of either PPI or NPI—but it is still *polarity sensitive* in that it yields a different semantic computation depending on polarity of its environment. In the positive context, *quite hot* can be paraphrased as *rather hot* or *moderately hot*, or an understated/attenuated way of saying “a noticeable degree of heat”/“definitely qualifies as hot, but let's downplay it” But in the negative context, the same paraphrases don't work: *The roof is not quite hot* $\neq$ *The roof is not { #rather, moderately } hot*. When negated, *quite* seems to convey that the minimum degree for the selected predicate to hold (e.g. the minimum temperature required for tin to qualify as “hot”) exceeds the current degree by a minor amount (and has not been met). Instead of the expected simple inversion of “it is not the case that the roof is moderately hot,” which would still be a true statement if the roof were “barely hot” or “only very slightly hot”, this example suggests that the roof is not hot at all.

- (A.4) a. + **Some** roofs are [$\oslash$ [extremely hot]].
 b. + **All** roofs are [$\oslash$ [extremely hot]].
 c. – **No** roofs are [$\oslash$ [extremely hot]].
- (A.5) a. + **Everybody** was [$\oslash$ [extremely smart]].
 b. – **Nobody** was [$\oslash$ [extremely smart]].
- (A.6) a. + **Something** particularly interesting [$\oslash$ [happened last week]].
 b. + **Everything** particularly interesting [$\oslash$ [happened last week]].
 c. + **Nothing** particularly interesting [$\oslash$ [happened last week]].

The end result is that the @*N* and @*P* bigram tokens are limited to those that have some kind of quantitative element in their preceding context. While it is possible that this feature will have no effect on the frequency distributions, it could affect the marginal frequencies of the Mirror Subset. Since the quantity restriction is present in *both* approximated polarity subsets—negative polarity approximation, @*N*, and positive polarity approximation, @*P*—it should not bias the environment-specific co-occurrence frequencies, but this potential confound needs to be considered when comparing the results of the **mirror** subset (@*N* vs. @*P*) to the Superset ($*N^+$ vs. $*C_{N+}$). See below for further discussion of the comparison of results from the different approximation methods.

Despite the nuances of these caveats, it is clear that the binary choice that the different analyses investigate is not strictly equivalent. Essentially, one—the superset method, which assigns tokens to the comparison subset by just the *absence of evidence* of privation/strong negative polarity—is a “choice” between strong negative environments (N^+) and anything else ($*C_{N+}$), while the other—the mirror subset method, which assigns tokens to the comparison subset only by the *presence of evidence* of positive polarity—is a “choice” between negative and positive environments. This difference impacts any conclusions that can be drawn based on the results.

The trigger-absent Superset— $*N^+$ vs. $*C_{N+}$ —can provide frequency evidence for how often a given bigram co-occurs with strong negative polarity environment vs. when it does not. In other words, this method’s results can help answer the question, “*What is sensitive to strong negative polarity environments?*”. In a practical sense, this means that *weak* negative polarity sensitivities should be minimized/less apparent in these results because both environment sets permit **wNPIs**²⁰².

²⁰²Strongly negative environments still license **wNPIs** and weakly negative environments are included in the “not strong” complement set

On the other hand, the Mirror Subset—@*N* vs. @*P*—provides evidence for how often a given bigram co-occurs with a negatively quantified environment versus how often it co-occurs with a positively quantified environment. It can best answer the question, “*What is sensitive to polarity sign (positive or negative) of the environment?*”. In this approach, wNPIs (*N*[−]) are excluded from the comparison class of @*P* (at least partially), while still permitted in @*N*. Despite there still being some chance that questions, conditionals, or comparatives still appear *alongside* the explicitly positive quantification operators in @*P*, the odds of weak NPI licensing are higher for @*N* than @*P*, and higher for @*N* relative to @*P* than for **N*⁺ relative to **C*_{*N*+}. The strong negative (*explicitly* negative) triggers were all directly excluded from @*P*, just as the selected positive operators were excluded from @*N* and there may be theoretical reasons to believe that, even if a *N*[−] licenser co-occurs with positive quantification, the polarity of the bigram itself will depend on which element is closer, and @*P* requires a *direct* dependency relation between the bigram and the positive node.

Thus a bigram showing a strong significant effect in the **mirror** approach is a good candidate for a PSI—PPI if more prevalent in @*P* or NPI if more prevalent in @*N*—but no specific distinction of negation strength can be made.

The association metrics will therefore indicate different things for each polarity approximation method. Under the absent negative trigger approach, where the comparison set is defined by the **absence** of a *strong negative trigger*, association metrics can be interpreted as follows for any given lexical item (bigram, adverb, or adjective). An increased/positive association with (strongly) negated environments, ↑₍₊₎ **N*⁺²⁰³, indicates it occurred disproportionately in these strongly negative contexts. This suggests not only that it has an increased likelihood of polarity sensitivity (in general), but also that it is a *strong* sensitivity, and will only be licensed in **strong** (*N*⁺)—but not *weak* (*N*[−])—negative polarity.

For the same data, an increased/positive association with the comparison environment set means the item occurred disproportionately often in a weak negative context or a positive context. Since both positive and negative polarity are possibilities/the distinction between them is not discernible from the given data, no **transparent** suggestion of polarity sensitivity can be made. Such a result could be evidence of a very strong association with positive (**P*) environments (a strong candidate for a PPI or perhaps a candidate for a *strong PPI*) that is apparent even with the dampening effect of the **N*[−] environment “pollution”. However, it could also easily be equally acceptable in either subset—positive *or* weakly negative. This latter potentiality would necessarily not meet the criteria for *polarity* sensitivity, though any such items might be evidence that the feature of environment

²⁰³or a significant *negative* association with the comparison set, ↓_(−) **C*_{*N*+}

polarity is not sufficiently explanatory for the given data.

Alternatively, under the **mirror** approach to approximating polarity, where the comparison set is defined instead by the **presence of a positive trigger**, an increased/positive association with negated environments, $\uparrow_{(+)} @N$, indicates it occurred much more often in the sample of negative contexts than in the sample of positive contexts. This suggests it has an increased likelihood of polarity sensitivity: It requires a negative environment, but there is no additional implication for the strength of the licensing environment.

Similarly, an increased/positive association with the positive environment set, $\uparrow_{(+)} @P$, indicates a disproportionate occurrence in positive contexts. Such an outcome suggests an increased likelihood of polarity sensitivity—specifically, an increased likelihood of being a *positive* polarity item.

Under both approaches to approximating polarity of the collected bigram tokens, an increased association with the comparison set (“positive” or “not negated”) is less likely to be found significant/show evidence of a strong effect, just due to the size discrepancy between the “negative” and its respective “comparison”.

That is, it is much easier to see evidence of “disproportionate” co-occurrence with the comparatively tiny set, because the expected frequency would be correspondingly low. It is much harder to see evidence of a “disproportionate” co-occurrence with the overwhelmingly dominant set, because the expected frequency would be correspondingly high (a much higher bar to meet).

Many bigrams will occur in both approximations, and comparing any given lexical form’s patterns in the two approaches may provide additional insight.

Any *contradicting* outcomes, “C~N” or “N~C” above, are not expected to happen at all. If they do, the particular case warrants a closer manual inspection—it could be due to different word senses being employed or semi-lexicalized/idiomatic/non-compositional phrasing that pattern differently under each approach’s constraints. However, if such an explanation cannot be found, the approximation specifications may be seriously flawed, indicating that any other association values are suspect as well.

Due to the increased likelihood of identifying **wNPI** sensitivities in the **mirror** approach, a bigram that exhibits a strong skew toward @N but a relatively weaker preference for N^+ , represented by the “=~N” column above, may be a **wNPI**. In a similar vein, if a bigram

Table A.1: Interpreting Outcomes for Polar Approximation Methods

Potential Outcomes for any given lexical unit (bigram or component word)

The outcome for each method are simplified to 3 categories:

“+N”: association with the negative polarity approximation (@N or *N⁺)

“+C”: association with the comparison polarity approximation (@P or *C_{N+})

“=”: no evidence of association (toward either)

approach	$N \sim N$	$N \sim =$	$N \sim C$	$= \sim =$	$= \sim N$	$= \sim C$	$C \sim C$	$C \sim =$	$C \sim N$
*E	+N	+N	+N	=	=	=	+C	+C	+C
@E	+N	=	+C	=	+N	+C	+C	=	+N

shows a strong preference for N^+ without a corresponding pattern with @N (the “N~=” column), has an increased likelihood of being an sNPI.²⁰⁴

However, both conclusions necessitate ascertaining that there is not some other factor of variation at work between the bigram baseline and the ANYmirror subset. For example, it could be an issue of the specific sample or frequencies—the specific bigram’s marginal frequency in each case, or perhaps a result of the change in the overall sample size—or there could be another, more predictive/predictable factor such as quantifier usage or syntactic constraints that prevent the same prevalence in @N as observed in N^+ . Recall that @N does not include the most common negation trigger, sentential/phrasal negation via the lemma *not*: if a bigram occurs commonly with *not* but rarely with any other negative trigger, while the mirror data would not contain the necessary information to identify it as an NPI, it could very well still be one. That is, a hypothetical bigram, *b*, with the following observed frequency distribution, the question would not be *Is b an NPI?*, but rather, **Why** does *b* disprefer the less common negation triggers—what features are at play that have not been identified, captured, and quantified?

token set	example triggers	<i>b</i> tokens
@N	<i>nobody, nothing, none</i>	100
@P	<i>somebody, everybody, something, everything</i>	50
$N^+ - @N$	<i>not, n't</i>	5,000
$CN^+ - @P$	[unmarked]	200

The negative association patterns can be complicated to interpret due to the *strength* of

²⁰⁴It would be worth investigating *why* it patterns differently, since sNPIS should be licensed by both *N⁺ and @N, and prohibited in both *C_{N+} and @P.

the given representation, interpreting the paired outcomes for the *comparison* environments becomes complicated for a different reason: The parameter of *strength* is not well defined for positive polarity, but the approximations constructed here can be conceptualized as different “strengths” of a sort. @*P* consists of explicitly positive contexts (i.e. with explicit positive quantity) while * $\mathcal{C}_{N+}$ contains these tokens *plus* other tokens that are either unmarked for polarity (and are thus positive usually), or are weakly negative, so, in a sense, these can represent strong and weak positive polarity respectively. A preference for the explicitly positive quantified contexts *and* a preference for that plus some cases that are “neutral” or weakly negative—the corroborated “ $C \sim C$ ” case—actually indicates a lower likelihood of PPI-hood than a preference for the explicitly positive cases of @*P* alone—the “ $= \sim C$ ” outcome pair.

Given that the sign of the polarity of * $\mathcal{C}_{N+}$ is undefined/inconsistent (these tokens could have positive or negative polarity), while the sign of @*P* is defined (positive polarity), the potential outcome pair of “ $C \sim =$ ” is difficult to interpret, as it would essentially mean that *diluting* the polarity consistency makes the set *more attractive* to the bigram or word in question. This would seem to suggest that polarity, or at least the sign of the polarity environment, is not the determining factor for the item’s distribution. There must either be some other systematic difference between the polarity approximations, or there is a random sampling effect for this particular item²⁰⁵.

A.2 Trigger Lemma vs. Case-Normalized Forms

Table A.2: Mirror Positive Trigger Lemma Frequencies By Lowercase Form

Trigger Lemma	Trigger Lower	lemma total	trigger lower total
something	something	309,632	309,483
	somethings	309,632	131
	somethin	309,632	18
or	or	304,971	304,970
	ors	304,971	1
all	all	243,962	243,951
	alls	243,962	10

Continued on next page

²⁰⁵but given the overall size of both datasets, this seems unlikely. Despite the **mirror** subset being less than 3% of the PBR superset, it is still a large sample of corpus data.

Table A.2: Mirror Positive Trigger Lemma Frequencies By Lowercase Form

Trigger Lemma	Trigger Lower	lemma total	trigger lower total
some	alleg	243,962	1
	some	107,634	107,631
	somes	107,634	2
	someing	107,634	1
often	often	82,539	82,536
	oftens	82,539	2
	oftenleft	82,539	1
always	always	82,238	82,238
sometimes	sometimes	64,150	64,150
both	both	49,318	49,318
many	many	40,204	40,203
	manies	40,204	1
everything	everything	37,848	37,831
	everythings	37,848	16
	everythiing	37,848	1
everyone	everyone	31,844	31,829
	everyones	31,844	15
someone	someone	27,628	27,618
	someones	27,628	9
	someon	27,628	1
everybody	everybody	7,894	7,889
	everybodys	7,894	4
	everbody	7,894	1
either	either	3,655	3,654
	eather	3,655	1
somebody	somebody	2,473	2,470
	somebodies	2,473	3
every	every	334	334

Table A.3: Superset Negative Trigger Lemma Frequencies By Lowercase Form

Trigger Lemma	Lemma Total	Lower Form	Form Total
		<i>not</i>	2,005,270
		<i>n't</i>	863,289
		<i>ain't</i>	2,052
		<i>nit</i>	608
Continued on next page			
<i>not</i>	2,871,807		

Table A.3: Superset Negative Trigger Lemma Frequencies By Lowercase Form

Trigger Lemma	Lemma Total	Lower Form	Form Total
		<i>aint</i>	270
		<i>nt</i>	155
		<i>doe</i>	32
		<i>n't</i>	30
		<i>na</i>	16
		<i>oud</i>	15
		<i>none</i>	14
		<i>OTHER</i> ²⁰⁶	56
never	111,089	<i>never</i>	111,089
		<i>nothing</i>	100,482
nothing	100,513	<i>nothings</i>	30
		<i>nothingis</i>	1
		<i>none</i>	31,422
none	31,453	<i>nones</i>	30
		<i>non-exalts</i>	1
nor	15,925	<i>nor</i>	15,925
without	14,825	<i>without</i>	14,814
		<i>w/o</i>	11
no	12,438	<i>no</i>	12,437
		<i>nys</i>	1
neither	6,528	<i>neither</i>	6,528
		<i>nobody</i>	6,357
nobody	6,365	<i>nobodies</i>	6
		<i>nobody's</i>	1
		<i>nobodys</i>	1
hardly	5,525	<i>hardly</i>	5,525
rarely	4,474	<i>rarely</i>	4,474
		<i>few</i>	4,015
few	4,289	<i>fewer</i>	273
		<i>fewest</i>	1
barely	1,231	<i>barely</i>	1,231
seldom	1,039	<i>seldom</i>	1,039
scarcely	796	<i>scarcely</i>	796

²⁰⁶*OTHER* category comprised of 11 distinct lowercase forms (with lemma *not*) which have a marginal frequency < 10: *n*, *nots*, *\t*, *nto*, *n-t*, *nat*, *n'y*, *n't-*, *n't*, *nod*, *ane*, *amp*, *dne*, *ima*, *n'a*, *n'l*, *itl*, *nodd*, *owo*

Table A.4: Mirror Negative Trigger Lemma Frequencies By Lowercase Form

Trigger Lemma	Trigger Lower	lemma total	trigger lower total
never	never	111,102	111,102
	nothing	100,044	100,012
nothing	nothings	100,044	31
	nothingis	100,044	1
none	none	31,248	31,215
	nones	31,248	32
	non-exalts	31,248	1
nor	nor	15,944	15,944
no	no	9,175	9,174
	nys	9,175	1
neither	neither	6,491	6,491
	nobody	6,361	6,353
nobody	nobodies	6,361	6
	nobodys	6,361	1
	nobody's	6,361	1
hardly	hardly	5,527	5,527
rarely	rarely	4,480	4,480
barely	barely	1,245	1,245
seldom	seldom	1,042	1,042
scarcely	scarcely	797	797

A.3 Association Measure Tables

A.3.1 Superset NEQ+ Context-Blind AMs

Table A.5: Top NEQ+ Superset Context–Blind Bigrams by Assorted AMs, by descending raw joint frequency (N=6,347,362)

#	Bigram	Top For	f	LRC	dPavg	dP1	P1	dP2	P2	G2	f1	f2	unexpF	unexp%
1	<i>as good</i>	f; unexpF	51,552	1.5	0.10	0.16	0.26	0.05	0.08	38,950	656,683	202,048	30,649	59.5
2	<i>so many</i>	f; G2; unexpF	50,211	3.4	0.26	0.43	0.51	0.08	0.09	119,575	566,272	97,776	41,488	82.6
3	<i>very good</i>	f; unexpF	49,486	1.6	0.11	0.15	0.25	0.06	0.08	39,822	606,530	202,048	30,179	61.0
4	<i>so much</i>	f; unexpF	44,925	2.8	0.20	0.33	0.42	0.07	0.08	85,188	566,272	107,549	35,330	78.6
5	<i>most important</i>	f	32,595	2.4	0.13	0.18	0.24	0.08	0.09	52,166	346,992	138,658	25,015	76.7
6	<i>too late</i>	dP1; P1; G2; unexpF	29,868	7.9	0.48	0.89	0.95	0.07	0.07	150,893	433,149	31,301	27,732	92.8
7	<i>immediately clear</i>	G2	24,476	5.4	0.35	0.29	0.29	0.42	0.42	141,538	57,730	83,958	23,712	96.9
8	<i>much better</i>	G2	22,093	5.2	0.29	0.43	0.44	0.16	0.17	107,450	131,845	49,936	21,056	95.3
9	<i>immediately available</i>	G2	21,281	5.2	0.31	0.25	0.26	0.36	0.37	115,832	57,730	81,972	20,535	96.5
10	<i>very first</i>	P1	6,431	7.0	0.43	0.85	0.95	0.01	0.01	27,479	606,530	6,798	5,781	89.9
11	<i>mutually exclusive</i>	dPavg	5,727	11.4	0.78	0.79	0.79	0.77	0.77	76,266	7,437	7,253	5,719	99.9
12	<i>too shabby</i>	dP1; P1	5,397	7.9	0.45	0.89	0.96	0.01	0.01	27,241	433,149	5,613	5,014	92.9
13	<i>very own</i>	dP1; P1	4,076	8.0	0.45	0.89	0.98	0.01	0.01	18,353	606,530	4,161	3,678	90.2
14	<i>very least</i>	dP1; P1	3,497	8.5	0.45	0.89	0.99	0.01	0.01	15,993	606,530	3,539	3,159	90.3
15	<i>partly cloudy</i>	LRC; dPavg; unexp%	2,509	12.6	0.75	0.82	0.82	0.68	0.68	36,822	3,710	3,048	2,507	99.9
16	<i>closely related</i>	dP2; P2	1,925	8.6	0.51	0.14	0.14	0.89	0.89	22,266	2,160	14,257	1,920	99.7
17	<i>wide open</i>	dP2; P2	1,343	8.4	0.50	0.09	0.09	0.91	0.91	15,348	1,469	15,884	1,339	99.7
18	<i>aesthetically pleasing</i>	unexp%	980	10.9	0.52	0.36	0.36	0.68	0.68	13,766	1,448	2,739	979	99.9
19	<i>terminally ill</i>	dP2	973	9.7	0.52	0.15	0.15	0.89	0.89	12,809	1,092	6,466	972	99.9
20	<i>long overdue</i>	LRC	912	11.7	0.49	0.82	0.82	0.16	0.16	11,929	5,576	1,117	911	99.9
21	<i>vitality important</i>	P2	772	5.1	0.44	0.01	0.01	0.87	0.89	5,317	866	138,658	753	97.6
22	<i>minimally invasive</i>	LRC; dPavg; unexp%	753	12.0	0.62	0.45	0.45	0.79	0.79	11,841	948	1,683	753	100.0

Continued on next page

Table A.5: Top NEQ+ Superset Context–Blind Bigrams by Assorted AMs, by descending raw joint frequency

#	Bigram	Top For	f	LRC	dPavg	dP1	P1	dP2	P2	G2	f1	f2	unexpF	unexp%
23	<i>legally binding</i>	LRC; dPavg; dP1	664	13.0	0.53	0.92	0.92	0.15	0.15	9,313	4,552	722	663	99.9
24	<i>crystal clear</i>	dP2; P2	406	5.8	0.49	0.01	0.01	0.97	0.98	3,436	414	83,958	401	98.7
25	<i>scantily clad</i>	LRC; dPavg; dP2; P2; unexp%	244	15.0	0.84	0.70	0.70	0.98	0.98	4,984	248	348	244	100.0
26	<i>independently wealthy</i>	unexp%	235	10.1	0.32	0.10	0.10	0.54	0.54	3,150	432	2,313	235	99.9

A.3.2 Adverb and Adjective AM tables corresponding to selection provided for frequency and versatility comparisons

Table A.6: *S ALL+ Polar AMs for Union of Top ADV Frequency and Versatility

Metric	f		P1		dP1		LRC		G2		f2
Polarity	neg	pos	neg	pos	neg	pos	neg	pos	neg	pos	
L2											
<i>very</i>	189,823	9,030,189	0.02	0.98	−0.03	0.03	−1.11	1.11	−167,026	167,026	9,220,012
<i>more</i>	161,916	8,497,762	0.02	0.98	−0.03	0.03	−1.26	1.26	−185,605	185,605	8,659,678
<i>most</i>	19,204	7,101,224	0.00	1.00	−0.05	0.05	−4.03	4.03	−525,721	525,721	7,120,428
<i>so</i>	344,146	4,829,392	0.07	0.93	0.02	−0.02	0.61	−0.61	58,503	−58,503	5,173,538
<i>as</i>	531,731	2,699,909	0.16	0.84	0.13	−0.13	2.08	−2.08	727,573	−727,573	3,231,640
<i>too</i>	309,669	2,667,071	0.10	0.90	0.06	−0.06	1.32	−1.32	197,292	−197,292	2,976,740
<i>really</i>	97,510	1,847,878	0.05	0.95	0.01	−0.01	0.17	−0.17	1,651	−1,651	1,945,388
<i>much</i>	53,022	1,707,596	0.03	0.97	−0.01	0.01	−0.53	0.53	−9,361	9,361	1,760,618
<i>pretty</i>	1,956	1,505,944	0.00	1.00	−0.04	0.04	−4.97	4.97	−119,539	119,539	1,507,900
<i>less</i>	13,450	1,125,336	0.01	0.99	−0.03	0.03	−1.88	1.88	−39,786	39,786	1,138,786
<i>also</i>	1,966	1,062,460	0.00	1.00	−0.04	0.04	−4.46	4.46	−80,113	80,113	1,064,426
<i>quite</i>	78,500	910,255	0.08	0.92	0.04	−0.04	0.87	−0.87	24,225	−24,225	988,755
<i>always</i>	103,883	500,214	0.17	0.83	0.13	−0.13	2.14	−2.14	141,065	−141,065	604,097
<i>just</i>	54,583	543,355	0.09	0.91	0.05	−0.05	1.09	−1.09	24,672	−24,672	597,938
<i>particularly</i>	55,527	456,207	0.11	0.89	0.06	−0.06	1.36	−1.36	36,700	−36,700	511,734
<i>almost</i>	397	433,210	0.00	1.00	−0.04	0.04	−5.25	5.25	−35,325	35,325	433,607
<i>only</i>	113,502	319,110	0.26	0.74	0.22	−0.22	2.92	−2.92	242,355	−242,355	432,612
<i>increasingly</i>	73	373,560	0.00	1.00	−0.04	0.04	−7.00	7.00	−32,850	32,850	373,633
<i>that</i>	164,768	42,033	0.80	0.20	0.75	−0.75	6.36	−6.36	831,670	−831,670	206,801
<i>yet</i>	51,867	42,888	0.55	0.45	0.50	−0.50	4.66	−4.66	197,895	−197,895	94,755
<i>immediately</i>	56,099	35,647	0.61	0.39	0.57	−0.57	5.04	−5.04	231,722	−231,722	91,746
<i>exactly</i>	43,813	12,296	0.78	0.22	0.74	−0.74	6.19	−6.19	216,170	−216,170	56,109
<i>necessarily</i>	42,595	6,046	0.88	0.12	0.83	−0.83	7.14	−7.14	230,466	−230,466	48,641
<i>albeit</i>	3	15,663	0.00	1.00	−0.04	0.04	−4.71	4.71	−1,374	1,374	15,666
<i>preferably</i>	1	4,781	0.00	1.00	−0.04	0.04	−3.30	3.30	−419	419	4,782
<i>legendarily</i>		1,007		1.00		0.04		1.26		91	1,007
<i>yearly</i>		2,474		1.00		0.04		2.56		223	2,474

Table A.7: *S ALL+ Polar AMs for Union of Top ADJ Frequency and Versatility

Metric	f		P1		dP1		LRC		G2		f2
Polarity	neg	pos	neg	pos	neg	pos	neg	pos	neg	pos	
L2											
<i>important</i>	49,097	1,952,845	0.02	0.98	-0.02	0.02	-0.84	0.84	-22,080	22,080	2,001,942
<i>many</i>	13,930	1,821,912	0.01	0.99	-0.04	0.04	-2.52	2.52	-89,076	89,076	1,835,842
<i>good</i>	130,954	1,539,168	0.08	0.92	0.04	-0.04	0.86	-0.86	39,311	-39,311	1,670,122
<i>much</i>	48,267	1,287,855	0.04	0.96	-0.01	0.01	-0.26	0.26	-2,183	2,183	1,336,122
<i>likely</i>	9,528	874,621	0.01	0.99	-0.03	0.03	-1.99	1.99	-33,382	33,382	884,149
<i>more</i>	8,417	806,911	0.01	0.99	-0.03	0.03	-2.05	2.05	-31,852	31,852	815,328
<i>different</i>	45,138	757,054	0.06	0.94	0.01	-0.01	0.33	-0.33	2,630	-2,630	802,192
<i>popular</i>	17,028	747,995	0.02	0.98	-0.02	0.02	-0.95	0.95	-10,607	10,607	765,023
<i>difficult</i>	29,381	700,004	0.04	0.96	0.00	0.00	-0.09	0.09	-262	262	729,385
<i>available</i>	55,203	584,259	0.09	0.91	0.04	-0.04	1.00	-1.00	21,603	-21,603	639,462
<i>easy</i>	86,698	491,343	0.15	0.85	0.11	-0.11	1.90	-1.90	98,128	-98,128	578,041
<i>high</i>	19,371	499,260	0.04	0.96	-0.01	0.01	-0.19	0.19	-595	595	518,631
<i>bad</i>	104,709	321,088	0.25	0.75	0.20	-0.20	2.79	-2.79	210,059	-210,059	425,797
<i>close</i>	29,099	378,440	0.07	0.93	0.03	-0.03	0.68	-0.68	6,151	-6,151	407,539
<i>clear</i>	71,543	274,861	0.21	0.79	0.16	-0.16	2.46	-2.46	119,592	-119,592	346,404
<i>great</i>	32,815	273,436	0.11	0.89	0.06	-0.06	1.33	-1.33	21,070	-21,070	306,251
<i>beautiful</i>	7,051	285,121	0.02	0.98	-0.02	0.02	-0.80	0.80	-3,299	3,299	292,172
<i>low</i>	2,832	274,557	0.01	0.99	-0.03	0.03	-2.00	2.00	-10,874	10,874	277,389
<i>interested</i>	23,428	236,299	0.09	0.91	0.05	-0.05	1.04	-1.04	10,215	-10,215	259,727
<i>sure</i>	128,062	128,755	0.50	0.50	0.46	-0.46	4.40	-4.40	459,579	-459,579	256,817
<i>possible</i>	20,216	218,496	0.08	0.92	0.04	-0.04	0.94	-0.94	7,452	-7,452	238,712
<i>aware</i>	19,298	209,325	0.08	0.92	0.04	-0.04	0.93	-0.93	7,047	-7,047	228,623
<i>able</i>	13,840	196,946	0.07	0.93	0.02	-0.02	0.53	-0.53	2,038	-2,038	210,786
<i>enough</i>	21,582	126,223	0.15	0.85	0.10	-0.10	1.83	-1.83	23,291	-23,291	147,805
<i>wrong</i>	15,175	131,262	0.10	0.90	0.06	-0.06	1.25	-1.25	9,074	-9,074	146,437
<i>funny</i>	9,896	112,250	0.08	0.92	0.04	-0.04	0.84	-0.84	3,202	-3,202	122,146
<i>shabby</i>	5,529	1,540	0.78	0.22	0.74	-0.74	6.04	-6.04	27,252	-27,252	7,069
<i>rosy</i>	2,073	3,282	0.39	0.61	0.34	-0.34	3.53	-3.53	6,090	-6,090	5,355
<i>dissimilar</i>	2,511	2,087	0.55	0.45	0.50	-0.50	4.45	-4.45	9,530	-9,530	4,598
<i>farfetched</i>	564	732	0.44	0.56	0.39	-0.39	3.57	-3.57	1,812	-1,812	1,296
<i>chopped</i>		1,553		1.00		0.04		1.78		140	1,553
<i>earliest</i>		1,929		1.00		0.04		2.10		174	1,929
<i>evolving</i>		2,953		1.00		0.04		2.71		266	2,953
<i>northerly</i>		1,792		1.00		0.04		1.99		162	1,792
<i>southerly</i>		1,339		1.00		0.04		1.57		121	1,339

A.3.3 Mirror Subset Adverb and Adjective AM tables

The following tables correspond to the union of the three strongest values across each of the given AMs for each polarity.

A.3.3.1 Adverb AMs

Table A.8: @S:ALL+ Polar Adverb AMs for Union of Top

Metric Polarity	f		P1		dP1		LRC		G2		f2
	neg	pos	neg	pos	neg	pos	neg	pos	neg	pos	
<i>more</i>	78,325	207,448	0.27	0.73	0.12	-0.12	0.81	-0.81	22,015	-22,015	285,773
<i>very</i>	8,914	175,004	0.05	0.95	-0.14	0.14	-1.96	1.96	-29,010	29,010	183,918
<i>too</i>	46,648	111,487	0.29	0.71	0.13	-0.13	0.95	-0.95	15,766	-15,766	158,135
<i>even</i>	2,838	58,341	0.05	0.95	-0.13	0.13	-1.96	1.96	-9,473	9,473	61,179
<i>pretty</i>	127	24,525	0.01	0.99	-0.17	0.17	-4.70	4.70	-8,277	8,277	24,652
<i>particularly</i>	9,243	3,703	0.71	0.29	0.54	-0.54	3.42	-3.42	18,490	-18,490	12,946
<i>rather</i>	32	8,364	0.00	1.00	-0.17	0.17	-4.58	4.58	-2,890	2,890	8,396
<i>plain</i>	17	5,025	0.00	1.00	-0.17	0.17	-4.38	4.38	-1,751	1,751	5,042
<i>ever</i>	4,709	333	0.93	0.07	0.76	-0.76	5.65	-5.65	14,227	-14,227	5,042
<i>somewhat</i>	16	4,475	0.00	1.00	-0.17	0.17	-4.26	4.26	-1,553	1,553	4,491
<i>maybe</i>	9	2,654	0.00	1.00	-0.17	0.17	-3.91	3.91	-924	924	2,663
<i>any</i>	1,066	103	0.91	0.09	0.74	-0.74	4.87	-4.87	3,079	-3,079	1,169
<i>longer</i>	821	89	0.90	0.10	0.73	-0.73	4.65	-4.65	2,328	-2,328	910

A.3.3.2 Adjective AMs

Table A.9: @S:NEQ+ Polar Adverb AMs for Union of Top

Metric	f		P1		dP1		LRC		G2		f2
Polarity	neg	pos	neg	pos	neg	pos	neg	pos	neg	pos	
<i>more</i>	78,325	43,523	0.64	0.36	0.18	-0.18	0.80	-0.80	12,706	-12,706	121,848
<i>too</i>	46,648	23,197	0.67	0.33	0.19	-0.19	0.95	-0.95	9,100	-9,100	69,845
<i>very</i>	8,914	36,972	0.19	0.81	-0.33	0.33	-1.96	1.96	-19,893	19,893	45,886
<i>even</i>	2,838	12,140	0.19	0.81	-0.32	0.32	-1.94	1.94	-6,374	6,374	14,978
<i>particularly</i>	9,243	777	0.92	0.08	0.43	-0.43	3.30	-3.30	8,550	-8,550	10,020
<i>pretty</i>	127	5,123	0.02	0.98	-0.48	0.48	-4.69	4.69	-6,125	6,125	5,250
<i>ever</i>	4,709	67	0.99	0.01	0.49	-0.49	5.28	-5.28	5,953	-5,953	4,776
<i>rather</i>	32	1,774	0.02	0.98	-0.48	0.48	-4.60	4.60	-2,187	2,187	1,806
<i>any</i>	1,066	17	0.98	0.02	0.49	-0.49	4.40	-4.40	1,328	-1,328	1,083
<i>plain</i>	17	1,051	0.02	0.98	-0.48	0.48	-4.38	4.38	-1,308	1,308	1,068
<i>somewhat</i>	16	914	0.02	0.98	-0.48	0.48	-4.22	4.22	-1,129	1,129	930
<i>maybe</i>	9	546	0.02	0.98	-0.48	0.48	-3.88	3.88	-678	678	555
<i>before</i>	289		1.00		0.50		4.10		401		289

Table A.10: @S:ALL+ Polar Adjective AMs for Union of Top

Metric	f		P1		dP1		LRC		G2		f2
Polarity	neg	pos	neg	pos	neg	pos	neg	pos	neg	pos	
<i>different</i>	1,493	34,359	0.04	0.96	-0.13	0.13	-2.06	2.06	-6,011	6,011	35,852
<i>late</i>	11,049	2,230	0.83	0.17	0.66	-0.66	4.37	-4.37	27,848	-27,848	13,279
<i>familiar</i>	204	12,064	0.02	0.98	-0.16	0.16	-3.09	3.09	-3,254	3,254	12,268
<i>similar</i>	177	6,789	0.03	0.97	-0.15	0.15	-2.43	2.43	-1,564	1,564	6,966
<i>sure</i>	5,772	964	0.86	0.14	0.69	-0.69	4.56	-4.56	15,139	-15,139	6,736
<i>early</i>	5,175	428	0.92	0.08	0.75	-0.75	5.45	-5.45	15,337	-15,337	5,603
<i>larger</i>	29	2,376	0.01	0.99	-0.16	0.16	-2.79	2.79	-694	694	2,405
<i>unable</i>	9	1,226	0.01	0.99	-0.17	0.17	-2.72	2.72	-393	393	1,235
<i>unrelated</i>	4	893	0.00	1.00	-0.17	0.17	-2.69	2.69	-303	303	897
<i>darker</i>	1	544	0.00	1.00	-0.17	0.17	-2.39	2.39	-196	196	545

Table A.11: @S:NEQ+ Polar Adjective AMs for Union of Top

Metric	f		P1		dP1		LRC		G2		f2
Polarity	neg	pos	neg	pos	neg	pos	neg	pos	neg	pos	
<i>late</i>	11,049	452	0.96	0.04	0.47	-0.47	4.24	-4.24	12,328	-12,328	11,501
<i>different</i>	1,493	7,164	0.17	0.83	-0.33	0.33	-2.04	2.04	-4,097	4,097	8,657
<i>sure</i>	5,772	211	0.96	0.04	0.47	-0.47	4.25	-4.25	6,522	-6,522	5,983
<i>early</i>	5,175	99	0.98	0.02	0.49	-0.49	4.96	-4.96	6,373	-6,373	5,274
<i>familiar</i>	204	2,526	0.07	0.93	-0.43	0.43	-3.08	3.08	-2,343	2,343	2,730
<i>similar</i>	177	1,420	0.11	0.89	-0.39	0.39	-2.40	2.40	-1,104	1,104	1,597
<i>larger</i>	29	503	0.05	0.95	-0.45	0.45	-2.78	2.78	-513	513	532
<i>greater</i>	33	446	0.07	0.93	-0.43	0.43	-2.48	2.48	-424	424	479
<i>unable</i>	9	257	0.03	0.97	-0.47	0.47	-2.68	2.68	-290	290	266

APPENDIX B

CHAPTER 3 APPENDIX

This appendix pertains to [Chapter 3](#)

Table B.1: Top LRC for **Positive** bigrams where $*S$ PBR bigram subtotal ≥ 40

38 unique adjectives where Positive *exactly*~ADJ $f \geq 10$
and overall *exactly*~ADJ $f \geq 40$
exactly marginal frequency = 12,296

Positive <i>exactly</i> ...	f	LRC	dPavg	G2	dP1	P1	dP2	P2	f2	unexp_f	unexp_r
<i>alike</i>	643	8.57	0.07	6,641	0.08	0.08	0.052	0.052	8,110	642	1
<i>opposite</i>	454	7.96	0.05	4,352	0.06	0.06	0.037	0.037	8,137	453	1
<i>right</i>	5,635	7.92	0.25	54,374	0.04	0.04	0.457	0.458	125,709	5,613	1
<i>zero</i>	292	7.00	0.03	2,457	0.03	0.03	0.024	0.024	9,250	290	0.99
<i>parallel</i>	159	6.50	0.02	1,271	0.03	0.03	0.013	0.013	6,173	158	0.99
<i>perpendicular</i>	40	6.07	0.02	341	0.03	0.03	0.003	0.003	1,185	40	0.99
<i>analogous</i>	68	6.01	0.01	537	0.02	0.02	0.005	0.006	2,759	68	0.99
<i>equal</i>	420	5.69	0.02	2,738	0.01	0.01	0.034	0.034	34,460	414	0.99
<i>correct</i>	448	5.18	0.02	2,605	0.01	0.01	0.036	0.036	52,545	439	0.98
<i>same</i>	386	5.07	0.02	2,199	0.01	0.01	0.031	0.031	47,931	377	0.98
<i>equivalent</i>	106	4.88	0.01	637	0.01	0.01	0.008	0.009	11,088	104	0.98
<i>contrary</i>	58	4.50	0.01	347	0.01	0.01	0.005	0.005	6,162	57	0.98
<i>identical</i>	198	4.08	0.01	913	<0.01	<0.01	0.015	0.016	42,684	190	0.96
<i>proportional</i>	41	3.83	0.01	224	0.01	0.01	0.003	0.003	5,692	40	0.98
<i>wrong</i>	391	3.63	0.02	1,477	<0.01	<0.01	0.030	0.032	131,262	368	0.94
<i>average</i>	67	3.57	<0.01	308	<0.01	<0.01	0.005	0.005	14,489	64	0.96
<i>square</i>	25	3.39	<0.01	139	0.01	0.01	0.002	0.002	3,282	24	0.98
<i>double</i>	47	3.13	<0.01	205	<0.01	<0.01	0.004	0.004	11,476	45	0.96
<i>enough</i>	211	2.63	0.01	570	<0.01	<0.01	0.015	0.017	126,223	188	0.89
<i>similar</i>	286	2.49	0.01	693	<0.01	<0.01	0.020	0.023	200,667	250	0.87
<i>perfect</i>	125	2.09	<0.01	285	<0.01	<0.01	0.009	0.010	94,808	108	0.86
<i>true</i>	200	1.85	0.01	360	<0.01	<0.01	0.013	0.016	201,597	164	0.82
<i>comparable</i>	23	1.51	<0.01	74	<0.01	<0.01	0.002	0.002	10,234	21	0.92
<i>sure</i>	46	<0.01	<0.01	18	<0.01	<0.01	0.002	0.004	128,755	23	0.50
<i>accurate</i>	42	<0.01	<0.01	10	<0.01	<0.01	0.001	0.003	137,619	17	0.41

B.1 Evaluating *Exactly*’s Potential Superset *S Bias

Table B.2 and Table B.3 evaluate *exactly*’s potential polarity approximation method bias.

Table B.2: *Polarity Approximation Exactly* Contingency Table

	12: exactly	$\neg$ exactly	<i>margin</i>
11: subset	f: 1,031	1,679,602	f1: 1,680,633
$\neg$ subset	55,078	70,225,662	70,280,740
margin	f2: 56,109	71,905,264	N: 71,961,373

Table B.3: @S*Exactly* Association

	unexp_f	LRC	dP1	dP2	P1	P2
subset~exactly	-279.4	-0.21	-0.005	<0.001	0.018	0.001

Table B.3 indicates there is only a weak *negative* (repellant) association between *exactly* and the mirror subset: It is 0.5 % **less** likely that a bigram containing *exactly* will be a member of the subset than it is for a bigram without *exactly*. However, it is significant, as the LRC value of $-0.206 \neq 0 (p \leq 0.001)$.

This could indicate that the additional syntactic selection restrictions imposed in the subset patterns (i.e., requiring explicit positive markers and excluding sentential negation) impacts the likelihood of *exactly* bigrams occurring *at all* more than it affects polarity distribution of those occurrences.

APPENDIX C

CHAPTER 4 APPENDIX

This appendix pertains to [Chapter 4](#). All source code and example data can be found in my (AndreaTheLinguist’s) [sanpi](#)²⁰⁷ GitHub repository.

C.1 Examples: Bigram Governed Operator Dependencies

Table C.1: Sample Hit from RBdirect/ALL-RBdirect_final
*N Bigram Governed—“*n’t especially glamorous*”

<i>not especially glamorous</i>	
hit_id	pcc_eng_val_3.02703_x38921_190:7-8-9
bigram	especially_glamorous
sent_text	<i>This is another job that isn’t especially glamorous, but it’s available (and will continue to be for the indeterminate future).</i>
neg_form	<i>n’t</i>
adv_form	<i>especially</i>
adj_form	<i>glamorous</i>
text_window	this is another job that is <u>n’t</u> especially glamorous , but it ’s available (
sent_id	pcc_eng_val_3.02703_x38921_190
match_id	7-8-9
bigram_id	pcc_eng_val_3.02703_x38921_190:8-9
token_str	This is another job that is <u>n’t</u> especially glamorous , but it ’s available (and will continue to be for the indeterminate future) .
lemma_str	this be another job that be not especially glamorous , but it be available (and will continue to be for the indeterminate future) .
neg_deprel	advmod
neg_head	ADJ
mod_deprel	advmod
mod_head	ADJ
neg_lemma	not
adv_lemma	especially

Continued on next page

²⁰⁷<https://github.com/AndreaTheLinguist/sanpi>

Table C.1: *N Bigram Governed—“*n’t especially glamorous*”, *cont’d*

<i>not especially glamorous</i>	
adj_lemma	glamorous
neg_index	6
adv_index	7
adj_index	8
dep_neg	{‘node’: ‘neg’, ‘contiguous’: False, ‘relation’: ‘advmod’, ‘head’: {‘lemma’: ‘glamorous’, ‘ix’: 8, ‘xpos’: ‘JJ’}, ‘target’: {‘lemma’: ‘not’, ‘ix’: 6, ‘xpos’: ‘RB’}}
dep_mod	{‘node’: ‘mod’, ‘contiguous’: True, ‘relation’: ‘advmod’, ‘head’: {‘lemma’: ‘glamorous’, ‘ix’: 8, ‘xpos’: ‘JJ’}, ‘target’: {‘lemma’: ‘especially’, ‘ix’: 7, ‘xpos’: ‘RB’}}
json_source	/share/compling/data/sanpi/1_json_grew-matches/ RBdirect/bigram-PccVa.direct-adj-head/ BIGRAM.pcc_eng_val-03.json
utt_len	26
pattern	direct-adj-head
category	RBdirect
adv_form_lower	especially
adj_form_lower	glamorous
bigram_lower	especially_glamorous
trigger_lower	n’t
trigger_lemma	not
all_forms_lower	n’t_especially_glamorous
dep_distance	1
window_len	15
quarantine	False
id_prefix	pcc_eng_val_3
part	PccVa

Table C.2: Sample Hit from NEGmirror/ALL-NEGmirror_final
@N Bigram Governed—“*never quite comfortable*”

<i>never quite comfortable</i>	
hit_id	pcc_eng_val_3.04925_x42562_13:08-09-10
bigram	quite_comfortable
sent_text	<i>For whatever reason, little Ted was <u>never</u> quite comfortable about being born.</i>
neg_form	<i>never</i>
adv_form	<i>quite</i>

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Table C.2: @N Bigram Governed—“*never quite comfortable*”, cont’d

<i>never quite comfortable</i>	
adj_form	<i>comfortable</i>
text_window	whatever reason , little ted was <u>never</u> quite comfortable about being born .
sent_id	pcc_eng_val_3.04925_x42562_13
match_id	08-09-10
bigram_id	pcc_eng_val_3.04925_x42562_13:09-10
token_str	For whatever reason , little Ted was <u>never</u> quite comfortable about being born .
lemma_str	for whatever reason , little Ted be never quite comfortable about be bear .
neg_deprel	advmod
neg_head	ADJ
mod_deprel	advmod
mod_head	ADJ
neg_lemma	never
adv_lemma	quite
adj_lemma	comfortable
neg_index	7
adv_index	8
adj_index	9
dep_neg	{‘node’: ‘neg’, ‘contiguous’: False, ‘relation’: ‘advmod’, ‘head’: {‘lemma’: ‘comfortable’, ‘ix’: 9, ‘xpos’: ‘JJ’}, ‘target’: {‘lemma’: ‘never’, ‘ix’: 7, ‘xpos’: ‘RB’}}
dep_mod	{‘node’: ‘mod’, ‘contiguous’: True, ‘relation’: ‘advmod’, ‘head’: {‘lemma’: ‘comfortable’, ‘ix’: 9, ‘xpos’: ‘JJ’}, ‘target’: {‘lemma’: ‘quite’, ‘ix’: 8, ‘xpos’: ‘RB’}}
adv_form_lower	quite
adj_form_lower	comfortable
utt_len	14
json_source	/share/compling/data/sanpi/1_json_grew-matches/ NEGmirror/bigram-PccVa.neg-mirror-R/BIGRAM.pcc_eng_val-03.json
bigram_lower	quite_comfortable
all_forms_lower	never_quite_comfortable
pattern	neg-mirror-R
category	NEGmirror
trigger_lower	never
trigger_lemma	never
dep_distance	1
window_len	13
quarantine	False
id_prefix	pcc_eng_val_3

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Table C.2: @N Bigram Governed—“*never quite comfortable*”, cont’d

<i>never quite comfortable</i>	
part	PccVa

Table C.3: Sample Hit from POSmirror/ALL-POSmirror_final
@P Bigram Governed—“*often very cheap*”

<i>often very cheap</i>	
hit_id	pcc_eng_val_3.02291_x38229_25:3-4-5
bigram	very_cheap
sent_text	<i>These are <u>often</u> very cheap and you can get them at walmart for around 65 cents.</i>
adv_form	<i>very</i>
adj_form	<i>cheap</i>
text_window	these are <u>often</u> very cheap and you can get them at
sent_id	pcc_eng_val_3.02291_x38229_25
match_id	3-4-5
bigram_id	pcc_eng_val_3.02291_x38229_25:4-5
token_str	These are <u>often</u> very cheap and you can get them at walmart for around 65 cents .
lemma_str	this be often very cheap and you can get they at walmart for around 65 cent .
mod_deprel	advmod
mod_head	ADJ
mir_deprel	advmod
mir_head	ADJ
mir_lemma	often
adv_lemma	very
adj_lemma	cheap
mir_form	<i>often</i>
mir_index	2
adv_index	3
adj_index	4
dep_mod	{‘node’: ‘mod’, ‘contiguous’: True, ‘relation’: ‘advmod’, ‘head’: {‘lemma’: ‘cheap’, ‘ix’: 4, ‘xpos’: ‘JJ’}, ‘target’: {‘lemma’: ‘very’, ‘ix’: 3, ‘xpos’: ‘RB’}}
dep_mir	{‘node’: ‘mir’, ‘contiguous’: False, ‘relation’: ‘advmod’, ‘head’: {‘lemma’: ‘cheap’, ‘ix’: 4, ‘xpos’: ‘JJ’}, ‘target’: {‘lemma’: ‘often’, ‘ix’: 2, ‘xpos’: ‘RB’}}

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Table C.3: @P Bigram Governed—“*often very cheap*”, cont’d

<i>often very cheap</i>	
adv_form_lower	very
adj_form_lower	cheap
utt_len	17
json_source	/share/compling/data/sanpi/1_json_grew-matches/ POSmirror/bigram-PccVa.pos-mirror-R/BIGRAM.pcc_eng_val-03.json
bigram_lower	very_cheap
all_forms_lower	often_very_cheap
pattern	pos-mirror-R
category	POSmirror
trigger_lower	often
trigger_lemma	often
dep_distance	1
window_len	11
quarantine	False
id_prefix	pcc_eng_val_3
part	PccVa

C.2 Examples: *Trigger* Governed Operator Dependencies

Table C.4: Sample Hit from RBdirect/ALL-RBdirect_final:
*N Trigger Governed—“*nothing particularly significant*”

<i>nothing particularly significant</i>	
hit_id	pcc_eng_val_3.09483_x49876_020:13-14-15
bigram	particularly_significant
sent_text	<i>Through the years there had been a few minor things, but <u>nothing</u> particularly significant.</i>
neg_form	<i>nothing</i>
adv_form	<i>particularly</i>
adj_form	<i>significant</i>
text_window	<i>a few minor things , but <u>nothing</u> particularly significant .</i>
sent_id	pcc_eng_val_3.09483_x49876_020
match_id	13-14-15
bigram_id	pcc_eng_val_3.09483_x49876_020:14-15

Continued on next page

Table C.4: *N Trigger Governed—“*nothing particularly significant*”, cont’d

<i>nothing particularly significant</i>	
token_str	Through the years there had been a few minor things , but <u>nothing particularly significant</u> .
lemma_str	through the year there have be a few minor thing , but <u>nothing particularly significant</u> .
neg_deprel	amod
neg_head	NEG
mod_deprel	advmod
mod_head	ADJ
neg_lemma	nothing
adv_lemma	particularly
adj_lemma	significant
neg_index	12
adv_index	13
adj_index	14
dep_neg	{‘node’: ‘neg’, ‘contiguous’: False, ‘relation’: ‘amod’, ‘head’: {‘lemma’: ‘nothing’, ‘ix’: 12, ‘xpos’: ‘NN’}, ‘target’: {‘lemma’: ‘significant’, ‘ix’: 14, ‘xpos’: ‘JJ’}}
dep_mod	{‘node’: ‘mod’, ‘contiguous’: True, ‘relation’: ‘advmod’, ‘head’: {‘lemma’: ‘significant’, ‘ix’: 14, ‘xpos’: ‘JJ’}, ‘target’: {‘lemma’: ‘particularly’, ‘ix’: 13, ‘xpos’: ‘RB’}}
json_source	/share/compling/data/sanpi/1_json_grew-matches/RBdirect/bigram-PccVa.direct-neg-head/BIGRAM.pcc_eng_val-03.json
utt_len	16
pattern	direct-neg-head
category	RBdirect
adv_form_lower	particularly
adj_form_lower	significant
bigram_lower	particularly_significant
trigger_lower	nothing
trigger_lemma	nothing
all_forms_lower	nothing_particularly_significant
dep_distance	1
window_len	10
quarantine	False
id_prefix	pcc_eng_val_3
part	PccVa

Table C.5: Sample Hit from RBdirect/ALL-NEGmirror_final
@N Trigger Governed—“*nothing intrinsically wrong*”

<i>nothing intrinsically wrong</i>	
hit_id	pcc_eng_val_1.8610_x13991_83:3-4-5
bigram	intrinsically_wrong
sent_text	<i>There is nothing intrinsically wrong with a project like this, just as there is nothing wrong with the themes the photographers choose to focus on.</i>
neg_form	nothing
adv_form	intrinsically
adj_form	wrong
text_window	<i>there is nothing intrinsically wrong with a project like this ,</i>
sent_id	pcc_eng_val_1.8610_x13991_83
match_id	3-4-5
bigram_id	pcc_eng_val_1.8610_x13991_83:4-5
token_str	<i>There is nothing intrinsically wrong with a project like this , just as there is nothing wrong with the themes the photographers choose to focus on .</i>
lemma_str	there be nothing intrinsically wrong with a project like this , just as there be nothing wrong with the theme the photographer choose to focus on .
neg_deprel	amod
neg_head	NEG
mod_deprel	advmod
mod_head	ADJ
neg_lemma	nothing
adv_lemma	intrinsically
adj_lemma	wrong
neg_index	2
adv_index	3
adj_index	4
dep_neg	{‘node’: ‘neg’, ‘contiguous’: False, ‘relation’: ‘amod’, ‘head’: {‘lemma’: ‘nothing’, ‘ix’: 2, ‘xpos’: ‘NN’}, ‘target’: {‘lemma’: ‘wrong’, ‘ix’: 4, ‘xpos’: ‘JJ’}}
dep_mod	{‘node’: ‘mod’, ‘contiguous’: True, ‘relation’: ‘advmod’, ‘head’: {‘lemma’: ‘wrong’, ‘ix’: 4, ‘xpos’: ‘JJ’}, ‘target’: {‘lemma’: ‘intrinsically’, ‘ix’: 3, ‘xpos’: ‘RB’}}
json_source	/share/compling/data/sanpi/1_json_grew-matches/ NEGmirror/bigram-PccVa.neg-mirror-L/BIGRAM.pcc_eng_val-01.json
utt_len	27
pattern	neg-mirror-L
category	NEGmirror
adv_form_lower	intrinsically
adj_form_lower	wrong

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Table C.5: @N Trigger Governed—“*nothing intrinsically wrong, cont’d*”

<i>nothing intrinsically wrong</i>	
bigram_lower	intrinsically_wrong
trigger_lower	nothing
trigger_lemma	nothing
all_forms_lower	nothing_intrinsically_wrong
dep_distance	1
window_len	11
quarantine	False
id_prefix	pcc_eng_val_1
part	PccVa

Table C.6: Sample Hit from POSmirror/ALL-POSmirror_final:
@P Trigger Governed—“*something fairly bad*”

<i>something fairly bad</i>	
hit_id	pcc_eng_val_1.9883_x16019_07:10-11-12
bigram	fairly_bad
sent_text	<i>It seemed clear to me that Santi had done something fairly bad to his knee.</i>
adv_form	<i>fairly</i>
adj_form	<i>bad</i>
text_window	to me that santi had done something fairly bad to his knee .
sent_id	pcc_eng_val_1.9883_x16019_07
match_id	10-11-12
bigram_id	pcc_eng_val_1.9883_x16019_07:11-12
token_str	It seemed clear to me that Santi had done something fairly bad to his knee .
lemma_str	it seem clear to I that Santi have do something fairly bad to he knee .
mod_deprel	advmod
mod_head	ADJ
mir_deprel	amod
mir_head	MIR
mir_lemma	something
adv_lemma	fairly
adj_lemma	bad
mir_form	something
mir_index	9
adv_index	10
adj_index	11

Continued on next page

Table C.6: @P Trigger Governed—“*something fairly bad*”, *cont’d*

<i>something fairly bad</i>	
	{‘node’: ‘mod’, ‘contiguous’: True, ‘relation’: ‘advmod’,
dep_mod	‘head’: {‘lemma’: ‘bad’, ‘ix’: 11, ‘xpos’: ‘JJ’},
	‘target’: {‘lemma’: ‘fairly’, ‘ix’: 10, ‘xpos’: ‘RB’}
	{‘node’: ‘mir’, ‘contiguous’: False, ‘relation’: ‘amod’,
dep_mir	‘head’: {‘lemma’: ‘something’, ‘ix’: 9, ‘xpos’: ‘NN’},
	‘target’: {‘lemma’: ‘bad’, ‘ix’: 11, ‘xpos’: ‘JJ’}
adv_form_lower	fairly
adj_form_lower	bad
utt_len	16
json_source	/share/compling/data/sanpi/1_json_grew-matches/ POSmirror/bigram-PccVa.pos-mirror-L/BIGRAM.pcc_eng_val-01.json
bigram_lower	fairly_bad
all_forms_lower	something_fairly_bad
pattern	pos-mirror-L
category	POSmirror
trigger_lower	something
trigger_lemma	something
dep_distance	1
window_len	13
quarantine	False
id_prefix	pcc_eng_val_1
part	PccVa

APPENDIX D

CHAPTER 5 APPENDIX

This appendix pertains to [Chapter 5](#). All source code and example data can be found in my ([AndreaTheLinguist](#)'s) `sanpi`²⁰⁸ GitHub repository.

D.1 Meta-Tools and Resources

- (D.1) server resources: `g2` cluster
- (D.2) shell/`bash` command line tools
 - a. `slurm`: resource manager software for running “jobs” on the `g2` server cluster ([Yoo *et al.* 2003](#))
 - b. `GNU parallel`: tool which enables parallel processing from the command line
used especially in producing bigram subset, the preliminarily restricted search area ([Tange 2023](#))
 - c. `comm`: to perform ID set comparisons
- (D.3) Core Data Collection & Processing Tools
 - a. `grew/grewpy`—<https://grew.fr>
 - i. `grew`: command line interface [Guillaume \(2019, 2021\)](#)
used to search `.conllu` files for specified tokens
 - ii. `grewpy`: the python module for accessing `grew` tools from within python programs ([Guillaume n.d.](#))
used to create the bigram subset
 - b. `Python` & various additional modules, but especially:
 - i. `pandas`: extensive tabular data processing toolkit ([McKinney 2010](#))
 - ii. `pyconll`: module used to iterate over and parse text with `CoNLL(-U)` formatting, i.e., the individual parsed corpus `.conllu` files ([Grioni n.d.](#))
 - iii. `multiprocessing`: module enabling parallel processing from within a `python` program ([noa n.d.b](#))

²⁰⁸<https://github.com/AndreaTheLinguist/sanpi>

- iv. **logging**: module to produce better meta-processing messages, which are vital for catching and deciphering errors or other problems during development and processing, particularly when multiprocessing has been employed (**noa n.d.a**)

D.2 Full Pattern Specifications for Mirror Subset, @S

[D.1](#) and [D.2](#) provide the full pattern specifications used for bigram-headed mirror patterns. [D.3](#) and [D.4](#) provide the full pattern specifications used for trigger-headed mirror patterns.

```

pattern {
  ADJ [xpos=re"JJ.?"];
  ADV [xpos=re"RB.?"];
  ADV < ADJ;
  mod: ADJ -[advmod]-> ADV;
  MIR [lemma="somebody"|"someone"|"everyone"|"everybody"|"something"
    |"everything"|"all"|"many"|"some"|"every"|"always"|"sometimes"
    |"often"|"either"|"both"|"or"];
  mir: ADJ -[~advcl|amod|discourse|prep|obl|obl:npm|parataxis
    |dislocated|rel|nsubjpass|attr|punct|csubj|ccomp
    |reparandum|obj|compound|expl|partmod|dep|conj]-> MIR;
  MIR << ADV
}

without {
  NEG [lemma="not"|"hardly"|"scarcely"|"never"|"rarely"|"barely"|"seldom"|"no"
    |"nothing"|"none"|"nobody"|"neither"|"without"|"few"|"nor"|"ain't"|"aint"];
  ADJ [-advcl|amod|discourse|prep|obl|obl:npm|parataxis|dislocated|rel]-> NEG;
}

without {
  NEG [lemma="not"|"hardly"|"scarcely"|"never"|"rarely"|"barely"|"seldom"|"no"
    |"nothing"|"none"|"nobody"|"neither"|"without"|"few"|"nor"|"ain't"|"aint"];
  NEG [-acl:relcl|advcl|appos|conj|mmod|parataxis]-> ADJ;
}

without {
  NEG [lemma="not"|"hardly"|"scarcely"|"never"|"rarely"|"barely"|"seldom"
    |"no"|"nothing"|"none"|"nobody"|"neither"|"without"|"few"|"nor"|"ain't"|"aint"];
  NR [lemma="think"|"believe"|"want"|"seem"|"suppose"|"expect"|"imagine"|"likely"|"probable"|"appear"|"look"
    |"intend"|"choose"|"plan"|"wish"|"suggest"|"advise"|"advisable"
    |"desirable"|"should"|"ought"|"better"|"most"|"usually"];
  NR -[re"[^E].*"]-> ADJ;
  NR -[re"[^E].*"]-> NEG;
  NEG << ADV;
}

without {
  NEG [lemma="not"|"hardly"|"scarcely"|"never"|"rarely"|"barely"|"seldom"
    |"no"|"nothing"|"none"|"nobody"|"neither"|"without"|"few"|"nor"|"ain't"|"aint"];
  NR [lemma="think"|"believe"|"want"|"seem"|"suppose"|"expect"|"imagine"|"likely"|"probable"|"appear"|"look"
    |"intend"|"choose"|"plan"|"wish"|"suggest"|"advise"|"advisable"
    |"desirable"|"should"|"ought"|"better"|"most"|"usually"];
  NR -[re"[^E].*"]-> ADJ;
  NR -[re"[^E].*"]-> RELAY;
  RELAY -[re"[^E].*"]-> NEG;
  NEG << RELAY;
  NEG << ADV;
}

without {
  NEG [lemma="not"|"hardly"|"scarcely"|"never"|"rarely"|"barely"|"seldom"|"no"
    |"nothing"|"none"|"nobody"|"neither"|"without"|"few"|"nor"|"ain't"|"aint"];
  ADJ -[re"[^E].*"]-> RELAY;
  RELAY -[re"[^E].*"]-> NEG;
  NEG << ADV;
}

```

Literal D.1: POSmirror/pos-mirror-R.pat

```

pattern {
  ADJ [xpos=re"JJ.?"];
  ADV [xpos=re"RB.?"];
  ADV < ADJ;
  mod: ADJ -[advmod]-> ADV;
  NEG [lemma="nobody"|"nothing"|"no"|"none"|"never"|"rarely"
    |"seldom"|"hardly"|"scarcely"|"barely"|"neither"|"nor"];
  neg: ADJ -[~advcl|amod|discourse|prep|obl|obl:npm|parataxis
    |dislocated|rel|nsubjpass|attr|punct|csubj|ccomp
    |reparandum|obj|compound|expl|partmod|dep|conj]-> NEG;
  NEG << ADV
}

without {
  NEG [lemma="not"|"without"|"few"|"ain't"|"aint"];
  ADJ -[~advcl|amod|discourse|prep|obl|obl:npm|parataxis]-> NEG;
}

without {
  NEG [lemma="not"|"without"|"few"|"ain't"|"aint"];
  NEG -[~acl:relcl|advcl|appos|conj|nmod|parataxis]-> ADJ;
}

without {
  NEG [lemma="not"|"hardly"|"scarcely"|"never"|"rarely"|"barely"|"seldom"
    |"no"|"nothing"|"none"|"nobody"|"neither"|"without"|"few"|"nor"|"ain't"|"aint"];
  NR [lemma="think"|"believe"|"want"|"seem"|"suppose"|"expect"|"imagine"|"likely"|"probable"|"appear"|"look"
    |"intend"|"choose"|"plan"|"wish"|"suggest"|"advise"|"advisable"
    |"desirable"|"should"|"ought"|"better"|"most"|"usually"];
  NR -[re"[^E].*"]-> ADJ;
  NR -[re"[^E].*"]-> NEG;
  NEG << ADV;
}

without {
  NEG [lemma="not"|"hardly"|"scarcely"|"never"|"rarely"|"barely"|"seldom"
    |"no"|"nothing"|"none"|"nobody"|"neither"|"without"|"few"|"nor"|"ain't"|"aint"];
  NR [lemma="think"|"believe"|"want"|"seem"|"suppose"|"expect"|"imagine"|"likely"|"probable"|"appear"|"look"
    |"intend"|"choose"|"plan"|"wish"|"suggest"|"advise"|"advisable"
    |"desirable"|"should"|"ought"|"better"|"most"|"usually"];
  NR -[re"[^E].*"]-> ADJ;
  NR -[re"[^E].*"]-> RELAY;
  RELAY -[re"[^E].*"]-> NEG;
  NEG << RELAY;
  NEG << ADV;
}

without {
  NEG [lemma="not"|"hardly"|"scarcely"|"never"|"rarely"|"barely"|"seldom"|"no"
    |"nothing"|"none"|"nobody"|"neither"|"without"|"few"|"nor"|"ain't"|"aint"];
  ADJ -[re"[^E].*"]-> RELAY;
  RELAY -[re"[^E].*"]-> NEG;
  NEG << ADV;
}

```

Literal D.2: NEGmirror/neg-mirror-R.pat

```

pattern {
  ADJ [xpos=re"JJ.?"];
  ADV [xpos=re"RB.?"];
  ADV < ADJ;
  mod: ADJ -[advmod]-> ADV;
  MIR [lemma="somebody"|"someone"|"everyone"|"everybody"|"something"
    |"everything"|"all"|"many"|"some"|"every"|"always"|"sometimes"
    |"often"|"either"|"both"|"or"];
  mir: MIR -[~acl:relcl|advcl|appos|conj|nmod|parataxis|acl|obl
    |advmod|acomp|nsubj|attr|dobj|list|partmod|prep]-> ADJ;
  MIR << ADV
}

% without direct ADJ head
without {
  NEG [lemma="not"|"hardly"|"scarcely"|"never"|"rarely"|"barely"|"seldom"|"no"
    |"nothing"|"none"|"nobody"|"neither"|"without"|"few"|"nor"|"ain't"|"aint"];
  ADJ -[~advcl|amod|discourse|prep|obl|obl:npm|parataxis|dislocated|rel]-> NEG;
}

% without direct NEG head
without {
  NEG [lemma="not"|"hardly"|"scarcely"|"never"|"rarely"|"barely"|"seldom"|"no"
    |"nothing"|"none"|"nobody"|"neither"|"without"|"few"|"nor"|"ain't"|"aint"];
  NEG -[~acl:relcl|advcl|appos|conj|nmod|parataxis]-> ADJ;
}

% without neg-raised negation
without {
  NEG [lemma="not"|"hardly"|"scarcely"|"never"|"rarely"|"barely"|"seldom"
    |"no"|"nothing"|"none"|"nobody"|"neither"|"without"|"few"|"nor"|"ain't"|"aint"];
  NR [lemma="think"|"believe"|"want"|"seem"|"suppose"|"expect"|"imagine"|"likely"|"probable"|"appear"|"look"
    |"intend"|"choose"|"plan"|"wish"|"suggest"|"advise"|"advisable"
    |"desirable"|"should"|"ought"|"better"|"most"|"usually"];
  NR -[re"[~E].*"]-> ADJ;
  NR -[re"[~E].*"]-> NEG;
  NEG << ADV;
}

% without
without {
  NEG [lemma="not"|"hardly"|"scarcely"|"never"|"rarely"|"barely"|"seldom"
    |"no"|"nothing"|"none"|"nobody"|"neither"|"without"|"few"|"nor"|"ain't"|"aint"];
  NR [lemma="think"|"believe"|"want"|"seem"|"suppose"|"expect"|"imagine"|"likely"|"probable"|"appear"|"look"
    |"intend"|"choose"|"plan"|"wish"|"suggest"|"advise"|"advisable"
    |"desirable"|"should"|"ought"|"better"|"most"|"usually"];
  NR -[re"[~E].*"]-> ADJ;
  NR -[re"[~E].*"]-> RELAY;
  RELAY -[re"[~E].*"]-> NEG;
  NEG << RELAY;
  NEG << ADV;
}

without {
  NEG [lemma="not"|"hardly"|"scarcely"|"never"|"rarely"|"barely"|"seldom"|"no"
    |"nothing"|"none"|"nobody"|"neither"|"without"|"few"|"nor"|"ain't"|"aint"];
  ADJ -[re"[~E].*"]-> RELAY;
  RELAY -[re"[~E].*"]-> NEG;
  NEG << ADV;
}

```

Literal D.3: POSmirror/pos-mirror-L.pat

```

pattern {
  ADJ [xpos=re"JJ.?"];
  ADV [xpos=re"RB.?"];
  ADV < ADJ;
  mod: ADJ -[advmod]-> ADV;
  NEG [lemma="nobody"|"nothing"|"no"|"none"|"never"|"rarely"
    |"seldom"|"hardly"|"scarcely"|"barely"|"neither"|"nor"];
  neg: NEG -[~acl:relcl|advcl|appos|conj|nmod|parataxis|acl|obl
    |advmod|acomp|nsubj|attr|dobj|list|partmod|prep]-> ADJ;
  NEG << ADV
}

% without direct ADJ head
without {
  NEG [lemma="not"|"without"|"few"|"ain't"|"aint"];
  ADJ -[~advcl|amod|discourse|prep|obl|obl:nmod|parataxis]-> NEG;
}

% without direct NEG head
without {
  NEG [lemma="not"|"without"|"few"|"ain't"|"aint"];
  NEG -[~acl:relcl|advcl|appos|conj|nmod|parataxis]-> ADJ;
}

% without neg-raised negation
without {
  NEG [lemma="not"|"hardly"|"scarcely"|"never"|"rarely"|"barely"|"seldom"
    |"no"|"nothing"|"none"|"nobody"|"neither"|"without"|"few"|"nor"|"ain't"|"aint"];
  NR [lemma="think"|"believe"|"want"|"seem"|"suppose"|"expect"|"imagine"|"likely"|"probable"|"appear"|"look"
    |"intend"|"choose"|"plan"|"wish"|"suggest"|"advise"|"advisable"
    |"desirable"|"should"|"ought"|"better"|"most"|"usually"];
  negraise: NR -[re"[^E].*"]-> ADJ;
  NR -[re"[^E].*"]-> NEG;
  NEG << ADV;
}

% without
without {
  NEG [lemma="not"|"hardly"|"scarcely"|"never"|"rarely"|"barely"|"seldom"
    |"no"|"nothing"|"none"|"nobody"|"neither"|"without"|"few"|"nor"|"ain't"|"aint"];
  NR [lemma="think"|"believe"|"want"|"seem"|"suppose"|"expect"|"imagine"|"likely"|"probable"|"appear"|"look"
    |"intend"|"choose"|"plan"|"wish"|"suggest"|"advise"|"advisable"
    |"desirable"|"should"|"ought"|"better"|"most"|"usually"];
  negraise: NR -[re"[^E].*"]-> ADJ;
  relay: NR -[re"[^E].*"]-> RELAY;
  RELAY -[re"[^E].*"]-> NEG;
  NEG << RELAY;
  NEG << ADV;
}

without {
  NEG [lemma="not"|"hardly"|"scarcely"|"never"|"rarely"|"barely"|"seldom"|"no"
    |"nothing"|"none"|"nobody"|"neither"|"without"|"few"|"nor"|"ain't"|"aint"];
  relay: ADJ -[re"[^E].*"]-> RELAY;
  RELAY -[re"[^E].*"]-> NEG;
  NEG << ADV;
}

```

Literal D.4: NEGmirror/neg-mirror-L.pat

APPENDIX E

CHAPTER 6 APPENDIX

This appendix pertains to [Chapter 6](#). All source code and example data can be found in my (AndreaTheLinguist’s) GitHub repositories, [puddin](#)²⁰⁹ and [sanpi](#)²¹⁰.

E.1 Making BiRC

E.1.1 As a slurm job

To create the subsets as a `slurm` job, use `mkbigram` alias like so: (from anywhere with `sanpi` `conda` env). Note that `DIR` must be an absolute path.

```
DIR=/share/compling/data/new_york_times; mkbigram
DIR=/share/compling/data/puddin/puddin_1; mkbigram
```

²⁰⁹<https://github.com/AndreaTheLinguist/puddin>

²¹⁰<https://github.com/AndreaTheLinguist/sanpi>

1. **mklog**: alias to generate output path for **stderr** & **stdout** based on the conllu directory (DIR) and subset name (NAME).

- (a) DIR: DIR=/full/path/to/conllu/dir/
- (b) NAME: NAME=[SUBSET_NAME]

```
# alias mklog='
(
    LOGSTEM="/share/compling/data/sanpi/logs/$(basename ${DIR})/"
    make_${NAME}_subset.$(date +%y-%m-%d_%H%M");
    mkdir -p $(dirname "${LOGSTEM}");
    echo "sending stdout & stderr to ${LOGSTEM}..."
) #'
```

2. **runselect**: alias to call the shell script, **select_for_subset.sh** with pre-designated values for DIR, NAME, and LOGSTEM

```
# alias runselect='
(
    echo "bash /share/compling/projects/sanpi/script/select_for_subset.sh
        '${DIR}' '${NAME}' '--requeue' 1>${LOGSTEM.out} 2>${LOGSTEM.err" ;
    ( bash /share/compling/projects/sanpi/script/select_for_subset.sh
        $DIR $NAME "--requeue"
        1>${LOGSTEM.out
        2>${LOGSTEM.err
    )
    && echo "Script run successful. Closed @ `date`"
) #'
```

Note: To request different (non-default) memory or time allotments (or other slurm parameters) add additional flags to **"--requeue"** string above (following **bash ...** command, not just **echo ...**),

e.g., **"--requeue"** $\Rightarrow$ **"--requeue --mem=15G -t1:00:00"**

3. **mkbigram**: command to run **mklog** and **runselect** together

```
## needs DIR!! (must be ABSOLUTE)
# alias mkbigram='
(NAME=bigram; echo -e
"\nInitiating ${NAME} subset processing for ..${DIR}#compling"/";
echo "== `date` ==";
mklog; runselect) #'
```

E.1.1.1 Call Order

1. `mkbigram` alias calls (`mklog` and) `runselect` alias, which calls `./script/select_for_subset.sh`
2. `./script/select_for_subset.sh` calls the slurm script `./slurm/grewpy_subset.slurm.sh`

I believe one of these shell scripts activates the `parallel-sanpi` env, so if that doesn't exist:

- (a) either create using the file `setup/parallel-sanpi_env.yml` following instructions in `setup/README.md`
 - (b) or tweak the shell scripts to get around it (use a different env or do not activate any from within the script)
3. The slurm script `grewpy_subset.slurm.sh` finally calls `./script/create_grewpy_subset.py`

```
CONLLU_PATH="/share/compling/data/puddin/PccTe.conll/pcc_eng_test-01.conllu"; \  
time python (  
    "/share/compling/projects/sanpi/script/create_grewpy_subset.py"  
    -n 'bigram' "${CONLLU_PATH}")
```

This is equivalent to:

```
...  
time python (  
    "/share/compling/projects/sanpi/script/create_grewpy_subset.py"  
    --subset_name 'bigram'  
    --pat_path "share/compling/projects/sanpi/Pat/RBXadj/rb-bigram.pat"  
    "${CONLLU_PATH}")
```

Literal E.2: Call `create_grewpy_subset.py` on `pcc_eng_test-01.conllu`

```

$ python script/create_grewpy_subset.py -h
usage: create_grewpy_subset.py [-h] [-p PAT_PATH] [-n SUBSET_NAME] conllu_path

positional arguments:
  conllu_path          path to `.conllu` file to get subset of

options:
  -h, --help            show this help message and exit
  -p PAT_PATH, --pat_path PAT_PATH
                        path to `.pat` file for pattern to match (default:
                        /share/compling/projects/sanpi/Pat/RBXadj/rb-bigram.pat)
  -n SUBSET_NAME, --subset_name SUBSET_NAME
                        optional string to use as output file label for subset. If
                        ↪ none given,
                        `.pat_path` filestem will be used. Output path template:
                        ↪ [conllu_path
                        parent]/subset_[label]/[label]:[conllu_path
                        ↪ stem].{context.psv, conllu}
                        (default: None)

```

Literal E.3: `-help` message for python script, `create_grewpy_subset.py`

E.1.2 Using the python script directly

To create the subset one conllu file at a time, just directly call it as indicated above.

E.1.3 Extra Post-Processing

To restructure storage of final outputs, I used this command:

```

(parallel-sanpi) arh234@compling-compute-02:~$ \
> C=/share/compling/data/sanpi/corpora_shortcuts/debug; \
> N=/share/compling/data/sanpi/subsets/bigram_$(basename $C); \
> echo $N; mkdir -p $N; cd $N; \
> ls $C/*conll/subset_bigram/*conllu \
> | parallel -k "echo -e '\nLinking {}'...; B=$(basename {});\
> echo \"      \"$(du -h --time {} | cut -d/ -f1)\$B\"; \"

```

```
> D=\$(basename \$(dirname \$(dirname {}))); \
> S=bigram-\${D%.conll}; \
> mkdir -p \${S}; echo \" linkname -> \${S}/\${B}\"; \
> ln -s {} \${S}/\${B} && ls */*conllu | head -3 \
> | parallel "echo; echo {}"; \
> echo '-----'; head {}"
```

/share/compling/data/sanpi/subsets/bigram_debug

Linking

```
→ /share/compling/data/sanpi/corpora_shortcuts/debug/apw.conll/subset_bigram/
BIGRAM.apw_eng_199911.conllu...
100K 2023-04-02 23:53 BIGRAM.apw_eng_199911.conllu
linkname -> bigram-apw/BIGRAM.apw_eng_199911.conllu
```

Linking

```
→ /share/compling/data/sanpi/corpora_shortcuts/debug/apw.conll/subset_bigram/
BIGRAM.apw_eng_200412.conllu...
4.5M 2023-03-31 19:49 BIGRAM.apw_eng_200412.conllu
linkname -> bigram-apw/BIGRAM.apw_eng_200412.conllu
```

Linking

```
→ /share/compling/data/sanpi/corpora_shortcuts/debug/nyt.conll/subset_bigram/
BIGRAM.nyt_eng_200402.conllu...
5.0M 2023-03-31 19:48 BIGRAM.nyt_eng_200402.conllu
linkname -> bigram-nyt/BIGRAM.nyt_eng_200402.conllu
```

Linking

```
→ /share/compling/data/sanpi/corpora_shortcuts/debug/nyt.conll/subset_bigram/
BIGRAM.nyt_eng_200405.conllu...
1.3M 2023-03-31 19:47 BIGRAM.nyt_eng_200405.conllu
linkname -> bigram-nyt/BIGRAM.nyt_eng_200405.conllu
```

bigram-apw/BIGRAM.apw_eng_199911.conllu

```
-----
# sent_id = apw_eng_19991101_0059_18
# text = Katz , an adviser to local governments and an expert on stadium
→ financing ,
# > has tried to position himself as a moderate best able to build on the
→ economic
# > recovery generated under popular two-term Mayor Edward G. Rendell .
1 Katz Katz - NNP - 16 nsubj 16:nsubj -
2 , , - , - 0 - 0:- -
3 an a - DT - 4 det 4:det -
4 adviser adviser - NN - 1 appos 1:appos -
5 to to - TO - 4 prep 4:prep -
```

```

6  local      local      -    JJ      -    7    amod      7:amod      -
7  governments government -    NNS     -    5    pobj      5:pobj      -
8  and        and        -    CC      -    4    cc        4:cc        -

bigram-apw/BIGRAM.apw_eng_200412.conllu
-----
# sent_id = apw_eng_20041227_0001_12
# text = Many worshippers said they did n't know about the earthquake until they
# > arrived at Mass because they had been too tired from Christmas activities to
# > watch television news .
1  many      many      -    JJ      -    2    amod      2:amod      -
2  worshippers worshipper -    NNS     -    3    nsubj     3:nsubj     -
3  said      say        -    VBD     -    0    root      0:root      -
4  they      they      -    PRP     -    7    nsubj     7:nsubj     -
5  did      did        -    AUXD    -    7    dep       7:dep       -
6  n't      not        -    RB      -    7    neg       7:neg       -
7  know     know      -    VB      -    3    ccomp     3:ccomp     -
8  about     about     -    IN      -    7    prep      7:prep      -

bigram-nyt/BIGRAM.nyt_eng_200402.conllu
-----
# sent_id = nyt_eng_20040229_0052_34
# text = Of the newcomers , Posada said he was most comfortable with Vazquez ,
# > a former Expo , because they are fellow Puerto Ricans
# > and played against each other in winter ball .
1  of      of      -    IN      -    6    prep      6:prep      -
2  the     the     -    DT      -    3    det       3:det       -
3  newcomers newcomer -    NNS     -    1    pobj      1:pobj      -
4  ,      ,      -    ,      -    0    -         0:-         -
5  Posada  Posada  -    NNP     -    6    nsubj     6:nsubj     -
6  said    say     -    VBD     -    0    root      0:root      -
7  he      he      -    PRP     -    10   nsubj     10:nsubj    -
8  was     be      -    VBD     -    10   cop       10:cop      -

```

E.2 Puddin

E.2.1 Processing Statistics

In early stages of development for this project, we attempted to run the parsing on a machine with lots of memory and CPUs, but no GPU available. However, **stanza**

has built-in GPU acceleration, and running the same code with access to a GPU increased processing speeds by a factor of 20—the presence of a single GPU yielded a $\sim 20\times$ increase in parsing speed.

E.2.2 Puddin Parse Example

```
# newdoc id = pcc_eng_val_1.0001_#00001
# sent_id = pcc_eng_val_1.0001_#00001_1
# text = A Podcast About Things
1 A a DET DT Definite=Ind|PronType=Art 2 det _ start_char=0|end_char=1
2 Podcast Podcast NOUN NN Number=Sing 0 root _ start_char=2|end_char=9
3 About about ADP IN _ 4 case _ start_char=10|end_char=15
4 Things thing NOUN NNS Number=Plur 2 nmod _ start_char=16|end_char=22

# sent_id = pcc_eng_val_1.0001_#00001_2
# text = Pod Tea'd 010: The Varying Levels Of Audio Quality
1 Pod Pod NOUN NN Number=Sing 0 root _ start_char=24|end_char=27
2 Tea'd Tea'd ADJ JJ Degree=Pos 1 appos _ start_char=28|end_char=33
3 010 010 NUM CD NumType=Card 1 nummod _ start_char=34|end_char=37
4 : : PUNCT : _ 7 punct _ start_char=37|end_char=38
5 The the DET DT Definite=Def|PronType=Art 7 det _ start_char=39|end_char=42
6 Varying vary VERB VBG VerbForm=Ger 7 amod _ start_char=43|end_char=50
7 Levels level NOUN NNS Number=Plur 1 appos _ start_char=51|end_char=57
8 Of of ADP IN _ 10 case _ start_char=58|end_char=60
9 Audio audio NOUN NN Number=Sing 10 compound _ start_char=61|end_char=66
10 Quality quality NOUN NN Number=Sing 7 nmod _ start_char=67|end_char=74

# sent_id = pcc_eng_val_1.0001_#00001_3
# text = RIP Headphone Users.
1 RIP rip VERB VB Mood=Imp|VerbForm=Fin 0 root _ start_char=76|end_char=79
2 Headphone headphone NOUN NN Number=Sing 3 compound _ start_char=80|end_char=89
3 Users user NOUN NNS Number=Plur 1 obj _ start_char=90|end_char=95
4 . . PUNCT . _ 1 punct _ start_char=95|end_char=96
```

Literal E.4: Puddin Representation of `pile/val.jsonl:20`, the first Pile-CC line in `val(itation set)`

```

# sent_id = pcc_eng_val.1.0001.z00001.4
# text = Recording from a phone or tablet mic is a very bad idea, and thus my audio quality is super duper loud at times.
1 Recording record VERB VBG VerbForm=Ger 12 csubj _ start_char=97|end_char=106
2 from from ADP IN _ 4 case _ start_char=107|end_char=111
3 a a DET DT Definite=Ind|PronType=Art 4 det _ start_char=112|end_char=113
4 phone phone NOUN NN Number=Sing 1 obl _ start_char=114|end_char=119
5 or or CONJ CC _ 7 cc _ start_char=120|end_char=122
6 tablet tablet NOUN NN Number=Sing 7 compound _ start_char=123|end_char=129
7 mic mic NOUN NN Number=Sing 4 conj _ start_char=130|end_char=133
8 is be AUX VBZ Mood=Ind|Number=Sing|Person=3|Tense=Pres|VerbForm=Fin 12 cop _ start_char=134|end_char=136
9 a a DET DT Definite=Ind|PronType=Art 12 det _ start_char=137|end_char=138
10 very very ADV RB _ 11 advmod _ start_char=139|end_char=143
11 bad bad ADJ JJ Degree=Pos 12 amod _ start_char=144|end_char=147
12 idea idea NOUN NN Number=Sing 0 root _ start_char=148|end_char=152
13 , , PUNCT , _ 22 punct _ start_char=152|end_char=153
14 and and CONJ CC _ 22 cc _ start_char=154|end_char=157
15 thus thus ADV RB _ 22 advmod _ start_char=158|end_char=162
16 my my PRON PRP$ Number=Sing|Person=1|Poss=Yes|PronType=Prs 18 nmod:poss _ start_char=163|end_char=165
17 audio audio NOUN NN Number=Sing 18 compound _ start_char=166|end_char=171
18 quality quality NOUN NN Number=Sing 22 nsubj _ start_char=172|end_char=179
19 is be AUX VBZ Mood=Ind|Number=Sing|Person=3|Tense=Pres|VerbForm=Fin 22 cop _ start_char=180|end_char=182
20 super super ADV RB _ 21 advmod _ start_char=183|end_char=188
21 duper duper ADJ JJ Degree=Pos 22 amod _ start_char=189|end_char=194
22 loud loud ADJ JJ Degree=Pos 12 conj _ start_char=195|end_char=199
23 at at ADP IN _ 24 case _ start_char=200|end_char=202
24 times time NOUN NNS Number=Plur 22 obl _ start_char=203|end_char=208
25 . . PUNCT . _ 12 punct _ start_char=208|end_char=209

# sent_id = pcc_eng_val.1.0001.z00001.5
# text = My apologies!
1 My my PRON PRP$ Number=Sing|Person=1|Poss=Yes|PronType=Prs 2 nmod:poss _ start_char=210|end_char=212
2 apologies apology NOUN NNS Number=Plur 0 root _ start_char=213|end_char=222
3 ! ! PUNCT . _ 2 punct _ start_char=222|end_char=223

# sent_id = pcc_eng_val.1.0001.z00001.6
# text = Anyway, we discussed things that happened.
1 Anyway anyway ADV RB _ 4 advmod _ start_char=224|end_char=230
2 , , PUNCT , _ 4 punct _ start_char=230|end_char=231
3 we we PRON PRP Case=Nom|Number=Plur|Person=1|PronType=Prs 4 nsubj _ start_char=232|end_char=234
4 discussed discuss VERB VBD Mood=Ind|Number=Plur|Person=1|Tense=Past|VerbForm=Fin 0 root _ start_char=235|end_char=244
5 things thing NOUN NNS Number=Plur 4 obj _ start_char=245|end_char=251
6 that that PRON WDT PronType=Rel 7 nsubj _ start_char=252|end_char=256
7 happened happen VERB VBD Mood=Ind|Tense=Past|VerbForm=Fin 5 acl:relcl _ start_char=257|end_char=265
8 . . PUNCT . _ 4 punct _ start_char=265|end_char=266

# sent_id = pcc_eng_val.1.0001.z00001.7
# text = I honestly don't remember what we talked about.
1 I I PRON PRP Case=Nom|Number=Sing|Person=1|PronType=Prs 5 nsubj _ start_char=267|end_char=268
2 honestly honestly ADV RB _ 5 advmod _ start_char=269|end_char=277
3 do do AUX VBP Mood=Ind|Tense=Pres|VerbForm=Fin 5 aux _ start_char=278|end_char=280
4 n't not PART RB _ 5 advmod _ start_char=280|end_char=283
5 remember remember VERB VB VerbForm=Inf 0 root _ start_char=284|end_char=292
6 what what PRON WP PronType=Int 5 obj _ start_char=293|end_char=297
7 we we PRON PRP Case=Nom|Number=Plur|Person=1|PronType=Prs 8 nsubj _ start_char=298|end_char=300
8 talked talk VERB VBD Mood=Ind|Tense=Past|VerbForm=Fin 6 acl:relcl _ start_char=301|end_char=307
9 about about ADP IN _ 6 case _ start_char=308|end_char=313
10 . . PUNCT . _ 5 punct _ start_char=313|end_char=314

# sent_id = pcc_eng_val.1.0001.z00001.8
# text = Just go listen already, would you?
1 Just just ADV RB _ 2 advmod _ start_char=315|end_char=319
2 go go VERB VB Mood=Imp|VerbForm=Fin 0 root _ start_char=320|end_char=322
3 listen listen VERB VB VerbForm=Inf 2 xcomp _ start_char=323|end_char=329
4 already already ADV RB _ 3 advmod _ start_char=330|end_char=337
5 , , PUNCT , _ 2 punct _ start_char=337|end_char=338
6 would would AUX MD VerbForm=Fin 2 parataxis _ start_char=339|end_char=344
7 you you PRON PRP Case=Nom|Person=2|PronType=Prs 6 nsubj _ start_char=345|end_char=348
8 ? ? PUNCT . _ 2 punct _ start_char=348|end_char=349

```

E.2.3 Retrieving *The Pile*

E.2.4 Distribution Across Parts

Table E.1: files by corpus part

corpus part	total .conllu files
Apw	178
Nyt2	127
Pcc29	109
Pcc08	108
Pcc00	108
Pcc12	108
Pcc11	108
Pcc10	108
Pcc09	108
Pcc13	108
Pcc07	108
Pcc05	108
Pcc04	108
Pcc03	108
Pcc02	108
Pcc01	108
Pcc06	108
Pcc14	108
Pcc16	108
Pcc28	108
Pcc22	108
Pcc17	108
Pcc18	108
Pcc19	108

Cont'd on next page

Table E.1: files by corpus part

corpus part	total .conllu files
Pcc20	108
Pcc21	108
Pcc24	108
Pcc23	108
Pcc15	108
Pcc25	108
Pcc26	108
Pcc27	108
Nyt1	65
PccTe	3
PccVa	3
TOTAL	3,617

E.2.5 Puddin Iterations

1. acquired The Pile (Gao *et al.* 2020)
The Pile can be downloaded from the hosting site: <https://pile.eleuther.ai/>. The files are large so this can take some time to complete. The shell tools `wget` (Nikšić n.d.) and `screen` (?) were used to prevent failures due to program closures or server connection issues. A shell script for this is included in the `puddin` repository.
2. Prepared data to feed into the `stanza` (Qi *et al.* 2020) parser:
 - (a) Converted original `jsonlines` formatted files into a `pandas` dataframe in python.
 - (b) added necessary meta data to the dataframe (e.g. document IDs)–The Pile only contains the raw text itself, and the name of the subset

3. Attempted to parse the all original text as stored in *The Pile* with **stanza** (Qi *et al.* 2020) (and add document ids, sentence ids, and sentence text meta information along the way)
4. Attempted to use dependency parse search tools—**grew** (Guillaume 2019, 2021)—on a sample of this simple parse output
5. Discovered that line breaks in the original text strings added as meta information comments resulted in improperly formatted conllu files which were unreadable by tools designed to work with this format
6. Attempted to fix this obvious and run-of-the-mill error by removing all line breaks from texts
7. While removing these line breaks from the meta comments was simple, confirming its success involved a more thorough and directed examination of the data (input and output) which illuminated other problems with texts or portions of texts that were not natural language or not English. These include but are not limited to:
 - weird formatting/punctuation,
 - formatting markup such as **wikitext** and **html**,
 - code blocks of various computer languages,
 - significant portions of texts not in English²¹¹,
 - and symbols with uninterpretable encodings
8. Attempted to clean all text that had predictable formatting markup, e.g. **html** or **wikitext/wikimedia**. This started with attempts to use existing tools for parsing these formats, but the existing tools for parsing **wikimedia** could not predict all the different variations.
9. Attempted to manually develop regular expression operations to convert the trouble cases myself, but soon (maybe not soon enough) realized not all cases could be fixed with a reasonable allocation of resources

²¹¹non-English data in and of itself is not an issue, but the language model being used to add the dependency parses was not expecting or able to yield accurate parses of other languages

10. Changed tactics and added exclusion filtering steps where the documents most prone to parsing failures were eliminated.

A note on the decision to exclude these data

It is true that the parser did not crash on these texts in the initial attempt on these texts, so they could have remained in the dataset as they were, in theory. However, “not crashing” or “yielding a parse” is not equivalent to “creating useful language data”. Since they had already been identified as potentially problematic in an effort to convert them, and they still had the *potential* of causing problems, even though they hadn’t yet, the most practical solution was to exclude them from this build of Puddin. They were *relocated* and stored with their initial text id—rather than simply dropped from the data structures entirely—to allow for the potential to parse them at a later date if desired.

11. Excluding the worst of it was an improvement, but the arduous recovery of even predictable formatting cases was not a reasonable allocation of resources due to unresolved errors in processing (i.e. `beautiful_soup` html processing; Richardson (2023)), and the sheer memory and time requirements. On the original server location, `beautiful_soup` was encountering opaque errors that did not occur in sample tests in the development environment. In retrospect, these opaque errors were likely related to/masking underlying “out of memory” errors. However, excluding these problem data consumed far less memory than attempting to process them. Additionally, the regular expression filters that identified problematic texts were initially written to be as precise as (i.e. as weak) possible in an effort to minimize the rate of exclusion. Unfortunately These complex patterns, also contributed to the resource overload and processing time. Since there wasn’t even a way to be certain that these filters were strong enough to serve their purpose of avoiding errors in the first place, the complexities were removed in favor of simpler, stronger filters that were much faster to process. They excluded any text that included even one match of the following patterns/cases:

```
pattern_type_dict = {
# json formatted texts
'json': re.compile(r'{"\w+":{"\w+":'}),
# constructions specific to formal languages
'code':
    re.compile(r'(=| [=!><] [=!><])\s?(self|true|false|\w+\.?\.?\w*)',
    re.IGNORECASE),
# words with underscores embedded, preceding, or following
'_word': re.compile(r'[\w]*?_[\w]+?'),
```

```
# words containing both letters and digits
'a0wrđ':
    re.compile(r'\d*[a-z]+\d+[a-z]*\d*[a-z]*'
               r'|\d{3:}[a-z]+[a-z]*\d*[a-z]*',
               re.IGNORECASE),
# words containing embedded punctuation (medial only)
'punc': re.compile(r'\b[a-z]+[^\w\s\-\\'/\&@]+?[a-zA-Z]+\b')
}
```

12. The original server housing the data and code, however, was still running into “OOM Killer” (out of memory) errors (opaque at first), so added many “checkpoint” save points and restructuring (slicing original raw input `jsonl` datasets into smaller subsets) to minimize data loss/processing repetition.
13. These measures did not suffice though, so the data and code were relocated to a server cluster: `g2`
14. Running on a cluster allowed dealt with the resource deficit, but required running the code using `slurm`: created `slurm sbatch` bash script for submitting jobs for parsing given data sets
15. Still ran into some errors in processing due to resource allocation request limitations and trying to balance getting OOM errors with the likelihood of having the job pre-empted by something with higher priority²¹²
16. manually started and resubmitted jobs until all (non-excluded) documents were parsed for the *Pile-CC* subset
17. validated the dataset—for quality assurance, raw/initial, intermediate, and final versions of full dataframes for each data group were saved, in addition to sliced dataframes corresponding to the output corpus `.conllu` files both before parsing commenced, and after it had finished.

E.2.6 Puddin Validation

The full validation process is described below:

²¹²In retrospect, making the parsing code parallel might have avoided some of the memory consumption problems, and/or allowed for resource allocations to be more variable.

1. cleaned processing/parsing info index output log file
2. added system path information: file size and last modified time
3. extrapolated additional values for processing time
4. removed records for output files that were replaced by later iterations
(put these in their own file)
5. used this parsing index to flag files with outlying elapsed time values—suspiciously slow or fast—and outlying output file size values—suspiciously large or small—and moved those data files to a purgatory location
6. re-ran `parse_pile.py` on the original data for the offending outputs²¹³
This illuminated some of the issues from the parsing procedure
7. (wrote and) ran an additional program to:
 - (a) go through all the text ids in the original data set (as loaded from source `jsonlines` files) and locate corresponding text ids in the final corpus files.
 - (b) pull in the stored exclusions data and integrated that with the parsing status output.
 - (c) flag texts (text ids) that were missing from both the conllu outputs and the exclusions; a.k.a. *missing texts*
 - (d) flag texts in the exclusions that actually *were* successfully parsed (found in a `.conllu` file); a.k.a. *invalid exclusions*
 - (e) compile all of this meta information into dataframes (1 per source data group/`.jsonl` file)
 - (f) compile all invalid exclusions into dataframes (1 per source data group/`.jsonl` file)
 - (g) report counts for successful parses and excluded texts for each data group

²¹³The program checks for existing files at various processing points, so only the data that lacked parsing outputs were redone.

E.2.6.1 Main Process Slurm stdout

```
1 JOB ID: 773159
2 JOB NAME: ValidatePuddin
3 started @ 2022-08-11 02:13:07 from /share/compling/projects/puddin/logs
4 . . . . .
5 running on compling-compute-02 with:
6 - 10 cores
7 - 25600 mem/cpu
8 * * * * *
9
10 2022-08-11 02:13AM: Starting Puddin Validation of ALL data groups...
11 2022-08-11 02:13AM: Loading processing meta info from
    ↪ /share/compling/data/puddin/info/completed-puddin_meta-index.pkl ...
12
13 -----
14 | 2022-08-11 02:13AM
15 | processing 32 inputs with 10 CPUs...
16
17 -----
18 | 2022-08-11 03:28AM
19 task | time  data group  successful  exclusions
20 -----|-----
21 01 | 1.25h 02          1080804    667197
22 02 | 1.28h 09          1082487    669367
23 03 | 1.38h 08          1078421    667840
24 04 | 1.40h 06          1079734    667303
25 05 | 1.42h 07          1081773    666772
26 06 | 1.45h 05          1080599    667271
27 07 | 1.62h 00          1081093    667632
28 08 | 1.66h 04          1080734    665253
29 09 | 1.97h 01          1081008    666245
30 10 | 2.07h 03          1079740    668730
31 11 | 1.28h 11          1078906    667341
32 12 | 1.42h 14          1081710    667029
33 13 | 1.53h 12          1081995    666851
34 14 | 1.70h 16          1079720    667409
35 15 | 1.92h 15          1081993    667570
36 16 | 1.75h 17          1081358    667054
37 17 | 1.70h 19          1081367    667466
38 18 | 1.83h 18          1079637    669858
39 19 | 2.55h 13          1081029    666006
40 20 | 2.84h 10          1080557    667205
```

```

41 21 | 1.79h 21 1082171 666888
42 22 | 2.83m test 32697 20093
43 23 | 3.11m val 32612 20180
44 24 | 2.30h 20 1081894 666688
45 25 | 1.96h 22 1079918 665890
46 26 | 1.23h 26 1080409 667073
47 27 | 1.07h 28 1080442 666737
48 28 | 1.68h 24 1080672 668030
49 29 | 1.28h 27 1080664 667664
50 30 | 1.82h 23 1079667 665915
51 31 | 1.74h 25 1079621 667772
52 32 | 1.07h 29 1083529 667711
53
54 -----
55 | 2022-08-11 07:22AM
56 | = 32 data groups validated in 5.16h
57 -----
58
59 Job Closed
60 Thu Aug 11 07:23:02 EDT 2022

```

E.2.6.2 Subprocesses, by Processor/PoolWorker

```

1 -----
2 | INFO: 2022-08-11 02:13AM
3 |
4 ↳ /share/compling/projects/puddin/script/validate_data_group.py:assess_data_group:76
5 | . . . . .
6 VALIDATION LOG: 04
7 PID: 1218760
8 -----
9 | INFO: 2022-08-11 02:13AM
10 |
11 ↳ /share/compling/projects/puddin/script/validate_data_group.py:_prep_raw_dataframe:194
12 | . . . . .
13 Loading initial/raw dataframe:
14 > /share/compling/data/puddin/pile_tables/raw/pile_04_Pile-CC_df.pkl.gz
15 ...
16 -----
17 | INFO: 2022-08-11 02:17AM
18 |
19 ↳ /share/compling/projects/puddin/script/validate_data_group.py:_check_conll_path:234

```

```

17 | . . . . .
18 directory of conllu files to be searched:
19 > /share/compling/data/puddin/Pcc04.conll
20 -----
21 | INFO: 2022-08-11 03:43AM
22 |
    ↪ /share/compling/projects/puddin/script/validate_data_group.py:_get_parse_status:269
23 | . . . . .
24 2022-08-11 -- 02:18AM pcc_eng_04-001.conllu total docs: 9999
25 2022-08-11 -- 02:18AM pcc_eng_04-002.conllu total docs: 9999
26 2022-08-11 -- 02:19AM pcc_eng_04-003.conllu total docs: 9999
27 2022-08-11 -- 02:19AM pcc_eng_04-004.conllu total docs: 9999
28 2022-08-11 -- 02:20AM pcc_eng_04-005.conllu total docs: 9999
29 2022-08-11 -- 02:20AM pcc_eng_04-006.conllu total docs: 9999
30 2022-08-11 -- 02:21AM pcc_eng_04-007.conllu total docs: 9998
31 2022-08-11 -- 02:22AM pcc_eng_04-008.conllu total docs: 9999
32 2022-08-11 -- 02:22AM pcc_eng_04-009.conllu total docs: 9998
33 2022-08-11 -- 02:23AM pcc_eng_04-010.conllu total docs: 9999
34 2022-08-11 -- 02:23AM pcc_eng_04-011.conllu total docs: 9999
35 2022-08-11 -- 02:24AM pcc_eng_04-012.conllu total docs: 9999
36 2022-08-11 -- 02:25AM pcc_eng_04-013.conllu total docs: 9999
37 2022-08-11 -- 02:25AM pcc_eng_04-014.conllu total docs: 9998
38 2022-08-11 -- 02:26AM pcc_eng_04-015.conllu total docs: 9998
39 2022-08-11 -- 02:27AM pcc_eng_04-016.conllu total docs: 9999
40 2022-08-11 -- 02:28AM pcc_eng_04-017.conllu total docs: 9999
41 2022-08-11 -- 02:28AM pcc_eng_04-018.conllu total docs: 9999
42 2022-08-11 -- 02:29AM pcc_eng_04-019.conllu total docs: 9998
43 2022-08-11 -- 02:30AM pcc_eng_04-020.conllu total docs: 9999
44 2022-08-11 -- 02:31AM pcc_eng_04-021.conllu total docs: 9999
45 2022-08-11 -- 02:32AM pcc_eng_04-022.conllu total docs: 9999
46 2022-08-11 -- 02:32AM pcc_eng_04-023.conllu total docs: 9999
47 2022-08-11 -- 02:33AM pcc_eng_04-024.conllu total docs: 9999
48 2022-08-11 -- 02:34AM pcc_eng_04-025.conllu total docs: 9999
49 2022-08-11 -- 02:35AM pcc_eng_04-026.conllu total docs: 9999
50 2022-08-11 -- 02:36AM pcc_eng_04-027.conllu total docs: 9999
51 2022-08-11 -- 02:37AM pcc_eng_04-028.conllu total docs: 9999
52 2022-08-11 -- 02:37AM pcc_eng_04-029.conllu total docs: 9998
53 2022-08-11 -- 02:38AM pcc_eng_04-030.conllu total docs: 9999
54 2022-08-11 -- 02:39AM pcc_eng_04-031.conllu total docs: 9998
55 2022-08-11 -- 02:40AM pcc_eng_04-032.conllu total docs: 9999
56 2022-08-11 -- 02:42AM pcc_eng_04-033.conllu total docs: 9999
57 2022-08-11 -- 02:44AM pcc_eng_04-034.conllu total docs: 9999
58 2022-08-11 -- 02:45AM pcc_eng_04-035.conllu total docs: 9999
59 2022-08-11 -- 02:46AM pcc_eng_04-036.conllu total docs: 9999

```


60	2022-08-11	--	02:47AM	pcc_eng_04-037.conllu	total docs: 9999
61	2022-08-11	--	02:48AM	pcc_eng_04-038.conllu	total docs: 9999
62	2022-08-11	--	02:49AM	pcc_eng_04-039.conllu	total docs: 9999
63	2022-08-11	--	02:50AM	pcc_eng_04-040.conllu	total docs: 9999
64	2022-08-11	--	02:51AM	pcc_eng_04-041.conllu	total docs: 9999
65	2022-08-11	--	02:52AM	pcc_eng_04-042.conllu	total docs: 9999
66	2022-08-11	--	02:53AM	pcc_eng_04-043.conllu	total docs: 9999
67	2022-08-11	--	02:54AM	pcc_eng_04-044.conllu	total docs: 9999
68	2022-08-11	--	02:54AM	pcc_eng_04-045.conllu	total docs: 9999
69	2022-08-11	--	02:56AM	pcc_eng_04-046.conllu	total docs: 9999
70	2022-08-11	--	02:57AM	pcc_eng_04-047.conllu	total docs: 9999
71	2022-08-11	--	02:58AM	pcc_eng_04-048.conllu	total docs: 9998
72	2022-08-11	--	02:59AM	pcc_eng_04-049.conllu	total docs: 9999
73	2022-08-11	--	02:59AM	pcc_eng_04-050.conllu	total docs: 9999
74	2022-08-11	--	03:00AM	pcc_eng_04-051.conllu	total docs: 9999
75	2022-08-11	--	03:01AM	pcc_eng_04-052.conllu	total docs: 9999
76	2022-08-11	--	03:02AM	pcc_eng_04-053.conllu	total docs: 9999
77	2022-08-11	--	03:03AM	pcc_eng_04-054.conllu	total docs: 9999
78	2022-08-11	--	03:04AM	pcc_eng_04-055.conllu	total docs: 9999
79	2022-08-11	--	03:06AM	pcc_eng_04-056.conllu	total docs: 9999
80	2022-08-11	--	03:07AM	pcc_eng_04-057.conllu	total docs: 9999
81	2022-08-11	--	03:08AM	pcc_eng_04-058.conllu	total docs: 9998
82	2022-08-11	--	03:09AM	pcc_eng_04-059.conllu	total docs: 9999
83	2022-08-11	--	03:09AM	pcc_eng_04-060.conllu	total docs: 9999
84	2022-08-11	--	03:11AM	pcc_eng_04-061.conllu	total docs: 9998
85	2022-08-11	--	03:12AM	pcc_eng_04-062.conllu	total docs: 9998
86	2022-08-11	--	03:13AM	pcc_eng_04-063.conllu	total docs: 9999
87	2022-08-11	--	03:14AM	pcc_eng_04-064.conllu	total docs: 9999
88	2022-08-11	--	03:15AM	pcc_eng_04-065.conllu	total docs: 9999
89	2022-08-11	--	03:16AM	pcc_eng_04-066.conllu	total docs: 9999
90	2022-08-11	--	03:16AM	pcc_eng_04-067.conllu	total docs: 9999
91	2022-08-11	--	03:17AM	pcc_eng_04-068.conllu	total docs: 9999
92	2022-08-11	--	03:18AM	pcc_eng_04-069.conllu	total docs: 9999
93	2022-08-11	--	03:19AM	pcc_eng_04-070.conllu	total docs: 9998
94	2022-08-11	--	03:20AM	pcc_eng_04-071.conllu	total docs: 9999
95	2022-08-11	--	03:21AM	pcc_eng_04-072.conllu	total docs: 9999
96	2022-08-11	--	03:22AM	pcc_eng_04-073.conllu	total docs: 9999
97	2022-08-11	--	03:23AM	pcc_eng_04-074.conllu	total docs: 9999
98	2022-08-11	--	03:23AM	pcc_eng_04-075.conllu	total docs: 9999
99	2022-08-11	--	03:24AM	pcc_eng_04-076.conllu	total docs: 9998
100	2022-08-11	--	03:25AM	pcc_eng_04-077.conllu	total docs: 9999
101	2022-08-11	--	03:26AM	pcc_eng_04-078.conllu	total docs: 9999
102	2022-08-11	--	03:26AM	pcc_eng_04-079.conllu	total docs: 9999
103	2022-08-11	--	03:27AM	pcc_eng_04-080.conllu	total docs: 9998

```

104 2022-08-11 -- 03:28AM pcc_eng_04-081.conllu total docs: 9999
105 2022-08-11 -- 03:29AM pcc_eng_04-082.conllu total docs: 9999
106 2022-08-11 -- 03:30AM pcc_eng_04-083.conllu total docs: 9999
107 2022-08-11 -- 03:30AM pcc_eng_04-084.conllu total docs: 9999
108 2022-08-11 -- 03:31AM pcc_eng_04-085.conllu total docs: 9999
109 2022-08-11 -- 03:32AM pcc_eng_04-086.conllu total docs: 9998
110 2022-08-11 -- 03:32AM pcc_eng_04-087.conllu total docs: 9999
111 2022-08-11 -- 03:32AM pcc_eng_04-088.conllu total docs: 9999
112 2022-08-11 -- 03:33AM pcc_eng_04-089.conllu total docs: 9999
113 2022-08-11 -- 03:33AM pcc_eng_04-090.conllu total docs: 9999
114 2022-08-11 -- 03:34AM pcc_eng_04-091.conllu total docs: 9998
115 2022-08-11 -- 03:35AM pcc_eng_04-092.conllu total docs: 9999
116 2022-08-11 -- 03:37AM pcc_eng_04-093.conllu total docs: 9998
117 2022-08-11 -- 03:37AM pcc_eng_04-094.conllu total docs: 9999
118 2022-08-11 -- 03:37AM pcc_eng_04-095.conllu total docs: 9999
119 2022-08-11 -- 03:38AM pcc_eng_04-096.conllu total docs: 9998
120 2022-08-11 -- 03:38AM pcc_eng_04-097.conllu total docs: 9998
121 2022-08-11 -- 03:38AM pcc_eng_04-098.conllu total docs: 9999
122 2022-08-11 -- 03:39AM pcc_eng_04-099.conllu total docs: 9999
123 2022-08-11 -- 03:39AM pcc_eng_04-100.conllu total docs: 9999
124 2022-08-11 -- 03:39AM pcc_eng_04-101.conllu total docs: 9998
125 2022-08-11 -- 03:40AM pcc_eng_04-102.conllu total docs: 9999
126 2022-08-11 -- 03:40AM pcc_eng_04-103.conllu total docs: 9999
127 2022-08-11 -- 03:40AM pcc_eng_04-104.conllu total docs: 9999
128 2022-08-11 -- 03:41AM pcc_eng_04-105.conllu total docs: 9999
129 2022-08-11 -- 03:41AM pcc_eng_04-106.conllu total docs: 9998
130 2022-08-11 -- 03:41AM pcc_eng_04-107.conllu total docs: 10430
131 2022-08-11 -- 03:43AM pcc_eng_04-108.conllu total docs: 10431
132 -----
133 | INFO: 2022-08-11 03:43AM
134 |
135 ↪ /share/compling/projects/puddin/script/validate_data_group.py:_load_exclusions:318
136 | . . . . .
137 Loading exclusions dataframe:
138 > /share/compling/data/puddin/pile_exclusions/pile_04_Pile-CC_excl.pkl.gz
139 -----
140 | INFO: 2022-08-11 03:52AM
141 |
142 ↪ /share/compling/projects/puddin/script/validate_data_group.py:_load_exclusions:345
143 | . . . . .
144 Exclusions loaded:
145
146 slice_id excl_type ... row_ix
147 ↪ slice_code
148
149 raw_id ...

```

```

145 pcc_04_0137260 <NA> wiki ... 0 <NA>
146 pcc_04_0179339 <NA> wiki ... 1 <NA>
147 pcc_04_0250146 <NA> wiki ... 2 <NA>
148 pcc_04_0286221 <NA> wiki ... 3 <NA>
149 pcc_04_0304692 <NA> wiki ... 4 <NA>
150 ... ... ...
151 pcc_04_1543586 pcc_eng_04_096.5333_x1543586 fail ... 665248 096
152 pcc_04_1561698 pcc_eng_04_097.6669_x1561698 fail ... 665249 097
153 pcc_04_1627756 pcc_eng_04_101.7524_x1627756 fail ... 665250 101
154 pcc_04_1696358 pcc_eng_04_106.0044_x1696358 fail ... 665251 106
155 pcc_04_1718191 pcc_eng_04_107.03550_x1718191 fail ... 665252 107
156
157 [665253 rows x 11 columns]
158 -----
159 | INFO: 2022-08-11 03:52AM
160 |
161 ↳ /share/compling/projects/puddin/script/validate_data_group.py:assess_data_group:132
162 | . . . . .
163 No original texts missing from 04 output ^_^
164 -----
165 | INFO: 2022-08-11 03:52AM
166 |
167 ↳ /share/compling/projects/puddin/script/validate_data_group.py:_identify_invalid_excl:491
168 | . . . . .
169 All 04 exclusions in are valid ^_^
170 -----
171 | INFO: 2022-08-11 03:52AM
172 |
173 ↳ /share/compling/projects/puddin/script/validate_data_group.py:_save_validated_excl:498
174 | . . . . .
175 Saving updated exclusions dataframe to
176 ↳ pile_exclusions/validated/pile_04_Pile-CC_excl.pkl.gz
177 -----
178 | INFO: 2022-08-11 03:52AM
179 |
180 ↳ /share/compling/projects/puddin/script/validate_data_group.py:assess_data_group:142
181 | . . . . .
182 ## `04` data assessment complete
183 2022-08-11 -- 03:52AM
184 status overview:
185 .....
186 excl_type known_fail success
187 a0wrdr False False 263170
188 punc False False 249027

```

184	code	False	False	96849
185	_wrld	False	False	50601
186	html	False	False	5055
187	wiki	False	False	445
188	json	False	False	84
189	fail	True	False	22
190				
191	slice	known_fail	success	
192	pcc_eng_04_108	False	True	10431
193	pcc_eng_04_107	False	True	10430
194	pcc_eng_04_064	False	True	9999
195	pcc_eng_04_074	False	True	9999
196	pcc_eng_04_073	False	True	9999
197	pcc_eng_04_072	False	True	9999
198	pcc_eng_04_071	False	True	9999
199	pcc_eng_04_069	False	True	9999
200	pcc_eng_04_068	False	True	9999
201	pcc_eng_04_067	False	True	9999
202	pcc_eng_04_066	False	True	9999
203	pcc_eng_04_065	False	True	9999
204	pcc_eng_04_063	False	True	9999
205	pcc_eng_04_077	False	True	9999
206	pcc_eng_04_060	False	True	9999
207	pcc_eng_04_059	False	True	9999
208	pcc_eng_04_057	False	True	9999
209	pcc_eng_04_056	False	True	9999
210	pcc_eng_04_002	False	True	9999
211	pcc_eng_04_054	False	True	9999
212	pcc_eng_04_053	False	True	9999
213	pcc_eng_04_052	False	True	9999
214	pcc_eng_04_051	False	True	9999
215	pcc_eng_04_075	False	True	9999
216	pcc_eng_04_078	False	True	9999
217	pcc_eng_04_049	False	True	9999
218	pcc_eng_04_079	False	True	9999
219	pcc_eng_04_105	False	True	9999
220	pcc_eng_04_104	False	True	9999
221	pcc_eng_04_103	False	True	9999
222	pcc_eng_04_102	False	True	9999
223	pcc_eng_04_100	False	True	9999
224	pcc_eng_04_099	False	True	9999
225	pcc_eng_04_098	False	True	9999
226	pcc_eng_04_095	False	True	9999
227	pcc_eng_04_094	False	True	9999

228	pcc_eng_04_092	False	True	9999
229	pcc_eng_04_090	False	True	9999
230	pcc_eng_04_089	False	True	9999
231	pcc_eng_04_088	False	True	9999
232	pcc_eng_04_087	False	True	9999
233	pcc_eng_04_085	False	True	9999
234	pcc_eng_04_084	False	True	9999
235	pcc_eng_04_083	False	True	9999
236	pcc_eng_04_082	False	True	9999
237	pcc_eng_04_081	False	True	9999
238	pcc_eng_04_050	False	True	9999
239	pcc_eng_04_001	False	True	9999
240	pcc_eng_04_023	False	True	9999
241	pcc_eng_04_028	False	True	9999
242	pcc_eng_04_026	False	True	9999
243	pcc_eng_04_025	False	True	9999
244	pcc_eng_04_024	False	True	9999
245	pcc_eng_04_047	False	True	9999
246	pcc_eng_04_022	False	True	9999
247	pcc_eng_04_021	False	True	9999
248	pcc_eng_04_020	False	True	9999
249	pcc_eng_04_018	False	True	9999
250	pcc_eng_04_017	False	True	9999
251	pcc_eng_04_016	False	True	9999
252	pcc_eng_04_013	False	True	9999
253	pcc_eng_04_012	False	True	9999
254	pcc_eng_04_011	False	True	9999
255	pcc_eng_04_010	False	True	9999
256	pcc_eng_04_008	False	True	9999
257	pcc_eng_04_006	False	True	9999
258	pcc_eng_04_005	False	True	9999
259	pcc_eng_04_004	False	True	9999
260	pcc_eng_04_003	False	True	9999
261	pcc_eng_04_027	False	True	9999
262	pcc_eng_04_055	False	True	9999
263	pcc_eng_04_037	False	True	9999
264	pcc_eng_04_035	False	True	9999
265	pcc_eng_04_042	False	True	9999
266	pcc_eng_04_041	False	True	9999
267	pcc_eng_04_039	False	True	9999
268	pcc_eng_04_043	False	True	9999
269	pcc_eng_04_040	False	True	9999
270	pcc_eng_04_036	False	True	9999
271	pcc_eng_04_044	False	True	9999

```

272 pcc_eng_04_030 False      True      9999
273 pcc_eng_04_034 False      True      9999
274 pcc_eng_04_033 False      True      9999
275 pcc_eng_04_032 False      True      9999
276 pcc_eng_04_045 False      True      9999
277 pcc_eng_04_046 False      True      9999
278 pcc_eng_04_038 False      True      9999
279 pcc_eng_04_097 False      True      9998
280 pcc_eng_04_029 False      True      9998
281 pcc_eng_04_009 False      True      9998
282 pcc_eng_04_101 False      True      9998
283 pcc_eng_04_007 False      True      9998
284 pcc_eng_04_106 False      True      9998
285 pcc_eng_04_096 False      True      9998
286 pcc_eng_04_014 False      True      9998
287 pcc_eng_04_058 False      True      9998
288 pcc_eng_04_093 False      True      9998
289 pcc_eng_04_091 False      True      9998
290 pcc_eng_04_015 False      True      9998
291 pcc_eng_04_061 False      True      9998
292 pcc_eng_04_086 False      True      9998
293 pcc_eng_04_062 False      True      9998
294 pcc_eng_04_019 False      True      9998
295 pcc_eng_04_080 False      True      9998
296 pcc_eng_04_076 False      True      9998
297 pcc_eng_04_031 False      True      9998
298 pcc_eng_04_070 False      True      9998
299 pcc_eng_04_048 False      True      9998
300 -----
301 | INFO: 2022-08-11 03:53AM
302 |
303   ↪ /share/compling/projects/puddin/script/validate_data_group.py:assess_data_group:179
304 | . . . . .
304 Data group `04` parsing status info and invalid exclusions saved as:
305 - /share/compling/data/puddin/info/validation_by_group/04-pcc_status.pkl.gz
306   last modified: Thu Aug 11 03:53:05 2022
307 - /share/compling/data/puddin/info/validation_by_group/04_invalid-excl.pkl.gz
308   last modified: Thu Aug 11 03:53:05 2022
309 -----
310 | INFO: 2022-08-11 03:53AM
311 |
311   ↪ /share/compling/projects/puddin/script/validate_data_group.py:assess_data_group:76
312 | . . . . .
313 VALIDATION LOG: 17

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314 PID: 1218760
315 -----
316 | INFO: 2022-08-11 03:53AM
317 |
318 ↪ /share/compling/projects/puddin/script/validate_data_group.py:_prep_raw_dataframe:194
319 | . . . . .
319 Loading initial/raw dataframe:
320 > /share/compling/data/puddin/pile_tables/raw/pile_17_Pile-CC_df.pkl.gz
321 ...
322 -----
323 | INFO: 2022-08-11 04:13AM
324 |
325 ↪ /share/compling/projects/puddin/script/validate_data_group.py:_check_conll_path:234
326 | . . . . .
326 directory of conllu files to be searched:
327 > /share/compling/data/puddin/Pcc17.conll
328 -----
329 | INFO: 2022-08-11 05:27AM
330 |
331 ↪ /share/compling/projects/puddin/script/validate_data_group.py:_get_parse_status:269
332 | . . . . .
332 2022-08-11 -- 04:14AM pcc_eng_17-001.conllu total docs: 9999
333 2022-08-11 -- 04:15AM pcc_eng_17-002.conllu total docs: 9999
334 2022-08-11 -- 04:16AM pcc_eng_17-003.conllu total docs: 9997
335 2022-08-11 -- 04:16AM pcc_eng_17-004.conllu total docs: 9999
336 2022-08-11 -- 04:17AM pcc_eng_17-005.conllu total docs: 9999
337 2022-08-11 -- 04:17AM pcc_eng_17-006.conllu total docs: 9999
338 2022-08-11 -- 04:18AM pcc_eng_17-007.conllu total docs: 9999
339 2022-08-11 -- 04:18AM pcc_eng_17-008.conllu total docs: 9999
340 2022-08-11 -- 04:19AM pcc_eng_17-009.conllu total docs: 9999
341 2022-08-11 -- 04:20AM pcc_eng_17-010.conllu total docs: 9999
342 2022-08-11 -- 04:20AM pcc_eng_17-011.conllu total docs: 9999
343 2022-08-11 -- 04:21AM pcc_eng_17-012.conllu total docs: 9999
344 2022-08-11 -- 04:21AM pcc_eng_17-013.conllu total docs: 9996
345 2022-08-11 -- 04:22AM pcc_eng_17-014.conllu total docs: 9999
346 2022-08-11 -- 04:22AM pcc_eng_17-015.conllu total docs: 9999
347 2022-08-11 -- 04:23AM pcc_eng_17-016.conllu total docs: 9999
348 2022-08-11 -- 04:23AM pcc_eng_17-017.conllu total docs: 9999
349 2022-08-11 -- 04:24AM pcc_eng_17-018.conllu total docs: 9999
350 2022-08-11 -- 04:24AM pcc_eng_17-019.conllu total docs: 9999
351 2022-08-11 -- 04:25AM pcc_eng_17-020.conllu total docs: 9999
352 2022-08-11 -- 04:25AM pcc_eng_17-021.conllu total docs: 9998
353 2022-08-11 -- 04:26AM pcc_eng_17-022.conllu total docs: 9999
354 2022-08-11 -- 04:26AM pcc_eng_17-023.conllu total docs: 9999

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355	2022-08-11	--	04:27AM	pcc_eng_17-024.conllu	total docs: 9999
356	2022-08-11	--	04:28AM	pcc_eng_17-025.conllu	total docs: 9999
357	2022-08-11	--	04:29AM	pcc_eng_17-026.conllu	total docs: 9999
358	2022-08-11	--	04:29AM	pcc_eng_17-027.conllu	total docs: 9999
359	2022-08-11	--	04:30AM	pcc_eng_17-028.conllu	total docs: 9998
360	2022-08-11	--	04:30AM	pcc_eng_17-029.conllu	total docs: 9999
361	2022-08-11	--	04:31AM	pcc_eng_17-030.conllu	total docs: 9999
362	2022-08-11	--	04:32AM	pcc_eng_17-031.conllu	total docs: 9999
363	2022-08-11	--	04:32AM	pcc_eng_17-032.conllu	total docs: 9998
364	2022-08-11	--	04:33AM	pcc_eng_17-033.conllu	total docs: 9999
365	2022-08-11	--	04:34AM	pcc_eng_17-034.conllu	total docs: 9999
366	2022-08-11	--	04:34AM	pcc_eng_17-035.conllu	total docs: 9999
367	2022-08-11	--	04:35AM	pcc_eng_17-036.conllu	total docs: 9999
368	2022-08-11	--	04:35AM	pcc_eng_17-037.conllu	total docs: 9998
369	2022-08-11	--	04:36AM	pcc_eng_17-038.conllu	total docs: 9999
370	2022-08-11	--	04:37AM	pcc_eng_17-039.conllu	total docs: 9999
371	2022-08-11	--	04:37AM	pcc_eng_17-040.conllu	total docs: 9998
372	2022-08-11	--	04:38AM	pcc_eng_17-041.conllu	total docs: 9999
373	2022-08-11	--	04:39AM	pcc_eng_17-042.conllu	total docs: 9998
374	2022-08-11	--	04:39AM	pcc_eng_17-043.conllu	total docs: 9999
375	2022-08-11	--	04:40AM	pcc_eng_17-044.conllu	total docs: 9999
376	2022-08-11	--	04:41AM	pcc_eng_17-045.conllu	total docs: 9999
377	2022-08-11	--	04:41AM	pcc_eng_17-046.conllu	total docs: 9999
378	2022-08-11	--	04:42AM	pcc_eng_17-047.conllu	total docs: 9999
379	2022-08-11	--	04:43AM	pcc_eng_17-048.conllu	total docs: 9997
380	2022-08-11	--	04:43AM	pcc_eng_17-049.conllu	total docs: 9999
381	2022-08-11	--	04:44AM	pcc_eng_17-050.conllu	total docs: 9998
382	2022-08-11	--	04:45AM	pcc_eng_17-051.conllu	total docs: 9999
383	2022-08-11	--	04:46AM	pcc_eng_17-052.conllu	total docs: 9998
384	2022-08-11	--	04:47AM	pcc_eng_17-053.conllu	total docs: 9998
385	2022-08-11	--	04:47AM	pcc_eng_17-054.conllu	total docs: 9998
386	2022-08-11	--	04:48AM	pcc_eng_17-055.conllu	total docs: 9999
387	2022-08-11	--	04:49AM	pcc_eng_17-056.conllu	total docs: 9998
388	2022-08-11	--	04:50AM	pcc_eng_17-057.conllu	total docs: 9999
389	2022-08-11	--	04:50AM	pcc_eng_17-058.conllu	total docs: 9999
390	2022-08-11	--	04:51AM	pcc_eng_17-059.conllu	total docs: 9999
391	2022-08-11	--	04:52AM	pcc_eng_17-060.conllu	total docs: 9999
392	2022-08-11	--	04:53AM	pcc_eng_17-061.conllu	total docs: 9999
393	2022-08-11	--	04:53AM	pcc_eng_17-062.conllu	total docs: 9999
394	2022-08-11	--	04:54AM	pcc_eng_17-063.conllu	total docs: 9999
395	2022-08-11	--	04:55AM	pcc_eng_17-064.conllu	total docs: 9999
396	2022-08-11	--	04:56AM	pcc_eng_17-065.conllu	total docs: 9998
397	2022-08-11	--	04:56AM	pcc_eng_17-066.conllu	total docs: 9999
398	2022-08-11	--	04:57AM	pcc_eng_17-067.conllu	total docs: 9999

399	2022-08-11	--	04:57AM	pcc_eng_17-068.conllu	total docs: 9999
400	2022-08-11	--	04:58AM	pcc_eng_17-069.conllu	total docs: 9999
401	2022-08-11	--	04:59AM	pcc_eng_17-070.conllu	total docs: 9999
402	2022-08-11	--	05:00AM	pcc_eng_17-071.conllu	total docs: 9997
403	2022-08-11	--	05:00AM	pcc_eng_17-072.conllu	total docs: 9998
404	2022-08-11	--	05:01AM	pcc_eng_17-073.conllu	total docs: 9999
405	2022-08-11	--	05:01AM	pcc_eng_17-074.conllu	total docs: 9999
406	2022-08-11	--	05:02AM	pcc_eng_17-075.conllu	total docs: 9998
407	2022-08-11	--	05:02AM	pcc_eng_17-076.conllu	total docs: 9998
408	2022-08-11	--	05:03AM	pcc_eng_17-077.conllu	total docs: 9999
409	2022-08-11	--	05:04AM	pcc_eng_17-078.conllu	total docs: 9999
410	2022-08-11	--	05:05AM	pcc_eng_17-079.conllu	total docs: 9999
411	2022-08-11	--	05:05AM	pcc_eng_17-080.conllu	total docs: 9999
412	2022-08-11	--	05:06AM	pcc_eng_17-081.conllu	total docs: 9999
413	2022-08-11	--	05:06AM	pcc_eng_17-082.conllu	total docs: 9999
414	2022-08-11	--	05:07AM	pcc_eng_17-083.conllu	total docs: 9999
415	2022-08-11	--	05:07AM	pcc_eng_17-084.conllu	total docs: 9999
416	2022-08-11	--	05:08AM	pcc_eng_17-085.conllu	total docs: 9997
417	2022-08-11	--	05:09AM	pcc_eng_17-086.conllu	total docs: 9999
418	2022-08-11	--	05:09AM	pcc_eng_17-087.conllu	total docs: 9999
419	2022-08-11	--	05:10AM	pcc_eng_17-088.conllu	total docs: 9998
420	2022-08-11	--	05:11AM	pcc_eng_17-089.conllu	total docs: 9999
421	2022-08-11	--	05:11AM	pcc_eng_17-090.conllu	total docs: 9999
422	2022-08-11	--	05:12AM	pcc_eng_17-091.conllu	total docs: 9999
423	2022-08-11	--	05:12AM	pcc_eng_17-092.conllu	total docs: 9998
424	2022-08-11	--	05:13AM	pcc_eng_17-093.conllu	total docs: 9998
425	2022-08-11	--	05:13AM	pcc_eng_17-094.conllu	total docs: 9999
426	2022-08-11	--	05:14AM	pcc_eng_17-095.conllu	total docs: 9998
427	2022-08-11	--	05:15AM	pcc_eng_17-096.conllu	total docs: 9999
428	2022-08-11	--	05:15AM	pcc_eng_17-097.conllu	total docs: 9999
429	2022-08-11	--	05:16AM	pcc_eng_17-098.conllu	total docs: 9998
430	2022-08-11	--	05:17AM	pcc_eng_17-099.conllu	total docs: 9999
431	2022-08-11	--	05:18AM	pcc_eng_17-100.conllu	total docs: 9998
432	2022-08-11	--	05:19AM	pcc_eng_17-101.conllu	total docs: 9999
433	2022-08-11	--	05:20AM	pcc_eng_17-102.conllu	total docs: 9999
434	2022-08-11	--	05:21AM	pcc_eng_17-103.conllu	total docs: 9999
435	2022-08-11	--	05:22AM	pcc_eng_17-104.conllu	total docs: 9999
436	2022-08-11	--	05:23AM	pcc_eng_17-105.conllu	total docs: 9998
437	2022-08-11	--	05:24AM	pcc_eng_17-106.conllu	total docs: 9999
438	2022-08-11	--	05:25AM	pcc_eng_17-107.conllu	total docs: 10748
439	2022-08-11	--	05:27AM	pcc_eng_17-108.conllu	total docs: 10749
440	-----				
441	INFO: 2022-08-11 05:27AM				

```

442 |
    ↳ /share/compling/projects/puddin/script/validate_data_group.py:_load_exclusions:318
443 | . . . . .
444 Loading exclusions dataframe:
445     > /share/compling/data/puddin/pile_exclusions/pile_17_Pile-CC_excl.pkl.gz
446 -----
447 | INFO: 2022-08-11 05:37AM
448 |
    ↳ /share/compling/projects/puddin/script/validate_data_group.py:_load_exclusions:345
449 | . . . . .
450 Exclusions loaded:
451
452 slice_id excl_type ... row_ix slice_code
453 raw_id
454 pcc_17_0076048 <NA> wiki ... 0 <NA>
455 pcc_17_0083079 <NA> wiki ... 1 <NA>
456 pcc_17_0166391 <NA> wiki ... 2 <NA>
457 pcc_17_0275568 <NA> wiki ... 3 <NA>
458 pcc_17_0426570 <NA> wiki ... 4 <NA>
459 ...
460 pcc_17_1491082 pcc_eng_17_093.2714_x1491082 fail ... 667049 093
461 pcc_17_1523757 pcc_eng_17_095.2766_x1523757 fail ... 667050 095
462 pcc_17_1571222 pcc_eng_17_098.2098_x1571222 fail ... 667051 098
463 pcc_17_1615251 pcc_eng_17_100.9327_x1615251 fail ... 667052 100
464 pcc_17_1687284 pcc_eng_17_105.3732_x1687284 fail ... 667053 105
465
466 [667054 rows x 11 columns]
467 -----
468 | INFO: 2022-08-11 05:37AM
469 |
    ↳ /share/compling/projects/puddin/script/validate_data_group.py:assess_data_group:132
470 | . . . . .
471 No original texts missing from 17 output ^_^
472 -----
473 | INFO: 2022-08-11 05:37AM
474 |
    ↳ /share/compling/projects/puddin/script/validate_data_group.py:_identify_invalid_excl:491
475 | . . . . .
476 All 17 exclusions in are valid ^_^
477 -----
478 | INFO: 2022-08-11 05:37AM
479 |
    ↳ /share/compling/projects/puddin/script/validate_data_group.py:_save_validated_excl:498

```

```

480 Saving updated exclusions dataframe to
    ↳ pile_exclusions/validated/pile_17_Pile-CC_excl.pkl.gz
481 -----
482 | INFO: 2022-08-11 05:37AM
483 |
    ↳ /share/compling/projects/puddin/script/validate_data_group.py:assess_data_group:142
484 | . . . . .
485 ## `17` data assessment complete
486 2022-08-11 -- 05:37AM
487 status overview:
488 .....
489 excl_type  known_fail  success
490 a0wrđ      False      False      264442
491 punc       False      False      249339
492 code       False      False      97123
493 _wrđ       False      False      50425
494 html       False      False      5171
495 wiki       False      False      451
496 json       False      False      70
497 fail       True       False      33
498 .....
499 slice      known_fail  success
500 pcc_eng_17_108 False      True      10749
501 pcc_eng_17_107 False      True      10748
502 pcc_eng_17_063 False      True      9999
503 pcc_eng_17_074 False      True      9999
504 pcc_eng_17_073 False      True      9999
505 pcc_eng_17_070 False      True      9999
506 pcc_eng_17_069 False      True      9999
507 pcc_eng_17_068 False      True      9999
508 pcc_eng_17_067 False      True      9999
509 pcc_eng_17_066 False      True      9999
510 pcc_eng_17_064 False      True      9999
511 pcc_eng_17_062 False      True      9999
512 pcc_eng_17_078 False      True      9999
513 pcc_eng_17_061 False      True      9999
514 pcc_eng_17_060 False      True      9999
515 pcc_eng_17_059 False      True      9999
516 pcc_eng_17_058 False      True      9999
517 pcc_eng_17_057 False      True      9999
518 pcc_eng_17_002 False      True      9999
519 pcc_eng_17_051 False      True      9999
520 pcc_eng_17_049 False      True      9999
521 pcc_eng_17_077 False      True      9999

```

522	pcc_eng_17_079	False	True	9999
523	pcc_eng_17_046	False	True	9999
524	pcc_eng_17_094	False	True	9999
525	pcc_eng_17_106	False	True	9999
526	pcc_eng_17_104	False	True	9999
527	pcc_eng_17_103	False	True	9999
528	pcc_eng_17_102	False	True	9999
529	pcc_eng_17_101	False	True	9999
530	pcc_eng_17_099	False	True	9999
531	pcc_eng_17_097	False	True	9999
532	pcc_eng_17_096	False	True	9999
533	pcc_eng_17_091	False	True	9999
534	pcc_eng_17_080	False	True	9999
535	pcc_eng_17_090	False	True	9999
536	pcc_eng_17_089	False	True	9999
537	pcc_eng_17_087	False	True	9999
538	pcc_eng_17_086	False	True	9999
539	pcc_eng_17_084	False	True	9999
540	pcc_eng_17_083	False	True	9999
541	pcc_eng_17_082	False	True	9999
542	pcc_eng_17_081	False	True	9999
543	pcc_eng_17_047	False	True	9999
544	pcc_eng_17_001	False	True	9999
545	pcc_eng_17_045	False	True	9999
546	pcc_eng_17_015	False	True	9999
547	pcc_eng_17_044	False	True	9999
548	pcc_eng_17_024	False	True	9999
549	pcc_eng_17_023	False	True	9999
550	pcc_eng_17_022	False	True	9999
551	pcc_eng_17_020	False	True	9999
552	pcc_eng_17_019	False	True	9999
553	pcc_eng_17_018	False	True	9999
554	pcc_eng_17_017	False	True	9999
555	pcc_eng_17_016	False	True	9999
556	pcc_eng_17_014	False	True	9999
557	pcc_eng_17_027	False	True	9999
558	pcc_eng_17_012	False	True	9999
559	pcc_eng_17_011	False	True	9999
560	pcc_eng_17_010	False	True	9999
561	pcc_eng_17_009	False	True	9999
562	pcc_eng_17_008	False	True	9999
563	pcc_eng_17_007	False	True	9999
564	pcc_eng_17_006	False	True	9999
565	pcc_eng_17_005	False	True	9999

566	pcc_eng_17_004	False	True	9999
567	pcc_eng_17_026	False	True	9999
568	pcc_eng_17_025	False	True	9999
569	pcc_eng_17_039	False	True	9999
570	pcc_eng_17_033	False	True	9999
571	pcc_eng_17_043	False	True	9999
572	pcc_eng_17_041	False	True	9999
573	pcc_eng_17_038	False	True	9999
574	pcc_eng_17_036	False	True	9999
575	pcc_eng_17_035	False	True	9999
576	pcc_eng_17_034	False	True	9999
577	pcc_eng_17_055	False	True	9999
578	pcc_eng_17_031	False	True	9999
579	pcc_eng_17_030	False	True	9999
580	pcc_eng_17_029	False	True	9999
581	pcc_eng_17_065	False	True	9998
582	pcc_eng_17_032	False	True	9998
583	pcc_eng_17_105	False	True	9998
584	pcc_eng_17_072	False	True	9998
585	pcc_eng_17_050	False	True	9998
586	pcc_eng_17_042	False	True	9998
587	pcc_eng_17_052	False	True	9998
588	pcc_eng_17_100	False	True	9998
589	pcc_eng_17_053	False	True	9998
590	pcc_eng_17_098	False	True	9998
591	pcc_eng_17_054	False	True	9998
592	pcc_eng_17_095	False	True	9998
593	pcc_eng_17_056	False	True	9998
594	pcc_eng_17_093	False	True	9998
595	pcc_eng_17_092	False	True	9998
596	pcc_eng_17_040	False	True	9998
597	pcc_eng_17_088	False	True	9998
598	pcc_eng_17_037	False	True	9998
599	pcc_eng_17_075	False	True	9998
600	pcc_eng_17_076	False	True	9998
601	pcc_eng_17_021	False	True	9998
602	pcc_eng_17_028	False	True	9998
603	pcc_eng_17_071	False	True	9997
604	pcc_eng_17_085	False	True	9997
605	pcc_eng_17_048	False	True	9997
606	pcc_eng_17_003	False	True	9997
607	pcc_eng_17_013	False	True	9996
608	-----			
609	INFO: 2022-08-11 05:38AM			

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610 |
    ↳ /share/compling/projects/puddin/script/validate_data_group.py:assess_data_group:179
611 | . . . . .
612 Data group `17` parsing status info and invalid exclusions saved as:
613 - /share/compling/data/puddin/info/validation_by_group/17-pcc_status.pkl.gz
614     last modified: Thu Aug 11 05:38:14 2022
615 - /share/compling/data/puddin/info/validation_by_group/17-invalid-excl.pkl.gz
616     last modified: Thu Aug 11 05:38:14 2022
617 -----
618 | INFO: 2022-08-11 05:38AM
619 |
    ↳ /share/compling/projects/puddin/script/validate_data_group.py:assess_data_group:76
620 | . . . . .
621 VALIDATION LOG: 25
622 PID: 1218760
623 -----
624 | INFO: 2022-08-11 05:38AM
625 |
    ↳ /share/compling/projects/puddin/script/validate_data_group.py:_prep_raw_dataframe:194
626 | . . . . .
627 Loading initial/raw dataframe:
628     > /share/compling/data/puddin/pile_tables/raw/pile_25_Pile-CC_df.pkl.gz
629     ...
630 -----
631 | INFO: 2022-08-11 05:54AM
632 |
    ↳ /share/compling/projects/puddin/script/validate_data_group.py:_check_conll_path:234
633 | . . . . .
634 directory of conllu files to be searched:
635     > /share/compling/data/puddin/Pcc25.conll
636 -----
637 | INFO: 2022-08-11 07:20AM
638 |
    ↳ /share/compling/projects/puddin/script/validate_data_group.py:_get_parse_status:269
639 | . . . . .
640 2022-08-11 -- 05:55AM    pcc_eng_25-001.conllu    total docs: 9999
641 2022-08-11 -- 05:55AM    pcc_eng_25-002.conllu    total docs: 9999
642 2022-08-11 -- 05:56AM    pcc_eng_25-003.conllu    total docs: 9999
643 2022-08-11 -- 05:56AM    pcc_eng_25-004.conllu    total docs: 9999
644 2022-08-11 -- 05:57AM    pcc_eng_25-005.conllu    total docs: 9999
645 2022-08-11 -- 05:57AM    pcc_eng_25-006.conllu    total docs: 9999
646 2022-08-11 -- 05:58AM    pcc_eng_25-007.conllu    total docs: 9999
647 2022-08-11 -- 05:58AM    pcc_eng_25-008.conllu    total docs: 9999
648 2022-08-11 -- 05:59AM    pcc_eng_25-009.conllu    total docs: 9999

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649	2022-08-11	--	05:59AM	pcc_eng_25-010.conllu	total docs: 9999
650	2022-08-11	--	05:59AM	pcc_eng_25-011.conllu	total docs: 9999
651	2022-08-11	--	06:00AM	pcc_eng_25-012.conllu	total docs: 9999
652	2022-08-11	--	06:00AM	pcc_eng_25-013.conllu	total docs: 9999
653	2022-08-11	--	06:01AM	pcc_eng_25-014.conllu	total docs: 9999
654	2022-08-11	--	06:02AM	pcc_eng_25-015.conllu	total docs: 9998
655	2022-08-11	--	06:02AM	pcc_eng_25-016.conllu	total docs: 9999
656	2022-08-11	--	06:03AM	pcc_eng_25-017.conllu	total docs: 9999
657	2022-08-11	--	06:03AM	pcc_eng_25-018.conllu	total docs: 9999
658	2022-08-11	--	06:04AM	pcc_eng_25-019.conllu	total docs: 9999
659	2022-08-11	--	06:04AM	pcc_eng_25-020.conllu	total docs: 9999
660	2022-08-11	--	06:05AM	pcc_eng_25-021.conllu	total docs: 9998
661	2022-08-11	--	06:05AM	pcc_eng_25-022.conllu	total docs: 9999
662	2022-08-11	--	06:06AM	pcc_eng_25-023.conllu	total docs: 9999
663	2022-08-11	--	06:07AM	pcc_eng_25-024.conllu	total docs: 9999
664	2022-08-11	--	06:07AM	pcc_eng_25-025.conllu	total docs: 9999
665	2022-08-11	--	06:07AM	pcc_eng_25-026.conllu	total docs: 9998
666	2022-08-11	--	06:08AM	pcc_eng_25-027.conllu	total docs: 9999
667	2022-08-11	--	06:08AM	pcc_eng_25-028.conllu	total docs: 9999
668	2022-08-11	--	06:09AM	pcc_eng_25-029.conllu	total docs: 9999
669	2022-08-11	--	06:09AM	pcc_eng_25-030.conllu	total docs: 9999
670	2022-08-11	--	06:10AM	pcc_eng_25-031.conllu	total docs: 9999
671	2022-08-11	--	06:10AM	pcc_eng_25-032.conllu	total docs: 9999
672	2022-08-11	--	06:10AM	pcc_eng_25-033.conllu	total docs: 9999
673	2022-08-11	--	06:11AM	pcc_eng_25-034.conllu	total docs: 9999
674	2022-08-11	--	06:11AM	pcc_eng_25-035.conllu	total docs: 9999
675	2022-08-11	--	06:12AM	pcc_eng_25-036.conllu	total docs: 9999
676	2022-08-11	--	06:13AM	pcc_eng_25-037.conllu	total docs: 9998
677	2022-08-11	--	06:13AM	pcc_eng_25-038.conllu	total docs: 9999
678	2022-08-11	--	06:14AM	pcc_eng_25-039.conllu	total docs: 9999
679	2022-08-11	--	06:14AM	pcc_eng_25-040.conllu	total docs: 9999
680	2022-08-11	--	06:15AM	pcc_eng_25-041.conllu	total docs: 9999
681	2022-08-11	--	06:15AM	pcc_eng_25-042.conllu	total docs: 9999
682	2022-08-11	--	06:16AM	pcc_eng_25-043.conllu	total docs: 9999
683	2022-08-11	--	06:16AM	pcc_eng_25-044.conllu	total docs: 9999
684	2022-08-11	--	06:17AM	pcc_eng_25-045.conllu	total docs: 9999
685	2022-08-11	--	06:18AM	pcc_eng_25-046.conllu	total docs: 9999
686	2022-08-11	--	06:18AM	pcc_eng_25-047.conllu	total docs: 9998
687	2022-08-11	--	06:19AM	pcc_eng_25-048.conllu	total docs: 9999
688	2022-08-11	--	06:20AM	pcc_eng_25-049.conllu	total docs: 9999
689	2022-08-11	--	06:22AM	pcc_eng_25-050.conllu	total docs: 9999
690	2022-08-11	--	06:23AM	pcc_eng_25-051.conllu	total docs: 9999
691	2022-08-11	--	06:24AM	pcc_eng_25-052.conllu	total docs: 9999
692	2022-08-11	--	06:25AM	pcc_eng_25-053.conllu	total docs: 9999

693	2022-08-11	--	06:27AM	pcc_eng_25-054.conllu	total docs: 9999
694	2022-08-11	--	06:28AM	pcc_eng_25-055.conllu	total docs: 9999
695	2022-08-11	--	06:29AM	pcc_eng_25-056.conllu	total docs: 9999
696	2022-08-11	--	06:31AM	pcc_eng_25-057.conllu	total docs: 9998
697	2022-08-11	--	06:32AM	pcc_eng_25-058.conllu	total docs: 9998
698	2022-08-11	--	06:32AM	pcc_eng_25-059.conllu	total docs: 9999
699	2022-08-11	--	06:34AM	pcc_eng_25-060.conllu	total docs: 9999
700	2022-08-11	--	06:36AM	pcc_eng_25-061.conllu	total docs: 9999
701	2022-08-11	--	06:37AM	pcc_eng_25-062.conllu	total docs: 9998
702	2022-08-11	--	06:38AM	pcc_eng_25-063.conllu	total docs: 9999
703	2022-08-11	--	06:40AM	pcc_eng_25-064.conllu	total docs: 9999
704	2022-08-11	--	06:42AM	pcc_eng_25-065.conllu	total docs: 9999
705	2022-08-11	--	06:43AM	pcc_eng_25-066.conllu	total docs: 9999
706	2022-08-11	--	06:45AM	pcc_eng_25-067.conllu	total docs: 9999
707	2022-08-11	--	06:47AM	pcc_eng_25-068.conllu	total docs: 9998
708	2022-08-11	--	06:49AM	pcc_eng_25-069.conllu	total docs: 9999
709	2022-08-11	--	06:54AM	pcc_eng_25-070.conllu	total docs: 9998
710	2022-08-11	--	06:57AM	pcc_eng_25-071.conllu	total docs: 9999
711	2022-08-11	--	06:59AM	pcc_eng_25-072.conllu	total docs: 9999
712	2022-08-11	--	07:01AM	pcc_eng_25-073.conllu	total docs: 9999
713	2022-08-11	--	07:04AM	pcc_eng_25-074.conllu	total docs: 9999
714	2022-08-11	--	07:05AM	pcc_eng_25-075.conllu	total docs: 9999
715	2022-08-11	--	07:06AM	pcc_eng_25-076.conllu	total docs: 9999
716	2022-08-11	--	07:08AM	pcc_eng_25-077.conllu	total docs: 9999
717	2022-08-11	--	07:09AM	pcc_eng_25-078.conllu	total docs: 9999
718	2022-08-11	--	07:10AM	pcc_eng_25-079.conllu	total docs: 9999
719	2022-08-11	--	07:11AM	pcc_eng_25-080.conllu	total docs: 9997
720	2022-08-11	--	07:11AM	pcc_eng_25-081.conllu	total docs: 9999
721	2022-08-11	--	07:12AM	pcc_eng_25-082.conllu	total docs: 9999
722	2022-08-11	--	07:12AM	pcc_eng_25-083.conllu	total docs: 9997
723	2022-08-11	--	07:12AM	pcc_eng_25-084.conllu	total docs: 9999
724	2022-08-11	--	07:13AM	pcc_eng_25-085.conllu	total docs: 9999
725	2022-08-11	--	07:14AM	pcc_eng_25-086.conllu	total docs: 9999
726	2022-08-11	--	07:15AM	pcc_eng_25-087.conllu	total docs: 9998
727	2022-08-11	--	07:15AM	pcc_eng_25-088.conllu	total docs: 9999
728	2022-08-11	--	07:16AM	pcc_eng_25-089.conllu	total docs: 9999
729	2022-08-11	--	07:16AM	pcc_eng_25-090.conllu	total docs: 9998
730	2022-08-11	--	07:16AM	pcc_eng_25-091.conllu	total docs: 9998
731	2022-08-11	--	07:17AM	pcc_eng_25-092.conllu	total docs: 9999
732	2022-08-11	--	07:17AM	pcc_eng_25-093.conllu	total docs: 9999
733	2022-08-11	--	07:17AM	pcc_eng_25-094.conllu	total docs: 9999
734	2022-08-11	--	07:17AM	pcc_eng_25-095.conllu	total docs: 9999
735	2022-08-11	--	07:18AM	pcc_eng_25-096.conllu	total docs: 9999
736	2022-08-11	--	07:18AM	pcc_eng_25-097.conllu	total docs: 9999


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737 2022-08-11 -- 07:18AM pcc_eng_25-098.conllu total docs: 9999
738 2022-08-11 -- 07:18AM pcc_eng_25-099.conllu total docs: 9999
739 2022-08-11 -- 07:19AM pcc_eng_25-100.conllu total docs: 9999
740 2022-08-11 -- 07:19AM pcc_eng_25-101.conllu total docs: 9999
741 2022-08-11 -- 07:19AM pcc_eng_25-102.conllu total docs: 9999
742 2022-08-11 -- 07:19AM pcc_eng_25-103.conllu total docs: 9999
743 2022-08-11 -- 07:19AM pcc_eng_25-104.conllu total docs: 9999
744 2022-08-11 -- 07:20AM pcc_eng_25-105.conllu total docs: 9999
745 2022-08-11 -- 07:20AM pcc_eng_25-106.conllu total docs: 9999
746 2022-08-11 -- 07:20AM pcc_eng_25-107.conllu total docs: 9873
747 2022-08-11 -- 07:20AM pcc_eng_25-108.conllu total docs: 9872
748 -----
749 | INFO: 2022-08-11 07:20AM
750 |
751 ↳ /share/compling/projects/puddin/script/validate_data_group.py:_load_exclusions:318
752 | . . . . .
753 Loading exclusions dataframe:
754 > /share/compling/data/puddin/pile_exclusions/pile_25_Pile-CC_excl.pkl.gz
755 -----
756 | INFO: 2022-08-11 07:21AM
757 |
758 ↳ /share/compling/projects/puddin/script/validate_data_group.py:_load_exclusions:345
759 | . . . . .
760 Exclusions loaded:
761
762 slice_id excl_type ... row_ix slice_code
763 raw_id
764 pcc_25_0173963 <NA> wiki ... 0 <NA>
765 pcc_25_0273195 <NA> wiki ... 1 <NA>
766 pcc_25_0277955 <NA> wiki ... 2 <NA>
767 pcc_25_0280309 <NA> wiki ... 3 <NA>
768 pcc_25_0329656 <NA> wiki ... 4 <NA>
769 ... ...
770 pcc_25_1335354 pcc_eng_25_083.5065_x1335354 fail ... 667767 083
771 pcc_25_1407428 pcc_eng_25_087.9656_x1407428 fail ... 667768 087
772 pcc_25_1452815 pcc_eng_25_090.7842_x1452815 fail ... 667769 090
773 pcc_25_1456387 pcc_eng_25_091.0056_x1456387 fail ... 667770 091
774 pcc_25_1736884 pcc_eng_25_108.3428_x1736884 fail ... 667771 108
775
776 [667772 rows x 11 columns]
777 -----
778 | INFO: 2022-08-11 07:22AM
779 |
780 ↳ /share/compling/projects/puddin/script/validate_data_group.py:assess_data_group:132
781 | . . . . .

```

```

778 No original texts missing from 25 output ^_^
779 -----
780 | INFO: 2022-08-11 07:22AM
781 |
782   ↪ /share/compling/projects/puddin/script/validate_data_group.py:_identify_invalid_excl:491
783 | . . . . .
784 All 25 exclusions in are valid ^_^
785 -----
786 | INFO: 2022-08-11 07:22AM
787 |
788   ↪ /share/compling/projects/puddin/script/validate_data_group.py:_save_validated_excl:498
789 | . . . . .
790 Saving updated exclusions dataframe to
791   ↪ pile_exclusions/validated/pile_25_Pile-CC_excl.pkl.gz
792 -----
793 | INFO: 2022-08-11 07:22AM
794 |
795   ↪ /share/compling/projects/puddin/script/validate_data_group.py:assess_data_group:142
796 | . . . . .
797 ## `25` data assessment complete
798 2022-08-11 -- 07:22AM
799 status overview:
800 .....
801 excl_type  known_fail  success
802 a0wrđ      False      False      264930
803 punc       False      False      249628
804 code       False      False      97148
805 _wrđ       False      False      50328
806 html       False      False      5227
807 wiki       False      False      421
808 json       False      False      71
809 fail       True       False      19
810 .....
811 slice      known_fail  success
812 pcc_eng_25_001  False      True      9999
813 pcc_eng_25_050  False      True      9999
814 pcc_eng_25_077  False      True      9999
815 pcc_eng_25_076  False      True      9999
816 pcc_eng_25_075  False      True      9999
817 pcc_eng_25_074  False      True      9999
818 pcc_eng_25_073  False      True      9999
819 pcc_eng_25_072  False      True      9999
820 pcc_eng_25_071  False      True      9999
821 pcc_eng_25_069  False      True      9999

```

818	pcc_eng_25_067	False	True	9999
819	pcc_eng_25_066	False	True	9999
820	pcc_eng_25_065	False	True	9999
821	pcc_eng_25_064	False	True	9999
822	pcc_eng_25_063	False	True	9999
823	pcc_eng_25_061	False	True	9999
824	pcc_eng_25_060	False	True	9999
825	pcc_eng_25_059	False	True	9999
826	pcc_eng_25_056	False	True	9999
827	pcc_eng_25_002	False	True	9999
828	pcc_eng_25_054	False	True	9999
829	pcc_eng_25_053	False	True	9999
830	pcc_eng_25_052	False	True	9999
831	pcc_eng_25_078	False	True	9999
832	pcc_eng_25_079	False	True	9999
833	pcc_eng_25_081	False	True	9999
834	pcc_eng_25_097	False	True	9999
835	pcc_eng_25_106	False	True	9999
836	pcc_eng_25_105	False	True	9999
837	pcc_eng_25_104	False	True	9999
838	pcc_eng_25_103	False	True	9999
839	pcc_eng_25_102	False	True	9999
840	pcc_eng_25_101	False	True	9999
841	pcc_eng_25_100	False	True	9999
842	pcc_eng_25_099	False	True	9999
843	pcc_eng_25_098	False	True	9999
844	pcc_eng_25_096	False	True	9999
845	pcc_eng_25_082	False	True	9999
846	pcc_eng_25_095	False	True	9999
847	pcc_eng_25_094	False	True	9999
848	pcc_eng_25_093	False	True	9999
849	pcc_eng_25_092	False	True	9999
850	pcc_eng_25_089	False	True	9999
851	pcc_eng_25_088	False	True	9999
852	pcc_eng_25_086	False	True	9999
853	pcc_eng_25_085	False	True	9999
854	pcc_eng_25_084	False	True	9999
855	pcc_eng_25_051	False	True	9999
856	pcc_eng_25_055	False	True	9999
857	pcc_eng_25_049	False	True	9999
858	pcc_eng_25_029	False	True	9999
859	pcc_eng_25_027	False	True	9999
860	pcc_eng_25_048	False	True	9999
861	pcc_eng_25_024	False	True	9999

862	pcc_eng_25_023	False	True	9999
863	pcc_eng_25_022	False	True	9999
864	pcc_eng_25_020	False	True	9999
865	pcc_eng_25_019	False	True	9999
866	pcc_eng_25_018	False	True	9999
867	pcc_eng_25_017	False	True	9999
868	pcc_eng_25_016	False	True	9999
869	pcc_eng_25_014	False	True	9999
870	pcc_eng_25_013	False	True	9999
871	pcc_eng_25_012	False	True	9999
872	pcc_eng_25_011	False	True	9999
873	pcc_eng_25_010	False	True	9999
874	pcc_eng_25_009	False	True	9999
875	pcc_eng_25_007	False	True	9999
876	pcc_eng_25_006	False	True	9999
877	pcc_eng_25_005	False	True	9999
878	pcc_eng_25_004	False	True	9999
879	pcc_eng_25_003	False	True	9999
880	pcc_eng_25_028	False	True	9999
881	pcc_eng_25_025	False	True	9999
882	pcc_eng_25_030	False	True	9999
883	pcc_eng_25_036	False	True	9999
884	pcc_eng_25_046	False	True	9999
885	pcc_eng_25_045	False	True	9999
886	pcc_eng_25_044	False	True	9999
887	pcc_eng_25_043	False	True	9999
888	pcc_eng_25_042	False	True	9999
889	pcc_eng_25_041	False	True	9999
890	pcc_eng_25_040	False	True	9999
891	pcc_eng_25_039	False	True	9999
892	pcc_eng_25_031	False	True	9999
893	pcc_eng_25_038	False	True	9999
894	pcc_eng_25_034	False	True	9999
895	pcc_eng_25_032	False	True	9999
896	pcc_eng_25_033	False	True	9999
897	pcc_eng_25_035	False	True	9999
898	pcc_eng_25_087	False	True	9998
899	pcc_eng_25_047	False	True	9998
900	pcc_eng_25_026	False	True	9998
901	pcc_eng_25_008	False	True	9998
902	pcc_eng_25_057	False	True	9998
903	pcc_eng_25_058	False	True	9998
904	pcc_eng_25_070	False	True	9998
905	pcc_eng_25_062	False	True	9998

```

906 pcc_eng_25_015 False      True      9998
907 pcc_eng_25_068 False      True      9998
908 pcc_eng_25_021 False      True      9998
909 pcc_eng_25_091 False      True      9998
910 pcc_eng_25_090 False      True      9998
911 pcc_eng_25_037 False      True      9998
912 pcc_eng_25_083 False      True      9997
913 pcc_eng_25_080 False      True      9997
914 pcc_eng_25_107 False      True      9873
915 pcc_eng_25_108 False      True      9872
916 -----
917 | INFO: 2022-08-11 07:22AM
918 |
919 | ↪ /share/compling/projects/puddin/script/validate_data_group.py:assess_data_group:179
920 | . . . . .
921 Data group `25` parsing status info and invalid exclusions saved as:
922 - /share/compling/data/puddin/info/validation_by_group/25-pcc_status.pkl.gz
923   last modified: Thu Aug 11 07:22:45 2022
924 - /share/compling/data/puddin/info/validation_by_group/25_invalid-excl.pkl.gz
925   last modified: Thu Aug 11 07:22:45 2022

```

```

#!/bin/bash
# start from data/pile/train/ dir:
wget https://mystic.the-eye.eu/public/AI/pile/train/00.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/01.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/02.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/03.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/04.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/05.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/06.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/07.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/08.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/09.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/10.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/11.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/12.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/13.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/14.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/15.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/16.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/17.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/18.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/19.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/20.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/21.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/22.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/23.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/24.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/25.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/26.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/27.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/28.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/train/29.jsonl.zst
unzstd 00.jsonl.zst
unzstd 01.jsonl.zst
unzstd 02.jsonl.zst
unzstd 03.jsonl.zst
unzstd 04.jsonl.zst
unzstd 05.jsonl.zst
unzstd 06.jsonl.zst
unzstd 07.jsonl.zst
unzstd 08.jsonl.zst
unzstd 09.jsonl.zst
unzstd 10.jsonl.zst
unzstd 11.jsonl.zst
unzstd 12.jsonl.zst
unzstd 13.jsonl.zst
unzstd 14.jsonl.zst
unzstd 15.jsonl.zst
unzstd 16.jsonl.zst
unzstd 17.jsonl.zst
unzstd 18.jsonl.zst
unzstd 19.jsonl.zst
unzstd 20.jsonl.zst
unzstd 21.jsonl.zst
unzstd 22.jsonl.zst
unzstd 23.jsonl.zst
unzstd 24.jsonl.zst
unzstd 25.jsonl.zst
unzstd 26.jsonl.zst
unzstd 27.jsonl.zst
unzstd 28.jsonl.zst
unzstd 29.jsonl.zst
cd ..
wget https://mystic.the-eye.eu/public/AI/pile/test.jsonl.zst
wget https://mystic.the-eye.eu/public/AI/pile/val.jsonl.zst
unzstd test.jsonl.zst
unzstd val.jsonl.zst
echo "finished at ${date}"

```

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